Ans. (d)

- Shear stress distribution for rectangular cross-section = $\frac{V}{2I}\left(\frac{d^2}{4}-Y^2\right)$

  Maximum will be at y = 0 (at N-A).

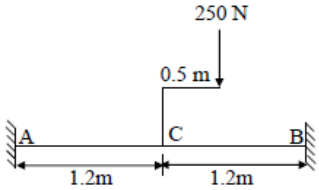
$$\left[\tau_{max}=\frac{V}{\frac{2bd^3}{12}}\times\frac{d^2}{4}=\frac{3}{2}\left(\frac{V}{bd}\right)\right]$$

- Average shear stress for a beam of cross section area (b×d) = $\frac{V}{bd}$

So, $\frac{\tau_{max}}{\tau_{avg}}=\frac{\frac{3}{2}\left(\frac{v}{bd}\right)}{\frac{v}{(bd)}}=\frac{3}{2}$

$$\boxed{\tau_{max}=1.5\,\tau_{avg}}$$

678.

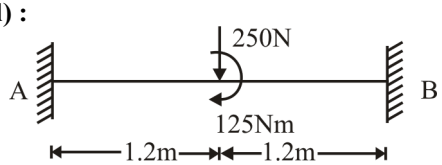


A horizontal bar of 40 mm diameter solid section is 2.40 m long and is rigidly held at both ends so that no angular rotation occurs axially or circumferentially at the ends (as shown in figure). The maximum tensile stress in the bar is nearly

(a) 12.2 N/mm^2 (b) 13.7 N/mm^2
(c) 15.2 N/mm^2 (d) 16.7 N/mm^2

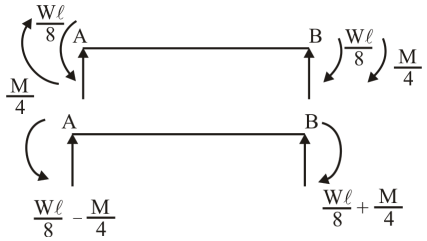
ESE 2018

Ans. (d) :



Fixed end moment at A & B due to vertical point load = $\frac{W\ell}{8}$

Fixed end moment at A & B due to applied moment = $\frac{M}{4}$



Maximum Bending moment will occur at B,

$$M_B=\frac{W\ell}{8}+\frac{M}{4}\Rightarrow\frac{250\times2.4\times10^3}{8}+\frac{125\times10^3}{4}$$

$M_B = 105 \times 10^3$ N–mm

$$\text{Bending stress}=\frac{M}{Z}=\frac{105\times10^3}{\frac{\pi\times40^3}{32}}\Rightarrow\frac{105}{2\pi}$$

$\sigma_b = 16.72$ N/mm^2

679. A steel wire of 20 mm diameter is bent into a circular shape of 10m radius. If E is the modulus of elasticity, is 2×10^6 kg/cm^2, then the maximum tensile stress induced in the wire is, nearly

(a) 2×10^3 kg/cm^2 (b) 4×10^3 kg/cm^2
(c) 2×10^4 kg/cm^2 (d) 4×10^4 kg/cm^2

ESE 2018

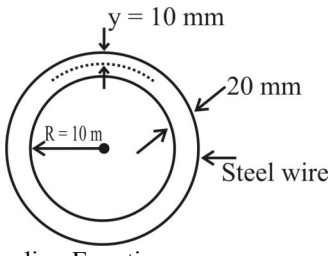
Ans. (a)

Given,

$$y=\frac{D}{2}=\frac{10}{10}=1$$

$R = 1000 = 10^3$

$E = 2 \times 10^6$ kg/cm^2



We know Bending Equation–

$$\left[\frac{M}{I}=\frac{\sigma}{Y}=\frac{E}{R}\right]$$

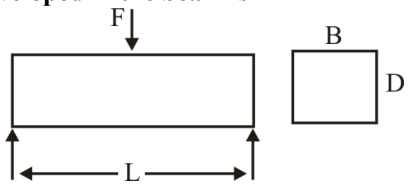
$$\therefore\frac{\sigma}{Y}=\frac{E}{R}$$

Maximum tensile stress induced in the wire is :

$$\sigma=\frac{E}{R}\times Y$$

$$\sigma=\frac{2\times10^6\times1}{10^3}=2\times10^3\,kg/cm^2$$

680. A homogeneous prismatic simply supported beam is subjected to a point load F. The load can be placed anywhere along the span of the beam. The very maximum flexural stress developed in the beam is



(a) $\frac{3FL}{2BD^2}$ (b) $\frac{3FL}{4BD^2}$

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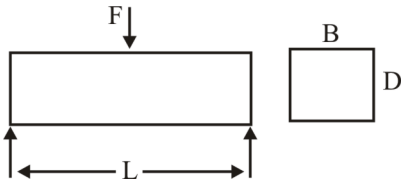
(c) $\frac{2FL}{3BD^2}$ (d) $\frac{4FL}{3BD^2}$

ESE 2017

Ans. (a)



Maximum bending moment occurs at mid span of beam.

Hence, $M_{max} = \frac{FL}{4}$

$\Rightarrow$ Section modulus 'Z' = $\frac{BD^2}{6}$

Bending Equation

$\left[\frac{M}{I} = \frac{\sigma}{Y}\right]$

$\left[\sigma = \frac{M}{Z}\right]$ where, $\left[Z = \frac{I}{Y}\right]$

$$\sigma = \frac{\frac{FL}{4}}{\frac{BD^2}{6}}$$

$$\boxed{\sigma = \frac{3FL}{2BD^2}}$$

681. The span of a cantilever beam is 2 m. The cross-section of the beam is a hollow square with external sides 100 mm; and its I = 4×10^5 mm^4. The safe bending stress for the beam material is 100 N/mm^2. The safe concentrated load at the free end would be

(a) 100 N (b) 200 N

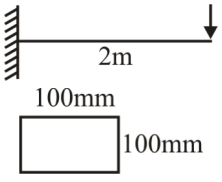
(c) 300 N (d) 400 N

ESE 2017

Ans. (d)

Span length 'L' = 2m

Hollow square 'a' = 100mm



$I = \frac{bd^3}{12} = 4\times10^5\,mm^4$

σ = 100MPa

- From bending equation

$\left[\frac{\sigma}{Y} = \frac{M}{I}\right]$

$\sigma = \frac{M}{I}(Y_{max})$

$100 = \frac{M}{(4\times10^5)}(50)$

$M = 4\times10^5 \times \frac{100}{50}$

$\boxed{M = 8\times10^5\,N-mm}$

For cantilever beam maximum bending moment

$[M = W\ell]$

$M = W\,(2000mm)$

$8\times10^5 = W\,(2000)$

$\boxed{W = 400N}$

682. A timber beam of rectangular section of length 8 m is simply supported. The beam carries a uniformly distributed load of 12 kN/m, over the entire length and a point load of 10 kN at 3 m from the left support. if the depth is two times the width and stress in the timber is not to exceed to 10 N/mm^2, what is the suitable depth of the section?

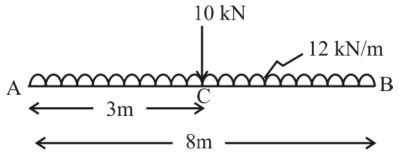
(a) 412 mm (b) 512 mm

(c) 612 mm (d) 712 mm

ISRO Scientist/Engineer 2020

Ans. (b) : l = 8 m

w = 12 kN/m

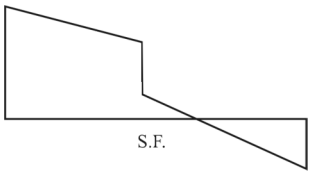


$R_A + R_B = 96 + 10 = 106$

$R_B \times 8 = 10\times3 + 12\times8\times\frac{8}{2}$

$R_B = 51.75kN$

$R_A = 54.25kN$



Maximum bending moment B and C become S.F. is zero between B and C

$BM_x = 51.75\,x - 12x \times\frac{x}{2}$

$\frac{\partial BM_x}{\partial x} = 0$

$51.75 - 12x = 0$

$x = 4.315$

Maximum bending moment =

$= 51.75\times4.3125 - \frac{12\times4.3175^2}{2}$

= 111.5859 kN-m

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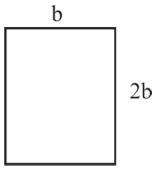


$$\frac{M}{I}=\frac{\sigma}{y}$$

$$\frac{111.5859\times10^3}{b(2b)^3}\times12=\frac{10\times10^6}{b}$$

b = 0.2557 meter

depth = 2b = 2 × 255.7 mm = 511.59 mm
= 512 mm

683. The proportion of second moment of area about the centroidal axis to second moment of area about the base of a rectangle will be

(a) 0.25 (b) 0.50
(c) 2.0 (d) 3.0

ISRO Scientist/Engineer 2018

Ans : (a) : Consider a rectangle of width b and depth d. Also consider a small strip of thickness dy at distance y from centroidal axis.

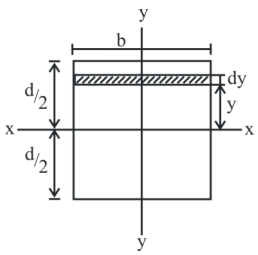
We know that second moment of area of strip about xx axis.

= da × y^2

Where da = area of small strip

For moment of Inertia of whole are, integrate between limits $-\frac{d}{2}$ and $\frac{d}{2}$

$$\therefore I_{xx}=\int_{-d/2}^{d/2} da.\,y^2$$



$$I_{xx}=\int_{-d/2}^{d/2}(b.dy)y^2 \quad \because da=b.dy$$

$$I_{xx}=b\int_{-d/2}^{d/2} y^2.\,dy \quad .....(i)$$

$$=b\left[\frac{y^3}{3}\right]_{-d/2}^{d/2}$$

$$I_{xx}=b\left[\frac{d^3}{24}-\left(\frac{-d^3}{24}\right)\right]$$

$$I_{xx}=\frac{bd^3}{12}$$ second moment about centroidal axix.

• **Second moment of area about base :**

$$I_{base}=\frac{bd^3}{3}$$

$$\text{Ratio}=\frac{\frac{bd^3}{12}}{\frac{bd^3}{3}}$$

$$\boxed{\text{Ratio}=\frac{1}{4}=0.25}$$

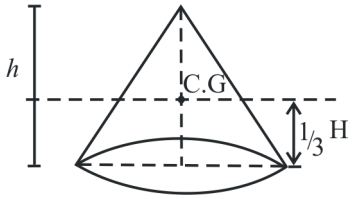
684. Centre of gravity of a thin hollow cone lies on the axis of symmetry at a height of

(a) One half of the total height above base
(b) One third of the total height above base
(c) One fourth of the total height above base
(d) At the centre of the base

ISRO Scientist/Engineer 2018

Ans : (b)

• Centre of the gravity of the thin hollow cone lies on the axis of the symmetry at the height of $\frac{'H'}{3}$ from base on its axis.



• The centre of the gravity of hollow cone lies on the $\frac{'2H'}{3}$ height from the top of the cone.

• Centre of the gravity of the solid hollow cone lies on the axis of the symmetry at the height of $\frac{'H'}{4}$ from the base of the solid cone.

685. If the depth is kept constant for a beam of uniform strength, then its width will vary in proportion to -------, where M is the bending moment,

(a) M (b) √M
(c) M^2 (d) M^3

ISRO Scientist/Engineer 2013

Ans : (a) Beam of uniform strength has the same maximum bending stress all along the length.

$$\left[\frac{M}{I}=\frac{\sigma}{Y}\right] \quad \therefore \quad Y=\frac{d}{2},\quad I=\frac{bd^3}{I_2}$$

$$\frac{M}{\frac{bd^3}{I_2}}=\frac{\sigma}{\frac{d}{2}}$$

$$\frac{\sigma bd^2}{6}=M$$

$$\left(\frac{\sigma d^2}{2}\right)b=M$$: (When depth is constant)

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- $\boxed{b \alpha M}$ width α M (Bending Moment)

  $\sqrt{\left(\frac{\sigma b}{2}\right)}\, d = \sqrt{M}$ (when width is constant)

- $\boxed{d \alpha \sqrt{M}}$

686. A simply supported beam carries a central concentrated load W. Area of cross section is 300 mm^2. The shear stress at the neutral axis layer is 5 N/mm^2. What is the magnitude of W in kN?

(a) 1000 (b) 2000
(c) 100 (d) 3000

APTRASCO 2017

Ans : (b)

Maximum shear force in beam, $V = \frac{W}{2}$ kN

Maximum shear stress lie in a beam at NA

$\tau_{max} = 1.5\tau_{avg}$

or $5 = 1.5 \times \frac{V}{A}$

or $5 = 1.5 \times \frac{W/2}{300}$

or $\boxed{W = 2000\,kN}$

687. A 60 mm diameter shaft is subjected to a torque of 6 kN-m, C = 8 × 10^4 N/mm^2. The maximum shear stress induced in the shaft will be :

(a) $\frac{8000}{9\pi}$ (b) $\frac{4000}{9\pi}$
(c) $\frac{12000}{9\pi}$ (d) $\frac{16000}{9\pi}$

APTRANSCO 2017

Ans : (b) : Maximum shear stress $(\tau_{max}) = \frac{16T}{\pi D^3}$

$T = 6kN - m = 6 \times 10^6\, N - mm$

$(\tau_{max}) = \frac{16 \times 6 \times 10^6}{\pi \times (60)^3} = \frac{4000}{9\pi}$

688. If the breadth of a beam is doubled and the depth remaining same, then moment of inertia increased by:

(a) 2 times (b) 3 times
(c) 4 times (d) 6 times

WBPSC Poly Lect. 2019

Ans. (a) :

$I_1 = \frac{bd^3}{12}$ ..... I_1 = initial moment of inertia of beam.

$I_2 = \frac{2bd^3}{12}$ ..... I_2 = final moment of inertia

$I_2 = 2(I_1)$

689. Shear stress on principal planes is :

(a) Zero
(b) Maximum
(c) Minimum
(d) Depends on axial forces

OPSC AEE 2019 Paper–I

Ans. (a) : Principal plane may be defined as, "the plane on which normal stress attains its maximum and minimum value." So these planes are also called as major principal plane and minor principal plane.
The shear stress on principal plane is zero.

690. The maximum shear stress caused due to a shear force in a beam of rectangular cross-section is how much more is percentage than its average value?

(a) 200% (b) 150%
(c) 100% (d) 50%

OPSC AEE 2019 Paper–I

Ans. (d) : for rectangular cross-section,

$\tau_{max} = \frac{3}{2}\tau_{avg}$

$\frac{\tau_{max}}{\tau_{avg}} = \frac{3}{2} = 1.5$

$\tau_{max} = \uparrow 50\%\ \tau_{avg}$

691. The Lueders' Lines in a material indicate that :

(a) The material is failing in flexure
(b) The material is failing due to fatigue
(c) The material is failing due to its crushing
(d) The material is failing in shear

OPSC AEE 2019 Paper–I

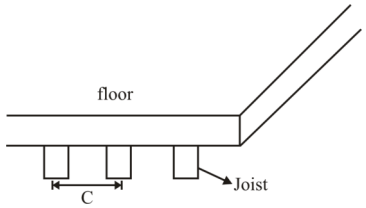
Ans. (d) : Lueder's lines are elongated surface markings or depressions, often visible to the unaided eye, that form along the length of a tension specimen at an angle of approximately 45^0 to the loading axis (i.e. the plane of maximum shear) it is caused by localized plastic deformation and result from discontinuous (in homogeneous) yielding it is also known as luedert band or stretcher strain.

692. A floor has to carry a load of 12 kN/m^2. The floor is supported on rectangular joists each 100 mm wide, 300 mm deep and 5 m long. If maximum stress in the joists should not exceed 8 MN/m^2, the centre to centre distance of joists will be :

(a) 430 mm (b) 400 mm
(c) 360 mm (d) 320 mm

ESE 2020

Ans. (d)



C → centre to centre distance between two joist

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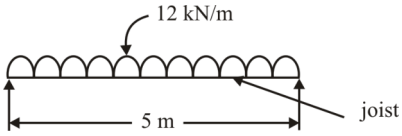


Hence, load on each joist will be UDL and intensity of udl (w) = 12 C kN/m

$B.M_{max}$ in joist = $\frac{W\ell^2}{8}$

$$B.M_{max} = \frac{12C\times(5\times10^3)^2}{8}$$

$$\boxed{B.M._{max} = 37.5C\ kN-m}$$

From Bending Equation,

$$\frac{B.M.}{I} = \frac{\sigma_{max}}{y_{max}}$$

$$\Rightarrow \sigma_{max} = \frac{B.M_{max}}{Z} = \frac{37.5C}{\frac{bd^2}{6}}$$

$$\Rightarrow \sigma_{max} = \frac{37.5C}{\frac{0.1\times(0.3)^2}{6}}$$

$$\Rightarrow 8\times10^3 = \frac{37.5C}{\frac{0.1\times(0.3)^2}{6}}$$

$$\Rightarrow \boxed{C = 0.32m} \text{ or } \boxed{C = 320mm}$$

693. Consider the following statements :
The theory of simple bending assumes that :

1. The material of the beam is homogeneous, isotropic and obeys Hooke's law.
2. The plane section remains plane after bending.
3. Each cross-section of the beam is symmetric about the loading plane.
4. Young's moduli are the same in tension and compression.

Of the above statements which are correct?

(a) 1 and 2 only (b) 1, 3 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

OPSC AEE 2019 Paper–I

Ans. (d) : In case of the theory of simple bending there are the following assumptions

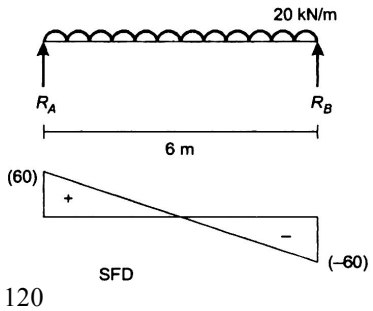
(1) Only pure bending can occur-there's no shear force, torsion nor axial load.
(ii) The material of the beam is homogeneous, isotropic and obeys Hooke's law.
(iii) The plane section remains plane after bending.
(iv) Each cross-section of the beam is symmetric about the loading plane.

694. A uniform beam of length 6 m carries ultimate load of 20 kN/m inclusive of self-weight, the design shear force is-

(a) 120 kN (b) 90 kN
(c) 60 kN (d) 30 kN

MPPSC AE 2017

Ans. (c) :



$R_A + R_B = 120$
$\Sigma M_A = 0$
$6R_B - 120 \times 3 = 0$
$[R_B = 60\ kN]$
$[R_A = 60\ kN]$
Maximum SF = 60 kN

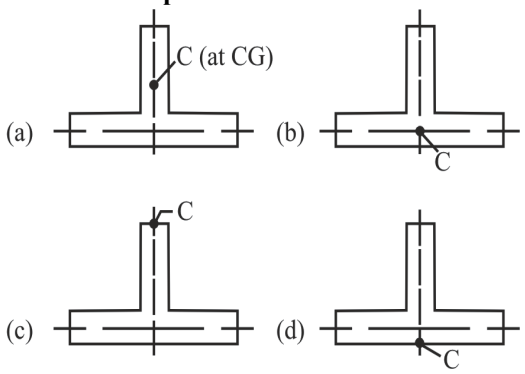
695. Rate of change of angular momentum is equal to-

(a) force (b) torque
(c) linear momentum (d) impulse

JPSC AE 2009

Ans. (b) : Torque can be defined as the rate of change of angular momentum, analogous to force. The net external force of any system is always equal to total torque on system.

696. In a thin–wall T–section, the shear centre C is located at the point shown in



RRB SSE (Bilaspur) Yellow Paper : 21.12.2014

Ans : (b)

697. The factor of safety in bending for rectangular beams in terms of yield point stress (σ_{yp}) and working stress (σ_W) is

(a) $1.5\frac{\sigma_{yp}}{\sigma_W}$ (b) $1.3\frac{\sigma_{yp}}{\sigma_W}$
(c) $1.33\frac{\sigma_{yp}}{\sigma_W}$ (d) $\frac{\sigma_{yp}}{\sigma_W}$

APPSC AEE MAINS 2016

Ans. (d) : the factor of safety in bending for rectangular beams in terms of yield point stress (σ_{yp}) and working stress (σ_W) is

$$= \frac{\sigma_{yp}}{\sigma_W}$$

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698. Flexural rigidity in usual notations is

(a) $\frac{M}{EI}$ (b) $\frac{EI}{R}$

(c) EI (d) MR

APPSC AEE MAINS 2016

Ans. (c) : EI

699. If the average shear stress in a rectangular section beam is 5 N/sq.mm, then maximum shear stress for the circular section of the equal area is

(a) 7.50 N/sq.mm (b) 6.65 N/sq.mm

(c) 10 N/sq.mm (d) 2.50 N/sq.mm

APPSC AEE MAINS 2016

Ans. (b) : For circular section

$$\tau_{max} = \frac{4}{3}\tau_{avg}$$

$$= \frac{4}{3} \times 5 = 6.65 N/sq.mm$$

700. Two prismatic beams A and B have same length. The one having larger of the following will be stronger in flexure :

(a) section modulus (b) area of cross section

(c) moment of inertia (d) modulus of rigidity

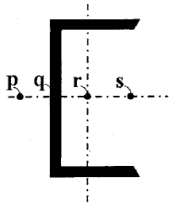
PSPCL AE 2012

Ans. (a) The strength of beam will be more whose section modulus values is large enough

$$\frac{\sigma}{y} = \frac{M}{I}, \quad \sigma = \frac{M}{z} \Rightarrow M = \sigma z$$

$$M \propto Z$$

701. The location of shear centre of the channel section shown below is

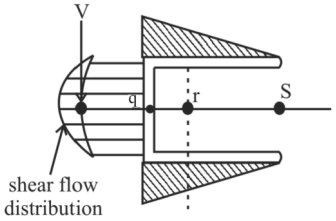


(a) p (b) q

(c) r (d) s

GPSC AE March 2018

Ans. (a) :



Shear centre - The point where a shear force can act without producing any twist in section. In general not the centroid but a point through which a force transverse to the axis of a beam section can act and not cause any twisting of beam section. In given figure at point P when force applied on twist occurs.

702. Two beams, one having a square cross-section and another having a circular cross-section, are subjected to the same amount of bending moment. If the cross-sectional area as well as the material of both the beams are same then

(a) maximum bending stress developed in both the beams is the same

(b) the circular beam experiences more bending stress than the square one

(c) the square beam experiences more bending stress than the circular one

(d) both the beams will experience the same deformation

GPSC AE March 2018

Ans. (b) : Since cross sectional area of square and circular cross sections are equal

Let d = diameter of the circular cross-sections are equal

a^2 = Side of Square

$$\therefore \frac{\pi}{4}d^2 = a^2$$

Now, σ for circular cross section

$$\sigma = \frac{My}{I} = \frac{M \times \frac{d}{2}}{\pi\frac{d^4}{64}} = \frac{32M}{\pi d^3}$$

σ for square cross - section

$$\sigma = \frac{My}{I} = \frac{M \times \frac{a}{2}}{\frac{a^4}{12}} = \frac{6M}{a^3}$$

$$\frac{\sigma \text{ for circular cross section}}{\sigma \text{ for square cross section}} = \frac{\frac{32M}{\pi d^3}}{\frac{6M}{a^3}}$$

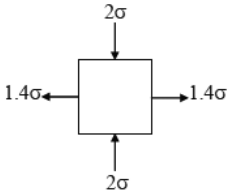
$$= \frac{32M}{\pi d^3} \times \frac{a^3}{6M} = \frac{32}{\pi}\left(\frac{a^3}{d^3}\right)$$

$$= \frac{32}{\pi}\left(\sqrt{\frac{\pi}{4}}\right)^3 \times \frac{d^3}{d^3} = 9.4$$

Since, $\left[a^2 = \frac{\pi}{4}d^2\right]$

$\therefore \sigma$ for circular cross section > σ for square cross section

703. The state of stress at a point in an elastic material, with yield stress of 200 MPa in simple tension, and Poisson's ratio 0.3, is as shown in the figure.



The permissible value of σ by maximum strain theory is

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(a) 75 MPa (b) 100 MPa
(c) 150 MPa (d) 200 MPa

ESE 2017

Ans. (b)

Yield Stress $f_y = 200MPa$

$\sigma_1 = 1.4\sigma$

$\sigma_2 = -2\sigma$ (because of compressive nature)

$\mu = 0.3$

Maximum Strain energy theory, the failure or yielding occurs at a point in a member when the strain energy per unit volume in a bi-axial stress system reaches the limiting strain energy per unit volume as determined from a simple tension test.

$$\frac{\sigma_1}{E} - \mu\frac{\sigma_2}{E} = \frac{f_y}{E}$$

$$\frac{1.4\sigma}{E} + 0.3\times\frac{2\sigma}{E} = \frac{200}{E}$$

$1.4\sigma + 0.6\sigma = 200$

$$\boxed{\sigma = 100MPa}$$

704. The principal stresses at a point in a stressed material are $\sigma_1 = 200$ N/mm^2, $\sigma_2 = 150$ N/mm^2, and $\sigma_3 = 200$ N/mm^2. E = 210 kN/mm^2 and μ = 0.3. The volumetric strain will be

(a) 8.954×10^{-4} (b) 8.954×10^{-2}
(c) 6.54×10^{-3} (d) 6.54×10^{-4}

ESE 2017

Ans. (*) : The principal stresses are–

$\sigma_1 = 200MPa$

$\sigma_2 = 150MPa$

$\sigma_3 = 200MPa$

$E = 210 \times 10^3$ MPa

$\mu = 0.3$

The volumetric strain is the unit change in volume i.e. the change in volume divided by the original volume.

$$\varepsilon_1 = \frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_3}{E}$$

$$\left[\varepsilon_1 = (200 - 0.3\times150 - 0.3\times200)\frac{1}{E} = \frac{95}{E}\right]$$

$$\varepsilon_2 = \frac{\sigma_2}{E} - \frac{\mu\sigma_1}{E} - \frac{\mu\sigma_3}{E}$$

$$\left[\varepsilon_2 = (150 - 0.3\times200 - 0.3\times200)\frac{1}{E} = \frac{30}{E}\right]$$

$$\varepsilon_3 = \frac{\sigma_3}{E} - \frac{\mu\sigma_1}{E} - \frac{\mu\sigma_2}{E}$$

$$\left[\varepsilon_3 = (200 - 0.3\times150 - 0.3\times200)\frac{1}{E} = \frac{95}{E}\right]$$

Volumetric strain $\varepsilon_v = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$

$$\varepsilon_V = \frac{95}{E} + \frac{30}{E} + \frac{95}{E}$$

$$\varepsilon_V = \frac{95+30+95}{210\times10^3}$$

$$\boxed{\varepsilon_V = 1.048\times10^{-3}}$$

Note– Commission has given no correct answer.

705. The values of displacements is {D} necessary to ensure the equilibrium of the joints are determined using the relation (displacement vector = {D}; stiffness matrix = [K]; and load vector ={P})-

(a) {P} + [K] {D} = 0 (b) {D} + [K] {P} = 0
(c) {P} {D} +[K] = 0 (d) [K] {P} = 0

MPPSC AE 2017

Ans. (a) : Stiffness is defined as the amount of force applied or moment applied to cause unit deformation.

For equilibrium of joints, net force = 0

Hence {P} + [K] {D} = 0

Where {P} = External loads

[K] {D} = Loads generated due to deformation

706. A rectangular beam carries a maximum bending moment of M. If its depth is doubled, its moment carrying capacity will be

(a) M (b) 2 M
(c) 3 M (d) 4 M

GPSC AE March 2018

Ans. (d) :

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$M = \frac{I\times\sigma}{y}$$

$$M = \frac{bd^3\times\sigma\times2}{12\times d}$$

$$M = \frac{bd^2\times\sigma}{6}$$

When depth is double–

$$M_2 = \frac{b(2d)^3}{12}\times\frac{\sigma\times2}{2d}$$

$$M_2 = \frac{b\times8d^3}{12}\times\frac{\sigma}{d}$$

$$M_2 = \frac{8bd^2}{12}\times\sigma$$

$$M_2 = \frac{2bd^2}{3}\times\sigma$$

$$M_2 = \frac{4bd^2}{6}\times\sigma$$

$$\boxed{M_2 = 4M}$$

707. A rectangular section 100 mm × 200 mm is subjected to moment of 20 kNm. The maximum bending stress is

(a) 30 N/mm^2 (b) 5/6 N/mm^2
(c) 10000 N/mm^2 (d) 300 N/mm^2

GPSC AE March 2018

Ans. (a) : 100 mm × 200 mm, moment (M) = 20 kNm

$$\frac{M}{I} = \frac{\sigma}{y}$$

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$$\sigma = \frac{My}{I} = \frac{(20\times1000)\left(\frac{200}{2}\right)\times1000}{\frac{100\times(200)^3}{12}}$$

$\boxed{I = \frac{bd^3}{12}}$ for rectangle.

$\sigma_{maximum} = 30\,N/mm^2$

708. A plate 100 mm wide, 10 mm thick is having a hole of diameter 10 mm symmetrical about the axis of the plate. The plate is subjected to a force of 9 kN. The maximum stress on a section passing through centre of the hole will be

(a) 10 N/mm^2 (b) > 10 N/mm^2
(c) < 9 N/mm^2 (d) 9 N/mm^2

GPSC AE March 2018

Ans. (a)

B = 100 mm, t = 10 mm,
plate hole d = 10 mm, P = 9 kN.
Area of plate = 100 × 10 = 1000 mm^2

Area of hole = $\frac{\pi}{4}\times10^2 = 78.54\,mm^2$

Net area = 1000 – 78.54
= 921.46 mm^2

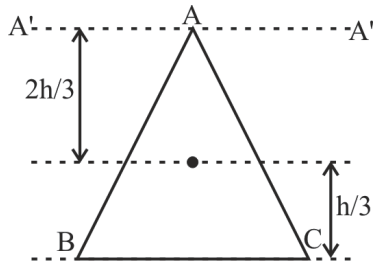
Maximum stress $= \frac{9\times10^3}{921.46}N/mm^2$
= 9.77 N/mm^2

709. Moment of inertia of a triangular section of base b and height h about an axis passing through the vertex and parallel to the base is ..... times that passing through its Centroid and parallel to its base

(a) Nine (b) Six
(c) Four (d) Two

GOA PSC Assistant Prof. 2020

Ans. (a) :



b

We know,

$I_{xx} = \frac{bh^3}{36}$

From 1 axis theorem

$I_{A'A'} = I_{xx} + A(d^2)$

$= I_{xx} + \frac{bh}{2}\left(\frac{2h}{3}\right)^2$

$= \frac{bh^3}{36} + \frac{4bh^3}{18}$

$= \frac{bh^3 + 8bh^3}{36}$

$I_{A'A'} = 9\left[\frac{bh^3}{36}\right]$

$\boxed{I_{A'A'} = 9(I_{xx})}$

710. A homogeneous simply supported prismatic beam of width B and depth D and span L is subjected to concentrated load of magnitude P. The load can be placed may where along the span of beam. The maximum flexural stress developed in beam is-

(a) $\frac{2PL}{3BD^2}$ (b) $\frac{3PL}{4BD^2}$
(c) $\frac{4PL}{3BD^2}$ (d) $\frac{3PL}{2BD^2}$

JPSC AE 2009

Ans. (d) :

Maximum flexural stress developed in beam

$= \frac{3}{2}\frac{PL}{BD^2}$

711. The maximum shear stress in a solid shaft of circular cross section having diameter d subjected to T is τ. If the torque is increased by four times and the diameter of the shaft is increased by two times, the maximum shear stress in shaft will be-

(a) 2τ (b) τ
(c) τ/2 (d) τ/4

JPSC AE 2009

Ans. (c) :

$\frac{T}{J} = \frac{\tau}{R}$

$\tau = \frac{\pi D^4}{32}$

$\boxed{\frac{T}{\frac{\pi D^3}{16}} = \tau}$ ...(1)

$\frac{4T}{\frac{\pi}{16}(2D)^3} = \tau_2$ ...(2)

Equation 2 / equation 1

$\frac{\tau_2}{\tau} = \frac{1}{2}$

$\boxed{\tau_2 = \frac{\tau}{2}}$

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712. Simple bending equation is :

(a) $\frac{M}{I}=\frac{R}{E}=\frac{\sigma}{y}$ (b) $\frac{M}{I}=\frac{E}{R}=\frac{\sigma}{y}$

(c) $\frac{I}{M}=\frac{E}{R}=\frac{y}{\sigma}$ (d) $\frac{M}{I}=\frac{R}{E}=\frac{y}{\sigma}$

OPSC Civil Services Pre 2011

Ans. (b) : Bending equation = $\frac{M}{I}=\frac{E}{R}=\frac{\sigma}{y}$

713. Which of the following is correct explanation of development of state of pure shear

(a) Tension in one direction and equal compression in perpendicular direction

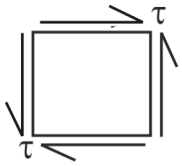
(b) Equal compression in two mutually perpendicular direction

(c) Equal tension in two mutually perpendicular direction

(d) None of the above

Sikkim PSC AE 2017

Ans. (d) : Pure shear (means no normal stress)
i.e.



714. A beam of triangular cross section is places such that one side of the triangle is horizontal and the apex of the triangle is below the horizontal side. Maximum shear intensity in the section at any particular location will be

(a) At the horizontal side

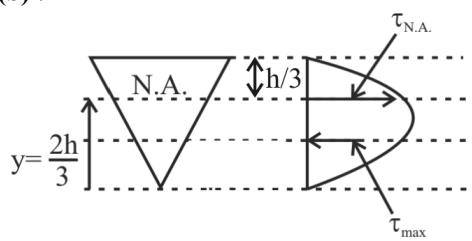
(b) Below the neutral axis

(c) Above the neutral axis

(d) At the neutral axis

Sikkim PSC AE 2017

Ans. (b) :



$\tau_{NA} = 1.33\ \tau_{avg}$

$\tau_{max} = 1.5\ \tau_{avg}$

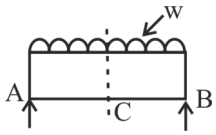
715. A beam is simply supported over a span L and it is carrying a total UDL of W over the whole span. If a prop at the center of the beam such that the top level of the prop is same as the level of end supports. The reaction of the prop will be

(a) W/2 (b) W/4

(c) 3 W/8 (d) 5 W/8

Sikkim PSC AE 2017

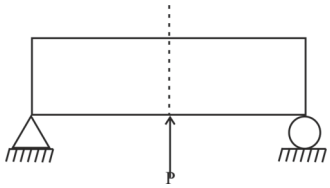
Ans. (c) :



C is center, due to UDL, deflection at center will be,

$$\Delta_c = \frac{5wl^4}{384EI} = \frac{5wl^3}{384EI}$$

If we put a prop at center, then it makes deflection zero at center.



Prop is P

Due to prop deflection at center will be $= \frac{Pl^3}{48EI}$, both deflection will be equal,

$$\therefore \quad \frac{5wl^3}{384EI} = \frac{Pl^3}{48EI}$$

$$\boxed{P = \frac{5}{8}w}$$

716. Shear stress across principal plane is :

(a) Twice the normal stress

(b) Equal to normal stress

(c) Half of normal stress

(d) Zero

TRB Poly. Lect. Tech. 2017

Ans. (d) : Shear stress across principal plane is zero,

717. The maximum stress of a shaft of 100 mm diameter and 2.7 m length subjected to a torque of 125 kNm with shear modulus of 75 GPa is

(a) $\frac{2000}{\pi}$ MPa (b) $\frac{200}{\pi}$ MPa

(c) $\frac{100}{\pi}$ MPa (d) $\frac{125}{\pi}$ MPa

TRB Poly. Tech. Lect. 2012

Ans. (a) : Maximum shear stress in the shaft due to torque (T)

$$\tau_{max} = \frac{16T}{\pi d^3} = \frac{16\times125\times10^6\,(\text{Nmm})}{\pi(100\,\text{mm})^3}$$

$$\boxed{\tau_{max} = \frac{2000}{\pi}\text{MPa}}$$

718. A steel beam is replaced by a corresponding aluminium beam of same cross sectional shape and dimension and is subjected to same loading. The maximum bending stress now will be

(a) unaltered (b) increase

(c) decrease

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(d) vary in proportion to their modulus of elasticity

TRB Poly. Tech. Lect. 2012

Ans. (a) : From the theory of pure Bending

we have $\frac{f}{y}=\frac{M}{I}$

$\Rightarrow \boxed{f=\frac{my}{I}}$

from the formula, It can be concluded that Bending stress (f) Depends on

(1) B.M. (Loading condition)

(2) MOI (dimensions)

(3) y $\rightarrow$ distance of any fiber from N.A.

$\therefore$ Max Bending stress will be unaltered.

719. A column of rectangular cross-section d units × 1 unit at top is subjected to a load P in addition to its self weight. Then the profile for uniform stress along its length is (ρ = density, σ = stress)

(a) $de\left(\frac{\rho x}{\sigma}\right)$ (b) $de\left(\frac{\sigma}{\rho x}\right)$

(c) $de\left(\frac{x}{\sigma}\right)$ (d) $de\left(\frac{\rho}{\sigma}\right)$

TRB Poly. Tech. Lect. 2012

Ans. (c) :

720. The moment of inertia of a triangular section of height h and base b about its base, is

(a) $bh^2/12$ (b) $b^2h/12$

(c) $bh^3/12$ (d) $b^3h/12$

UKPSC AE (Urban) 2011

Ans. (c) : Moment of inertia of a triangular area of an axis xx parallel to the base.

• MOI of base $\boxed{I=\frac{bh^3}{12}}$

721. The width of the strongest beam of rectangular sections that can be cut from a cylindrical log of diameter 40 cm, would be.

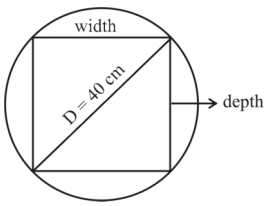
(a) 10.41 cm (b) 23.09 cm

(c) 16.19 cm (d) 27.53 cm

UJVNL AE 2016

Ans. (b) : Given that :

Dia of circular beam = 40 cm



Strongest rectangular beam

width of rectangular beam = $\frac{D}{\sqrt{3}}=\frac{40}{\sqrt{3}}=23.09$ cm

depth of rectangular beam = $\frac{D\sqrt{2}}{\sqrt{3}}=\frac{40\sqrt{2}}{\sqrt{3}}=32.65$ cm

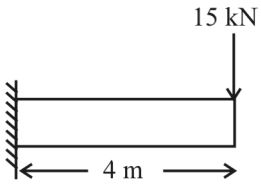
722. A cantilever beam 50 mm wide and 150 mm deep having length 4m is failed by applying a force of 15-kN at the free end. The bending stress at the failure is given by.

(a) 480 N/mm^2 (b) 160 N/mm^2

(c) 320 N/mm^2 (d) zero

UJVNL AE 2016

Ans. (c) : Given that :



B = 50 mm

D = 150 mm

Length = 4 m

Maximum bending moment = Wl

= 15 × 4 = 60 kNm

Now,

from flexure formula

$\frac{\sigma}{y}=\frac{M}{I}\Rightarrow\frac{\sigma}{D/2}=\frac{60\times10^6}{\frac{BD^3}{12}}$

$\sigma=\frac{60\times6\times10^6}{B\times D^2}$

$\sigma=\frac{60\times6\times10^6}{50\times150\times150}$

σ = 320 N/mm^2

723. A rectangular beam 10 cm wide, is subjected to a maximum shear force of 50000 N, the corresponding maximum shear stress being 3 N/mm^2. The depth of the beam is

(a) 25 cm (b) 22 cm

(c) 16.67 cm (d) 30 cm

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (a) : $\tau_{max}=\frac{3}{2}\frac{V}{bd}$ in rectangular beam

$3=\frac{3}{2}\times\frac{50000}{100\times d}$

d = 250 mm

or d = 25 cm

724. The unit of modulus of section is:

(a) kg-m^2 (b) m^3 or cm^3

(c) cm^4 or m^4 (d) cm^2 or m^2

UJVNL AE 2012

Ans. (b) : The unit of modulus of section is m^3 or cm^3

725. A beam of square cross-section with side 100 mm is placed with one diagonal, vertical. If the shear force acting on the section is 10 kN, the maximum shear stress is

(a) $1\frac{N}{mm^2}$ (b) $1.125\frac{N}{mm^2}$

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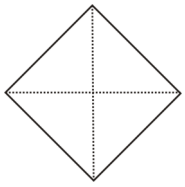
(c) $2\frac{N}{mm^2}$ (d) $2.25\frac{N}{mm^2}$

APPSC AEE 2012

Ans. (b): Given:
Shear force = 10 kN
P = 10 kN = 10000N
Area (A) = 100 × 100 = 10000 mm^2



Max. shear stress $(z_{max}) = \frac{9}{8} z_{avg.}$

$$z_{max} = \frac{9}{8} \times \frac{P}{A} = \frac{9}{8} \times \frac{10000}{10000}$$

$$= \frac{9}{8} = 1.125 \frac{N}{mm^2}$$

726. Total shear force/total area in shear =
(a) average shear stress
(b) maximum shear stress
(c) effective shear stress
(d) shear stress

BPSC AE Aug. 2019 Paper-V

Ans. (a) : $\frac{\text{Total shear force}}{\text{Total area}} = \text{Average shear stress}$

$$\tau = \frac{V}{A}$$

727. What will be the depth of the beam subjected to a maximum shearing stress of 5 MPa, while carrying 50kN load with its width being 100mm?
(a) 100 (b) 150
(c) 200 (d) 250

UPRVUNL AE 2016

Ans.: (b) Given
$\tau_{max} = 5MPa$
$P = 50kN = 50 \times 10^3 N$
$b = 100mm$
$d = ?$
$\because \tau_{max} = 1.5\ \tau_{avg}$

$$\therefore \tau_{avg} = \left(\frac{\tau_{max}}{1.5}\right) = \left(\frac{5}{1.5}\right) MPa$$

$$\tau_{avg} = \left(\frac{p}{bd}\right)$$

$$\frac{5}{1.5} = \frac{50 \times 10^3}{100 \times d}$$

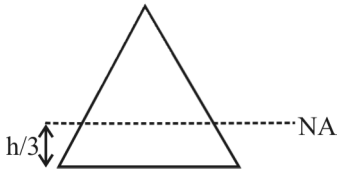
$$\therefore d = \left(\frac{50 \times 1.5}{5 \times 100}\right) \times 1000$$

$d = 150mm$

728. A beam of triangular cross-section is placed with its base horizontal. The maximum shear stress intensity in the section will be
(a) At the neutral axis
(b) At the base
(c) Above the neutral axis
(d) below the neutral axis

GPSC AE June 2019

Ans. (c) :



The maximum shear stress intensity in the section will be above the neutral axis.

τ_{max} will maximum at $\frac{h}{2}$ height.

729. A beam of uniform strength has at every cross-section same
(a) Bending moment (b) Bending stress
(c) Deflection (d) Stiffness

GPSC AE June 2019

Ans. (b) : A beam of uniform strength has at every cross-section same bending stress.

730. Radius of gyration of a circular section with diameter D is
(a) $\frac{D}{2}$ (b) $\frac{D}{4}$
(c) $\frac{D}{3}$ (d) $\frac{D}{\sqrt{3}}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : For circular section $I = \frac{\pi}{64} d^4, A = \frac{\pi}{4} d^2$

Radius of gyration $= k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64} d^4}{\frac{\pi}{4} d^2}} = \frac{d}{4}$

731. A rectangular beam section with depth 400 mm and width 300 mm is subjected to a bending moment of 60 kN/m. The maximum bending stress in the section is
(a) 7.50 MPa (b) 2.50 MPa
(c) 1.56 MPa (d) 0.42 MPa

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Maximum bending stress in the beam

$$f_{max} = \frac{M}{Z} = \frac{M}{\left[\frac{bd^2}{6}\right]} = \frac{60 \times 10^6}{\left[\frac{300 \times 400^2}{6}\right]} = 7.5 \text{ MPa}$$

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732. The relation governing the simple bending of beam is:

(a) $\frac{\sigma}{y}=\frac{M}{E}=\frac{I}{R}$ (b) $\frac{\sigma}{y}=\frac{M}{R}=\frac{E}{I}$

(c) $\frac{\sigma}{E}=\frac{M}{I}=\frac{y}{R}$ (d) $\frac{\sigma}{y}=\frac{M}{I}=\frac{E}{R}$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) :

$$\frac{\sigma}{y}=\frac{M}{I}=\frac{E}{R} \Rightarrow \text{Equation of bending}$$

σ = bending stress
m = bending moment
R = radius of curvature
I = second moment of area of C/S about neutral axis
E = Young's modulus of elasticity

733. For the shafts having square cross sections, the maximum shear stress occurs at a point

(a) on the edge of the cross section that is closest to the centre axis of the shaft
(b) on the diagonal of the cross section that is far to the centre axis of the shaft
(c) on the diagonal of the cross section, at half the edge length from the centre axis of the shaft
(d) on the diagonal of the cross section, at one-fourth of diagonal length from the centre axis of the shaft

GPSC AE Dec 2018

Ans. (a) :

$$\frac{T}{J}=\frac{\tau}{r}$$

$$\tau=\frac{Tr}{J}$$

Shear stress is maximum near the section closest to the centre axis of shaft.

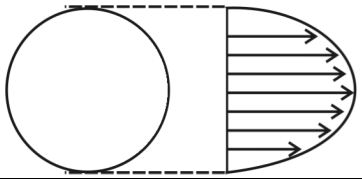
734. Ratio of maximum shear stress to average shear stress is 4/3 in a ..... section

(a) rectangular (b) triangular
(c) circular (d) hexagonal

GPSC AE Dec 2018

Ans. (c) : In circular section,

$$\frac{\tau_{max}}{\tau_{avg}}=\frac{4}{3}$$

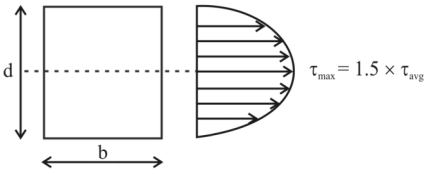


735. The nature of shear stress distribution in a rectangular beam is

(a) elliptic (b) parabolic
(c) linear (d) uniform

GPSC AE Dec 2018

Ans. (b) : The nature of shear stress distribution in a rectangular beam is parabolic.



736. A homogeneous simply supported prismatic beam of width B, depth D and span L is subjected to a concentrated load of magnitude 100 kN. The load can be placed anywhere along the span of beam. The maximum flexural stress developed in the beam is :

(a) $\frac{300L}{BD^2}$ (b) $\frac{150L}{BD^2}$

(c) $\frac{25L}{2BD^2}$ (d) $\frac{75L}{BD^2}$

BHEL ET 2019

Ans. (b) : Maximum B.M. due to point load

$$=\frac{PL}{4}$$

$$=\frac{100\times L}{4}=25\ L$$

From Flexural formula

$$\frac{\sigma}{y}=\frac{M}{I}$$

$$\frac{\sigma}{\left(\frac{D}{2}\right)}=\frac{25L}{\left(\frac{BD^3}{12}\right)}$$

$$\boxed{\sigma=\frac{150L}{BD^2}}$$

737. Radius of curvature of the beam is equal to

(a) $\frac{ME}{I}$ (b) $\frac{M}{EI}$

(c) $\frac{EI}{M}$ (d) $\frac{MI}{E}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Bending equation

$$\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R.}$$

$$\frac{M}{I}=\frac{E}{R.}$$

Radius of curvature of the $(R)=\frac{EI}{M}$

738. The ratio of flexural strength of a square section with its two sides horizontal to its diagonal horizontal is

(a) $\sqrt{2}$ (b) 2

(c) $2\sqrt{2}$ (d) $\frac{\sqrt{2}}{5}$

APPSC AEE (Civil/Mechanical) 2012

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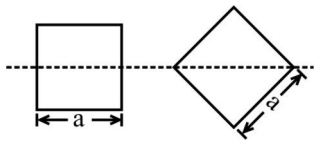
Ans. (a)



a= Side of the square beam
M_1= Moment of resistance of Section 1
M_2= Moment of resistance of Section 2

$$Z_1 = \frac{bd^2}{6} = \frac{a \times a^2}{6} = \frac{a^3}{6} \quad \text{.....(i)}$$

$$I_2 = 2 \times \frac{bh^3}{12} = 2 \times \frac{a\sqrt{2} \times \left(\frac{a}{\sqrt{2}}\right)^3}{12} = \frac{a^4}{12}$$

$$y_{max} = \frac{a}{\sqrt{2}}$$

$$Z_2 = \frac{I}{y_{max}} = \frac{\frac{a^4}{12}}{\frac{a}{\sqrt{2}}} = \frac{a^3}{6\sqrt{2}}$$

Moment of resistance of a section is directly proportional to their section modulus.

$$\frac{M_1}{M_2} = \frac{Z_1}{Z_2} = \frac{\frac{a^3}{6}}{\frac{a^3}{6\sqrt{2}}} = \sqrt{2}$$

739. The nature of distribution of horizontal shear stress in a rectangular beam is

(a) linear (b) parabolic
(c) hyperbolic (d) elliptic

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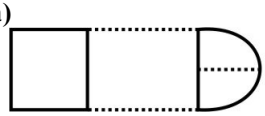
Ans. (a)



$\tau_{max} = 1.5\tau_{avg}$

Shape of shearing stress distribution across rectangular cross section is parabolic only.

740. Section modulus of a circular section about an axis through its centre of gravity is

(a) $\frac{\pi}{32}d^3$ (b) $\frac{\pi}{16}d^3$
(c) $\frac{\pi}{8}d^3$ (d) $\frac{\pi}{64}d^3$

APPSC AEE (Civil/Mechanical) 2012

Ans. (a) Section modulus (z) = I/y

I = Moment of Inertia $= \frac{\pi d^4}{64}$

$y = \frac{d}{2}$

$$\boxed{z = \frac{\pi d^3}{32}}$$

741. A steel plate 50mm wide and 100mm thick is to be bent into a circular arc of radius 10m. If $E = 2 \times 10^5\ N/mm^2$, then the maximum bending stress induced will be

(a) 200 N/mm^2 (b) 100 N/mm^2
(c) 10,000 N/mm^2 (d) 1000 N/mm^2

APPSC AEE (Civil/Mechanical) 2012

Ans. (d) b=50mm, d= 100mm, R = 10m
E= 2×10^5 N/mm^2
Bending equation :-

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

y = 50mm

$$\frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \times y$$

$$\sigma = \frac{2 \times 10^5}{10 \times 10^3} \times 50$$

$$\boxed{\sigma = 1000\ N/mm^2}$$

742. A beam of square section is placed with one diagonal placed horizontally. The maximum shear stress occurs at

(a) the N.A. (b) $\frac{3}{8}$ d from top
(c) $\frac{3}{8}$ from N.A. (d) at the extreme fibres

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) $\frac{3}{8}$ d from top

743. Section modulus of a beam is defined as

(a) IY (b) $\frac{Y}{I}$
(c) $\frac{I}{Y_{max}}$ (d) Y^2I

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Section modulus:–

$$\frac{M}{I} = \frac{\sigma}{y} \text{ OR } M = \sigma \times \frac{I}{y}$$

$$M = \sigma_{max} \times \frac{I}{y_{max}} = \sigma_{max} \times Z$$

Where $Z = \frac{I}{Y_{max}}$. The term Z is known as modulus of section or section modulus.

744. A beam of uniform strength is one, which has same

(a) bending moment throughout the section
(b) shearing force throughout the section
(c) deflection throughout the bean
(d) bending stress at every section

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Ans. (d) A beam of uniform strength is one which has same bending stress at every section.

745. Neutral axis of a beam is the axis at which

(a) the shear force is zero
(b) the section modulus is zero
(c) the bending stress is maximum
(d) the bending stress is zero

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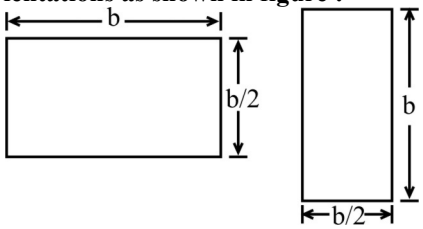
Ans. (d) Neutral axis of a beam is the axis at which the bending stress is zero.

746. A beam cross-section is used in two different orientations as shown in figure :



Bending moments applied in both causes are same. The maximum bending stresses induced in cases (A) and (B) are related as

(a) $\sigma_A = \sigma_B$ (b) $\sigma_A = 2\sigma_B$

(c) $\sigma_A = \frac{\sigma_B}{2}$ (d) $\sigma_A = \frac{\sigma_B}{4}$

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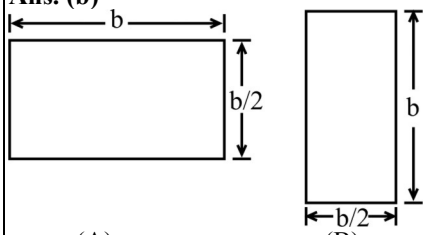
Ans. (b)



(A) (B)

For first case:–

$$I_A = \frac{1}{12} \times b\left(\frac{b}{2}\right)^3 = \frac{b^4}{96}$$

$$\frac{M}{I_A} = \frac{\sigma_A}{y} \Rightarrow \frac{M \times 12 \times 8}{4 \times b^2}$$

$$\boxed{\sigma_A = \frac{24M}{b^3}} \quad \ldots\ldots\ldots (i)$$

For second case

$$I_B = \left(\frac{b}{2}\right) \times (b)^3 \times \frac{1}{12}$$

$$I_B = \frac{b^4}{24}$$

$$\frac{M}{I_B} = \frac{\sigma_B}{y}$$

$$\frac{M}{\frac{b^4}{12 \times 2}} = \frac{\sigma_B}{\frac{b}{2}}$$

$$\boxed{\sigma_B = \frac{12M}{b^3}} \quad \ldots\ldots\ldots (ii)$$

For equation first and second $\boxed{\sigma_A = 2\sigma_B}$

747. The neutral axis of the cross-section a beam is that axis at which the bending stress is

(a) minimum (b) maximum
(c) zero (d) none of the above

GPSC Assistant Prof. 2016

Ans. (c) : The neutral axis of x-section a beam is that axis at which the bending stress is zero.

748. Product EI is called

(a) Flexural rigidity
(b) Torsional rigidity
(c) Second moment of area
(d) Relative stiffness

GPSC Assistant Prof. 2016

Ans. (a) : Product EI is called Flexural rigidity.

749. At point of contraflexure in a beam:

(a) Bending moment is minimum
(b) Shear force is minimum
(c) Bending moment is maximum
(d) Bending moment is either zero or changes sign

PTCUL AE 2016

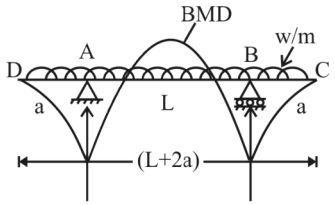
Ans. (d) : Bending moment is either zero or changes sign at point of contraflexure in a beam.

750. In a simply supported beam (L + 2a) with equal over hangs (a) and carrying a uniformly distributed load over its entire length, B.M. at the middle point of the beam will be zero if:

(a) L = 2a (b) L = 4a
(c) L < 2a (d) L > 1

PTCUL AE 2016

Ans. (a) :



$R_A = \frac{w}{2}(L + 2a)$ $R_B = \frac{w}{2}(L + 2a)$

If L = 2a then B.M at middle point of the beam will be zero.

751. If all dimensions of a simply supported beam is increased 'n' times, the ratio by which maximum bending stress will increase under its self weight is –

(a) n : 1 (b) (1/n) : 1
(c) 1 : (1/n) (d) 1: n

UPPCL AE 2014

Ans. (a) : $\sigma_b = \frac{My}{I}$

$\sigma_b \propto y$ (distance of outer fiber from N.A.)

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$$\frac{(\sigma_b)_{initial}}{(\sigma_b)_{final}} = \frac{y_i}{y_f} = \frac{\frac{b}{2}}{\frac{nb}{2}} = \frac{1}{n}$$

$$(\sigma_b)_{final} = n(\sigma_b)_{initial}$$

$$\frac{(\sigma_b)_{final}}{(\sigma_b)_{initial}} = \frac{n}{1}$$

752. The intensity of longitudinal stress in the cross section of a beam at any distance'd' from the neutral axis, is proportional to –

(a) d (b) $1/d^3$

(c) $1/d^2$ (d) 1/d

UPPCL AE 2014

Ans. (a) : In beam under bending

$$\sigma = \frac{M}{I} \times y$$

$$\boxed{\sigma \propto y}$$

$\therefore$ longitudinal stress is proportional to d.

753. A square section with side 'X' of a beam is subjected to a shear force 'S'. The magnitude of shear stress at the top edge of the square is

(a) $\frac{1.5S}{X^2}$ (b) $\frac{S}{X^2}$

(c) $\frac{0.5S}{X^2}$ (d) Zero

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Ans. (d) : At the top edge of the square, the magnitude of the shear stress should be zero. Because shear stress is a surface phenomina.

754. The strength of the beam mainly depends on

(a) Bending moment (b) C.G. of the section

(c) Section modulus (d) Its weight

APPSC (PHE) AE Paper-2 2012

Ans. (c) : In case of rectangular section –

$\tau_{max} = \frac{3}{2}\tau_{mean}$

where ; τ_{max} = Maximum shear stress

τ_{mean} = Mean shear stress

755. The ratio of maximum shear stress to average shear stress is 1.5 in a beam of

(a) Circle (b) Rectangle

(c) Triangle (d) Any cross-section

APPSC (PHE) AE Paper-2 2012

Ans. (b) : The maximum shear stress developed in a beam of rectangular cross-section is

$\tau_{max} = 1.5\ \tau_{avg}$

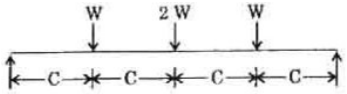
$\frac{\tau_{max}}{\tau_{avg}} = 1.5$

where;

τ_{max} = Maximum shear stress

τ_{avg} = Average shear stress = $\frac{\text{Load}}{\text{Area}}$

756. A simply supported beam is loaded as shown in fig. The maximum shear force in the beam will be



(a) Zero (b) W

(c) 2 W (d) 4 W

APPSC (PHE) AE Paper-2 2012

Ans. (c) : A simply supported beam–

From the figure –

Reaction at A and B in the beam is R_A and R_B respectively due to symmetrical loading.

$R_A = R_B$ .....(i)

& $R_A + R_B = 4$ W .....(ii)

$\therefore$ from equation (i) and (ii)

$R_A = R_B = 2$ W

therefore, from SFD; maximum shear force = 2W

757. The equivalent bending moment under combined action of bending moment M and torque T is

(a) $\sqrt{M^2, T^2}$ (b) $\frac{1}{2}\sqrt{M^2 + T^2}$

(c) $M + \sqrt{M^2 + T^2}$ (d) $\frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$

APPSC (PHE) AE Paper-2 2012

Ans. (d) : Combined effect of bending and Torsion torque :

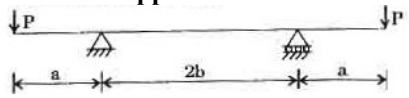
Equivalent bending moment (M_e)

$$= \frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$$

Where; M = Bending moment

T = Torque or Twisting moment

758. For the beam shown is the fig. the elastic curve between the supports B and C will be



(a) circular (b) parabolic

(c) elliptic (d) a straight line

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Ans. (b) :

759. Which of the following stresses are associated with the design of pins in bushed pin type flexible coupling?

(i) Bearing stress (ii) Bending stress

(iii) Axial tensile (iv) Transverse shear

Code:

(a) (i), (iii) and (iv) (b) (ii), (iii) and (iv)

(c) (i), (ii) and (iii) (d) (i), (ii) and (iv)

APPSC (PHE) AE Paper-2 2012

Ans. (d) : Stresses are associated with the design of pins in bushed pin type flexible coupling is –

1- Bearing stress

2- Bending stress

3- Transverse shear

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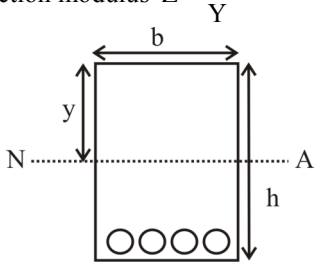
760. Section modulus (Z) for a rectangular cross section is

(a) $bh^3/12$ (b) $bh^3/6$

(c) $bh^2/12$ (d) $bh^2/6$

APPSC AEE 2016

Ans. (d) : Section modulus $Z = \frac{I}{Y}$



$$Z = \frac{\frac{bh^3}{12}}{\frac{h}{2}} = \frac{bh^2}{6}$$

761. Assumption made in the theory of bending is

(a) Radius of curvature is small

(b) Radius of curvature is large

(c) Transverse sections of the beam do not remain plane after bending

(d) Doesn't follow Hook's law in bending

APPSC AEE 2016

Ans. (b) : Assumptions is theory of Bending,

1. The material of beam is homogeneous and isotropic.
2. The value of young's modulus of plasticity is same in tension and compression.
3. The transverse sections which were plane before bending, remain plane after bending also.
4. The radius of curvature is large as compared to dimensions of the cross section.
5. Each layer of the beam is free to expand or contract, independently of the layer, above or below it.

762. In a beam of rectangular cross section, the ratio of the maximum shear stress to the average shear stress is

(a) 2.50 (b) 2.00

(c) 1.50 (d) 1.33

APPSC AEE 2016

Ans. (c) : In rectangular beam section

$$\frac{\tau_{max}}{\tau_{avg.}} = \frac{3}{2}$$

in circular section

$$\frac{\tau_{max}}{\tau_{avg.}} = \frac{4}{3}$$

763. In simply supported beam of rectangular cross section, the bending stress is ..... and shear stress is ..... at neutral axis.

(a) Zero, Maximum

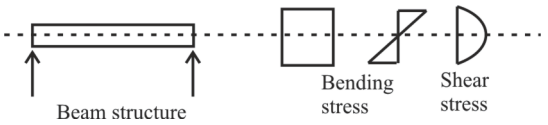
(b) Maximum, Zero

(c) Zero, Zero

(d) Maximum, Maximum

GPSC AE January 2018

Ans. (a) :



So, for simply supported beam, bending stress is zero at neutral axis and shear stress is maximum at neutral axis.

764. The equation of pure bending in usual notation is

(a) $\frac{M}{R} = \frac{E}{Y} = \frac{f}{I}$ (b) $\frac{M}{Y} = \frac{E}{I} = \frac{f}{R}$

(c) $\frac{M}{I} = \frac{E}{R} = \frac{f}{Y}$ (d) $\frac{M}{R} = \frac{E}{I} = \frac{f}{Y}$

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Ans. (c) : For pure bending-

According to Bending equation-

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

M = bending moment

I = moment of inertia

σ = bending stress

y = distance of extreme fibre from NA.

E = Young's modulus

R = radius of circle

765. For rectangular section of simply supported beam the maximum shear stress (T_{max}) is ..... times shear stress (T_{avg}).

(a) 1.3 (b) 1.0

(c) 1.5 (d) 2.0

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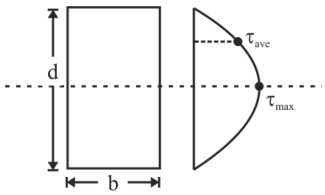
Ans. (c) : For a rectangular section of beam.

$$\tau_{max} = \frac{3}{2}\tau_{avg} = 1.5\tau_{avg}$$



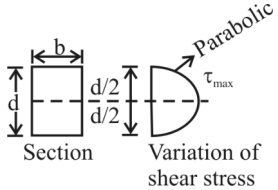
766. What is the nature of distribution of shear stress in a rectangular beam?

(a) Linear (b) Parabolic

(c) Hyperbolic (d) Elliptic

UPRVUNL AE 2015

Ans. (b) :



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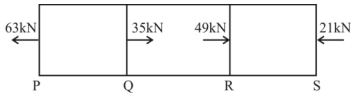
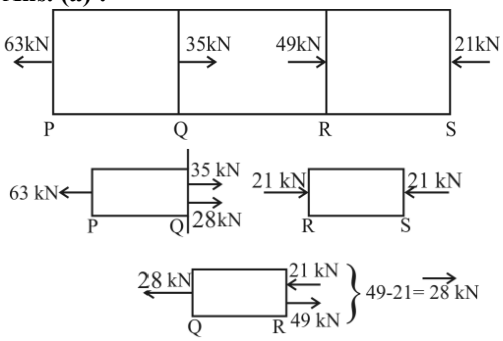
767. A bar having a cross-sectional area of 700mm^2 is subjected to axial loads at the positions indicated. The value of stress in the segment QR is



(a) 40 MPa (b) 50 MPa
(c) 70 MPa (d) 120 MPa

UPRVUNL AE 2015

Ans. (a) :



Stress in the segment QR is :
Load = 28kN = 28000N
A = 700 mm^2

$$\text{Stress} = \frac{P}{A} = \frac{28000}{700} = 40 N/mm^2$$

768. The ratio of maximum shear stress to average shear stress for rectangular section is :

(a) 1.5 (b) 2.5
(c) 1 (d) 2

UPPCL AE 2019

Ans. (a)

For rectangular section

$$\tau_{max} = 1.5\ \tau_{avg}$$

$$\tau_{max} = \frac{3}{2}\frac{F}{bd}$$

$$\tau_{avg} = \frac{F}{bd}$$

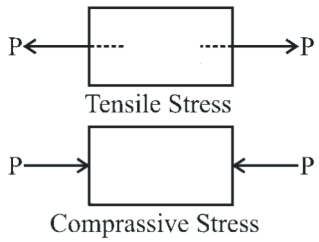
$$\frac{\tau_{max}}{\tau_{avg}} = 1.5$$

769. Tensile stress and compressive stress are the examples of :

(a) Shear stress (b) Direct stress
(c) Bending stress (d) Twisting stress

LMRC AM 2018

Ans. (b) :

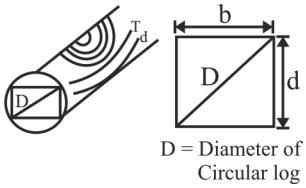


770. For the strongest rectangular beam cut from a circular log, the ratio of the width and depth is:

(a) 0.505 (b) 0.707
(c) 0.606 (d) 0.404
(e) 0.303

Haryana PSC AE 2018 Paper–I

Ans. (b) :



For max. M.O.R., its Z should be maximum

$$Z = \frac{bd^2}{6} \Rightarrow b^2 + d^2 = D^2$$

$$\Rightarrow \quad d^2 = D^2 - b^2$$

$$z = \frac{b(D^2 - b^2)}{6}$$

For z to be maximum

$$\frac{dz}{db} = 0$$

$$\frac{d}{b} = \sqrt{2} \quad \Rightarrow 0.707 = \frac{b}{d}$$

771. The yield moment of a cross-section is defined as the moment that will just produce the yield stress in:

(a) The outermost fibre of the section
(b) The innermost fibre of the section
(c) The neutral fibre of the section
(d) The fibres everywhere
(e) None of the above options

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Ans. (a) : Max stresses are on the outermost fibre of the section.

772. The ratio of moments of inertia of a triangular section about its base and about a centroidal axis parallel to its base is:

(a) 3 (b) 4
(c) 1 (d) 2
(e) 5

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Ans. (a) : Moment of inertia about base $= \frac{bh^3}{12}$

Moment of inertia about centroidal axis $= \frac{bh^3}{36}$

Ratio $= \frac{bh^3}{12} \times \frac{36}{bh^3} = 3$

773. The locus of the moment of inertia about inclined axes of the principal axis is:

(a) Straight line (b) Parabola
(c) Circle (d) Ellipse
(e) Hyperbola

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Ans. (d) : The locus of the moment of inertia about inclined axes of the principal axis is Ellipse.

774. The maximum shear stress (q_{max}) in a rectangular beam is:

(a) 1.25 times the average
(b) 1.5 times the average
(c) 1.75 times the average
(d) 2.0 times the average
(e) 2.5 times the average

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Ans. (b) : $q_{max} = 1.33\ q_{avg}$ - for solid circular section
$q_{max} = 1.5\ q_{avg}$ - for Rectangular section
$q_{max} = 1.33\ q_{avg}$ - for triangular section
$q_{max} = 1.25\ q_{avg}$ - for Diamond section
$\therefore\ q_{max}$ = Max shear stress
$\therefore\ q_{avg}$ = Average shear stress

775. For a beam of T-cross-section subjected to a shear force of F, the maximum shear force occurs at:

(a) Top of the section
(b) Bottom of the section
(c) Neutral axis of the section
(d) Junction of web and flange
(e) None of these options

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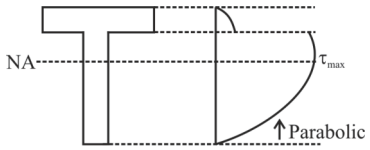
Ans. (c) :



Shear stress distribution in T-beam

776. A beam is said to be of uniform strength if:

(a) Bending moment is same throughout the beam
(b) Shear stress is same throughout the beam
(c) Deflection is same throughout the beam
(d) Bending stress is same at every section along its longitudinal axis
(e) All of the above

Haryana PSC AE 2018 Paper–I

Ans. (d) : $M = \sigma \times z$

$$\sigma = \frac{M}{Z}$$

$\sigma\ \alpha\ M$
σ = Bending stress
M = Bending Moment

777. Maximum permissible bending stress is depend on :

(i) $\frac{T_w}{t}$ **(ii)** $\frac{t}{t_w}$ **(iii)** $\frac{t}{t_y}$

(a) (i) (ii)
(b) (i) (iii)
(c) (ii) (iii)
(d) (i) (ii) (iii)

UP Jal Nigam AE 2016

Ans. (d) : Maximum permissible bending stress is depend on :

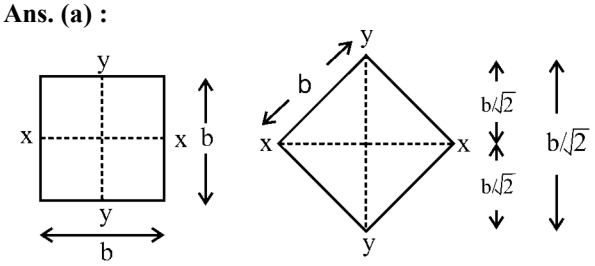
(i) $\frac{T_w}{t}$, (ii) $\frac{t}{t_w}$, (iii) $\frac{t}{t_y}$

778. The ratio of the moments of resistance of square beam when square section is placed with two sides horizontal with a diagonal horizontal is :

(a) 1.414
(b) 0.707
(c) 1.712
(d) 0.87

UP Jal Nigam AE 2016

Ans. (a) :



For a given stress, the ratio of moments of resistance of resistance of square beam when section is placed with two sides horizontal to that when the beam is placed with a diagonal horizontal is $\sqrt{2}$.

$$\frac{M_1}{M_2} = \frac{\sigma_b Z_1}{\sigma_b Z_2} = \frac{b^3/6}{b^3/6\sqrt{2}}$$

$$\frac{Z_1}{Z_2} = \sqrt{2} = 1.414$$

779. If the area of cross-section of a circular section beam is made four times, keeping the loads, length, support condition and material of the beam is unchanged, then the quantities (List I) will change through different factors (List II). Match the List I with List II and select the correct answer using the code given the lists:

List I	List II
A. Maximum bending moment	**1. 8**
B. Deflection	**2. 1**
C. Bending stress	**3. 1/8**
D. Section modulus	**4. 1/16**

(a) A-2, B-1, C-2, D-4
(b) A-2, B-4, C-3, D-1
(c) A-3, B-4, C-2, D-1
(d) A-2, B-1, C-3, D-4

UPRVUNL AE 2014

Ans. (b) :

(a) Maximum BM	1
(b) Deflection	1/16
(c) Bending stress	1/8
(d) Section modulus	8

780. A cantilever beam with rectangular cross-section is subjected to uniformly distributed load. The deflection at the tip is δ_1. If the width and depth of the beam are doubled then deflection at tip is δ_2. Then $\frac{\delta_2}{\delta_1}$ is

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(a) 0.0625 (b) 16
(c) 0.5 (d) 2

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Deflection $\propto \frac{1}{I}$

$$\therefore \quad \frac{\delta_2}{\delta_1} = \frac{I_1}{I_2} = \frac{\left(\frac{bd^3}{12}\right)}{\frac{(2b)(2d)^3}{12}} = \frac{1}{16} = 0.0625$$

781. A rectangular beam section with depth 400 mm and width 300 mm is subjected to a bending moment of 60 kN/m. The maximum bending stress in the section is

(a) 7.50 MPa (b) 2.50 MPa
(c) 1.56 MPa (d) 0.42 MPa

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Maximum bending stress in the beam

$$f_{max} = \frac{M}{Z} = \frac{M}{\left[\frac{bd^2}{6}\right]} = \frac{60\times10^6}{\left[\frac{300\times400^2}{6}\right]} = 7.5 \text{ MPa}$$

782. In case of pure bending, the beam will bend into an arc of a/an

(a) parabola (b) hyperbola
(c) circle (d) ellipse

APPSC AEE (CIVIL/MECHANICAL) 2019

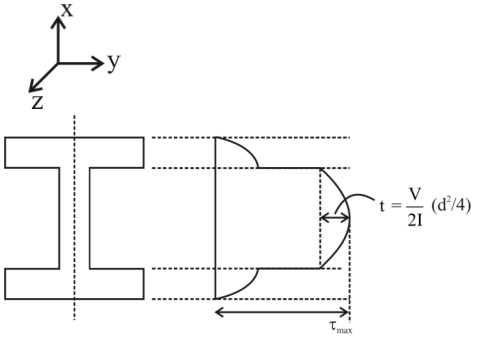
Ans. (c) : If a beam is subjected to pure bending the elastic curve is a circular arc with constant radius.

783. A beam of I-section, is loaded in yz-plane (with y-axis along the length), find which statement is correct :

(a) variation of τ_{xz} is linear and that of τ_{xy} is parabolic
(b) variation of τ_{xy} is linear and that of τ_{yz} is parabolic
(c) variation of τ_{xz} is parabolic and that of τ_{xy} is linear
(d) variation of τ_{xy} is parabolic and that of τ_{yz} is linear

HPPSC Poly. Lect. 2016

Ans : (a)



Variation of τ_{xz} is linear and that of τ_{xy} is parabolic.

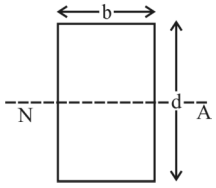
784. A simply supported rectangular beam having width 'b' and depth 'd' carries a central load W and undergoes deflection δ at the center. If the width and depth are interchanged, the deflection at the center of the beam would attain the value :

(a) $\frac{d}{b}\delta$ (b) $\left(\frac{d}{b}\right)^2\delta$
(c) $\left(\frac{d}{b}\right)^3\delta$ (d) $\left(\frac{d}{b}\right)^{\frac{3}{2}}\delta$

HPPSC Poly. Lect. 2016

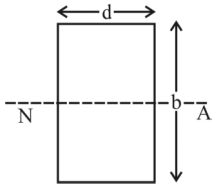
Ans : (b) According to question

Case-I



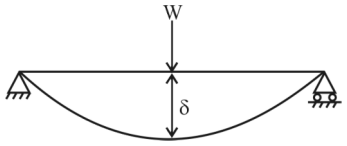
Moment of inertia $(I_1) = \frac{bd^3}{12}$

Case-II



Moment of inertia $(I_2) = \frac{db^3}{12}$

Max. deflection at centre of beam in simply supported beam–



$$\delta = \frac{wl^3}{48EI}$$

$$\frac{wl^3}{48E} = \text{constant}$$

$$\delta \propto \frac{1}{I}$$

$$\frac{\delta_2}{\delta_1} = \frac{I_1}{I_2} = \left(\frac{\frac{bd^3}{12}}{\frac{db^3}{12}}\right), \quad \boxed{\delta_2 = \left(\frac{d}{b}\right)^2.\delta_1}$$

785. Moment of inertia for a channel section is:

(a) $I_{xx} = \frac{\left(BD^3 - BD\right)}{12}$

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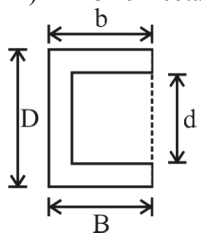
(b) $I_{xx} = \frac{(BD^3 - bd)}{12}$

(c) $I_{xx} = \frac{(BD^3)}{12}$

(d) $I_{xx} = \frac{(BD^3 - bd^3)}{12}$

IRCON AE 2017

Ans. (d) : Moment of inertia of channel section = MOI of rectangle of (BXD) – MOI of rectangle of (bxd)



$$= \frac{BD^3}{12} - \frac{bd^3}{12} = \frac{(BD^3 - bd^3)}{12}$$

786. Maximum shear stress in I section at neutral axis is given by :

(a) $\tau_{max=} = \frac{F}{l}\left(\frac{B}{8}(D^2 - d^2) + \frac{bd^2}{6}\right)$

(b) $\tau_{max=} = \frac{F}{lb}\left(\frac{B}{8}(D^2 - d^2) + \frac{bd^2}{8}\right)$

(c) $\tau_{max=} = \frac{F}{lb}\left(\frac{B}{6}(D^2 - d^2) + \frac{b}{8}\right)$

(d) $\tau_{max=} = \frac{F}{b}\left(\frac{B}{6}(D^2 - d^2) + \frac{bd^2}{8}\right)$

GPHC AE 2018

Ans : (b) Maximum shear stress in I-section at neutral axis :

$$T_{max} = \frac{F}{lb}\left[\frac{B}{8}(D^2 - d^2) + \frac{bd^2}{8}\right]$$

787. A rectangular beam is to be cut from a circular log of wood of diameter D. The depth of the strongest section is

(a) D/3 (b) $D/\sqrt{3}$

(c) $2D/\sqrt{3}$ (d) $D\sqrt{\frac{2}{3}}$

WBPSC AE 2014

Ans. (d) : For strongest beam, depth of strongest section is $\Rightarrow \sqrt{\frac{2}{3}}D$

Breadth of strongest section $= \frac{D}{\sqrt{3}}$

Section modulus $= \frac{D^3}{9\sqrt{3}}$

788. Two simply supported beams B_1 and B_2 have spans I and 2I respectively. Beam B_1 has a cross-section of 1 × 1 units and beam B_2 has a cross-section of 2 × 2 units. These beams are subjected to concentrated load W at the center of their span. The ratio of the maximum flexural stress in these beams is

(a) 4 (b) 2

(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

WBPSC AE 2007

Ans. (a) : Flexural stress, $F_B = \frac{M}{Z}$

$M \propto$ length

$Z \propto$ width × depth^2

$$\frac{F_{B_1}}{F_{B_2}} = \frac{\frac{1}{1\times1\times1}}{\frac{2}{2\times2\times2}} = 4$$

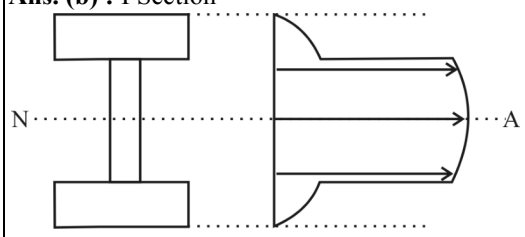
789. The variation of the horizontal shear stress in the flange of an I section is

(a) Linear (b) Parabola

(c) Elliptical (d) Cubic parabola

WBPSC AE 2007

Ans. (b) : I Section



Maximum shear stress on N.A.

780. A simply supported beam of rectangular cross-section 100 mm × 200 mm carrying a concentrated load at mid-span. If the maximum shear stress developed in the beam is 3 MPa, what is the magnitude of concentrated load?

(a) 20 kN (b) 40 kN

(c) 80 kN (d) 160 kN

TSPSC AEE 2017

Ans. (b) :

$$\tau_{max} = \frac{3}{2}\tau_{avg}$$

$$3 = \frac{3}{2} \times \frac{P}{bd}$$

$$3 = \frac{3}{2} \times \frac{P}{200 \times 100}$$

$$P = 40\,KN$$

781. The moment of inertia of a square of side (a) about an axis through its center of gravity is

(a) $a^4/4$ (b) $a^4/8$

(c) $a^4/12$ (d) $a^4/36$

Chandigarh SDE 2017

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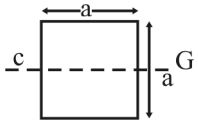
Ans. (c) :



Moment of inertia of square section of about

C.G. $= \frac{a^4}{12}$

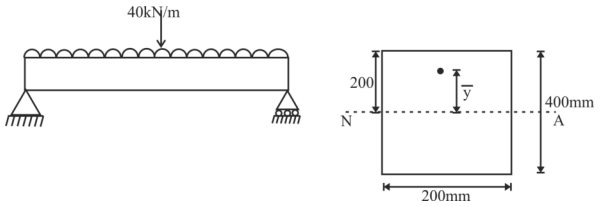
782. A rectangular beam 200 mm wide and 400 mm deep over a distance of 4 meters and carries a uniformly distributed load 40 kN/m. Determine the maximum shear stress acting on the beam cross section

(a) 1.5 N/mm^2 (b) 1.0 N/mm^2

(c) 3.0 N/mm^2 (d) 2.0 N/mm^2

TSGENCO_2015

Ans. (a) :



Shear force = $(S.F.)_{max} = \frac{wl}{2} = \frac{40 \times 4}{2} = 80$ kN

Shear stress = $\tau_{max} = \frac{V_A \bar{y}}{I_b} = \frac{80 \times 10^3 \times 200 \times 200 \times 10}{\left(\frac{(400)^3 \times 200}{12}\right) \times 200}$

$\tau_{max} = 1.5$ N/mm^2

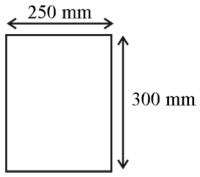
783. A beam of rectangular cross-sectional dimensions 250 mm × 300 mm is subjected to a shear force of 75 kN. The intensity of maximum shear stress in the cross-section is

(a) 1.5 N/mm^2 (b) 2.0 N/mm^2

(c) 2.2 N/mm^2 (d) 2.5 N/mm^2

UPSC JWM 2017

Ans. (a) :



$\tau_{avg} = \frac{75 \times 10^3}{250 \times 300} = 1 \text{ N/mm}^2$

$\tau_{max} = 1.5\ \tau_{avg}$

$\tau_{max} = 1.5 \times 1$

$\tau_{max} = 1.5$ N/mm^2

784. The shape of the shear stress distribution diagram for a rectangular beam is

(a) Parabola (b) Rectangle

(c) Hyperbola (d) Triangular

APPSC AEE MAINS 2016

Ans. (a) : shear force for rectangular cross section-

$= \frac{6F}{bd^3}\left[\left(\frac{d}{2}\right)^2 - y^2\right]$

This shows that the variation is parabolic.

785. The maximum shearing stress produced by shrinkfit between two cylinders (mounted one inside the other) of E = 2×10^6 kg(f)/m^2 and shrinkage factor = 0.002, is equal to

(a) 2000 kg(f)/m^2 (b) 2500 kg(f)/m^2

(c) 4000 kg(f)/m^2 (d) 1000 kg(f)/m^2

APPSC AEE MAINS 2016

Ans. (a) : For maximum shearing stress

$= \frac{2 \times 10^6 \times 0.002}{2}$

$= 2000 \text{kg(f)/m}^2$

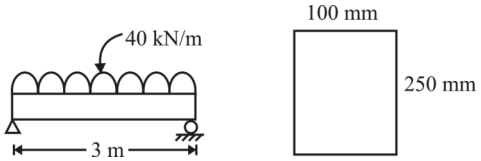
786. A simply supported wooden beam 100 mm wide, 250 mm deep and 3 m long is carrying a uniformly distributed load of 40 kN/m. The maximum shear stress will be :

(a) 2.4 MPa (b) 2.8 MPa

(c) 3.2 MPa (d) 3.6 MPa

ESE 2020

Ans. (d)



Maximum shear force (V_{max})

$= \frac{W\ell}{2} = \frac{40 \times 3}{2}$

$\boxed{V_{max} = 60 \text{ kN}}$ ..... (i)

Maximum shear stress for rectangular beam

$(\tau_{max}) = \frac{3}{2}\tau_{avg}$

$\tau_{max} = \frac{3}{2}\left(\frac{V_{max}}{bd}\right) = \frac{3}{2}\left(\frac{60 \times 10^3}{100 \times 250}\right) = 3.6 \text{ N/mm}^2$

787. A circular beam of 100 mm diameter is subjected to a shear force of 30 kN. The maximum shear stress will be nearly :

(a) 5.1 MPa (b) 6.3 MPa

(c) 7.5 MPa (d) 8.7 MPa

ESE 2020

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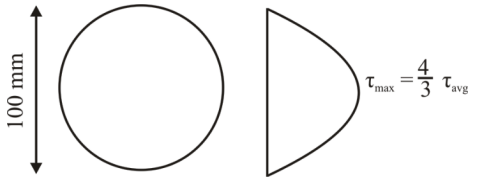
Ans. (a)



For solid circular shaft,

$$\tau_{max} = \frac{4}{3}\tau_{avg}$$

$$= \frac{4}{3}\left(\frac{F}{A}\right) = \frac{4}{3}\times\left(\frac{30\times10^3}{\frac{\pi}{4}\times100^2}\right)$$

$$\boxed{\tau_{max} = 5.09\,MPa}$$

788. A beam of rectangular cross-section is 100 mm wide and 200 mm deep. If the section is subjected to a shear force of 20 kN, then the maximum shear stress in the section is :

(a) 1.25 N/mm^2 (b) 1.5 N/mm^2

(c) 1.6 N/mm^2 (d) 1.75 N/mm^2

OPSC AEE 2019 Paper–I

Ans. (b)

$$S = 20\ kN = 20\times10^3\ N$$

$$\tau_{max} = \frac{3}{2}(\tau_{avg})$$

$$\tau_{max} = 3/2\left(\frac{S}{bd}\right)$$

$$= \frac{3}{2}\left(\frac{20\times10^3}{100\times200}\right)$$

$$\tau_{max} = 1.5\ N/mm^2$$

789. A steel wire of 20 mm diameter is bent into a circular shape of 10 m radius, then the maximum stress induced in the wire is

[Take E = 2 × 10^6 kg/cm^2]

(a) 1 × 10^3 kg/cm^2 (b) 2 × 10^3 kg/cm^2

(c) 4 × 10^3 kg/cm^2 (d) 6 × 10^3 kg/cm^2

OPSC AEE 2019 Paper–I

Ans. (b)

d = 20 mm = 2 cm

$$y_{max} = \frac{d}{2} = \frac{2}{2} = 1cm$$

R = 10 m = 10 × 10^3 mm = 10^3 cm

E = 2 × 10^6 kg/cm^2

From bending equation, $F_{max} = \frac{E}{R}y_{max}\ f$

$$= \frac{2\times10^6}{10^3}\times1 = 2\times10^3\ kg/cm^2$$

790. According to the law of moments, if a number of coplanar forces acting on a particle are in equilibrium, then

(a) their algebraic sum of zero

(b) their liners of action are at equal distances

(c) The algebraic sum of their moments about any points in their plane is zero

(d) the algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point

Chandigarh SDE 2017

Ans. (c) : According to the law of moments is a number of coplaner forces acting on a particle are in equilibrium then

(I) Their lines of action are as equal distances.

(II) The algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point.

791. A beam of rectangular cross-section is 100 mm wide and 200 mm deep. If the section is subjected to a shear force of 20 kN, then the maximum shear stress in the section is

(a) 1 N/mm^2 (b) 1.125 N/mm^2

(c) 1.33 N/mm^2 (d) 1.5 N/mm^2

Chandigarh SDE 2017

Ans. (d) :

$$\tau_{avg} = \frac{F}{A}$$

$$A = 100\times200 = 20000\,mm^2$$

$$F = 20\,KN$$

$$\tau_{avg} = \frac{20\times1000}{20000} = 1$$

$$\tau_{max} = 1.5\,\tau_{avg}$$

$$\tau_{max} = 1.5\times1$$

$$\tau_{max} = 1.5\ N/mm^2$$

792. A portion of a beam between two sections is said to be in pure bending when there is .....

(a) Constant bending moment and zero shear force

(b) Constant shear force and zero bending moment

(c) Constant bending moment and constant shear force

(d) None of the above

RPSC POLY. TECH. LECT. 2014

Ans : (a) Pure bending condition is when there is constant bending moment and zero shear force.

shear force $V = \frac{dM}{dx}$; for constant M

$$\boxed{V = 0}$$

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4. Combined, Direct & Bending Stress

793. The ratio of M.I. of a section about its centroidal axis to the distance of the extreme fibre from the centroidal axis is called
(a) radius of gyration (b) rigidity modulus
(c) section modulus (d) None of the above

BPSC AE Aug. 2019 Paper-V

Ans. (c) : Section modulus is a geometric property for a given cross section used in design of beams or flexural members.

$$\text{Section modulus } (z) = \frac{I}{y}$$

I : moment of inertia of section about centroidal axis.
y : distance of extreme fibre from centroidal axis.

794. Two beams, one of circular x-section and other of square x-section, have equal areas of x-section. If subjected to bending
(a) circular section is more economical
(b) square section is more economical
(c) both sections are equally economical
(d) both sections are equally stiff

Punjab PSC Sub divisional Engineer 2014

Ans. (b) : Two beam one circular cross-section and other square cross section and same area of cross-section in bending condition the square section is more economical.

795. Consider the following statements about flitched beams :
1. A flitched beam has a composite section made of two or more materials joined together in such a manner that they behave as a unit piece and each material bends to the same radius of curvature.
2. The total moment of resistance of a flitched beam is equal to the sum of the moments of resistance of individual sections.
3. Flitched beams are used when a beam of one material, if used alone, would require quite a large cross-sectional area.

(a) 1, 2 and 3 are correct (b) 1 and 3 are correct
(c) 2 and 3 are correct (d) 1 and 2 are correct

OPSC AEE 2019 Paper–I

Ans. (a)
(1) A flitched beam has a composite section made of two or more materials joined together in such a manner that they behave as a unit piece and each material bends to the same radius of curvature.
(2) The total moment of resistance of a flitched beam is equal to the sum of the moments of resistance of individual sections.
(3) Flitched beams are used when a beam of one material, if used alone, would require quite a large cross-sectional area.

796. In combined stress theory, kern of a section is that area of the cross-section where upon loading there will be no tension anywhere in the cross-section. So, the kern of an I-section is
(a) Rectangular in shape
(b) Square in shape
(c) Circular in shape
(d) Rhombus in shape

GPSC AE (CLASS 1 & 2) 2019

Ans : (d) : Shape of kern for different sections–
(1) Rectangle → Rhombus
(2) Square → Square
(3) Circle → Circle (Solid/hollow)
(4) I-section→Rhombus

797. The moment of inertia of an area is always least with respect to :
(a) Vertical axis (b) Bottom most axis
(c) Radius of gyration (d) Central axis

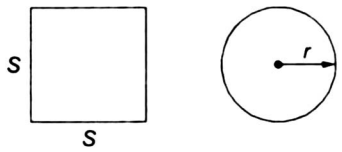
GPSC AE Class (1&2) Paper - 2 2017

Ans. (d) : The MOI (moment of inertia) is least from the central axis & maximum from base if taken (bottom).

798. What is true if the areas of cross-section of square and circular beams, subjected to equal bending moments are the same?
(a) circular beam is more economical
(b) square beam is more economical
(c) both the beams are equally strong
(d) both the beams are equally economical

OPSC AEE 2019 Paper–I
Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) :



For same area,

$$S^2 = \frac{\pi}{4}(2r)^2$$

$$S = r\sqrt{\pi}$$

$$Z_{square} = \frac{S^3}{6} = \frac{r^3(\sqrt{\pi})^3}{6} = 0.928r^3$$

$$Z_{circle} = \frac{\pi}{32}d^3$$

$$= \frac{\pi}{32} \times 8r^3 = \frac{\pi r^3}{4} = 0.785r^3$$

Since, $Z_{square} > Z_{circle}$,
So square section is more economical.

799. Core is a zone in which stresses are purely
(a) compressive (b) tensile
(c) shear (d) bending

(BPSC AE Paper V 2001)

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Ans : **(a)** In core or kernel zone when loading is applied then the stresses in column is always compressive.

800. A vertical wall of a circular bunker is subjected to horizontal pressure due to coal stored therein. The wall of the bunker is designed for
(a) Axial tension
(b) Hoop tension and shear force
(c) Hoop tension and bending moment
(d) All are correct

MPPSC AE 2016

Ans : (c) Vertical wall of circular bunker is subjected to horizontal pressure due to which hoop stress is developed and as the bunker is circular resultant forces act outside the wall, producing bending stresses. Therefore it should be designed for hoop tension and bending moment.

801. The core of hollow circular section of internal and external diameters of 60 mm and 80 mm respectively is circle of radius (in mm)
(a) 40 (b) 70
(c) 80 (d) None of these

(BPSC AE Paper V 2001)

Ans : (d) Core of hollow circular section

$$= \frac{D_1^2 + D_2^2}{8D_1} = \frac{80^2 + 60^2}{8 \times 80} = 15.62 \text{ mm}$$

802. During tensile test of flat bar of low-carbon steel with polished surfaces, the Lueder's lines on the flat sides of the specimen indicate that the failure is due to
(a) Shear (b) Torsion
(c) Direct tension (d) Rupture

(BPSC AE Paper V 2006)

Ans : (a) During tensile test of flat bar of low carbon steel with polished surfaces, the Lueder's lines on the flat sides of the specimen indicate that the failure is due to shear. Cup and cone formation on the flat surface indicate failure due to direct stress in tension.

803. Section modulus is defined as the :
(a) ratio of bending stress to moment
(b) ratio of bending moment to shear force
(c) ratio of shear force to the area
(d) ratio of bending moment to stress

Rajasthan AE (Nagar Nigam) 2016

Ans. (d) :
■ Section modulus is the ratio of bending moment and bending stress

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\boxed{\frac{M}{\sigma} = \frac{I}{y} = z}$$

Where, I = moment of inertia
y = distance from N.A.
M = bending moment
σ = stress

804. The middle third rule in the stability analysis of retaining walls ensures the condition of no....
(a) Overturning (b) Tension
(c) Compression (d) Crushing

RPSC POLY. TECH. LECT. 2014

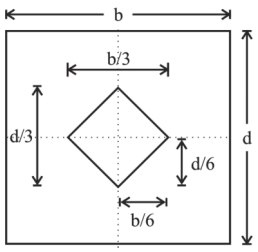
Ans : (b) Retaining walls are made of concrete and tensile strength of concrete is assumed to zero. So, eccentricity of load should be within such limits so that it does not produce any tension.

$f_d - f_b \geq 0$; $f_d \rightarrow$ direct stress; $f_b \rightarrow$ bending stress

$$\therefore \quad \frac{F}{bd} - \frac{F \times e}{Z_{xx}} \geq 0$$

$$e \leq \frac{Z_{xx}}{bd} = \frac{bd^2}{6bd} \Rightarrow \boxed{e \leq d/6}$$

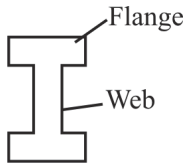
Similarly ; $\boxed{e \leq b/6}$



805. In case of an I–beam, major percentage of the shear force at a section is resisted by :
(a) Top flange
(b) Bottom flange
(c) Top and bottom flanges together
(d) Web

TNPSC AE 2015

Ans. (d)



Flange – resisting compression stress
Web – resisting shear stress
In case of an I–beam, major percentage of the shear force at a section is resisted by Web.

806. In a circular section of diameter D, if 'D' is doubled, then its polar M.I. will be increased by :
(a) 4 times (b) 8 times
(c) 16 times (d) 32 times

TNPSC AE 2015

Ans. (c) : Polar moment of inertia of circular section of diameter D

$$J = \frac{\pi D^4}{32}$$

when D = 2D

then $J = \frac{\pi (2D)^4}{32} = 16 \times \frac{\pi D^4}{32} = 16$ times

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807. For a solid circular beam of 40 mm dia, the section modulus is :

(a) $\pi \times 10^3\ mm^3$ (b) $2\pi \times 10^3 mm^3$

(c) $4\pi \times 10^3\ mm^3$ (d) $8\pi \times 10^3\ mm^3$

TNPSC AE 2015

Ans. (b) : $\text{Section modulus}(z) = \dfrac{\text{moment of inertia}(I)}{\text{distance from N.A.}(y)}$

For circular section,

$$I = \frac{\pi D^4}{64} = \frac{\pi \times (40)^4}{64} = 4\pi \times 10^4\ mm^4$$

$$y = \frac{D}{2} = \frac{40}{2} = 20\ mm$$

$$z = \frac{4\pi \times 10^4}{20} = 2\pi \times 10^3\ mm^3$$

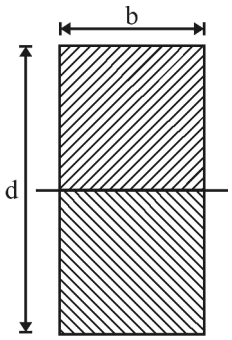
808. The plastic section modulus for a rectangular section of width b and depth d is

(a) $\dfrac{bd^3}{3}$ (b) $\dfrac{bd^2}{6}$

(c) $\dfrac{bd^2}{4}$ (d) $\dfrac{bd^2}{12}$

ASSAM PSC CCE (PRE) 2015

Ans : (c) :



Plastic section modulus

$$Z_P = \frac{A}{2}(y_t + y_c)$$

$$A = b.d$$

$$y_t = \frac{d}{4}$$

$$y_c = \frac{d}{4}$$

$$Z_P = \frac{bd}{2}\left(\frac{d}{4} + \frac{d}{4}\right)$$

$$\boxed{Z_P = \frac{bd^2}{4}}$$

809. A rectangular timber beam ($b \times d$) is cut out of a cylindrical log of diameter 'D'. The width (b) of the strongest timber beam will be-

(a) $\sqrt{3}D$ (b) $\dfrac{D}{\sqrt{3}}$

(c) $\sqrt{2}D$ (d) $\dfrac{D}{\sqrt{2}}$

Maharashtra PSC (Mains) 2015 Paper–I

OR

The ratio of width to depth of a strongest beam that can be out of a cylindrical log of wood with homogeneous and isotropic properties is

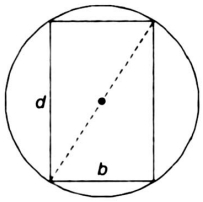
(a) $\dfrac{1}{\sqrt{5}}$ (b) $\dfrac{1}{\sqrt{3}}$

(c) $\dfrac{1}{\sqrt{4}}$ (d) $\dfrac{1}{\sqrt{2}}$

OPSC AEE 2019 Paper–I

GPSC AE June 2019

Ans. (b) :



$$(b^2 + d^2) = D^2$$

$$Z = \frac{bd^2}{6} = \frac{b(D^2 - b^2)}{6}$$

For the beam cross-section to be strongest. Z should be maximum.

$$\Rightarrow \quad \frac{dZ}{db} = 0$$

$$\Rightarrow \quad D^2 - 3b^2 = 0$$

$$b = \frac{D}{\sqrt{3}}$$

Note :

$$\Rightarrow \quad d = \sqrt{D^2 - \frac{D^2}{3}} = \sqrt{\frac{2}{3}}D$$

$$\Rightarrow \text{Section modulus } Z_{max} = \left[\frac{\frac{D}{\sqrt{3}} \times \frac{2D^2}{3}}{6}\right] = \frac{D^3}{9\sqrt{3}}$$

$$\Rightarrow \quad \boxed{\frac{d}{b} = \sqrt{2}} = \boxed{\frac{b}{d} = \frac{1}{\sqrt{2}}}$$

810. The ratio of the flexural strengths of two beams of square cross section, the first beam being placed with its top and bottom sides horizontally and second beam being placed with one diagonal horizontally, is-

(a) $\sqrt{3}$ (b) $\dfrac{1}{\sqrt{3}}$

(c) $\dfrac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

Maharashtra PSC (Mains) 2015 Paper–I

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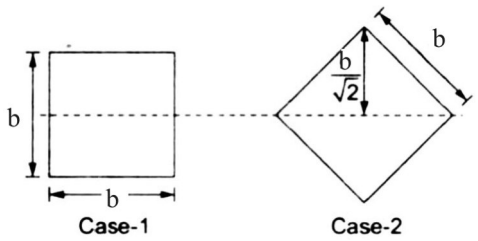
Ans. (d) :



$$\text{MOR } \alpha\, Z = \left(\frac{I}{y}\right)$$

$$\frac{MOR_1}{MOR_2} = \frac{Z_1}{Z_2} = \frac{I_1}{I_2} \times \frac{y_2}{y_1}$$

$$I_1 = \frac{bd^3}{12} = \frac{b^4}{12} \qquad (b = d)$$

$$I_2 = \frac{b^4}{12}$$

$$y_1 = \frac{b}{2};\ y_2 = \frac{b}{\sqrt{2}}$$

Hence, $\frac{MOR_1}{MOR_2} = \frac{b^4}{12} \times \frac{12}{b^4} \times \frac{b}{\sqrt{2}} \times \frac{2}{b} = \sqrt{2}$

811. A is a shaft of diameter *d* and *B* is a shaft of diameter 2*d*. The ratio of polar modulus of section *B* to section *A* is-

(a) 2 (b) 6
(c) 8 (d) 16

BPSC AE Pre 2018

Ans. (c) Polar modulus of section $= \frac{\pi}{16} D^3$

So,

$$\frac{\text{Polar modulus of shaft B}}{\text{Polar modulus of shaft A}} = \frac{\frac{\pi D_B^3}{16}}{\frac{\pi D_A^3}{16}}$$

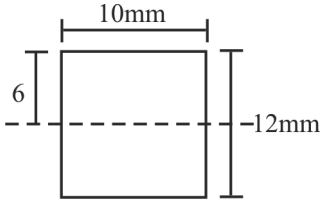
$$= \frac{D_B^3}{D_A^3} = \frac{(2d)^3}{(d)^3} \Rightarrow \frac{Z_B}{Z_A} = 8$$

812. The plastic section modulus of a rectangular section of width 10 mm and depth 12 mm is

TNPSC AE 2018

(a) 100 mm^3 (b) 180 mm^3
(c) 240 mm^3 (d) 360 mm^3

Ans. (d) :



Plastic section modulus is movement of area about equal area axis. For rectangular section:-

$$Z_P = \left(b \times \frac{d}{2} \times \frac{d}{4}\right) \times 2 = \frac{bd^2}{4} = \frac{10 \times 12^2}{4} = 360 mm^3$$

Note : Plastic section modulus $= \frac{bd^2}{4}$

Elastic section modulus $= \frac{bd^2}{6}$

Shape factor = 1.5

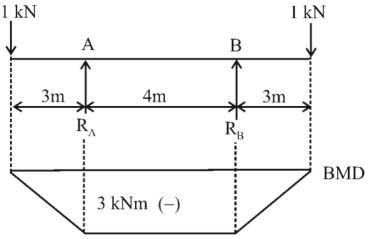
Plastic modulus and elastic modulus are different. So, shape factor comes into picture here

813. The cross section of the beam has $I_{xx} = 3 \times 10^6$ mm^4 and it is subjected to the loading shown in the figure. The neutral axis is situated at 70 mm from the extreme fibre. The maximum tensile stress in the beam would be

(a) 70 MPa (b) 7 GPa
(c) 7 MPa (d) 70 kN/mm^2
(e) 70 GPa

CGPSC AE 2014 Shift–I

Ans. (a) :



$M_{max} = 3kNm = 3 \times 10^6$ Nmm

B.M. is hogging through out, so top flange will be subjected to tension and bottom flange will be subjected to compression

$$\sigma_{t_{max}} = \frac{(M_{max})\, y_{max}}{I}$$

$$= \frac{3 \times 10^6 \times 70}{3 \times 10^6} = 70 \text{ N/mm}^2$$

$\boxed{\sigma_{t_{max}} = 70MPa}$

814. The relationship between maximum bending stress (s) offered by a section, moment of resistance (M) and section modulus (Z) is given by

(a) $M = \frac{\sigma}{Z}$ (b) $M = \frac{Z}{\sigma}$
(c) $M = \sigma.Z$ (d) $M = \frac{1}{\sigma.Z}$

TNPSC AE 2018

Ans. (c) : Bending equation of beam

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$\sigma = \frac{My}{I}$; $\frac{I}{y} = Z$ = section modulus

$\sigma = \frac{M}{Z} \Rightarrow$ $\boxed{M = \sigma \times Z}$

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815. The greatest eccentricity which a load w can have without tension on the cross-section of a short column of external diameter D and internal diameter d, is

(a) $\frac{D^2+d^2}{8D}$ (b) $\frac{D^2-d^2}{8D}$

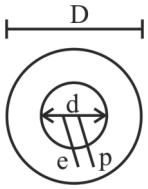
(c) $\frac{\pi\left(D^2+d^2\right)}{8D}$ (d) $\frac{D^2+d^2}{8\pi D}$

TNPSC AE 2018

Ans. (a) : Let the load be applied at eccentricity. Direct stress $=\frac{P.e.}{I}y$ for zero tension on outer fiber

P – applied load

e – Eccentricity



$$\sigma_A = 0 = \frac{P}{A} - \frac{Pe\times\frac{D}{2}}{\pi\left(\frac{D^4-d^4}{64}\right)}$$

$$\frac{64PeD}{2\pi(D^4-d^4)} = \frac{P}{A}$$

$$\frac{64PeD}{2\pi(D^4-d^4)} = \frac{P}{\pi\left(\frac{D^2-d^2}{4}\right)}$$

$$\Rightarrow e = \frac{D^2+d^2}{8D}$$

816. A short hollow cast iron column has a cross section of 300 cm^2 and the section modulus Z = 10 × 10^5 mm^3 has

(i) Axial load of 250 kN

(ii) A load of 50 kN on a bracket attached to column whose load line is at 200 mm from the axis of the column

The maximum and minimum stress intensities respectively are

(a) 10 MPa and 10 MPa (b) 10 MPa and 0 MPa

(c) 0 MPa and 10 MPa (d) None of the above

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : Area of cross section A = 300 cm^2

Section modulus Z = 10 × 10^5 mm^3

Axial load P = 250 kN

Eccentric load = 50 kN

$$\boxed{\frac{M}{I}=\frac{\sigma}{y}} \qquad Z=\frac{I}{y}$$

(i) Stress = $\sigma = \frac{M}{Z} = \frac{WR}{Z} = \frac{250\times0}{10\times10^5} = 0$

(ii) $\sigma = \frac{WR}{Z} = \frac{50\times10^3\times200}{10\times10^5} = \frac{100}{10} = 10$

So, 0 is minimum and 10 is maximum

817. At a certain point in a structural member, there are perpendicular stresses 80 N/mm^2 and 20 N/mm^2, both tensile. What is the equivalent stress tension, according to the maximum principal strain theory? (Poisson's ratio = 0.25)

(a) Zero (b) 20 N/mm^2

(c) 60 N/mm^2 (d) 75 N/mm^2

ISRO Scientist/Engineer 2020

Ans. (d) $\frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} = \frac{\sigma_{eq}}{E}$

$$\sigma_1 - \mu\sigma_2 = \sigma_{eq}$$

$80 - (0.25)\,(20) = \sigma_{eq}$

$\sigma_{eq} = 75$MPa

818. Under axial loading, a material will fail in shear, if the ratio of shear strength to tensile strength is less than

(a) 1 (b) 3/4

(c) 1/2 (d) 1/4

ISRO Scientist/Engineer 2014

Ans : (c) $\tau = \frac{\sigma_t}{2} \Rightarrow \frac{\tau}{\sigma_t} = \frac{1}{2}$

819. Bending equation is

(a) M/I = R/E = F/Y (b) I/M = E/R = Y/R

(c) M/I = E/R = σ/Y (d) M/I = R/E = Y/F

GPSC AE (CLASS 1 & 2) 2019

ISRO Scientist/Engineer 2013

Ans : (c)

- Bending Equation or Ewer Bernoulli's Equation–

$$\left[\frac{\sigma}{Y}=\frac{M}{I}=\frac{E}{R}\right]$$

Where,

σ = Bending Stress

M = Bending Moment

I = Moment of Inertia

R = Radius of Curvature

Assumptions to calculate bending stress–

- Beam is initially straight, and has a constant cross-section
- Beam is made of homogeneous material and the beam has a longitudinal plane of symmetry.
- Resultant of the applied loads lies in the plane of symmetry.
- Plane cross-sections remains plane before and after bending.

820. For a prismatic beam of length L & moment of inertia I, the stiffness factor is

(a) EI/L (b) 2EI/L

(c) 3EI/L (d) 4EI/L

ISRO Scientist/Engineer 2013

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Ans : (a) Stiffness factor : It is the moment required to rotate the end while acting on it through a unit rotation, without translation of the far end of beam.

$$\left[\text{Stiffness Factor 'k'} = \frac{EI}{L}\right]$$

821. A beam of uniform strength means the quantity.....is constant throughout its length.

(a) P/A (b) FAy/Ib
(c) M/Z (d) TR/J

APTRASCO 2017

Ans : (c) For uniform strength beam stress is constant throughout.

$$\frac{M}{I} = \frac{\sigma}{y},\ \sigma = \frac{My}{I} = \frac{M}{Z}$$

822. The relation between the radius of curvature, R, and bending moment, M, and flexural rigidity EI, is given by:

(a) $R = \frac{M}{EI}$ (b) $M = \frac{EI}{R}$
(c) $EI = \frac{R}{M}$ (d) $E = \frac{MI}{R}$

WBPSC Poly Lect. 2019

Ans. (b) : Flexural formula

$$\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$$

$$\frac{M}{I} = \frac{E}{R}$$

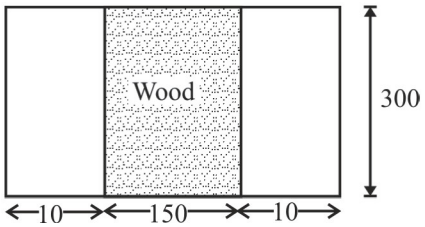
$$M = \frac{EI}{R}$$

823. A flitched beam consists of a wooden joist 150 mm wide and 300 mm deep strengthened by steel plates 10 mm thick and 300 mm deep one on either side of the joist. If modulus of elasticity of steel is 20 times that of wood, then the width of equivalent wooden section will be :

(a) 150 mm (b) 350 mm
(c) 500 mm (d) 550 mm

OPSC AEE 2019 Paper–I

Ans. (d)



$$M = \frac{E_s}{E_w} = 20$$

$$b_{ev} = b + 2mt$$

$$= 150 + 2 \times 20 \times 10 = 550 \text{ mm}$$

824. The diameter of the core of solid circular column of diameter "D", where stress induced due to a normal concentrated load of any eccentricity with respect to the centre of the column is :

(a) 0.20D (b) 0.25D
(c) 0.33D (d) 0.5D

OPSC AEE 2019 Paper–I

Ans. (b) : Condition for no tension is $\sigma_{min} = 0$

$$\frac{P}{A} - \frac{M}{Z} = 0$$

$$\frac{P}{A} = \frac{M}{Z}$$

$$\frac{P}{\frac{\pi}{4}D^2} = \frac{P \times e}{\frac{\pi}{32}D^3}$$

$$\therefore e = \frac{D}{8}$$

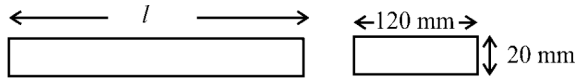
$$\therefore \text{ Diameter of core} = 2 \times \frac{D}{8} = \frac{D}{4} = 0.25D$$

825. A steel plate of width 120 mm and of thickness 20 mm is bent into a circular arc of radius 10 m. If Elastic modulus of the section is 200 GPa, the maximum stress induced in the beam is

(a) 200 MPa (b) 20 GPa
(c) 20 GPa (d) 180 MPa
(e) 110 MPa

CGPSC AE 2014 Shift–I

Ans. (a) :



$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$$\sigma = \frac{My}{I} = \frac{Ey}{R}$$

$$\sigma = \frac{Ey}{R} = \frac{200 \times 10^3 \times 20/2}{10 \times 1000}$$

$$\boxed{\sigma = 200 \text{ MPa}}$$

826. A simply supported beam of span L and constant width b carries a point load W at mid span. The depth of the beam required at the mid span for maximum extreme fiber stress p

(a) $d = \frac{3WL}{2bp}$ (b) $d = \sqrt{\frac{3WL}{2bp}}$
(c) $d^3 = \frac{3WL}{2bp}$ (d) $d = \frac{3WL}{2bp^2}$

APPSC AEE MAINS 2016

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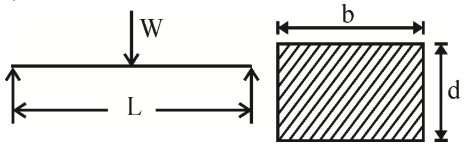
Ans. (b) :



Constant width b

$$M = \frac{WL}{4}$$

$$\frac{M}{I} = \frac{p}{y}$$

$$\frac{WL}{4 \times \frac{1}{2} bd^3} = \frac{p}{\frac{d}{2}}$$

$$\frac{3WL}{bd^2} = 2p$$

$$d = \sqrt{\frac{3WL}{2bp}}$$

827. A simply supported beam of constant with and varying depth and uniform strength is subjected to a central concentrated load. The depth of the beam d_x at a distance x from one of the supports is proportional to-

(a) $x^{1/2}$ (b) $x^{1/3}$

(c) x (d) x^2

JPSC AE 2009

Ans. (a) : By bending equation-

$$\frac{M}{I} = \frac{\sigma}{y}$$

Uniform strength is given, so σ is constant.

$$M = \frac{W}{2} \times x$$

W = concentrated load

$$\frac{I}{y} = \frac{bd^2}{6}$$

On substituting

$$\frac{Wx}{2\sigma} = \frac{bd^2}{6}$$

So, $d^2 \propto x \rightarrow \boxed{d \propto x^{\frac{1}{2}}}$

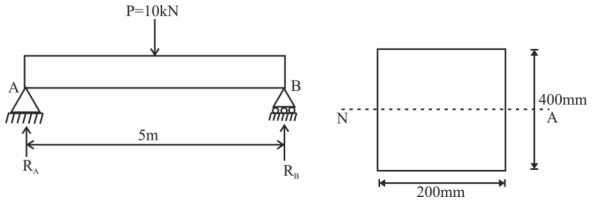
828. A beam having rectangular cross section 200 mm wide and 400 mm deep is simply supported over a span of 5 m. It is carrying a concentrated load of 10 kN at the centre of the span. The maximum bending stress developed at quarter span of the beam is

(a) 2.34 N/mm^2 (b) 1.25 N/mm^2

(c) 1.17 N/mm^2 (d) 2.0 N/mm^2

TSGENCO_2015

Ans. (c) :



$$R_A = R_B = \frac{P}{2}$$

$R_A = R_B = 5$ kN

$$\text{Moment} = M_{at}x = \frac{L}{4} = R_A \times \frac{L}{4} = \frac{5 \times 5}{4} = 6.25kN - m$$

$$\text{Bending stress} = \sigma_b = \frac{mY}{I} = \frac{6.25 \times 10^6 \times 200}{\frac{(400)^3 \times 200}{12}}$$

Bending stress = σ_b = 1.17 N/mm^2.

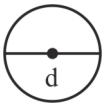
829. For a circular section, the elastic modulus (Z_e) is : (d = diameter)

(a) $\frac{\pi}{32}d^3$ (b) $\frac{\pi}{16}d^3$

(c) $\frac{\pi}{32}d^2$ (d) $\frac{\pi}{16}d^2$

SJVNL ET 2019

Ans. (a) :



We know that $I = \frac{\pi}{64}d^4$, $y = d/2$

Elastic modulus (z_e) = $\frac{I}{y}$

put this value in eq^n

$$z_e = \frac{\frac{\pi}{64}d^4}{\frac{d}{2}}$$

$$z_e = \frac{\pi}{32}d^3$$

830. What are the dimensions of flexural rigidity of a beam element?

(a) MT (b) MT^{-2}

(c) ML^3T^{-2} (d) MLT^{-2}

UPPCL AE 2015

Ans. (c) : Dimensions of flexural rigidity–

$EI = N/m^2\ m^4$

$= N - m^2$

$= [MLT^{-2}]\ [L^2]$

$\boxed{EI = [ML^3T^{-2}]}$

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831. The resultant cuts the base of the circular column of diameter 'd' with an eccentricity equal to 'd/4'. The ratio between the maximum compressive stress and maximum tensile stress is

(a) 3 (b) 4
(c) 5 (d) infinite

GPSC AE June 2019

Ans. (a) : For solid circular section, the maximum eccentricity for tension is d/8

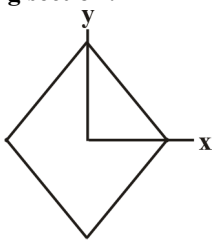
$$\sigma_{c_i t} = \frac{4P}{\pi d^2}\left[1 \pm \frac{8e}{d}\right]$$

$$\text{for } e = d/4$$

$$\frac{\sigma_c}{\sigma_t} = \frac{\left(1+8\,e/d\right)}{\left(1-\frac{8e}{d}\right)} = \frac{3}{-1}$$

The minus sign is dve to opposite nature of force.

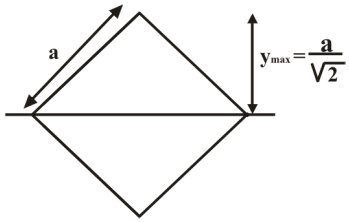
832. A square section of side 'a' is oriented as shown in the figure. Determine the section modulus of the following section?



(a) $\frac{a^4}{12\sqrt{2}}$ (b) $\frac{a^3}{12\sqrt{2}}$
(c) $\frac{a^4}{6\sqrt{2}}$ (d) $\frac{a^3}{6\sqrt{2}}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) :



The section modulus is

$$Z = \frac{I}{y_{max}} = \frac{\left[\frac{a.a^3}{12}\right]}{\left[\frac{a}{\sqrt{2}}\right]} = \frac{a^3}{6\sqrt{2}}$$

833. A mild steel flat of width 120mm and thickness 10mm is bent into an arc of a circle of radius 10m, by applying a moment 'M'. If 'E' is 2×10^5 N/mm^2, magnitude of 'M' will be:-

(a) 0.2×10^5 N.mm (b) 2×10^5 N.mm
(c) 2×10^6 N.mm (d) 0.2×10^4 N.mm

UPPCL AE 2014

Ans. (b) : $\frac{M}{I} = \frac{E}{R}$

$$I = \frac{bd^3}{12} = \frac{120\times(10)^3}{12}\ (mm)^4$$

$E = 2 \times 10^5$ N/mm^2

$R = 10m = 10^4$ mm

$$M = \frac{EI}{R} = \frac{2\times10^5\times120\times10^3}{10^4\times12}$$

$M = 2 \times 10^5$ N-mm

834. Beam of uniform strength means

(a) A beam with uniform cross section
(b) A beam subjected to same moment throughout
(c) A beam in which maximum bending stress is same at all sections
(d) A beam with same maximum shear stress at all section

UPRVUNL AE 2015

Ans. (c) : Beam section should be plane before and after the bending.

835. The relationship between bending moment M, flexural rigidity EI and radius of curvature R is given by -

(a) EI = MR (b) EI = 1/MR
(c) EI = M/R (d) EI = R/M

Rajasthan VP ITI 2018

Ans. (a) According to bending equation –

$$\boxed{\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}}$$

Where,
M = Bending moment of section
I = Moment of inertia of section
E = Young modulus
R = Radius of curvature
σ = Maximum bending moment of section
y = Distance between natural axis to extreme fibre.
From equation EI=MR

836. In case of circular section the section modulus is given by :

(a) $\frac{\pi d^2}{16}$ (b) $\frac{\pi d^3}{32}$
(c) $\frac{\pi d^2}{16}$ (d) $\frac{\pi d^4}{64}$

UP Jal Nigam AE 2016

Ans. (b) :

Section modulus = $\frac{I}{y} = \frac{\pi d^4}{64}\times\frac{2}{d} = \frac{\pi d^3}{32}$

Section modulus of circular section = $\frac{\pi d^3}{32}$

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837. Parallel axis theorem :

(a) $I_{AB} = I_a + hA^2$ (b) $I_{AB} = I_G + Ah$

(c) $I_{AB} = I_a + Ah^2$ (d) $I_{GG} = I_{AB} + Ah^2$

UP Jal Nigam AE 2016

Ans. (c) : Parallel axis theorem

$I_{AB} = I_G + Ah^2$

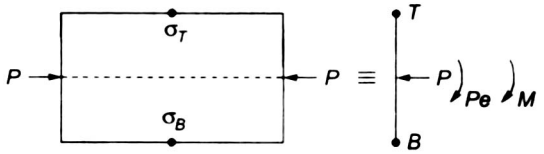
838. When the tendons are placed at an eccentricity of e, the extreme fibre stresses in beam are-

(a) $\frac{P}{A} \mp \frac{Pe}{Z} \mp \frac{M}{Z}$ (b) $\frac{P}{A} \mp \frac{Pe}{A} \mp \frac{M}{Z}$

(c) $\frac{P}{Z} \mp \frac{Pe}{Z} \mp \frac{M}{Z}$ (d) $\frac{P}{A} \mp \frac{Pe}{A} \mp \frac{M}{A}$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (a) :



$$\sigma_T = \frac{P}{A} - \frac{Pe}{Z} - \frac{M}{Z}$$

$$\sigma_B = \frac{P}{A} + \frac{Pe}{Z} + \frac{M}{Z}$$

839. Two beams with equal cross-sections, made of the same materials, having the same support conditions, and each are loaded with equal uniform load per unit length. One beam is twice as long as the other. The maximum bending stress in the longer beam is larger by a factor of :

(a) 1.25 (b) 2

(c) 3 (d) 4

HPPSC Poly. Lect. 2016

Ans : (b) We know that the bending equation is given by–

$$\boxed{\frac{M}{I} = \frac{\sigma}{y}} = \frac{E}{R}, \text{ [where same material is given]}$$

So, $\sigma \propto M$

then, $\frac{\sigma_1}{\sigma_2} = \frac{M_1}{M_2}$

$$\frac{\sigma_2}{\sigma_1} = \frac{2M}{M}$$

$$\boxed{\frac{\sigma_2}{\sigma_1} = 2}$$

840. A beam of rectangular cross-section of width 200 mm and depth 500 mm is subjected to a bending moment of 6 kN-m. The bending stress at a distance of 250 mm from bottom fiber is :

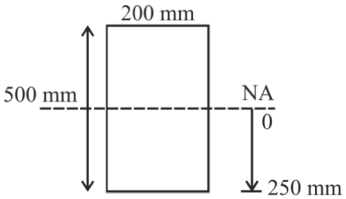
(a) 50 N/mm^2 (b) 100 N/mm^2

(c) 150 N/mm^2 (d) None of these

HPPSC Poly. Lect. 2016

Ans : (d) Given,

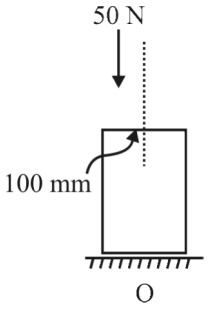
BM = 6 kN-m



$\bar{y} = 0$

$$\frac{M}{I} = \frac{\sigma}{y} \qquad \boxed{\sigma = 0}$$

841. The equivalent force couple system for given lamina at point O will be:



(a) Moment 5000 Nmm

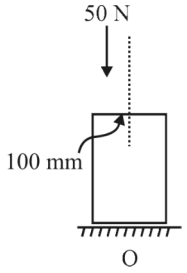
(b) Force 50 N

(c) Force 50 N and Moment 5000 Nmm

(d) Force 50 N and Moment 5000 Nmm

IRCON AE 2017

Ans. (d) : $\varepsilon F_y = 0$



$R_0 + 50 = 0$

$R_0 = -50$ kN (upward)

$M_0 = 50 \times 100 = 5000$ Nmm

(Bending moment at point O)

842. The eccentricity in a hollow circular column of external diameter 25 cm, internal diameter 15 cm for a eccentric load of 1000 KN" for no-tension condition is

(a) 2.75 cm (b) 3.00 cm

(c) 3.50 cm (d) 4.25 cm

WBPSC AE 2016

Ans. (d) :

$$e = \frac{D^2 + d^2}{8D} = \frac{25^2 + 15^2}{8 \times 25}$$

$$= \frac{625 + 225}{8 \times 25} = \frac{850}{200} = 4.25\,\text{cm}$$

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5. Slope and Deflection

843. In a simply supported beam carrying a uniformly distributed load over its entire span, slope is maximum at
(a) 1/4 from either end
(b) 1/3 from either end
(c) mid-span supported ends
(d) supported ends

Chandigarh AE 2017

Ans. (d) : In a simply supported beam carrying a uniformly distributed load over its entire span, slope is maximum at supported ends.

844. Castigliano's first theorem is applicable
(a) for elastic structure
(b) for all statically determinate structures
(c) only when principle of super position is valid
(d) for all indeterminate structure

Arunachal Pradesh Poly. Tech. Lect. 2019
APPSC AEE 2012

Ans. (c) Castigliano's first theorem – When forces act one elastic systems subject to small displacement, the displacement corresponding to any force in direction of the force is equal to the partial derivative of the total strain energy with respect to that force.

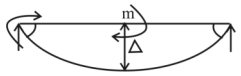
$$\delta_i = \frac{\partial U}{\partial F_i}$$

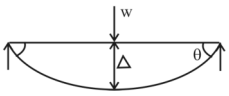
This theorem is valid only when principle of superposition is valid.

845. Unit load method used in the structural analysis is :
(a) application only to statically indeterminate structures
(b) another name for stiffness method
(c) an extension of Maxwells Betti Theorem
(d) based on Catigliano's Theorem

PSPCL AE 2012

Ans. (d) :





Unit load method is designed for determine the deflection or rotation at any point in the section if deflection is caused by loading (moment or load)
– It is further derive of Castigliano's theorem.
ie, based on it.

$$w\Delta = M$$

846. If the strain energy of a deformed elastic body is represented as a function of the displacement d1, d2,....., a partial derivation of that function with respect to any chosen displacement gives the corresponding force. This statement/ principle is called as :
(a) Principle of strain energy
(b) Maxwell's law of reciprocal deflections
(c) Conjugate beam / truss principle
(d) First theorem of Castigliano
(e) Bette's law

CGPSC AE 2014 Shift–I

Ans. (d) : Castigliano's first theorem

$$\boxed{\frac{\partial U}{\partial P} = \Delta} \quad \& \quad \boxed{\frac{\partial U}{\partial M_i} = \theta_i}$$

Where,
$U \rightarrow$ Total strain energy
$\Delta \rightarrow$ Displacement in the direction of force P
$\theta \rightarrow$ Rotation in the direction of moment M

847. If E is the modulus of elasticity and I is the moment of inertia, then the flexural rigidity or flexural stiffness of the member is
(a) $\frac{E}{I}$ (b) EI
(c) $\frac{I}{E}$ (d) EI^2

TNPSC AE 2018

Ans. (b) : The product EI is an index of the bending (flexural) strength of an element and is called flexural rigidity of the element. Differential equation of beam is

$$\boxed{M = EI\frac{d^2y}{dx^2}}$$

848. Moment area method can be easily adopted for the following case :
(a) Simply supported beam with eccentric point load
(b) Simply supported beam with point loads & UDL
(c) Simply supported beam with two symmetrically placed unequal point loads
(d) Cantilever with point loads and UDL

TNPSC AE 2015

Ans. (d) Moment area Method (Mohr's Method):-
-This method determines slope and deflection due to bending only.
-This method is use-full particularly for beams which is subjected to series of concentrated loads or having segments with different M.I.
According to Mohr's Method-
- Change in slope is the area of $\frac{M}{EI}$ diagram between the two points.
- Deflection of any point is equal to moment of area of $\frac{M}{EI}$ diagram between two points.
- Calculating area of $\frac{M}{EI}$ diagram of the cantilever beam with UDL or Point Load is easy, as they require standard results for calculations.

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849. When the left support of a fixed beam rotates, the fixed end moment at right support is:

(a) 3 EI θ_L/L (b) 4 EI θ_L/L

(c) 2 EI θ_L/L (d) 3 EI θ_L/L

APPSC Poly. Tech. Lect. 2020

Ans. (a) : Allowing left end or support rotates by an amount θ_A, the fixed end moment will be (at right support)–

$$FEM = \frac{3EI\theta_L}{L}$$

850. The.....at any section in a given beam is equal to.....at corresponding section in conjugate beam.

(a) slope, shear force

(b) deflection, shear force

(c) slope, bending moment

(d) slope, deflection

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (a) The slope at any section in a given beam is equal to shear force at corresponding section in conjugate beam

851. A two span continuous beam having equal spans each of length *l* is subjected to u.d.l. of w per unit run over the whole beam. The beam has constant EI. The bending moment at the middle support is

(a) $\frac{wl^2}{4}$ (b) $\frac{wl^2}{8}$

(c) $\frac{wl^2}{12}$ (d) $\frac{wl^2}{16}$

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (b) Theorem of three moment

$$M_A l_1 + 2M_B(l_1 + l_2) + M_C l_2 = \frac{w_1 l_1^3}{4} + \frac{w_2 l_2^3}{4}$$

Given that,

$M_A = M_C = 0$, $l_1 = l_2 = l$, $w_1 = w_2 = w$

Put the value

$$0 \times l + 2M_B(l + l) + 0 \times l = \frac{wl^3}{4} + \frac{wl^3}{4}$$

$$2M_B \times 2l = \frac{2wl^3}{4}$$

$$\boxed{M_B = \frac{wl^2}{8}}$$

852. The slope of the bending moment diagram of any section of loaded beam is

(a) Torsion at the section

(b) Maximum shear force of the beam

(c) Shear force at that section

(d) Maximum bending moment of the beam

RPSC POLY. TECH. LECT. 2014

Ans : (c) Slope of BMD at any section gives shear force at that section.

$$S_x = \frac{dM}{dx}$$

Similarly, slope of shear force diagram shows load acted at that point.

$$w = \frac{dS}{dx}$$

853. Radius of curvature is

(a) $\frac{d^2y}{dx^2}$ (b) $\frac{M}{E}$

(c) $\frac{M}{I}$ (d) $\frac{My}{EI}$

TRB Poly. Tech. Lect. 2012

Ans. (a) :

We know $\boxed{\frac{M}{EI} = \frac{1}{R} = \frac{d^2y}{dx^2}}$

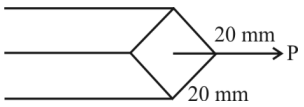
854. A square bar of size 20 mm × 20 mm is subjected to direct tensile force. What is the largest tensile force the bar can sustain if its shear strength is 100 MPa?

(a) 100 MPa (b) 50 MPa

(c) 200 MPa (d) 150 MPa

TRB Poly. Tech. Lect. 2012

Ans. (c) :



τ_{max} = 100 MPa (shear stress/Strength)

$$\sigma_1 = \frac{P}{20 \times 20}, \sigma_2 = 0$$

$$\therefore \tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

$$\Rightarrow \sigma_1 = 2 \times 100$$

$$\boxed{\sigma_1 = 200 MPa}$$

855. Match the List-I with List-II and select the correct answer using the codes given below the lists :

Real Beam (support conditions)		Corresponding conjugate beam (support conditions)	
A.	Fixed support	1.	Interior support
B.	Free end	2.	Hinge
C.	Interior support	3.	Fixed support
D.	Hinge	4.	Free end

Codes :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	3	4	1	2
(d)	4	3	2	1

OPSC Civil Services Pre 2011

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Ans. (d) :

	Real Beam (support conditions)	Conjugate beam (support conditions)
A	Fixed support	Free End
B	Free end	Fixed support
C	Interior support	Hinge
D	Hinge	Interior support

856. Consider the following statements:

1. **The deformation of structures subjected to several simultaneously applied loads may be obtained by algebraically summing the deformations due to the individual loads.**
2. **Such superposition holds both for linear and nonlinear systems.**

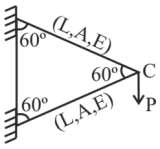
Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

UPSC JWM 2017

Ans. (a) : According to Castigliano's theorem sum of deformation of individuals load equal to deformation of summation of force applied.

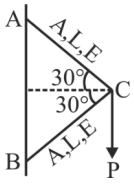
857. The vertical deflection of joint C of the frame shown below:



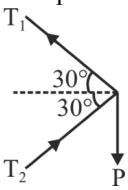
(a) PL/AE (b) 2PL/AE
(c) PL/2AE (d) 3PL/AE

Haryana PSC AE 2017, APPSC AEE Mains 2016

Ans. (b) :



Considering equilibrium at point 'C'



$\sum F_x = 0$

$T_1 \cos 30° = T_2 \cos 30°$

$T_1 = T_2 = T$

$\sum F_y = 0$

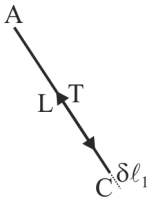
$T_1 \sin 30° + T_2 \sin 30° = P$

$\frac{T_1}{2} + \frac{T_2}{2} = P$

$T_1 = T_2 = T = P$ .....(1)

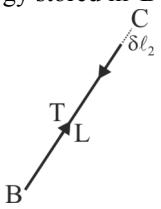
Let assume elongation in 'AC' is $\delta\ell_1$ and contraction in 'BC' is $\delta\ell_2$

So strain energy stored in 'AC'



$$\frac{1}{2} T \times \delta\ell = \frac{1}{2} T \times \frac{TL}{AE} = \frac{1}{2AE} T^2 \ell$$

How total strain energy stored in 'BC'



$$\frac{1}{2} T \times \delta\ell_2 = \frac{1}{2AE} T^2 \ell$$

Total strain energy stored in body = workdone by load P

Suppose vertical direction at 'C' is 'x'

The workdone by P is $= \frac{1}{2} P \times x$

$$\frac{1}{2AE} T^2 \ell + \frac{1}{2} \frac{T^2 \ell}{AE} = \frac{1}{2} P \times x$$

$$\frac{T^2 \ell}{AE} = \frac{1}{2} P \times x$$

$$x = \frac{2T^2}{P} \frac{L}{AE}$$

From equation (1) $T = P$

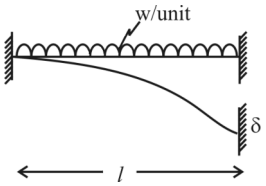
Then $\boxed{x = \frac{2PL}{AE}}$

858. A fixed beam AB of span L carries a uniformly distributed load w per unit length. During loading, the support B sinks downwards by an amount δ. If $\delta = wL^4/(72\,EI)$ what is the fixing moment at B?

(a) $wL^2/12$ (b) $wL^2/6$
(c) $6EI\delta/L^2$ (d) zero

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (d) :



Fixed end moment at B due to UDL $= \frac{wL^2}{12}$

Fixed end moment at B due to sink support $= \frac{6EI\delta}{L^2}$ (anti clock wise)

$\therefore$ Final moment A to B $= \frac{wL^2}{12} - \frac{wL^2}{12} = 0$

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859. If E is Young's modulus and I is moment of inertia, then the expression $EI\frac{d^3y}{dx^3}$ at any section for a beam is equal to

(a) load intensity is equal to
(b) shear force at the section
(c) bending moment at the section
(d) slope at the section

TNPSC AE 2018

Ans. (b) :

Differential equation of beam

$$EI\frac{d^2y}{dx^2} = M;\ \text{differentiate w.r.t } x$$

$$EI\frac{d^3y}{dx^3} = \frac{dM}{dx} = V = \text{shear force at the section.}$$

$$EI\frac{d^3y}{dx^3} = V$$

860. If E is young's modulus and I is moment of inertia, then the expression $EI\frac{d^4y}{dx^4}$ at any section for a beam is equal to

(a) Load intensity at the section
(b) Shear force at the section
(c) Bending moment at the section
(d) Slope at the section

TNPSC AE 2019
UP Jal Nigam AE 2016

Ans. (a) $EI\frac{d^4y}{dx^4} \rightarrow \text{load intensity}$

$$EI\frac{d^4y}{dx^4} \rightarrow q(\text{load intensity})$$

$$EI\frac{d^3y}{dx^3} \rightarrow \text{Shear force}$$

$$EI\frac{d^2y}{dx^2} \rightarrow \text{Moment}$$

$$EI\frac{dy}{dx} \rightarrow \text{Slope}\,\theta$$

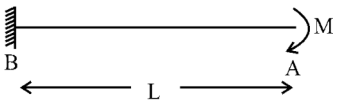
$$EIy \rightarrow \text{deflection}$$

861. A cantilever beam of length L (EI being constant throughout the section) is subjected to a couple M at the free end. The slope and deflection at the free end will be given by

(a) ML/EI & $ML^2/2EI$ (b) $2ML/EI$ & $ML^2/2EI$
(c) ML/EI & $ML^2/4EI$ (d) ML/EI & $ML^2/8EI$
(e) ML^3/EI & $ML^2/2EI$

CGPSC AE 2014 Shift–I

Ans. (a) :



$$\text{Slope, } \theta_A = \frac{ML}{EI}$$

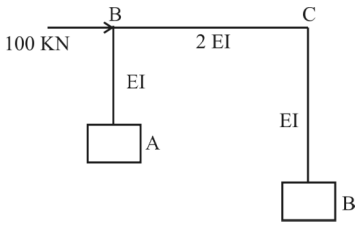
$$\text{Deflection, } \delta_A = \frac{ML^2}{2EI}$$

862. The slope deflection equation at B of element BC in the following frame is (AB = 6 m and BC = CD = 8m)

(a) $M_{BC} = 0.5\ EI\ (2\theta_C - \theta_B)$
(b) $M_{BC} = 0.5\ EI\ (2\theta_B - \theta_C)$
(c) $M_{BC} = 0.5\ EI\ (2\theta_B + \theta_C)$
(d) $M_{BC} = 0.5\ EI\ (2\theta_C - \theta_B)$
(e) $M_{BC} = 0.33\ EI\ (2\theta_C - \theta_B)$

CGPSC AE 2014 Shift–I

Ans. (c) :



Slope deflection equation

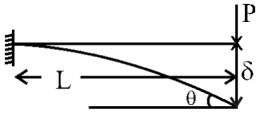
$$M_{BC} = \bar{M}_{BC} + \frac{2EI}{L}(2\theta_B + \theta_C)$$

Since member BC having no external loading fixed and moment $\bar{M}_{BC} = 0$

$$\therefore\ M_{BC} = \frac{2EI}{8} \times 2(2\theta_B + \theta_C)$$

$$M_{BC} = 0.5\ EI(2\theta_B + \theta_C)$$

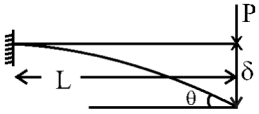
863. The slope and deflection of the free-end of a cantilever beam shown in the figure respectively are



(a) $PL^2/2EI$, $PL^3/3EI$ (b) $PL^2/4EI$, $PL^3/3EI$
(c) $PL^2/2EI$, $PL^3/6EI$ (d) $PL^2/4EI$, $PL^3/6EI$
(e) $PL^2/4EI$, $PL^3/9EI$

LMRC AM 2015
CGPSC AE 2014 Shift–I

Ans. (a) :



$$\text{Direct Result } \theta = \frac{PL^2}{2EI},\ \delta = \frac{PL^3}{3EI}$$

864. The differential equation of the deflection curve generated due to the temperature difference between the bottom and top fibers for the beam loaded as shown in the following figure can be given as (assume α as the coefficient of thermal expansion) :

(a) $\alpha(T_1 - T_2 \times T_1)/h$ (b) $\alpha(T_1 - T_2)/h$
(c) $\alpha(T_1 \times T_2 - T_2)/h$ (d) $\alpha(T_2 - T_1)/h$
(e) $\alpha(T_1 \times T_2 - T_2)/(T_1 \times h)$

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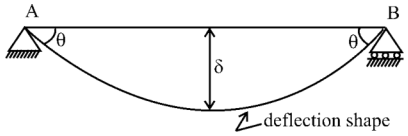
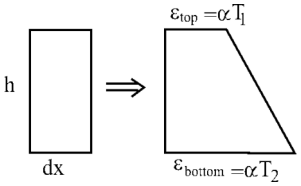
Ans. (d) :



$T_2 > T_1 \Rightarrow$ Compression of top & Tension at bottom

Now strain diagram



Differential equation

$$\frac{d^2y}{dx^2} = \frac{M}{EI} = \frac{1}{R} = K(\text{curvature})$$

$\therefore$ We have to find curvature (K)

we know,

$\varepsilon_{mean} = K.h$

$$K = \frac{\varepsilon_{mean}}{h} = \frac{\alpha(T_2 - T_1)}{h}$$

$$\therefore \boxed{\frac{d^2y}{dx^2} = \frac{\alpha(T_2 - T_1)}{h}}$$

865. A 6.4 m long simply supported wrought iron beam carries a point load of 48 kN at its mid-point. The moment of inertia is 80×10^6 mm^4. If the elastic modulus is 192 GPa, the deflection at the mid -point of the beam is :

(a) 15.06 (b) 17.3

(c) 17.8 (d) 17.06

(e) 17.06

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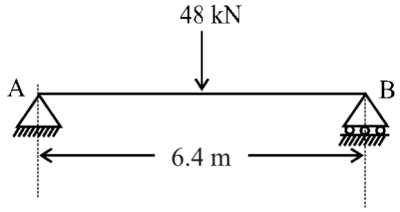
Ans. (e) :



$$\Delta_C = \frac{PL^3}{48EI}$$

$$= \frac{48\times10^3 \times (6.4\times10^3)^3}{48\times192\times10^3\times80\times10^6}$$

$$\boxed{\Delta_C = 17.06\,\text{mm}}$$

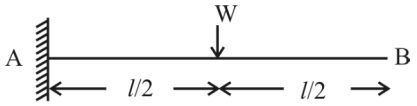
866. A cantilever AB is subjected to a concentrated load at the free end, the slope and deflection at the free end are $WL^2/2EI$ and $WL^3/3EI$. If the same load is applied at mid-span point the deflection at the free end will be

(a) $\frac{5}{384}\frac{Wl^3}{EI}$ (b) $\frac{5}{48}\frac{Wl^3}{EI}$

(c) $\frac{Wl^3}{6EI}$ (d) $\frac{Wl^3}{16EI}$

GPSC AE March 2018, GPSC AE Dec 2018
Maharashtra PSC Mains 2018 Paper-1
GPSC Assistant Prof. 2016, NPSC AE 2012

Ans. (b) :



$$Y_B = \frac{Wl_1^3}{3EI} + (l - l_1)\frac{Wl_1^2}{2EI}$$

$$Y_B = \frac{W(l/2)^3}{3EI} + (l - l/2)\frac{W(l/2)^2}{2EI}$$

$$Y_B = \frac{Wl^3}{24EI} + \frac{Wl^3}{16EI} = \frac{5}{48}\frac{Wl^3}{EI}$$

867. Maximum deflection in a simply supported beam subjected to pure bending is given by :

or

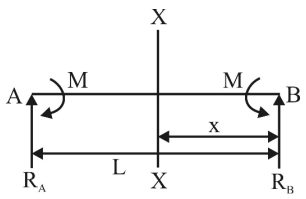
A beam simply supported at both the ends, of length L carries two equal unlike couples M at two ends. If the flexural rigidity EI = constant, then the central deflection of beam is given by

(a) $\delta = \frac{ML}{2EI}$ (b) $\delta = \frac{ML^2}{8EI}$

(c) $\delta = \frac{ML^2}{12EI}$ (d) $\delta = \frac{5}{48}\frac{ML^2}{EI}$

Arunachal Pradesh Poly. Tech. Lect. 2019
***ESE 1995,* TNPSC AE 2015**

Ans. (b) :



for section X–X,

$$EI\frac{d^2y}{dx^2} = -M$$

$$EI\frac{dy}{dx} = -Mx + C_1 \quad \text{.....(i)}$$

$$\frac{dy}{dx} = \text{slope}$$

if $x = L/2$ then $\theta = 0$

$$EI\times0 = -M\times\frac{L}{2} + C_1$$

$$C_1 = \frac{ML}{2}$$

from eq^n (i)

$$EIy = \frac{-Mx^2}{2} + \frac{ML}{2}.x + C_2$$

if $x = 0$ then $y = 0$

$C_2 = 0$

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deflection at x = L/2

$$EIy = \frac{-M(L/2)^2}{2} + \frac{ML}{2} \times L/2$$

$$EIy = \frac{ML^2}{8}$$

$$\boxed{y = \frac{ML^2}{8EI}}$$

868. Other parameters being unchanged, if the span of a cantilever carrying end point load is doubled, the maximum slope is increased by :

(a) 2 times (b) 4 times
(c) 8 times (d) 16 times

TNPSC AE 2015

Ans. (b) :

At point B

$$\text{slope } (\theta) = \frac{WL^2}{2EI}$$

$$\text{deflection } (\delta) = \frac{WL^3}{3EI}$$

If length is twice

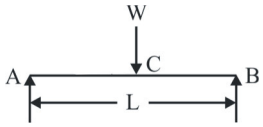
$$\text{slope } (\theta) = \frac{W(2L)^2}{2EI} = 4 \times \frac{WL^2}{2EI} = 4 \text{ times}$$

869. If maximum slope of a simply supported beam of span 'L' with midpoint load is 1^0, then maximum deflection is :

(a) πL/240 (b) πL/270
(c) πL/540 (d) πL/576

TNPSC AE 2015

Ans. (c) :



$$\text{slope at point C } (\theta_c) = \frac{WL^2}{16EI} = 1^0$$

$$\theta_c = \frac{\pi}{180}$$

$$\text{Deflection at point C } (y_c) = \frac{WL^3}{48EI}$$

$$y_c = \frac{WL^2}{16EI} \times \frac{L}{3}$$

$$y_c = \frac{\pi}{180} \times \frac{L}{3}$$

$$\boxed{y_c = \frac{\pi L}{540}}$$

870. A cantilever beam PQ of length L m and cross-section B×D mm^2 is subjected to UDL of w kN/m throughout the span. Another cantilever beam RS (made-up of same material) of length 2L metre and cross-section 2B×2D mm^2 is subjected to same amount of UDL throughout the span. What is the ratio of deflection at the 'free end' for beam PQ to that of beam RS?

(a) 1 (b) 8
(c) 1/2 (d) 1/8

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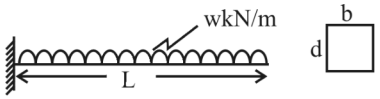
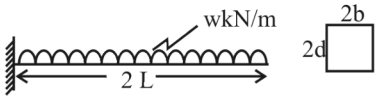
Ans. (a) :

For beam PQ, deflection at free end $= \frac{wL^4}{8EI}$, $I = \frac{bd^3}{12}$



cantilever beam PQ



cantilever beam RS

For beam RS, deflection at free end $= \frac{w \times (2L)^4}{8EI}$

$$I = \frac{2b \times (2d)^3}{12}$$

$$\frac{PQ}{RS} = \frac{wL^4/bd^3}{w(2L)^4/2b \times (2d)^3}$$

$$\frac{PQ}{RS} = 1$$

871. An overhanging beam, having same overhangs of length L/2 on both the sides, is subjected to clockwise moment M on the left side free end and anticlockwise moment M on the right side free end. What is the deflection at midpoint of the beam, if distance between the supports is L? (Take Flexural Rigidity of beam = EI)

(a) Zero (b) $ML^2/2EI$
(c) $ML^2/8EI$ (d) $ML^2/9\sqrt{3}EI$

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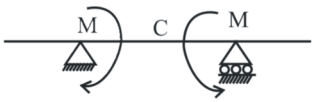
Ans. (c) :



$ML^2/8EI$ (standard result)

$$EI\frac{d^2y}{dx^2} = M$$

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$$\text{or, } EI\frac{dy}{dx} = Mx + C$$

$$\text{or, } EI \times y = M\frac{x^2}{2} + C_1x + C_2$$

At x = 0, y=0

x = L, y = 0

$C_2 = 0$

$$EI \times 0 = \frac{ML^2}{2} + C_1 L \quad \text{at } x = L, y = 0$$

or, $$C_1 = \frac{Mx^2}{2} - \frac{ML}{2}.x$$

$$(EIY)_{x=\frac{l}{2}} = \frac{ML^2}{8} - \frac{ML^2}{4} = -\frac{ML^2}{8}$$

$$\boxed{y = \frac{ML^2}{8EI}}$$

872. When one of the supports of a beam is at a lower level as compared to the other, it will cause a moment at both ends. The magnitude of this moment introduced in slope deflection equation is

(a) $-\frac{3EI\delta}{l^2}$ (b) $-\frac{4EI\delta}{l^2}$

(c) $-\frac{6EI\delta}{l^2}$ (d) $-\frac{2EI\delta}{l^2}$

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Ans : (c) When one support of a beam is at lower level or sinks, the moment generated will be = $-\frac{6EI\delta}{l^2}$

873. A cable is supported at both ends at the same level and is subjected to U.D.L. over the entire span. If y_c is the central dip and 'l' is the span of a cable, then horizontal thrust developed at the support is

(a) $\frac{wl^3}{8y_c}$ (b) $\frac{wl^2}{3y_c}$

(c) $\frac{wl^2}{8y_c}$ (d) $\frac{wl^2}{2y_c}$

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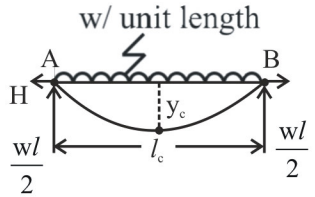
Ans : (c)



Taking moment about c

$$-H_A \times y_c + V_A \times \frac{l}{2} - \frac{wl}{2} \times \frac{l}{4} = 0$$

$$H_A.y_c = \frac{wl}{2} \times \frac{l}{2} - \frac{wl^2}{8}$$

$$= \frac{wl^2}{4} - \frac{wl^2}{8}$$

$$H_A.y_c = \frac{wl^2}{8}$$

$$\boxed{H_A = \frac{wl^2}{8y_c}}$$

874. A two span continuous beam having equal spans each of length l is subjected to u.d.l. w per unit length. The beam has constant EI. The reaction at the middle support is

(a) wl (b) $\frac{5wl}{2}$

(c) $\frac{5wl}{4}$ (d) $\frac{5wl}{8}$

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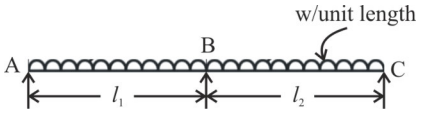
Ans : (c)



Given that,

$l_1 = l_2 = l$, $w_1 = w_2 = w$, $M_A = M_L = 0$

Three moment theorem,

$$M_A l_1 + 2M_B(l_1 + l_2) + M_C l_2 = \frac{w_1 l_1^2}{4} + \frac{w_2 l_2^2}{4}$$

$$0 \times l_1 + 2M_B(l + l) + 0 \times l_2 = \frac{wl^2}{4} + \frac{wl^2}{4}$$

$$\boxed{M_B = \frac{wl^2}{8}}$$

Consider about half part of beam

$$-2M_B = R_A l - \frac{wl^2}{2}$$

$$R_A l = \frac{wl^2}{2} - M_B = \frac{wl^2}{2} - \frac{wl^2}{8} = \frac{3wl^2}{8}$$

$$R_A = \frac{3wl}{8}$$

$$R_A = R_C = \frac{3wl}{8}$$

now $\Sigma V = 0$

$$R_A + R_B + R_C = w(l + l) = 2wl$$

$$R_B = 2wl - (R_A + R)$$

$$= 2wl - \left(\frac{3wl}{8} + \frac{wl^2}{8}\right)$$

$$= \frac{8wl - 3wl}{4} = \frac{5wl}{4}$$

$$\boxed{R_B = \frac{5wl}{4}}$$

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875. A simply supported beam subjected to uniformly varying load of intensity W at left end (A) reduces to zero at right end (B). Then slope at end B is

(a) $\frac{5WL^3}{360EI}$ (b) $\frac{9WL^3}{360EI}$

(c) $\frac{3WL^3}{360EI}$ (d) $\frac{7WL^3}{360EI}$

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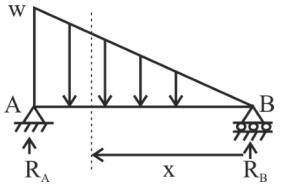
Ans : (d)



(i) Total load $= \frac{0+W}{2} \times L = \frac{WL}{2}$

(ii) Reaction $R_A = \frac{WL}{3}$, $R_B = \frac{Wl}{6}$

(iii) $M_x = \left(\frac{0+Wx}{2}\right) x \times \frac{x}{3} - \frac{Wl}{6} \times x = \frac{Wx^3}{6L} - \frac{WLx}{6}$

$M_x = \frac{WLx}{6}\left(1 - \frac{x^2}{L^2}\right)$

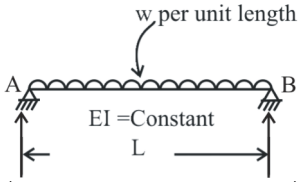
maximum shear force at

$x = \frac{L}{\sqrt{3}}$

Slope at edge of 'B'

$\theta_B = \frac{7WL^3}{360EI}$

876. The deflection at the mid span of beam AB unit load method will be–



(a) $wL^4/128EI$ (b) $wL^4/384EI$

(c) $5wL^4/128L$ (d) None of above

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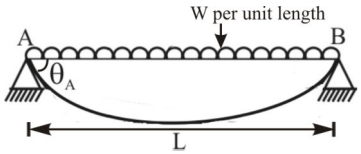
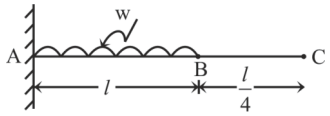
Ans : (d)



$\theta_A = \left|\frac{w\ell^3}{24EI}\right|$; $\theta_B = -\frac{w\ell^3}{24EI}$

Deflection at mid point– $y = \frac{5w\ell^4}{384EI}$

877. A cantilever beam 'AC' of uniform cross-section carries a uniformly distributed load over the portion 'AB' of length '*l*' as shown. Slope at free end 'C' will be



(a) $\frac{wl^3}{6EI}$ (b) $\frac{5wl^2}{96EI}$

(c) $\frac{5wl^3}{48EI}$ (d) $\frac{wl^2}{2EI}$

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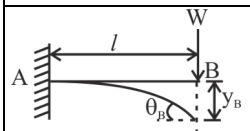
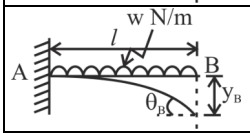
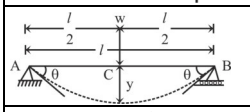
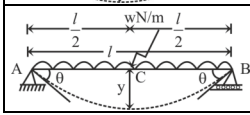
Ans : (a) Slope and Deflection for different types of loaded beam.

Figure	Max. Slope	Max. Deflection
	$\theta_B = \frac{Wl^2}{2EI}$	$y_B = \frac{Wl^3}{3EI}$
	$\theta_B = \frac{Wl^3}{6EI}$	$y_B = \frac{Wl^4}{8EI}$
	$\theta_A = \theta_B = \frac{Wl^2}{16EI}$	$y_C = \frac{Wl^3}{48EI}$
	$\theta_A = \theta_B = \frac{Wl^3}{24EI}$	$y = \frac{5Wl^4}{384EI}$

878. A cantilever beam is having Young's modulus, $E = 2\times10^5$ N/mm^2 Moment of Inertia $I = 8 \times 10^7$ mm^4 and length = 2500 mm. What is the uniformly distributed load which the beam can carry in order to produce a deflection of 5 mm at the free end ?

(a) 16.384 kN/m (b) 25.344 kN/m

(c) 14.543 kN/m (d) 20.384 kN/m

Rajasthan AE (Nagar Nigam) 2016

Ans. (a) :

$E = 2 \times 10^5$ N/mm^2

$I = 8 \times 10^7$ mm^4

$l = 2500$ mm

$\delta = 5$ mm

$w = ?$

We know that the UDL act on cantilever beam then deflection on free end.

$\delta = \frac{w\ell^4}{8EI}$

$5 = \frac{w \times 2500^4}{8 \times 2 \times 10^5 \times 8 \times 10^7}$

$w = 16.3894$ N/mm

$w = 16.389$ kN/m

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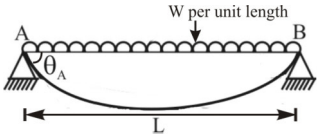
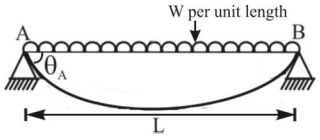
878. The ratio θ_A and θ_B for beam as shown will be–



UKPSC AE 2013 Paper–I

(a) 1:2 (b) 2:1
(c) 1:1 (d) 1:3

Ans : (c)



$$\theta_A = \frac{w\ell^3}{24EI}\Bigg|; \quad \theta_B = -\frac{w\ell^3}{24EI}$$

Ratio of $\left|\frac{\theta_A}{\theta_B}\right| = \left|\frac{\frac{w\ell^3}{24EI}}{\frac{w\ell^3}{24EI}}\right| = \frac{1}{1}$

Deflection at mid point– $y = \frac{5w\ell^4}{384EI}$

880. The slope at the free end of a cantilever beam of length 'L' subjected to moment 'M' at the free end is given by (EI is flexural rigidity)

(a) ML^2/EI (b) ML/EI
(c) $ML/2EI$ (d) $ML/3EI$

(RPSC A.En. 2013)
UKPSC AE 2012 Paper–I

Ans : (b)

$$\theta_{max} = \frac{ML}{EI}$$

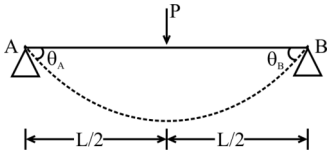
$$y_{max} = \frac{ML^2}{2EI}$$

881. A simply supported beam is acted upon by a concentrated load at its centre. It causes a maximum deflection of 10mm and slope at its ends of 0.002 radians. The span of the beam is

(a) 10m (b) 12m
(c) 15m (d) 20m

[UPPSC State Eng. A.E. 2011]
[UPPSC State Eng. A.E. 2007 (II)]

Ans : (c)



$$\theta_A = \theta_B = \frac{PL^2}{16EI} \Rightarrow 0.002 \quad \text{..... (i)}$$

$$\delta_C = \frac{PL^3}{48EI} \Rightarrow 10\text{mm}$$

$$\left(\frac{PL^2}{16EI}\right) \times \frac{L}{3} = 10$$

From equation (i)

$$(0.002) \times \frac{L}{3} = 10$$

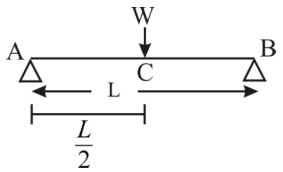
$L = 15000$ mm $= 15$m

882. If a simply supported beam of span ℓ carries a point load W at the mid span, then downward deflection under the load will be

(a) $W\ell^3/3EI$ (b) $W\ell^3/8EI$
(c) $W\ell^3/48EI$ (d) $\frac{5}{384}\frac{W\ell^3}{3EI}$

TNPSC AE 2019, GPSC AE March 2018
UPPCL AE 2018, TSPSC Deputy Engineer 2015
WBPSC AE 2014, CGPSC AE 2014 Shift–I
***RPSC GWD* A.En 2014, Uttrakhand AE 2013**
UK Combined AE Paper– I, 2012,
APPSC (PHE) AE Paper-2 2012
UKPSC AE (Paper–I) 2007,
WBPSC AE (Common) 1998

Ans : (c)



$$(y_c) = \frac{WL^3}{48EI}$$

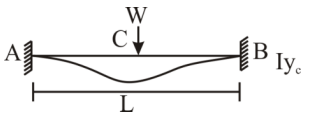
$$(\theta_A = \theta_B) = \frac{WL^2}{16EI}$$

883. The maximum deflection of fixed beam with central point load W is given as

(a) $\frac{WL^4}{192EI}$ (b) $\frac{WL^3}{192EI}$
(c) $\frac{WL^3}{384EI}$ (d) $\frac{WL^4}{384EI}$

RRB SSE (Bilaspur) Yellow Paper : 21.12.2014
RPSC A En. 2013, ESE 1996

Ans : (b)



$$\Delta_f = \frac{\Delta_s}{4}$$

$$= \frac{WL^3}{4 \times 48EI}$$

$$\Delta_f = \frac{WL^3}{192EI}$$

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884. A cantilever beam of rectangular cross–section is subjected to a concentrated load W at its free end. If the width of the beam is doubled the deflection at the free end as compared to the earlier case will be

(a) 16 times (b) 8 times
(c) 2 times (d) Half

RPSC AE 2013

Ans : (d) Deflection due to concentrated load in Cantilever $y_1 = \frac{wL^3}{3EI_1}$

$$I_1 = \frac{bd^3}{12}$$

if, b' = 2b; $I_2 = \frac{bd^3}{6} = 2I_1$

$$y' = \frac{WL^3}{3EI_2}$$

$$y' = \frac{WL^3}{3EI_2} = \frac{WL^3}{3E \times 2I} = \frac{WL^3}{6EI} = \frac{y_1}{2}$$

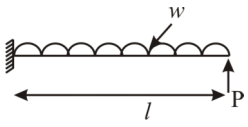
$$\boxed{y' = \frac{y_1}{2}}$$

885. A cantilever beam of span 3 m is subjected to uniformly distributed load of 80 kN/m throughout the span. Determine the value of point load that can be applied in upward direction at free end in order to gave the resulting deflection at free end as zero.

(a) 30 kN (b) 90 kN
(c) 60 kN (d) 80 kN

RPSC VPITI 2012

Ans : (b)



As deflection at free end is zero

$$\frac{w\ell^4}{8EI} - \frac{p\ell^3}{3EI} = 0$$

$$P = \frac{3w\ell}{8} = \frac{3 \times 80 \times 3}{8}$$

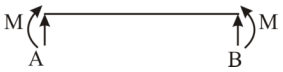
$$\boxed{P = 90\,kN}$$

886. A simply-supported beam of length L carries two equal unlike couples M at two ends. If the flexural rigidity EI=constant, then the central deflection of the beam is given by

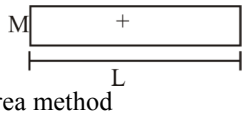
(a) $\frac{MI^2}{4EI}$ (b) $\frac{M\ell^2}{64EI}$
(c) $\frac{ML^2}{16EI}$ (d) $\frac{ML^2}{8EI}$

OPSC AEE 2015 PAPER–I
JPSC Combined AE Paper V 2013
BPSC AE 2012 Paper–V

Ans : (d)



BMD

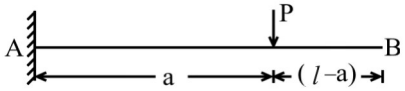


From moment area method

$$y_c = \frac{A\bar{y}}{EI} = \frac{M \times \frac{L}{2} \times \frac{L}{4}}{EI}$$

$$\boxed{y_c = \frac{ML^2}{8EI}}$$

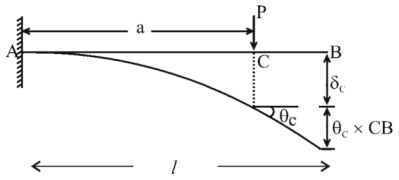
887. In a cantilever beam of uniform flexural rigidity EI and loaded as shown in the figure, the deflection at free end 'B' would be :



(a) $\frac{Pa(3\ell^2 - a^2)}{6EI}$ (b) $\frac{Pa^2(\ell - a)}{6AE}$
(c) $\frac{P(\ell - a)^2}{6EI}$ (d) $\frac{Pa^2(3\ell - a)}{6EI}$

UPPSC State Eng. A.E. 2007 (I)
UPPSC State Eng. A.E. 2003

Ans : (d)



$$\delta_B = \delta_C + \theta_C \times CB$$

$$\delta_B = \frac{Pa^3}{3EI} + \frac{Pa^2}{2EI}(l - a)$$

$$= \frac{Pa^2}{EI}\left(\frac{a}{3} + \frac{l}{2} - \frac{a}{2}\right) = \frac{Pa^2}{EI}\left(\frac{2a - 3a + 3l}{6}\right)$$

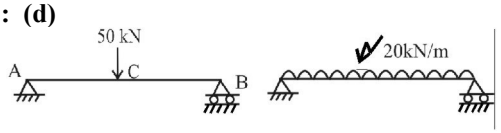
$$\delta_B = \frac{Pa^2(3l - a)}{6EI}$$

888. A point load of 50kN acting at mid-span of a simply supported beam produces the same maximum deflection in the beam as caused by a uniformly distributed load of 20kN/m over the whole span. The span of the beam is

(a) 10 m (b) 8 m
(c) 6 m (d) 4 m

(JPSC Combined AE Paper V 2013)

Ans : (d)



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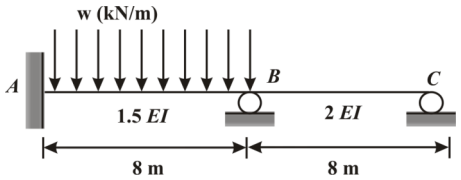
Central deflection $\delta_1 = \frac{W\ell^3}{48EI}$ $\delta_2 = \frac{5w\ell^4}{384EI}$

$\frac{W\ell^3}{48EI} = \frac{5w\ell^4}{384EI} \Rightarrow \frac{50\times\ell^3}{48EI} = \frac{5\times20\times\ell^4}{384EI} \Rightarrow \boxed{\ell = 4m}$

889. Using slope-deflection method, the stiffness K_{AB} and K_{BC} of a beam shown below respectively are-



(a) $\frac{1.5EI}{8}$ and $\frac{2EI}{8}$ (b) $\frac{EI}{2}$ and $\frac{3EI}{8}$

(c) $\frac{3EI}{8}$ and $\frac{EI}{2}$ (d) $\frac{1.5EI}{8}$ and $\frac{EI}{4}$

MPPSC AE 2017

Ans. (c) : In slope deflection method, stiffness is given as $\frac{2EI}{l}$ for fixed end

$$K_{AB} = \frac{2(1.5EI)}{8} = \frac{3EI}{8}$$

$$K_{BC} = \frac{2(2EI)}{8} = \frac{4EI}{8} = \frac{EI}{2}$$

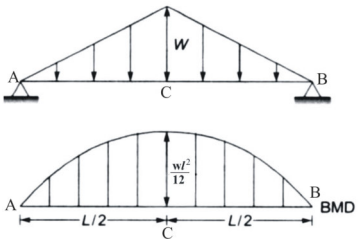
890. What is the area of bending moment diagram of a simply supported beam subjected to triangular load of intensity 'W' kN/m at centre of span (L) and zero at both ends using Simpson's rule, assuming an equal interval of $\left(\frac{L}{2}\right)$?

(a) $\frac{WL^3}{8}$ (b) $\frac{WL^3}{12}$

(c) $\frac{WL^3}{16}$ (d) $\frac{WL^3}{18}$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) :



$R_A = \frac{1}{2}\times\frac{WL}{2} = \frac{WL}{4}$

Moment at C

$M_C = \frac{WL}{4}\times\frac{L}{2} - \frac{1}{2}\times\frac{L}{2}\times W\frac{1}{3}\times\frac{L}{2}$

$= \frac{3WL^2 - WL^2}{24} = \frac{WL^2}{12}$

Area by Simpson's rule,

$= \frac{h}{3}[y_0 + 4y_1 + y_2]$

Area = $\frac{1}{3}\times\frac{L}{2}\left(4\times\frac{WL^2}{12}\right) = \frac{WL^3}{18}$

891. A continuous beam has AB as one of the spans. It is subjected to fixed end moments at end A and B as FEM_A and FEM_B respectively under given loads. It has length 'L' and flexural rigidity EI. It rotates at end A and B by θ_A and θ_B respectively. The member axis rotates through clockwise angle $R = \frac{\Delta}{L}$. The moment at end A is given by-

(a) $M_A = FEM_B + \frac{4EI}{L}(2\theta_A + \theta_B - 3R)$

(b) $M_A = FEM_A + \frac{2EI}{L}(2\theta_A + \theta_B - 3R)$

(c) $M_A = FEM_A + \frac{4EI}{L}(2\theta_A + \theta_B + 3R)$

(d) None of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) : $M_A = FEM_A + \frac{2EI}{L}(2\theta_A + \theta_B - 3R)$

It is the slope deflection equation at end A.

892. A continuous beam ABCD has end A as fixed and support B and C are simple supports while end D is free. The fixed end moments for beam AB are FEM_{AB} = -19.2 kNm and FEM_{BA} = 28.8 kNm. The moment equation for slope deflection method for beam AB can be written as-

(a) $M_{ab} = \frac{2EI}{L}(2\theta_A + \theta_B) + 28.8$

$M_{ba} = \frac{2EI}{L}(\theta_A + 2\theta_B) - 19.2$

(b) $M_{ab} = \frac{2EI}{L}(\theta_A) - 19.2$

$M_{ba} = \frac{4EI}{L}(\theta_A) + 28.8$

(c) $M_{ab} = \frac{4EI}{L}(\theta_A) - 19.2$

$M_{ba} = \frac{2EI}{L}(\theta_A) - 28.8$

(d) $M_{ab} = \frac{2EI}{L}(\theta_B) - 19.2$

$M_{ba} = \frac{4EI}{L}(\theta_B) + 28.8$

Maharashtra PSC (Mains) 2015 Paper–I

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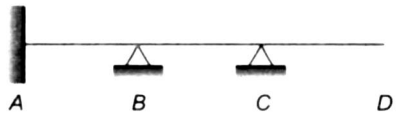
Ans. (d) :



$M_{FAB} = -19.2$ kNm

$M_{FBA} = 28.8$ kNm

Now, for span *AB*,

$$M_{AB} = M_{FAB} + \frac{2EI}{L}\left(2\theta_A + \theta_B - \frac{3\delta}{L}\right)$$

$\delta = 0$ *and* $\theta_A = 0$

$$M_{AB} = -19.2 + \frac{2EI}{L}\theta_B$$

$$M_{BA} = M_{FBA} + \frac{2EI}{L}\left(2\theta_B + \theta_A - \frac{3\delta}{L}\right)$$

$\theta_A = 0; \delta = 0$

$$M_{BA} = 28.8 + \frac{4EI}{L}\theta_B$$

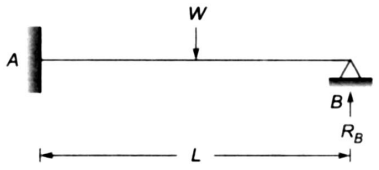
893. A propped cantilever of span '*L*' fixed at *A* and simply supported at '*B*' is subjected to concentrated load '*W*' at centre, reaction at *B*-

(a) $\frac{3}{16}W$ (b) $\frac{W}{4}$

(c) $\frac{5}{16}W$ (d) $\frac{7}{16}W$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) :



$\delta_B = 0$

$\Rightarrow$ $\delta_{B \text{ due to load W}} + \delta_{B \text{ due to } R_B} = 0$

$$\frac{W\left(\frac{L}{2}\right)^3}{3EI} + \frac{W\left(\frac{L}{2}\right)^2}{2EI} \times \frac{L}{2} - \frac{R_B (L)^3}{3EI}$$

$$R_B = \left(+\frac{3W}{8}\right)\left(\frac{1}{2} + \frac{1}{3}\right) = \left(+\frac{5W}{16}\right)$$

894. A propped cantilever is subjected to a concentrated load of 16 kN at the centre of the span. The length of beam is 4 m. The flexural rigidity *EI* = 4 × 10^2 kN/m^2. The reactions at the fixed end and simply supported end are-

(a) 9.87 kN at fixed end and 6.13 kN at simply supported end

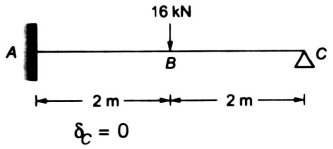
(b) 8 kN each at fixed end and simply supported end

(c) 11 kN at fixed end and 5 kN at simply supported end

(d) 16 kN at fixed end and zero kN at simply supported end

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) :



$\delta_C = 0$

Using compatibility equation

$\delta + \delta_{R_C} = 0$

By using formula

$$\frac{wL^3}{3EI} + \frac{wL^2}{2EI} \times L - \frac{R_c \times L^3}{3EI}$$

$$\frac{16 \times (2)^3}{3EI} + \frac{16 \times 2^2}{2EI} \times 2 - \frac{R_C \times (4)^3}{3EI} = 0$$

$$\frac{16 \times 8}{3} + 64 = \frac{R_C \times 64}{3}$$

$R_C = 5$ kN

Now $\Sigma V = 0$

$\Rightarrow$ $R_A + R_C = 16$ kN

$R_A = 16 - 5 = 11$ kN

895. Fill in the blanks-

The primary unknowns in slope deflections methods are _____ and obtained by solving _____ equations-

(a) Rotational and translational displacements, joint force equilibrium

(b) Moment and forces, compatibility

(c) Stiffness and flexibility, differential

(d) None of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (a) : Slope deflection method is a displacement method in which displacements are taken as unknowns and are obtained by solving joint force equilibrium equation.

896. In the moment distribution method, the carry over moment is equal to-

(a) double of its corresponding distributed moment and has same sign

(b) one-half of its corresponding distributed moment and has same sign

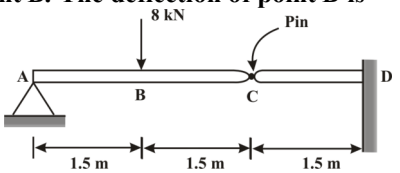
(c) one-half of its corresponding distributed moment and has opposite sign

(d) None of the above

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : One-half of its corresponding distributed moment and has same sign.

897. The beam *AC* simply supported at *A* and at *C* is pinned to a cantilever beam *CD* as shown in figure. Both beam have identical flexural rigidities *EI*. The vertical load of 8 kN acts at point *B*. The deflection of point *B* is-



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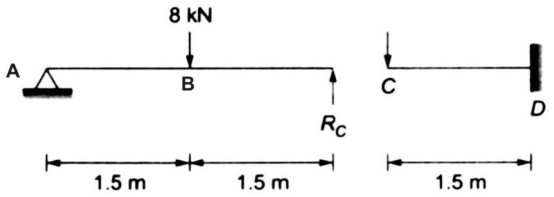
(a) $\frac{2.25}{EI}$ (b) $\frac{4.5}{EI}$

(c) $\frac{6.75}{EI}$ (d) $\frac{8}{EI}$

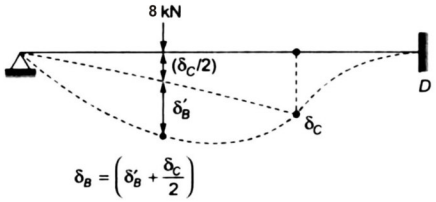
Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) : For problems involving internal hinge, make FBD of different parts after separating hinge



Beam AC acts as a S.S.B.

Hence $R_A = R_C = 4$ kN, now deflected shape can be drawn as



δ'_B = deflection of AC due to 8 kN load

$$\delta'_B = \frac{Pl^3}{48EI} = \frac{8\times 3^3}{48EI} = \frac{4.5}{EI}$$

$$\frac{\delta_C}{2} = \left[\frac{1}{2}\times\frac{Pl^3}{3EI}\right]$$

$$= \frac{1}{2}\times\frac{4\times(1.5)^3}{3EI} = \frac{2.25}{EI}$$

$$\delta_B = \frac{4.5+2.25}{EI} = \frac{6.75}{EI}$$

898. While using three moments equation in the beam analysis, a fixed end of a continuous is replaced by an addition span of

(a) zero length

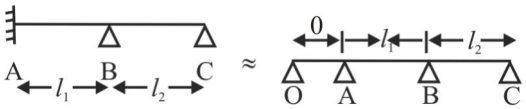
(b) zero moment of inertia

(c) infinite length

(d) infinite moment of inertia

TNPSC AE 2018

Ans. (a) :



Fixed support A is assumed as continuous support OA with length zero.

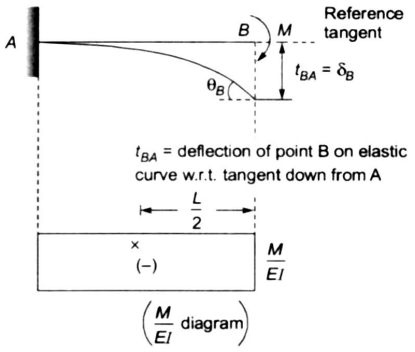
899. Maximum deflection of a cantilever due to pure moment '*M*' at its free end, is-

(a) $\frac{ML^2}{3EI}$ (b) $\frac{ML^2}{4EI}$

(c) $\frac{ML^2}{2EI}$ (d) $\frac{ML^2}{6EI}$

OPSC AEE 2019, 2015 Paper–I
Maharashtra PSC (Mains) 2015 Paper–I
ESE 2012, UK Combined AE Paper– I, 2012
JPSC AE 2009, UKPSC AE (Paper–I) 2007

Ans. (c) : Let us find δ_B (deflection at B) using moment area method :



$t_{BA} = \delta_B$ = moment of area under M/EI diagram between A and B about B.

$$\delta_B = -\left(\frac{M}{EI}\times L\right)\times\frac{L}{2} = -\frac{ML^2}{2EI}$$

$$\delta_B = \frac{ML^2}{2EI}(\downarrow)$$

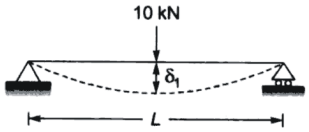
900. A simply supported beam deflects by 5 mm when it is subjected to a concentrated load of 10 kN at its centre. What will be the deflection in a $\frac{1}{10}$ model of the beam, if the model is subjected to a 1 kN load at its centre?

(a) 5 mm (b) 0.5 mm

(c) 0.05 mm (d) 0.005 mm

BPSC AE 2019 Paper (V) Section-I

Ans. (a) :



We know that

$$\delta = \frac{WL^3}{48EI}$$

If W = 10kN, then

$$5 = \frac{10\times L^3}{48EI}$$

$$\Rightarrow \quad \frac{L^3}{EI} = \frac{5\times 48}{10} \text{.....(i)}$$

Now, As per question,

$$P = 1\text{ kN},\ E_1 = E_2,\ I' = \frac{I}{10^4},\ L' = \frac{L}{10}$$

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So,

$$\delta_2 = \frac{1\times L'^3}{48EI'}$$

$$\delta_2 = \frac{10}{48}\times\frac{L^3}{EI}$$

$$\delta_2 = \frac{10}{48}\times\frac{5\times 48}{10} \ldots\ldots\ldots [\text{from eq}^n (i)]$$

$$\boxed{\delta_2 = 5\text{mm}}$$

901. The moment distribution method is formulated by

(a) Thomas Young (b) Kani

(c) Mohr (d) Hardy Cross

TNPSC AE 2018

Ans. (d) : Moment distribution method was formulated by Hardy Cross.

902. Maximum deflection at the mid–span of a simply–supported beam of span *l*, with uniformly distributed load (w) all over the beam span, and flexural rigidity EI, is (modulus of elasticity = E; moment of inertia of beam = I)

Maharashtra PSC (Mains) 2019 Paper–I

GPSC AE March 2018, TNPSC AE 2018

Rajasthan AE (Nagar Nigam) 2016, MPPSC AE 2016

OR

A simply supported beam of span *l* carries a uniformly distributed load of w per unit length on whole span. Flexural rigidity being EI. The maximum deflection of beam is given by:

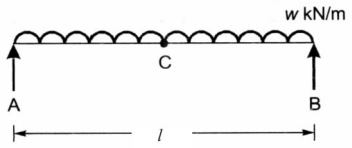
(a) $\frac{5wl^4}{48EI}$ (b) $\frac{5wl^4}{384EI}$

(c) $\frac{wl^3}{48EI}$ (d) $\frac{wl^3}{3EI}$

GPHC AE 2018, *UPPSC State Eng. A.E. 2011*

UPPSC State Eng. A.E. 2007 (II)

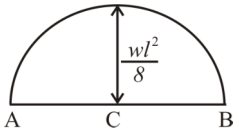
Ans. (b) Maximum downward deflection will be at mid point i.e. at C.



$$R_A = R_B = \frac{wl}{2}$$

Moment at distance x from beam

$$M_x = R_a.x - \frac{wx^2}{2} = \frac{wl}{2}x - wx^2$$



Maximum deflection at centre

$$C = \frac{1}{EI}\times\frac{2}{3}\times\frac{l}{2}\times\frac{wl^2}{8}\times\frac{5}{8}\times\frac{l}{2} = \delta_c = \frac{5wl^4}{384EI}$$

903. Which one of the following method is convenient for determining deflection of beam of non uniform flexural rigidity?

(a) Macaulay's method

(b) Conjugate beam method

(c) Moment area method

(d) Double integration method

TNPSC AE 2018

Ans. (b) In conjugate beam method (i) bending moment diagram (M/EI) is assumed as loading diagram on the beam and a new BMD is constructed from this, the diagram will be deflection curve. If the span has varying flexural rigidity the also we can easily plot BMD and determine deflection. In remaining three methods beam is of uniform flexural rigidity.

904. Match the List-I with List-II and select your correct answer using the codes given below:

List-I (Name of the associated with the methods)	List-II (Method)
A. G.N. Maney	1. Moment distribution
B. Hardy cross	2. Slope deflection method
C. Euler	3. Theorem of three moments
D. Clapeyron	4. Crippling load on column

TNPSC AE 2018

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	3	4
(c)	1	2	4	3
(d)	2	1	3	4

Ans. (a) :

- G.N. Money developed slope deflection method.
- Hardy cross developed moment distribution method.
- Euler gave formula for crippling load on a column.
- Claypeyron found three moment equation for continuous beams.

905. If E is the Young's modulus and I is the moment of Inertia. The maximum deflection of a cantilever beam of length *l* carrying a concentrated load W at free end is

(a) $\frac{Wl^3}{3EI}$ (b) $\frac{Wl^4}{3EI}$

(c) $\frac{Wl^3}{8EI}$ (d) $\frac{Wl^4}{8EI}$

TNPSC AE 2018, LMRC AE 2017 I–Shift

Karnataka PSC AE 2017 Paper - II

UTTR. AE 2013, UPPSC State Eng. A.E. 2011

UPPSC State Eng. A.E. 2007 (II)

UKPSC AE (Paper–I) 2007

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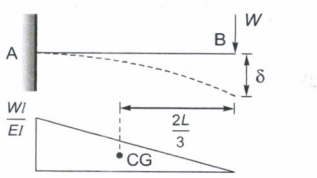
Ans. (a) :



(M/EI) diagram

Deflection at B δ = Moment of area of (M/EI) diagram about point B

$$\delta = \frac{1}{2} \times \frac{Wl}{EI} \times l \times \frac{2}{3} l = \frac{Wl^3}{3EI}$$

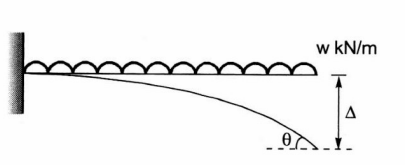
906. If the deflection at the free end of uniformly loaded cantilever beam is 15 mm and the slope of deflection curve at free end is 0.02 radian, then the length of the beam is :

(a) 0.8 m (b) 1.0 m

(c) 1.13 m (d) 1.5 m

OPSC AEE 2019 Paper–I, ***HPPSC 2016-II, ESE 2015***

Ans. (b) :



Deflection at free end, $\Delta = \frac{wL^4}{8EI} = 0.015 - (i)$

Slope of free end, $\theta = \frac{wL^3}{6EI} = 0.02 - (2)$

Dividing eqn (1) by eqn (2)

$\frac{3}{4}L = \frac{1.5}{2}$

$\Rightarrow L = 1$ m

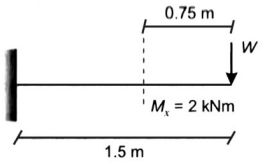
907. A cantilever of length 1.5 m is loaded with a concentrated load W at the unsupported end. The bending moment at the centre of the beam is 2 kNm. What is the magnitude of the load 'W'?

(a) 11.333 kN (b) 3 kN

(c) 2.666 kN (d) Zero

[HPPSC AE 2016]

Ans. (c) :



Moment at distance x from free and

$M_x = Wx = 2$ kNm

x = 0.75 m

$\therefore W = \frac{2 \text{ kNm}}{0.75 \text{ m}} = 2.666$ kN

908. In the conjugate beam method, the fixed support in actual beam is considered as ..... support in the conjugate beam.

(a) Free (b) Hinge

(c) Fixed (d) Roller

Karnataka PSC AE 2017 Paper - II

Ans. (a) :

$$\frac{dV}{dx} = w \text{ or } \frac{d^2M}{dx^2} = w$$

and $\frac{d\theta}{dx} = \frac{M}{EI}$ or $\frac{d^2y}{dx^2} = \frac{M}{EI}$

On integration

$$V = \int w dx \quad M = \int \left[\int w dx \right] dx$$

$$\theta = \int \frac{M}{EI} dx \quad y = \int \left[\int \frac{M}{EI} dx \right] dx$$

Therefore

- Slope at any point in real beam = shear at that point in conjugate beam.
- Deflection at any point in real beam=bending moment at that point in conjugate beam for fixed support, both shear and BM exist and hence in conjugate beam both slope and deflection short exist which is the case of free support.

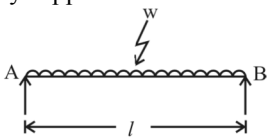
909. What is the ratio of peak deflection of simply supported beam subjected to uniformly distributed load to corresponding cantilever beam with same parameters ?

(a) $\frac{1}{10}$ (b) $\frac{1}{8}$

(c) $\frac{1}{6}$ (d) $\frac{1}{4}$

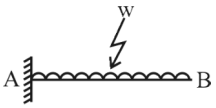
Rajasthan JEN (Degree) Shift - II 2016

Ans. (a) : Simply supported beam–



slope $(\theta) = \frac{w\ell^3}{24EI}$, Deflection $(\Delta_1) = \frac{5w\ell^4}{384EI}$

Cantilever beam



Slope $(\theta) = \frac{w\ell^3}{6EI}$

Deflection $(\Delta_2) = \frac{w\ell^4}{8EI}$

Ratio, $\frac{\Delta_1}{\Delta_2} = \frac{\frac{5w\ell^4}{384EI}}{\frac{w\ell^4}{8EI}}$

$\frac{\Delta_1}{\Delta_2} = \frac{1}{9.6} \simeq \frac{1}{10}$

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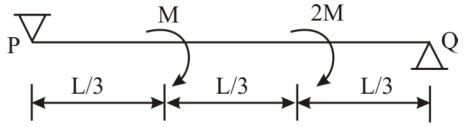
910. The figure shows a simply supported beam PQ of uniform flexural rigidity EI carrying two moments M and 2 M



The slope at P will be :

(a) 0 (b) ML/(9EI)
(c) ML/(6EI) (d) ML/(3EI)

JPSC AE (Pre) 2019

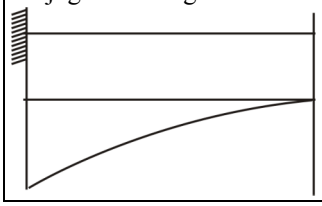
Ans. (c) :
For this,
Slope at support 'P' of real beam
= shear force at support 'P' of conjugate beam
Or, slope at support 'P' of real
= reaction at 'P' of conjugate beam
So, slope at supportf 'P' of real beam = ML/6EI

911. Bending moment at any section in a conjugate beam gives ..... in the actual beam :

(a) Slope (b) Curvature
(c) Deflection (d) Frictional resistance

WBPSC Poly Lect. 2019
GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) : Bending moment at any section in a conjugate beam give deflection in a actual beam

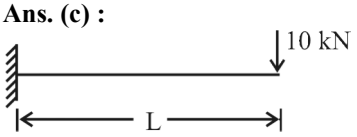


912. A free end of a cantilever beam rotates by 0.001 radians under a point load 10 kN. Then deflection at the free end due to a moment of 100 kN-m is :

(a) 40 mm (b) 20 mm
(c) 10 mm (d) 25 mm

Rajasthan JEn (Degree) 2016

Ans. (c) :

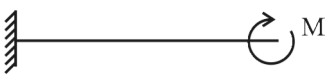


$$\theta = \frac{PL^2}{2EI}$$

$$0.001 = \frac{10L^2}{2EI}$$

$$\frac{0.001}{10} = \left(\frac{L^2}{2EI}\right) \text{.....(i)}$$

and



$$y = \frac{ML^2}{2EI}$$

$$y_{max} = 100 \times 1000\left(\frac{L^2}{2EI}\right)$$

from equation (i)

$$y_{max} = 100 \times 1000 \times \frac{0.001}{10}$$

$$\boxed{y_{max} = 10\text{mm}}$$

913. What will be deflection at the centre of a simply supported beam of rectangular cross-section if the depth is doubled, for the same load W?

(a) $\frac{1}{2}$ of first case (b) $\frac{1}{6}$ of first case
(c) $\frac{1}{8}$ of first case (d) $\frac{1}{4}$ of first case

Kerala PSC AE 2015

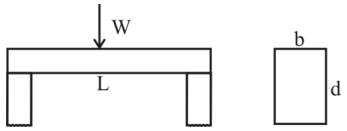
OR

If the depth of a simply supported beam carrying an isolated load at its centre, is doubled, the deflection of the beam at the centre will be changed by a factor of

(a) 2 (b) 1/2
(c) 1/8 (d) 8

ISRO Scientist/Engineer 2015
WBPSC AE (Common) 1998

Ans. (c) : Deflection at centre of a s/s beam is given as



$$\delta = \frac{WL^3}{48EI} = \frac{WL^3 \times 12}{48Ebd^3} = \frac{WL^3}{4Ebd^3}$$

$$\delta_1 = \frac{WL^3}{4bEd^3}$$

For $d_2 = 2d_1$

$$\delta_2 = \frac{WL^3}{4bEd_2^3} = \frac{WL^3}{4bE(2d_1)^3} = \frac{\delta_1}{8}$$

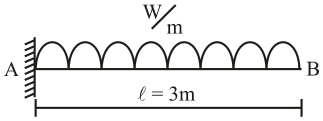
914. A cantilever beam, 3 m long, carries a uniformly distributed load over the entire length. If the slope at the free end is 1^0, the deflection at the free end is

(a) 49.27 mm (b) 39.27 mm
(c) 30.27 mm (d) 20.27 mm

ESE 2017

Ans. (b)

- Length of cantilever beam 'ℓ' = 3m
- Slope at the free end $\theta_B = 1^0 = \frac{\pi}{180}$



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- Slope at free end, $\theta_B = \dfrac{W\ell^3}{6EI}$
- Deflection at free end

i.e. $\delta_B = \dfrac{W\ell^4}{8EI}$

$\delta_B = \dfrac{W\ell^3}{6EI} \times \dfrac{6\ell}{8}$

$\delta_B = \dfrac{\pi}{180^0} \times \dfrac{6 \times 3000}{8}$

$\boxed{\delta_B = 39.27\text{mm}}$

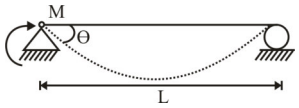
915. The ratio $\left(\dfrac{s}{t}\right)$ of, (s) stiffness of a beam (of constant EI) at the near end when the far end is hinged, to (t) the stiffness of the same beam at the near end when the far end is fixed, is

(a) $\dfrac{1}{2}$ (b) $\dfrac{3}{4}$

(c) $\dfrac{1}{1}$ (d) $\dfrac{4}{3}$

ESE 2017

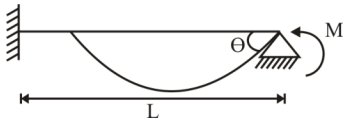
Ans. (b)



$\theta = \dfrac{ML}{3EI}$

$\left[\dfrac{M}{\theta} = \dfrac{3EI}{L} = \text{stiffness 'S'}\right]$

Stiffness of a beam at the near end when far end is hinged.



$\theta = \dfrac{ML}{4EI}$

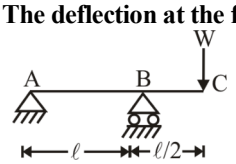
$\left[\dfrac{M}{\theta} = \dfrac{4EI}{L} = t\right]$

Stiffness of the same beam at the near end when far end is fixed.

So, Ratio $\dfrac{\text{'S'}}{t} = \dfrac{3EI}{\ell} \Big/ \dfrac{4EI}{\ell}$

$\boxed{\dfrac{S}{t} = \dfrac{3}{4}}$

916. An overhanging beam of uniform EI is loaded as shown below. The deflection at the free end is



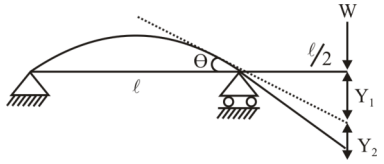
(a) $\dfrac{W\ell^3}{81EI}$ (b) $\dfrac{W\ell^3}{8EI}$

(c) $\dfrac{W\ell^3}{27EI}$ (d) $\dfrac{2W\ell^3}{27EI}$

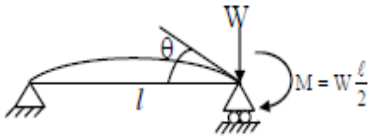
ESE 2017
UPPSC AE 2003

Ans. (b) : The deflection at 'C' due to rotation of support C is.

$Y_1 = \dfrac{\theta\ell}{2} = \dfrac{M\ell}{3EI}\left(\dfrac{\ell}{2}\right) = \dfrac{W\ell^2}{6EI}\left(\dfrac{\ell}{2}\right) = \dfrac{W\ell^3}{12EI}$



The deflection at C due to cantilever effect at over hang BC is.



$Y_2 = \dfrac{W\left(\dfrac{\ell}{2}\right)^3}{3EI}$

$Y = Y_1 + Y_2$

$Y = \dfrac{W\ell^3}{12EI} + \dfrac{W\left(\dfrac{\ell}{2}\right)^3}{3EI}$

$Y = \dfrac{W\ell^3}{EI}\left[\dfrac{1}{12} + \dfrac{1}{24}\right]$

$\dfrac{W\ell^3}{EI}\left[\dfrac{2+1}{24}\right] = \dfrac{3W\ell^3}{24EI} = \dfrac{W\ell^3}{8EI}$

$\left[\text{The deflection at the free end 'Y'} = \dfrac{W\ell^3}{8EI}\right]$

917. A propped cantilever beam of length L is subjected to a moment M at the propped end. The support moment at the fixed end will be:

(a) M (b) M/2

(c) M/3 (d) 2 M

ISRO Scientist/Engineer 2020

Ans. (b) : In propped cantilever beam, the carry over moment is $\dfrac{M}{2}$, with same direction.

918. Two simply supported beams are made up of the same material and are of the same cross section. Both beams carry uniformly distribution loads of equal intensities. One beam is 2 m long and the other is 4 m long. The 2 m long beam shows a central deflection of 1

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mm. What is the central deflection of the 4m long beam?

(a) 16 mm (b) 2 mm
(c) 8 mm (d) 1 mm

ISRO Scientist/Engineer 2020
ESE 2015

Ans. (a) :

Method (1)–

Deflection of S.S.B. under UDL is given as–

$$\delta = \frac{5wl^4}{384EI}$$

For 2 m beam–

$$\delta_1 = \frac{5w2^4}{384EI} = 1\text{mm}$$

$$\delta_1 = \frac{5w}{384EI} = \frac{1}{2^4} \quad \text{.....(i)}$$

For 4 m beam

$$\delta_2 = \frac{5w4^4}{384EI}$$

by eqn. (i) $\delta_2 = \frac{1}{2^4} \times 4^4 = 16\,\text{mm}$

Method (2)–

We know deflection of S.S.B. under UDL is given as–

$$\delta = \frac{5wl^4}{384EI}$$

So, $\delta \propto l^4$

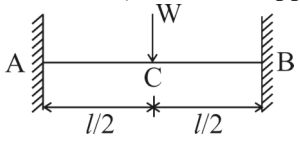
$$\frac{\delta_1}{\delta_2} = \left(\frac{l_1}{l_2}\right)^4 \Rightarrow \boxed{\delta_2 = 16\text{ mm}}$$

919. Two beams A and B of equal span carries same central point load. Beam 'A' has fixed end supports and Beam 'B' has simply supported ends. The deflection of Beam 'A' will be _____ as compared to that of Beam 'B'.

(a) One eighth (b) One fourth
(c) One half (d) Double

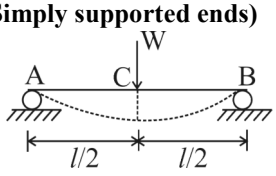
ISRO Scientist/Engineer 2019

Ans : (b) : BEAM 'A' - (Fixed end Support)



• Deflection $(\Delta c) = \frac{Wl^3}{192EI}$

BEAM 'B' - (Simply supported ends)



• Deflection s $\left(\Delta_C^A\right) = \frac{Wl^3}{48EI}$

So, $\frac{\text{Deflection of Beam A}}{\text{Deflection of Beam B}} = \frac{Wl^3/192EI}{Wl^3/48EI}$

$\left[\text{Deflection of Beam A} = \frac{1}{4}(\text{Deflection of Beam B})\right]$

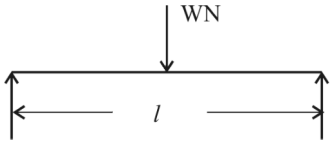
920. A simply supported beam A carries a point load at its mid span. Another identical beam B carries the same load but uniformly distributed over the entire span. The ratio of the maximum deflection of the beams A and B will be.

(a) 8/5 (b) 5/8
(c) 3/5 (d) 5/3

ISRO Scientist/Engineer 2017
GPSC R&B AE 2016, ESE 2001

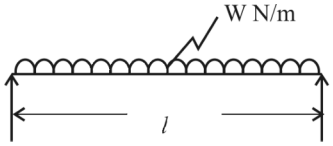
Ans : (a) :

Case - 1



$$y_1 = \frac{Wl^3}{48EI}$$

Case - 2



$$y_2 = \frac{5wl^4}{384EI} = \frac{5Wl^3}{384EI} \quad \text{(given } w \times l = W\text{)}$$

$$\frac{y_1}{y_2} = \frac{Wl^3}{48EI} \times \frac{384EI}{5Wl^3} = \frac{8}{5}$$

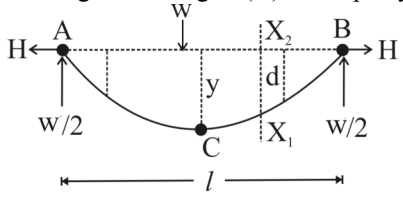
921. If *l* is the span of a light suspension bridge whose each cable carries total weight (w) and the central dip is y, the horizontal pull at each support, is

(a) *wl*/4y (b) *wl*/8y
(c) *wl*/2y (d) *wl*

ISRO Scientist/Engineer 2015

Ans : (b) :

• Expression for horizontal pull of suspension bridge having total weight (w) and dip is y.



• $Hy = \frac{x}{l}\left[\sum M_B - \sum M_x\right]$ Taking moment about 'C' and equating.

$$\sum M_B = w\frac{l}{2} = \frac{wl}{2}$$

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$$\sum M_x = w \times x/2 = \frac{wx}{2}$$

$$\therefore Hy = \frac{x}{l} \times \left[\frac{wl}{2} - \frac{wx}{2}\right]$$

$$Hy = \frac{wx}{l}\left[\frac{l}{2} - \frac{l}{4}\right] \quad \therefore \left[x = \frac{l}{2}\right]$$

$$Hy = \frac{wl}{2l}\left[\frac{l}{4}\right]$$

$$Hy = \frac{wl}{8}$$

$$\left[H = \frac{wl}{8y}\right]$$

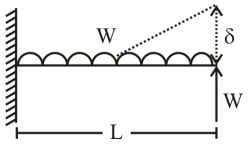
922. A cantilever beam carries a uniformly distributed total load W over its whole length and a concentrated upward load W at its free end. The net vertical deflection at the free end is

(a) Zero (b) $\frac{5.WL^3}{24\ EI}$ downwards

(c) $\frac{5.WL^3}{24\ EI}$ upwards (d) $\frac{5.WL^3}{48\ EI}$ downwards

WBPSC AE 2016, ESE 2012
ISRO Scientist/Engineer 2013
OPSC Civil Services Pre 2011

Ans : (c)



Deflection due to UDL = $\frac{wl^4}{8EI}$

But, $w \times L = W$

So, deflection due to UDL $= \frac{Wl^3}{8EI}$

Deflection due to concentrated load $= \frac{Wl^3}{3EI}$

net deflection $= \frac{Wl^3}{8EI} - \frac{Wl^3}{3EI} \Rightarrow -\frac{5Wl^3}{24EI}$

The net deflection at free end–

$$\left[\delta = \frac{5WL^3}{24EI}\ (\text{upwards})\right]$$

923. A uniform cantilever beam has a span of 2 m and carries a point load of 6kN at free end. The magnitude of moment to be aplied at free end for zero vertical deflection at that point is (neglect self-wt. of the beam).

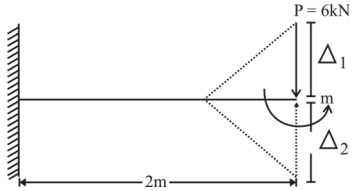
(a) 5 kN.m (b) 10 kN.m

(c) 11 kN.m (d) 8 kN.m

ISRO Scientist/Engineer 2013

Ans : (d)

■ **Cantilever Beam**



■ Moment applied to counter act the deflection due to point load in opposite direction.

Deflection due to moment $'\Delta_1' = \frac{ML^2}{2EI}$

Deflection due to point load $'\Delta_2' = \frac{PL^3}{3EI}$

Now, $\Delta_1 = \Delta_2$

$$\frac{ML^2}{2EI} = \frac{PL^3}{3EI}$$

$$\frac{M}{2} = \frac{PL}{3}$$

$$M = \frac{2PL}{3}$$

$$M = \frac{2 \times 6 \times 2}{3}$$

$$\boxed{M = 8kN.m}$$

924. A fixed beam settles at one end by δ, if EI is the flexural rigidity of the beam, the fixed end moments at the ends be

(a) $EI\ \delta/L^2$ (b) $2EI\ \delta/L^2$

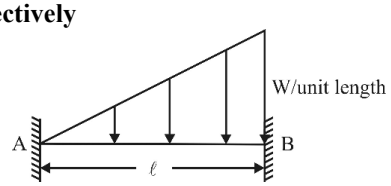
(c) $4EI\ \delta/L^2$ (d) $6EI\ \delta/L^2$

APTRASCO 2017

Ans : (d) When a fixed beam settles at one end by δ, then the moment generated at other ends is –

$$\boxed{M_\delta = \frac{6EI\delta}{L^2}}$$

925. Fixed end moments at A and B for the fixed beam shown in the figure, subjected to the indicated uniformly varying load, are respectively



(a) $W\ell^2/30$ and $W\ell^2/20$

(b) $W\ell^2/20$ and $W\ell^2/30$

(c) $W\ell^2/12$ and $W\ell^2/8$

(d) $W\ell^2/8$ and $W\ell^2/12$

APTRANSCO 2017
Maharashtra PSC (Mains) 2015 Paper–I

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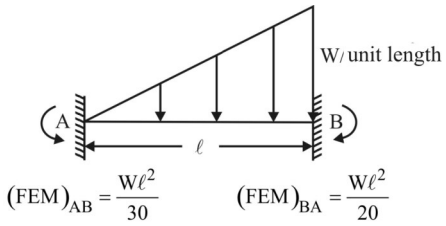
Ans : (a) : Moment at fixed end of fixed beam for U.V.L. respectively $\frac{W\ell^2}{30}$ and $\frac{W\ell^2}{20}$

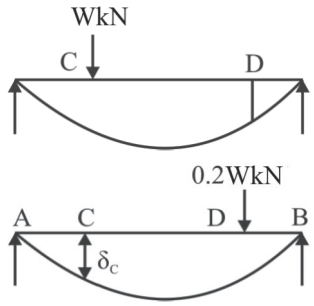


926. A simple supported beam AB of length "L = 3 m". If the displacement at one-third length from the right hand support, at point D, due to load "W" at 1.0 m from left hand support, at point C is 5 mm, then the displacement at C due to a load of 0.2 W at D will be

(a) 1 mm (b) 2 mm
(c) 5 mm (d) 25 mm

OPSC AEE 2019 Paper–I

Ans. (a)



Maxwell Betties Theorem :

Virtual work done by the 1st loading due to displacement caused by IInd loading is equal to virtual work done by the IInd loading due to displacement caused by Ist loading

$W \times \delta_C = 0.2\ W \times 5$

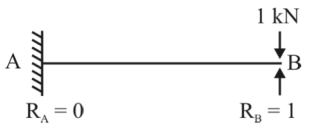
$\delta_C = 1mm$

927. A propped cantilever AB is fixed at A and simply supported at B. If a 1 kN concentrated load is acting at B, what is the vertical reaction at support A?

(a) 0 kN (b) 1 kN
(c) 0.5 kN (d) 2 kN

OPSC AEE 2019 Paper–I

Ans. (a)



928. The principle of virtual work can be applied to elastic system by considering the virtual work of :

(a) Internal force only
(b) External force only
(c) Internal as well as external forces
(d) None of these

OPSC AEE 2019 Paper–I

Ans. (c) : The principle of virtual work can be applied to elastic system by considering the virtual work of Internal as well as external forces.

929. In the slope deflection equations, the deformations are considered to be caused by

(a) Shear force
(b) Bending moment
(c) Axial force
(d) Bending moment and shear force both

OPSC AEE 2019 Paper–I

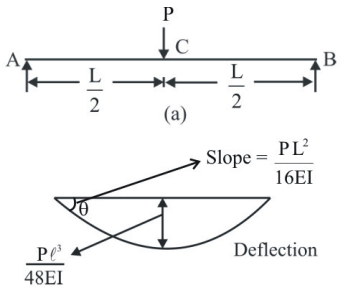
Ans. (b) : It is assumed that deformation are caused due to bending moment only and axial deformation are neglected.

930. The slope at the end of a simple supported beam of length (L), subjected to concentrated load (P), mement of inertia of its cross-section (I) and Young's modulus of the material is (E) is

(a) $\frac{PL^2}{16EI}$ (b) $\frac{PL^2}{24EI}$
(c) $\frac{PL^2}{48EI}$ (d) $\frac{PL^2}{8EI}$

OPSC AEE 2019 Paper–I
APPSC AEE MAINS 2016, TNPSC AE 2012

Ans. (a)

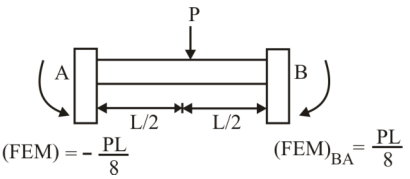


931. What is the value of the bending moment at built-in ends of a prismatic beam of length L, carrying single concentrated load P at the middle?

(a) $\frac{PL}{4}$ (b) $\frac{PL}{8}$
(c) $\frac{3PL}{8}$ (d) $\frac{PL}{2}$

OPSC AEE 2019 Paper–I

Ans. (b)



FEM = fixed end moment

932. The identical bars, one simply supported and other fixed at ends, are acted upon by equal loads applied at the midpoints. The ratio of strain energy stored in the simply supported beam and the fixed ended beam is :

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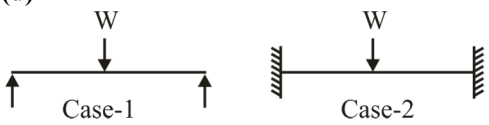
(a) 1 (b) 2
(c) 3 (d) 4

OPSC AEE 2019 Paper–I

Ans. (d)



We know, work done = Strain energy

$$U_{SSB} = \frac{1}{2}W.\delta_{max} = \frac{1}{2}.W.\frac{W\ell^3}{48EI} = \frac{W^2\ell^3}{96EI}$$

$$U_{fixed} = \frac{1}{2}.W.\ y_{max} = \frac{1}{2}W.\frac{W\ell^3}{192EI} = \frac{W^2\ell^3}{384EI}$$

$$\frac{U_{SSB}}{U_{fixed}} = \frac{\frac{W^2\ell^3}{96EI}}{\frac{W^2\ell^3}{384EI}} = 4$$

y_{max} = maximum deflection in fixed beam due to point load (W) act at the mid of span.

δ_{max} = maximum deflection in simply supported beam due to point load (W) act at mid of span.

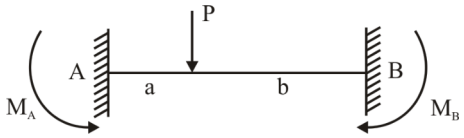
933. A beam of length L is fixed at its both ends and carries two concentrated loads of W each at a distance of $\frac{1}{3}$ from both ends. The fixed end moment at A will be :

(a) $\frac{-WL}{3}$ (b) $\frac{-2WL}{9}$

(c) $\frac{-6WL}{15}$ (d) $\frac{-4WL}{27}$

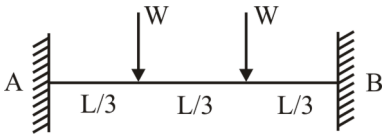
ESE 2020

Ans. (b) : We know that,

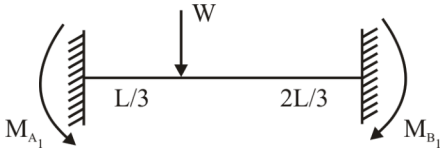


$$M_A = \frac{Pab^2}{(a+b)^2} \ \& \ M_B = \frac{Pa^2b}{(a+b)^2}$$

Now, as per question,



Applying principle of superposition,



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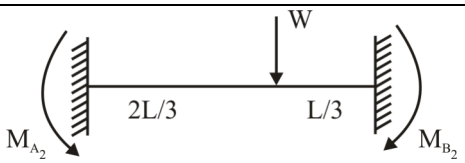


Now,

$$M_{A_1} = \frac{W\left(\frac{L}{3}\right)\left(\frac{2L}{3}\right)^2}{\left(\frac{L}{3}+\frac{2L}{3}\right)^2} = \frac{4WL}{27}$$

$$M_{A_2} = \frac{W\left(\frac{2L}{3}\right)\left(\frac{L}{3}\right)^2}{\left(\frac{L}{3}+\frac{2L}{3}\right)^2} = \frac{2WL}{27}$$

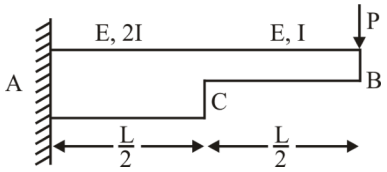
Now,

F.E.M. at A i.e., $M_A = M_{A_1} + M_{A_2}$

$$= \frac{4WL}{27} + \frac{2WL}{27}$$

$$\boxed{M_A = \frac{2WL}{9}(\text{Hogging})}$$

934. A cantilever beam ACB has end A fixed and subjected to a point load P at free end B. The point C is mid-point of AB and the moment of inertia of AC is twice that of CB. The deflection at the free end will be :

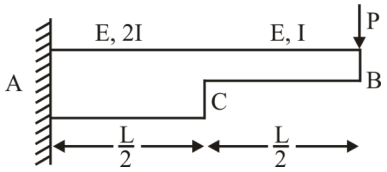


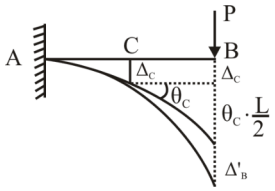
(a) $\frac{PL^3}{3EI}$ (b) $\frac{PL^3}{48EI}$

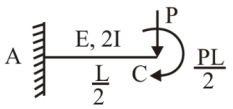
(c) $\frac{5PL^3}{96EI}$ (d) $\frac{9PL^3}{48EI}$

ESE 2020

Ans. (d)







$$\theta_C = \frac{ML}{EI} + \frac{WL^2}{2EI}$$

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$$= \frac{\frac{PL}{2}\times\frac{L}{2}}{2EI} + \frac{P\times\left(\frac{L}{2}\right)^2}{2E\times 2I} = \frac{PL^2}{8EI} + \frac{PL^2}{16EI}$$

$$\boxed{\theta_C = \frac{3PL^2}{16EI}}$$

$$\Delta_C = \frac{WL^3}{3EI} + \frac{ML^2}{2EI}$$

$$= \frac{P\times\left(\frac{L}{2}\right)^3}{3E\times 2I} + \frac{P\times\frac{L}{2}\times\left(\frac{L}{2}\right)^2}{2E\times 2I} = \frac{PL^3}{48EI} + \frac{PL^3}{32EI}$$

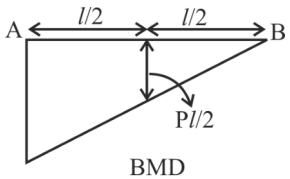
$$\boxed{\Delta_C = \frac{5PL^3}{96EI}}$$

$$\Delta_B = \Delta_C + \theta_C \cdot \frac{L}{2} + \Delta'_B$$

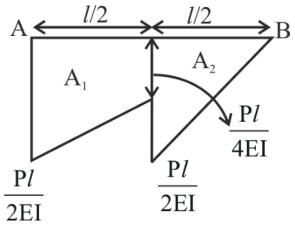
$$= \frac{5PL^3}{96EI} + \frac{3PL^3}{32EI} + \frac{PL^3}{24EI}$$

$$\boxed{\Delta_B = \frac{9PL^3}{48EI}}$$

Method (2)–



BMD



$\frac{M}{EI}$ Diagram

From area moment theorem–

$$t_{BA} = \delta = A_1\bar{X}_1 + A_2\bar{X}_2$$

$$\bar{X}_1 = \frac{2}{3}\times\frac{l}{2} = \frac{l}{3} \quad A_2 = \frac{1}{2}\times\frac{Pl}{2EI}\times\frac{l}{2} = \frac{Pl^2}{8EI}$$

$$\bar{X}_1 = \frac{l}{2} + \left[\frac{2\times\frac{Pl}{2EI} + \frac{Pl}{4EI}}{\frac{Pl}{2EI} + \frac{Pl}{4EI}}\right]\times\left[\frac{1}{3}\times\frac{l}{2}\right] = \frac{7}{9}l$$

$$A_1 = \frac{1}{2}\left[\frac{Pl}{2EI} + \frac{Pl}{4EI}\right]\times\frac{l}{2} = \frac{3}{16}\frac{Pl^2}{EI}$$

$$\delta = \frac{3}{16}\frac{Pl}{EI}\times\frac{7}{9}l + \frac{Pl^2}{8EI}\times\frac{l}{3}$$

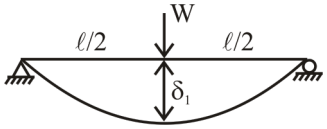
$$\delta = \frac{9Pl^3}{48EI}$$

935. A beam of uniform cross-section, simply supported at ends carries a concentrated load W at mid span. If the ends of the beam are fixed and only load P is applied at the mid span such that the deflection at the centre remains the same, the value of the load P will be :

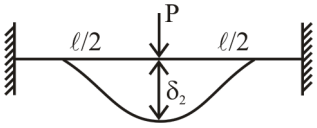
(a) 6W (b) 4W
(c) 2W (d) W

ESE 2020, APPSC AEE (Mains) 2016

Ans. (b)



When both ends are fixed



As per the question

$\delta_1 = \delta_2$

$$\frac{W\ell^3}{48EI} = \frac{P\ell^3}{192EI}$$

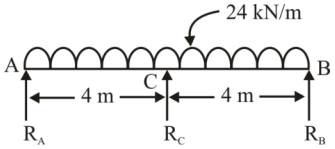
$$\boxed{P = 4W}$$

936. A simply supported beam of span 8 m carries a uniformly distributed load of 24 kN/m run over the whole span. The beam is propped at the middle of the span. The values of $E = 200 \times 10^6$ kN/m^2 and $I = 20 \times 10^{-5}$ m^4. The amount by which the prop should yield in order to make all three reactions equal will be nearly :

(a) 20 mm (b) 15 mm
(c) 10 mm (d) 5 mm

ESE 2020

Ans. (b)



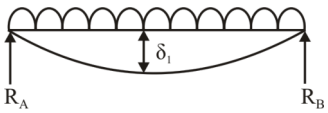
Given, $R_A = R_B = R_C = R$

Now, $\Sigma f_y = 0$

$R_A + R_B + R_C = 24 \times 8$

$3R = 24 \times 8$

$$R = \frac{24\times 8}{3} = 64\,\text{kN}$$



Assume middle point is removed, then

$$\delta_1 = \frac{5w\ell^4}{384EI}$$

$$= \frac{5\times 24\times 8^4}{384\times 200\times 10^6\times 20\times 10^{-5}}$$

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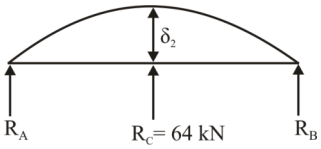
$\boxed{\delta_1 = 0.032\text{m}(\downarrow)}$



Apply middle reaction (R_c) as the point load (up)

$$\delta_2 = \frac{w\ell^3}{48EI}$$

$$= \frac{64\times(8)^3}{48\times200\times10^6\times20\times10^{-5}}$$

$\boxed{\delta_2 = 0.017\text{m}(\uparrow)}$

The yield/settlement of support C = 0.032–0.017

$\delta = 0.015$ m($\downarrow$)

or $\boxed{\delta = 15\text{mm}(\downarrow)}$

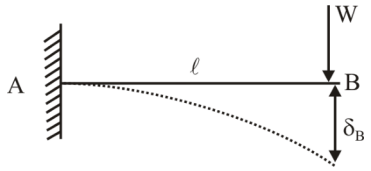
937. A cantilever beam AB as shown in figure is subjected to a point load of 12 kN over a span of 6 m with E = 2 × 10^5 N/mm^2 and I_{XX} = 6 ×10^7 mm^4. The deflection at the free end will be :

(a) 80 mm (b) 72 mm

(c) 64 mm (d) 56 mm

ESE 2020

Ans. (b)



Maximum deflection at free end (δ_B) = $\frac{W\ell^3}{3EI}$

$$= \frac{12\times10^3\times(6\times10^3)^3}{3\times2\times10^5\times6\times10^7}$$

$\boxed{\delta_B = 72\text{mm}}$

938. A simple supported steel I beam is replaced by a C-channel made with aluminum. Both sections have similar moment of inertia and are subjected to same loading. The following will remain true,

(a) The maximum bending stress and deflection will remain unaltered

(b) Only the maximum bending stress will remain unaltered

(c) Only the maximum deflection will remain unaltered

(d) None of the above

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : Steel-I

Steel channel (Al) given → I_1 (moment of inertia) = I_2 moment of inertia W_1 loading = W_2 loading

We know,

$M = \sigma z$, $M = \frac{wl}{4}$

$M = \sigma\frac{I}{Y}$ $\boxed{y_1 = y_2 = y/2}$

$\boxed{\sigma_1 = \sigma_2}$

for deflection at centre

$\Delta = \frac{wl^3}{48EI}$

$I_1 = I_2 = I$

$\boxed{E_{Al} = \frac{1}{3}E_{Steel}}$

$E_2 \neq E_1$

$\boxed{\Delta_1 \neq \Delta_2}$

deflection get altered

$\sigma_1 = \sigma_2$ As per above condition given

939. The elastic curve of a beam means

(a) BMD (b) SFD

(c) Deflection curve (d) Stress-strain curve

APPSC AEE MAINS 2016

Ans. (c) : The elastic curve of a beam means deflection.

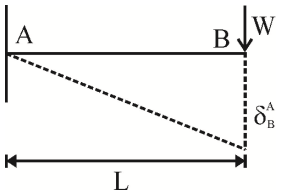
940. A cantilever beam of length L is subjected to a point load W at free end. Then the slope at the free end is equal to

(a) $\frac{WL^3}{3EI}$ (b) $\frac{WL^2}{24EI}$

(c) $\frac{WL^3}{60EI}$ (d) $\frac{WL^2}{2EI}$

APPSC AEE MAINS 2016

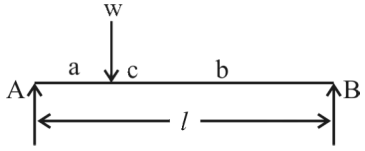
Ans. (d) : A cantilever beam of length L is subjected to a point load W at free end-



Slope $\theta_B^A = \frac{WL^2}{2EI}$

Deflection $\left(\delta_B^A\right) = \frac{WL^3}{3EI}$

941. A simply supported beam of length '*l*' carries a point load 'W' at point 'C' as shown in the figure. The maximum deflection lies at



(a) Point A (b) Point B

(c) Point C (d) Between points B and C

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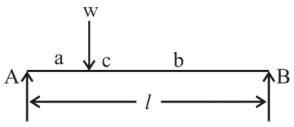
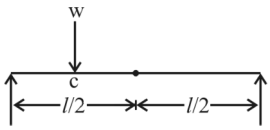
Ans. (d) :



For SSB, under unsymmetrical loading maximum deflection occurs between point of application of load and mid span



Region of zero slope (or) maximum deflection (i.e. in between points C and B)

942. The deflection of beams may be decreased by

(a) Increasing the depth of beam
(b) Increase the span
(c) Decreasing the depth of beam
(d) Increasing the width of beam

Maharashtra PSC Mains 2018 Paper-1

Ans. (a) : For simply supported beam deflection at centre $(\delta) = \frac{5Wl^4}{384EI}$

$$\delta = \frac{1}{I}$$

$$\delta \propto \frac{1}{D^4}$$

So as depth increases deflection is decreases.

943. Unit load method is based on

(a) internal strain energy
(b) theorem of minimum potential energy
(c) theorem of minimum deflection
(d) Castigliano's theorem

GPSC AE March 2018

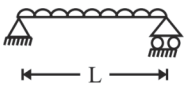
Ans. (a) : In unit load method, a virtual unit load is applied at the point of deflection (δ) and in the direction of deflection. It is internal strain energy based method.

944. A simply supported beam of span L carrying a uniformly distributed load registers a deflection of y cm at the centre. If the span of the beam is doubled, the deflection at the centre for the same uniformly distributed load would be

(a) 2y (b) 4y
(c) 8y (d) 16y

GPSC AE March 2018

Ans. (d) :



For simply supported beam having UDL:

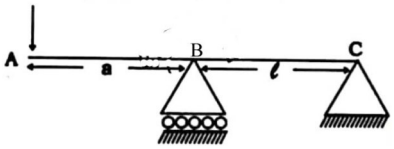
Deflection at centre $= \frac{5}{384}\frac{WL^4}{EI}$

$$\delta \propto L^4$$

If L' = 2L

So, δ' = 16δ or 16 y

945. For this given figure, slope at end B is



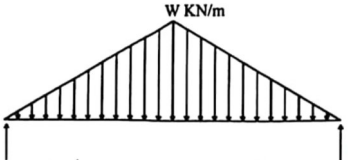
(a) $\frac{(Pa)l}{6EI}$ (b) $\frac{(Pa(l)}{3EI}$

(c) $\frac{(Pa)(l-2A)}{8EI}$ (d) $\frac{Pl^2}{6EI}$

Vizag steel MT 2017

Ans. (b) : Slope at end B is $\frac{(Pa(l)}{3EI}$

946. Maximum deflection for the given beam is.



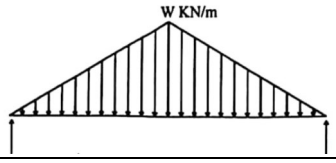
(a) $\frac{5WL^4}{384EI}$ (b) $\frac{WL^4}{120EI}$

(c) $\frac{WL^4}{48EI}$ (d) $\frac{3WL^4}{98EI}$

Vizag steel MT 2017

Ans. (b) : Maximum deflection for the given beam is

$$\frac{WL^4}{120EI}$$

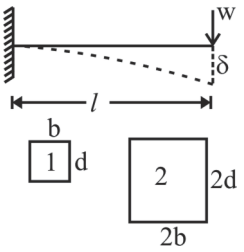


947. A cantilever beam (A) with rectangular cross section is subjected to concentrated load at its free end. If width and depth of another cantilever beam (B) are twice those of (A) then deflection at free end of beam B as compared to (A) will be in %

(a) 6.25 (b) 14
(c) 24.8 (d) 6.27

Arunachal Pradesh Poly. Tech. Lect. 2019
Vizag steel MT 2017

Ans. (a) :



Deflection at free end $(\delta) = \frac{wl^3}{3EI}$

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i.e. $\delta \propto \frac{1}{I}$

So, $\frac{\delta_1}{\delta_2} = \frac{I_2}{I_1} = \frac{2b \times (2d)^3/12}{b.d^3/12} = 16$

$\delta_1 = 16\delta_2$ *i.e.*, $\frac{\delta_2}{\delta_1} = \frac{1}{16} = 0.0625$

$\boxed{\frac{\delta_2}{\delta_1} = 6.25\%}$

948. Conjugate beam of an indeterminate structure will be

(a) Unstable (b) Indeterminate
(c) Stable (d) Can't say

Vizag steel MT 2017

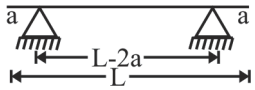
Ans. (a) : Neglective axial forces, statically determinate real beams have statically determinate conjugate beams. And statically indeterminate conjugate beams have unstable conjugate beams.

949. A beam of overall length L rests on two simple supports with equal overhangs on both sides. Two equal loads act at the free ends. If deflection at the centre of beam is same as at the either end then the length of either overhang is

(a) 0.152 L (b) 0.207 L
(c) 0.252 L (d) 0.277 L

Vizag steel MT 2017

Ans. (b) :



$a = \frac{(L-2a)}{2\sqrt{2}}$

$a(2\sqrt{2}+2) = L$

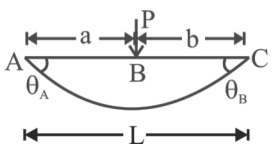
$\boxed{a = 0.207\ L}$

950. A simply supported beam ABC of length L carries a concentrated load P at an intermediate point B. If the slope at A is 0.75 times the slope at C then the length of portion AB is equal to

(a) 3/4 L (b) 4/7 L
(c) 5/7 L (d) 2/7 L

Vizag steel MT 2017

Ans. (c) :



$\theta_A = \frac{Pab(b+L)}{6EI}$

$\theta_C = \frac{Pab(a+L)}{6EI}$

Now, $\theta_A = 0.75\ \theta_C$

$\frac{Pab(b+L)}{6EI} = \frac{3}{4}\frac{Pab(a+L)}{6EI}$

$4L + 4b = 3a + 3L$

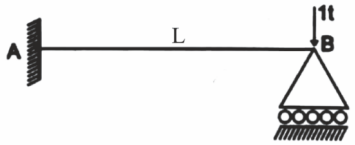
$L + 4b - 3a = 0$

$L + 4(L - a) - 3a = 0$

$5L - 7a = 0$

$\boxed{a = \frac{5L}{7}}$

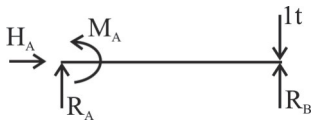
951. Reaction at support A for given propped cantilever.



(a) 0 t (b) 1 t
(c) 0.5 t (d) 2 t

Vizag steel MT 2017

Ans. (a) :



$\sum V = 0, R_A + R_B = 1t$

$\sum M_A = 0,\ R_B.L - 1t(L) = 0$

$R_B = 1t$

Now deflection i.e. $A_B = 0 = R_B = 1t$, $R_A = 1t - R_B = 0$

At B = o, $R_A = 0$

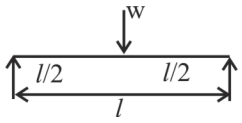
952. If the length of a simply supported beam carrying concentrated load at the centre is doubled, the deflection at the centre will become :

(a) two times (b) four times
(c) eight times (d) sixteen times

Punjab MCP SDO 2015

Ans. (c) :

$y = \frac{wl^3}{48EI}$ $\frac{y_1}{y_2} = \frac{l_1^3}{l_2^3}$



$\frac{y_1}{y_2} = \frac{l_1^3}{(2l_1)^3}$

$\frac{y_1 \times 8 \times l_1^3}{y_2 \times l_1^3} = y_1 \times 8 = y_2$

$\boxed{y_2 = 8y_1}$

953. The maximum deflection of a cantilever beam of length 'l' with a uniformly distributed load of 'W' per unit length is .....

(a) $\frac{Wl^3}{3EI}$ (b) $\frac{Wl^3}{8EI}$

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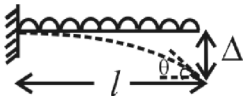
(c) $\frac{Wl^3}{16EI}$ (d) $\frac{Wl^3}{48EI}$

GPSC R&B AE 2016

Ans. (b) : Deflection $= \frac{wl^4}{8EI}$,



w = UDL Total load on beam W =wL

$\Delta = \frac{Wl^3}{8EI}$ $\theta = \frac{wl^3}{6EI} = \frac{WL^2}{6EI}$

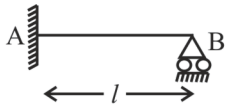
954. A propped cantilever of span *l* carries a uniformly distributed load of w per unit run over its entire span. The value of prop reaction to keep the beam horizontal is-

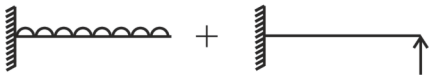
(a) $\frac{wl}{3}$ (b) $\frac{3}{8}wl$

(c) $\frac{wl}{2}$ (d) $\frac{5}{8}wl$

TNPSC AE 2018, APPSC (PHE) AE Paper-2 2012
TRB Poly. Tech. Lect. 2012, JPSC AE 2009
APPSC AEE (Civil/Mechanical) 2012
Maharashtra PSC AE (Mains) 2012 Paper–I

Ans. (b) : A propped cantilever beam A as fixed and B is propped. Deflection at B = 0.





So, deflection at B for cantilever beam with UDL is equal to deflection at B for cantilever beam with upward reaction.

So, $\frac{wl^4}{8EI} = (R_b)\frac{l^3}{3EI}$ ...(1)

$R_A + R_B = wl$ ...(2)

By equation (1)

$$\boxed{R_B = \frac{3wl}{8}}$$

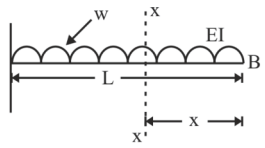
955. The deflection at the free end of a cantilever beam of span 'L' and flexural rigidity EI subjected to UDL(w / unit run) is

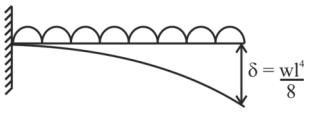
(a) $wL^2/2EI$ (b) $wL^3/3EI$

(c) $wL^4/8EI$ (d) $wL^3/6EI$

Rajasthan AE Nagar Nigam Shift-3, 2016
APPSC AEE 2016, *TSGENCO_2015*
ASSAM PSC CCE (PRE) 2015
J&K PSC Pre 2013
Maharashtra PSC AE (Mains) 2012 Paper–I
UKPSC AE Paper–I 2007, WBPSC AE 2003

Ans. (c) :





$$m_x = \frac{-wx^2}{2}$$

$$EI\frac{d^2y}{dx^2} = \frac{-wx^2}{2}$$

$$EI\frac{dy}{dx} = \frac{-wx^3}{6} + C_1$$

At x = l, $\frac{dy}{dx} = 0$

$$C_1 = \frac{wl^3}{6}$$

$$EI\frac{dy}{dx} = \frac{-wx^3}{6} + \frac{wl^3}{6}$$

$$EI\ y = \frac{-wx^4}{24} + \frac{wl^3x}{6} + C_2$$

At x = 1, y = 0, $C_2 = \frac{wl^4}{8}$

$$EI\ y = \frac{-wx^4}{24} + \frac{wl^2x}{6} - \frac{wl^4}{8}$$

At x = 0, y = y_{max} $\boxed{l = L}$

$$\boxed{y_{max} = \frac{-wL^4}{8}}$$

956. If maximum deflection of a beam is given by, δ = kWL^3/EI then what are the values of k if the beam is (a) cantilever and (b) both and fixed (udl with total load W and length = L, flexural rigidity = EI

(a) 5/384, 1/384 (b) 1/384, 5/384

(c) 1/8, 1/384 (d) 1/3, 5/384

TANGEDCO AE 2018

Ans. (c) :

$$\delta = \frac{KWL^3}{EI}$$

(a) For cantilever beam, for udl loading

$$\delta_{max} = \frac{WL^3}{8EI}$$ where W = wL

(b) For simply supported beam, for udl

$$\delta_{max} = \frac{5WL^3}{384EI}$$

where, W = wL

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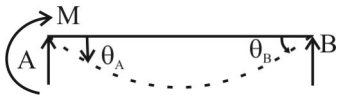
957. A uniform simply supported beam is subjected to a clockwise moment M at the left end. The moment required at the right end of the beam so that the rotation of right end is zero is equal to

(a) 2 M (b) M

(c) $\frac{M}{2}$ (d) $\frac{M}{3}$

TRB Poly. Tech. Lect. 2012

Ans. (c) :

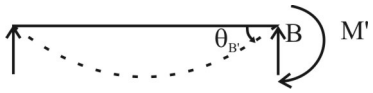


$\theta_A = \frac{ML}{3EI}$

Due to moment at A, $\theta_B = \frac{M_0 L}{6EI}$ ———(1)

Moment at B, M' applied to make

$\theta_B = 0$



$\therefore \theta_B' = \frac{M_O' L}{3EI}$ ———(2)

Hence equating both to make zero

$\boxed{M_o' = \frac{M_o}{2}}$

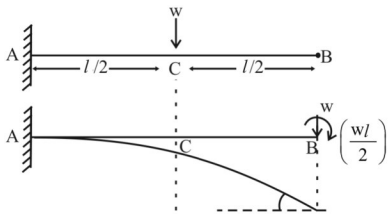
958. A cantilever beam of span *l* and flexural rigidity EI is subjected to a concentrated load W at mid-span. The slope at the free end is

(a) $\frac{Wl^2}{2EI}$ (b) $\frac{Wl^2}{4EI}$

(c) $\frac{Wl^2}{8EI}$ (d) $\frac{Wl^2}{3EI}$

TRB Poly. Tech. Lect. 2012

Ans. (a) :



From direct standard

Result $\theta_c = \theta_b = \frac{wl^2}{2EI}$ (for point load)

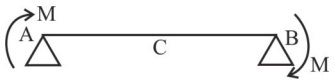
959. Clockwise moments M are acting at both the ends of a uniform simply supported beam. The ratio of slope at the end to the slope at centre will be

(a) 0.5 (b) 1

(c) 2 (d) 3

TRB Poly. Tech. Lect. 2012

Ans. (c) :

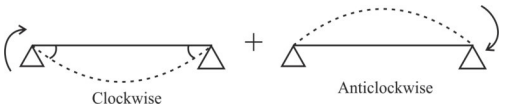


As we know,

$\theta_A = \frac{+ML}{3EI}$ (clockwise)

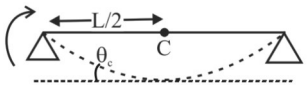
$\theta_B = \frac{-ML}{6EI}$ (anticlockwise)

Hence,



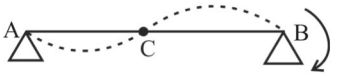
$= \frac{ML}{3EI} - \frac{ML}{6EI} = \boxed{\frac{ML}{6EI} = \theta_A = \text{clockwise}}$

As we know,



$\theta_c = \frac{ML}{24EI}$ (clockwise)

When both end moment applied then rotation will be like.



Both have same nature so

$\theta_c = \frac{ML}{24EI} + \frac{ML}{24EI} = \frac{ML}{12EI}$

$\boxed{\frac{\theta_A}{\theta_C} = \frac{ML}{6EI} \div \frac{ML}{12EI} = 2}$

960. Deflection can be controlled in a beam by using the appropriate

(a) water cement ratio (b) span depth ratio

(c) aspect ratio (d) modular ratio

UKPSC AE (Urban) 2011

Ans. (c) : Maximum ratio of lift to deflection work are shown for untapered controls of high aspect ratio.

961. In the moment area method, the deflection of a point A from the tangent at B is equal to the

(a) Area of M/EI diagram between A and B

(b) Moment of M/EI diagram between A and B about A

(c) Moment of M/EI diagram between A and B about B

(d) 0.5x area of M/EI diagram between A and B

Punjab PSC SDE 2016

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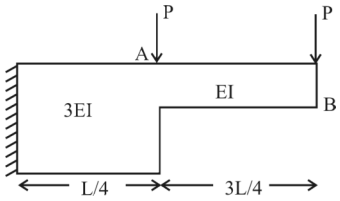
Ans. (b) : Deflection of any point on the elastic curve wrt tangent drawn at another point is equals to moment of area under $\frac{M}{EI}$ diagram between two points, about point first.
Slope between two points on the elastic curve equals to the area of $\frac{M}{EI}$ diagram between those two points.

962. The cantilever beam shown in figure has load P acting at points A and B. The deflection at B is Δ, when the load at B is removed. When the load at A is removed, the deflection at A will be.



(a) $\Delta/4$ (b) $2\Delta/3$
(c) $\Delta/2$ (d) Δ

UJVNL AE 2016

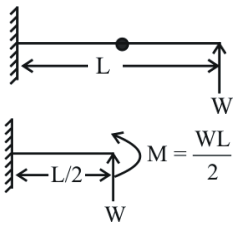
Ans. (d) : From Maxwell reciprocal theorem
$\Delta_A \times P = \Delta_B \times P$
$\Delta_A = \Delta_B = \Delta$

963. A cantilever beam carries a concentrated load W at its free end acts upward. The deflection at mid span of beam will be.

(a) $\frac{7WL^3}{48EI}$ (b) $\frac{WL^3}{6EI}$
(c) $\frac{WL^3}{48EI}$ (d) $\frac{WL^3}{3EI}$

UJVNL AE 2016

Ans. (c)

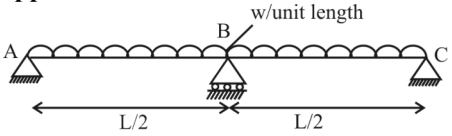


Deflection of mid span = $\frac{W\ell^3}{3EI} - \frac{M\ell^2}{2EI}$ $\left(\ell = \frac{L}{2}\right)$

$= \frac{WL^3}{24EI} - \frac{WL \times L^2}{2 \times 2 \times 4EI}$

$= \frac{WL^3}{24EI} - \frac{WL^3}{16EI} = \frac{WL^3}{48EI}$

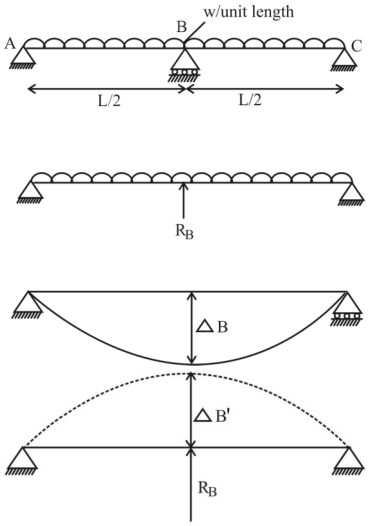
964. For the beam shown below, the reaction at support B



(a) $\frac{3wL}{8}$ (b) $\frac{wL}{6}$
(c) $\frac{5wL}{8}$ (d) $\frac{wL}{4}$

UJVNL AE 2016

Ans. (c) :



There is no deflection at support B
So, $\Delta B = \Delta B'$

$\frac{5wL^4}{384EI} = \frac{R_B L^3}{48EI}$

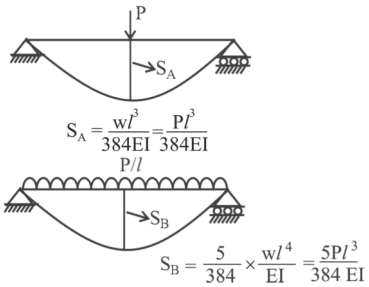
$R_B = \frac{5}{8}wL$

965. System A is simply supported beam with a load P at midspan. System B is the same beam, but the load is replaced by uniformly distributed load of intensity P/L, wherein L is the span. The midspan deflection of system B will

(a) be the same as that of system A at midspan
(b) be less than that of system A at midspan
(c) be more than that of system A at midspan
(d) bear no relation to that of system

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (b) :



$S_A = \frac{wl^3}{384EI} = \frac{Pl^3}{384EI}$

$S_B = \frac{5}{384} \times \frac{wl^4}{EI} = \frac{5Pl^3}{384\,EI}$

$\therefore S_A > S_B$

The mid span of system B deflection is less than that of system A at midspan.

966. The ratio of maximum deflection of a beam to its span is called

(a) section modulus of the beam

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(b) modulus rigidity of the beam
(c) stiffness of the beam
(d) bulk modulus of the beam

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Ans. (c) : The stiffness of a beam is a measure of it's resistance against deflection.
ie the ratio of maximum deflection of a beam to its span can be termed as stiffness of the beam.

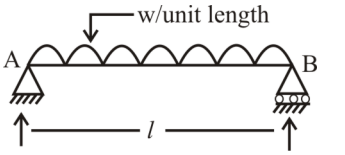
967. Slope at the end of the simply supported beam of span *l* with uniformly distributed load w/unit length over the entire span is given by

(a) $\frac{wl^2}{16EI}$ (b) $\frac{wl^3}{16EI}$

(c) $\frac{wl^3}{24EI}$ (d) $\frac{wl^2}{24EI}$

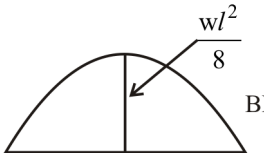
GPSC AE Dec. 2018, APPSC AEE 2012
***RPSC ACF 2011*, UKPSC AE (Paper–I) 2007**
UPPSC State Eng. A.E. 2007 (II)

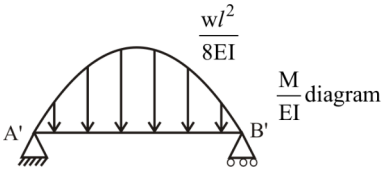
Ans. (c) :



$R_A = \frac{w\ell}{2}$ $R_B = \frac{w\ell}{2}$

By conjugate method





$R'_A = R'_B = \theta_A = \theta_B = \frac{1}{2} \times \frac{2}{3} \times l \times \frac{wl^2}{8EI}$

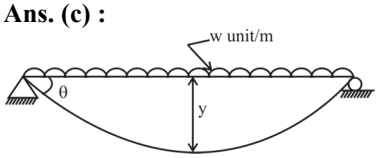
$\theta_B = \frac{wl^3}{24EI}$

968. A simply supported beam is of rectangular section. It carries a uniformly distributed load over the whole span. The deflection at the centre is 'y'. If depth of the beam is doubled, the deflection at centre will be :

(a) 2y (b) 4y
(c) y/8 (d) y/2

TRB Poly. Lect. Tech. 2017

Ans. (c) :



$y = \frac{5}{384}\frac{Wl^4}{EI}$ $\theta = \frac{Wl^3}{24EI}$

$y = \frac{5}{384}\frac{Wl^4}{E\left(\frac{bd^3}{12}\right)}$ when, depth = d

Now,

$y_1 = \frac{5}{384}\frac{WL^4}{E\left(\frac{b(2d)^3}{12}\right)} = \frac{5}{384}\frac{WL^4}{E\left(\frac{8bd^3}{12}\right)}$

$= \frac{5}{384}\frac{WL^4}{E\frac{bd^3}{12}} \times \frac{1}{8}$

$y_1 = y/8$

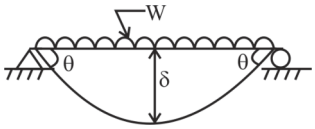
969. Slope at the support end of a simply supported beam subjected to uniformly distributed load W throughout its length *l* is equal to (where elasticity E and moment of inertia I are constant) :

(a) $\frac{Wl^2}{24EI}$ (b) $\frac{Wl^3}{4EI}$

(c) $\frac{Wl^3}{8EI}$ (d) $\frac{Wl^2}{48EI}$

SJVNL ET 2019

Ans. (a) : W = wl



$\theta = \frac{wl^3}{24EI}$ (standard result)

$\theta = \frac{Wl^2}{24EI}$

970. The bending moment at any section in a conjugate beam gives in the actual beam

(a) bending moment (b) slope
(c) curvature (d) deflection

BPSC Poly. Tech. Lect. 2016

Ans. (d) : The bending moment at any section in a conjugate beam gives the deflection in the actual beam i.e., by second moment of area.
Likewise, by first moment of area, the shear force in a conjugate beam gives the slope in the actual beam.

971. A beam of length L fixed by both ends carries a uniformly distributed load of W per unit length. The deflection at the midspan is

(a) $\frac{WL^4}{384EI}$ (b) $\frac{5WL^4}{384EI}$

(c) $\frac{3WL^4}{384EI}$ (d) None of the above

EI is rigidity of the beam.

BPSC Poly. Tech. Lect. 2016
APPSC AEE (Civil/Mechanical) 2012

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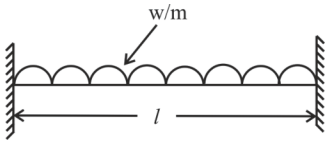
Ans. (a) :



$\Delta = \frac{1}{5} \times$ deflection in simple supported beam carrying udl.

$= \frac{1}{5} \times \frac{5}{384} \frac{wl^4}{EI}$

$\boxed{\Delta_{at\ centre} = \frac{1}{384} \frac{wl^4}{EI}}$

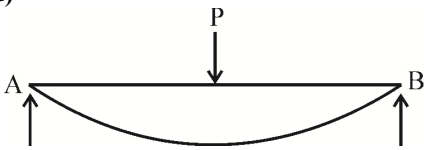
972. An isotropic, prismatic beam of span length (s) made up of a heterogeneous material (elastic modulus E and MI, I) and carries a mid-point load (P). The deflection under the load will be:

(a) $Ps^3/3EI$, downward (b) $Ps^3/3EI$, upward

(c) $Ps^3/12EI$, downward (d) $Ps^3/48EI$,downward

UPRVUNL AE 2016

Ans.: (d)



An isotropic, prismatic beam of span lengths and carries a mid- point load (p) then the maximum deflection will be.

$$\delta_{max} = \frac{Ps^3}{48EI}$$

And maximum slope will be.

$$\theta_{max} = \frac{Ps^2}{16EI}$$

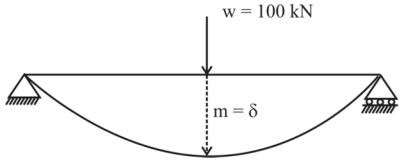
973. The maximum deflection in 'm' in a simply supported beam of span 10m subjected to a central point load of 100 kN is :
($EI = 2 \times 10^4$ kN – m^2)

(a) $\frac{10}{48}$ (b) $\frac{10}{96}$

(c) $\frac{10}{192}$ (d) $\frac{10}{16}$

TRB Poly. Lect. Tech. 2017

Ans. (b) :



Maximum deflection m,

$\delta = m = \frac{wL^3}{48EI} = \frac{100 \times 10^3}{48 \times 2 \times 10^4} = \frac{10}{48 \times 2} = \frac{10}{96}$

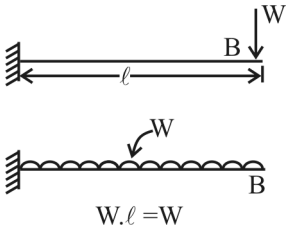
974. The ratio of maximum deflection of a cantilever beam of span 'I' with (i) a load of 'W' at the free end & (ii) a uniformly distributed load over entire length of total 'W' is given by

(a) 8/5 (b) 8/3

(c) 3/8 (d) –5/8

GPSC AE June 2019

Ans. (b) :



$W.\ell = W$

$(\delta_B)_1 = \frac{W\ell^3}{3EI}$

$(\delta_B)_2 = \frac{W\ell^3}{8EI}$

$\boxed{\frac{(\delta_B)_1}{(\delta_B)_2} = \frac{8}{3}}$

975. "The deflection of a point in the real beam is numerically equal to the bending moment at the corresponding point in the conjugate beam". This theorem is used for analysis of beams in which of the following methods?

(a) Castigliano's method

(b) Moment area method

(c) Conjugate beam method

(d) None of the above

GPSC AE June 2019

Ans. (c) : 'The deflection of a point in real beam is numerically equal to bending moment at the corresponding point in conjugate beam'. This theorem is used for analysis of beam by conjugate beam method.

976. A cantilever beam with rectangular cross-section is subjected to uniformly distributed load. The deflection at the tip is δ_1. If the width and depth of the beam are doubled then deflection at tip is δ_2. Then $\frac{\delta_2}{\delta_1}$ is

(a) 0.0625 (b) 16

(c) 0.5 (d) 2

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Deflection $\propto \frac{1}{I}$

$\therefore \quad \frac{\delta_2}{\delta_1} = \frac{I_1}{I_2} = \frac{\left(\frac{bd^3}{12}\right)}{\frac{(2b)(2d)^3}{12}} = \frac{1}{16} = 0.0625$

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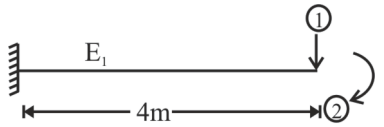
977. Displacement coordinators for the beam are as shown in figure. The flexibility matrix is given by:

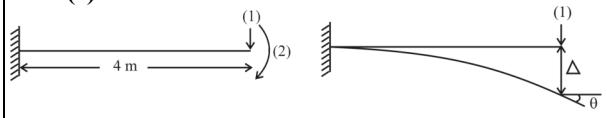


(a) $\frac{1}{EI}\begin{bmatrix} 64/3 & -8 \\ -8 & 64 \end{bmatrix}$ (b) $\frac{1}{EI}\begin{bmatrix} 64/3 & 8 \\ 8 & -64/3 \end{bmatrix}$

(c) $\frac{1}{EI}\begin{bmatrix} 64/3 & 8 \\ 8 & 4 \end{bmatrix}$ (d) $\frac{1}{EI}\begin{bmatrix} 4 & -8 \\ -8 & 64/3 \end{bmatrix}$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (c)



$f_{11} = \Delta = \frac{wL^3}{3EI} = \frac{1\times 4^3}{3EI} = \frac{64}{3EI}$

$f_{12} = f_{21} = \frac{wL^2}{2EI} = \frac{L\times 16}{2EI} = \frac{8}{EI}$

$f_{22} = \theta = \frac{ML}{EI} = \frac{1\times 4}{EI} = \frac{4}{EI}$

$f = \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} = \begin{bmatrix} \frac{64}{3EI} & \frac{8}{EI} \\ \frac{8}{EI} & \frac{4}{EI} \end{bmatrix} = \frac{1}{EI}\begin{bmatrix} 64/3 & 8 \\ 8 & 4 \end{bmatrix}$

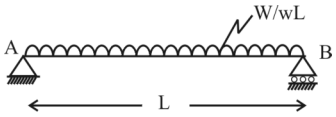
978. Maximum deflection of a simply supported beam with the total uniformly distribute load 'W' is:

(a) $\frac{Wl^3}{384EI}$ (b) $\frac{5Wl^3}{384EI}$

(c) $\frac{Wl^3}{48EI}$ (d) $\frac{5Wl^3}{48EI}$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (b)



Slope =

$\theta_{A,B} = \frac{wL^3}{24EI} = \frac{WL^2}{24\ EI}$

$W = wl, \quad w = \frac{W}{l}$

Deflection = $y_C = \frac{5WL^3}{384\ EI} = \frac{5wL^4}{384\ EI}$

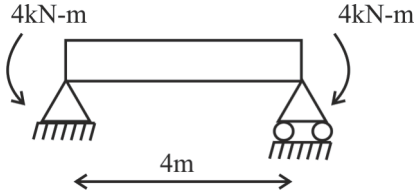
979. If a simply supported beam of span 4m is subjected to terminal couple of 4 kN-m at both the ends, then the magnitude of the central deflection would be

(a) 4/EI (b) 8/EI

(c) 2/EI (d) 16/EI

GPSC AE Dec 2018, ESE 2009

Ans. (b) :



$\Delta_{central} = \frac{ML^2}{8EI} = \frac{4\times(4)^2}{8EI} = \frac{8}{EI}$

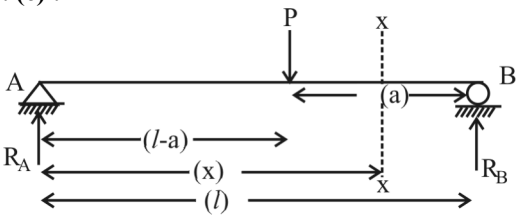
980. The location of the maximum deflection for a simply supported beam of span *l* having point load P at a distance from the right hand end would be

(a) Below point load P (b) $\sqrt{\frac{l^2-a^2}{3}}$

(c) $\sqrt{\frac{l^2-(l-a)^2}{3}}$ (d) $\sqrt{\frac{a^2-(l-a)^2}{3}}$

GPSC AE Dec 2018

Ans. (c) :



$\sum M_A = 0 \Rightarrow R_B \times l = P\times(l-a)$

$R_B = P\left(1-\frac{a}{l}\right)$

$R_A = \frac{Pa}{l}$

$M_x = R_A x - P\times(x-l+a)$

$M_x = \frac{EId^2y}{dx^2} - \frac{Pax}{l} - P(x-l+a)$

$\frac{EIdy}{dx} = \frac{Pax^2}{2l} - \frac{P(x-l+a)^2}{2} + c$

For maximum deflection, $\frac{dy}{dx} = 0$

$x = \sqrt{\frac{l^2-(l-a)^2}{3}}$

981. Which of the following represents the bending at a section of the beam?

(a) $EI\frac{d^4y}{dx^4}$ (b) $EI\frac{d^3y}{dx^3}$

(c) $EI\frac{d^2y}{dx^2}$ (d) $EI\frac{dy}{dx}$

GPSC AE Dec 2018

APPSC AEE (Civil/Mechanical) 2012

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Ans. (c) :

$$\frac{M}{EI}=\frac{d^2y}{dx^2}$$

EIY = Deflection

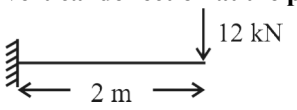
$EI\frac{dy}{dx}$ = Slope

$EI\frac{d^2y}{dx^2}$ = Bending moment

$EI\frac{d^3y}{dx^3}$ = Shear force

$EI\frac{d^4y}{dx^4}$ = Load intensity

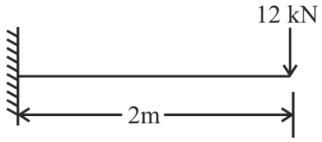
982. A cantilever beam of span 2 m is shown in the figure. The moment to be applied at the fee end for zero vertical deflection at the point is :



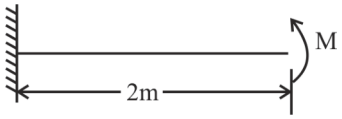
(a) 12 kN-m (anticlockwise)
(b) 16 kN-m (anticlockwise)
(c) 6 kN-m (anticlockwise)
(d) 24 kN-m (anticlockwise)

BHEL ET 2019

Ans. (b) :



$$\Delta_1=\frac{PL^3}{3EI}=\frac{12\times(2)^3}{3EI}$$



When M is anticlockwise

$$\Delta_2=\frac{ML^2}{2EI}$$

$$\Delta_2=\frac{M\times(2)^2}{2EI}$$

$\Delta_1=\Delta_2$ (for zero vertical deflection)

$$\Rightarrow \quad \frac{12\times(2)^3}{3EI}=\frac{M\times(2)}{EI}$$

M = 8 × 2 = 16 kN-m

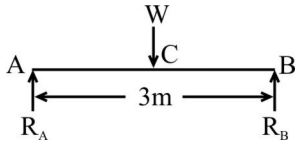
983. A simply supported becam span 3m, is subjected to a central point load of 5kN, then the slope at the mid span is equal to

(a) $\frac{25}{24EI}$ (b) $\frac{256}{EI}$

(c) $\frac{40}{48EI}$ (d) Zero

APPSC AEE (Civil/Mechanical) 2012

Ans. (d)



Slope $=\frac{w\ell^2}{16EI}$, Deflection $=\frac{w\ell^3}{48EI}$

W = 5 KN (Central point load.)

l = 3 m (beam span)

Slope at the mid span = 0

Deflection at the mid span = Maximum.

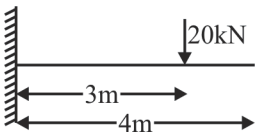
984. A rolled steel beam having a span of 4m carries a point load of 20kN at 3m from the left support. If the moment of inertia of the section is $12\times10^7\,mm^4$, E=200 kN/mm^2 then the deflection of the beam under the point load is equal to

(a) 25 mm (b) 7.5 mm

(c) 13.33 mm (d) 50 mm

APPSC AEE (Civil/Mechanical) 2012

Ans. (b)



l = 4m

a = 3m

E = 200 kN/mm^2

I = 12×10^7 mm^4

The deflection of the beam under the point load.

$$=\frac{Wa^3}{3EI}=\frac{(20\times1000)\times(3\times10^3)^3}{3\times200\times10^3\times12\times10^7}$$

$$=\frac{2\times10^4\times27\times10^9}{3\times2\times12\times10^5\times10^7}=\frac{270}{36}=7.5\text{ mm}$$

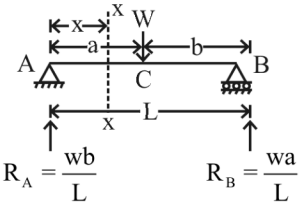
985. For a simply supported beam AB of length L carries a point load W at point C at a distance a from A and b from B. The deflection at C is:

(a) $\frac{WL^3}{48EI}$ (b) $\frac{Wa^2b^2}{48EI}$

(c) $\frac{5}{384}\frac{WL^4}{EI}$ (d) None of these

PTCUL AE 2016

Ans. (d) :



$M_1=\frac{wb}{L}.x$

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$$\frac{EId^2y}{dx^2} = \frac{wb}{L}.x$$ Integrating both side

$$\frac{EIdy}{dx} = \frac{wb}{L}.\frac{x^2}{2} + C_1 .....(i)$$

Again integrating both sides

$$EI\,y = \frac{wb}{L}.\frac{x^3}{6} + C_1x + C_2 \quad ..... (ii)$$

Applying boundary conditions in (ii) x = 0, y = 0

$$\boxed{C_2 = 0}$$

x = L, y = 0

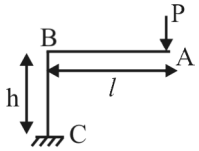
$$\frac{-wbL^2}{6} = C_1.L$$

$$C_1 = \frac{-wbL}{6}$$

The deflection at C

$$\frac{wba^3}{6EIL} - \frac{wbaL}{6EI}$$

986. A rigid cantilever frame ABC is fixed at C. It carries a load P at A. Neglecting axial deformation the vertical deflection of the point A is given by:

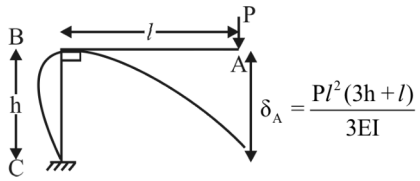


(a) $\frac{Pl^2h}{3EI}$ (b) $\frac{Pl^2h(2h+l)}{2EI}$

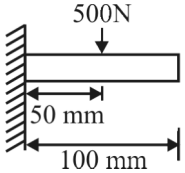
(c) $\frac{Pl^2h}{2EI}$ (d) $\frac{Pl^2(3h+l)}{3EI}$

PTCUL AE 2016

Ans. (d) :



987. A cantilever beam with flexural rigidity of 200 Nm^2 is loaded as shown in the figure. The deflection (in mm) at the tip of the beam is:

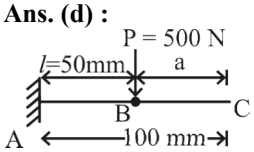


(a) 0 (b) 0.16

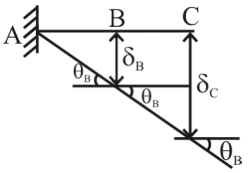
(c) 1.20 (d) 0.26

PTCUL AE 2016

Ans. (d) :



EI = 200 N-m^2



$\delta_C = \delta_B + \theta_B.a$

$$\delta_B = \frac{pl^3}{3EI},\ \theta_B = \frac{pl^2}{2EI}$$

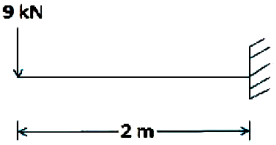
$$\boxed{\delta_C = \frac{pl^3}{3EI} + \frac{pl^2.a}{2EI}}$$

$$\delta_C = \frac{500\times(0.05)^3}{3\times200} + \frac{500\times(0.05)^2\times(0.05)}{2\times200}$$

$\delta_C = 2.6 \times 10^{-4}$ m

$\boxed{\delta_C = 0.26 \text{ mm}}$

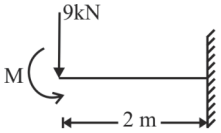
988. For the cantilever beam shown below, the moment to be applied at free end for zero vertical deflection at that point is –?



(a) 9 kN.m anti clockwise
(b) 9 kN.m clockwise
(c) 12 kN.m clockwise
(d) 12 kN.m anti clockwise

UPPCL AE 2014

Ans. (d) :



$$\delta_1 = \frac{PL^3}{3EI}$$

$$\delta_2 = \frac{-ML^2}{2EI}$$

$$\delta_1 + \delta_2 = 0$$

$$\frac{PL^3}{3EI} = \frac{ML^2}{2EI}$$

$$M = \frac{2}{3}\times9\times2 = 12 \text{ kN-m}$$

$\boxed{\text{M = 12 kN-m anticlockwise}}$

989. Ratio of deflections of free end of cantilever due to a concentrated load at 1/3rd and 2/3rd of the span is –

(a) 3/7 (b) 4/7

(c) 2/7 (d) 1/7

UPPCL AE 2014

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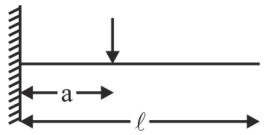
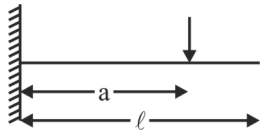
Ans. (c) : $y = \frac{Wa^2}{2EI} \times \left(\ell - \frac{a}{3}\right)$

Case-I : $a = \frac{1}{3} \times \ell$



$$y_1 = \frac{W \times \ell^2}{2EI \times 9} \times \left(\ell - \frac{\ell}{3 \times 3}\right) = \frac{W\ell^3}{18EI} \times \frac{8}{9}$$

Case-II : $a = \frac{2}{3}\ell$



$$y_2 = \frac{W \times 4 \times \ell^2}{2EI \times 9} \times \left(\ell - \frac{2\ell}{3 \times 3}\right) = \frac{4W\ell^3}{18EI} \times \frac{7}{9}$$

$$\frac{y_1}{y_2} = \frac{2}{7}$$

990. A beam length 4 m, fixed at both ends carries a point load of 120 kN at the centre. If EI for the beam is 2000 kN m^2, deflection at the centre of beam is

(a) 1.0 mm (b) 2.0 mm
(c) 5.0 mm (d) 10.0 mm

APPSC (PHE) AE Paper-2 2012

Ans. (b) Let, δ_c = deflection at the centre (c) of beam is fixed at both ends $= \frac{wl^3}{192EI}$

Given,

EI = 2000 kN-m^2, w=120 kN, l = 4m

$$= \frac{120\ \text{kN} \times (4)^3\ \text{m}^3}{192 \times 2000\ \text{kN} - \text{m}^2} = 0.02\ \text{m}$$

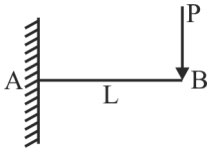
= 20 mm [Nearly option is correct]

991. Maximum deflection for a cantilever of span L loaded at the free end by P is given by

(a) $\frac{PL^2}{3EI}$ (b) $\frac{PL^2}{6EI}$
(c) $\frac{PL^3}{8EI}$ (d) $\frac{PL^3}{3EI}$

SJVNL ET 2019
Punjab PSC SDE 2016
APPSC (PHE) AE Paper-2 2012

Ans. (d) :



Let, AB is cantilever beam of span L, loaded at the free end by P.

At point 'A' there is no deflection

Let, a section X-X at a distance x form LHS from point A

Bending equation –

$M_x = -R(L-X)$

By curvature equation

$$EI\frac{d^2y}{dx^2} = +Mx - P(L-x)$$

$$EI\frac{dy}{dx} = -\int P(L-X)dx + G = -P\left(Lx - \frac{x^2}{2}\right) + C$$

Boundary condition for all cantilever beam–

(i) at x = 0 ; $\frac{dy}{dx} = 0$

From equation (1)

$EI \times 0 = -P(L \times 0 - 0) + C$

$C_s = 0$ New, putting this value in equation .....(1)

$$EI\frac{dt}{dx} = -P\left(Lx - \frac{x^2}{2}\right) + 0 \qquad \text{.....(2)}$$

equation represents slope equation

Therefore, solving equation (2)

$$EI \times y = -P\left(\frac{Lx^2}{2} - \frac{x^3}{6} + C_1x + C_2\right) \qquad \text{..... (3)}$$

$\therefore$ Boundary condition–

x = 0; y = 0

$$\therefore\ EI \times B = -P\ L \times 0 - \frac{0}{6} + C_1 \times 0 + C_2$$

$C_C = 0$

and also, C_1=0

Therefore';

$$EI.y = -P\left(\frac{Lx^2}{Z} - \frac{X^3}{6}\right) \qquad \text{.....(4)}$$

Equation (4) represents the deflection equation.

$\therefore$ Deflection at free end (B) of the cantilever beam is

i.e, at x = L Deflection (y) = y_B

$$EI.y_B = -P\left(\frac{L^3}{2} - \frac{L^3}{6}\right)$$

$$y_B = = \frac{-PL^3}{3EI}$$

Negative sign represents the deflection is downward direction.

$$y_B = \frac{PL^3}{3EI}$$

992. In the procedure of Macaulay's method, for getting the deflection equation

(a) The differential equation of flexure is integrated for once
(b) The differential equation of flexure is integrated for twice
(c) The differential equation of flexure is integrated for thrice
(d) None of the above

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Ans. (b) : In Macaulay's method

- 1st Integration of equation of flexure is getting slope equation.
- 2nd Integration of equation of flexure is getting deflection equation.

993. The Castigliano's second theorem can be used compute deflections

(a) In statistically determinate structures only
(b) At the point under the load only
(c) for beams and frames only
(d) For any type of structure

APPSC AEE 2016

Ans. (d) : Castigliano's second theorem:- The first partial derivatives of the total energy in a structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action. It is applicable for all types of structure.

994. The principle of virtual work is applied to elastic system by considering virtual work done by

(a) Internal forces only
(b) External forces only
(c) Internal as well as external forces
(d) None of the above

APPSC AEE 2016

Ans. (c) : The principle of virtual work done states that in equilibrium the virtual work of the forces applied to a system is zero (external as well as internal forces).

995. The method in which a single equation is formed for all loads on the beam and the equation constructed in such a way that the integral constants apply to all the sections of the beam is

(a) Moment area method
(b) Conjugate beam method
(c) Macaulay's method
(d) 6method of super position

APPSC AEE 2016

Ans. (c) : In Macaulay's method a single equation is formed for all loads on the beam and the equation constructed is such a way that the integral constants apply to all the sections of beam.
The starting point is the relation from Euler Bernoulli beam theory.

$$EI\frac{d^2y}{dx^2} = M$$

996. A simply supported beam of 10 m span is carrying a load of 4.8 kN at mid span. If Young's modulus of elasticity (E) is 2×10^8 kN/m^2 and moment of inertia (I) is 20 cm^2, then the maximum deflection will be

(a) 5.00 mm (b) 2.50 mm
(c) 0.50 mm (d) 0.25 mm

APPSC AEE 2016

Ans. (b) : When loading at center is SSB then deflection $\Delta = \frac{wl^3}{48EI}$

$$= \frac{4.8\times\left(10^3\right)^3}{48\times2\times10^8\times10^{-6}\times20\times10^4}$$

$\Delta = 2.5$mm

997. The differential equation of flexure is

(a) $EI\frac{d^2y}{dx^2} = -M$ (b) $EI\frac{dy}{dx} = M^2$

(c) $EI\frac{d^2y}{dx^2} = M^2$ (d) $M\frac{d^2y}{dx^2} = -EI$

APPSC AEE 2016

Ans. (a) : Differential equation of flexure is

$$EI\frac{d^2y}{dx^2} = -M$$

998. In moment area method the slope of real beam at a section is represented by

(a) Bending moment of M/EI diagram
(b) Shear force of M/EI diagram
(c) Total area of M/EI diagram
(d) Area of shear force diagram of real beam

GPSC AE January 2018

Ans. (b) : In moment area method, the area of bending moment diagram is utilized (cos M/EI diagram) for computing the slope.

999. A fixed or clamped support in real beam becomes ..... in conjugate beam.

(a) Free (b) Remains fixed
(c) Hinge (d) Roller

GPSC AE January 2018

Ans. (a) :

For real beam	For conjugate beam
Fixed	free
Free	fixed
Pinned or Hinged	Pinned or hinged
Roller	Roller
Intermediate roller	Intermediate hinged

1000. Which of the following statements is not applicable to Unit Load Method?

(a) External and internal forces are in equilibrium
(b) The structure is determinate
(c) The material is strained well within elastic limit
(d) The structure is redundant

GPSC AE January 2018

Ans. (d) : • In unit load method, the shape of the structure after it has managed to equilibriate the applied loads.
• The structure is in determinate form,
• The material is strained well within elastic limit.
• Deflection and rotation of determinate structure is calculated.

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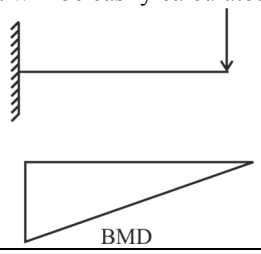
1001. Moment area method is best suitable for finding

(a) Slope and deflection of cantilever beam
(b) Slope and deflection of continuous beam
(c) Deflection of simply supported beam
(d) Slope of fixed beam

GPSC AE January 2018, ESE 2001

Ans. (a) : Moment area method is best suited for determining slope and deflection for cantilever beam because bending moment diagram of cantilever beam is ease and its area will be easily calculated.



1002. Match List-I (deflection) with List-II (expression) for deflection in different types of beams of span 'I' subjected to total load 'w' and select the correct answer using the loads given below the list :

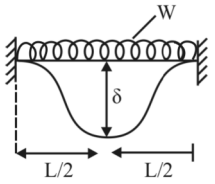
List-I	List-II
A. Central deflection in a fixed beam subject to uniformly distributed load	**1.** $\frac{Wl^3}{3EI}$
B. Central deflection in a simply supported beam subject to uniformly distributed load	**2.** $\frac{Wl^3}{48EI}$
C. Central deflection in a simply supported beam subject to concentrate load at mid-span	**3.** $\frac{5Wl^3}{384EI}$
D. Deflection at free end of a cantilever beam subject to concentrated load at free end	**4.** $\frac{Wl^3}{384EI}$

Code :

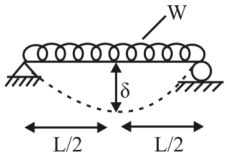
	A	B	C	D
(a)	4	3	1	2
(b)	3	4	2	1
(c)	3	4	1	2
(d)	4	3	2	1

MPPSC AE 2014

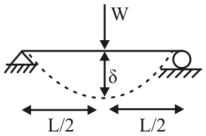
Ans. (d)

(1) 

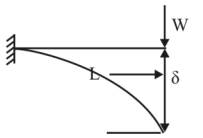
$\rightarrow \delta = \frac{WL^3}{384\,EI}$

(2) 

$\rightarrow \delta = \frac{5WL^3}{384\,EI}$

(3) 

$\rightarrow \delta = \frac{WL^3}{48\,EI}$

(4) 

$\rightarrow \delta = \frac{WL^3}{3\,EI}$

1003. The first theorem of Castigliano states that in a beam or truss subjected to any load system, the deflection at any point y is given by the partial differential coefficient of X w.r.t. a force P, acting at the point y in the direction in which the deflection is desired, where X is:

(a) angular deflection
(b) moment of inertia
(c) load with respect to position
(d) total strain energy

LMRC AM 2015

Ans (d) "If there is any elastic system in equilibrium under the action of a set of forces W_1, W_2, W_3.....W_n and corresponding displacements δ_1, δ_2, δ_3.....δn and a set of moments M_1, M_2, M_3.....M_n and the corresponding rotations θ_1, θ_2, θ_3.....θ_n, then the patial derivative of the total strain energy U with respect to one of the forces or moments, taken individually would yield its corresponding displacement in its direction of action".

Expressed mathematically

$$\frac{\partial U}{\partial W_1} = \delta_1 \text{ and } \frac{\partial U}{\partial M_1} = \theta_1$$

The above theorem of castigliano can be used for computting the deflection and slope at any point of a beam. The procedure is discussed below. Consider a beam subjected to loads W_1, W_2.....W_n It is desired to find deflection under load W_1

The strain energy of bending is given by

$$U = \int_0^L \frac{M^2 dx}{2EI} \left(\text{where M is a function of } W_1\right)$$

$$\delta_1 = \frac{\partial U}{\partial W_1} = \int_0^L M\left(\frac{\partial M}{\partial W_1}\right)\frac{dx}{EI}$$

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1004. Which of the following method is NOT used for finding deflection of determinate beams?

(a) Moment area method
(b) Macaulay's method
(c) Displacement method
(d) Castingliano's method

CIL MT 2020

Ans. (c) : In the stiffness material displacements (rather than forces) are taken as the unknown quantities for this reson the method is also called the displacement method.
Example –
(i) Slope deflection method
(ii) Moment distribution method
(iii) minimum potential energy method .

1005. Which of the following method is NOT used to determine slope and deflection at a section in loaded beam?

(a) Macaulay's method
(b) Moment area method
(c) Double integration method
(d) Rankine's method

CIL MT 2020

Ans. (d) : Method used to determine slope and deflection are :
(1) Macaulay's Method
(2) Double integration method
(3) Moment are method
(4) Conjugate beam method
(5) Castigliano's method
Rankine's method is not used to determine slope and deflection at a section in loaded beam.

1006. Macaulay's method is also known as :

(a) slope deflection method
(b) moment distribution method
(c) method of singularity functions
(d) Kani's method
(e) moment area method

CGPSC AE 2017

Ans. (c) Macaulay's method is also known as method of singularity functions.

1007. Conjugate beam method can be directly used only for :

(a) a cantilever beam
(b) a simply supported beam
(c) a continuous beam
(d) an overhanging beam
(e) either a cantilever beam or a continuous beam

CGPSC AE 2017

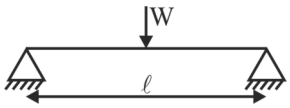
Ans. (b) Conjugate beam method can be directly used only for a simply supported beam.

1008. The ratio of the maximum deflection of beam simply supported at its ends with an isolated central load and that with a uniformly distributed load over its entire length is

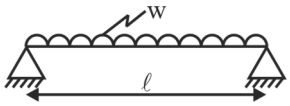
(a) 1:1 (b) 24/15:1
(c) 15/25:1 (d) 2/3:14

Haryana PSC AE 2010

Ans. (b) :



$$y_1 = \frac{W\ell^3}{48EI}$$



$$y_2 = \frac{5w\ell^4}{384EI} = \frac{5W\ell^3}{384EI}$$

$$\frac{y_1}{y_2} = \frac{W\ell^3 \times 384EI}{48EI \times 5W\ell^3} = \frac{\frac{24}{15}}{1} = \frac{24}{15} : 1$$

1009. A simply supported RCC beam carrying uniform load has deflection of 10 mm at the centre. If both ends of the beam are now fixed, the deflection at the centre would be

(a) 25 mm (b) 10 mm
(c) 8 mm (d) 2 mm

Haryana PSC AE 2010

Ans. (d) : Simply Supported Beam (SSB)
Deflection, $y_1 = 10$ mm
Fixed Beam (FB)
Deflection, $y_2 = \frac{1}{5} \times S.S.B = \frac{1}{5} \times 10 = 2$ mm

1010. The deflection at any point of a perfect time can be obtained by applying a unit load at the joint in:

(a) Vertical direction
(b) Horizontal direction
(c) Inclined direction
(d) The direction in which the deflection is required

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(e) All of the above questions

Haryana PSC AE 2018 Paper–I

Ans. (d) : The direction in which the deflection is required, a unit load is applied to obtain the deflection at that joint.

1011. If width of a simply-supported beam carrying a concentrated load at its centre-span is doubled, the deflection of the beam at the centre-span is changed by

(a) 1/2 times (b) 2 times

(c) 3 times (d) 1/4 times

(e) 8 times

Haryana PSC AE 2018 Paper–I

Ans. (a) : Simply supported beam under central point

deflection $(y_1) = \frac{WL^3}{48EI}$ $\frac{WL^3}{48 \times E \times \frac{bd^3}{12}}$

width is twice = 2b

then Deflection $(y_2) = \frac{WL^3}{48 \times E \times \frac{2bd^3}{12}}$

$$\frac{y_1}{y_2} = \frac{1}{2}$$

$\therefore$ Hence Deflection is $\frac{1}{2}$ width is doubled.

1012. For a rectangular simply-supported beam, the deflection is:

(a) Directly proportional to its weight

(b) Inversely proportional to its width

(c) Directly proportional to the cube of its length

(d) Inversely proportional to the cube of its depth

(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (c) : For a rectangular simply-supported beam, the deflection is

$$\Delta = \frac{WL^3}{48EI}$$

Hence, deflection $(\Delta) \propto L^3$

1013. The deflection can be controlled by using the appropriate:

(a) Aspect ratio (b) Modular ratio

(c) Span/width ratio (d) Water/Cement ratio

Haryana PSC AE 2017

Ans. (c) : The deflection can be controlled by using the appropriate span/width ratio.

1014. Prismatic cantilever beam of length L carries a uniformly distributed load throughout its length. If at the free end of the beam, vertical deflection is 18 mm and slope of the deflection curve is 0.02 rad, the length of beam is -

(a) 0.8 m (b) 1.0 m

(c) 1.2 m (d) 1.5 m

Rajasthan VP ITI 2018

Ans. (c) : Given that, Due to uniformly distanced load, deflection free end of cantilever beam = 18 mm

& Slope = 0.02 Radian

Slope $\theta_B = \frac{w\ell^3}{6EI} = 0.02$

Deflection $y_B = \frac{w\ell^4}{8EI} = \frac{w\ell^3}{6EI} \times \frac{6\ell}{8}$

$$18 = \frac{w\ell^4}{6EI} \times \frac{6\ell}{8}$$

$$18 = 0.02 \times \frac{6\ell}{8}$$

$$\ell = \frac{18 \times 8}{6 \times 0.02} = 1200\text{mm}$$

$$\boxed{\ell = 1.2\text{m}}$$

1015. Slope and deflection at a section in a loaded beam can be found out by which of the following method :

(a) Double integration method

(b) Macaulay's method

(c) Moment area method

(d) All of the above

UP Jal Nigam AE 2016

Ans. (d) : Slope and deflection at section in a loaded beam can be found out by following methods :

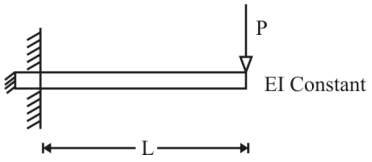
(1) Double integration method

(2) Macaulay's method

(3) Moment area method

(4) Conjugate beam method

1016. Consider following given figure and calculate the strain due to bending in the cantilever beam.



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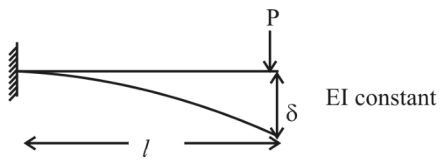
(Where P is the point load, E is Young's modulus, L is the length of the cantilever beam and I is the moment of inertia)

(a) PL/3EI (b) $P^2L^3/6EI$

(c) P^3L^3/EI (d) $P^3L^3/2EI$

UPRVUNL AE 2014

Ans. (b) :



$$\text{Deflection}(\delta) = \frac{P^2L^3}{6EI}$$

1017. Match the following given lists and choose the correct answer from the options given below the lists:

List I (Actual beam)	List II (Conjugate beam)
A. Curvature	1. Shear
B. Deflection	2. Load
C. Slope	3. Moment

(a) A-2, B-1, C-3 (b) A-2, B-3, C-1

(c) A-3, B-1, C-2 (d) A-1, B-3, C-2

UPRVUNL AE 2014, ESE 2000

Ans. (b) :

Actual beam	Conjugate beam
(A) Curvature	Load
(B) Deflection	Moment
(C) Slope	Shear

1018. Which of the following represents the shear force at a section of the beam ?

(a) $EI\frac{d^4y}{dx^4}$ (b) $EI\frac{d^3y}{dx^3}$

(c) $EI\frac{d^2y}{dx^2}$ (d) $EI\frac{dy}{dx}$

GPSC AE March 2018

Ans. (b) :

$$\frac{d^2y}{dx^2} = \frac{M}{EI} \qquad \because \text{Shear force} = \frac{dM}{dx}$$

So, $\boxed{\text{Shear force} = \left(\frac{d^3y}{dx^3}\right)EI}$

1019. Virtual work means

1. **work done by real forces due to virtual displacement**
2. **work done by virtual forces during real displacement**

(a) Both 1 and 2 are correct

(b) Both 1 and 2 are wrong

(c) 1 is correct and 2 is wrong

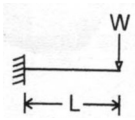
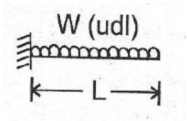
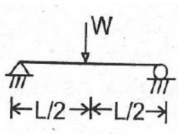
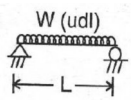
(d) 2 is correct and 1 is wrong

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Virtual work = virtual force × real displacement (or)

virtual displacement × real force

1020. Match List-I with List-II and select the correct answer using the codes given the lists :

List -I (Loaded members)	List- II (Maximum deflection)
A 	1. $\frac{WL^3}{8EI}$
B 	2. $\frac{WL^3}{48EI}$
C 	3. $\frac{WL^3}{3EI}$
D 	4. $\frac{5WL^3}{384EI}$

Codes :

	A	B	C	D
(a)	3	1	2	4
(b)	1	2	3	4
(c)	2	3	4	1
(d)	4	2	3	1

OPSC Civil Services Pre 2011

Ans. (a) :

1021. In the slope deflection equation the deformation is considered to be caused by

(a) Bending moment (b) Shear force

(c) Axial force (d) All of the above

WBPSC AE 2012

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Ans. (a) : In the slope deflection equation the deformation is considered to be caused by bending moment.

1022. The ratio of maximum deflection of a simply supporter beam with a central load W and of a cantilever of beam length and with a load w at free end is

(a) $\frac{1}{2}$ (b) $\frac{1}{10}$

(c) $\frac{1}{16}$ (d) $\frac{1}{12}$

WBPSC AE (MCD) 1998

Ans. (c) :

Maximum deflection at center for simply supported beam $= \frac{WL^3}{48EI}$

Maximum deflection at the free end for cantilever beam by same span $= \frac{WL^3}{3EI}$

Ratio $= \frac{\frac{WL^3}{48EI}}{\frac{WL^3}{3EI}} = \frac{1}{16}$

1023. To calculate by 'conjugate beam method' the slope and deflection at any section of a beam for any loading the conjugate beam is assumed to be loaded by

(a) bending moment diagram of the beam

(b) M/I diagram of the beam

(c) M/EI diagram of the beam

(d) actual loading

WBPSC AE (Common) 1998

Ans. (c) : Slope on real beam = shear on conjugate beam

Deflection on real beam = moment on conjugate beam

The load on the conjugate beam is the $\frac{M}{EI}$

Diagram of the loads on the actual beam.

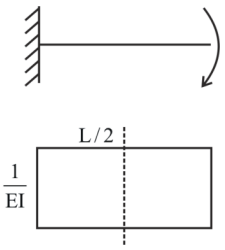
1024. A cantilever beam of span L, uniform flexural rigidity EI is subjected to a unit couple at its free end. The deflection at the centre of the beam is:

(a) $L^2/2EI$ (b) $L^2/8EI$

(c) $L^2/4EI$ (d) $L^2/16EI$

TSPSC Deputy Engineer 2015

Ans. (b) :



$$\delta = \frac{1}{EI} \times \frac{L}{2} \times \frac{L}{4} = \frac{L^2}{8EI}$$

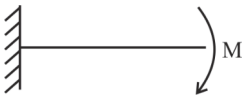
1025. A cantilever beam of span 'L' is subjected to a clockwise moment at free end. What is the ratio of deflection to rotation at the free end?

(a) 2L (b) 3L/2

(c) L (d) L/2

TSPSC AEE 2017

Ans. (d) :



Deflection at free end $\delta = \frac{ML^2}{2EI}$

Reaction at free end $\theta = \frac{ML}{EI}$

$$\frac{\delta}{\theta} = \frac{\frac{ML^2}{2EI}}{\frac{ML}{EI}} = \frac{L}{2}$$

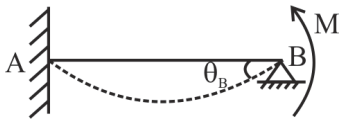
1026. Moment required to rotate near end of prismatic beam through unit angle, with translation, the far end being fixed, will be

(a) EI/L (b) 2 EI/L

(c) 3 EI/L (d) 4 EI/L

Chandigarh AE 2017

Ans. (d) :



$$\theta_B = \frac{ML}{4EI}$$

$$M = \frac{4EI}{L}\theta_B$$

given $\theta_B = 1$

$\therefore \quad M = \frac{4EI}{L}$

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6. Principal Stress & Principal Planes, Theory of Failure

1027. When a member is subjected to like principal stresses then the greatest shear stress will occur on mutually perpendicular planes which will be:

(a) 2 in number (b) 4 in number
(c) 3 in number (d) Non of the above

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) :
We know,

$$\tan 2\theta = \frac{-(\sigma_x - \sigma_y)}{2\tau_{xy}}$$

→ By solving Above eq. we get two value of θ, $(\theta s)^2$ $(90 + \theta s)$
One of them Angle corresponds to major plane & other will corresponds to minor plane

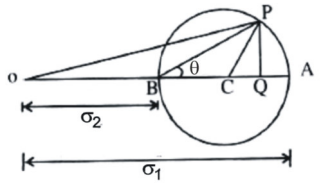
1028. A plane which is not subjected to shear stress (tangential stress) in known as :

(a) Compound plane (b) Simple plane
(c) Principal plane (d) Non-shear plane

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : Principal plane : Plane on which normal stresses are maximum or minimum are called principal plane & shear stress on these planes are zero.

1029. The radius of Mohr's circle in figure is equal to:



OR

What is the radius of Mohr's circle in case of biaxial state of stress?

(a) Sum of two principal stresses
(b) Difference of two principal stresses
(c) Half the sum of two principal stresses
(d) Half the difference of two principal stresses

Nagaland PSC CTSE 2017 Paper–I
ESE 2009

Ans. (d) :

$$\text{Radius} = \tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

$\tau_{max} \rightarrow$ max shear stress
$\sigma_1 \rightarrow$ major principal stress
$\sigma_2 \rightarrow$ minor principal stress

1030. The maximum shear stress of the Mohr's circle is equal to :

(a) Radius (b) Diameter
(c) Circumference (d) Area

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : Max. shear stress of Mohr's circle is equal to Radius of the Mohr's circle

i.e $\text{Radius} = \tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

1031. In Mohr's circle, the centre of circle from Y-axis is taken as :

(a) $\frac{\sigma_x - \sigma_y}{2}$ (b) $\frac{\sigma_x - \sigma_y}{2} + \tau$
(c) $\frac{\sigma_x + \sigma_y}{2}$ (d) $\frac{\sigma_x + \sigma_y}{2} + \tau$

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) :

Centre of circle from Y axis is taken as $= \frac{\sigma_x + \sigma_y}{2} + \tau$

1032. The extremeties of any diameter on Mohr's circle represent :

(a) Principal stresses
(b) Normal stresses on plane at 45^0
(c) Shear stress on planes at 45^0
(d) Normal and shear stresses on a plane

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : Mohr's circle is the locus of points representing magnitude of normal & shear stress at various plane in a given stress dement.

1033. The plane of maximum obiliquity is inclined to the major principal plane at an angle.

(a) $\frac{\pi}{2} + \frac{\theta_{max}}{4}$ (b) $\frac{\pi}{4} + \frac{\theta_{max}}{2}$
(c) $\frac{\pi}{4} + \frac{\theta_{max}}{4}$ (d) $\frac{\pi}{4} + \theta_{max}$

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : The plane of maximum obliquity is inclined to the major principal plane at an angle of $\left(\frac{\pi}{4} + \frac{\theta_{max}}{2}\right)$

1034. On two perpendicular planes passing through a point there are complimentary shear stresses ±100 N/mm². The normal stress on these planes are Zero's then maximum principal stress at the point is

(a) 200 N/mm² (b) Zero
(c) 100 N/mm² (d) 50 N/mm²

APTRASCO 2017

Ans : (c) : Maximum principal stress on two perpendicular planes passing through a point

$$\sigma = \frac{\sigma_x - \sigma_y}{2} \pm \sqrt{\left[\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2\right]}$$

$$= \frac{0}{2} \pm \sqrt{0 + 100^2}$$

$$= \pm 100\,N/mm^2$$

Maximum principal stress $(\sigma_1) = 100\,N/mm^2$

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1035. The principal stresses at a point in a bar are 160 N/mm^2 (tensile) and 80 N/mm^2 (compressive) the accompanying maximum shear stress intensity is

(a) 100 N/mm^2 (b) 110 N/mm^2
(c) 120 N/mm^2 (d) 140 N/mm^2

APTRANSCO 2017
ESE 2014

Ans : (c)

$$\text{Maximum shear stress } (\tau_{max}) = \frac{\sigma_{max} - \sigma_{min}}{2}$$

$$= \frac{160 - (-80)}{2}$$

$$\boxed{(\tau_{max}) = 120 \text{N/mm}^2}$$

1036. In plane stress problem there are normal tensile stresses σ_x and σ_y accompanied by shear stress τ_{xy} at a point along orthogonal Cartesian coordinates X and Y respectively, If it observed that the minimum principle stress on a certain plane is zero then :

(a) $\tau_{xy} = \sqrt{(\sigma_x + \sigma_y)}$ (b) $\tau_{xy} = \sqrt{(\sigma_x - \sigma_y)}$

(c) $\tau_{xy} = \sqrt{(\sigma_x \times \sigma_y)}$ (d) $\tau_{xy} = \sqrt{(\sigma_x / \sigma_y)}$

OPSC AEE 2019 Paper–I
ESE 2012, 1998

Ans. (c)

$$\sigma_2 = 0 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\frac{\sigma_x + \sigma_y}{2} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\Rightarrow \left(\frac{\sigma_x + \sigma_y}{2}\right)^2 - \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 = \tau_{xy}^2$$

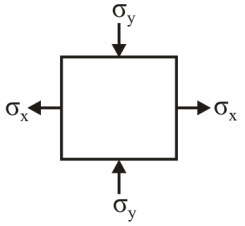
$$\tau_{xy} = \sqrt{\sigma_x . \sigma_y}$$

1037. Select in which case of the following biaxial stress, pure shear condition prevails :

(a) $\sigma_x = 2\,\sigma_y$ (b) $2\,\sigma_x = \sigma_y$
(c) $\sigma_x = \sigma_y$ (d) $\sigma_x = -\sigma_y$

OPSC AEE 2019 Paper–I

Ans. (d) : When normal stresses acting on perpendicular planes are equal in magnitude and opposite in direction is known as pure shear condition



$\sigma_x = -\sigma_y$

1038. A circle is marked on a mild steel plate and then it is subjected to two normal stresses is a mutually perpendicular direction along with simple shear. After the loading, the circle

(a) assumes the shape of a ellipse
(b) assumes the shape of a cycloid
(c) remains as a circle
(d) assumes the shape of a square

GPSC AE March 2018

Ans. (a) : If a circle is marked on a mild steel plate, two stresses are provided perpendicular 'h' with simple shear, the circular shape will be assumed to convert into an ellipse.

1039. State true or false :

1. **Any two orthogonal surfaces are sufficient to completely specify the principal stresses for a biaxial state of stress.**
2. **Only one surface is required to specify the maximum shear stress completely.**

(a) 1 is true but 2 is false
(b) 1 is false, but 2 is true
(c) both 1 and 2 are true
(d) both 1 and 2 are false

GPSC AE March 2018

Ans. (c) : Both statements are true as any two orthogonal surfaces are sufficient for completely specify the principal stresses for a biaxial state of stresses.
Only one surface is required to specify maximum shear stress completely.

1040. Which of the following yield criteria are suitable for ductile and isotropic material ?

1. **Maximum normal stress theory**
2. **Maximum shear stress theory**
3. **Maximum energy distortion theory**
4. **Maximum compressive theory**

(a) 1 and 2 (b) 2 and 3
(c) 1, 2 and 3 (d) 1, 2, 3 and 4

GPSC AE March 2018

Ans. (b) : Maximum stress theory and maximum energy distortion theory both have criteria of ductile and isotropic material, but maximum normal stress and maximum compressive theory has anisotropic material.

1041. Which one of the following is conservative failure theory for brittle material ?

(a) Maximum normal stress theory
(b) Maximum shear theory
(c) Coulomb-Mohr theory
(d) St. Venant theory

GPSC AE March 2018

Ans. (c) : Coulomb-Mohr's theory is often used in predicting the failure of brittle material and is applied to cases of 2D stress.

1042. A rectangular block is subjected to bi-axial principal stresses of intensity σ_x and σ_y, both tensile, with $\sigma_y > \sigma_x$. Which of the following will give the radius of the Mohr's circle?

(a) Minimum shear stress

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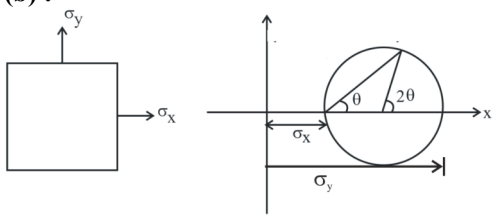
(b) Maximum shear stress
(c) Minimum Normal stress
(d) Maximum Normal stress

UPSC JWM 2017

Ans. (b) :



Radius = $\frac{\sigma_y - \sigma_x}{2}$ when $\theta = 45^0$

Maximum shear stress = $\frac{\sigma_y - \sigma_x}{2}$

Maximum shear stress = Radius of Mohr's circle

1043. The radius of a Mohr's circle for stresses at a point in a body to $\sigma_x = -160N/mm^2$, $\sigma_y = 350N/mm^2$, $\tau_{xy} = 0$ is

(a) 255 (b) 190
(c) 350 (d) 160

TRB Poly. Tech. Lect. 2012

Ans. (a) : We know,

τ_{max} = radius of a Mohr's circle

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \text{(given)}$$

$$= \sqrt{\left(\frac{-160 - 350}{2}\right)^2 + 0^2}$$

$\{\sigma_x = -160\ N/mm^2,\ \sigma_y = 350\ N/mm^2\}$

$\Rightarrow$ $\boxed{\tau_{max} = 255N/mm^2}$

1044. Design of shaft made of brittle materials is based on

(a) Guest's theory (b) St. Venant's theory
(c) Rankine's theory (d) Von Mises theory

TNPSC AE 2018

Ans. (c) : Maximum principal stress theory (Rankine theory, Lame's theory). It is most suitable for brittle materials.
Maximum shear strain energy theory is suitable for ductile material.

1045. If Rankine's criteria is applied for failure of brittle material, then which of the following will be necessary condition?

(a) Maximum shear forces
(b) Maximum strain energy
(c) Maximum principal stress
(d) Maximum shear strain energy

CIL MT 2017

Ans. (c) : If Rankine's criteria is applied for failure of brittle material, then Maximum principal stress will be necessary condition

1046. The state of two dimensional stress acting on a concrete lamina consists of a direct tensile stress of 6 MPa and shear stress of 4 MPa, which cause cracking of concrete. Then the tensile strength of the concrete in MPa is

(a) 6 MPa (b) 8 MPa
(c) 10 MPa (d) 12 MPa

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : $\sigma_{p_1} = \frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau_{xy}^2}$

$$= \frac{6+0}{2} \pm \sqrt{\left(\frac{6}{2}\right)^2 + 4^2}$$

$$= 3 \pm \sqrt{9+16}$$

$$= 3 \pm 5$$

$\sigma_{1/2} \Rightarrow$

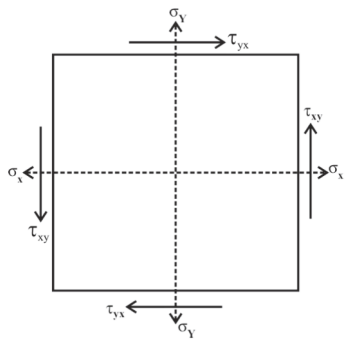
$\boxed{\sigma_{p_1} = 8}$ $\boxed{\sigma_{p_2} = -2}$

1047. The symmetry of the stress tensor at a point in a body when at equilibrium is obtained from :

(a) Conservation of energy
(b) Moment equilibrium equations
(c) Force equilibrium equations
(d) None of these

JPSC AE (Pre) 2019

Ans. (b) :



Taking moment equilibrium about the centre, we get

$$\tau_{yx} \times \frac{d}{2} + \tau_{yx} \times \frac{d}{2} = \tau_{xy} \times \frac{d}{2} + \tau_{xy} \times \frac{d}{2}$$

$\boxed{\tau_{xy} = \tau_{yx}}$

1048. The maximum shear stress occurs on

(a) Principal planes
(b) 45^0 with principal planes
(c) 90^0 with principal planes
(d) Independent of the principal planes

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : Plane of maximum shear stress is inclined at 45^0 with the principal planes.

1049. When a member is subjected to axial tensile load, the greatest normal stress is equal to

(a) Half the maximum shear stress
(b) Maximum shear stress
(c) Twice the maximum shear stress

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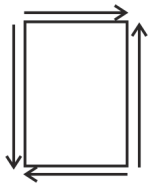
(d) Three times the maximum shear stress

TNPSC AE 2019

Ans. (c) $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$ Here for axial loading

$\tau_{max} = \frac{\sigma_1 - 0}{2} \Rightarrow \sigma_n = 2\tau_{max}$

1050. Stage of stress in plane element is shown if figure. Which one of the followings figures shown is the correct sketch of Mohr's circle of the state of stress?



(a)

(b)

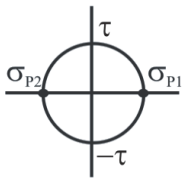
(c)

(d)

TNPSC AE 2019

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) Given $\sigma_x = \sigma_y = 0$



$\sigma_{P_1} / \sigma_{P_2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= 0 \pm 0\sqrt{\tau^2} = \pm\tau$

$\sigma_{P_1} = +\tau \;\&\; \sigma_{P_2} = -\tau$

1051. if σ_1, σ_2 and σ_3 are principal stresses and σ_e is the allowable stress, the equation

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2}{m}(\sigma_1 \times \sigma_2 + \sigma_2 \times \sigma_3 + \sigma_3 \times \sigma_1) = \sigma_e^2$$

belongs to which failure theory ($\frac{1}{m}$ is the Poisson's ratio).

(a) Strain energy theory
(b) Shear strain energy theory
(c) Maximum principal strain theory
(d) Maximum shear stress theory

TNPSC AE 2019

Ans. (a) Maximum strain energy method:-

$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2}{m}(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \le \sigma_e^2$

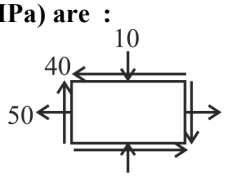
1052. In a Mohr's diagram, a point above Mohr's envelope indicates-

(a) imaginary condition
(b) safe condition
(c) imminent failure condition
(d) condition of maximum obliquity

GUJRAT PSC AE - II 2017

Ans : (a) In a Mohr's diagram, a point above Mohr's envelope indicates imaginary condition.

1053. The state of plane stress at a point is represented by the stress element below. Assuming tensile force positive and compressive force negative, the principal stresses (MPa) are :



(a) 50, -50
(b) 70, -30
(c) -70, -30
(d) 70, 30
(e) 30, 70

CGPSC AE 2014 Shift–I

Ans. (b) :

Principal stresses,

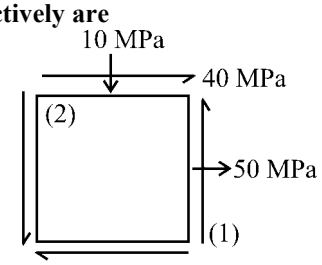
$\sigma_{max/min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$

$= \frac{50 + (-10)}{2} \pm \sqrt{\left(\frac{50 - (-10)}{2}\right)^2 + (40)^2}$

$= 20 \pm 50$

$\boxed{\sigma_{max} = 70MPa,\ \sigma_{min} = -30MPa}$

1054. For the state of stress shown in the following figure, if Mohr's circle is to be plotted. If the diameter of the Mohr's circle is d and the principal stresses are p and q (magnitudes only), the values of d, p & q (in MPa units) respectively are



(a) 50, 70, 30
(b) 100, 70, 30
(c) 90, 70 , 30
(d) 100, 100, 30
(e) 100, 40, 0

CGPSC AE 2014 Shift–I

Ans. (b) :

Given as,

$\sigma_x = 50$ MPa

$\sigma_y = -10$ MPa

$\tau_{xy} = 40$ MPa

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Radius of Mohr's circle (R)

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{50-(-10)}{2}\right)^2 + 40^2} = 50 \text{ MPa}$$

So, diameter d = 2 R = 2 × 50 = 100 MPa

$$p/q = \left(\frac{\sigma_x + \sigma_y}{2}\right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{50-10}{2} \pm \sqrt{\left(\frac{50-(-10)}{2}\right)^2 - 40^2}$$

$$= 20 \pm 50$$

p = 70 MPa (T)

q = 30 MPa (C)

1055. In a plane strain problem in xy plane, the shear strain = 15×10^{-6} and the normal strain in x and y direction is zero For this strain, what is the diameter of the Mohr's circle of strain?

(a) 5×10^{-6} (b) 7.5×10^{-6}

(c) 15×10^{-6} (d) 30×10^{-6}

APPSC Poly. Tech. Lect. 2020
ESE 2005

Ans. (c) :

$$\text{Radius of Mohr's circle} = \frac{\phi_{max}}{2}$$

$$= \frac{15\times10^{-6}}{2}$$

$$= 7.5\times10^{-6}$$

So, diameter of Mohr's circle strain $= 15\times10^{-6}$

1056. If Mohr's circle for two dimensional stress system has zero radius, both principal stresses are

(a) of equal magnitude and of same sign

(b) of equal magnitude and of opposite sign

(c) equal to zero and shear stress is non-zero

(d) equal to zero and shear stress is also equal to zero

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) : If Mohr's circle for two dimensional stress system has zero radius, both principal stresses are of equal magnitude and of same sign.

1057. If the ratio of two principal stresses is 1/2, what is the ratio of minimum principal stress to maximum shear stress?

(a) 1/2 (b) 1

(c) 2 (d) 4

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) : $\frac{\sigma_x}{\sigma_y} = \frac{1}{2}(\text{given})$

$$\sigma_{min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2}$$

$$\tau_{max} = \frac{\sigma_x - \sigma_y}{2}$$

$$\frac{\sigma_{min}}{\tau_{max}} = \frac{\left(\frac{\sigma_x + \sigma_y}{2}\right) - \left(\frac{\sigma_x - \sigma_y}{2}\right)}{\frac{\sigma_x - \sigma_y}{2}}$$

$$= \frac{2\sigma_y}{2} \times \frac{2}{\sigma_x - \sigma_y}$$

$$= \frac{2\sigma_y/\sigma_y}{\frac{\sigma_x}{\sigma_y} - \frac{\sigma_y}{\sigma_y}} = \frac{2}{\frac{1}{2} - 1} = 4$$

1058. For an element subjected to pure shear stress τ, the maximum principal stress will be

(a) 2τ (b) $\frac{\tau}{2}$

(c) τ (d) 4τ

ASSAM PSC CCE (PRE) 2015

Ans : (c)

$$\text{Maximum principal stress } (\sigma_{max}) = \frac{\sigma}{2} + \sqrt{\frac{\sigma^2}{4} + \tau^2}$$

For pure shear stress $\boxed{\sigma = 0}$

$$\sigma_{max} = \frac{0}{2} + \sqrt{\frac{0}{4} + \tau^2}$$

Maximum principal stress $\boxed{\sigma_{max} = \tau}$

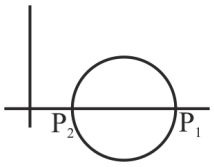
1059. When two unequal like principal stresses p_1 and p_2 act, then the radius of Mohr's circle will be

(a) $\frac{p_1 + p_2}{2}\cos\theta$ (b) $\frac{p_1 - p_2}{2}\cos\theta$

(c) $\frac{p_1 + p_2}{2}$ (d) $\frac{p_1 - p_2}{2}$

ASSAM PSC CCE (PRE) 2015

Ans : (d)



$$\text{Radius} = \frac{P_1 - P_2}{2}$$

$$\text{Center} = \frac{P_1 + P_2}{2}$$

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1060. A rectangular bar is subjected to an axial tensile load producing a tensile stress p on a section normal to the axis of loading. The tangential stress induced on any oblique plane at an angle θ to the cross-section is

OR

If a bar of cross section area 'A' is subjected to a tensile force 'P', resultant shear stress on a oblique plane inclined at an angle θ to its axis is

(a) $p\cos^2\theta$ (b) $\frac{p}{2}\cos 2\theta$

(c) $\frac{p}{2}\sin 2\theta$ (d) $\frac{p}{2}\sin^2\theta$

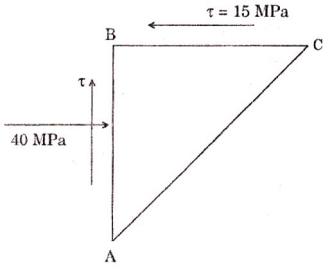
ASSAM PSC CCE (PRE) 2015
UPPCL AE 2014, ESE 2002

Ans : (c) Due to tensile stress 'p' the tangential stress induced on any oblique plane at an angle θ to the cross-section is $\tau = \frac{P}{2A}\sin 2\theta$

$$\boxed{\tau = \frac{p}{2}\sin 2\theta}$$

Normal stress $\boxed{\sigma_n = p\cos^2\theta}$

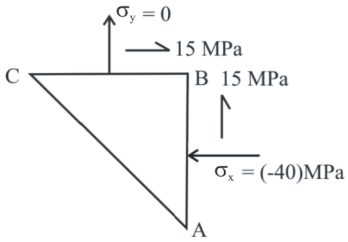
1061. If AC is principal plane, then magnitude of principal tensile stresses will be



(a) 15 MPa (b) 5 MPa
(c) 45 MPa (d) Zero

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (b) Section is put in this form



$$\sigma_1 = \left(\frac{\sigma_x + \sigma_y}{2}\right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$\sigma_x = -40$ MPa

$\tau = 15$ MPa

$$\sigma_1 = \frac{-40}{2} \pm \sqrt{\left(-\frac{40}{2}\right)^2 + 15^2}$$

$$= -20 \pm \sqrt{400 + 225}$$

$$= -20 \pm \sqrt{625} = -20 \pm 25$$

+ sign

$\therefore$ $-20 + 25 = 5$ MPa (Tensile)

$\because$ – sign

$-20 - 25$

$\because$ $= -45$ MPa (Comprassive)

1062. If an element is subjected to pure shearing stress T_{xy}, the maximum principal stress is equal to :

(a) $2\,T_{xy}$ (b) $\frac{T_{xy}}{2}$

(c) T_{xy} (d) $\sqrt{1-(T_{xy})^2}$

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : Principal stress developed due to bending stress (σ) & shear stress (τ)

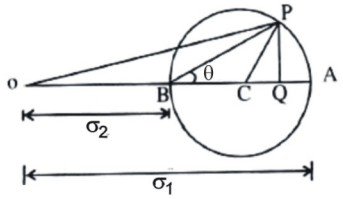
$$\sigma_1/\sigma_2 = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

→ But in question element have only pure shear stress (T_{xy}), means $\sigma = 0$.

$$\therefore \quad \sigma_1/\sigma_2 = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} = \pm\sqrt{0 + (T_{xy})^2}$$

$$\boxed{\sigma_1/\sigma_2 = \pm T_{xy}}$$

1063. The resultant stress on an inclined plane which is inclined at angle θ to the normal cross-section of a body which is subjected to a direct tensile stress (σ) in one plane is :



(a) σ sin θ (b) σ sin2 θ
(c) σ cosθ (d) σ cos2θ

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) :

$$\sigma_{AE} = \frac{F'}{A} = \frac{F\cos\theta}{\frac{A}{\cos\theta}} = \frac{F}{A}\cos^2\theta = \sigma\cos^2\theta = \sigma\cos^2\theta$$

$$\tau_{AE} = \frac{F}{A'} = \frac{F\sin\theta}{\frac{A}{\cos\theta}} = \sigma\sin\theta\cos\theta$$

$$\therefore \quad \sigma_R = \sqrt{(\sigma_{AE})^2 + (\tau_{AE})^2}$$

$$= \sqrt{(\sigma\cos^2\theta)^2 + (\sigma\sin\theta\cos\theta)^2}$$

$$= \sqrt{\sigma^2\cos^4\theta + \sigma^2\sin^2\theta\cos^2\theta} = \sigma\cos\theta\sqrt{\cos^2\theta + \sin^2\theta}$$

$(\because \sin^2\theta + \cos^2\theta = 1)$

$$\boxed{\sigma_R = \sigma\cos\theta}$$

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1064. When a body is subjected to a direct tensile stress (σ) in one plane, then normal stress on an oblique section of the body inclined at an angle θ to the normal of the section is :

(a) $\sigma \cos \theta$ (b) $\sigma \cos^2\theta$

(c) $\sigma \sin\theta$ (d) $\sigma \sin^2 \theta$

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) :

$\sigma_n = \sigma_1 \cos^2 \theta + \sigma_2 \sin^2 \theta$

(Tensile stress in one plane $\sigma_2 = 0$)

$\boxed{\sigma_n = \sigma \cos^2 \theta}$

1065. In a Mohr's circle of σ – τ plane (σ = normal stress, τ = shear stress), the vertical diameter represents

(a) Maximum shear stress

(b) Maximum normal stress

(c) Principal stress

(d) Minimum normal stress

RPSC AE (GWD) 2014

Ans : (a) In Mohr's circle of σ – τ plane, the vertical diameter represents maximum shear stress.

1066. Mohr's circle for the state of stress defined by $\begin{bmatrix} 30 & 0 \\ 0 & 30 \end{bmatrix}$ **MPa is a circle with**

(a) Center at (0, 0) and radius 30 MPa

(b) Center at (0, 0) and zero radius

(c) Center at (30, 0) and radius 30 MPa

(d) Center at (30, 0) and zero radius

RPSC POLY. TECH. LECT. 2014

Ans : (d) Comparing given matrix with $\begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{yx} & \sigma_{yy} \end{bmatrix}$

$\tau_{xy} = \tau_{yx} = 0$ and $\sigma_{xx} = \sigma_{yy} = 30$ MPa

Centre of Mohr's circle $C = \left(\frac{\sigma_{xx} + \sigma_{yy}}{2}, 0\right)$

$= \left(\frac{30+30}{2}, 0\right) = (30, 0)$

Radius of Mohr's circle $r = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \left(\tau_{xy}\right)^2}$

$= \sqrt{\left(\frac{30-30}{2}\right)^2 + (0)^2}$

$r = 0$

1067. The normal stresses on two perpendicular planes through a point in a stressed material are 80 MPa (tensile) and 20 MPa (compressive) respectively value of maximum shear stress at the point

(a) 50 MPa (b) Zero

(c) 40 MPa (d) 60 MPa

RPSC AEN 2013

Ans : (a) Given, $\sigma_x = 80$MPa, $\sigma_y = -20$MPa

$\tau_{max} = \frac{\pm(\sigma_x - \sigma_y)}{2} = \frac{\pm(80-(-20))}{2}$

$\boxed{\tau_{max} = \pm 50\text{MPa}}$

1068. If an element of a stressed body is in the state of pure shear with magnitude of shear stress as 80 N/mm² then the principal stress at that elements

(a) 40 N/mm² (b) 120 N/mm²

(c) 148 N/mm² (d) 80 N/mm²

RPSC POLY. TECH. LECT. 2014

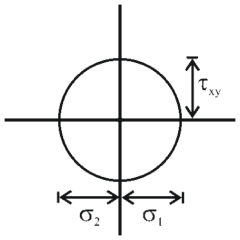
Ans : (d) Given, $\tau = 80$ MPa, $\sigma_x = \sigma_y = 0$

Centre of mohr's circle = (0, 0)

Radius of mohr circle = 80 MPa

Principal stress $\sigma_{1/2} = \pm \tau_{xy}$

$= \pm$ 80 MPa



1069. The maximum shear stress occurs on :

(a) Principal plane

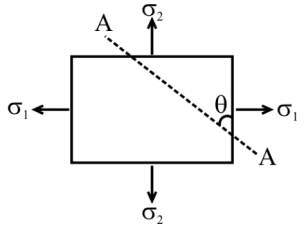
(b) 45^0 with principal plane

(c) 90^0 with principal plane

(d) Independent of principal plane

[UPPSC State Eng. A.E. 2007 (II)]

Ans : (b)



Shear stress on the plane of section A - A

$\tau = -\frac{(\sigma_1 - \sigma_2)}{2} \sin 2\theta$

For shear stress to be maximum $\sin 2\theta = \pm 1$

then $\theta = 45^0$ or 135^0

$\therefore \quad \tau_{max} = \left|\frac{(\sigma_1 - \sigma_2)}{2}\right|$

Hence τ_{max} will occur on a plane which it inclined at 45^0 with principal planes.

1070. A body is subjected to a stress of 3 N/mm² in compression and a tensile stress of 5 N/mm² both acting perpendicular to each other. The maximum shear stress in the body will be

(a) 2N/mm² (b) 3 N/mm²

(c) 4N/mm² (d) 5 N/mm²

[UPPSC State Eng. A.E. 2007(I), 2004(I)]

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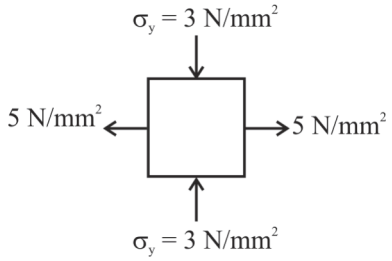
Ans : (c)



$\sigma_x = 5\text{N/mm}^2,\ \sigma_y = -3\text{N/mm}^2$

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \left(\tau_{xy}\right)^2}$$

$$\tau_{max} = \frac{\left(\sigma_x - \sigma_y\right)}{2}$$

$$= \frac{\left(5-(-3)\right)}{2}$$

$$= \frac{(5+3)}{2} = 4\,\text{N/mm}^2$$

1071. Mohr's circle is drawn for

(1) a point
(2) a square block
(3) a rectangular block
(4) isotropic material

The correct answer is-

(a) 1, 2 and 3
(b) 2 and 4
(c) 1 and 4
(d) 2, 3 and 4

BPSC AE 2012 Paper–V

Ans : (c) Mohr's circle is drawn for a body of homogenous material at a point subjected to stresses.

1072. In a plane stress problem-

(1) normal stress in the third direction is zero
(2) strain in the third direction is zero
(3) strain in all direction is present
(4) normal stresses in all directions are present

The correct answer is-

(a) 1 and 4 (b) 2 and 4
(c) 1 and 3 (d) 2 and 3

BPSC AE 2012 Paper–V

Ans : (c) In plane stress problem, stress in third direction is zero and strain is present in all direction.

1073. An element submitted to the action of biaxial stress

(a) $\overline{\sigma_x} = 2\phi y$ (b) $\overline{\sigma_x} = \sigma_y$
(c) $\overline{2\sigma_x} = \sigma_y$ (d) $\overline{\sigma_x} = -\sigma_y$

(BPSC AE Paper V 2006)

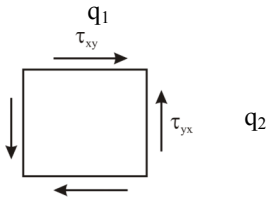
Ans : (b) An element subjected to biaxial stress can be represented by $\overline{\sigma}_x = \sigma_y$.

1074. In an element the complimentary shear stresses are always

(a) Equal in sign
(b) Equal in both magnitude and sign
(c) Equal in magnitude but opposite in sign
(d) None of these

(BPSC AE Paper V 2006)

Ans : (c)



Magnitude of complimentary shear stresses is same and they are acted in opposite rotations.

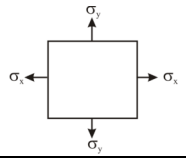
$$\tau_{xy} = -\tau_{yx}$$

1075. For an element under biaxial stress the sum of complementary normal stresses are

(a) Equal to maximum shear stress
(b) Half of the maximum shear stress
(c) Always constant
(d) Sometime constant

(BPSC AE Paper V 2006)

Ans : (c) For biaxial stress system sum of complimentary normal stresses are always constant.



$$\sigma_x + \sigma_y = \sigma_\theta + \sigma_{90+\theta}$$

1076. Principal planes are subject to :

(a) Normal stress only
(b) Tangential stress only
(c) Both normal and tangential stress
(d) None of the above

(UKPSC AE (Paper–I) 2007)

Ans : (a) Principal plane is a plane on which shear stress is zero and normal stresses are maximum.

1077. If a prismatic bar of cross section = A is subjected to a tensile force = P. the shear stress on a plane inclined at an angle 45^0 with the axis is given by-

(a) $\frac{1}{2}P/A$ (b) $\frac{3}{2}P/A$
(c) $\frac{2}{3}P/A$ (d) $2P/A$

(BPSC AE Paper V 2006)

Ans : (a) $\sigma_x = \frac{P}{A}$

$$\tau_{45^o} = \frac{\sigma_x + \sigma_y}{2}\sin 2\theta = = \frac{P}{2A}\sin 90^0 = \frac{P}{2A}$$

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1078. Principal stresses at a point in a plane stressed element are $\sigma_x = \sigma_y = 500\ N/mm^2$. Normal stress on the plane inclined at 45^o to the *x*-axis will be-

(a) Zero (b) 500 N/mm^2

(c) 1000 N/mm^2 (d) 707 N/mm^2

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : $\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta$

$= \frac{500+500}{2} + \frac{500-500}{2}\cos 90 = 500 N/mm^2$

1079. In case of biaxial stresses, the maximum value of shear stress is-

(a) Difference of normal stresses

(b) Half the difference of normal stresses

(c) Sum of normal stresses

(d) Half the sum of normal stresses

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : For biaxial stress condition

$\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta$

if $\theta = 45^0$, $\sin 2\theta = 1$

$\tau_{45^0} = \frac{\sigma_x - \sigma_y}{2}$

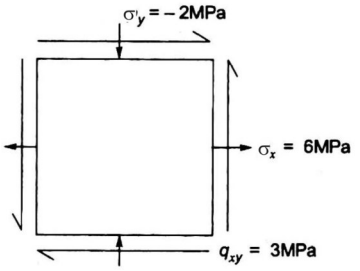
1080. At a point '*P*', the state of stress is σ_x = 6 MPa, σ_y = -2 MPa and q_{xy} = 3 MPa, the magnitude of principal stresses for this state of stress will be-

(a) 9 MPa and – 1 MPa (b) 7 MPa and – 3 MPa

(c) 7 MPa and – 1 MPa (d) 8 MPa and – 3 MPa

OPSC AEE 2015 PAPER–I

Ans. (b) :



$\sigma_{max/min} = \left(\frac{\sigma_x + \sigma_y}{2}\right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$

$= \left(\frac{6+(-2)}{2}\right) \pm \sqrt{\left(\frac{6-(-2)}{2}\right)^2 + 3^2}$

$= 2 \pm 5$

σ_{max} = 7 MPa; σ_{min} = –3 MPa

1081. Principal plane is defined as a plane on which the shear stress is-

(a) Maximum (b) Half of the normal

(c) Zero (d) None of these

TNPSC AE 2018, RPSC ACF 2011

OPSC AEE 2015 PAPER–I

Ans. (c) : According of Mohr's circle the principle stress is plotted on x-axis and shear stress is plotted on y-axis so principal plane will be on x-axis and if we take a point on x-axis then y will be zero (y=0). So the shear stress will be zero on principal plane.

1082. In a rectangular element subjected to like principal tensile stresses P_1 and P_2 in two mutually perpendicular directions *x* and *y*, the maximum shear stress would occur along the-

(a) Plane normal to *y*-axis

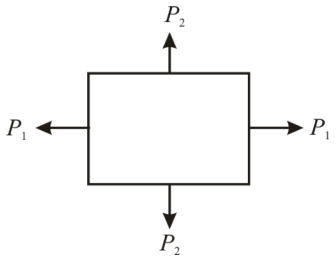
(b) Plane normal to *x*-axis

(c) Plane at 45^o and 135^o to the *y*-direction

(d) Plane at 45^o to the *y*-direction

OPSC AEE 2015 PAPER–I, ESE 1995

Ans. (c) :



$\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$

$\tau_{xy} = 0$

$\tau_{\theta max}$ at $\theta = 45^0$

$= \frac{P_1 - P_2}{2}$

For maximum shear stress–

$\tan 2\theta = -\left(\frac{\sigma_x - \sigma_y}{2\tau_{xy}}\right)$

$\because\ \theta = 45^o$ or 135^o

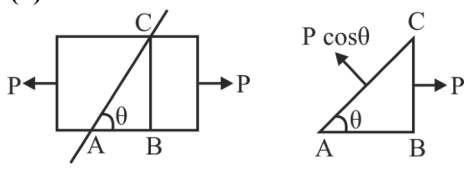
1083. If the normal cross-section A of a member is subjected to tensile force P, the resulting normal stress in an oblique plane inclined at angle 'θ' to the transverse plane will be

(a) $\frac{P}{A}\sin^2\theta$ (b) $\frac{P}{A}\cos^2\theta$

(c) $\frac{P}{2A}\sin^2\theta$ (d) $\frac{P}{2A}\cos^2\theta$

TNPSC AE 2018

Ans. (b) :



$\sigma_{AC} = \frac{P\cos\theta}{A'} = \frac{P\cos\theta}{\frac{A}{\cos\theta}} = \frac{P\cos^2\theta}{A}$

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1084. A thin plate having stress components as σ_x = 40 MPa, σ_y = –20 MPa and τ_{xy} = 10 MPa. What will be the yield strength in simple tensile as per Von-mises criterion?

(a) Y = 3100 MPa (b) Y = 55.67 MPa
(c) Y = 54.3 MPa (d) Y = 1500 MPa

[HPPSC 2016-II]

Ans. (b) : Principal stress is given by

$$\sigma_{1/2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2_{xy}}$$

$$= \frac{40-20}{2} \pm \sqrt{\left(\frac{40+20}{2}\right)^2 + 10^2}$$

$= 10 \pm 31.62$

$\sigma_1 = 41.62$ MPa; $\sigma_2 = -21.62$ MPa

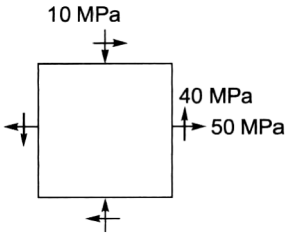
As per Von-Mises criteria :

$$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 \le Y^2$$

$$(41.62)^2 + (-21.62)^2 - 41.62(-21.62) \le Y^2$$

Y = 55.67 MPa

1085. A material yields under the following state of plane stress shown in figure, as per Von Mises criterion, the yield stress of the material is:



(a) 20 MPa (b) 74.16 MPa
(c) 50 MPa (d) 88.88 MPa

[HPPSC AE 2016]

Ans. (d) : Von Mises comprise of maximum distortion energy criteria.

$$\frac{1}{12G}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right] \le \left(\frac{f_y^2}{6G \times FOS}\right)$$

$$\Rightarrow \sigma_{1/2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

$= 20 \pm 50 = 70, -30$

$= \sigma_1 = 70$ MPa, $\sigma_2 = -30$ MPa; $\sigma_3 = 0$

$$\therefore \frac{1}{12G} \cdot \left[\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2 + \sigma_2^2 + \sigma_1^2\right] \le \frac{f_y^2}{6G}$$

$$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 \le f_y^2$$

$$f_y^2 \ge 7900$$

$$f_y \ge 88.88 \text{ MPa}$$

1086. At a point in a steel member, the major principal stress is 200 N/mm^2 (tensile) and the minor principal stress is compressive. If the uniaxial tensile yield stress 250 N/mm^2, then according to the maximum shear stress theory, the magnitude of the minor principal stress (compressive) at which yielding will commence is :

(a) 250 N/mm^2 (b) 150 N/mm^2
(c) 50 N/mm^2 (d) 200 N/mm^2

JPSC AE (Pre) 2019, ESE 1998

Ans. (c) :

Max. shear stress

$$= \frac{\sigma_1 - \sigma_3}{2} [\sigma_3 = \text{ minor principal stress}]$$

$$= \frac{200 - (-\sigma_3)}{2} = \frac{250}{2}$$

$\sigma_3 = 50$MPa

1087. Principal stress are :

(a) Parallel to principal planes
(b) Normal to principal planes
(c) Parallel to each other
(d) Opposite to each other

GPSC AE Class (1&2) Paper - 2 2017

Ans. (b) : The maximum stress induced in a plane is called the principal stress and the plane at which the maximum stress induced referred to the principal plane where the shear stress is considered zero.

1088. A rectangular block is subjected to normal stresses on two orthogonal faces 600 MPa and 400 MPa of same nature. What shall be maximum shear stress produced on a plane ?

(a) 100 MPa (b) 500 MPa
(c) 300 MPa (d) 250 MPa

Rajasthan JEn (Degree) 2016

Ans. (a) : Maximum shear stress $(\tau_{max}) = \frac{\sigma_1 - \sigma_2}{2}$

$$\tau_{max} = \frac{600 - 400}{2}$$

$$\boxed{\tau_{max} = 100 \text{MPa}}$$

1089. A plane which is not subjected to shear stress is known as:

(a) Compound plane (b) Simple plane
(c) Non-shear plane (d) Principal plane

KPSC AE 2016

Ans. (d) : The maximum stress induced in a plane is called the principal stress and the plane at which the maximum stress induced referred to the principal plane where the shear stress is considered zero.

1090. Which of the following theory gives best result for ductile materials?

(a) Rankine's theory
(b) St. Venant's theory
(c) Mises-Henky Theory
(d) Guest or Tresca's Theory

APPSC Poly. Tech. Lect. 2020

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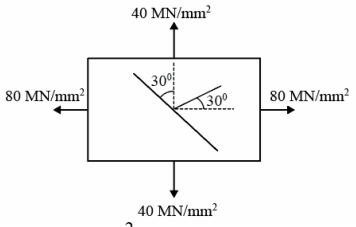
Ans. (c) : Maximum shear strain energy theory or maximum Distortion energy theory (von mises-henky theory)–This assumes that failure occur when shear strain energy (distortion energy) in the complex system is equal to that at the yield point in tensile compression test.

– Mises-Henky theory does not apply to brittle material for which elastic limit stress in tension and compression is quite different. Failure of most ductile material is ground by the distortion energy criteria or von mises theory.

Maximum principal stress theory (Rankine's theory)–In it is will suited to brittle materials. maximum shear stress theory (Guest-Tresca). This assumes that maximum shear stress in the complex stress system becomes equal to that at the yield point in simple tensile test. This theory holds good for ductile materials.

1091. The principal stresses in the wall of a container are 40 MN/mm^2 and 80 MN/mm^2. The normal makes an angle of 30^0 with a direction of maximum principal stress. The resultant stresses (in magnitude) in the plane will be nearly

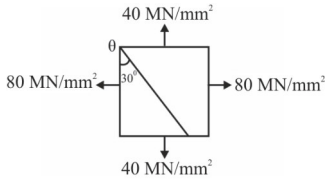


(a) 84 MN/mm^2 (b) 72 MN/mm^2
(c) 64 MN/mm^2 (d) 58 MN/mm^2

ESE 2019

Ans. (b)

$[\theta = 30^0]$



$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(2\theta) + \tau_{xy}\sin 2\theta$$

$$\sigma_\theta = \frac{80+40}{2} + \frac{80-40}{2}\cos(2\times 30) + 0 \quad \text{------ } [\tau_{xy} = 0]$$

$$\sigma_\theta = 60 + 20\cos 60^0$$

$$\left[\sigma_\theta = 60 + 20\times\frac{1}{2} = 70 MN/mm^2\right]$$

$$\tau_\theta = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\tau_\theta = -\left(\frac{80-40}{2}\right)\sin(2\times 30) - 0$$

$$\tau_\theta = -20\sin 60 = -20\times\frac{\sqrt{3}}{2}$$

$$\left[\tau_\theta = -10\sqrt{3}\right]$$

The resultant stresses in the plane;

$$\sigma_R = \sqrt{\sigma_\theta^2 + \tau_\theta^2}$$

$$\sigma_R = \sqrt{(70)^2 + \left(10\sqrt{3}\right)^2}$$

$$\sigma_R = \sqrt{5200}$$

$$\boxed{\sigma_R = 72.11 MN/mm^2}$$

1092. At a point in a material, the stresses acting on two planes at right angles to each other are σ_z = 120 MPa and σ_y = – 200 MPa and τ_{zy} = – 80 MPa. The maximum shear stress on the element will be nearly

(a) 142 MPa (b) 155 MPa
(c) 167 MPa (d) 179 MPa

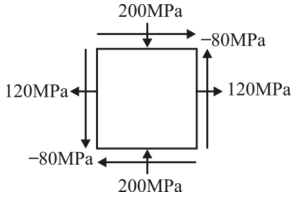
ESE 2019

Ans. (d)

$\sigma_z = 120$ MPa

$\sigma_y = -200$ MPa

$\tau_{zy} = -80$ MPa



The maximum shear stress;

$$\tau_{max} = \sqrt{\left(\frac{\sigma_z - \sigma_y}{2}\right)^2 + \tau_{zy}^2}$$

$$\tau_{max} = \sqrt{\left(\frac{120-(-200)}{2}\right)^2 + (-80)^2}$$

$$\tau_{max} = \sqrt{\left(\frac{320}{2}\right)^2 + (-80)^2}$$

$$\tau_{max} = \sqrt{25600 + 6400}$$

$$\tau_{max} = \sqrt{32000}$$

$$\tau_{max} = 178.89 MPa$$

$$\boxed{\tau_{max} \simeq 179 MPa}$$

1093. The normal stresses on the two mutually perpendicular planes at a point are 120 MPa (Tensile) and 60 MPa (Tensile). If the shear stress across these planes is 30 MPa, the principal stresses will be nearly

(a) 124 MPa (Tensile) and 24 MPa (Compressive)
(b) 132 MPa (Tensile) and 24 MPa (Compressive)
(c) 124 MPa (Tensile) and 48 MPa (Tensile)
(d) 132 MPa (Tensile) and 48 MPa (Tensile)

ESE 2019

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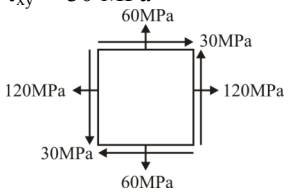
Ans. (d)

σ_x = 120 MPa (Tensile)

σ_y = 60 MPa (Tensile)

Shear stress 'τ_{xy}' = 30 MPa



Principal stresses

$$\sigma_{1/2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{120+60}{2} \pm \sqrt{\left(\frac{120-60}{2}\right)^2 + (30)^2}$$

$$= \frac{180}{2} \pm \sqrt{\left(\frac{60}{2}\right)^2 + (30)^2}$$

$$= 90 \pm \sqrt{(30)^2 + (30)^2}$$

$$= 90 \pm \sqrt{1800}$$

$$= 90 \pm 42.43$$

Major stress σ_1 = 132 MPa (Tensile)

Minor stress σ_2 = 48 MPa (Tensile)

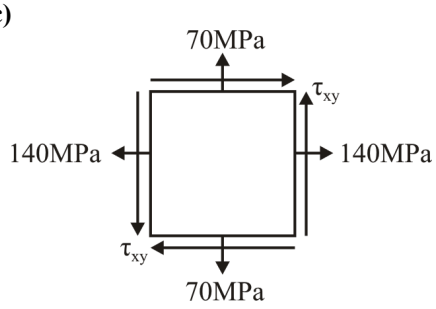
1094. The normal stresses on two mutually perpendicular planes are 140 N/mm^2 (Tensile) and 70 N/mm^2 (Tensile). If the maximum shear stress is 45 N/mm^2, the shear stress on these planes will be nearly

(a) 20.9 N/mm^2 (b) 24.6 N/mm^2

(c) 28.3 N/mm^2 (d) 32.0 N/mm^2

ESE 2019

Ans. (c)



σ_x = 140 MPa

σ_y = 70 MPa

τ_{max} = 45 MPa

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$(45)^2 = \left(\frac{140-70}{2}\right)^2 + \tau_{xy}^2$$

$$2025 = \left(\frac{70}{2}\right)^2 + \tau_{xy}^2$$

$$\tau_{xy}^2 = 2025 - (35)^2$$

$$\tau_{xy}^2 = 2025 - 1225$$

$$\tau_{xy}^2 = 800$$

$$\boxed{\tau_{xy} = 28.282 \text{MPa}} \quad \text{or } 28.3 \text{ N/mm}^2$$

1095. Principal stresses at a point in an elastic material are 1.5σ (tensile), σ (tensile) and 0.5σ (compressive). The elastic limit in tension is 210 MPa and μ = 0.3. The value of σ at failure when computed by maximum principal strain theory is, nearly

(a) 140.5 MPa (b) 145.5 MPa

(c) 150.5 MPa (d) 155.5 MPa

ESE 2018

Ans. (d) : Principal stresses at a point in an elastic material are–

$\sigma_1 = 1.5\sigma$ (T)

$\sigma_2 = \sigma$ (T), f_y = 210 MPa

$\sigma_3 = 0.5\sigma$ (C)

$\mu = 0.3$

Now, From Maximum principal strain theory.

$$\frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_3}{E} \le \frac{fy}{E}$$

$$\frac{1.5\sigma}{E} - \frac{0.3\sigma}{E} - 0.3\left(\frac{-0.5\sigma}{E}\right) \le \frac{f_y}{E}$$

$$1.5\sigma - 0.3\sigma + 0.15\sigma \le 210$$

$$\boxed{\sigma = 155.5 \text{MPa}}$$

1096. The state of stress at a point is given by : σ_x = 80 MPa, σ_y = 100 MPa and τ_{xy} = 60 MPa. If the yield strength for the material is 150 MPa, as determined in a uniaxial test, then the maximum shear stress is, nearly

(a) 150.8 MPa (b) 127.4 MPa

(c) 119.3 MPa (d) 104.0 MPa

ESE 2018

Ans. (*) : Given,

Stress, σ_x = 80MPa

σ_y = 100 MPa

τ_{xy} = 60MPa

and, yield strength = 150 MPa

$$\sigma_{1,}\sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 = \frac{80+100}{2} + \sqrt{\left(\frac{80-100}{2}\right)^2 + 60^2}$$

$$[\sigma_1 = 150.83 \text{MPa}]$$

$$\sigma_2 = \frac{80+100}{2} - \sqrt{\left(\frac{80-100}{2}\right)^2 + 60^2} = 29.17 \text{MPa}$$

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So, τ_{max} = maximum $\left[\left|\frac{\sigma_1-\sigma_2}{2}\right|, \left|\frac{\sigma_1}{2}\right|, \left|\frac{\sigma_2}{2}\right|\right]$

$= \frac{150.83-29.18}{2}$

= 60.82 MPa

Note– Commission has given no correct option.

1097. If strains on a piece of metal are $\varepsilon_x = -120\mu$ m/m, $\varepsilon_y = -30\ \mu$ m/m, and $\gamma = 120\ \mu$ m/m, what is the maximum principal strain?

(a) 0 (b) 50 μ m/m

(c) 75 μ m/m (d) 150 μ m/m

ESE 2018

Ans. (d) Given,

Strain on piece of metal are

$\varepsilon_x = -120\mu$ m/m

$\varepsilon_y = -30\mu$ m/m

$\gamma = 120\mu$ m/m

Maximum principal strain–

$$\left[\varepsilon_{max}, \varepsilon_{min} = \frac{\varepsilon_x+\varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x-\varepsilon_y}{2}\right)^2 + \left(\frac{\gamma}{2}\right)^2}\right]$$

$$\varepsilon_{max}, \varepsilon_{min} = \frac{-120\mu-30\mu}{2} \pm \sqrt{\left(\frac{-120\mu+30\mu}{2}\right)^2 + \left(\frac{120\mu}{2}\right)^2}$$

$$\varepsilon_{max}, \varepsilon_{min} = \frac{-150\mu}{2} \pm \sqrt{(-45\mu)^2 + (60\mu)^2}$$

$$\varepsilon_{max}, \varepsilon_{min} = -75\mu \pm \sqrt{5625\mu^2}$$

$$\varepsilon_{max}, \varepsilon_{min} = -75\mu \pm 75\mu$$

$\boxed{\varepsilon_{max} = -150\mu\,m/m}$ and $\boxed{\varepsilon_{min} = 0}$

Maximum magnitude value of strain will be considered.

1098. A principal plane contains

(a) maximum shear stress

(b) no shear stress

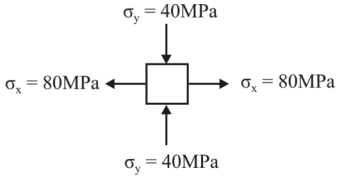
(c) no normal stress

(d) maximum normal stress and minimum normal stress

WBPSC AE 2014

Ans. (b) : Principal plane : Plane on which normal stresses are maximum or minimum are called principal plane & shear stress on these planes are zero.

1099. The state of stress at a certain point in a stressed body is as shown in the figure. Normal stress in x-direction is 80 MPa (Tensile) and in y-direction is 40 MPa (Compressive). The radius of the Mohr's circle for this state of stress will be

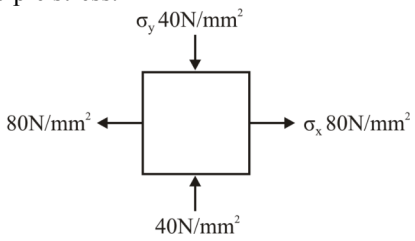


(a) 60 MPa (b) 40 MPa

(c) 20 MPa (d) 10 MPa

ESE 2017

Ans. (a) As there is no shear stress then given stresses are principle stress.



The given principal stresses are–

$\sigma_1 = \sigma_x = 80$ MPa

$\sigma_2 = \sigma_y = -40$MPa

Radius of mohr circle = $\frac{\sigma_1-\sigma_2}{2}$

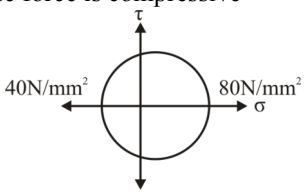
Radius = $\frac{80-(-40)}{2} = \frac{120}{2}$

[Radius = 60 MPa]

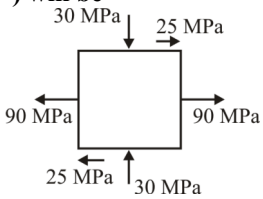
Draw I[st] co-ordinate = (80, 0)

And II[nd] co-ordinate = (–40, 0)

(–) because force is compressive



1100. For the state of stress shown in the figure, the maximum and minimum principal stresses (taking tensile stress as +, and compressive stress as –) will be



(a) 95 MPa and (–35) MPa

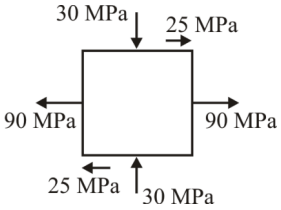
(b) 60 MPa and 30MPa

(c) 95 MPa and (–30) MPa

(d) 60 MPa and 35 MPa

ESE 2017

Ans. (a)



Stress 'σ_x' = 90 MPa

'σ_y' = –30 MPa

Shear stress 'τ_{xy}' = 25 MPa

■ Maximum and Minimum principal stress are–

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$$\sigma_{max/min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{max/min} = \frac{90-30}{2} \pm \sqrt{\left(\frac{90+30}{2}\right)^2 + (25)^2}$$

$$\sigma_{max/min} = 30 \pm \sqrt{(60)^2 + (25)^2}$$

$\sigma_{max} = 30 + 65$

$\boxed{\sigma_{max} = 95 MPa}$

$\sigma_{min} = 30 - 65$

$\boxed{\sigma_{min} = -35 MPa}$

1101. The principal stresses at a point in a bar are 160 N/mm^2 (tensile) and 80 N/mm^2 (compressive) the accompanying maximum shear stress intensity is

(a) 100 N/mm^2 (b) 110 N/mm^2

(c) 120 N/mm^2 (d) 140 N/mm^2

AP TRANSCO AEE 2017

Ans. (c) : Given,

$\sigma_x = 160$ N/mm^2 (tensile)

$\sigma_y = -80$ N/mm^2 (compressive)

$$\tau_{max} = \frac{\sigma_x - \sigma_y}{2}$$

$$\tau_{max} = \frac{160 - (-80)}{2}$$

$\tau_{max} = 120$ N/mm^2

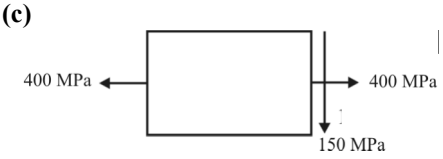
1102. A body is subjected to a direct tensile stress of 400 MPa in one plane accompanied by a simple shear stress of 150 MPa. The maximum normal stress will be

(a) -100 MPa (b) 150 MPa

(c) 450 MPa (d) 550 MPa

ISRO Scientist/Engineer 2019

Ans : (c)



Tensile stress $\sigma_x = 400$ MPa

Shear stress $\tau_x = 150$ MPa

For Maximum Normal Stress–

$$\sigma_n = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \left(\tau_{xy}\right)^2}$$

$$\sigma_n = \frac{400}{2} + \sqrt{\left(\frac{400}{2}\right)^2 + (150)^2}$$

$$\sigma_n = 200 + \sqrt{(200)^2 + (150)^2}$$

$$\sigma_n = 200 + \sqrt{62500}$$

$\sigma_n = 200 + 250$

$\boxed{\sigma_{n(max)} = 450 MPa}$

1103. A body is subjected to a direct tensile stress of 300 MPa in one plane accompanied by a simple shear stress of 200 MPa. The maximum normal stress will be

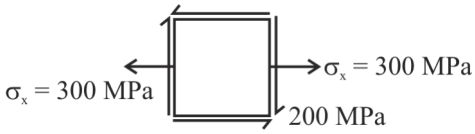
(a) –100 MPa (b) 250 MPa

(c) 300 MPa (d) 400 MPa

ISRO Scientist/Engineer 2017

Ans : (d)

- $\sigma = 300$MPa
- $\tau_{xy} = 200$MPa



Maximum normal stress–

$$\sigma_{max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{max} = \frac{300}{2} + \sqrt{\left(\frac{300}{2}\right)^2 + (200)^2} \quad \because \{\sigma_y = 0\}$$

$$\sigma_{max} = 150 + \sqrt{(150)^2 + (200)^2}$$

$\sigma_{max} = 150 + 250$

$\boxed{\sigma_{max} = 400 MPa}$

1104. Along the principal plane subjected to maximum principal stresses

(a) Maximum shear stress acts

(b) Minimum shear stress acts

(c) No shear stress acts

(d) Partial shear stress acts

ISRO Scientist/Engineer 2018

Ans : (c) : At principal plane shear stress is zero. There is maximum direct or normal stress but no shear stress at principal plane-

The equation for shear stress in terms of co-ordinate stress is-

$$-\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta - \tau_{x.y}(\cos 2\theta)$$

Divide above equation by cos2 θ

$$\frac{-\tau_\theta}{\cos 2\theta} = \left(\frac{\sigma_x - \sigma_y}{2}\right)\frac{\sin 2\theta}{\cos 2\theta} - \tau_{xy}$$

$$-\tau_{\theta P} = \cos 2\theta_P \left[\left(\frac{\sigma_x - \sigma_y}{2}\right)\tan 2\theta_P - \tau_{xy}\right]$$

$$-\tau_{\theta P} = \cos 2\theta_P \left[\left(\frac{\sigma_x - \sigma_y}{2}\right)\left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right) - \tau_{xy}\right]$$

$$\because \tan 2\theta_P = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$-\tau_{\theta P} = \cos 2\theta_P (\tau_{xy} - \tau_{xy})$$

$$-\tau_{\theta P} = \cos 2\theta_P (0)$$

$\boxed{-\tau_{\theta P} = 0}$

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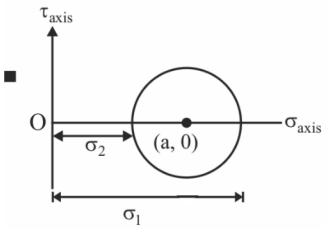
1105. In two dimensional stress system, the radius of the Mohr's circles represents

(a) Maximum normal stress
(b) Minimum normal stress
(c) Minimum shear stress
(d) Maximum shear stress

ISRO Scientist/Engineer 2013
ESE 2012

Ans : (d)

- Mohr's circles represents the state of stress at a points.



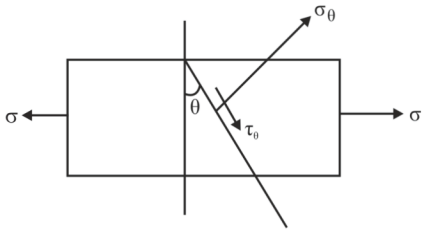
- ⇒ Radius of Mohr's circle 'r' = $\frac{\sigma_1 - \sigma_2}{2}$.
- Radius of Mohr circle represent the value of maximum shear stress.
- Normal stress on the plane of maximum shear stress is represented by co-ordinate of centre of Mohr circle.

1106. A prismatic bar is subjected to axial tension. What is the aspect angle (ϕ) which defines an oblique section on which normal and shearing stresses are equal?

(a) 30^0 (b) 45^0
(c) 60^0 (d) 90^0

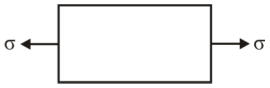
OPSC AEE 2019 Paper–I

Ans. (b)



$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_\theta = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$



$\sigma_x = \sigma$
$\sigma_y = 0$
$\tau_{xy} = 0$

$$\therefore\ \sigma_\theta = \frac{\sigma}{2} + \frac{\sigma}{2}\cos 2\theta$$

$$\tau_\theta = -\frac{\sigma}{2}\sin 2\theta$$

$\sigma_\theta = \tau_\theta$ (given)

$$\frac{\sigma}{2} + \frac{\sigma}{2}\cos 2\theta = -\frac{\sigma}{2}\sin 2\theta$$

$$\frac{\sigma}{2} = -\frac{\sigma}{2}(\sin 2\theta + \cos 2\theta)$$

$\cos 2\theta + \sin 2\theta = -1$
$(\cos 2\theta + 1) = -\sin 2\theta$
$2\cos^2\theta = -2\sin\theta\cos\theta$
$\tan\theta = -1$
$\theta = 135^0$
$\theta_1 = \theta - 90^o$
$= 135^o - 90^o$
$= 45^o$

1107. In a material the principal stresses are 60 MN/m^2, 48 MN/m^2 and –36 MN/m^2 and $\frac{1}{m} = 0.3$, the total strain energy per unit volume will be nearly :

(a) 43.5 kNm/m^3 (b) 35.5 kNm/m^3
(c) 27.5 kNm/m^3 (d) 19.5 kNm/m^3

ESE 2020

Ans. (d) : Given, $\sigma_1 = 60$ MN/m^2, $\sigma_2 = 48$ MN/m^2
$\sigma_3 = -36$ MN/m^2, $\mu = 0.3$

Strain Energy per unit volume

$$= \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

$$= \frac{10^{12}}{2\times 200\times 10^9}\left[\begin{matrix}(60)^2 + (48)^2 + (-36)^2 - 2\times 0.3(60\\ \times 48 - 48\times 36 - 60\times 36)\end{matrix}\right]$$

$$= \frac{10^{12}}{4\times 10^{11}}\left[7200 - 0.6(2880 - 1728 - 2160)\right]$$

$$= \frac{10}{4}\left[7200 + 604.8\right]$$

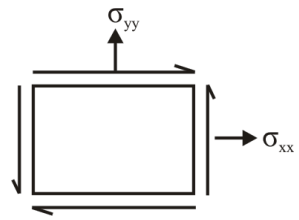
$= 19.51$ kN/m^3

1108. At a point in a two dimensional stress system, the normal stress on two mutually perpendicular planes are σ_{xx} and σ_{yy} and shear stress is τ_{xy}. One of the principal stresses will become zero when the value of shear stress τ_{xy} is :

(a) $\pm(\sigma_{xx}\sigma_{yy})$ (b) $\pm\sqrt{\sigma_{xx} - \sigma_{yy}}$
(c) $\pm\sqrt{\sigma_{xx} + \sigma_{yy}}$ (d) $\pm\sqrt{\sigma_{xx}\sigma_{yy}}$

ESE 2020

Ans. (d)



$$\sigma_1 = \frac{\sigma_{xx} + \sigma_{yy}}{2} + \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2} = 0$$

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$$\Rightarrow \frac{\sigma_{xx}+\sigma_{yy}}{2} = -\sqrt{\left(\frac{\sigma_{xx}-\sigma_{yy}}{2}\right)^2+\tau_{xy}^2}$$

Taking square on both side

$$\Rightarrow \left(\frac{\sigma_{xx}+\sigma_{yy}}{2}\right)^2 = \left(\frac{\sigma_{xx}-\sigma_{yy}}{2}\right)^2+\tau_{xy}^2$$

$$\Rightarrow \frac{\sigma_{xx}^2}{4}+\frac{\sigma_{yy}^2}{4}+\frac{\sigma_{xx}.\sigma_{yy}}{2} = \frac{\sigma_{xx}^2}{4}+\frac{\sigma_{yy}^2}{4}-\frac{\sigma_{xx}.\sigma_{yy}}{2}+\tau_{xy}^2$$

$$\Rightarrow \sigma_{xx}.\sigma_{yy} = \tau_{xy}^2$$

$$\Rightarrow \boxed{\tau_{xy} = \pm\sqrt{\sigma_{xx}.\sigma_{yy}}}$$

1109. "The longitudinal strain is uniform across the cross-section of the cylinder". This assumption is reasonably good for a large portion except near the ends. The theory based on this assumption is called :

(a) Coulomb's Theory
(b) St. Vecant's Theory
(c) Maximum Strain Energy Theory
(d) Lame's Theory

SJVNL ET 2019

Ans. (d) : The longitudinal strain is uniform across the cross-section of the cylinder. This assumption is reasonably good for a large portion except near the ends. The theory based on this assumption is called Lame's theory.

1110. A bar is subjected to fluctuating tensile load from 20 kN to 100 kN. The material has yield strength of 240 MPa and endurance limit in reversed bending is 160 MPa. According to the Soderberg principle, the area of cross-section in mm^2 of the bar for a factor of safety of 2 is:

(a) 400 (b) 600
(c) 750 (d) 1000

PTCUL AE 2016

Ans. (d) $\frac{\sigma_m}{\sigma_y}+\frac{\sigma_v}{\sigma_e} = \frac{1}{F.S.}$

$$1000\left(\frac{60}{A\times 240}+\frac{40}{A\times 160}\right) = \frac{1}{2}$$

$$1000\left(\frac{1}{4A}+\frac{1}{4A}\right) = \frac{1}{2}$$

$$1000\left(\frac{2}{4A}\right) = \frac{1}{2}$$

$A = 1000\ mm^2$

1111. The theory which is in good agreement with experiments for ductile materials is

(a) Maximum stress theory
(b) Maximum strain theory
(c) Maximum shear theory
(d) All the given answers

APPSC AEE MAINS 2016

Ans. (c) : The theory which is in good agreement with experiments for ductile materials is maximum shear stress theory.

1112. An element is subjected to p_x = 35 MPa (tensile) and p_y = 20 MPa (tensile) and shear stress q = 7.5 MPa. Then the direction of principal stresses is

(a) 45° (b) 22.5°
(c) 30° (d) 15°

APPSC AEE MAINS 2016

Ans. (b) : $\tau = -\left(\frac{\sigma_x-\sigma_y}{2}\right)\sin 2\theta+\tau_{xy}\cos 2\theta$

$$\tau = 0$$

$$0 = -\left(\frac{35-20}{2}\right)\sin 2\theta_p+7.5\cos 2\theta_p$$

$$7.5\sin 2\theta_p = 7.5\cos 2\theta_p$$

$$\tan 2\theta_p = 1$$

$$\theta_p = \frac{45}{2} = 22.5^\circ$$

1113. Shearing stress on principal planes will be

(a) Equal to normal stresses
(b) Half the normal stresses
(c) Greater than normal stresses
(d) zero

APPSC AEE MAINS 2016

Ans. (d) : Shearing stress on principal planes will be zero.

1114. If normal stresses of same nature p_x and p_y and shear stress q are acting on two perpendicular planes and $q = (p_x p_y)^{1/2}$, then the major and minor principal stresses respectively are

(a) $p_x + p_y$ and $p_x - p_y$
(b) p_x and $p_x - p_y$
(c) $0.5(p_x + p_y)$ and $0.5(p_x - p_y)$
(d) $p_x + p_y$ and zero

APPSC AEE MAINS 2016

Ans. (d) : $\sigma_x = p_x, \sigma_y = p_y, \tau_{xy} = \sqrt{p_x p_y}$

$$\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2}\pm\sqrt{\left(\frac{p_x-p_y}{2}\right)^2+\left(\sqrt{p_x p_y}\right)^2}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2}\pm\sqrt{\frac{p_x^2}{4}-\frac{2p_x p_y}{4}+\frac{p_y^2}{4}+p_x p_y}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2}\pm\sqrt{\frac{1}{4}\left(p_x^2+2p_x p_y+p_y^2\right)}$$

$$\sigma_{1,2} = \frac{p_x+p_y}{2}\pm\frac{p_x+p_y}{2}, \sigma_1 = p_x+p_y, \sigma_2 = 0$$

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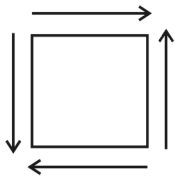
1115. An element in a strained body is subjected to only shear stress of intensity 50 MPa tending to rotate the body in clockwise direction. What is the magnitude of principal stresses?

(a) ± 50 MPa
(b) + 50 MPa, – 25 MPa
(c) ± 25 MPa, – 50 MPa
(d) ± 25 MPa

Maharashtra PSC Mains 2018 Paper-1

Ans. (a) :



under pure shear condition

$\sigma_1 = (-)\ \sigma_2 = \tau = 50$ MPa

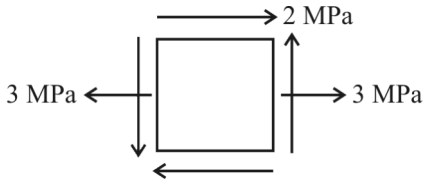
Principal stress = ± 50 MPa

1116. At a point in the web of a girder the bending stress (σ_x) is 3 MPa (tensile) and the shearing stress (τ) at the same point is 2 MPa, then the maximum shear stress is

(a) 1.5 MPa (b) 4 MPa
(c) 2.5 MPa (d) 1 MPa

Maharashtra PSC Mains 2018 Paper-1

Ans. (c) :



$\sigma_x = 3$ MPa $\quad \sigma_y = 0 \quad \tau_{xy} = 2$ MPa

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{3-0}{2}\right)^2 + 2^2}$$

$$= \sqrt{(1.5)^2 + (2)^2}$$

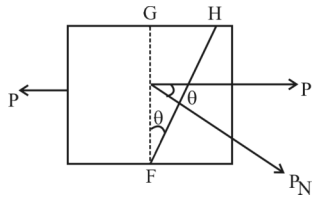
$= 2.5$ MPa

1117. In a linear strain system, strain in a direction θ with X-direction (where ε_x is the strain in X-direction) is

(a) $\varepsilon_x \cos\theta$ (b) $\varepsilon_x \sin\theta$
(c) $\varepsilon_x \cos^2\theta$ (d) $\varepsilon_x \sin^2\theta$

UPSC JWM 2017

Ans. (c) :



Normal stress ON FH = $\frac{P_n}{A_{FH}}$

$$= \frac{P\cos\theta}{A_{FH}/\cos\theta} = \frac{P}{A_{FG}}\cos^2\theta = \sigma\cos^2\theta$$

$$(\text{strain})_x = \frac{\sigma}{E}\cos^2\theta$$

$$(\text{strain})_x = \varepsilon_x \cos^2\theta$$

1118. In Principal planes,

(a) shear stress is maximum
(b) shear stress is zero
(c) shear stress is minimum
(d) none of above

GPSC R&B AE 2016

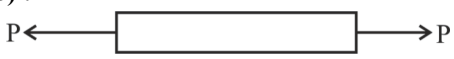
Ans. (b) : In principal planes, shear stress is zero and normal stress is maximum.
Shear stress will be maximum at plane having 45° angle from principal plane.

1119. If a prismatic member with area of cross-section A is subjected to a tensile load P, then the maximum shear stress and its inclination with the direction of load respectively-

(a) $\frac{P}{A}$ *and* 45° (b) $\frac{P}{2A}$ *and* 45°
(c) $\frac{P}{A}$ *and* 60° (d) $\frac{P}{A}$ *and* 30°

RPSC AE 2018, JPSC AE 2009, ESE 2000

Ans. (b) :



Normal stress in direction of load $= \frac{P}{A}$

In case of uniaxial stress maximum and minimum principal stress, $\sigma_1 = \frac{P}{A}, \sigma_2 = 0$

Maximum shear stress $= \frac{\sigma_1 - \sigma_2}{2} = \frac{\frac{P}{A} - 0}{2}$

$$\boxed{\tau_{max} = \frac{P}{2A}} \qquad \tan 2\theta = \frac{-(\sigma_x - \sigma_y)}{2\tau_{xy}}$$

$\boxed{\theta_{max} = 45^\circ}$

1120. σ_x, σ_y and τ_{xy} are normal and shear stresses on x and y faces. What is the radius of Mohr's circle in terms of these stresses?

(a) $\frac{\sigma_x - \sigma_y}{2}$ (b) $\frac{\sigma_x - \sigma_y}{2} + \tau_{xy}$
(c) $\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$ (d) $\sqrt{\frac{\sigma_x - \sigma_y}{2} - \tau_{xy}^2}$

JPSC AE 2009, ESE 2004

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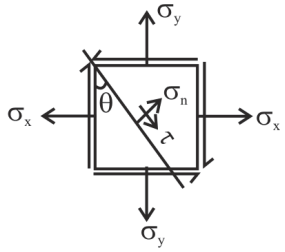
Ans. (c) :



$$\sigma_n = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$R = \sqrt{\left[\frac{1}{2}(\sigma_x - \sigma_y)\right]^2 + \tau_{xy}^2}$$

$$\sigma_{avg} = \frac{1}{2}(\sigma_1 + \sigma_2)$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

1121. A circular bar is subjected to an axial force and shear force, the difference between two principal stresses is 120 MPa. Based on maximum shear stress theory what is the factor of safety, if elastic limit of the bar is 300 MPa?

(a) 0.5 (b) 2
(c) 2.5 (d) 3

TANGEDCO AE 2018, PTCUL AE 2016

Ans. (c) : According to maximum shear theory,

$$\tau_{max} \le \left(\frac{\sigma_y}{2}\right)$$

$$\sigma_1 - \sigma_2 \le \frac{\sigma_y}{FOS}$$

$$120 \le \frac{300}{FOS}$$

$$\text{FOS} = \frac{300}{120} = 2.5$$

1122. The maximum shear stress is equal to the :

(a) sum of the principal stresses
(b) difference of the principal stresses
(c) half of the difference of the principal stresses
(d) half of the sum of principal stresses

TRB Poly. Lect. Tech. 2017

Ans. (c) : The maximum shear stress is equal to the half of the difference of the principal stress.

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2}$$

The maximum and minimum normal stresses are called principal stresses.

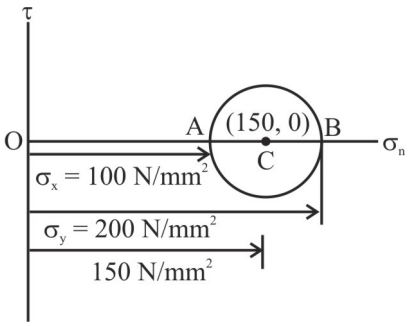
1123. A two dimensional stress system has like stresses $\sigma_x = 100$ N/mm^2 and $\sigma_y = 200$ N/mm^2 in two mutually perpendicular directions. The xy coordinates of the centre of the Mohr's circle are

(a) 100, 50 (b) –50, 0
(c) 150, 0 (d) 0, 150

TRB Poly. Tech. Lect. 2012
ESE 2010

Ans. (c) : Given $\sigma_x = 100$ N/mm^2, $\sigma_y = 200$ N/mm^2

$$OC = \frac{\sigma_x + \sigma_y}{2} = \frac{100 + 200}{2} = 150\,\text{N/mm}^2$$



1124. Principal planes are subject to

(a) normal stresses only
(b) tangential stresses only
(c) both normal stresses and tangential stresses
(d) none of these

UKPSC AE (Urban) 2011

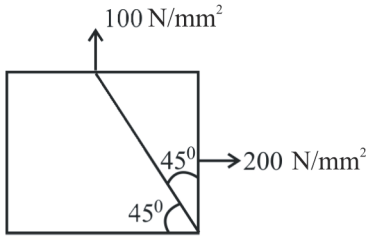
Ans. (a) : In every object, there are three planes which are mutually perpendicular to each other.

1125. At a point in a strained material, if two mutually perpendicular tensile stresses of 200 N/mm^2 and 10 N/mm^2 are acting then the intensity of tangential stress on a plane inclined a 45^0 to the axis of the minor stress will be.

(a) 100 N/mm^2 (b) 75 N/mm^2
(c) 50 N/mm^2 (d) 125 N/mm^2

UJVNL AE 2016

Ans. (c) : Given,
$\sigma_x = 200$ N/mm^2
$\sigma_y = 100$ N/mm^2
$\theta = 45^0$



Tangential stress, $\tau = \left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta - \tau_{xy}\cos 2\theta$

$$\tau = \left(\frac{200 - 100}{2}\right)\sin 90^0 - 0$$

$\tau = 50$ N/mm^2

1126. The determine of stresses on oblique sections of strained material is an important step in the design of

(a) framed structures

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(b) reveted connections
(c) welded connections
(d) nut and bolt jointed connections

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (a) : The determination of stresses on oblique sections of strained material is an important step in the design of framed structure.

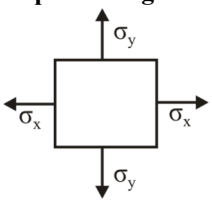
1127. Principal planes are the planes on which the resultant stress is the-

(a) Shear stress (b) Normal stress
(c) Tangential stress (d) None of the above

UJVNL AE 2012

Ans. (b) : Principal planes are the planes on which the resultant stress is the Normal stress.

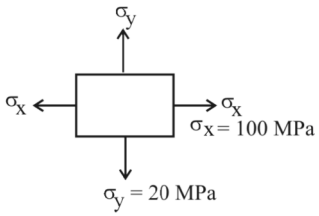
1128. Fig shows the state of stress at a certain point in a stressed body. The magnitudes of normal stresses in the x and y directions are 100 MPa and 20 MPa respectively. The radius of Mohr's stress circle representing this state of stress is :



(a) 120 (b) 80
(c) 60 (d) 40

TRB Poly. Lect. Tech. 2017

Ans. (d) :



Radius of mohr's circle ,

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$R = \sqrt{\left(\frac{100 - 20}{2}\right)^2 + 0}$$

$R = \sqrt{1600}$

$R = 40$ MPa

1129. Which geometrical method is used to find principal planes, maximum shear stress and their planes?

(a) Mohr's Circle of Stress
(b) Macaulay's Method
(c) Moment Area Theorem
(d) SFD and BMD

SJVNL ET 2019

Ans. (a) : Mohr's Circle of Stress method is used to find principal planes, maximum shear stress and their planes.

1130. For ductile materials the most appropriate failure theory is

(a) Maximum shear stress theory
(b) Maximum principal stress theory
(c) Maximum principal strain theory
(d) Shear strain energy theory

GPSC AE June 2019

Ans. (a) : Commonly used failure criteria for ductile materials are the Tresca or maximum shear stress theory.

1131. The plane stresses on x and y faces are 4.2 N/mm^2 and 9.6 N/mm^2 respectively. The value of normal stress at point where shear stress is maximum, would be

(a) 6.9 N/mm^2 (b) 2.7 N/mm^2
(c) 5.4 N/mm^2 (d) 13.8 N/mm^2

GPSC AE June 2019

Ans. (a) : Maximum shear stress $= \frac{\sigma_x - \sigma_y}{2}$

$= \frac{4.2 + 9.6}{2}$

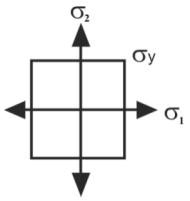
$= 6.9$ N/mm^2

1132. According to the maximum principal stress theory, the yield locus is a/an

(a) square (b) circle
(c) hexagon (d) ellipse

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : As per Maximum principal stress theory



yielding locus is a square.

1133. Let σ_1, σ_2 and σ_3 are the principal stresses at a material point. If the yield stress of the material is σ_y, then according to Von Mises theory yielding will not occur if

(a) $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 < 2(\sigma_y)^2$
(b) $\max[(\sigma_1 - \sigma_2), (\sigma_2 - \sigma_3), (\sigma_3 - \sigma_1)] < \sigma_y$
(c) $(\sigma_1)^2 + (\sigma_2)^2 + (\sigma_3)^2 < (\sigma_y)^2$
(d) $(\sigma_1)^2 + (\sigma_2)^2 + (\sigma_3)^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) < (\sigma_y)^2$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : VonMises Yielding Failure theory

$$\sigma_y \geq \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \leq 2\sigma_y^2$$

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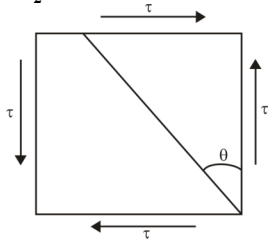
1134. The state stress at a point is shown below. θ represents the principal plane corresponding to principal stress σ_1 and σ_2 ($\sigma_1 > \sigma_2$). Values of θ, σ_1, and σ_2 are



(a) 0°, 90°, τ and -τ
(b) 30°, 120°, τ and -τ
(c) 45°, 135°, τ and -τ
(d) 45°, 135°, $\frac{\tau}{2}$ and $-\frac{\tau}{2}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : Due to pure shear diagonal tension ($\sigma_1 = +\tau$) and diagonal compression ($\sigma_2 = -\tau$) develops.
$\therefore \quad \theta_1 = 45°, \theta_2 = 135°, \sigma_1 = \tau, \sigma_2 = -\tau$

1135. A point in a strained material is subjected to two mutually perpendicular stresses of 150 MPa (tensile) and 50 MPa (compressive), then what will be the magnitude of maximum shear stress in the component?

(a) 50 MPa (b) 100 MPa
(c) 150 MPa (d) 200 MPa

Maharashtra PSC (Mains) 2019 Paper–I
ESE 2003

Ans. (b) : $\sigma_1 = 150$ MPa
$\sigma_2 = -50$ MPa

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

$$\tau_{max} = \frac{150 - (-50)}{2}$$

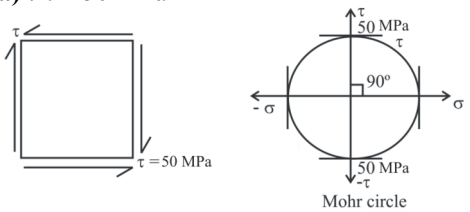
$$\tau_{max} = \frac{200}{2} = 100 \text{ MPa}$$

1136. A plane element is subjected to a shearing stresses of 50 MPa. The principal stresses existing in this element and the directions of the planes on which they occur would be ..... at ..... respectively.

(a) 50 MPa at 45° (b) 50 MPa at 90°
(c) 7 MPa at 45° (d) 7 MPa at 90°

GPSC AE Dec 2018

Ans. (a) : τ = 50 MPa



MPa

Principal stress $\sigma = -\sigma = 50$ MPa
Direction = 45°

1137. Saint-Venant's principal states that

(a) the difference between the effects of two different but statically equivalent loads become very small at sufficiently large distances from load
(b) if a part is separated or isolated from body or surroundings, appropriate reactions and/or internal forces must be added
(c) when an elastic body is acted upon by some external forces on various points, then the resultant effect on the body is the vector sum of the effects caused by each of the loads acting independently on the respective points of the body
(d) a body has a definite shape and dimensions before loading

GPSC AE Dec 2018

Ans. (a) : Saint-Venant's principle states that the difference between the effects of two different but statically equivalent loads become very small at sufficiently large distances from load.

1138. The maximum tangential stress acting on a plane inclined to the direction of the axial stress (p) in a tensile bar is equal to

(a) p (b) 2p
(c) p/2 (d) 3p/2

APPSC AEE MAINS 2016

Ans. (c) : $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

$$= \frac{p - 0}{2}$$

$$\tau_{max} = \frac{p}{2}$$

1139. The center of the Mohr's Circle for pure shear case lies in

(a) at τ distance from origin
(b) at $\frac{\tau}{2}$ distance from origin
(c) at origin
(d) at $\frac{\tau}{\sqrt{2}}$ distance from origin

APTRANSCO 2017

Ans : (c) : For pure shear case, $\sigma_x = \sigma_y = 0$ and only τ_{xy} will be acting as principal stress.

1140. Consider the following statements:

1. The maximum shear stress is one half of the normal stress in the case of uniaxial stress field
2. In a biaxial stress field, acted upon by normal stresses unaccompanied by shear stresses, the maximum shear stress is any one of the normal stresses.
3. The Mohr's stress circle will be tangential to the vertical axis in the case of uniaxial stress field.

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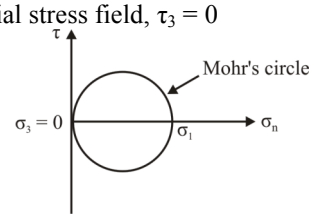
Which of the above statements are *correct*?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

ESE 2017

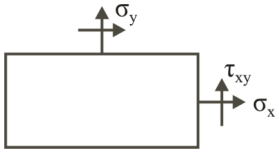
Ans. (d)

- Maximum shear stress induced in a body subjected to uniaxial stress is half of normal stress and is inclined at angle of 45^0 with respect to reference axis (with vertical).
- In Uniaxial stress field, $\tau_3 = 0$



$$\left[\tau_{max} = \frac{\sigma_{12} - \sigma_3}{2} = \frac{\sigma_1 - 0}{2} = \frac{\sigma_1}{2}\right]$$

- In Biaxial stress field



$$\left[\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}\right]$$

1141. The shear on principal plane is

(a) minimum (b) maximum
(c) zero (d) infinity

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) The shear on principal plane is zero.

1142. If a body carries two unlike principal stresses, the maximum shear stress is given by

(a) sum of the principal stresses
(b) difference of the principal stresses
(c) half the difference of the principal stresses
(d) half the sum of the principal stresses

APPSC AEE (Civil/Mechanical) 2012
ESE 2008

Ans. (c & d) If a body carries two unlike principal stresses, the maximum shear stress is given by half the difference of the principal stresses.

Or

If a body carries two unlike principal stresses the maximum shear stress is given by half the sum of the principal stresses.

OPTION 'C' is more appropriate

1143. The radius of Mohr's circle for two unlike principal stresses of magnitude σ is

(a) $\frac{\sigma}{2}$ (b) σ
(c) $\frac{\sigma}{4}$ (d) Zero

APPSC AEE (Civil/Mechanical) 2012

Ans. (b)

1144. A solid circular shaft is subjected to a maximum shearing stress of 140MPa. The magnitude of maximum normal stress developed in the shaft is

(a) 140 MPa (b) 80MPa
(c) 70MPa (d) 60MPa

APPSC AEE (Civil/Mechanical) 2012

Ans. (a) 140 Mpa

1145. A square element is subjected to bi-axial tensile stresses of equal magnitudes. The shear stress on an inclined plane is equal to

(a) the magnitude of the tensile stress
(b) sum of the tensile stresses
(c) zero
(d) half the sum of the tensile stresses

WBPSC AE (Common) 1998

Ans. (c) :

1146. If the principal stresses at a point in a strained body are σ_x and $\sigma_y (\sigma_x > \sigma_y)$, resultant stress on a plane carrying the maximum shear stress is equal to

(a) $\sqrt{\sigma_x^2 + \sigma_y^2}$ (b) $\sqrt{\sigma_x^2 - \sigma_y^2}$
(c) $\frac{\sqrt{\sigma_x^2 + \sigma_y^2}}{2}$ (d) $\frac{\sqrt{\sigma_x^2 - \sigma_y^2}}{2}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) If the principal stresses at a point in strained body are σ_x and σ_y then the resultant stress on a plane the maximum shear stress is equal to $\frac{\sqrt{\sigma_x^2 + \sigma_y^2}}{2}$

1147. A body is subjected to two normal stresses $20kN/m^2$ (tensile) and $10kN/m^2$ (compressive) acting mutually perpendicular to each other. The maximum shear stress is

(a) 30 kN/m^2 (b) 25 kN/m^2
(c) 15 kN/m^2 (d) 10 kN/m^2

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) For two mutually perpendicular stress

$$\tau_{max} = \frac{\sigma_x - \sigma_y}{2} = \frac{20 - (-10)}{2} = 15 KN/m^2$$

1148. Principal planes will be free of

(a) normal stress
(b) shear stress
(c) both normal and shear stresses
(d) None of these

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) Principal planes will be free of shear stress.

1149. On two perpendicular planes there are normal stresses, σ_1 and σ_2 and shear stress q. If $q^2 = \sigma_1\sigma_2$, the major and minor principal stresses respectively are

(a) $\sigma_1 + \sigma_2$ and zero

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(b) $\sigma_1 + \sigma_2$ and $\sigma_1 - \sigma_2$

(c) Zero and $\sigma_1 - \sigma_2$

(d) $\sigma_1 - \sigma_2$ and $\sigma_1 + \sigma_2$

APPSC AEE (Civil/Mechanical) 2012

Ans. (a)

$$\sigma_{major/minor} = \frac{\sigma_1 + \sigma_2}{2} \pm \frac{1}{2}\sqrt{(\sigma_1 - \sigma_2)^2 + 4q^2}$$

$$= \frac{\sigma_1 + \sigma_2}{2} \pm \frac{1}{2}\sqrt{(\sigma_1 - \sigma_2)^2 + 4\sigma_1\sigma_2}$$

$$= \frac{\sigma_1 + \sigma_2}{2} \pm \frac{\sigma_1 + \sigma_2}{2}$$

$$\sigma_{major} = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 + \sigma_2}{2} = \sigma_1 + \sigma_2$$

$$\sigma_{minor} = \frac{\sigma_1 + \sigma_2}{2} - \frac{\sigma_1 + \sigma_2}{2} = 0$$

1150. Angle between the principal planes is

(a) 270^0 (b) 180^0

(c) 90^0 (d) 45^0

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Angle between the principal plane is 90^0.

1151. For a two-dimensional stress system the coordinates of the centre of Mohr's circle are

(a) $\left[\frac{\sigma_x - \sigma_y}{2}, 0\right]$

(b) $\left[\frac{\sigma_x + \sigma_y}{2}, 0\right]$

(c) $\left[0, \frac{\sigma_x - \sigma_y}{2}\right]$

(d) $\left[0, \frac{\sigma_x + \sigma_y}{2}\right]$

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) For a two - dimensionl stress system the Co-ordinates of the centre of Mohr's circle are $\left[\frac{\sigma_x + \sigma_y}{2}, 0\right]$

The general equation of mohr's circles

$$\left(\sigma_0 - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_0^2 = \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2$$

$$\text{Centre} = \left(\frac{\sigma_x + \sigma_y}{2}, 0\right)$$

$$\text{Radius} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

1152. Principal stress at a point in plane stressed element are $\sigma_x = \sigma_y = 500$ kPa. Normal stress on the plane inclined at 45 degree to x-axis will be

(a) 1000 kPa (b) 750 kPa

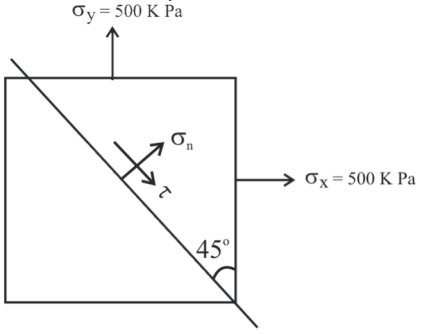
(c) 500 kPa (d) 250 kPa

GPSC Assistant Prof. 2016

Ans. (c) :

Given,

$\theta = 45^0$ from x – axis, $\tau_{xy} = 0$



$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\therefore \quad \sigma_n = \frac{500 + 500}{2} + \frac{500 - 500}{2}\cos 90 + 0$$

$$\sigma_n = 500 \text{ kPa}$$

1153. The maximum shear stress that can be induced in a member, which is subjected to an axial load is equal to

(a) maximum normal stress

(b) half of maximum normal stress

(c) twice the maximum normal stress

(d) thrice the maximum normal stress

GPSC Assistant Prof. 2016

Ans. (b) : $\tau_{max} = \frac{P}{2A}$

$$(\sigma_n)_{max} = \frac{P}{A}$$

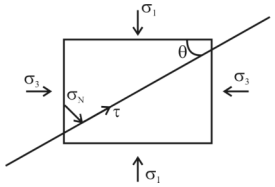
1154. The major and minor principal stresses are σ_1 and σ_3 respectively, acting within a body. A plane is inclined at an angle θ to the direction of the minor principal stress σ_3. then $\left(\frac{\sigma_1 - \sigma_3}{2}\right)$ sin 2θ is the:

(a) Compressive stress (b) Normal stress

(c) Shear stress (d) None of the above

PTCUL AE 2016

Ans. (c) : shear stress $(\tau) = \left(\frac{\sigma_1 - \sigma_3}{2}\right)\sin 2\theta$



1155. Principal stress of 30 MPa and –70 MPa acts on a material which has elastic limit stress in simple tension and compression as 60 MPa and 200 MPa respectively. Specify the reason for failure of the material assuming maximum principle stress theory.

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(a) Compression (b) Tension
(c) Unpredictable (d) None o the above

PTCUL AE 2016

Ans. (b) : $\sigma_1 = 70$ MPa $\sigma_3 = 30$ MPa
elastic limit $\rightarrow$ tension = 60 MPa
Compression = 200 MPa
According to maximum principal stress theory and given elastic limit, failure will be due to tension because $\sigma_1 > 60$ MPa

1156. In Mohr's diagram, a point above Mohr's envelope indicates
(a) Imaginary condition
(b) Imminent failure condition
(c) Safe condition
(d) Condition of maximum obliquity

UJVNL AE 2016

Ans. (a) : Point above Mohr's envelope indicate imaginary condition because failure had happened earlier of that.

1157. A shaft is subjected to maximum compressive stress of 20 N/mm^2 and maximum shear stress of 80 N/mm^2 due to torque. The maximum induced tensile stress is
(a) 70 N/mm^2 (b) 50 N/mm^2
(c) 60 N/mm^2 (d) 140 N/mm^2

UJVNL AE 2016

Ans. (d) : Maximum shear stress $\tau_{max} = 80$ N/mm^2
Compressive Stress (σ_2)
$= -20$N/mm^2 (due to compressive)

$$\tau_{max} = \left(\frac{\sigma_1 - \sigma_2}{2}\right)$$

$$80 = \frac{\sigma_1 - (-20)}{2}$$

$$80 = \frac{\sigma_1 + 20}{2}$$

$\sigma_1 = 140$ N/mm^2

1158. If an element is subjected to pure shearing stress τ_{xy}, then the maximum principal stress is equal to
(a) $\frac{\tau_{xy}}{2}$ (b) τ_{xy}
(c) $\sqrt{1-(\tau_{xy})^2}$ (d) $2\tau_{xy}$

UJVNL AE 2016

Ans. (b) : $\sigma_x = \sigma_y = 0$

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 = 0 + \sqrt{0 + \tau_{xy}^2}$$

$\sigma_1 = \tau_{xy}$

1159. The center of the Mohr's Circle for pure shear stress case lies
(a) at τ distance from origin
(b) at $\tau/2$ distance from origin
(c) at origin
(d) at $\tau\sqrt{2}$ distance from origin

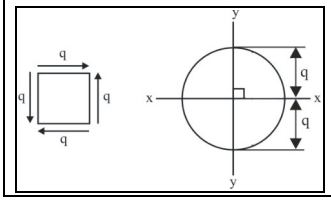
AP TRANSCO AEE 2017

Ans. (c) : For pure shear stress case, the centre of Mohr's circle lies at origin.

1160. Consider the following statements :
If the planes at right angle carry only shear stress of magnitude q in a certain instance, then the
(1) Diameter of Mohr's circle would be equal to 2q
(2) Center of Mohr's circle would lie at the origin
(3) Principal stresses are unlike and are of magnitude q each
(4) Angle between the principal plane and the plane of maximum shear would be 45^o
(a) 1, 2 and 3 only (b) 1, 2, and 4 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

AP TRANSCO AEE 2017,
APTRANSCO 2017, ESE 2014

Ans. (d) : If the planes at right angle carry only shear stress of magnitude q in a certain instance, then –
(i) Diameter of Mohr's circle would be equal to 2q
(ii) Center of Mohr's circle would lie at the origin
(iii) Principal stresses are unlike and are of magnitude q each
(iv) Angle between the principal plane and the plane of maximum shear would be 45^o



1161. A rectangular bar of cross sectional area A is subjected to an axial tensile load of P. The maximum shear stress will occur on a plane at X^0 to any normal cross-section. Where X^0
(a) 90^0 (b) 27^0
(c) 180^0 (d) 45^0

AP TRANSCO AEE 2017
APTRANSCO 2017, ESE 2013

Ans. (d) : Plane of maximum shear stress inclined 45^0 with the principal plane.

1162. If σ_1 and σ_2 are principal stresses, the shear stress on the principal planes is given by
(a) $\frac{\sigma_1 - \sigma_2}{2}$ (b) 0
(c) $\frac{\sigma_1 + \sigma_2}{2}$ (d) $\sigma_1 - \sigma_2$

APPSC (PHE) AE Paper-2 2012

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Ans. (b) : Principal plane are those planes where shear stress (τ_s) is zero. If principal stresses σ_1 and σ_2 are given

i.e, $\boxed{\tau_s = 0}$

1163. A plane stressed element is subjected to the state of stress given by $\sigma_x = \tau_{xy} = 100\ \text{kgf/cm}^2$ and $\sigma_y = 0$. Maximum shear stress in the element is equal to

(a) $50\sqrt{3}\ \text{kgf/cm}^2$ (b) $100\ \text{kgf/cm}^2$

(c) $50\sqrt{5}\ \text{kgf/cm}^2$ (d) $150\ \text{kgf/cm}^2$

APPSC (PHE) AE Paper-2 2012

Ans. (c) : Given ; $\sigma_x = \tau_{xy} = 100\ \text{kgf/cm}^2$

& $\sigma_y = 0$

Let, Maximum shear stress = τ_{max}

We know that in Mohr circle maximum stress (τ_{max}) is equal to the radius (R) of Mohr circle.

Hence,

$$\tau_{max} = R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{100-0}{2}\right)^2 + (100)^2} = \sqrt{(50)^2 + (100)^2}$$

$$= 50\sqrt{1+4} = 50\sqrt{5}\ \text{kgf/cm}^2$$

1164. The radius of Mohr's circle represents

(a) Minimum normal stress

(b) Maximum normal stress

(c) Minimum shear stress

(d) Maximum shear stress

APPSC AEE 2016

Ans. (d) : Radius of Mohr's circle represents the value of maximum shear stress.

• Normal stress on the plane of maximum shear stress is represented by coordinate of center of Mohr's circle.

1165. For an inclined plane in a rectangular block subjected to two mutually perpendicular normal stresses 1000 MPa and 400 MPa and shear stresses 400 MPa, the maximum normal stress will be

(a) 1200 MPa (b) 700 MPa

(c) 600 MPa (d) 200 MPa

APPSC AEE 2016

Ans. (a) : use of max. principal stress formula,

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{1000+400}{2} + \sqrt{\left(\frac{1000-400}{2}\right)^2 + 400^2}$$

$$= 700 + \sqrt{300^2 + 400^2} = 700 + 500 = 1200\ \text{MPa}$$

1166. At any point in a stressed body, the principal planes can be located by identifying a set of orthogonal planes passing through the point and having shear stress on it as

(a) zero (b) maximum

(c) minimum (d) anticlockwise

GPSC AE January 2018

Ans. (a) : At any point in a stressed body, the set of orthogonal planes passing through point is observed and according to these planes, principle planes can be located and principle planes have zero shear stress.

1167. Complimentary shear stresses are the shear stresses acting on a set of planes which are

(a) having zero normal stress

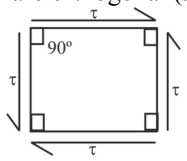
(b) having maximum normal stress

(c) at an angle of 45° to each other

(d) orthogonal to each other

GPSC AE January 2018

Ans. (d) : Complementary shear stress will be acting on a set of plane which are orthogonal (90°) to each other.



τ = shear stress

1168. Which theory of failure will you use for aluminium components under steady loading?

(a) Principal stress theory

(b) Principal strain theory

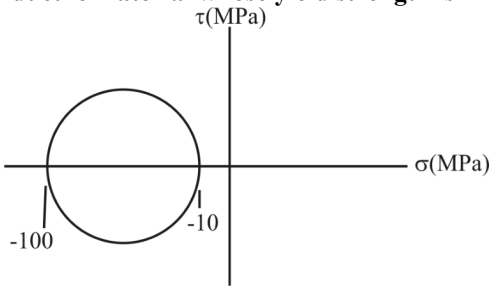
(c) Strain energy theory

(d) Maximum shear stress theory

UPRVUNL AE 2015

Ans. (d) : This theory is well justified for ductile material.

1169. The Mohr's circle of plane stress for a point in a body is shown. The design is to be done on the basis of the maximum shear stress theory for yielding will just begin if the designer chooses a ductile material whose yield strength is



(a) 45 MPa (b) 50 MPa

(c) 90 MPa (d) 100 MPa

UPRVUNL AE 2015

Ans. (c) : Yield strength of ductile material :

$\left[\sigma_1 - \sigma_2\right] \le \left[\frac{\sigma_y}{\text{F.O.S}}\right]$ for design on the basis of maximum stress theory. σ_1 and σ_2 are maximum and minimum principal stresses respectively.

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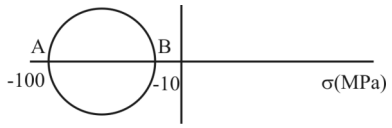

$\sigma_1 = -100$ MPa
$\sigma_2 = -10$ MPa

$$\frac{\sigma_y}{FOS} \geq (\sigma_1 - \sigma_2)$$
$$\geq (-100 - (-10)) = -90\,MPa = 90\,MPa\,(C)$$

1170. In a two-dimensional problem, if the state of stress at a point indicates the principal stresses as σ_1=10 MPa and σ_2=10 MPa, then the maximum shear stress at that point is _____.
(a) 10 MPa (b) 15 MPa
(c) 5 MPa (d) Zero

UPPCL AE 2018

Ans. (d) : Principal stress, $\sigma_1 = 10$ MPa
$\sigma_2 = 10$ MPa

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{10-10}{2} = 0$$

1171. At an internal hinge _____.
(a) slope is always discontinuous
(b) deflection is always discontinuous
(c) bending moment is always discontinuous
(d) shear force is always discontinuous

UPPCL AE 2018

Ans. (a) At an internal hinge slope is always discontinuous since slope changes its sign.

1172. Maximum principal stress theory is applicable to :
(a) Both ductile and brittle materials
(b) Elasto plastic materials
(c) Ductile materials only
(d) Brittle materials only

UPPCL AE 2019

Ans. (d) Maximum principal stress theory is applicable is Brittle material only.
Maximum shear stress theory is application to ductile material.

1173. In the Mohr's circle, the maximum shear stress τ_{max} is equal to $\frac{\sigma_1 - \sigma_2}{2}$, and it occurs in plane with an angle of α as:
(a) 60° (b) 45°
(c) 90° (d) 30°

LMRC AM 2018

Ans. (b) : $\because \tau = \left(\frac{\sigma_1 - \sigma_2}{2}\right)\sin 2\theta$ [Formula]

For maximum value of shear stress
$1 = \sin 2\theta$
$\sin 90° = \sin 2\theta$
$2\theta = 90°$
$\boxed{\theta = 45°}$

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

1174. For a sandy soil, the angle of internal friction is 30°. If the major principal stress at failure is 50 kN/m^2, then the corresponding minor principal stress (in kN/m^2) is
(a) 12.2 (b) 16.66
(c) 20.8 (d) 27.2

WBPSC AE 2016

Ans. (b) :

1175. The major and minor principal stresses on a two dimensional element are 10 MPa and - 6 MPa respectively. The maximum shear stress on the element is:
(a) 8 MPa (b) 11.7 MPa
(c) 10 MPa (d) 16 MPa

CIL MT 2020

Ans. (a) :
$\sigma_1 = 10$ MPa, $\sigma_2 = -6$ MPa

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$
$$= \frac{10 - (-6)}{2} = \frac{16}{2}$$
$$\tau_{max} = 8\ MPa$$

1176. The plane which pass through the point in such a manner that the resultant stress across them is totally a normal stress is called :
(a) Minor axis plane (b) Principal plane
(c) Major stress plane (d) Major axis plane
(e) All of the other options

CGPSC AE 2017

Ans. (b) The plane which pass through the point in such a manner that the resultant stress across them is totally a normal stress is called Principal plane.

1177. According to Mohr's circle, the shear stress is maximum when :
(a) $\theta = 45^0$ (b) $\theta = 90^0$
(c) $\theta = 180^0$ (d) $\theta = 0^0$
(e) $\theta = 120^0$

CGPSC AE 2017

Ans. (a) According to Mohr's circle, the shear stress is maximum when $\theta = 45^0$.

1178. In theories of failure, which theory states that no shearing stresses and shearing strains will be present anywhere in the block but only the volume changes?
(a) Guest theory
(b) St Venant theory
(c) Prandl's theory
(d) Distortion Energy theory
(e) Bohr's theory

CGPSC AE 2017

Ans. (d) Distortion energy theory (mises - henky theory) states that no shearing stresses and shearing strains will be present anywhere in the block but only the volume changes.
Mises : Henkey theory does not apply to brittle material.

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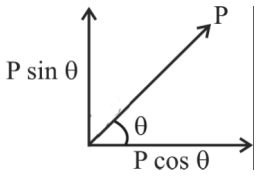
1179. The normal component of a forced inclined through θ is obtained by multiplying the force by:

(a) sin θ (b) cos θ
(c) tan θ (d) sin θ cos θ
(e) sec θ

Haryana PSC AE 2018 Paper–I

Ans. (d) :



Normal component can be obtained by sinθ cos θ.

1180. Maximum strain theory for the failure of a material at the elastic limit is known as:

(a) Haig's theory
(b) Guest's or Tresca's theory
(c) Rankine's theory
(d) Saint Venant's theory
(e) Newmark's theory

Haryana PSC AE 2018 Paper–I

Ans. (d) :

Theory	Given by	Remark
Maximum Principal Stress or Maximum Normal Stress	Rankine	Suitable for Brittle
Maximum Principal Strain	St. Venant	Neither accurate for Brittle nor Ductile
Maximum shear Stress	Guest and Treseca	Suitable for Ductile
Maximum Strain Energy	Haigh and Beltrami	Ductile
Maximum shear Strain Energy	Vonmises and Henky	Ductile

1181. What does the Williot - Mohr diagram yield?

(a) Forces in members of a truss
(b) Moments in a fixed beam
(c) Reactions at the supports
(d) Joint displacement of a pin jointed frame

Haryana PSC AE 2017

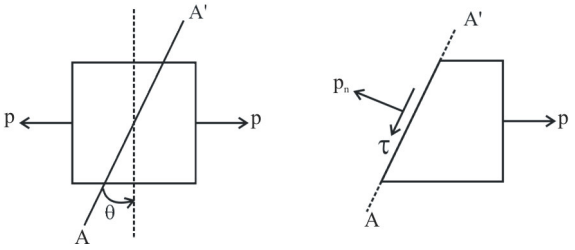
Ans. (d) : The Williot - Mohr diagram yield the resultant deflection of all joints by a single solution.

1182. A body is subjected to a tensile stress (p), what is the value of the normal stress on an oblique plane at an angle θ to the cross-section of the body?

(a) $p \sin 2\theta/2$ (b) $p \cos\theta$
(c) $p \cos^2\theta$ (d) $p \sin^2\theta$

UPRVUNL AE 2014

Ans. (c) :



Normal stress at Plane A-A'

$$p_n = p \cos^2\theta$$

Shear stress at Plane A-A'

$$\tau = \frac{p}{2} \sin 2\theta$$

1183. For ductile material, the suitable failure theory is

(a) Maximum shear strain energy theory
(b) Maximum shear stress theory
(c) Both (a) and (b)
(d) None of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : Failure Criterion Theory
(1) For brittle material
(i) Maximum principal stress **OR** Rankine's theory
(ii) Maximum principal strain **OR** St. Venant's theory
(iii) Maximum strain energy density
(2) Ductile material
(i) Maximum shear stress **OR** Guest and Tresca's theory
(ii) Maximum distortion energy density **OR** Vonmises and Hencky's theory

1184. A plate in plane stress condition at the onset of yielding has stress components as σ_x = 80 MPa, σ_y = (–) 40MPa and τ_{xy} = 25 MPa. Yield strength in simple tension test as per Mises criterion is :

(a) Y = 65 MPa (b) Y = 73.65 MPa
(c) Y = 81.65 MPa (d) Y = 114.3 MPa

HPPSC Poly. Lect. 2016

Ans : (a) Given,
$\sigma_x = 80$ MPa, $\sigma_y = -40$ MPa
$\tau_{xy} = 25$ MPa

$$\text{Maximum shearing stress} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{80-(-40)}{2}\right)^2 + 25^2} = 65 \text{ MPa}$$

1185. The maximum shear stress on an element under plane stress is 10 MPa and the maximum principal stress is 10 MPa. The minimum principal stress is :

(a) 10 MPa (b) 20 MPa
(c) –10 MPa (d) –20 MPa

HPPSC Poly. Lect. 2016

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Ans : (c) $\tau_{max} = \dfrac{\sigma_{major} - \sigma_{minor}}{2}$

$10 = \dfrac{10 - \sigma_{minor}}{2}$

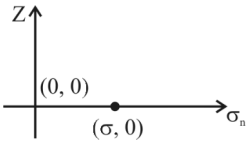
$\sigma_{minor} = -10\,MPa$

1186. If the mohr's circle for stresses reduces to a point, a body is subjected to :

(a) pure shear stress
(b) uniaxial stress only
(c) equal axial stresses on two mutually perpendicular planes and the planes are free of shear stresses
(d) equal and opposite axial stresses on two mutually perpendicular planes and planes are free of shear stresses

HPPSC Poly. Lect. 2016

Ans : (c) If the mohr's circle for stresses reduce to a point, a body is subjected to equal axial stresses on two mutually perpendicular planes and planes are free of shear stresses.



1187. A cross-section of a body is acted upon by a stress, intensity 'p'. The tangential stress 'pt' acting one a plane inclined at an angle 0 to the direction of applied stress is given by

(a) $pt = p.\sin^2\theta$ (b) $pt = p.\dfrac{\sin^2\theta}{2}$

(c) $pt = p.\sin^2\theta$ (d) $pt = p.\dfrac{\sin 2\theta}{2}$

WBPSC AE (Common) 1998
WBPSC AE 2007

Ans. (d) : A cross-section of a body is acted upon by a stress, intensity 'p'. The tangential stress 'pt' acting one a plane inclined at an angle θ to the direction of applied stress is given by $pt = p.\dfrac{\sin 2\theta}{2}$.

1188. The principal plane of a stressed body is the plane where

(a) normal stress and tangential stress the maximum
(b) there is no normal stress
(c) there is no tangential stress
(d) magnitude of normal stress is equal to that of tangential stress

WBPSC AE (Common) 1998

Ans. (c) : The principal plane of a stressed body is the plane where there is no tangential stress.

1189. At a point in elastic medium normal stresses in two mutually perpendicular directions are 120 MPa, 40 MPa (both tensile) associated with a tangential stress of 30 MPa. The principal stresses at the locations are _____.

(a) 120 MPa, 40 MPa (both tensile)
(b) 130 MPa, 30 MPa (both tensile)
(c) 130 MPa, 30 MPa (both compressive)
(d) 130 MPa (tensile), 30 MPa (compressive)

TSPSC Deputy Engineer 2015

Ans. (b) : $\sigma_x = 120$ MPa, $\sigma_y = 40$ MPa, $\sigma_{xy} = 30$ MPa

$$\sigma_{1/2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

$$= \frac{120 + 40}{2} \pm \sqrt{\left(\frac{120 - 40}{2}\right)^2 + (30)^2}$$

$$= 80 \pm \sqrt{(40)^2 + (30)^2}$$

$$= 80 \pm 50$$

$\sigma_1 = 130$ MPa

$\sigma_2 = 30$ MPa

1190. The radius of Mohr's circle of stress is 35 MPa and minor principle stress is –20 MPa. The major principal stress is:

(a) 15 MPa (b) 50 MPa
(c) 60 MPa (d) 95 MPa

TSPSC AEE 2017

Ans. (b) : $R = \tau_{max} = 30$ MPa, $\sigma_2 = -20$ MPa

$R = \tau_{max} = \dfrac{\sigma_1 - \sigma_2}{2}$

$35 = \dfrac{\sigma_1 - (-20)}{2}$

$70 = \sigma_1 + 20$

$\sigma_1 = 50$ MPa

1191. According to the maximum shear stress failure criterion, yielding in material occurs when

(a) Maximum shear stress = yield stress
(b) Maximum shear stress = half of yield stress
(c) Maximum shear stress = 2 × yield stress
(d) Maximum shear stress just exceeds the yield stress

GPSC AE (CLASS 1 & 2) 2019
ESE 1995

Ans : (b) : $\tau_{max} \simeq \dfrac{\sigma_y}{2}$ (Maximum shear stress theory) or (Guest and Tresca theory)

$\dfrac{\sigma_1 - \sigma_2}{2} \le \dfrac{\sigma_1}{2}$

1192. The angle between maximum shear strain axis and principal strain axis is

(a) 0^0 (b) 30^0
(c) 45^0 (d) 90^0

GPSC AE June 2019
ESE 2007

Ans. (c) : The angle between maximum shear strain axis and principal strain axis is 45º.

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1193. Maximum strain energy theory of failure was postulated by

(a) Haigh (b) Rankine
(c) Tresca (d) St. Venant

GPSC AE Dec 2018

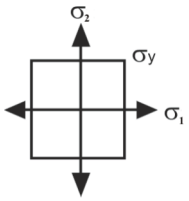
Ans. (a) : Maximum strain energy theory of failure was postulated by Haigh. According to this theory, a body under complex stress fails when the total strain energy on the body is equal to the strain energy at elastic limit in simple tension.

1194. According to the maximum principal stress theory, the yield locus is a/an

(a) square (b) circle
(c) hexagon (d) ellipse

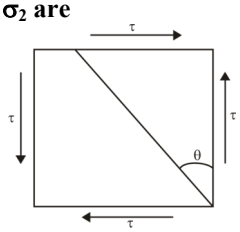
APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : As per Maximum principal stress theory



yielding locus is a square.

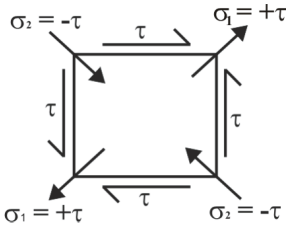
1195. The state stress at a point is shown below. θ represents the principal plane corresponding to principal stress σ_1 and σ_2 ($\sigma_1 > \sigma_2$). Values of θ, σ_1, and σ_2 are



(a) 0°, 90°, τ and -τ (b) 30°, 120°, τ and -τ
(c) 45°, 135°, τ and -τ (d) 45°, 135°, $\frac{\tau}{2}$ and $-\frac{\tau}{2}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) :



Due to pure shear diagonal tension ($\sigma_1 = +\tau$) and diagonal compression ($\sigma_2 = -\tau$) develops.
The angle between principal planes is 90°.
$\therefore$ $\theta_1 = 45°, \theta_2 = 135°, \sigma_1 = \tau, \sigma_2 = -\tau$

1196. At a material point the principal stresses are σ_1 = 100 MPa and σ_2 = 20 MPa. If the elastic limit is 200 MPa, what is the factor of safety based on maximum shear stress theory?

(a) 1.5 (b) 2
(c) 2.5 (d) 3

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : $\sigma_1 = 100$ MPa
$\sigma_2 = 20$ MPa
$\sigma_3 = 0$ (Minimum principal stress)
$\sigma_y = 200$ MPa
According to Maximum Shear Stress theory

$$\tau_{max} = \frac{\sigma_y}{2FOS}$$

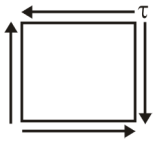
$$\Rightarrow \quad \frac{\sigma_1 - \sigma_3}{2} = \frac{\sigma_y}{2FOS}$$

$$\Rightarrow \quad \sigma_1 - \sigma_3 = \frac{\sigma_y}{FOS}$$

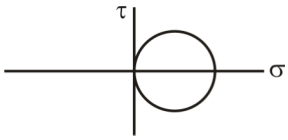
$$\Rightarrow \quad 100 - 0 = \frac{200}{FOS}$$

$$FOS = \frac{200}{100} = 2$$

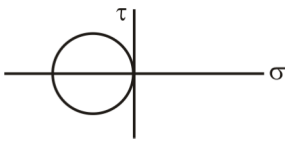
1197. Which of the following represents the Mohr's circle for the state of stress shown below?



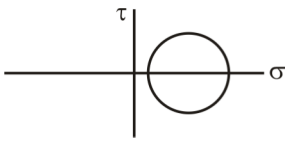
(a)



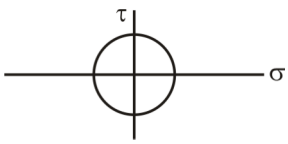
(b)



(c)



(d)



APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Under pure shear condition, centre of Mohr's circle coincides with origin.

1198. The tangential (or) shear stress on an oblique plane at an angle (θ) to the cross section of a body which is subjected to a direct tensile stress (σ) is equal to :

(a) $\sigma \cos^2 \theta$ (b) $\sigma \cos \theta$

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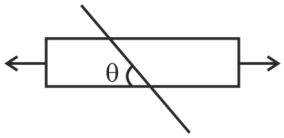
(c) $\frac{\sigma}{2}\sin 2\theta$ (d) $\sigma \sin^2 \theta$

TRB Poly. Lect. Tech. 2017

Ans. (c) :



Tangential stress on a plane inclined at an angle θ from major principle plane $= \frac{\sigma}{2}\sin 2\theta$

1199. Under uniaxial loading, the maximum shear stress is ..... times the uniaxial stress

(a) 0.5 (b) 2
(c) 1.5 (d) 3

GPSC AE Dec 2018

Ans. (a) : For uniaxial loading, $\sigma_2 = 0,\ \tau_{xy} = 0$

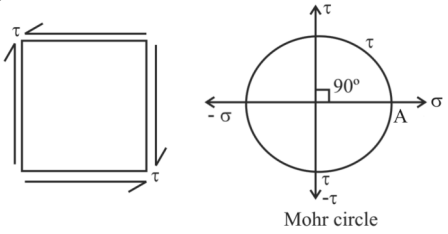
$$\tau_{max} = \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau_{xy}^2} \Rightarrow \boxed{\tau_{max} = \frac{\sigma}{2}}$$

1200. If a body is acted upon by pure shear stresses on two perpendicular planes, the planes inclined at 45° are subjected to no ..... stress.

(a) tensile (b) compressive
(c) shear (d) bending

GPSC AE Dec 2018

Ans. (c) :



Mohr circle

Plane A is a plane inclined by 45° and here shear stress is zero.

1201. Normal stress on a oblique plane inclined at angle 45° to the axis of a bar of square cross-section of side S when acted upon by a tensile force P is

(a) P/S^2 (b) $2P/S^2$
(c) $P/2S^2$ (d) $P/4S^2$

GPSC AE Dec 2018

Ans. (c) :

$$\sigma_n = \left(\frac{\sigma_1 + \sigma_2}{2}\right) + \left(\frac{\sigma_1 - \sigma_2}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

For uniaxial, $\sigma_2 = 0,\ \tau_{xy} = 0,\ \sigma_1 = \frac{P}{S^2}$

$$\sigma_n = \frac{P/S^2}{2}$$

$$\boxed{\sigma_n = \frac{P}{2S^2}}$$

1202. The 2 D stress at a point is given by the matrix $\begin{bmatrix}\sigma_{xx} & \tau_{xy}\\ \tau_{yx} & \sigma_{yy}\end{bmatrix} = \begin{bmatrix}50 & 15\\ 15 & 10\end{bmatrix}$ MPa. The maximum shear stress in (MPa) units is :

(a) 45 (b) 55
(c) 25 (d) 30

BHEL ET 2019

Ans. (c) :

$$\begin{bmatrix}\sigma_{xx} & \tau_{xy}\\ \tau_{yx} & \sigma_{yy}\end{bmatrix} = \begin{bmatrix}50 & 15\\ 15 & 10\end{bmatrix}\text{MPa}$$

$\sigma_{xx} = 50\,\text{MPa},\quad \sigma_{yy} = 10\,\text{MPa}$

$$\tau_{max} = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \left(\tau_{xy}\right)^2}$$

$$= \sqrt{\left(\frac{50-10}{2}\right)^2 + (15)^2}$$

$$= \sqrt{400 + 225}$$

$$= \sqrt{625} = 25\,\text{MPa}$$

1203. Factor of safety is defined as the ratio of

(a) $\frac{\text{working stress}}{\text{ultimate stress}}$ (b) $\frac{\text{ultimatel load}}{\text{design safe load}}$

(c) $\frac{\text{ultimate stress}}{\text{working stress}}$ (d) $\frac{\text{design safe load}}{\text{ultimateload}}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Factor of safety $= \frac{\text{uttimate stress}}{\text{working stress}}$

1204. The ratio of ultimate stress to the permissible stress is called .....

(a) Factor of safety (b) Design factor
(c) Stress factor (d) Load factor

GPSC AE January 2018

Ans. (a) :

$$\text{Factor of safety} = \frac{\text{Ultimate stress}}{\text{Permissible stress}}$$

$$\boxed{FOS = \frac{\sigma_u}{\sigma}}$$

1205. Shear stress energy is called as:

A. Distortion theory
B. Von Mises theory

(a) Only A (b) Neither A nor B
(c) Both A and B (d) Only B

UPPCL AE 2019

Ans. (c) : Shear strain energy theory is also known by the following names respectively.
(i) Distortion theory
(ii) Van-miser theory
(iii) Henky theory
This theory is suitable for mild steel or ductile material.

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7. Columns and Struts

1206. Effective length of a column of length 'L' with one end fixed and other end hinged is given by:

(a) $Le = L$ (b) $Le = L/2$

(c) $Le = L/\sqrt{2}$ (d) $Le = 2L$

TNPSC AE 2015

Ans. (c) :

end condition	**effective length**
Both ends are fixed	L/2
One end fixed and other hinge	$L/\sqrt{2}$
Both ends are hinge	L
One end fixed and other free	2L

1207. During earthquakes, the corner and edge of columns may be subject to:

(a) uniaxial bending
(b) biaxial bending
(c) combined biaxial bending and tension
(d) combined biaxial bending and torsion

APPSC Poly. Tech. Lect. 2020

Ans. (d) : During earthquakes, the corner and edge columns may be subject to combined biaxial bending and torsion.

1208. A column may be called as short column if the slenderness ratio is :

(a) Between 13 and 18 (b) more than 24

(c) Between 20 and 24 (d) Less than 12

Rajasthan AE (Nagar Nigam) 2016

Ans. (d) : A column may be called as short column if the slenderness ratio is < 12.

1209. The product of slenderness ratio and least radius of gyration gives_____.

(a) Length of column
(b) Width of column in X direction
(c) Area of column
(d) width of column in Y direction

LMRC AM 2019

Ans. (a) The product of slenderness ratio and least radius of gyration gives length of column.

slenderness ratio $(\lambda) = \frac{l_{eff}}{\gamma_{min}}$

Where,

l_{eff} = Effective length of column

γ_{min} = Minimum radius of gyration

1210. Slenderness ratio of a member is defined as the ratio of the effective length to the _____. of a section:

(a) moment of inertia (b) radius of gyration

(c) thickness (d) area

KPSC AE 2020

Ans. (b) : Slenderness ratio of a member is defined as the ratio of the effective length to the radius of gyration of a section.

1211. $P = \frac{\pi^2 EI}{L^2}$ is the equation for Euler's crippling load for a column if its

(a) both ends are fixed
(b) both ends are hinged
(c) one end is fixed and other end is free
(d) none of the above

UKPSC AE 2012 Paper–I

Ans : (b) Euler's crippling equation is used for columns with slenderness ratio between 30 to 120.

Euler's formula for different end conditions

(1) Both ends fixed $= \frac{4\pi^2 EI}{L^2}$

(2) Both ends hinged $= \frac{\pi^2 EI}{L^2}$

(3) One end free, other fixed $= \frac{\pi^2 EI}{4L^2}$

(4) One end fixed, other hinged $= \frac{2\pi^2 EI}{L^2}$

1212. The load at which an element, a member or a structure as a whole, either collapses in service or buckles in a load test and develops excessive lateral (out of plane) deformation or instability is called as :

(a) Buckling load (b) Yielding load

(c) Eccentric load (d) Failure load

RRB SSE (Shift-I) Online 03.09.2015

Ans : (a) The load at which an element, a member or a structure as a whole, either collapses in service or buckles in a load test and develops excessive lateral deformation or instability is called as buckling load

Euler's buckling load $\boxed{(P_e) = \frac{\pi^2 EI}{L_{eff}^2}}$

E = Modulus of elasticity

I = Moment of inertia

L_e = Equivalent length

1213. Rankine's formula is used for the analysis of steel compression member if the slenderness ratio $\left(\frac{l}{r}\right)$ is–

(a) $120 < \frac{l}{r} < 200$ (b) $200 < \frac{l}{r} < 250$

(c) $100 < \frac{l}{r} < 150$ (d) None of above

UKPSC AE 2013 Paper–I

Ans : (d) Rankine formula is used for both short and long column. So it can have any slenderness ratio.

1214. Column or strut is a compression member, the effective length of which exceeds X times the least lateral dimension: where X is

(a) 12 (b) 6

(c) 3 (d) 9

Ans : (c) Column or strut are compression member whose effective length is 3 times the least lateral dimension.

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1215. The slenderness ratio of column is zero when

(a) Its length is exactly equal to the radius of gyration
(b) Its length is half the radius of gyration
(c) Total load carried is less than half the dead weight
(d) If the length is supported on all sides throughout its length

RPSC POLY. TECH. LECT. 2011

Ans : (d) Slenderness ratio of column is zero if it is supported on all side through out its length.

1216. Four vertical columns of the same material, height and weight have the same end conditions. The buckling load will be the largest for column having the cross-section of:

(a) Solid square (b) Thin hollow circle
(c) Solid circle (d) H-section

UPPSC State Eng. A.E.2003 & 2008
UPPSC AE 12.04.2016 Paper–I

Ans : (b) Buckling load $(P_e) \cdot \frac{\pi^2 EI}{L_{eff}^2}$

Thin hollow circular sections are most efficient in resisting compression because their moment of inertia is maximum.

1217. The failure of a column depends upon-

(a) weight of a column
(b) length of column
(c) slenderness ratio
(d) cross sectional area of column

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) : Failure of columns generally occurs due to buckling. If the slenderness ratio is high, chances of buckling increases.

1218. If the load on a column is increased to a value that on its removal the deflection remains, the load is known as-

(a) Critical load (b) Crippling load
(c) Buckling load (d) All of these

Maharashtra PSC (Mains) 2015 Paper–I

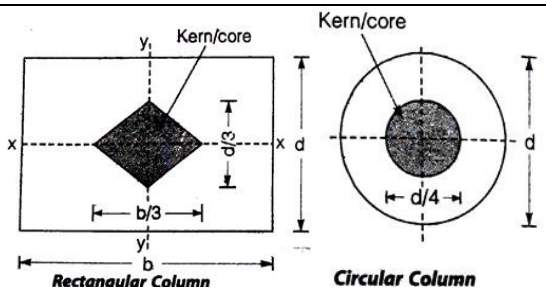
Ans. (d) : If on removal of the load, deflection still remains, it means that the column has already yielded and hence the load is critical load. It may also be crippling or buckling load since they also corresponds to failure of the column.

1219. Kern of rectangular columns is having _____ shape-

(a) rectangular (b) square
(c) diamond (d) triangular

BPSC AE Pre 2018

Ans. (c) : Kern – The kern of a section is the region in which compressive point load may be applied without producing any tensile stress on the cross section



To prevent any kind of stress reversal, force applied should be within an area near the cross section called as CORE or KERN. Shape of kern for rectangular and I-section is Rhombus and for square section shape is square for circular section shape is circular.

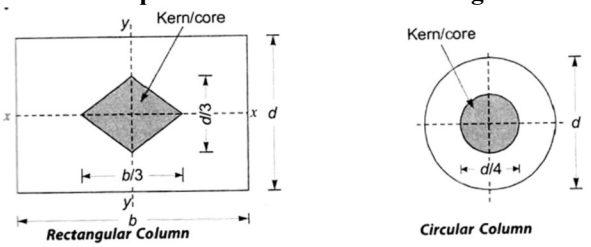
1220. Which one of the following is kern diameter of a circular column of diameter, d ?

(a) d/2 (b) d/4
(c) d/6 (d) d/8

Rajasthan JEN (Degree) Shift - II 2016

Ans. (b) : Kern – The kern of a section is the region in which compressive point load may be applied without producing any tensile stress on the cross section

Shape of kern in eccentric loadings



To prevent any kind of stress reversal, force applied should be within an area near the cross section called as CORE or KERN.

1221. Consider following for a short column :

(i) fails primarily due to crushing
(ii) Ultimate load capacity depends upon its length
(iii) Yields stress governs crushing strength
Which one of the following is correct answer ?

(a) (i) and (iii) only (b) (ii) and (iii) only
(c) (i) only (d) (i) and (ii) only

Rajasthan JEn (Degree) 2016

Ans. (a) Short column have slenderness ratio <12. They fail due to crippling not by buckling and there crushing strength is govern by their yield stress.

1222. A column that fails essentially by direct crushing at ultimate load is called

(a) Euler's column
(b) long column
(c) short column
(d) None of the given answers

APPSC AEE MAINS 2016

Ans. (c) : A column that fails essentially by direct crushing at ultimate load is called short column

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1223. Euler's buckling formula is applicable to columns :
(a) Subjected to eccentric load
(b) having initial curvature
(c) initially straight and subjected to only axial loads
(d) all of the above

PSPCL AE 2012

Ans. (c) : The Euler's buckling formula is applicable –
- When material isotropic, homogenous & linear elastic in which hook's law is applied.
- Before loading column is perfectly straight
- The column is long one not applicable in short column as short column fails in crushing
- Column fails only by buckling
- The load is perfectly axial.

1224. Euler's formula for a mild steel long column hinged at both ends is not valid for slenderness ratio
(a) greater than 80 (b) less than 80
(c) greater than 180 (d) greater than 120

Punjab PSC Sub divisional Engineer 2014

Ans. (b) : Euler's formula use for long column and not valid for slenderness ratio less than 80.

1225. Column of given length, cross-section and material property has different values of buckling loads for different end condition. The strongest column is one whose :
(a) One end if fixed and the other end is hinged
(b) Both the ends are hinged
(c) One end is fixed and the other end is entirely free
(d) Both the ends are fixed

OPSC Civil Services Pre 2011

Ans. (d) : For same length, cross-section and material property both ends are fixed is strongest column because buckling load according to Euler's formula is maximum for both end fixed condition.

1226. Slenderness ratio of a member is defined as the ratio of
(a) effective length to radius of gyration
(b) effective length to section modulus
(c) effective length to young's modulus
(d) effective length to radius of curvature

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (a) : Slenderness ratio is the ratio of length of a column and the least radius of gyration of its cross section.

1227. Strut is a
(a) tension member
(b) pin-jointed rigid frame
(c) compression member
(d) bending member

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (c) : Strut is a compression member use to maintain rigidity of structure and fails due to buckling with high slenderness ratio.

1228. The Perry-Robertson's formula considers :
(a) crushing mode
(b) initial curvature
(c) initial eccentricity and curvature
(d) buckling mode

APPSC Poly. Tech. Lect. 2020

Ans. (c) : The Perry-Robertson proof is based on the assumption that any imperfections in the strut, through faulty material or eccentricity of loading, can be allowed for by giving the strut an initial curvature.

1229. Cross sections which can develop their plastic moment resistance, but have inadequate plastic hinge rotation capacity because of local buckling called as:
(a) Semi compact section (b) Plastic section
(c) Compact section (d) Slender section

IRCON AE 2017

Ans. (c) : If a beam has very small slenderness ratio and the section with capacity of attaining its plastic moment at time of loading, is known as compact section.

1230. Base of the chimney subject :
(a) Direct stress due self-weight
(b) bending stress due wind load
(c) both (a) and (b)
(d) bending stress due self-weight direct stress due wind load

UP Jal Nigam AE 2016

Ans. (c) : Base of the chimney subject :
(a) Direct stress due self-weight
(b) Bending stress due wind load

1231. Short column fails primarily by -
(a) buckling or excessive lateral bending
(b) crushing without buckling
(c) buckling or crushing
(d) combination of crushing and buckling

Rajasthan VP ITI 2018

Ans. (b) : Short column fails primarily by crushing without buckling.

1232. The slenderness ratio of a column is zero when its length:
(a) Is zero
(b) Is equal to its radius of gyration
(c) Is supported on all side throughout its length
(d) Is between the points zero moments
(e) None of the above options

Haryana PSC AE 2018 Paper–I

Ans. (c) : Slenderness ratio of column is zero when its length is supported on all side throughout its length.

1233. If the depth of the section of an upper column is smaller than the lower column:
(a) Filler plates are provided with column splice
(b) Bearing plates are provided with column splice
(c) Neither filler plates nor bearing plates are provided with column splice
(d) Filler plates and bearing plates are provided with column splice
(e) None of the above options

Haryana PSC AE 2018 Paper–I

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Ans. (e) : If the depth of the section of an upper column is smaller than the lower column, filler plates and bearing plates are provided with column splice.

1234. Slenderness ratio of a long (slender) column is:

(a) Area of cross-section divided by radius of gyration
(b) Area of cross-section divided by least radius of gyration
(c) Radius of gyration divided by area of cross-section
(d) Length of column divided by area of cross-section
(e) Length of column divided by least radius of gyration

Haryana PSC AE 2018 Paper–I

Ans. (e) : Slenderness ratio $(\lambda) = \frac{l_{eff}}{r_{min}}$

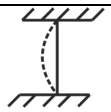
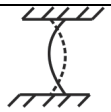
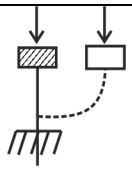
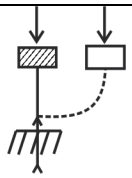
l_{eff} = Effective length of column.
r_{min} = least radius of gyration

1235. The effective length (*l*) for an equivalent column of overall length L fixed at one end and free at the other end is:

(a) 0.5 L (b) 0.7 L
(c) L (d) 1.5 L
(e) 2 L

Haryana PSC AE 2018 Paper–I

Ans. (e) :

Degree of End Restraint of compression member	Effect Length	Symbol
Effectively held in position and restrained against rotation at both ends	0.65 L	
Effectively held in position at both ends, but not restrained against rotation	1.00 L	
Effectively held in position and restrained against rotation at one end and at the other end partially restrained against rotation but not held in position	1.50 L	
Effectively held in position at one end but not restrained against rotation and at the other end restrained against rotation but not held in position	2.00 L	

1236. Which of the following types of stress is induced in the foundation bolts while fixing a column to its footing?

(a) Bearing (b) Pure compression
(c) Pure tension (d) Bending

UPRVUNL AE 2014

Ans. (c) : Pure tension stress is induced in the foundation bolts while fixing a column to its footing.

1237. Identify the correctly matched pair among the following

(a) Truss - Bending (b) Beam - Twisting
(c) Column - Buckling (d) Shaft - Shortening

UPRVUNL AE 2014

Ans. (c) :
- Truss carries axial, load and that can cause shortening.
- Beam carries force by bending
- Column resist force as, axial load which can cause buckling of column.
- Shaft carries force by twisting.

1238. The load at which an element, a member or a structure as a whole, either collapse in service or buckles in a load test and develops excessive lateral (out of plane) deformation or instability is called as:

(a) Yield load (b) Maximum load
(c) Buckling load (d) Plastic load

LMRC AM 2015

Ans (c) The web in a rolled steel section behaves like a column when placed under concentrated loads. The web is quite thin and is, therefore, subjected to buckling. The buckling of the column web is very much in fluenced by the restraints provided for flanges.

1239. Among the different modes of failures given, identify the one which is not applicable to a steel tension member:

(a) Yielding (b) Crippling
(c) Rupture (d) Block shear

CIL MT 2020

Ans. (b) : Crippling modes of failure one which is not application to a steel tension moments.

1240. Euler's formula is applicable for

(a) short column (b) long column
(c) medium column (d) All of the above

APPSC (PHE) AE Paper-2 2012

Ans. (b) : Euler's formula is applicable for long column.

1241. The ratio of equivalent length of the column to the maximum radius of gyration is called

(a) Poisson's ratio (b) Buckling factor
(c) Factor of safety (d) None of the above

APPSC (PHE) AE Paper-2 2012

Ans. (b) : Buckling factor is define as the ratio of equivalent length (L_e) of column to the maximum radius of gyration. It is also called slenderness ratio (S)

$S = \frac{L_e}{K}$

Where, K = radius of gyration

1242. Slenderness ratio of a column may be defined as the ratio of its effective length to the

(a) radius of column
(b) minimum radius of gyration
(c) maximum radius of gyration
(d) area of the cross-section

APPSC AEE (Civil/Mechanical) 2012

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Ans. (b) Slenderness Ratio (λ) : - Slenderness ratio of a compression member is defined as the ratio of its effective length to radius of gyration

$$\lambda = \frac{Le}{K}$$

L_e = Effective length

K = Least radius of gyration

$$K = \sqrt{\frac{I_{min}}{A}}$$

I_{min} = Minimum moment of Inertia about centroidal axis.

1243. Crippling load for a long column will be maximum when

(a) Both ends are hinged
(b) One end is hinged and other is fixed
(c) One end is fixed and other is free
(d) Both ends are fixed

Sikkim PSC AE 2017

Ans. (d) : As,

$$\boxed{P_e = \frac{\pi^2 EI_{min}}{le^2}}$$

L_e = min in case of when both end fixed, i.e.,

$$\boxed{L_e = \frac{L}{2}}$$

1244. The slenderness ratio of a vertical column of square cross-section of 2.5 cm sides and 300 cm effective length, is-

(a) 200 (b) 360
(c) 240 (d) 416

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (d) : Slenderness ratio $\lambda = \frac{l_{eff}}{r}$

$$r = \sqrt{\frac{I}{A}}; I = \frac{bd^3}{12} = \frac{2.5 \times 2.5^3}{12} = 3.255 \text{ cm}^4$$

$A = 2.5^2 = 6.25 \text{ cm}^2$

$$r = \sqrt{\frac{3.255}{6.25}} = 0.7216 \text{ cm}$$

$$\lambda = \frac{l_{eff}}{r} = \frac{300}{0.7216} = 415.74 \simeq 416$$

1245. The Euler's crippling load for a 2 m long slender steel rod of uniform cross-section hinged at both the ends is 1 kN. The Euler's crippling load for a 1 m long the long steel rod of the same cross-section and hinged at both the ends will be-

(a) 0.25 kN (b) 0.5 kN
(c) 2 kN (d) 4 kN

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (d) : $P_l = \frac{\pi^2 EI}{l^2}$

$$P_l \propto \frac{1}{l^2}$$

$$P_{l_2} = P_{l_1} \times \frac{l_1^2}{l_2^2}$$

$l_{effective} = 2\text{m}$

$p_l = 1\text{kN}$

$l_{effective} = 1 \text{ m}$

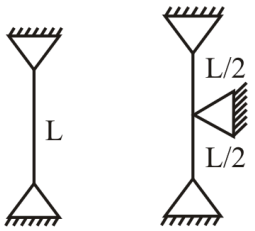
$$P_l = 1 \times \frac{2^2}{1^2} = 4 \text{ kN}$$

1246. A steel column is pinned at both ends and has a buckling load of 200 kN. If the column is restrained against lateral movement at its mid-height, its buckling load will be

(a) 100 kN (b) 200 kN
(c) 400 kN (d) 800 kN

ESE 2018

Ans. (d)



Pe = 200 kN, $\ell = L$ Pe = ?, $\ell = L/2$

From Euler's Formula

$$P_e = \frac{\pi^2}{L^2} EI \text{ and } P_e \propto \frac{1}{\ell^2}$$

$$\frac{(P_e)_2}{(P_e)_1} = \frac{(\ell_1)^2}{(\ell_2)^2} = \frac{(\ell)^2}{\left(\frac{\ell}{2}\right)^2}$$

$$\frac{(P_e)_2}{(P_e)_1} = \frac{4L^2}{L^2} = 4$$

$$(P_e)_2 = 4(P_e)_1$$

$$\boxed{(P_e)_2 = 4(200) = 800\text{kN}}$$

1247. A strut of length 5 m is fixed at one end and the other end is free. The Euler's buckling load is 50 kN. If both the ends of the strut are fixed, what will be the Euler's buckling load in kN?

(a) 500 kN (b) 250 kN
(c) 800 kN (d) 1000 kN

APTRASCO 2017

Ans : (c) : Euler's formula for crippling load

$$P = \frac{\pi^2 EI}{\ell_{eff}^2}$$

Case 1 – when one end fixed and other is free

$$P_1 = \frac{\pi^2 EI}{(2 \times 5)^2} = \frac{\pi^2 EI}{100} \text{kN}$$

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Case 2 – when both end fixed

$$P_2 = \frac{\pi^2 EI}{\left(\frac{5}{2}\right)^2} = \frac{4\pi^2 EI}{25}$$

$$\frac{P_1}{P_2} = \frac{25}{400} = \frac{1}{16}$$

$$P_2 = 16P_1 = 16 \times 50$$

$$\boxed{P_2 = 800\,kN}$$

1248. Radius of gyration of a circular section is

(a) $\frac{d}{\sqrt{12}}$ (b) $\frac{d}{\sqrt{18}}$

(c) $\frac{d}{\sqrt{24}}$ (d) $\frac{d}{4}$

TNPSC AE 2012

Ans. (d) : Radius of gyration (r) = $\sqrt{\frac{I}{A}}$

$$I = \frac{\pi d^4}{64}$$

$$A = \frac{\pi}{4} \times d^2$$

$$= \sqrt{\frac{\pi d^4}{64} \times \frac{4}{\pi d^2}} = \sqrt{\frac{d^2}{16}}$$

$$\boxed{\text{radius of gyration} = \frac{d}{4}}$$

1249. Euler's Buckling load for a column whose both ends are fixed, having a length of *l*, modulus of elasticity E and moment of inertia I is

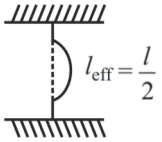
(a) $2\pi^2 EI/l^2$ (b) $4\pi^2 EI/l^2$

(c) $\pi^2 EI/4l^2$ (d) $\pi^2 EI/l^2$

TNPSC AE 2012, 19

ASSAM PSC CCE (PRE) 2015

Ans. (b)



P_e = Buckling loading = $\frac{\pi^2 EI}{l_e^2} = \frac{\pi^2 \times EI}{(l/2)^2}$

$$P_e = 4\pi^2 EI/l^2$$

1250. To avoid tension in a short column the load must be

(a) at outside of the cross section

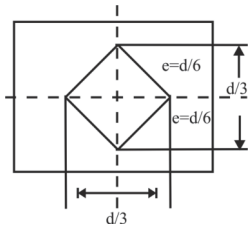
(b) within the middle-third of the cross-section

(c) within the one-third of the cross-section

(d) outside the middle-third of the cross-section

TNPSC AE 2019

Ans. (b)



$D = D_d + D_{bs} > 0$

$$\frac{P}{bd} - \frac{P.e}{\frac{Bd^2}{12}} \times \frac{d}{2} > 0$$

$$1 - \frac{e}{d/6} > 0 \Rightarrow e < d/6$$

1251. For keeping the stress wholly compressive the load may be applied on a circular column any where within a concentric circle of a diameter

(a) d/2 (b) d/3

(c) d/4 (d) d/8

TNPSC AE 2019

Ans. (c) We want tensile stress not to be happen in Column section or any compressive section because it very weak in tension so when any load is applied, it should within an eccentricity so that no tensile stress could occurs due to bending

$\sigma = \sigma_{direct} \pm \sigma_{bending} > 0$

$$= \frac{p}{\frac{\pi}{4}d^2} - \frac{p \times e_x}{\frac{\pi}{64} \times d^4} \times d/2 \Rightarrow 1 - \frac{e_x}{d/8} > 0$$

$$1 > \frac{e_x}{d/8}$$

$e_x < d/8$ (Radius)

Diameter = d/4

1252. If the crushing load of a column is 3000 kN and the Euler load is 2000 kN, them the Rankine load is :

(a) 12000 kN (b) 1200 kN

(c) 3000 kN (d) 2000 kN

(e) 1000 kN

CGPSC AE 2014 Shift–I

Ans. (b) : Rankine formula :

$$\frac{1}{P_R} = \frac{1}{P_C} + \frac{1}{P_E}$$ (Rankine's equation for failure)

$$= \frac{1}{3000} + \frac{1}{2000}$$

$$= \boxed{P_R = 1200\,kN}$$

1253. The buckling load of a column will be maximum, if its end conditions are as

(a) one end is fixed and other end is hinged

(b) both ends are pin-jointed

(c) one end is fixed and other end is free

(d) both ends are fixed

TNPSC AE 2012

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Ans. (d) :

$$P_e = \frac{\pi^2 EI}{(l_{eff})^2} = \frac{4\pi^2 EI}{l^2}$$

As per the formula buckling load inversely depends upon effective length of column. Shorter the effective length, higher will be the buckling load. For both end fixed beam effective length is $\frac{l}{2}$ for which buckling load will be maximum for any other given case.

1254. If the Euler load for a column is 1000 kN and crushing load is 1200 kN, then the Rankine load will approximately be :

(a) 900 (b) 1200
(c) 545 (d) 600

CIL MT 2017

Ans. (c) : Rankine's formula = $\frac{1}{P_R} = \frac{1}{P_C} + \frac{1}{P_E}$

Euler load (P_e) = 1000 kN
Crushing load (P_e) = 1200 kN
Rankine's load (P_R) = ?

$$\frac{1}{P_R} = \frac{1}{1200} + \frac{1}{1000}$$

$$\frac{1}{P_R} = \frac{1+1.2}{1200} = \frac{2.2}{1200}$$

P_R = 545 kN

1255. An allowable axial load 3m long column with hinged ends is 30 kN. Another column of same material. same cross-section and same length- but one end fixed and other end hinged-suffers buckling. What is the buckling load for the column?

(a) 60 kN (b) 30 kN
(c) 300 kN (d) 600 kN

APPSC Poly. Tech. Lect. 2020

Ans. (a) :
case-1

When both ends hinged

$$P_E = \frac{\pi^2 EI}{L^2} = 30kN$$

case-2

When one end fixed and other end hinged

$$P_E = 2\ \frac{\pi^2 EI}{L^2} = 2 \times 30kN$$

$$P_E = 60kN$$

1256. An electric pole 5m high is fixed into the foundation. It carries a wire at the top and is free to move sideways. The effective length of the pole is–

(a) 3.25 m (b) 4 m
(c) 5 m (d) 10 m

Gujarat PSC AE (N.W.R.) 2020
APPSC AEE (Mains) 2016

Ans. (d) : Case of one end fixed and another end free in air
So, effective length = 2L = 2 × 5 = 10 m

1257. A column, hinged at both ends is having an unsupported length of 3m, second moment of inertia I_{XX} = 400 cm^4, I_{YY} = 120 cm^4 and area of cross-section 30 cm^2. The slenderness ratio of the column is equal to :

(a) 150 (b) 120
(c) 130 (d) 82.2

KPSC AE 2020

Ans. (a) :
Given,

I_{xx} = 400 cm^4, I_{yy} = 120 cm^4, A = 30 cm^2

We know,

$$\lambda = \frac{\text{Effective length}}{\text{Least radius of gyration}} = \frac{l_e}{\gamma_{min}}$$

∴ The column is hinged at both ends

$l = l_{eff}$

l = 3m

$$\gamma_{min} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{I_{yy}}{A}} = \sqrt{\frac{120}{30}}$$

$$\gamma_{min} = \sqrt{4} = 2cm$$

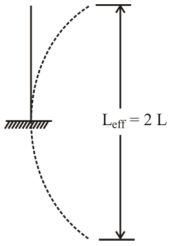
$$\lambda = \frac{l_e}{\gamma_{min}} = \frac{300}{2} = 150$$

1258. The effective slenderness ratio of a cantilever column is:

(a) $\frac{0.5L}{R}$ (b) $\frac{L}{R}$
(c) $\frac{\sqrt{2L}}{r}$ (d) $\frac{2L}{r}$

UPPSC AE 12.04.2016 Paper–I

Ans : (d) Cantilever column–



Slenderness Ratio (K) = $\frac{l_e}{r_{min}}$

$$\left[K = \frac{2l}{r}\right]$$

1259. The effective length of a column is the length between the points of :

(a) Supports
(b) maximum bending moments
(c) zero bending moments
(d) zero shear force

UPPSC State Eng. A.E. 2008

Ans : (c) The effective length of a column is the length between the points of zero bending moments, or the distance between points of contraflexure.

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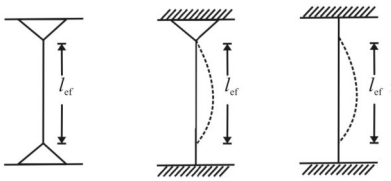


1260. Buckling load of a column fixed at lower end is given by :

(a) $\frac{\pi^2 EI}{4L^2}$ (b) $\frac{\pi^2 EI}{3L^2}$

(c) $\frac{\pi^2 EI}{2L^2}$ (d) $\frac{\pi^2 EI}{L^2}$

[UPPSC State Eng. A.E. 2007 (II)]

OR

What will be the Euler buckling load of a column when it is fixed at one and free at other end.

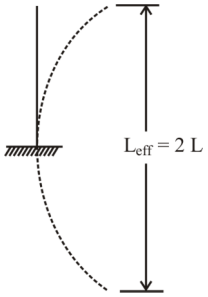
(a) $\pi^2 El/4l^2$ (b) $4\pi^2 El/l^2$

(c) $\pi^2 El/l^2$ (d) $2\pi^2 El/l^2$

DSSSB AE 2019

Rajasthan JEn (Degree) 2016

Ans : (a)



According to Euler's theory, the buckling load of a column is given by

$$P = \frac{\pi^2 EI}{l_{eff}^2}$$

We knwon that

$\therefore$ buckling load, $P = \frac{\pi^2 EI}{(2L)^2} = \frac{\pi^2 EI}{4L^2}$

1261. Which one of the following pairs is not correctly matched

Boundary Conditions of Column	**Euler's Buckling Load**
(a) Both pinned	(1) $\frac{\pi^2 EI}{L^2}$
(b) Both fixed	(2) $\frac{4\pi^2 EI}{L^2}$
(c) One fixed & one free	(3) $\frac{0.25\pi^2 EI}{L^2}$
(d) One fixed & one pinned	(4) $\frac{\sqrt{2}\pi^2 EI}{L^2}$

(UPPSC State Eng. A.E. 2003)

ESE 2002, 1995

Ans : (d)

Boundary condition	**Euler buckling load** $P = \frac{\pi^2 EI}{(L_{effective})^2}$
(i) Both pinned	$P = \frac{\pi^2 EI}{(L)^2} = \frac{\pi^2 EI}{L^2}$
(ii) Both fixed	$P = \frac{\pi^2 EI}{(L/2)^2} = \frac{4\pi^2 EI}{L^2}$
(iii) One fixed and one free	$P = \frac{\pi^2 EI}{(2L)^2} = \frac{0.25\pi^2 EI}{L^2}$
(iv) One fixed and one pinned	$P = \frac{\pi^2 EI}{\left(\frac{L}{\sqrt{2}}\right)^2} = \frac{2\pi^2 EI}{L^2}$

1262. If the end conditions of a long column are changed from both the ends hinged to both ends fixed the strength column increases by... times.

(a) 4 (b) 2

(c) 8 (d) None of these

(BPSC AE Paper V 2001)

Ans : (a)

Strength of column $P_e = \frac{\pi^2 EI}{l_e^2}$; l_e = effective length

For both ends hinged, $l_e = l$; $P_1 = \frac{\pi^2 EI}{l^2}$

For both ends fixed; $l_e = \frac{l}{2}$; $P_2 = \frac{4\pi^2 EI}{l^2}$

$\boxed{P_2 = 4P_1}$

1263. A long column of length 'L' and flexural rigidity 'EI' has pinned ends. The critical load is given by :

(a) $\frac{4\pi^2 EI}{L^2}$ (b) $\frac{L^2}{4\pi^2 EI}$

(c) $\frac{\pi^2 EI}{4L^2}$ (d) $\frac{\pi^2 EI}{L^2}$

(BPSC AE Paper V 2006)

Ans : (d) Critical load $P_e = \frac{\pi^2 EI}{L_e^2}$

For both ends pinned; $l_e = L$

$\therefore$ $\boxed{P_e = \frac{\pi^2 EI}{L^2}}$

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1264. In which form is the permissible stress in columns, f by Merchant – Rankine's formula?

(a) $f \propto \frac{\pi^2 E}{\lambda^2}$

(b) $f \propto \frac{f_y}{1+0.2\sec\left(c.\lambda.\sqrt{f}\right)}$

(c) $\frac{1}{(f)^n} \propto \frac{1}{(f_y)^n} + \frac{1}{(f_{cr})^n}$

(d) $f \propto \frac{f_y}{\phi+\sqrt{\phi^2+\lambda^2}}$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) : According to Merchant-Rankine's formula,

$$\frac{1}{f^n} \propto \frac{1}{(f_y)^n} + \frac{1}{(f_{cr})^n}$$

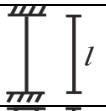
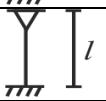
1265. If *l* is the length of the column, then the equivalent length of the column for one end fixed and other end free is

(a) l (b) $2l$

(c) $\frac{l}{\sqrt{2}}$ (d) $\frac{l}{2}$

TNPSC AE 2018, 2012
CGPSC AE 2014 Shift–I
Uttrakhand AE 2013, UKPSC AE (Paper–I) 2007

Ans. (b) :

End condition		Figure	Equivalent length
(i)	Both end fixed		$leq = \frac{l}{2}$
(ii)	One fixed and one hinged		$leq = \frac{l}{\sqrt{2}}$
(iii)	Both hinged		$leq = l$
(iv)	One fixed and one free		$leq = 2l$

1266. Euler's critical load of a column whose effective length L is given by :

(a) $\left(\frac{\pi EI}{L^2}\right)$ (b) $\left(\frac{\pi^2 EI}{L^2}\right)$

(c) $\left(\frac{\pi EI^2}{L^2}\right)$ (d) $\left(\frac{\pi E^2 I}{L^2}\right)$

Karnataka PSC AE 2017, 2015 Paper - II

Ans. (b) : Euler's critical load $= \frac{\pi^2 E}{\lambda^2} \times A$

$$= \frac{\pi^2 E}{\left(\frac{\ell}{r}\right)^2} \times A = \frac{\pi^2 E}{\ell^2} \times Ar^2 = \frac{\pi^2 EI}{\ell^2}$$

1267. Critical load for a column fixed at both the ends is P_{cr} If top end of the column allowed to laterally shift by small distance with rotation, then critical load of the column will be :

(a) $4P_{cr}$ (b) $2P_{cr}$

(c) $\frac{1}{2}P_{cr}$ (d) $\frac{1}{4}P_{cr}$

Rajasthan JEN (Degree) Shift - II 2016

Ans. (c) : $P_C = \frac{\pi^2 EI}{\ell^2}$

$P_C \propto \frac{1}{\ell^2}$

When both ends are fixed $(\ell) = \frac{\ell}{2}$

When one end is free from rotation $(\ell_1) = \frac{\ell}{\sqrt{2}}$

According to question–

$$\frac{P_{cr_1}}{P_{cr}} = \frac{\frac{\ell^2}{4}}{\frac{\ell^2}{2}}$$

$$\boxed{P_{cr_1} = \frac{1}{2}P_{cr}}$$

1268. A 1.5 m long column has a circular cross-section of 50 mm diameter. Consider Rankine's formula with values of σ_c = 560 N/mm², $\alpha = \frac{1}{1600}$ for pinned ends and factor of safety 3. If one end of the column is fixed and the other end is free, the safe load will be

(a) 9948 N (b) 9906 N

(c) 9864 N (d) 9822 N

ESE 2019

Ans. (b)

Length 'L' = 1.5m

d = 50mm

$\alpha = \frac{1}{1600}$

σ_c = 560 N/mm²

Factor of safety F.O.S. = 3.

$\left[K^2 = \frac{I}{A};\ \ell_{eff} = 2L\right]$

$$P_{rankine} = \frac{\sigma_c A}{1+\alpha\left(\frac{\ell_e}{k}\right)^2}$$

$$P_{rankine} = \frac{(560)\frac{\pi}{4}(50)^2}{1+\left[\frac{1}{1600}\times\frac{\left(2\times1.5\times10^3\right)^2}{\frac{\pi}{64}\times(50)^4}\frac{\pi}{4}(50)^2\right]}$$

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$$P_{rankine} = \frac{1099557.428}{1+\left[\frac{1}{1600}\times\frac{9000000}{306796.157}\times 1963.495\right]}$$

$$P_{rankine} = \frac{1099557.428}{1+\frac{17671455000}{490873851.2}} = \frac{1099557.428}{37}$$

$$\left[P_{rankine} = 29718N\right]$$

$$\text{So, Safe Load} = \frac{P}{F.O.S} = \frac{29718}{3}$$

$$\boxed{\text{Safe Load}=9906N}$$

1269. The two long columns of steel P of length L and Q of length 2L have the same cross section, strength characteristics and end conditions. The ratio of buckling load of column P to that of column Q is

(a) 0.5 (b) 1.0
(c) 2.0 (d) 4.0

ISRO Scientist/Engineer 2019

Ans : (d) : Column P : Length = L **Column Q** : Length = 2L

Column P and Q have same cross-sectional area and end conditions.

Bulking Load, $\left[P_{cr} = \frac{\pi^2 EI}{L_{eff}^2}\right]$

$$\Rightarrow \frac{(P_{cr})\text{column P}}{(P_{cr})\text{column Q}} = \frac{\left(\frac{\pi^2 EI}{L_{eff}^2}\right)_P}{\left(\frac{\pi^2 EI}{L_{eff}^2}\right)_Q}$$

$$\Rightarrow \frac{(P_{cr})\text{column P}}{(P_{cr})\text{column Q}} = \frac{(2L)^2}{(L^2)} = \frac{4L^2}{L^2}$$

$$\boxed{\frac{(P_{cr})\text{column P}}{(P_{cr})\text{column Q}} = 4}$$

1270. In beam-columns or eccentric loaded columns, an elastic critical stress in compression f_{cc} is :

(a) $\frac{\pi E}{\lambda}$ (b) $\frac{\pi^2 E}{\lambda^2}$
(c) $\frac{\pi E}{\lambda^2}$ (d) $\frac{\pi^2 E}{\lambda}$

where : E = Modulus of elasticity of steel
λ = Slenderness ratio in the plane of bending.

ESE 2020

Ans. (b) : From Euler's formula,

$$P_c = \frac{\pi^2 EI_{min}}{\ell^2}$$

Dividing by A on both side, we get,

$$\frac{P_C}{A} = \frac{\pi^2 EI_{min}}{\ell^2.A}$$

$$f_{cc} = \frac{\pi^2 E}{\ell^2}.(r_{min})^2 \quad \ldots\ldots\ldots\ldots \left[r_{min} = \sqrt{\frac{I}{A}}\right]$$

$$f_{cc} = \frac{\pi^2 E}{\left(\frac{\ell}{r_{min}}\right)^2}$$

$$\boxed{f_{cc} = \frac{\pi^2 E}{\lambda^2}} \quad \ldots\ldots\ldots\ldots \left[\lambda = \frac{\ell}{r_{min}}\right]$$

where, r_{min} = minimum radius of gyration.

1271. The strength of a column depends on which of the following imperfections ?

1. **The material being isotropic and homogeneous**
2. **Geometric variations of columns**
3. **Eccentricity of load**

(a) 1, 2 and 3 (b) 2 and 3 only
(c) 1 and 3 only (d) 1 and 2 only

ESE 2020

Ans. (b) : The following factors that influences the strength of a columns.

- Eccentricity of loads.
- Initial out of straightness (column curvature).
- Non homogeneity of material.
- Lateral loads.
- Shape of cross-section.
- End restraints.

Hence, right or correct answer is (b).

1272. Which of the following assumptions are made with respect to Euler's theory applied to columns ?

1. **The section of the column is uniform**
2. **The length of the column is very large compared to the lateral dimensions**
3. **The direct stress is large when compared with the bending stress**

(a) 1, 2 and 3 (b) 1 and 3 only
(c) 2 and 3 only (d) 1 and 2 only

ESE 2020

Ans. (d) : Assumptions of Euler's theory :

(1) It is valid for long column, where length of the column is very large when compared to the lateral dimension.
(2) The section of column is uniform and material is isotropic.
(3) Load is purely axial.
(4) Axis of the shaft is perfectly straight when unloaded.

→ The direct stress is small as compared to bending stress.

1273. Euler's buckling load for both ends fixed condition is equal to (in usual notations)

(a) $\pi^2 EI/l^2$ (b) $4\pi^2 EI/l^2$
(c) $2\pi^2 EI/l^4$ (d) $0.25\pi^2 EI/l^5$

APPSC AEE MAINS 2016

Ans. (b) : Case –1

Both ends fixed

$$P = \frac{4\pi EI}{L^2}$$

$$Le = \frac{L}{2}$$

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Case – 2 Both ends hinged

$$P = \frac{\pi^2 EI}{L^2}$$

$Le = L$

Case– 3 One end fixed and other end free

$$P = \frac{\pi^2 EI}{4L^2}$$

$Le = 2L$

Case– 4 One end fixed and other ends hinged

$$P = \frac{2\pi^2 EI}{L^2}$$

$$Le = \frac{L}{\sqrt{2}}$$

1274. The Rankine's constant for a cast iron column with both ends hinged

(a) 1/7500 (b) 1/1600
(c) 1/9000 (d) 0.00016

APPSC AEE MAINS 2016

Ans. (b) : The value of Rankin's constant are as follows

(a) For wrought iron $(a) = \frac{1}{9000}$

(b) For mild steel $(a) = \frac{1}{7500}$

(c) For cast iron $(a) = \frac{1}{1600}$

(d) For timber $(a) = \frac{1}{750}$

1275. Rankine's formula proposed for crippling load (P_r) to cover all cases from short to very long column in terms of ultimate load for short column (P_s) and Euler's crippling load for long column (P_c) is

(a) $P_r = P_s + P_c$ (b) $P_r = \frac{P_s + P_c}{2}$

(c) $\frac{1}{P_r} = \frac{1}{P_s} + \frac{1}{P_c}$ (d) $P_r = \frac{P_s - P_c}{P_s P_c}$

APPSC AEE MAINS 2016

Ans. (c) : Rankin's formula

$$\frac{1}{P_r} = \frac{1}{P_s} + \frac{1}{P_c}$$

Rankin's load = P_r

Buckling load $P_e = \frac{\pi^2 E I_{min}}{L_e^2}$

$$P_r = \frac{A\sigma_c}{1 + a\left(\frac{L_e}{k}\right)^2}$$

Here, A = Area of column

$a = \frac{\sigma_c}{\pi^2 E}$ = Rankine's constant

- This formula is valid for both long and short column.
- Effect of crushing and buckling is considered in this formula.

1276. For Euler formula to be valid for mild steel struts which has yield stress of 3200 kg(f)/sq.cm, the slenderness ratio should not be less than

(a) 80.48 (b) 40.28
(c) 160.96 (d) 53.65

APPSC AEE MAINS 2016

Ans. (a) :

$$\sigma = \frac{\pi^2 E}{\lambda^2}$$

$$\frac{3200 \times 9.81}{100} = \frac{\pi^2 \times 2.1 \times 10^5}{\lambda^2}$$

$\lambda = 81.25 \approx 80.48$

1277. A column of length 'L' is fixed at bottom and hinged at top then the crippling load on column is

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{2\pi^2 EI}{L^2}$

(c) $\frac{4\pi^2 EI}{L^2}$ (d) $\frac{4\pi^2 EI}{L}$

GPSC Poly. Tech. Lect. 2016

Ans. (b) : According to Euler's column theory, the crippling load on the column for different type of end condition is as follows:

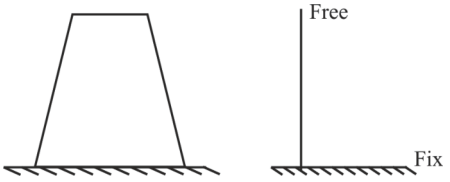
S.N.	End condition	Crippling load	Relation between equivalent length and actual length
1	Both end hinged	$\frac{\pi^2 EI}{l^2}$	$L = l$
2	One end fixed and other end free	$\frac{\pi^2 EI}{4l^2}$	$L = 2l$
3	Both end fixed	$\frac{4\pi^2 EI}{l^2}$	$L = \frac{l}{2}$
4	One end fixed and the other end hinged	$\frac{2\pi^2 EI}{l^2}$	$L = \frac{l}{\sqrt{2}}$

1278. The effective length of a chimney of 20 m height is taken as :

(a) 10 m (b) 20 m
(c) 28.28 m (d) 40 m

Punjab MCP SDO 2015

Ans. (d) : The effective length of a chimney of 20 m height is 40 m.



L = 20m, L = 2L = 2 × 20 = 40 m.

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1279. Slenderness ratio of 5 m long column hinged at both ends and having a circular cross-section with diameter 160 mm is :

(a) 31.25 (b) 62.50
(c) 100 (d) 125

Punjab MCP SDO 2015

Ans. (d) : L_{actual} = 5m = 5000 mm
Both endsd hinged L = l

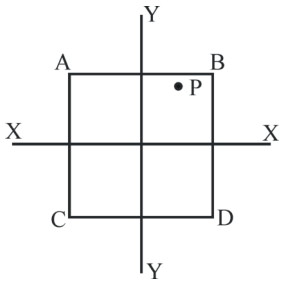
$$ROG = (r) = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi d^4}{64}}{\frac{\pi d^2}{4}}}$$

$$= \sqrt{\frac{d^2}{16}} = \frac{d}{4} = \frac{160}{4} = 40 \text{ mm}$$

r = 40 mm

$$S.R. = \frac{5000}{40} = 125$$

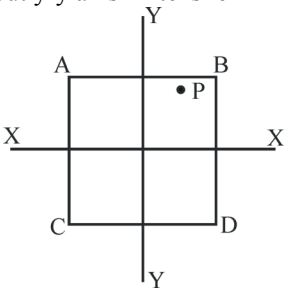
1280. A masonry pier ABCD as shown in figure, supports a vertical load W at a point P. The nature of stresses at A due to eccentricity of load about x-x axis and y-y axis respectively are:



(a) compressive and compressive
(b) tensile and tensile
(c) compressive and tensile
(d) tensile and compressive

Punjab MCP SDO 2015

Ans. (c) : Case Ist → about x-x axis → compressive
Case Ist → about y-y axis → tensile



1281. According to Euler's theory, the crippling load for a column length (l) hinged at both ends is:

(a) $\frac{\pi^2 EI}{l^2}$ (b) $\frac{\pi^2 EI}{4l^2}$

(c) $\frac{2\pi^2 EI}{l^2}$ (d) $\frac{4\pi^2 EI}{l^2}$

GPSC R&B AE 2016

Ans. (a) : By Euler's theory:
Crippling load or buckling load

$$P_C = \frac{\pi^2 EI}{(l_{eff})^2}$$

For hinged at both ends $L_{eff} = l$

$$\boxed{PC = \frac{\pi^2 EI}{l^2}}$$

1282. Which one of the following pairs is not correctly matched?

(a) Lame's constant - Thick cylinder
(b) Macaulay's method - Deflection of beam
(c) Euler's method - Theory of column
(d) Eddy's theorem - Torsion of shaft

JPSC AE 2009, ESE 1996

Ans. (d) : Euler's method is used for analyzing crippling load of column.
Lames theorem gives the solution to thick cylinder problem.
Eddy's theorem for analysis of arc.
Macaulay's theorem is used for determining deflection of beam.

1283. For calculating the allowable stress of long columns. The empirical :

formula $\sigma_0 = \frac{\sigma_y}{n}\left(1 - a\frac{1}{r}\right)$, is known as :

(a) Straight line formula (b) Parabolic formula
(c) Perry's formula (d) Rankine's formula

KPTCL AE 2016

Ans. (a) For calculating the allowable stress of long columns. The empirical Formula $\sigma_0 = \frac{\sigma_y}{n}\left(1 - a\frac{1}{r}\right)$, is known as straight line formula.

1284. The ratio of the effective lengths of two columns, one effectively held in position at one end and partially restrained at the other end, and the other effectively held in position at one end and free at the other end, is :

(a) 1/2 (b) 3/4
(c) 2/3 (d) 3/2

KPTCL AE 2016

Ans. (b) : (1) Effectively held in portion at one end and partially restrained at other end = 1.5 L
(2) Held in position at one end and free at the other end = 2 L

$$\therefore \text{ ratio} = \frac{1.5L}{2L} = \frac{3}{4}$$

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1285. Length of a square vertical column is 300 cm and its side is 2.5 cm. What is the slenderness ratio of the column.

(a) 360 (b) 240
(c) 416 (d) 574

Sikkim PSC AE 2017

Ans. (c) : L_e = 300 cm
a (side of square × section) = 2.5 cm
λ (slenderness ratio) = ?

$$\boxed{\lambda = \frac{L_{eff}}{r_{min}}} \qquad r_{min} = \sqrt{\frac{I_{min}}{A}}$$

A = 3.25, I = 3.25

$$\lambda = \frac{300}{\sqrt{\frac{3.25}{6.25}}} = 416$$

1286. A compression member has one end hinged and other end rigidly fixed against rotation and sway. Its Euler's buckling load is 120 kN. What will be its buckling load if its both ends are fixed against sway and rotation?

(a) 60 kN (b) 120 kN
(c) 240 kN (d) $120\sqrt{2}$ kN

TRB Poly. Tech. Lect. 2012

Ans. (c) : We know,

$$\boxed{P_e = \frac{\pi^2 E I_{min}}{L_{eff}^2}}$$

given, when one end hinged & other end fixed,

(1) $P_e = 120\,kN$, $L_{eff} = \frac{L}{\sqrt{2}}$

$$\therefore P_e = 120\,kN = \frac{\pi^2 E I_{min}}{\left(L/\sqrt{2}\right)^2} = \frac{\pi^2 E I_{min} \times 2}{L^2} \text{ ——(1)}$$

(2) When both end fixed

$L_{eff} = L/2$

$$P_{e_1} = \frac{\pi^2 E I_{min}}{\left(L/2\right)^2} = \frac{\pi^2 E I_{min} \times 4}{L^2}$$

$$\Rightarrow P_{e_1} = \frac{\pi^2 E I_{min} \times 2 \times 2}{L^2}$$

$P_{e_1} = 120 \times 2$

$$\boxed{P_{e_1} = 240\,kN}$$

1287. The most economical section for a column is

(a) Hexagonal (b) Rectangular
(c) Tubular section (d) Solid round

APPSC AEE 2012, TNPSC AE 2012

Ans. (c) : In tubular section, the steel tube will act both as longitudinal and transverse reinforcement. So, this section of column is economical.

1288. Slenderness ratio of compression member is

(a) $\frac{\text{Moment of Inertia}}{\text{Radius of gyration}}$
(b) $\frac{\text{Effective length}}{\text{Radius of gyration}}$
(c) $\frac{\text{Radius of gyration}}{\text{Area of cross-section}}$
(d) $\frac{\text{Moment of Inertia}}{\text{Area of cross -section}}$

APPSC AEE 2012

Ans. (b): Slenderness ratio of compression member

$$= \frac{\text{Effective length}}{\text{Radius of gyration}}$$

$$\lambda = \frac{l}{r}$$

1289. The critical load for a column of length '*l*' hinged at both ends and having flexural rigidity EI, is given by

(a) $P_c = \frac{\pi^2 EI}{l^3}$ (b) $P_c = \frac{\pi^2 EI}{l^2}$
(c) $P_c = \frac{\pi (EI)^2}{l^2}$ (d) None of these

APPSC AEE 2012

Ans. (b) : Critical load for a column hinged at both ends

$$\Rightarrow P_c = \frac{\pi^2 EI}{l^2}$$

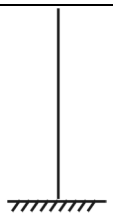
EI = flexural rigidity
l = length of column.

1290. The crippling load taken by a column with both ends hinged is 100 kN. The crippling load taken by the same column with one end fixed and other end free will be :

(a) 400 kN (b) 100 kN
(c) 50 kN (d) 25 kN

TRB Poly. Lect. Tech. 2017

Ans. (d) : Euler's formula for crippling load, $P = \frac{\pi^2 EI}{l_{eff}^2}$.

Both end hinged column	One end fixed and other end free column
$l_{eff} = l$ $P_1 = \frac{\pi^2 EI}{l^2}$ $P_1 = 100$ $= \frac{\pi^2 EI}{l^2}$	 $l_{eff} = 2l$ $P_2 = \frac{\pi^2 EI}{(2l)^2}$ $P_2 = \frac{\pi^2 EI}{4l^2}$ $P_2 = \frac{P_1}{4}$ $P_2 = \frac{100}{4}$ $= 25$ kN

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1291. Who was the first Swiss Mathematician to derive theoretical expression for buckling load?

(a) Rankine (b) Coulomb
(c) St. Venant (d) Leonard Euler

SJVNL ET 2019

Ans. (d) : Leonard Euler was the first Swiss Mathematician to derive theoretical expression for buckling load.

1292. For a column whose both ends are hinged, the crippling/critical load is calculated using

(a) $\frac{\pi^2 EI}{2l^2}$ (b) $\frac{\pi^2 EI}{4l^2}$
(c) $\frac{\pi^2 EI}{l^2}$ (d) $\frac{2\pi^2 EI}{l^2}$

BPSC AE Aug. 2019 Paper-V

Ans. (c) : By Euler's beam eq^n

crippling load $P_c = \frac{\pi^2 EI}{(kl)^2}$ $[le = l]$

for both ends hinged k = 1

$$\boxed{P_c = \frac{\pi^2 EI}{l^2}}$$

1993. Euler's formula gives 5% to 10% error in crippling load as compared to experimental result in practice because

(a) effect of direct stress is neglected
(b) pin joins are not free from friction
(c) assumptions are not correct
(d) materials do not behave in an ideal elastic way

BPSC AE Aug. 2019 Paper-V

Ans. (a) : The crippling load or Buckling load is the load over which a column prefer to deform laterally rather than compressing itself. Euler's beam equation gives crippling load as

$$P = \frac{\pi^2 EI}{(KL)^2}$$

The effect of direct stress in Euler's formula is neglected. So that 5-10% of error in crippling load is occurred

1294. Which of the following statements is correct?

(a) Euller buckling load increases with increase in effective length.
(b) Buckling load of a column does not depend on its cross-section
(c) If free end of a cantilever column is propped then the buckling load increases
(d) Two geometrically identical columns made of different material have same buckling load.

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : If the free end is supported by a prop, the effective length of column decreases. The load carrying capacity increase.

1295. Rankine-Gordon formula is applicable for

(a) Short columns (b) Long columns
(c) Both (a) and (b) (d) None of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : Rankine - Gordon Formula accounts both direct load effect and buckling effect.
∴ Applicable for both short and long columns.

1296. A column c/s 300 mm × 400 mm, 2250 mm long fixe at one end and free at other end. The ratio of effective length to the least lateral dimension is:

(a) 7.5 (b) 15
(c) 11.25 (d) 9

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (b) : $L_{eff} = 2l = 4500$ mm
$D_{min} = 300$ mm
$\therefore \frac{L_{eff}}{D_{min}} = \frac{4500}{300} = 15$

1297. Euler's formula for buckling of column does not hold good if slenderness ratio $\left(\frac{l_e}{K}\right)$ is ..... for mild steel column.

(a) Less than 80 (b) Greater than 90
(c) 120-160 (d) 90-120

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a) : Euler's formula for buckling of column does not hold good if slenderness ratio $\frac{l_e}{K}$ is less than 80 for mild steel column.

1298. The theoretical critical buckling load for a column with pinned ends is 100 kN. What is the theoretical critical buckling load of another column with the same dimensions and material, with fixed ends ?

(a) 25 kN (b) 50 kN
(c) 200 kN (d) 400 kN

BHEL ET 2019

Ans. (d) : $P_{cr_1} = \frac{\pi^2 EI}{L^2} = 100$

$P_{cr_2} = \frac{\pi^2 EI}{(L/2)^2}$

$= \frac{4\pi^2 EI}{L^2} = 4 \times 100 = 400$ kN

1299. The least radius of gyration for solid circular column is

(a) d (b) $\frac{d}{2}$
(c) $\frac{d}{4}$ (d) $\frac{d}{3}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) The least radius of gyration for solid circular colum is d/4.

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1300. In a mild steel tube 4 m long, the flexural rigidity of the tube is $1.2\times10^4\,N-m^2$. The tube is used as a strut with both ends hinged. The crippling load kN is given by

(a) 14.80 (b) 7.40
(c) 29.60 (d) 1.85

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) Tube length = 4 m
Flexural rigidity (E×I) = 1.2×10^4 N – m^2
End condition = Both ends hinged
Effective length = l

$$\text{Crippling load} = \frac{\pi^2 EI}{\ell^2}$$

$$P_e = \frac{\pi^2 \times 1.2\times10^4}{4^2}$$

$$\boxed{P_e = 7.40\ kN}$$

1301. The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is based on

(a) Johnson's parabolic formula
(b) Straight line formula
(c) Perry's formula
(d) Secant formula

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is based on Perry's formula.

1302. The Rankine constant (a) in Rankine's formula is equal to

(a) $\frac{\pi^2 E}{\sigma_C}$ (b) $\frac{\sigma_C}{\pi^2 E}$
(c) $\frac{\pi^2}{\sigma_C E}$ (d) $\frac{E\sigma_C}{\pi^2}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) Rankine's formula :-

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_e}$$

P_R = Rankine load

$P_c = \sigma_c \times A$ = crushing load

$$P_R = \frac{\sigma_c \times A}{1 + \frac{\sigma_c}{\pi^2 E}\left(\frac{Le}{K}\right)^2}$$

A = Area of column

$$a = \frac{\sigma_c}{\pi^2 E} = \text{Rankin's Constant.}$$

1303. When both ends of the column are pinned, then the formula for crippling load (P) is equal to

(a) $P = \frac{\pi^2 EI}{\ell^2}$ (b) $P = \frac{4\pi^2 EI}{\ell^2}$
(c) $P = \frac{2\pi^2 EI}{\ell^2}$ (d) $P = \frac{\pi^2 EI}{l^2}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (a) Crippling load of a column both end hinged

$$P_e = \frac{\pi^2 EI}{L^2}$$

E = Modulus of Elasticity
I = Moment of Inertia about centroidal axis.
l = Effective length.

1304. In Rankine's formula, the material constant for mild steel is

(a) $\frac{1}{9000}$ (b) $\frac{1}{5000}$
(c) $\frac{1}{1600}$ (d) $\frac{1}{7500}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (d)

Material	Value of 'a'
Wrought iron	1/9000
Cast iron	1/1600
Mild Steel	1/7500
Timber	1/750

1305. If the flexural rigidity of the column is doubled, then the strength of the column is increased by

(a) 16 (b) 8
(c) 2 (d) 4

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Case 1st :-

$$\text{Strength of column } (P_1) = \frac{\pi^2 EI}{L^2}$$

Case 2st :-
Flexural rigidity of column is doubled.

$$\text{strength of column } (P_2) = \frac{\pi^2 \times 2EI}{L^2}$$

$$P_2 = 2 \times P_1$$

In the flexural rigidity of the column is doubled, then the strength of the column is increased by two times.

1306. The crippling load of a column with one end fixed and other end hinged is

(a) $\sqrt{2}$ times that of a both ends hinged colum
(b) Two times that of a both ends hinged column
(c) Four times that of a both ends hinged column
(d) Eight times that of a both ends hinged column

APPSC AEE (Civil/Mechanical) 2012

Ans. (b)

End Conditions	Effective length
Both end hinged	L
One fixed other free	2L
Both end fixed	L/2
One end fixed and other hinged	$L/\sqrt{2}$

(i) Crippling load of a column with one end fixed and other end hinged.

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$$P_1 = \frac{\pi^2 EI}{(L/\sqrt{2})^2} = \frac{2\pi^2 EI}{L^2} \quad \ldots\ldots\ldots\ldots \text{(i)}$$

(ii) Crippling load of a column both end hinged

$$P_2 = \frac{\pi^2 EI}{L^2} \quad \ldots\ldots\ldots\ldots\text{(ii)}$$

$$\boxed{P_1 = 2 \times P_1}$$

1307. The ratio of average shear to maximum shear stress for a circular section is:

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$

(c) $\frac{3}{2}$ (d) None of the above

PTCUL AE 2016

Ans. (a) :

- Circular section:

average shear stress = $\left(\frac{3}{4}\right)$ maximum shear stress

- For rectangular section

average shear stress = $\left(\frac{2}{3}\right)$ maximum shear stress

1308. If the Euler load for a steel column is 1000 kN and crushing load is 1500 kN, the Rankine load is equal to

(a) 2500 kN (b) 1500 kN

(c) 1000 kN (d) 600 kN

AP TRANSCO AEE 2017
APTRANSCO 2017, ESE 2010

Ans. (d) : Given :

(P_E) Euler load = 1000 kN

(P_{CS}) crushing load = 1500 kN

$$\frac{1}{P_R} = \frac{1}{P_C} + \frac{1}{P_E}$$

$$\frac{1}{P_R} = \frac{1}{1500} + \frac{1}{1000}$$

P_R = 600 kN

1309. The slenderness ratio of a column with length 'L' and radius of gyration 'r' is defined as

(a) $\frac{2L}{r}$ (b) $\frac{r}{L}$

(c) $\frac{L}{r}$ (d) $\frac{r}{2L}$

AP TRANSCO AEE 2017

Ans. (c) : For a column :

$$\text{Slenderness ratio} = \frac{\text{Effective lenght of column}}{\text{Radius of gyration}}$$

$$\text{S.R.} = \frac{L}{r}, \quad L = \text{effective length}$$

1310. Effective length of a column effectively held in position and restrained in directions at both ends is:-

(a) L (b) 1.5 L

(c) 0.85 L (d) 0.67 L

UPPCL AE 2014

Ans. (d) :

$L_{effective} = 0.67L$

1311. If diameter of a long column is reduced by 20%, the percentage of reduction is Euler buckling load is

(a) 4 (b) 36

(c) 49 (d) 59

APPSC (PHE) AE Paper-2 2012

Ans. (d) : Let, D_1 = initial diameter of column

D_2 = Reduced diameter of column = 0.8 D_1

$$\because P_e = \frac{\pi^2 E I_{min}}{L_e^2}$$

$$\because \frac{(Pe)_2}{(Pe)_1} = \frac{I_2}{I_1} = \frac{(0.8D_1)^4}{D_1^4}$$

$$\frac{(Pe)_2}{(Pe)_1} \approx 0.4096 \approx 0.41$$

$(Pe)_2 = 0.41\ (Pe)_1$

$\therefore$ Percentage reduction = 59%

1312. A member under tension is called

(a) strut (b) tie

(c) strut-tie (d) column

APPSC (PHE) AE Paper-2 2012

Ans. (b) : A member under tension is called tie.

1313. The safe compressive load on a hollow cast iron column (one end fixed other hinged) of 150 mm external diameter, 100 mm internal diameter and 10 m length is (Use Euler's formula with a factor of safety of 5, and E = 95 GN/m^2)

(a) 74.8 kN (b) 149.6 kN

(c) 374 kN (d) 299.2 kN

APPSC (PHE) AE Paper-2 2012

Ans. (a) : For safe compressive load – Applying Euler's formula–

$$P_e = \frac{\pi^2 E I_{min}}{L_e^2} \ ;$$

Given; E = 95 GN/m^2 = 95 × 10^9 N/m^2

Le = Effective length of the column,

For one end fixed and one end hinged column

Le = L/$\sqrt{2}$; L = length of column = 10 m (given)

For hollow cast iron column

$$I_{min} = \frac{\pi}{64}\left[D^4 - d^4\right];$$

D = External diameter

= 150 mm = 0.15 m

d = Internal diameter

= 100 mm = 0.1 m

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$I_{min} = \frac{\pi}{64} \times \left[(0.15)^4 - (0.1)^4\right]$

$= 1.994 \times 10^{-5}\ m^4$

If $l_e = L/\sqrt{2}$

$\therefore P_e = \frac{2\pi^2 E I_{min}}{L^2}$

$= \frac{2\pi^2 \times 95 \times 10^9 \times 1.994 \times 10^{-5}}{(10)^2}$

$= 37.39 \times 10^4\ kN = 374\ kN$

1314. The equivalent length of a column fixed at both ends is

(a) 0.7 l (b) 0.5 l

(c) l (d) 2 l

APPSC (PHE) AE Paper-2 2012

Ans. (b) : Let,

The equivalent length of offer column fixed at both the end = L_e

$L_e = \frac{l}{2} = 0.5l$

where;

l = Length of fixed column

1315. Match List I with List II and select the correct answer using the codes given below the lists:

List I	List II
A. Both ends hinged	(i) L
B. One end fixed and other end free	(ii) $\frac{L}{\sqrt{2}}$
C. One end fixed and the other pin-joined	(iii) L/2
D. Both ends fixed	(iv) 2L

Code:

	A	B	C	D
(a)	i	iv	ii	iii
(b)	i	iii	ii	iv
(c)	iii	i	ii	iv
(d)	iii	i	iv	ii

APPSC (PHE) AE Paper-2 2012

ESE 2003

Ans. (a) :

A. Both ends hinged – L

B. One end fixed and other end free – 2L

C. One end fixed and the other pin-joined – $L/\sqrt{2}$

D. Both ends fixed – L/2

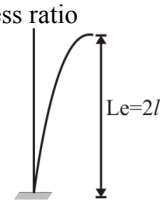
1316. The slenderness ratio of free standing column of length 4 m and 40mm×40mm section is

(a) 115.2 (b) 230.4

(c) 692.8 (d) 346.8

APPSC AEE 2016

Ans. (c) : Slenderness ratio



$(\lambda) = \frac{l_e}{k_{min}} = \frac{2 \times 4000}{\sqrt{\frac{I}{A}}}$

$= \frac{8000}{\sqrt{\frac{(40)^4}{12 \times (40)^2}}} = 692.64$

1317. Euler's crippling load for a column of length l with both ends hinged and flexural rigidity EI is given by

(a) $P=2\pi^2EI/I^2$ (b) $P=\pi^2EI/I^2$

(c) $P=4\pi^2EI/I^2$ (d) $P=\pi^2EI/(4I^2)$

APPSC AEE 2016

Ans. (b) : When both ends hinged then Le = l

Euler's crippling load $P = \frac{\pi^2 EI}{l^2}$

- When both fixed

  Le = $l/2$
- When one end free and one fixed

  Le = $2l$
- One end hinged and one fixed

  $Le = \frac{l}{\sqrt{2}}$

1318. Euler critical bucking load for a prismatic column of length L, flexural rigidity EI, hinged at both ends is _____.

(a) $\frac{2\pi EI}{L^2}$ (b) $\frac{4\pi^2 EI}{L^2}$

(c) $\frac{\pi^2 EI}{4L^2}$ (d) $\frac{\pi^2 EI}{L^2}$

UPPCL AE 2018

Ans. (d) : Effective length of column with both ends pinned is

Le = L

Euler load is, $P_{cr} = \frac{\pi^2 EI}{L_{eq}^2}$

$= \frac{\pi^2 EI}{L^2}$

1319. What will be the Euler buckling load of a column when it is fixed at one and free at other end?

(a) $\frac{\pi^2 EI}{\ell^2}$ (b) $\frac{\pi^2 EI}{4\ell^2}$

(c) $\frac{2\pi^2 EI}{\ell^2}$ (d) $\frac{4\pi^2 EI}{\ell^2}$

(e) $\frac{4\pi^2 EI}{2\ell^2}$

CGPSC AE 2017

Ans. (b) The Euler buckling load of a column when it is fixed at one end free at other end

$P_{cr} = \frac{\pi^2 EI}{4l^2}$

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(i) Both ends hinged $P_{cr} = \frac{\pi^2 EI}{L^2}$

(ii) Both ends fixed $P_{cr} = \frac{4\pi^2 EI}{L^2}$

(iii) One end fixed and other end hinged

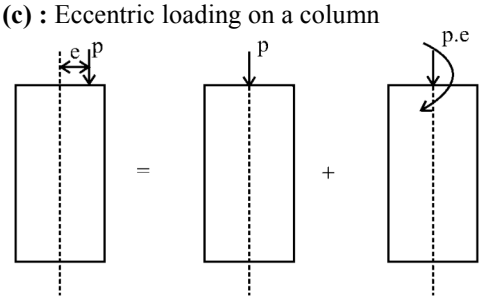
$$P_{cr} = \frac{2\pi^2 EI}{L^2}$$

1320. An eccentric load W, with eccentricity e is equivalent to :

(a) An axial load W
(b) A moment equivalent equal to W.e
(c) Both (a) and (b)
(d) None of the above

UP Jal Nigam AE 2016

Ans. (c) : Eccentric loading on a column



Eccentric loading = load along cetnroid axis (p) + couple (p.e)

1321. A compression member has a centre to centre length of 4.0m. It is fixed at one end and hinged at the other end. The effective length of the column is :

(a) 4.0 m (b) 3.2 m
(c) 2.8 m (d) 2.6 m

UPRVUNL AE 2014

Ans. (c) : L = 4 m

effective length $Le = \frac{l}{\sqrt{2}}$; when end fixed and other end hinged.

$Le = \frac{4}{\sqrt{2}}$

Le = 2.8 m

1322. A pin ended column is loaded with a compressive load P eccentrically. Determine the critical buckling load P_{cr} if modulus of elasticity is E and second moment of cross sectional area is I.

(a) $P_{cr} = EI/\pi^2 L^2$ (b) $P_{cr} = \pi^2 EI/3L^2$
(c) $P_{cr} = \pi EI/L^2$ (d) $P_{cr} = \pi^2 EI/L^2$

UPRVUNL AE 2014

Ans. (d) : For critical buckling load (P_{cr})

Depending upon various combinations of restraints, these may be the following four cases of end condition.

Case : 1 Both ends hinged / pin

$$Pcr = \frac{\pi^2 EI}{L^2}$$

Le = L Le = effective length

Case : 2 One end fixed and other end free

$$P_{cr} = \frac{\pi^2 EI}{4L^2}$$

Le = 2 L

Case : 3 Both ends fixed

$$P_{cr} = \frac{4\pi^2 EI}{L^2}$$

Le = L/2

Case : 4 One end fixed and other end hinged.

$$Le = \frac{L}{\sqrt{2}},\ P_{cr} = \frac{2\pi^2 EI}{L^2}$$

1323. Euler buckling load for one end fixed and the other hinged is given by:

(a) $\frac{\pi^2 EI}{l^2}$ (b) $\frac{2\pi^2 EI}{l^2}$
(c) $\frac{4\pi^2 EI}{l^2}$ (d) $\frac{\pi^2 EI}{4l^2}$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (b) : Eular buckling load for one end fixed and the other hinged is given by $\frac{2\pi^2 EI}{l^2}$ and $L_e = \frac{L}{\sqrt{2}}$.

There may be the following cases of end condition for Euler buckling load.

(1) Both ends hinged

$P_E = \frac{\pi^2 EI}{L^2}$ L_e = Effective length

$L_e = L$

(2) One end fixed and other end fee

$P_E = \frac{\pi^2 EI}{4L^2}$ P_E = Euler buckling load

$L_e = 2L$

(3) Both ends fixed

$P_E = \frac{4\pi^2 EI}{L^2}$

$L_e = L/2$

1324. Radius of gyration of a circular section with diameter D is

(a) $\frac{D}{2}$ (b) $\frac{D}{4}$
(c) $\frac{D}{3}$ (d) $\frac{D}{\sqrt{3}}$

APPSC AEE (CIVIL/MECHANICAL) 2019
APPSC (PHE) AE Paper-2 2012

Ans. (b) : For circular section $I = \frac{\pi}{64}d^4,\ A = \frac{\pi}{4}d^2$

Radius of gyration $= k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64}d^4}{\frac{\pi}{4}d^2}} = \frac{d}{4}$

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1325. The critical load for long columns can be calculated for one end fixed and other hinged condition as per Euler's formula

(a) $\frac{\pi^2 EI}{L^2}$ (b) $\frac{2\pi^2 EI}{L^2}$

(c) $\frac{4\pi^2 EI}{L^2}$ (d) $\frac{\pi^2 EI}{4L^2}$

WBPSC AE 2007

Ans. (b) :

$\because \quad P_{cr} = \frac{\pi^2 EI}{L_e^2}$

If

- Both ends fixed, $L_e = \frac{L}{2}$
- Both ends hinged, $L_e = L$
- One end fixed and other hinged $L_e = \frac{L}{\sqrt{2}}$
- One end fixed other free $L_e = 2L$

1326. The buckling load for a column fixed at both ends is 50 kN. If the ends of same column are changed to hinged condition, then the buckling load will be:

(a) 37.5 kN (b) 25 kN

(c) 12.5 kN (d) 100 kN

TSPSC AEE 2017

Ans. (c) : Given :

$(P_c)_1 = 50$ kN, $(P_c)_2 = ?$

$(L_e)_1 = \frac{L}{2} \quad (L_e)_2 = L$

$\because \quad P_c \propto \frac{1}{L_e^2}$

$\therefore \quad \frac{(P_c)_2}{(P_c)_1} = \left(\frac{(L_e)_1}{(L_e)_2}\right)^2$

$\therefore \quad (P_c)_2 = (P_c)_1 \times \left(\frac{\frac{L}{2}}{L}\right)^2$

$\therefore \quad (P_c)_2 = 50 \times \frac{1}{4} = 12.5\,\text{kN}$

1327. A short hollow cylindrical column has 200 mm external diameter and 100 mm internal diameter. What is the diameter of core for the column?

(a) 62.5 mm (b) 25 mm

(c) 50 mm (d) 33.3 mm

TSPSC AEE 2017

Ans. (a) : Given :

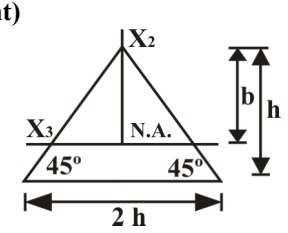
D = 200 mm, d = 100 mm

$\because$ core diameter of hallow cylindrical column

$= \frac{D^2 + d^2}{4D}$

$= \frac{(200)^2 + (100)^2}{4 \times 200} = 62.5\text{mm}$

1328. The beam of triangular cross-section as shown in the figure below, is subjected to pure bending. If a plastic hinge develops at a section, determine the location of neutral axis (distance b from top) at that section. The beam material is elastic-perfectly plastic (i.e., yield stress is constant)

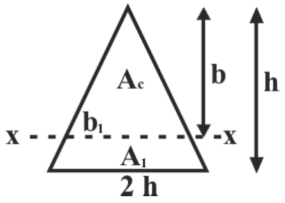


(a) $\frac{h}{\sqrt{3}}$ (b) $\frac{h}{2}$

(c) $\frac{h}{\sqrt{2}}$ (d) $\frac{h}{3}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) :



$A_c = A_t$

$\frac{1}{2} \times b \times b_1 = \frac{1}{2}\left[\frac{1}{2} \times 2h \times h\right]$

$b_1 = \frac{h^2}{b}$

from similar triangles

$b \to b_1$

$h \to 2h$

$b \times 2h = b_1 h$

$b = \frac{b_1}{2}$

$b = \frac{h^2}{b \times 2}$

$b^2 = \frac{h^2}{2}$

$b = \frac{h}{\sqrt{2}}$

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8. Torsion and Shear Centre

1329. In a rectangular shaft subjected to torsion the maximum shear stress occurs at :
(a) Centre
(b) Corners
(c) Middle of smaller side
(d) Middle of longer side

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : The shear stresses are zero at the corners & maximum at the middle of the longer side in the case of rectangular shaft.

1330. Polar modulus of a section is measure of strength of section in
a) Bending (b) Shear
(c) Torsion (d) None of the above

RPSC POLY. TECH. LECT. 2011

Ans : (c) Polar modulus of section is measure of strength of section in torsion. for circular section, polar moment of inertia $\boxed{J = \frac{\pi d^4}{32}}$

1331. The type of stress developed in circular shaft under pure torque is
(a) shear stress (b) torsional stress
(c) twisting (d) bending

(BPSC AE Paper V 2001)

Ans : (a) Shafts under pure torque develop shear stress only.

1332. Magnitude of shear stress induced in a shaft due to applied torque varies from-
(a) Maximum at centre to zero at circumference
(b) Maximum at centre to minimum (Not Zero) at the circumference
(c) Zero at centre to maximum at circumference
(d) Minimum (Not Zero) at centre to maximum at circumference

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) : Torsion formula,

$$\frac{T}{J} = \frac{\tau}{r}$$

$$\Rightarrow \quad \tau = \left(\frac{T}{J}\right) \times r$$

Since T and J are constant for a given case,

$$\boxed{\tau \propto r}$$

i.e., shear stress varies linearly with radius

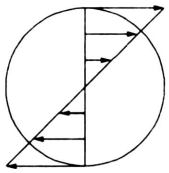


Figure : Shear stress variation in shaft.

1333. What would be the shape of the failure surface of a standard cast iron specimen subjected to torque?
(a) Cup and cone shape
(b) Plane surface perpendicular to the axis
(c) Pyramid type wedge shaped
(d) Helicoidal surface

GPSC AE (CLASS 1 & 2) 2019
ESE 2006

Ans : (d) :

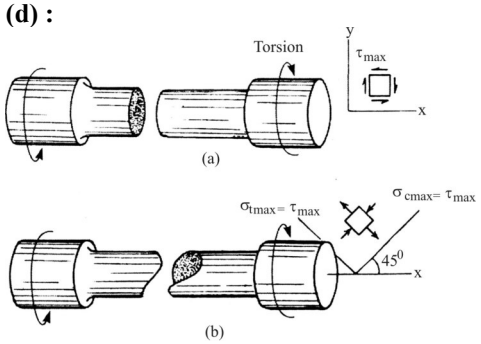


Figure : Fracture of (a) mild steel (b) cast iron is torsion

Brittle material (Ex - cost iron, glass etc.) fails due to normal stress which is max at 45^0 to the axis in case of tension.

Ductile material (Ex - Steel, Copper etc.) fails due to shear, which, in case of torsion occurs at 90^0 to the axis.

The shape of failure surface of

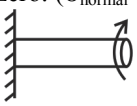
- Standard cast from is Helicoidal surface.
- Mild steel is conical shape.

1334. A long shaft of diameter d is subjected to twisting moment T at its ends. The maximum normal stress acting at its cross-section is equal to

(a) Zero (b) $16T/\pi d^3$
(c) $32T/\pi d^3$ (d) $64T/\pi d^3$

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : For a long shaft subjected twisting moment T. Then only shear stress (τ_{xy}) acts at its cross-section normal stress will be zero. ($\sigma_{normal} = 0$)



1335. The unit of torque in SI units is:
(a) kg/cm (b) Newton-metre (N-m)
(c) kg/cm^2 (d) None

KPSC AE 2016

Ans. (b) : Torque is the turning effect produced by a force on an object pivoted at a point. It is basically the force require to create rotation on an object which is pivoted at a point.

Torque = F × Perpendicular distance
= N × m (As SI unit of force is Newton and of distance is meter)

Therefore SI unit of Torque is N-m

1336. Statement (I): Hollow shafts are preferred in propeller shafts of airplanes.
Statement (II): Use of hollow shafts affords considerable reduction in the weight of the shaft for equal performance.
(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.

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(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I
(c) Statement-I is true but Statement-II is false.
(d) Statement-I is false but Statement-II is true.

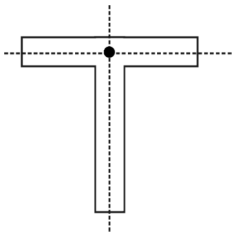
ESE 2017

Ans. (a)
- The advantage of hollow shaft over solid shaft is it have greater value of strength and higher value of rigidity. Because of the weight is low as compared to solid shaft it is more useful.
- Material strength of hollow circular shaft is properly utilized.

1337. For the T section the shear centre is located at
(a) Centre of the horizontal flange
(b) Centre of the vertical web
(c) At the centroid of the section
(d) Outside the T section

APTRASCO 2017

Ans : (a)



The shear centre for T-section is located at the intersection of lines drawn to middle of flange and web.
• Shear centre is a point where, when load acts then there will be only bending no twisting.

1338. The shear centre of a section is defined as that point :
(a) Through which load must be applied to produce zero twisting moment on the section
(b) At which shear force is zero
(c) At which shear force is maximum
(d) At which shear force is minimum

OPSC AEE 2019 Paper–I
ESE 1997

Ans. (a) : Shear Centre– Shear centre of a section is defined as that point through which the load must be applied to produce zero twisting moment on the section. Generally, if a beam has two axes of symmetry, then shear centre coincides with the centroid.

1339. For no torsion, the plane of bending should
(a) Be parallel to one of the principal axes
(b) Pass through shear centre of section
(c) Pass through neutral axis of the section
(d) Pass through centre of gravity of the section

OPSC AEE 2019 Paper–I

Ans. (b) : Shear center of a section is the centroid of all the shear forces in the section that's why if load is applied at shear center it produce zero twisting moment.

1340. If an area has any axis of symmetry then product of inertia will be:
(a) Zero (b) Positive
(c) Negative (d) Either positive or negative

APPSC Poly. Tech. Lect. 2020

Ans. (a) : If an area has any axis of symmetry then product of inertia will be zero.

1341. Which one of the following pairs is not correctly matched ?
(a) Lame's constants : Thick cylinder
(b) Macaulay's method : deflection of beams
(c) Euler's method : Theory of columns
(d) Eddy's theorem : Torsion of shafts

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : Eddy's theorem states that "The Bending moment at any section of an arc is proportional to the vertical intercept between the linear arch (or theoretical arch) and the centre line of the Actual arch"

1342. When a shaft is subjected to a twisting moment every cross-section of the shaft will be under.
(a) Tensile stress (b) Compressive stress
(c) Shear stress (d) Bending stress

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : When a shaft is subjected to a twisting moment every cross-section of the shaft will be under shear stress.

1343. In shafts with keyways the allowable stresses are usually in what % ?
(a) 25 (b) 50
(c) 75 (d) 95

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : The maximum permissible (Design) stresses in Bending (tension or compressive) may be taken as 112 N/mm^2 for shaft with allowance for keyways in about 75%.

1344. Which pattern of the drilling is not used for shafts?
(a) Central wedge cut (b) End wedge cut
(c) Vertical wedge cut (d) Alternate wedge cut

Maharashtra PSC Mains 2018 Paper-2

Ans. (c) : For shafts:
• The most commonly used drilling patterns is central wedge cut for square and rectangular sections.
• For circular sections, sometimes wedge cut (or) period cut is also used (end cut).
• When shafts are very large, drilling may be done in alternate wedge cut.

1345. If the load does NOT passes through shear centre of the section of the beam, then there will be
(a) No bending of the beam
(b) Only bending of the beam
(c) Only twisting of the beam
(d) Both bending and twisting of the beam

TSPSC AEE 2017

Ans. (d) : If the load does NOT passes through shear centre of the section of the beam, then there will be bending and twisting of the beam.

1346. The shear center in case of equal angle section is
(a) at the centroid of the section
(b) at the neutral axis of the section
(c) at the junction of two legs
(d) None of the above

WBPSC AE 2016

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Ans. (c) : The shear center in case of equal angle section is at the junction of two legs.

1347. Which of the section should be used preferably at places where torsion occurs?
(a) Angle (b) Channel
(c) Box section (d) Tee section

WBPSC AE 2016

Ans. (c) : Box section should be used at places where torsion occurs.

1348. The shear centre for an angle section is located _____.
(a) at the tip of the flange
(b) at the intersection of flanges
(c) at the centroid of the angle
(d) at the rigidity centre of the angle

TSPSC Deputy Engineer 2015

Ans. (b) : The shear centre for an angle section is located at the intersection of flanges.

1349. In case of a torsional problem the assumption- "Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having
(a) circular cross-section
(b) elliptical cross-section
(c) circular and elliptical cross-section
(d) any cross-section

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : In case of a torsional problem the assumption-
"Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having circular cross-section only. It is not valid for other shape of cross-section.

1350. Under pure shear the shape of the body gets distorted. What is the inference about its volume?
(a) shear and volume of body are not related
(b) remains the same
(c) decreases
(d) increases

CIL MT 2020

Ans. (b) : Volume remains the same under pure shear when the shape of the body gets distorted.

1351. The shear stress at a point in a shaft subjected to a torque is
(a) Directly proportional to the polar moment of inertia and to the distance of the point from the axis
(b) Directly proportional to the applied torque and inversely proportional to the polar moment of inertia
(c) Directly proportional to the applied torque and polar moment of inertia
(d) Inversely proportional to the applied torque and the polar moment of inertia

UPRVUNL AE 2015

Ans. (b) : By torsion equation

$$\frac{\tau}{r}=\frac{T}{J}=\frac{G\theta}{L}$$

We have $\left[\frac{\tau}{r}=\frac{T}{J}\right]$

1352. Consider the following statements:
A splined shaft is used for
(i) transmitting power
(ii) holding a flywheel rigidly in position
(iii) moving axially the gear wheels mounted on it.
Code:
(a) (ii) and (iii) are correct
(b) (i) and (iv) are correct
(c) (ii) and (iv) are correct
(d) (i) and (iii) are correct

APPSC (PHE) AE Paper-2 2012

Ans. (a) : A splined shaft is used for
Holding a flywheel rigidly in position.
Moving axially the gear wheels mounted on it.

1353. Stress developed in cylindrical bars subjected to pure torsion is
(a) normal stress
(b) shear stress
(c) both normal and shear stress
(d) no stress

BPSC AE Aug. 2019 Paper-V

Ans. (b) : A uniform circular shaft is subjected to a torque it can be shown that every section of the shaft is subjected to a state of pure shear similarly, pure torsion subjected shaft has only shear stress.

1354. Shear centre is :
(a) The point at which torque acts
(b) The same as centroidal axis
(c) The point at which bending moment acts
(d) The point at which resultant of internal forces passes

UPPCL AE 2019

Ans. (d) : Shear centre is the point at which resultant of internal forces passes.

1355. For the same material, length and given torque, a hollow shaft weights _____a solid shaft.
(a) less than (b) more than
(c) equal to (d) None of the above

UP Jal Nigam AE 2016
APPSC (PHE) AE Paper-2 2012

Ans. (a) : Solid shaft is completely filled with material but in case of hallow two concentric circle are enclosed and also some material is neglected that is why for some material length and given Torque a hollow shaft weight less than a solid shaft.

1356. If two shafts of the same length, one of which is hollow, transmit equal torque and have equal maximum stress, then they should have equal
(a) polar moment of inertia
(b) diameter
(c) polar modulus of section
(d) angle of twist

APPSC (PHE) AE Paper-2 2012

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Ans. (c) : Given; solid and hallow shafts have the same length let us assume shafts have the same materials.
Given, torque applied to both shafts is same, and the maximum shear stress is equal.

i.e, $T_S = T_H \qquad \because T = \frac{\pi}{16} d_3 . \tau$

$$\tau . \frac{\pi}{16} D_s^3 = \tau \frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$$

$$\frac{\pi}{16} D_S^3 = \frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$$

Where;

$\frac{\pi}{16} D_S^3 =$ Polar section modulus of solid shaft

& $\frac{\pi}{16}\left[\frac{D_H^4 - d_H^4}{D_H}\right]$ Polar section modulus of hollow shaft

Hence, polar section modulus in both the case is also same.

1357. The ratio of maximum shear stress developed in a solid shaft of diameter D and a hollow shaft of external diameter D and internal diameter d for the same torque is given by

(a) $\frac{\left(D^2 + d^2\right)}{D^2}$ (b) $\frac{\left(D^2 - d^2\right)}{D^2}$

(c) $\frac{\left(D^4 + d^4\right)}{D^4}$ (d) $\frac{\left(D^4 - d^4\right)}{D^4}$

OPSC AEE 2019, 2015 Paper–I

Ans. (d)

$$(\tau_{max})_{Solid} = \frac{T}{Z_P} = \frac{16T}{\pi D^3}$$

$$(\tau_{max})_{Hollow} = \frac{16.TD}{\pi\left(D^4 - d^4\right)}$$

$\because \; T_{Hollow} = T_{Solid}$

$$\frac{(\tau_{max})_{solid}}{(\tau_{max})_{Hollow}} = \frac{\frac{16T}{\pi D^3}}{\frac{16TD}{\pi\left(D^4 - d^4\right)}} = \frac{D^4 - d^4}{D^4}$$

1358. Two shaft of different diameter d_1 and d_2 are made from same material and are of same length under the action of same torque T the ratio of strain energy V_1 / V_2

(a) $\left[\frac{d_2}{d_1}\right]^3$ (b) $\left[\frac{d_1}{d_2}\right]^2$

(c) $\left[\frac{d_2}{d_1}\right]^2$ (d) $\left[\frac{d_2}{d_1}\right]^4$

Gujarat PSC AE 2017
ESE 1998

Ans : (d)

Strain energy (U) $= \frac{1}{2} T.\theta$ (In case of Torsion)

By using torsion equation

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$$

$$\theta = \frac{TL}{GJ}$$

Where, $J = \frac{\pi d^4}{32}$

Given,

$T_1 = T_2 = T$
$L_1 = L_2 = L$
$G_1 = G_2 = G$

$$U = \frac{1}{2}\frac{T^2 L}{GJ}$$

$$U \propto \frac{1}{J}$$

$$\frac{U_1}{U_2} = \frac{J_2}{J_1} = \frac{\frac{\pi}{32}(d_2)^4}{\frac{\pi}{32}(d_1)^4}$$

$$\frac{U_1}{U_2} = \frac{(d_2)^4}{(d_1)^4}$$

1359. In the torsion equation

$\frac{T}{J} = \frac{\tau_{max}}{R} = \frac{G\theta}{\ell}$ **the term,** $\frac{J}{R}$ **is called**

(a) shear modulus (b) section modulus
(c) polar modulus (d) none of these

Maharashtra PSC Mains 2018 Paper-1
TNPSC AE 2012

Ans. (c) : $\frac{T}{J} = \frac{\tau_{max}}{R} = \frac{G\theta}{\ell}$

$$\frac{J}{R} = \frac{T}{\tau_{max}}$$

$\frac{J}{R}$ = polar section modulus

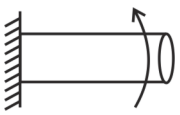
Section modulus = I/y

1360. Every cross section of a shaft, which is subjected to a twisting moment, is under

(a) Compressive stress (b) Tensile stress
(c) Shear stress (d) Bending stress

TNPSC AE 2019

Ans. (c) Shaft when the torque is applied it gives shear stress.



$$\boxed{\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}}$$

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1361. A solid circular shaft of diameter 8 mm is exposed to a maximum torque of 64 N-m. What is shear stress in the shaft?

(a) 2000 π (b) 2000/π
(c) 2/π (d) 20/π
(e) 2π

CGPSC AE 2014 Shift–I

Ans. (b) : Maximum shear stress in the shaft due to torque 'T'

$$\tau_{max} = \frac{T \times R}{J}$$

$$R = \frac{D}{2}$$

$$J = \frac{\pi D^4}{32}$$

$$\tau_{max} = \frac{16T}{\pi D^3} = \frac{16 \times 64 \times 10^3}{\pi (8)^3}$$

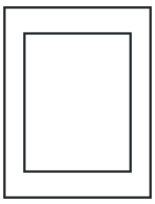
$$\boxed{\tau_{max} = \frac{2000}{\pi} \left(N/mm^2\right)}$$

1362. Which of the following sections should be preferably be used at a place where torsion occurs?

(a) Angle section (b) Box section
(c) Channel (d) I-section

[UPPSC State Eng. A.E. 2003 IInd paper]

Ans : (b)



Hollow rectangle box section

The box section are preferred where torsion occurs. The box sections are very strong in resisting torsion.

1363. A Circular shaft of length *"l"* subjected to a Torque "T", "G" is rigidity modulus and "J" is polar moment of inertia, then the total angle of twist is given by

(a) $\frac{Tl}{GJ}$ (b) $\frac{TJ}{Gl}$
(c) $\frac{TG}{Jl}$ (d) $\frac{GJ}{Tl}$

TNPSC AE 2019

Ans. (a)

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{l}$$

$$\boxed{\theta = \frac{Tl}{GJ}}$$

1364. A solid circular shaft of 60 mm diameter transmits a torque of 1600 Nm. The value of maximum shear stress developed is

(a) 37.72 MPa (b) 47.72 MPa
(c) 57.72 MPa (d) 67.72 MPa

(UPRVUNL AE 2015)

Ans : (a) Torsion equation $\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$

Given, $R = \frac{60}{2} = 30\,mm;$

$$T = 1600 \times 10^3\ N - mm$$

$$J = \frac{\pi D^4}{32} = \frac{\pi \times 60^4}{32} = 1272.345 \times 10^3\ mm^4$$

$$\tau = \frac{TR}{J}$$

$$\tau = \frac{1600 \times 10^3 \times 30}{1272.30 \times 10^3} = 37.73\ MPa$$

1365. A solid circular shaft of diameter d and Length L is fixed at one end and free at the other end. A torque T is applied at the free end. The shear modulus of the material is G, the angle of twist at the free end is

(a) 16 TL/ $\pi d^4 G$ (b) 32 TL/ $\pi d^4 G$
(c) 64 TL/ $\pi d^4 G$ (d) 128 TL/ $\pi d^4 G$

RPSC POLY. TECH. LECT. 2014

Ans : (b) $\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}$

$$\Rightarrow \quad \theta = \frac{TL}{GJ} = \frac{TL}{G \times \frac{\pi d^4}{32}}$$

$$\therefore \quad \boxed{\theta = \frac{32TL}{\pi d^4 G}}$$

1366. The strain energy due to torsion is (torsion = *T*; modulus of elasticity = *E*; moment of inertia = *I*; shear modulus = *G*; polar moment of area = *J*)-

(a) $\int \frac{T^2 dx}{2EI}$ (b) $\int \frac{T^2 dx}{2GJ}$
(c) $\int \frac{Tdx}{2EI}$ (d) $\int \frac{Tdx}{2GJ}$

MPPSC AE 2017

Ans. (b) : Strain energy in torsion : The torsional strain energy of a shaft is equal to the work done in twisting.

$$U = \frac{1}{2}.T.\theta\ ;\ \theta = \frac{Tl}{GJ}$$

Where, T = Applied torque
θ = Angle of twisting

$$\Rightarrow \quad \int_0^L \frac{T^2 dx}{2GJ} = \text{Torsional strain energy}$$

If *T* and *J* are constant i.e. prismatic bar under constant torque

$$U = \frac{T^2 L}{2GJ}$$

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1367. If a solid shaft is subjected to a torque (T) at its end such that maximum shear stress does not exceed f_s, the diameter of the shaft will be-

(a) $\frac{16T}{\pi f_s}$ (b) $\left(\frac{16T}{\pi f_s}\right)^{\frac{1}{2}}$

(c) $\left(\frac{16T}{\pi f_s}\right)^{\frac{1}{3}}$ (d) None of these

Maharashtra PSC (Mains) 2015 Paper–I
ESE 1995

OR

If a solid shaft is subjected to a torque T at its end such that maximum shear stress does not exceed τ, the diameter of shaft will be

(a) $\frac{17T}{\pi\tau}$ (b) $\left[\frac{16T}{\pi\tau}\right]^{1/2}$

(c) $\left[\frac{16T}{\pi\tau}\right]^{1/3}$ (d) $\left[\frac{16T}{\pi\tau}\right]^{1/4}$

TNPSC AE 2019
ESE 2015, 2012

Ans. (c) : From torsion formula

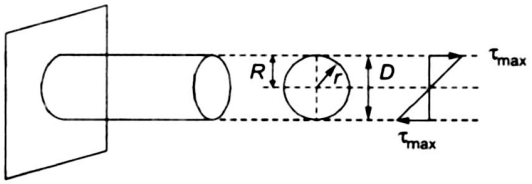
$$\frac{T}{J}=\frac{\tau}{r}=\frac{G\theta}{L}$$

T = Torque, J = Polar moment of inertia

τ = Shear stress, r = Distance from centre of shaft

G = Shear modulus, θ = Angle of twist

L = Length of shaft



Given, $\tau_{max} = f_s$

At $r = R = \frac{D}{2}; \tau = \tau_{max} = f_s$

$$J = \frac{\pi D^4}{32}$$

$$\frac{T}{\left(\frac{\pi D^4}{32}\right)} = \frac{f_s}{\left(\frac{D}{2}\right)}$$

$$\frac{32T}{2\pi D^3} = f_s$$

$$D^3 = \frac{16T}{\pi f_s}$$

$$\boxed{D = \left(\frac{16T}{\pi f_s}\right)^{\frac{1}{3}}}$$

1368. A circular shaft was initially subjected to bending moment and then was subjected to torsion. If the magnitude of bending moment is found to be the same as that of the torque, then the ratio of maximum bending stress to shear stress would be-

(a) 0.25 (b) 0.50

(c) 2.0 (d) 4.0

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (c) :

$$\sigma_B = \frac{32M}{\pi D^3}, \quad \tau_T = \frac{16T}{\pi D^3}$$

$$\frac{\sigma_B}{\tau_T} = \frac{2M}{T}$$

$$\frac{\sigma_{max}}{\tau_{max}} = \frac{2M}{T} = \frac{2M}{M} = 2 \qquad \text{(as Given m = T)}$$

1369. If a solid circular shaft is simultaneously subjected to a torque 'T' and a bending stress due to moment M the maximum torsional shearing stress is given by-

(a) $\frac{M}{T}$ (b) $\frac{T}{M}$

(c) $\frac{2M}{T}$ (d) $\frac{2T}{M}$

CIL MT 2020, OPSC AEE 2019 Paper–I
GPSC AE June 2019, Dec. 2018
Maharashtra PSC (Mains) 2017 Paper–I
TSGENCO 2015, APPSC AEE 2012

Ans. (c) : Bending stress $\sigma_y = \frac{My}{I}$

y is maximum for outer fibres

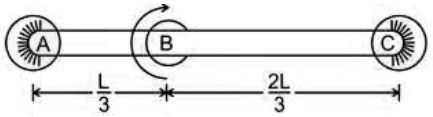
$$\sigma_{max} = \frac{M \times d/2}{\frac{\pi d^4}{64}} = \frac{32M}{\pi d^3}$$

similarly, $\tau = \frac{T.r}{J}$

$$\tau_{max} = \frac{T \times d/2}{\pi d^4/32} = \frac{16T}{\pi d^3}$$

$$\frac{\sigma_{max}}{\tau_{max}} = \frac{32M}{16T} = \frac{2M}{T}$$

1370. A bar AC shown in the figure given below is fixed at both ends and is subjected to a torque t at B where $AB = \frac{l}{3}$. Then torque at end A is-



(a) $\frac{t}{3}$ (b) $\frac{t}{2}$

(c) $\frac{2t}{3}$ (d) t

BPSC AE Pre 2018

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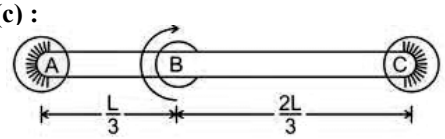
Ans. (c) :



Here, $\theta_A = \theta_C$ and, $t = t_A + t_C$

$$\therefore \theta = \frac{tL}{GJ} \Rightarrow \frac{t_A \times \left(\frac{L}{3}\right)}{G \times J} = \frac{t_B \times \left(\frac{2L}{3}\right)}{G \times J}$$

$$\Rightarrow t_A = 2t_B$$

Now, $t = t_A + t_B = t_A + \frac{1}{2}t_A$

$$\Rightarrow t = \frac{3}{2}t_A$$

$$\Rightarrow t_A = \frac{2}{3}t$$

1371. If a circular shaft is subjected to both torque T and bending moment M. Then the equivalent bending moment M_e is given by

(a) $M_e = \frac{M + \sqrt{M^2 + T^2}}{2}$

(b) $M_e = M + \sqrt{\frac{M^2 + T^2}{2}}$

(c) $M_e = M - \sqrt{\frac{M^2 + T^2}{2}}$

(d) $M_e = \frac{M - \sqrt{M^2 + T^2}}{2}$

TNPSC AE 2018, GPSC AE March 2018
PTCUL AE 2016, ESE 1995

Ans. (a) : Bending stress $(\sigma_b) = \frac{32M}{\pi D^3}$

Shear stress $(\tau) = \frac{16T}{\pi D^3}$,

Maximum principle stress $(\sigma) = \frac{1}{2}\sigma_b + \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$

$$= \frac{16M}{\pi D^3} + \frac{1}{2}\sqrt{\left(\frac{32M}{\pi D^3}\right)^2 + 4\left(\frac{16T}{\pi D^3}\right)^2}$$

$$= \frac{16}{\pi D^3}\left(M + \sqrt{M^2 + T^2}\right)$$

Therefore equivalent bending moment

$$M_e = \frac{M + \sqrt{M^2 + T^2}}{2}$$

1372. If T is the mean or average torque in non applied and N is the number of revolutions per minute. Then power (P) transmitted by the shaft in kW is

(a) $\frac{2\pi NT}{600}$ (b) $\frac{2\pi NT}{750}$

(c) $\frac{2\pi NT}{60000}$ (d) $\frac{2\pi NT}{45000}$

TNPSC AE 2018

Ans. (c) : Power = $T \times \omega = T \times \frac{2\pi N}{60}$ (in watt)

Power (in KW) $= \frac{2\pi NT}{60} \times 10^{-3} = \frac{2\pi NT}{60000}$

1373. When a shaft is subjected to torsion, the relation between maximum shear stress (τ), modulus of rigidity (C) and angle of twist (θ) is given by

(a) $\frac{C\theta}{\tau} = \frac{R}{L}$ (b) $\frac{C\theta}{L} = \frac{\tau}{R}$

(c) $\frac{C\theta}{R} = \frac{\tau}{L}$ (d) $\frac{C}{L\theta} = \frac{\tau}{R}$

where, L = length shaft, R = Radius of shaft

TNPSC AE 2018

Ans. (b) : Torsion formula

$$\frac{T}{J} = \frac{\tau}{r} = \frac{C\theta}{L}$$

1374. In a circular shaft subjected to pure twisting moment, the principal stress at a point close to the outer periphery of shaft act on a plane :

(a) 90^0 to the axis of the shaft
(b) 45^0 to the axis of the shaft
(c) 30^0 to the axis of the shaft
(d) Parallel to the axis of the shaft

[HPPSC 2016-II]

Ans. (b) : For pure twisting, $\sigma_x = \sigma_y = 0$ and only τ_{xy} acts

$$\tan 2\theta = \frac{2\tau}{\sigma_x - \sigma_y} = \infty = \tan 90^0 \text{ or } \tan 270^0$$

$\Rightarrow 2\theta = 90^\circ$ or 270°

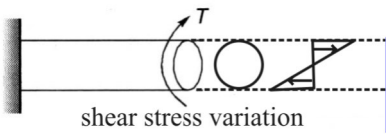
$\therefore \theta = 45^\circ$ or 135°

1375. Pure torsion of a shaft produces :

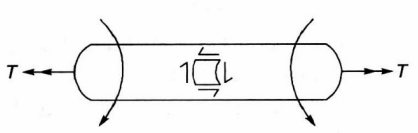
(a) longitudinal normal stress in shaft
(b) only direct shear stress in the transverse section of the shaft
(c) circumferential shear stress on surface element of shaft
(d) a longitudinal shear stress and a circumferential shear stress on a surface element of shaft

[HPPSC 2016-II]

Ans. (d) :



Torsion produces shear stress in the section as shown in figure:-



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1376. A solid shaft of diameter 100 mm, length 1000 mm is subjected to a twisting moment of M_t. The maximum shear stress developed in shaft is 60 N/mm^2. A hole of 50 mm diameter is now drilled throughout the shaft. To develop a maximum shear stress of 60 N/mm^2 in hollow shaft, the torque must be reduced by :

(a) $\frac{M_t}{4}$ (b) $\frac{M_t}{8}$

(c) $\frac{M_t}{12}$ (d) $\frac{M_t}{16}$

[HPPSC 2016-II]

Ans. (d) : Torsion equation:-

$$\frac{T}{J}=\frac{\tau}{R}=\frac{G\theta}{L}$$

$$\therefore \tau=\frac{T.R}{J}\Rightarrow 60\ \left(N/mm^2\right)=\frac{M_t\times\frac{100}{2}(mm)}{\frac{\pi\times100^4}{32}\left(mm^4\right)}$$

$M_t = 11780972.45$ Nmm

When 50 mm hole is drilled, $\tau_{max} = 60 N/mm^2$

$$\Rightarrow 60N/mm^2=\frac{M_h\times\frac{100}{2}mm}{\frac{\pi}{32}\left(100^4-50^4\right)mm^4}$$

$M_h = 11044661.67$ Nmm

$$\therefore \frac{M_h}{M_t}=0.9375$$

$\therefore$ Reduction in torque

$$=\left(M_t-M_h\right)=0.0625M_t=\frac{M_t}{16}$$

1377. If a circular shaft is subjected to a torque T and a bending moment M, the ratio of the maximum shear stress to the maximum bending stress is given by :

(a) 2M/T (b) 2T/M

(c) T/2M (d) M/T

JPSC AE (Pre) 2019
ESE 1997

Ans. (c) :

$$\tau=\frac{TR}{J};\ \sigma_b=\frac{My}{I}$$

For $$\frac{\tau_{max}}{\sigma_{max}}=\frac{\left(\frac{TR}{J}\right)}{\left(\frac{My}{I}\right)} \quad .....(i)$$

Here $$R=\frac{D}{2};y=\frac{D}{2}$$

$$J=\frac{\pi}{32}\times D^4, I=\frac{\pi}{64}\times D^4$$

putting these in eqn. (i)–

$$\boxed{\frac{\tau_{max}}{\sigma_{max}}=\frac{T}{2M}}$$

1378. A circular shaft is subjected to a twisting moment T and bending moment M. The ratio of maximum bending stress to maximum shear stress is given by :

(a) 2M/T (b) M/T

(c) 2T/M (d) M/2T

GPSC AE Class (1&2) Paper - 2 2017
ESE 1996, 2001

Ans. (a) : $\sigma_{maximum}=\frac{32M}{\pi d^3}$

$$\tau_{maximum}=\frac{16T}{\pi D^3}$$

$$\frac{\sigma_{max}}{\tau_{max}}=\frac{32M/\pi d^3}{16T/\pi d^3}=\frac{2M}{T}$$

1379. A shaft of length L having torsional rigidity GJ is subjected to a torque T. The corresponding stiffness shall be :

(a) $\frac{GJ}{L}$ (b) $\frac{TG}{L}$

(c) $\frac{TJ}{GL}$ (d) $\frac{TJ}{L}$

Rajasthan JEn (Degree) 2016

Ans. (a) : We know that torsion equation

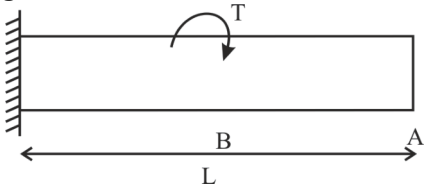
$$\frac{T}{J}=\frac{\tau}{R}=\frac{G\theta}{L}$$

$$\frac{T}{J}=\frac{G\theta}{L}$$

$$\boxed{\frac{T}{\theta}=\frac{GJ}{L}}$$

$$\left(\text{Stiffness}=\frac{T}{\theta}\right)$$

1380. A bar of circular cross-section of diameter D is subject to a torque T at B as shown in figure given below.



What is the angle of twist at A?

(a) Same as that at B (b) Zero

(c) Twice as that at B (d) Half as that at B

GPSC AE (CLASS 1 & 2) 2019
ESE 2005

Ans : (a) The angle of twist is same all over the length of shaft.

1381. The maximum shear stress induced in a solid circular shaft of diameter 15 cm, when the shaft transmits 150 kW power at 180 rpm, will be

(a) 16 N/mm^2 (b) 14 N/mm^2

(c) 12 N/mm^2 (d) 10 N/mm^2

ESE 2019

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Ans. (c)

Diameter 'd' = 15cm.

Shaft transmits 'P' = 150 kW.

N = 180 rpm.

Power transmitted 'P' = $\frac{2\pi NT}{60}$

$$150 = \frac{2\pi(180)T}{60}$$

$$T = \frac{150}{6\pi}$$

[T = 7.95 kN–m]

The maximum shear stress for solid circular section;

$$\tau_{max} = \frac{16T}{\pi d^3}$$

$$\tau_{max} = \frac{16\times(7.95\times10^6)}{\pi(150)^3}$$

$$\tau_{max} = \frac{127.2\times10^6}{10.6\times10^6}$$

$$\tau_{max} = 12MPa$$

$$\boxed{\tau_{max} = 12\ N/mm^2}$$

1382. Which one of the following statements is correct for the rotating shafts transmitting power?

(a) Lower the frequency of shaft lower will be the torque

(b) Higher the frequency of shaft lower will be the torque

(c) Frequency of the shaft does not influence the torque

(d) Higher the frequency of shaft higher will be the torque

ESE 2019

Ans. (b) : The main purpose of a shaft is to transmit its power to another member.

Let N be the revolutions per minute (rpm), T be the average torque N and ω be the angular speed of the shaft.

Work done per minute = Force × Distance

= (Average Torque) × (Angular displacement).

Power, $P = T \times \frac{2\pi N}{60}$

or $\left[P = \frac{2\pi NT}{60}\right]$

Frequency, $\left[N \propto \frac{1}{T}\right]$

So, Higher the frequency of shaft lower will be the torque.

1383. The required diameter for a solid shaft to transmit 400 kW at 150 rpm, with the working shear stress not to exceed 80 MN/m^2, is nearly

(a) 125 mm (b) 121 mm

(c) 117 mm (d) 113 mm

ESE 2018

Ans. (c) : Solid Shaft having power (P) = 400 kW

N = 150 rpm

Working shear stress (τ) = 80 MPa

Now, power transmission by the shaft are–

$$P = \frac{2\pi NT}{60}$$

$$T = \frac{60P}{2\pi N} = \frac{60\times400\times1000}{2\pi\times150}$$

[T = 25464.79 N-m]

$$\Rightarrow \frac{T}{J} = \frac{\tau}{r}$$

$$\Rightarrow \frac{25464.79}{\frac{\pi}{32}\times(D)^4} = \frac{80}{\left(\frac{D}{2}\right)}$$

$$D^3 = \left(\frac{25464.79\times16\times10^3}{\pi\times80}\right)$$

$$\boxed{D = 117.48mm}$$

Or

Max shear stress (τ_{max}) = $\frac{16T}{\pi D^3}$

$$D = \left(\frac{16T}{\pi\tau_{max}}\right)^{1/3} = \left(\frac{16\times25464.79}{\pi\times80}\right)^{\frac{1}{3}}$$

$$\boxed{D = 117.5m}$$

1384. A solid shaft *A* of diameter *D* and length *L* is subjected to a torque *T*; another shaft *B* of the same material and of the same length, but half the diameter, is also subjected to the same torque *T*. The ratio between the angles of twist of shaft *B* to that of shaft *A* is

(a) 32 (b) 16

(c) 8 (d) 4

ESE 2018

Ans. (b)

Shaft A	Shaft B
Dia = D	Dia = $\frac{D}{2}$
Length = L	Length = L
Torque = T	Torque = T

We know Torque Equation

$$\left[\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}\right]$$

Now, For the angle of twist–

$$\theta = \frac{TL}{GJ},$$

$$\left[\theta \propto \frac{1}{J}\right] \quad .....as\left[J = \frac{\pi D^4}{32}\right]$$

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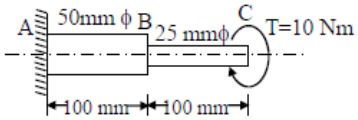
For Shaft A and B,

$$\frac{\theta_B}{\theta_A}=\frac{J_A}{J_B}=\frac{\pi D^4}{\frac{32}{\frac{\pi}{32}\left(\frac{D}{2}\right)^4}}$$

$$\boxed{\frac{\theta_B}{\theta_A}=16}$$

1385. A stepped steel shaft is subjected to a clockwise torque of 10 Nm at its free end. Shear modulus of steel is 80 GPa. The strain energy stored in the shaft is



(a) 1.73 Nmm (b) 2.53 Nmm
(c) 3.46 Nmm (d) 4.12 Nmm

ESE 2017

Ans. (a)
Torque 'T' = 10 Nm
Shear Modulus 'G' = 80 GPa
The strain energy due to torsion

$$U=\frac{T^2L}{2GJ}$$

$$=\frac{\left(10\times10^3\right)^2(100)}{2\times80\times10^3}\left[\frac{32}{\pi\times25^4}+\frac{32}{\pi\times50^4}\right]$$

$$U=\frac{10\times10^9}{160\times10^3}\ [0.00002607+0.000001629]$$

$$U=\frac{276997.466}{160\times10^3}$$

$$\boxed{U=1.73N-mm}$$

1386. Consider the following statements:

1. The shear stress distribution across the section of a circular shaft subjected to twisting varies parabolically.
2. The shear stress at the centre of a circular shaft under twisting moment is zero.
3. The shear stress at the extreme fibres of a circular shaft under twisting moment is maximum.

Which of the above statements is/are *correct?*

(a) 1, 2 and 3 (b) 1 only
(c) 2 only (d) 3 only

ESE 2017

Ans. (c)
- A shaft subjected to torsion, shear stress varies linearly (not in a parabolic manner)
- The torsional shear stress is zero at the centre of shaft
- The torsional shear stress is zero on extreme surface plane. The shear stress develops on the cross-sectional plane.

1387. If,
A = Cross sectional area
E = Young's modulus of elasticity
G = Modulus of rigidity
J = Polar moment of inertia
Then Torsional rigidity is given by,

(a) AE (b) GE
(c) EI (d) GJ

ISRO Scientist/Engineer 2013, ESE 2012

Ans : (d) Torsional Rigidity : The minimum force required to deform an object by twisting through a unit dimension.
It is the product of modulus of rigidity and polar moment of Inertia.
i.e., [K=GJ] Unit = Nm^2

$$\frac{T}{J}=\frac{G\theta}{L}$$

$$T=GJ\frac{\theta}{L}$$

$$\therefore\left\{\frac{\theta}{L}=1\right\}$$

$$\boxed{T=K=GJ}$$

1388. A circular shaft of diameter 'D' is made of a material for which Young's Modulus of Elasticity is 'E' and Poisson's ratio is 'μ'. The ratio of flexural rigidity to torsional rigidity for the shaft is

(a) $(1+2\mu)$ (b) $(1+\mu)$
(c) $(1-\mu)$ (d) $(1-2\mu)$

APTRANSCO 2017
AP TRANSCO AEE 2017, ESE 2015

Ans : (b) : Flexural rigidity = EI
Torsional rigidity = GJ

$$\frac{\text{Flexural rigidity}(EI)}{\text{Torsional rigidity}(GJ)}=\frac{2G(1+\mu)\times\frac{\pi D^4}{64}}{G\frac{\pi D^4}{32}}=(1+\mu)$$

1389. If the diameter of a shaft, subjected to a torque alone, is doubled, then the horse power "P" can be increased to :

(a) 16P (b) 8P
(c) 4P (d) 2P

OPSC AEE 2019 Paper–I
APPSC AEE 2012, ESE 1997

Ans. (b) : We know that, $P=\frac{2\pi NT}{60}$ and $\tau=\frac{16T}{\pi d^3}$

By both equation –

$$P=\frac{2\pi N}{60}\cdot\frac{\tau\pi d^3}{16}$$

So , $P\propto d^3$

$$\frac{P_2}{P_1}=\left(\frac{d_2}{d_1}\right)^3\Rightarrow\frac{P_2}{P_1}=\left(\frac{2d}{d}\right)^3$$

(Given, d_1 =d and d2 = $2d_1$ = 2d)

$$\boxed{P_2=8P_1}$$

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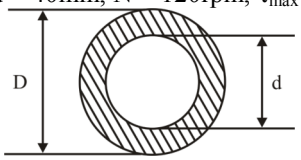
1390. A hollow shaft of external and internal diameters as 100 mm and 40 mm respectively is transmitting power at 120 rpm. If the shearing stress is not to exceed 50 MPa, the power the shaft can transmit will be :

(a) 100 kW (b) 120 kW
(c) 140 kW (d) 160 kW

ESE 2020

Ans. (b) : Given,
D = 100mm, d = 40mm, N = 120rpm, τ_{max} = 50 MPa



From torsion equation,

$$\tau_{allowable} = \frac{T \times \frac{d}{2}}{\frac{\pi}{32}(D^4 - d^4)}$$

$$50 = \frac{T \times \left(\frac{100}{2}\right)}{\frac{\pi}{32}(100^4 - 40^4)}$$

$$\boxed{T = 9.566 kN - m}$$

Maximum power which can be transmitted

$$P = \frac{2\pi NT}{60}$$

$$P = \frac{2\pi \times 120 \times 9.56}{60}$$

P = 120.13 kN.m/sec
P = 120.13 kW

$$\boxed{P \simeq 120 kW}$$

1391. If circular shaft is subjected to both Torque T and Bending Moment M. then the equivalent torque to is given by

(a) $T_e = M^2 + T^2$ (b) $T_e = \sqrt{M^2 + T^2}$
(c) $T_e = M^3 + T^3$ (d) $T_e = \sqrt{M^3 + T^3}$

TNPSC AE 2019

Ans. (b) The circular shaft is subjected to both Torque (T) and Bending Moment (M) has equivalent torque of–

$$\boxed{T_e = \sqrt{M^2 + T^2}}$$

1392. Two shafts, one solid and the other hollow, are made of the same materials and are having same length and weight. The hollow shaft as compared to solid shaft is

(a) Less strong (b) More strong
(c) Same strength (d) Can't be comparable

TNPSC AE 2019

Ans. (b) As given solid & hollow section have same material, weight & length
∴ Weight of solid section = weight of hollow section
γ×volume of Solid section=γ×volume of hollow section

$$\gamma \times \frac{\pi}{4} \times (d_s)^2 \times l = \gamma \times \frac{\pi}{4} \times \left(d_o^2 - d_i^2\right) \times l$$

$$d_s^2 = d_o^2 - d_i^2$$

∴ We know $\frac{\tau}{J} = \frac{I}{R} = \frac{G\theta}{L}$

Torsional strength is represented by torsional stiffness

i.e. , $K = \frac{GJ}{L} = \frac{T}{\theta}$

$$\frac{K_{hollow}}{K_{solid}} = \frac{\pi/32\left(d_o^4 - d_i^4\right)}{\pi/32} \frac{}{d_s^4} = \frac{\left(d_o^2\right)^2 - \left(d_i^2\right)^2}{d_s^4}$$

$$\frac{K_{hollow}}{K_{solid}} = \frac{\left(d_o^2 - d_i^2\right)\left(d_o^2 + d_i^2\right)}{\left(d_o^2 - d_i^2\right)^2}$$

$$\frac{K_{hollow}}{K_{solid}} = \frac{d_o^2 + d_i^2}{d_o^2 - d_i^2} > 1$$

$K_{hollow} > K_{Solid}$
So hollow is more stronger than solid section.

1393. A solid circular shaft is to transmit a power, P kW when turning N revolutions per min. For a given maximum shear stress, the shaft diameter will be proportional to :

(a) $(P/N)^{0.31}$ (b) $(P/N)^{0.33}$
(c) $(P/N)^{0.67}$ (d) $(P/N)^3$
(e) $(P/N)^{0.5}$

CGPSC AE 2014 Shift–I

Ans. (b) $P = T \times \omega$

$$P = T \times \frac{2\pi N}{60}$$

$$T = \frac{P \times 60}{2\pi N}$$

$$\tau_{max} = \frac{T \times r}{J} = \frac{Td}{2 \times J}$$

$$= \frac{P \times 60 \times d}{2\pi N \times 2\frac{\pi d^4}{32}}$$

$$d^3 \propto \frac{P}{N} \Rightarrow d \propto \left(\frac{P}{N}\right)^{1/3} \Rightarrow \boxed{d \propto \left(\frac{P}{N}\right)^{0.33}}$$

1394. If a shaft is simultaneously subjected to torque T/2 and a bending moment of 2M, then the ratio of maximum bending stress to maximum shear stress will be–

(a) 4M/T (b) 8M/T
(c) 4T/M (d) 8T/M

Gujarat PSC AE (N.W.R.) 2020

Ans. (b) : $\sigma_{max} = \frac{32M}{\pi d^3}$

$$\tau_{max} = \frac{16T}{\pi d^3}$$

$$\frac{\sigma}{\tau} = \frac{2M}{T} = \frac{2 \times 2M}{T/2} \Rightarrow \frac{\sigma}{\tau} = \frac{8M}{T}$$

Given,

$$\text{Torque} = \frac{T}{2}$$

Bending moment = 2 M

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1395. A solid shaft 125 mm in diameter transmits 120 kW at 160 r.p.m. The mean torque to which the shaft is subjected is approximately

(a) 7162 N-m (b) 7.2 N-m
(c) 93.75 N-m (d) 15 N-m

ASSAM PSC CCE (PRE) 2015

Ans : (a) : Given that

$P = 120\ kW = 120 \times 10^3\ W$

$N = 160\ rpm$

$$\text{Power } (P) = \frac{2\pi NT}{60}$$

$$\text{Torque } (T) = \frac{60 \times P}{2\pi N} = \frac{60 \times 120 \times 10^3}{2 \times 3.14 \times 160}$$

$$\boxed{T = 7162\ N-m}$$

1396. A 60 mm diameter shaft is subjected to a torque of 4 kN-m. The maximum shear stress induced in the shaft is

(a) 47.15 N/mm^2 (b) 94.30 N/mm^2
(c) 23.60 N/mm^2 (d) 69.08 N/mm^2

ASSAM PSC CCE (PRE) 2015

Ans : (b) : Given that,

$D = 60\ mm$

$T = 4\ kN-m$

$= 4 \times 10^6\ N-mm$

$$\text{Maximum shear stress } (\tau) = \frac{16T}{\pi D^3}$$

$$\boxed{\tau = 94.31\ N/mm^2}$$

1397. If a shaft is simultaneously subjected to a torque T and bending moment M, the maximum shear stress is

(a) $\frac{16}{\pi D^3}(M+T)$

(b) $\frac{16}{\pi D^3}(M+\sqrt{M^2+T^2})$

(c) $\frac{16}{\pi D^3}(M-\sqrt{M^2+T^2})$

(d) $\frac{16}{\pi D^3}\sqrt{M^2+T^2}$

APPSC AEE MAINS 2016

Ans. (d) : Combined effect of bending and torsion

- Principal stress $= \frac{16}{\pi D^3}\left[M \pm \sqrt{M^2+T^2}\right]$
- Maximum shear stress $= \frac{16}{\pi D^3}\sqrt{M^2+T^2}$
- Equivalent Bending moment $= \frac{1}{2}\left[M+\sqrt{M^2+T^2}\right]$
- Equivalent torque $= \sqrt{M^2+T^2}$

1398. The torque that produces a twist of one radian in a shaft per unit length is called

(a) Modulus of elasticity (b) Bulk modulus
(c) Torsion (d) Torsional rigidity

APPSC AEE MAINS 2016
ESE 2012, 2013

Ans. (d) : By torsion equation

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

$$\frac{T}{J} = \frac{G\theta}{L}$$

$$T = \frac{\theta}{L}.GJ$$

Here, given, $\theta = 1$ Radian and Length is unity.

So, $\boxed{T = GJ}$

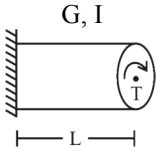
GJ is known as torsional rigidity.

1399. Strain energy stored in a solid shaft due to application of Torque 'T' at free end while other end is fixed, if G is shear modulus, J is polar moment of inertia, and L is the length of shaft is/will be

(a) $\frac{TL^2}{GJ}$ (b) $\frac{T^2L^2}{2GJ}$
(c) $\frac{2TL^2}{GJ}$ (d) $\frac{T^2L}{2GJ}$

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) : G, I



$$U = \frac{1}{2}T\theta = \frac{1}{2}T\left(\frac{TL}{GJ}\right) = \frac{T^2L}{2GJ}$$

1400. Two solid shafts 'A' and 'B' are made of the same material. The shaft 'A' is of 50 mm diameter and shaft 'B' is of 100 mm diameter. The strength of shaft 'B' is _____ of that of shaft 'A'.

(a) one-half (b) double
(c) four times (d) eight times

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) :

Shaft A	**Shaft B**
G	G
$d_A = 50$ min	$d_B = 100$ min
T_A	T_B

For tension equation $T = Z.Z_P$

$T \propto Z_P$

$$Z_P = \frac{\pi}{16}d^3$$

$$\frac{T_B}{T_A} = \frac{(Z_P)_B}{(Z_P)_A} = \frac{d_B^3}{d_A^3} = \left(\frac{100}{50}\right)^3 = 8$$

1401. Arrange the following sections in increasing torsional stiffness :

1. Open ring section
2. Close ring section
3. L-section
4. Circular disk section

(a) 1, 2, 3, 4 (b) 3, 1, 2, 4
(c) 3, 2, 1, 4 (d) 4, 3, 1, 2

GPSC AE March 2018

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Ans. (b) : Torsional stiffness = k = $\frac{GI_p}{L}$

By comparing I_p of section.

$k_{L-\text{section}} < k_{open\,ring} < k_{close\,ring} < k_{circular\,disk}$

1402. Torsional failure surface of ductile material occurs at

(a) transverse plane
(b) 60° to the transverse plane
(c) 45° to the transverse plane
(d) any random plane

GPSC AE March 2018

Ans. (a) : Ductile material (Ex - Steel, Copper etc.) fails due to shear, which, in case of torsion occurs at 90^0 to the axis.

1403. The maximum shear stress produced in a shaft is 5 N/mm^2. The shaft is of 40 mm diameter. The value of twisting moment is

(a) 628 Nm (b) 62.8 Nm
(c) 125.6 Nm (d) 1256 Nm

GPSC AE March 2018
ESE 2004, 2009

Ans. (b) : τ_{max} = 5 N/mm^2, d = 40 mm

$$\frac{T}{J} = \frac{\tau}{r}$$

$$J(\text{Polar moment}) = \frac{\pi d^4}{32}$$

$$\text{Twisting moment } T = \frac{\tau J}{r} = \frac{\tau \pi d^4}{32\left(\frac{d}{2}\right)}$$

$$T = 5(\pi)\frac{(40)^3}{16}$$

T = 62.8 N–m

1404. If a shaft is subjected to pure twisting moment, an element on the surface is subjected to

(a) normal tensile stress
(b) normal compressive stress
(c) pure shear stress
(d) bending stress

GPSC AE March 2018

Ans. (c) : Pure shear is a three-dimensional, homogeneous flattering of a body. Pure shear is different from simple shear. As pure shear involves or rigid body rotation as pure twisting moment.

1405. A circular shaft, when subjected to torsion, undergoes a twist of 1^0 in a length of 1.2 m. The modulus of rigidity of the shaft is 0.8×10^5 N/mm^2. If the maximum shear stress induced in the shaft is 80 N/mm^2, the radius of the shaft should be :

(a) $\frac{108}{\pi}$ (b) $\frac{144}{\pi}$
(c) $\frac{216}{\pi}$ (d) $\frac{266}{\pi}$

UPSC JWM 2017

Ans. (c) : $\theta = 1^0 \times \frac{\pi}{180}$ Radian

G = 0.8×10^5 N/mm^2

τ_{max} = 80 N/mm^2

L = 1.2×10^3 mm

$$\frac{\tau_{max}}{r} = \frac{G\theta}{L}$$

$$\frac{80}{r} = \frac{0.8\times10^5 \times \pi}{180\times1.2\times10^3}$$

$$r = \frac{216}{\pi}\text{mm}$$

1406. If 'CI_p' is torsional rigidity and 'l' is length of shaft, then torsional stiffness is .....

(a) $\frac{CI_p}{l}$ (b) $\frac{l}{CI_p}$
(c) CI_p (d) $\frac{1}{CI_p}$

GPSC R&B AE 2016

Ans. (a) : Torsion rigidity is the torque that can be transmitted by a shaft while undergoing unit angular twist in unit length (CI_P).

Torsional stiffness is the torque that can be transmitted by a shaft while undergoing angular twist.

$$\text{Torsional stiffness} = \frac{CI_p}{l}$$

l = length of shaft

1407. Polar modulus of shaft is ratio of:

(a) $\frac{\text{Polar moment of inertia}}{\text{Maximum radius}}$
(b) $\frac{\text{Polar moment of inertia}}{\text{Maximum diameter}}$
(c) $\frac{\text{Maximum diameter}}{\text{Polar moment of inertia}}$
(d) $\frac{\text{Maximum radius}}{\text{Polar moment of inertia}}$

GPSC R&B AE 2016

Ans. (a) : Polar modulus is basically defined as the ratio of polar moment of inertia (J) to the radius of shaft (R).

Polar modulus $(Z_p) = \frac{J}{R}$

1408. The ratio of the torsional moment of resistance of a solid circular shaft of diameter D and a hollow circular shaft of external diameter d with same material is :

(a) $\frac{D^4}{D^4 - d^4}$ (b) $\frac{D^3}{D^3 - d^3}$
(c) $\frac{D^4 - d^4}{D^4}$ (d) $\frac{D^3 - d^3}{D^3}$

OPSC Civil Services Pre 2011
ESE 2012, 2002

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Ans. (a) :
The polar modulus for a solid circular shaft

$$Z_1 = \frac{\pi D^3}{16}$$

The polar modulus for a hollow circular shaft

$$Z_2 = \pi\frac{\left(D^4 - d^4\right)}{16D}$$

$$\frac{Z_1}{Z_2} = \frac{\pi D^3}{16} \times \frac{16D}{\pi\left(D^4 - d^4\right)} = \frac{D^4}{D^4 - d^4}$$

1409. The weight of a hollow shaft shall be calculated using :

(a) $\rho g\left[\frac{\pi}{4}(D_0^2 - D_i^2) \times L\right]$

(b) $\rho g\left[\frac{\pi}{4}(D_0^2 + D_i^2) \times L\right]$

(c) $\frac{\rho g}{L}\left[\frac{\pi}{4}(D_0^2 - D_i^2)\right]$

(d) $\frac{\rho g}{L}\left[\frac{\pi}{4}(D_0^2 + D_i^2)\right]$

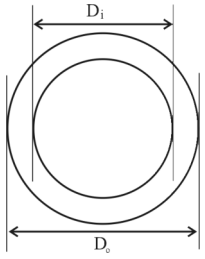
TRB Poly. Lect. Tech. 2017

Ans. (a) :

Weight of shaft = m × g

$= \rho\, vg = \rho \times (A \times L) \times g$

$= \rho g\left[\frac{\pi}{4}\left(D_0^2 - D_i^2\right)L\right]$



1410. Angular twist in a solid circular shaft of diameter 'd' and length 'L' subjected to torque "T" is θ. Angular twist, in another similar shaft of half of the length subjected to same Torque 'T', is :

(a) 0.25 θ (b) 0.5 θ

(c) θ (d) 2 θ

TRB Poly. Lect. Tech. 2017

Ans. (b) : $\frac{T}{J} = \frac{N\theta}{L}$

$$\frac{T}{\pi d^4/32} = \frac{N\theta}{L}$$

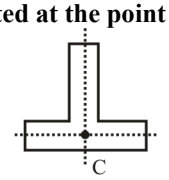
$$\theta = \frac{TL \times 32}{\pi d^4 N}$$

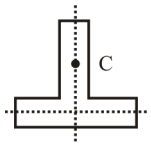
$$\theta' = \frac{T(L/2) \times 32}{\pi d^4 N}$$

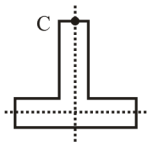
$$\theta' = \frac{\theta}{2} = 0.5\theta$$

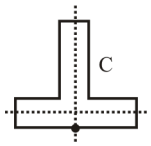
Angular twist is directly proportional to length of shaft and independent of radius, if Torque is same and length is halved then twist will be also halved.

1411. In a thin - walled T-section, the shear centre is located at the point shown is :

(a) 

(b) 

(c) 

(d) 

TRB Poly. Lect. Tech. 2017
ESE 1995

Ans. (a) : Shear centre of sections consisting of two intersecting narrow rectangles always lies at the intersection of centre lines of two rectangles.

1412. The angular twist of a shaft of radius R mm and length *l* mm subjected to a torque of 70 kN mm (G = 70 GPa) is

(a) $\frac{10l}{\pi R^4}$ (b) $\frac{2l}{\pi R^4}$

(c) $\frac{4l}{\pi R^4}$ (d) $\frac{15l}{\pi R^4}$

TRB Poly. Tech. Lect. 2012

Ans. (b) : We know that,

$$\frac{\tau}{r} = \frac{T}{J} = \frac{G.\theta}{L}$$

$$\theta = \frac{T.L}{J.G}$$

$$= \frac{70 \times 10^3 \times \ell}{\frac{\pi}{32}(2R)^4 \times 70 \times 10^3} = \frac{32\ell}{\pi R^4 \times 16}$$

$$\boxed{\theta = \frac{2\ell}{\pi R^4}}$$

1413. When a circular shaft is subjected to torsion, shear stress induced in the shaft at its centre shall be

TSGENCO_2015

OR

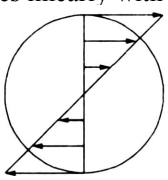
The shear stress at the centre of the circular shaft under torsion is :

(a) Zero (b) Minimum

(c) Maximum (d) Infinity

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : The shear stress at the centre of circular shaft under torsion is zero.
i.e., shear stress varies linearly with radius



Shear stress variation in shaft.

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1414. A shaft turns at 150 rpm under a torque of 1500 Nm. Power transmitted is

(a) 15π kW (b) 10π kW

(c) 7.5π kW (d) 5π kW

APPSC AEE 2012
ESE 1999

Ans. (c) : Given:
N = 150 rpm, T = 1500 Nm

$$\text{Power} = \frac{2\pi NT}{60} = 2\pi \times 150 \times 1500 = 2\pi \times 225000$$

Power = 7.5π kW

1415. The maximum shear stress occurs at the outside radius and equals :

(a) $\tau_{max} = \frac{2R}{\pi T^2}$ (b) $\tau_{max} = \frac{2R}{\pi T^3}$

(c) $\tau_{max} = \frac{2T}{\pi R^2}$ (d) $\tau_{max} = \frac{2T}{\pi R^3}$

TRB Poly. Lect. Tech. 2017

Ans. (d) : Effect of pure torsion on shaft –

$$\tau_{max} = \frac{16T}{\pi D^3}$$

$$\tau_{max} = \frac{16T}{\pi (2R)^3}$$

$$\tau_{max} = \frac{16T}{\pi (8R^3)}$$

$$\tau_{max} = \frac{2T}{\pi R^2}$$

τ_{max} = Maximum shear stress
T = Torque
D = Diameter of shaft

1416. While designing a mechanical system, a rectangular bar of length (*l*), width (b) and depth (d), where b>d, is subjected to torsion. The maximum shear stress will occur at:

(a) The corner
(b) Along the diagonal
(c) The center
(d) Middle of longer side

UPRVUNL AE 2016

Ans.: (d) While designing a mechanical system, a rectangular bar of length (*l*), width (b) and depth (d) when b > d, is subjected to torsion . Then the maximum shear stress will occur at middle of longer side.

1417. What is the polar moment of inertia (in cm^4) for a solid circular shaft of radius 10 cm?

(a) 7850 (b) 15700

(c) 3925 (d) 18500

GPSC AE June 2019

Ans. (b) : Polar moment of inertia for a solid circular shaft

$$= \frac{\pi D^4}{32} = \frac{\pi (20)^4}{32}$$ (Given R=10 cm, so D=20cm).

= 15707 cm^4

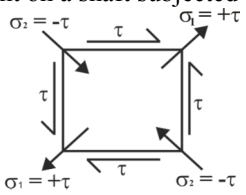
1418. A solid circular shaft of length L, cross-section A, second polar moment of area J and shear modulus G has one end fixed. A torsion T is applied at the other end. The volumetric strain at an arbitrary point on the surface will be

(a) 0 (b) $\frac{T}{(GJ)}$

(c) $\frac{TL}{(GJ)}$ (d) $\frac{T}{(AL)}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Element on a shaft subjected to torsion is



From figure principal stress

$\sigma_1 = +\tau$
$\sigma_2 = -\tau$
$\sigma_3 = 0$

Volumetric strain, $\in_v = \frac{\sigma_1 + \sigma_2 + \sigma_3}{E}(1-2\mu)$

$= \frac{(+\tau)+(-\tau)+0}{E}(1-2\mu)$

$\in_v = 0$

1419. A shaft turns at 200 rpm under a torque of 1800 Nm. The power transmitted is

(a) 6π kW (b) 12π kW

(c) 24π kW (d) 36π kW

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : $P = \frac{2\pi NT}{60}$

$= \frac{2\pi(200)(1800)}{60}$

$= 12000\ (\pi)$

$= (12\ \pi)$ kW

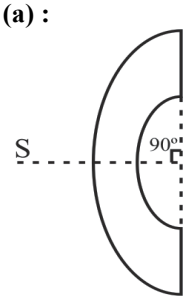
1420. Shear centre of a semi-circular arc of radius r is at

(a) $4r/\pi$ (b) $3r/\pi$

(c) $2r/\pi$ (d) r/π

GPSC AE Dec 2018
ESE 1998

Ans. (a) :



$$e = \frac{4r}{\pi}$$

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1421. Shear centre is the point in or outside a section through which the shear force applied produces .....

(a) only twisting
(b) only bending
(c) twisting and bending
(d) no twisting and no bending

GPSC AE Dec 2018

Ans. (b) : Shear centre is the point in or outside a section through which the shear force applied produces only bending.

1422. Torsional rigidity of a shaft is given by

(a) $\frac{T}{\ell}$ (b) $\frac{T}{J}$

(c) $\frac{T}{\theta}$ (d) $\frac{T}{r}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Torsion Equation

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{r}$$

T = Torque in N-mm
l = length of the shaft in mm
R = Radius of the circular shaft in mm
G = Modulus of rigidity shaft material in N/mm^2

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{r}$$

$$GJ=\frac{T\ell}{\theta}$$

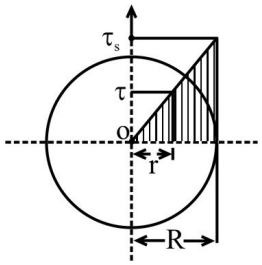
GJ is known as torsional rigidity of the shaft. It is important to note that the relative stiffness of two shafts is measured by the inverse ratio of the angles of twist is equal length of shafts when subjected to equal torques.

1423. Shear stress for a circular shaft due to torque varies

(a) from surface to centre parabolically
(b) from surface to centre linearly
(c) from centre to surface parabolically
(d) from centre to surface linearly

APPSC AEE (Civil/Mechanical) 2012

Ans. (d) Shear stress for a circular shaft due to torque varies from centre to surface linearly.



$$\frac{T}{J}=\frac{G\theta}{l}=\frac{\tau}{r}$$

The above relation states that the intensity of shear stress at any point in the cross-section of a shaft subjected to pure torsion is proportional to its distance from the centre.

1424. If two shafts of the same length, one of which is hollow, transmit equal torques and have equal maximum stress, then they should have equal

(a) angle of twist
(b) polar modulus of section
(c) polar moment of inertia
(d) diameter

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) It two shafts of the same strength, one of which is hollow, transmit equal torque and have equal maximum stress, then they should have equal Polar modulus of section.

$$\frac{T}{J}=\frac{\tau}{r}$$

$$T=J\left(\frac{\tau}{r}\right)$$

$$T=\left(\frac{J}{r}\right).\tau$$

If torque and Shear stress is maximum then Polar modulus section 'equal.

1425. A circular shaft subjected to torsion undergoes a twist of 1^0 in a length of 1 m. If the maximum shear stress induced is 120 MPa and the rigidity modulus is 0.8×10^5 MPa, the radius of the shaft in mm should be

ESE 2009

OR

A circular shaft subjected to torsion undergoes a twist of 1^0 in a length of 1.2 m. If the maximum shear stress induced is 100 MPa and the rigidity modulus is 0.8×10^5 MPa, the radius of the shaft in mm should be

(a) $\frac{270}{\pi}$ (b) $\frac{\pi}{270}$

(c) $\frac{180}{\pi}$ (d) $\frac{\pi}{180}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (a) Twist angle $(\theta)=1^0$

length of shaft = 1.2 m
Maximum shear stress = 100 MPa
Modulus of rigidity $(G)=0.8\times10^5$ MPa.

Torsion equation $\frac{T}{J}=\frac{G\theta}{l}=\frac{\tau}{r}$

$$\frac{100\times10^6}{r}=\frac{0.8\times10^5\times\pi\times10^6}{180\times1.2}$$

$$\boxed{r=\frac{270}{\pi}\text{mm}}$$

1426. Two shafts are of same length and same material. The diameter and maximum shear stress of the second shaft is twice that of the first shaft. Then the ratio of power developed between the first and second shaft is

(a) 16 (b) $\frac{16}{3\sqrt{3}}$

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(c) $\frac{1}{16}$ (d) $\frac{3}{16}$

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Power develop (P) = Tω

Case 1st:-

$$T = \frac{\pi d^3 \tau}{16}$$

$$P_1 = \frac{\pi d^3 \tau}{16} \times \omega$$

Case 2nd :-

$$P_2 = \frac{\pi (2d)^3 \times 2\tau}{16} \times \omega$$

assume, both case angular speed same then

Ratio of power developed $\left(\frac{P_1}{P_2}\right) = \frac{1}{16}$

$$\boxed{\frac{P_1}{P_2} = \frac{1}{16}}$$

1427. In a circular shaft subjected to pure torsion, the maximum shear stress occurs at

(a) the axis of the shaft
(b) between surface and axis
(c) shear stress is uniform throughout the cross section
(d) the surface

GPSC Assistant Prof. 2016

Ans. (d) : ∴ Maximum shear stress (τ_{max}) occur at the surface of circular shaft subjected to pure torsion.

1428. If K_h is the torque resisting capacity of a hollow shaft and K_s is that of a solid shaft, of the same material, length and weight. Then:

(a) $K_h > K_s$ (b) $K_h < K_s$
(c) $K_h = K_s$ (d) None of the above

PTCUL AE 2016

Ans. (a) : Torque resting capacity–

for solid shaft = $\frac{\pi}{16}\tau d^3$

for hallow shaft = $\frac{\pi}{16}\tau\left(\frac{d_0^4 - d_i^4}{d_0}\right)$

On comparing formula–

$T_{hollow} > T_{solid}$ shaft

1429. Consider the following statements:

(1) Shape factor
(2) Length of beam
(3) Type of loading
(4) Yield strength of material

Which of these affect the influence length of plastic hinge in a beam:

(a) 1 only (b) 1 and 3
(c) 2 and 3 (d) 3 and 4

PTCUL AE 2016

Ans. (b) : Shape factor and type of loading affect the influence length of plastic hinge in beam.

Direction : Match list I with list II and select the correct answer using code given below the two list in each question.

1430.

List I	List II
A. Shear center	1. Tension
B. Principal plane	2. Slope
C. Fixed end	3. Shear stress
D. Middle third rule	4. Twisting

Code:

	A	B	C	D
(a)	4	3	2	1
(b)	3	1	4	2
(c)	4	1	2	3
(d)	4	2	3	1

PTCUL AE 2016
ESE 1996

Ans. (a) :

Shear center	The load on this point does not cause twisting
Principal plane	Shear stress at this plane is zero
Fixed End	Slope at fixed end is zero.
Middle third rule	Middle third rule can be used to apply load such that there is no tension on a rectangular cross-section

Direction : Match list I with list II and select the correct answer using code given below the two list in each question.

1431.

List I (Type of section)	List II (Shape of factor)
A. Rectangle	1. 1.27
B. Circle	2. 1.15
C. I-section	3. 1.5
D. Thin ring	4. 1.7

Code:

	A	B	C	D
(a)	4	3	1	2
(b)	1	4	3	2
(c)	3	4	1	2
(d)	3	4	2	1

PTCUL AE 2016

Ans. (d) :

Type of section		shape factor
Rectangle	–	1.5
Circle	–	1.69 (≈ 1.7)
I-Section	–	1.15
Thin ring	–	1.27

1432. A circular shaft subjected to torsion undergoes a twist of 1^0 in a length of 120 cm, if maximum shear stress induced is limited to 1000 kg/cm^2 and modulus of rigidity = 0.8×10^6 kg/cm^2, then the radius of the shaft should be

(a) π/18 (b) 18/π
(c) π/27 (d) 27/π

UJVNL AE 2016

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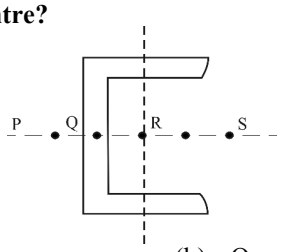
Ans. (d) : $\theta = 1^0$, length = 120 cm
$\tau_{max} = 1000\ kg/cm^2$
$G = 0.8 \times 10^6\ kg/cm^2$
using torsion equation = $\frac{\tau}{R} = \frac{G\theta}{L}$
$\frac{1000}{R} = \frac{0.8\times10^6\times1\times\pi}{180\times120}$
$R = 27/\pi$

1433. A solid shaft can resist a bending moment of 3.0 kN-m and a twisting moment of 4.0 kN-m together, then the maximum torque that can be applied is
(a) 7.0 kN-m (b) 4.5 kN-m
(c) 3.5 kN-m (d) 5.0 kN-m
UJVNL AE 2016

Ans. (d) : Bending moment = 3 kN-m
Twisting moment = 4 kN-m
Maximum Torque $= \sqrt{(3)^3 + (4)^2} = 5$ kN-m

1434. The possible location of the shear centre of the channel section, shown below :
APTRANSCO 2017
AP TRANSCO AEE 2017
OR
In the symmetrical channel section shown in the figure below, which point is likely to be the shear centre?



(a) P (b) Q
(c) R (d) S
ESE 2002

Ans : (a) The possible location of the shear centre of the channel section is at point P.

2. A 60 mm diameter shaft is subjected to torque of 6 kN-m, C= 8 × 10^4 N/mm^2. The maximum shear stress included in the shaft will be
(a) $\frac{8000}{9\pi}$ (b) $\frac{4000}{9\pi}$
(c) $\frac{12000}{9\pi}$ (d) $\frac{16000}{9\pi}$
AP TRANSCO AEE 2017
ESE 2013

Ans. (b) : Given,
Diameter of shaft = 60 mm
Torque = 6 kN-m
$C = 8 \times 10^4\ N/mm^2$
$\frac{T}{J} = \frac{\tau}{R}$
$\tau = \frac{6\times10^6\times60/2}{\frac{\pi}{32}(60)^4} = \frac{4000}{9\pi}$

1435. A shaft rotating at 150rpm is subjected to a torque of 1500 N.m. Horse power transmitted by the shaft is –
(a) p^2 (b) p
(c) 10 p (d) 1/10 p
UPPCL AE 2014

Ans. (c) : $P = T \times \omega$
$= \frac{1500\times2\times r\times150}{60}$
$P = 7500\ r$ watt
Horse power $= \frac{7500r}{745.7}$
$= 10.5\ r$
$= 10\ r$

1436. In shafts with keyways, the allowable stresses are usually_____ of the value given.
(a) 25% (b) 50%
(c) 75% (d) 95%
APPSC (PHE) AE Paper-2 2012

Ans. (c) : In shafts with keyway, the allowable stress are usually 75%

1437. A solid circular shaft is subjected to pure torsion. The ratio of maximum shear to maximum normal stress at any point would be
(a) 1:1 (b) 1:2
(c) 2:1 (d) 2:3
APPSC (PHE) AE Paper-2 2012
ESE 2013

Ans. (a) :
D = Dia of solid circular shaft
T = Torque or twisting moment
Maximum shear stress $(\tau_{max}) = \frac{16T}{\pi D^3}$ .....(i)
Maximum normal stress in case of shaft subjected to combined bending and torsion.
$(\sigma_b)_{max} = \frac{16\pi}{D^3}\left[M + \sqrt{M^2 + T^2}\right]$
In case of pure torsion;
Bending moment (M) = 0
$(\sigma_b)_{max}$ = Maximum normal stress
$\therefore (\sigma_b)_{max} = \frac{16T}{\pi D^3}$ ..... (ii)
$\frac{\tau_{max}}{(\sigma_b)_{max}} = \frac{16T/\pi D^3}{16T/\pi D^3} = \frac{1}{1}$
$\tau_{max} : (\sigma_b)_{max} = 1:1$

1438. A solid circular shaft is subjected to a maximum shearing stress of 140 MPa. The magnitude of the maximum normal stress developed in the shaft is
(a) 140 MPa (b) 80 MPa
(c) 70 MPa (d) 60 MPa
APPSC (PHE) AE Paper-2 2012

Ans. (a)
Maximum normal stress = Maximum shear stress
= 140 MPa

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1439. For shaft in torsion

(a) $\frac{q}{r}=\frac{T}{J}$ (b) $\frac{C\theta}{l}=\frac{T}{J}$

(c) $\frac{q}{r}=\frac{C\theta}{l}$ (d) All the above

APPSC AEE 2016

Ans. (d) : Torsion equation

$$\frac{q_{max}}{R}=\frac{q}{r}=\frac{T}{J}=\frac{C\theta}{L}$$

q = shear stress
C = shear modulus
r = distance from center of shaft
T = torque
L = length of shaft
θ = angle of twist
J = polar moment of Inertia

1440. The ratio of maximum shear stress of a solid shaft of diameter D to that of a hollow shaft having external diameter D and internal diameter 0.5D is

(a) 0.50 (b) 1.00
(c) 1.50 (d) 0.94

APPSC AEE 2016

Ans. (d) : max shear stress for solid shaft

$$\tau_{max_1}=\frac{16T}{\pi D^3}$$

max. shear stress for hollow shaft

$$\tau_{max_2}=\frac{16T}{\pi D^3\left[1-\left(\frac{d}{D}\right)^4\right]}$$

$$\text{Ratio}=\frac{\tau_{max_1}}{\tau_{max_2}}=\frac{\frac{16T}{\pi D^3}}{\frac{16T}{\pi D^3\left[1-\left(\frac{.5D}{D}\right)^4\right]}}$$

$= 0.94$

1441. Hoop stress for a wooden wheel with steel flat tyre in terms of outside diameter of the wheel *D*, inside diameter of the tyre *d* and Young's modulus *E* is

(a) E(D-d)/d (b) E(D/d)
(c) E(D/(D-d)) (d) E(d/D)

APPSC AEE 2016

Ans. (a) : Hoop stress for wooden wheel

$$=\frac{E(D-d)}{d}$$

E = modulus of elasticity
D = External diameter of wheel
d = Internal diameter of wheel

1442. A solid circular shaft of 60 mm diameter transmits a torque of 1600 Nm. The value of maximum shear stress developed is

(a) 37.72 MPa (b) 47.72 MPa
(c) 57.72 MPa (d) 67.72 MPa

UPRVUNL AE 2015

Ans. (a) : Explanation give that :
d = 60 mm, T = 1600 Nm = 1600 × 10^3 Nmm
Maximum shear stress = ?

$$\tau_{max}=\frac{16T}{\pi D^3}$$

$$=\frac{16\times1600\times10^3}{60^3\times77}$$

$$\boxed{\tau_{max}=37.72\,MPa}$$

1443. The power transmitted by the shaft is :

(a) $P=\frac{2\pi NT}{60}\times1000$ (b) $P=\frac{2\pi NT}{30}\times1000$

(c) $P=\frac{2\pi NT}{30}$ (d) $P=\frac{2\pi NT}{1000}$

(e) $P=\frac{2\pi NT}{2000}$

CGPSC AE 2017
APPSC AEE MAINS 2016
ESE 2012

Ans. (a) Power transmitted (P) by the shaft is

$$P=\frac{2\pi NT}{60}\times1000 \text{ watt}$$

P = Torque × angle turned per second
N = Rotation per minute (rpm)
T = Torque

1444. A shaft of diameter d subjected to a bending moment of M and a torque of T experiences:

(a) Maximum bending stress = $\frac{32M}{\pi d^3}$

(b) Maximum shear stress = $\frac{16T}{\pi d^3}$

(c) Maximum bending stress = $\frac{16T}{\pi d^3}$

(d) Both (a) and (b)
(e) Neither (a) nor (b)

Haryana PSC AE 2018 Paper–I

Ans. (d) : Torsional formula

$$\frac{\tau}{R}=\frac{T}{I_P},\ \tau=\frac{TR}{I_p},\ I_p=\frac{\pi}{32}D^4$$

$$\tau_{max}=\frac{TR}{\frac{\pi}{32}D^4}=\frac{16T}{\pi D^3}$$

Maximum bending formula:-

$$\frac{M}{I}=\frac{\sigma}{y},\ \sigma=\frac{My}{I},\ I=\frac{\pi}{64}D^4$$

$y = D/2$

$$\sigma_{max}=\frac{My}{\frac{\pi}{64}D^4}=\frac{32M}{\pi D^3}$$

1445. Strain energy in torsion of a shaft per unit volume is given by (q is maximum shear stress, E is modulus of elasticity and G is modulus of rigidity):

(a) $q^2/2G$ (b) q^2/G

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(c) $q^2/4G$ (d) $q^2/2E$

Haryana PSC AE 2017
ESE 2001

Ans. (c) :

$$(U) = \frac{1}{2}T\theta = \frac{1}{2}\frac{T^2L}{GJ} = \frac{T^2}{4G} \times \text{Vol. of shaft}$$

$$= \text{S.E/volume} = \frac{\tau^2}{4G} i.e. \frac{q^2}{4G} (\tau = q)$$

1446. A shaft turns at 200 rpm under a torque of 1800 Nm. The power transmitted is

(a) 6π kW (b) 12π kW
(c) 24π kW (d) 36π kW

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : $P = \frac{2\pi NT}{60} = \frac{2\pi(200)(1800)}{60}$

$= 12000\ (\pi) = (12\ \pi)$ kW

1447. A hollow shaft having internal diameter of 8 mm and external diameter of 9 mm and solid shaft having diameter of 6 mm are twisted by same twisting moment M_t. The two shafts are made from same material and have equal lengths. The ratio of torsional stiffness of hollow shaft to solid shaft is :

(a) 0.95 (b) 3.80
(c) 1.90 (d) 1.00

HPPSC Poly. Lect. 2016

Ans : (c) $\boxed{\frac{T}{J} = \frac{\tau}{R} = \frac{C\theta}{l}}$

$\frac{C\theta}{l} = \text{constant}$

Then, $\boxed{T \propto J}$

$$\frac{T_{Hollow}}{T_{Solid}} = \frac{\frac{\pi}{32}\left(D_H^4 - d_H^4\right)}{\frac{\pi}{32} \times D_s^4}$$

$$\frac{T_{Hollow}}{T_{Solid}} = \frac{9^4 - 8^4}{6^4} = 1.902$$

1448. When a member is subjected to twisting moment the stress in any elemental ring will be proportional to:

(a) its radial distance from center
(b) Modulus of Rigidity of Material
(c) Young's Modulus of Material
(d) the area of cross-section

JUVNL AE 2016

Ans. (a) : For twisting moment–

torsional stress $\tau = \frac{Tr}{J} = \frac{Tr}{\frac{\pi r^4}{2}}$

So, $\tau \propto \frac{1}{r^3}$

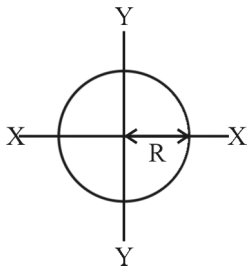
r = radial distance from centre

1449. Polar moment of inertia of a circular section with radius R is given by (Symbols and notations carry their usual meaning):

(a) $\frac{\pi R^3}{32}$ (b) $\frac{\pi R^4}{2}$
(c) $\frac{\pi R^4}{16}$ (d) $\frac{\pi R^4}{32}$

JUVNL AE 2016

Ans. (b) :



Polar moment of Inertia = $J = I_{zz} = I_{xx} + I_{yy}$

$$= \frac{\pi D^4}{64} + \frac{\pi D^4}{64} = \frac{\pi D^4}{32}$$

$$\boxed{J = \frac{\pi R^4}{2}}$$

1450. The polar moment of inertia of a hollow shaft of outer diameter (D) and inner diameter (d) is

(a) $\frac{\pi}{16}(D^4 - d^4)$ (b) $\frac{\pi}{32}(D^4 - d^4)$
(c) $\frac{\pi}{64}(D^4 - d^4)$ (d) none of these

TNPSC AE 2012

Ans. (b) Given, D = outer diameter
d = inner diameter

Moment of inertia $(I_{xx}) = \frac{\pi}{64}\left(D^4 - d^4\right)$

Moment of inertia $(I_{yy}) = \frac{\pi}{64}\left(D^4 - d^4\right)$

Polar moment of inertia (Izz) = $I_{xx} + I_{yy}$

$$= \frac{\pi}{64}\left(D^4 - d^4\right) + \frac{\pi}{64}\left(D^4 - d^4\right)$$

$$I_{ZZ} = \frac{\pi}{32}\left(D^4 - d^4\right)$$

1451. Let N be the revolutions per minutes (rpm) and T be the average torque (N-m), then power P transmitted by the shaft is :

(a) $P = \frac{2\pi T}{60N}$ W (b) $P = \frac{\pi T}{60N}$ W
(c) $P = \frac{2\pi N}{60T}$ W (d) $P = \frac{2\pi NT}{60}$ W

DMRC AM 2020

Ans. (d) : Power transmitted by the shaft

$$P = \frac{2\pi NT}{60} \text{ W}$$

N= rpm, T = torque (N-m)

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1452. Polar moment of inertia of a circular area is

(a) $\frac{\pi}{32}d^4$ (b) $\frac{\pi}{64}d^4$

(c) $\frac{\pi}{4}d^4$ (d) $\frac{\pi}{4}d^3$

APPSC (PHE) AE Paper-2 2012

Ans. (a) : d = Dia of circular area
For a circular area –

Moment of Inertia $(I_{xx}) = \frac{\pi}{64}d^4$

Moment of Inertia $(I_{yy}) = \frac{\pi}{64}d^4$

$\therefore$ Polar moment of inertia $(I_{zz}) = I_{xx} + I_{yy}$

$$= \frac{\pi}{64}d^4 + \frac{\pi}{64}d^4$$

$$I_{zz} = \frac{\pi}{32}d^4$$

9. Thick and Thin Cylinders and Springs

1453. The deformation of a spring produced by a unit load method is called-

(a) Flexibility (b) Stiffness
(c) Unit strain (d) None of these

OPSC AEE 2015 PAPER–I

The deformation of a spring produced by a unit load is called

(a) flexibility (b) stiffness
(c) influence coefficient (d) unit strain

BPSC AE 2019 Paper (V) Section-I

Ans. (a) : Flexibility is the inverse of stiffness–

$$\left[\text{stiffness} \propto \frac{1}{\text{flexibility}}\right]$$

Stiffness of a spring is defined as force required to cause unit deflection.
Flexibility of a spring is defined as deflection produced by an unit load.

1454. The proof load of a leaf spring is:

(a) deflection per unit load
(b) Load required flatten the spring
(c) load required to cause 2% permanent deflection
(d) load per unit deflection

APPSC Poly. Tech. Lect. 2020

Ans. (b) : The proof load of a leaf spring is load required to flatten the spring.
The load at which the plates become straight is called proof load.

1455. Pick the correct statement about the maximum bending stress in the various leafs of laminated spring, assuming it has been designed ideally

(a) It increase uniformly from the shortest leaf to the longest leaf
(b) It is largest in the longest leaf
(c) It is largest in the shortest leaf
(d) It is same in all leaf

RPSC A.En. 2013

Ans : (d) Bending stress in laminated leaf spring is as follows
w = Load acting
l = span of the spring
n = no. of plates
b = width of plates
t = thickness of plates

$$\boxed{f = \frac{3}{2}\frac{WL}{nbt^2}}$$

It is same for all leaves.

1456. The hoop stress induced in a thick cylinders due to radial pressure will be :

(a) Tensile (b) Compressive
(c) Shear (d) Bond

GPSC AE Class (1&2) Paper - 2 2017

Ans. (a) : Hoop stress is always be tensile in nature it also known as circumferential stress-
–Longitudinal stress will be tensile in nature
–Radial stress will be compressive in nature.

1457. If a steel tyre is heated and struck on a rigid wheel, after cooling the tyre will be subjected to

(a) Bending (b) Torsion
(c) Hoop stress (d) Compression

APPSC AEE MAINS 2016

Ans. (c) : If a steel tyre is heated and struck on a rigid wheel offer cooling the tyre will be subjected to hoop stress.

1458. In a thick cylinder subjected to internal pressure, the hoop stress is ..... at the outer surface and is ..... at the inner surface

(a) min, min (b) max, max
(c) max, min (d) min, max

APPSC AEE MAINS 2016

Ans. (d) : In a thick cylinder subject to internal pressure the hoop stress in minimum at the outer surface and is maximum at the inner surface.

1459. A thin cylinder is subjected to an external fluid pressure. The hoop stress will be

(a) Compressive (b) Tensile
(c) Bending stress (d) Zero

Punjab PSC SDE 2016

Ans. (a) : The hoop stress will be compressive outside and inside the thin and in thick cylinder hoop stress will be tensile.

1460. The stiffness of the spring is the

(a) Load required for breaking the spring
(b) Load required to compress the spring up to shearing proportional limit
(c) Load required to produce unit deflection
(d) Load per coil of the spring

Punjab PSC SDE 2016

Ans. (c) : Stiffness of spring or spring constant is load per unit deflection. It is expressed in N/m or kN/m.

1461. The intensity of circumferential stress is

(a) equal to the longitudinal stress
(b) half of the longitudinal stress

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(c) double the longitudinal stress
(d) triple the longitudinal stress

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (c) : For thin cylinder,

Longitudinal stress = $\sigma_l = \frac{Pd}{4t}$

circumferential stress / hoop stress = $\sigma_h = \frac{Pd}{2t}$

The intensity of circumferential stress = $2 \times \frac{Pd}{4t}$

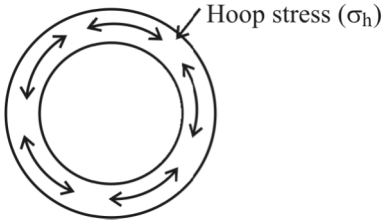
1462. If p is internal pressure in a thin cylinder of diameter d and thickness t, the developed hoop stress is :

(a) $\frac{pd}{t}$ (b) $\frac{pd}{2t}$

(c) $\frac{pd}{3t}$ (d) $\frac{pd}{4t}$

OPSC Civil Services Pre 2011

Ans. (b)



$H_P = \frac{Pd}{2t}$

Hoop stress : Hoop stress is the stress induced in the hallow cylinder or hollow sphere which acts tangentially to the circumference of the cylinder and sphere in direction due to the action internal fluid pressure. It is tensile in nature.

1463. Thickness of cylindrical shell is designed on the basis of:

(a) Longitudinal stress (b) Hoop stress
(c) Circumferential stress (d) Tensile stress

PTCUL AE 2016

Ans. (b) : Thickness of cylindrical shell is designed on the basis of hoop stress.

1464. Pressure vessels are made of

(a) non-ferrous materials
(b) sheet metal (steel)
(c) cast iron
(d) Any of the above

APPSC (PHE) AE Paper-2 2012

Ans. (d) : Pressure vesseles are made of cast iron, shear metal (steel) and non-ferrous materials.

1465. A thin cylindrical shell of diameter 'd', length 'l' and thickness 't' is subjected to internal pressure p. Poisson's ratio of the material is μ. The ratio of longitudinal strain to hoop strain is:

(a) $\frac{pd}{2t}$ (b) $\left(\frac{\mu-2}{2\mu-1}\right)$

(c) $\frac{pd}{2t}(1-\mu)$ (d) $\left(\frac{2\mu-1}{\mu-2}\right)$

UJVNL AE 2016

Ans. (d) :

Ratio of longitudinal strain to hoop strain $= \frac{2\mu-1}{\mu-2}$

Hoop stress $(\sigma_h) = \frac{pd}{2t}$

Longitudinal stress /axial stress $(\sigma_l = \sigma_z) = \frac{pd}{4t}$

1466. In case of laminated springs, the load at which the plates become straight is called.

(a) Working load (b) Safe load
(c) Proof load (d) None of the above

APPSC (PHE) AE Paper-2 2012

Ans. (c) : In case of laminated springs, the load at which the plates become straight is called proof load.

1467. Thin cylinders are frequently required to operate under pressure upto

(a) 5 MN/m^2 (b) 15 MN/m^2
(c) 30 MN/m^2 (d) 250 MN/m^2

APPSC (PHE) AE Paper-2 2012

Ans. (c) : Thin cylinder are frequently required to operate under pressure up to 30 MN/m^2.

1468. In a thick cylinder pressurized from inside, the hoop stress is maximum at

(a) the centre of the wall thickness
(b) the outer radius
(c) the inner radius
(d) both inner and outer radii

APPSC (PHE) AE Paper-2 2012

Ans. (c) : When a thick cylinder pressurized from inside, the hoop stress is maximum at the inner radius.

1469. Autofrettage is the method of

(a) joining thick cylinders
(b) calculating stresses in thick cylinders
(c) prestressing thick cylinders
(d) increasing the life of thick cylinders

APPSC (PHE) AE Paper-2 2012

Ans. (c) : Autofrettage is the method of prestressing thick cylinders.

1470. In thick cylinders, the circumferential stress :

(a) is zero
(b) varies along the thickness
(c) does not vary along the thickness
(d) is maximum at the centre
(e) is related to none of the other options

CGPSC AE 2017

Ans. (b) In thick cylinders, the circumferential stress varies along the thickness.

1471. The design of thin cylindrical shells is primarily based on:

(a) Hoop stress
(b) Arithmetic mean of the hoop and the longitudinal stress
(c) Geometric mean of the hoop and longitudinal stress

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(d) Longitudinal stress
(e) All of the above

Haryana PSC AE 2018 Paper–I

Ans. (a) : The hoop stress should be design criteria for thin cylindrical shall subjected to internal pressure and designed minimum wall thickness can be estimated from design equation based on allowable value of Hoop stress.

1472. For a Poisson's ratio of 0.3, the ratio of volumetric strain to hoop strain in this spherical cell is

(a) 3 (b) 0.33
(c) 9.5 (d) None of the above

GPSC AE June 2019

Ans. (a) : In a thin spherical cell

hoop strain $\in_n = \frac{Pd}{4tE}(1-\mu)$

Volumetric strain $\in_V = \frac{3Pd}{4tE}(1-\mu)$

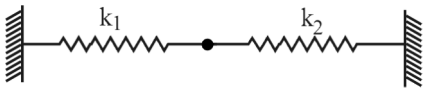
$$\boxed{\frac{\varepsilon_v}{\varepsilon_h} = 3}$$

1473. If two springs of stiffness k_1 and k_2 are connected in series, the stiffness of the combined spring is

(a) $\frac{(k_1+k_2)}{k_1k_2}$ (b) $\frac{k_1k_2}{(k_1+k_2)}$
(c) k_1k_2 (d) k_1+k_2

APPSC AEE (CIVIL/MECHANICAL) 2019
GPSC Assistant Prof. 2016
APPSC (PHE) AE Paper-2 2012
Arunachal Pradesh Poly. Tech. Lect. 2019
ESE 1995

Ans. (b) :



Given springs are connected in series,

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\therefore \quad k_{eq} = \frac{k_1k_2}{k_1+k_2}$$

1474. A closely - coiled helical spring of round steel wire 5 mm in diameter having 12 complete coils of 50 mm mean diameter is subjected to an axial load of 100 N. If modulus of rigidity is 80 GPa, the deflection of the spring will be :

(a) 36 mm (b) 32 mm
(c) 28 mm (d) 24 mm

ESE 2020, 2015

Ans. (d) : Given,

d = 5 mm, n = 12, R = $\frac{50}{2}$ = 25 mm

W = 100 N, G = C = 80 × 10^3 N/mm^2

We know that,

Deflection of the closed-coil helical spring

$$\delta = \frac{64WR^3n}{Cd^4}$$

$$= \frac{64\times100\times(25)^3\times12}{80\times10^3\times(5)^4}$$

$$\boxed{\delta = 24\text{mm}}$$

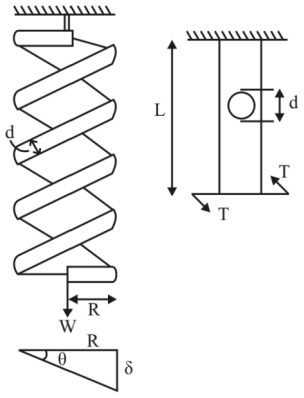
1475. The deflection δ of the closed coil helical spring is :

(a) $\frac{WR^2n}{8Cd^3}$ (b) $\frac{64WR^3n}{Cd^4}$
(c) $\frac{128WR^3n}{Cd^2}$ (d) $\frac{64WR^2n}{Cd^2}$

where :
W is the axial load
R is radius of the coil
n is the number of turns of coil
C is the modulus of rigidity
d is the diameter of the wire of the coil

ESE 2020

Ans. (b)



We know that,

$$\frac{T}{I_P} = \frac{C\theta}{L} = \frac{\tau}{r} \quad \text{..... (i)}$$

From above equation,

$$\theta = \frac{T\times L}{C\times I_p}$$

$$\theta = \frac{(W\times R)\times(2\pi Rn)}{C\times\frac{\pi d^4}{32}} \qquad [\because L = 2\pi Rn]$$

$$\boxed{\theta = \frac{64WR^2n}{Cd^4}} \quad \text{..... (ii)}$$

From above figure,

$\delta = R\theta$

putting value of from eq^n (ii)

$$\boxed{\delta = \frac{64WR^3n}{Cd^4}}$$

1476. If mass of the vibrating system in vertical modes is m and K is the spring constant at the site, then critical damping constant at the site will be:

(a) $\sqrt{Km}$ (b) $2\sqrt{Km}$

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(c) $\sqrt{\frac{K}{m}}$ (d) $2\sqrt{\frac{K}{m}}$

WBPSC Poly Lect. 2019

Ans. (b) : Critical damping constant = $2\sqrt{Km}$
m = mass of vibrating system
K = spring constant

1477. To double the period of oscillation of a simple pendulum with small swings
(a) Its length should be quadrupled
(b) The mass of the bob should be quadrupled
(c) Its length should be doubled
(d) The mass of the bob should be doubled

ISRO Scientist/Engineer 2019

Ans : (a) : We know that–

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T \propto \sqrt{l}$$

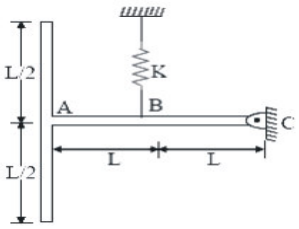
$T \rightarrow$ Period of oscillation
$l \rightarrow$ Length
So, to double the period of oscillation the length should be quadrupled (i.e. 4 times)

1478. The deformation of a spring produced by a unit load is called
(a) Stiffness (b) Flexibility
(c) Unit strain (d) None of these

ISRO Scientist/Engineer 2015
WBPSC AE 2012, APPSC AEE 2012

Ans. (b) : The deformation of spring produced by a unit load is called flexibility.

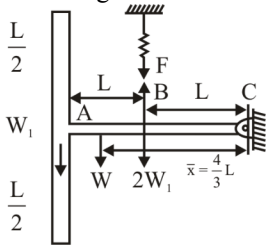
1479. A uniform T-shaped arm of weight W, pinned about a horizontal point C, is supported by a vertical spring of stiffness K. The extension of the spring is



(a) $\frac{3W}{4K}$ (b) $\frac{4W}{3K}$
(c) $\frac{3K}{4W}$ (d) $\frac{4K}{3W}$

ESE 2017

Ans. (b) : Assume weight of vertical member is W_1 also weight of horizontal member is $2W_1$.
Total composite bar weight is W.



Taking moments of weights about C.

$$\bar{x} = \frac{W_1(2L) + 2W_1(L)}{W_1 + 2W_1}$$

$$\boxed{\bar{x} = \frac{4}{3}L}$$

Total weight of composite bar acts at $\bar{x} = \frac{4}{3}L$ from C

$\Sigma M_C = 0$ (on FBD of composite bar)

$$F(L) = W\left(\frac{4}{3}L\right)$$

$$F = \frac{4}{3}W$$

$$\boxed{\delta = \frac{F}{K} = \frac{4}{3}\frac{W}{K}}$$

1480. A closely coiled helical spring made of 10 mm diameter steel wire has 15 coils of 100 mm mean diameter. The spring is subjected to an axial load of 100 N. For a modulus of rigidity of 8.16×10^4 N/mm^2, the stiffness of the spring will be nearly
(a) 5.9 N/mm (b) 6.8 N/mm
(c) 7.7 N/mm (d) 8.8 N/mm

ESE 2019

Ans. (b)
Diameter 'd' = 10mm
n = 15
R = 50mm -------- [D = 100mm]
P = 100N
$G = 8.16\times10^4$ N/mm^2

So, stiffness of spring, 'K' = $\frac{Gd^4}{64R^3n}$

$$K = \frac{\left(8.16\times10^4\right)\left(10^4\right)}{64\times(50)^3\times15}$$

$$K = \frac{81.6\times10^7}{12\times10^7}$$

$$\boxed{K = 6.8 N/mm}$$

1481. If two springs k_1 and k_2 are placed in parallel, the combined stiffness of spring is
(a) $k_1 + k_2$ (b) $k_1 + k_2 / (k_1 k_2)$
(c) $(k_1 k_2)/k_1 + k_2$ (d) $k_1 k_2$

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : Spring in parallel = $k_1 + k_2$

Spring in series = $\frac{1}{k_1} + \frac{1}{k_2}$

1482. A structural system consists of two springs, each of stiffness k, in parallel in- turn connected to third spring of same stiffness in series. A mass m is connected at the one end of the system and other end is fixed. What shall be the natural frequency of the system ?

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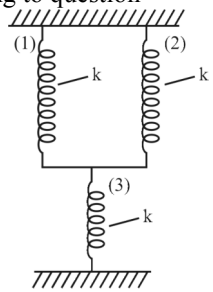
(a) $\sqrt{\frac{k}{m}}$ (b) $\sqrt{\frac{3k}{m}}$

(c) $\sqrt{\frac{3k}{2m}}$ (d) $\sqrt{\frac{2k}{3m}}$

Rajasthan JEN (Degree) Shift - II 2016

Ans. (d) According to question-



Spring (1) and (2) are parallel combined to each other–

$k_1 = k + k$

$k_1 = 2k$

k_1 and k are combined in series–

$$\frac{1}{k_{eq}} = \frac{1}{2k} + \frac{1}{k}$$

$$\boxed{k_{eq} = \frac{2}{3}k}$$

Natural frequency of system $(f) = \sqrt{\frac{k_{eq}}{m}}$

$$(f) = \sqrt{\frac{2k}{3m}}$$

1483. In the analysis of thin cylinders, assumptions made are-

(i) radial stress is neglected

(ii) the hoop and longitudinal stress distribution across section is uniform

Which statement/statements is/are correct?

(a) Both (i) and (ii) are true

(b) (i) is true and (ii) is false

(c) (i) is false and (ii) is true

(d) Both (i) and (ii) are false

BPSC AE Pre 2018

Ans. (a) : The following assumptions are made in order to derive the expressions for the analysis of thin cylinder –

(i) The diameter of the cylinder is more than 20 times the thickness of the shell.

(ii) The stresses are uniformly distributed through the thickness of the wall.

(iii) The ends of the cylindrical shell are not supported from sides.

(iv) The radial stresses are neglected.

(v) The weight of the cylinder and that of the fluid contained inside are not taken into account.

(vi) The atmospheric pressure is taken as the reference pressure.

1484. Maximum shear stress in a thin cylinder of thickness t and diameter d subjected to an internal fluid pressure p is equal to

(a) $\frac{pd}{8t}$ (b) $\frac{pd}{4t}$

(c) $\frac{pd}{2t}$ (d) None of these

(BPSC AE Paper V 2001)

Ans : (a) σ_l = longitudinal stress

$$\sigma_l \times \pi d \times t = \frac{\pi}{4} d^2 \times p$$

$$\therefore \boxed{\sigma_\ell = \frac{pd}{4t}}$$

Circumferential stress σ_c

$$\sigma_c \times 2tl = d \times l \times p$$

$$\boxed{\sigma_c = \frac{pd}{2t}}$$

If p is neglected;

Maximum shear stress in plane of σ_c and $\sigma_\ell = \frac{\sigma_c - \sigma_\ell}{2}$

$$\tau_{max} = \frac{\frac{pd}{2t} - -\frac{pd}{4t}}{2}$$

$$\tau_{max} = \frac{pd}{8t}$$

1485. In a thin cylindrical shell (D/t) ratio is 50. It is subjected to an internal fluid pressure of 2.5 MPa. What is the maximum hoop tension developed in the cylinder?

(a) 62.5 (b) 50

(c) 75 (d) 25

APTRASCO 2017

Ans : (a) Maximum hoop tension developed in the cylinder.

$$\sigma_h = \frac{pD}{2t} = \frac{2.5}{2} \times \frac{D}{t}$$

$$= \frac{2.5}{2} \times 50$$

$$= 62.5 \text{MPa}$$

1486. A thin walled cylindrical pressure vessel having radius 0.6 m and wall thickness 24 mm is subjected to an internal pressure of 1000 kP. The hoop stress (MPa) developed is

(a) 2.5 (b) 0.25

(c) 25 (d) 0.025

(e) 22.5

CGPSC AE 2014 Shift–I

Ans. (c) :

r = 0.6 m

t = 24 mm

P = 1000 kPa

$= 10^6$ Pa

$$\boxed{P = 10^6 \text{N/m}^2 = 1\text{MPa}}$$

$$\sigma_H = \frac{Pd}{2t} = \frac{1(\text{MPa}) \times 0.6 \times 2 \times 1000(\text{mm})}{2 \times 24}$$

$$\boxed{\sigma_H = 25\text{MPa}}$$

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1487. A water main of 180 cm diameter contains water at pressure head of 300m. The thickness of the shell required for water main, given that maximum permissible stressing metal is 600 kg/cm^2.will be

(a) 4.5 cm (b) 3.5 cm
(c) 2.5 cm (d) 1.5 cm

APPSC Poly. Tech. Lect. 2020

Ans. (a) : 4.5 cm

Given,

$$\sigma_h = 6000 kg/cm^2$$

$$d = 180cm$$

$$\frac{P}{\rho} = 300 \times 10cm$$

$$P = 300 \times 10^2 g/cm^2$$

$$P = 30 kg/cm^2$$

$$\sigma_h = \frac{Pd}{2t}$$

$$t = \frac{30 \times 180}{600 \times 2}$$

$$t = 4.5cm$$

1488. The ratio of circumferential stress to longitudinal stress in a thin cylinder subjected to internal hydrostatic pressure is

(a) 1/2 (b) 1
(c) 2 (d) 4

Gujarat PSC AE (N.W.R.) 2020
ASSAM PSC CCE (PRE) 2015
APPSC (PHE) AE Paper-2 2012

Ans. (c) :

$$\frac{\text{Circum perential stress}}{\text{Longitudinal stress}} = \frac{\sigma_C}{\sigma_L} = ?$$

When a thin cylinder subjected to internal hydrostatic pressure then there will be in the cylinder circumferential stress, Longitudinal stress and longitudinal stress is acting. But in case in thin cylinder, tangential stress is negligible.

We know that;

$$\sigma_C = \frac{Pd}{2t}$$

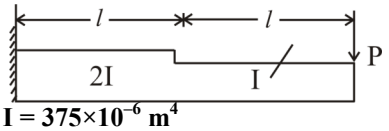
$$\sigma_L = \frac{Pd}{4t}$$

Where P = Internal pressure
d = Dia of cylinder
t = thickness of cylinder

therefore;

$$\frac{\sigma_C}{\sigma_L} = \frac{Pd/2t}{Pd/4t} = 2$$

1489.



I = 375×10^{-6} m^4
***l* = 0.5 m**
E = 200 GPa

Determine the stiffness of the beam shown in the above fig.

(a) 12×10^{10} N/m (b) 10×10^{10} N/m
(c) 4×10^{10} N/m (d) 8×10^{10} N/m

APPSC (PHE) AE Paper-2 2012

Ans. (a) : Given ;

$I = 375 \times 10^{-6}$ m^4

$l = 0.5$m

$E = 200$ GPa $= 200 \times 10^9$ Pa

$\therefore$ Bending moment $(M_x) = Px$

$$U = \int_0^l \frac{(Px)^2}{2EI} dx + \int_l^{2l} \frac{(Px)^2}{2E(2I)} dx$$

$$= \left[\frac{P^2x^3}{3 \times 2EI}\right]_0^l + \left[\frac{P^2x^3}{3 \times 2E \times 2I}\right]_l^{2l}$$

$$= \frac{P^2l^3}{6EI} + \frac{P^2(8l^3 - l^3)}{12\ EI} = \frac{P^2l^3}{6EI} + \frac{7P^2l^3}{12EI}$$

$$U = \frac{9P^2l^3}{12EI} = \frac{3P^2l^3}{4EI} \quad \text{..... (i)}$$

$$\because U = \frac{1}{2} P \times \delta = \frac{1}{2}\frac{P^2}{K} \quad \text{.....(ii)}$$

From equation (1) and (2)

$$\frac{1}{2}\frac{P^2}{K} = \frac{3P^2l^3}{4EI} \Rightarrow K = \frac{2EI}{3l^3}$$

$$K = \frac{2 \times 200 \times 10^9 \times 375 \times 10^{-6}}{3 \times (0.5)^3} = 4 \times 10^8\ N/m$$

No any option matching with the solution.

1490. Maximum value of shear stress in thin cylinders is equal to

(a) hoop stress (b) 2(hoop stress)
(c) 0.5(hoop stress) (d) 0.25(hoop stress)

APPSC AEE MAINS 2016

Ans. (c) : $\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{\frac{PD}{2t} - 0}{2} = \frac{PD}{4t}$

hoop stress $= \frac{PD}{2t}$

$$\tau_{max} = \frac{PD}{4t}$$

1491. If the hoop stress in a thin cylinder is 24 N/sq.mm, then its longitudinal stress is equal to

(a) 36 N/sq.mm (b) 24 N/sq.mm
(c) 12 N/sq.mm (d) 6 N/sq.mm

APPSC AEE MAINS 2016

Ans. (c) : Hoop stress $(\sigma_h) = \frac{Pd}{2t}$

Long. stress $(\sigma_\ell) = \frac{Pd}{4t} = \frac{Pd}{(2t)}\frac{1}{2}$

$$= \frac{24}{2} = 12 N/mm^2$$

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1492. A simply supported beam AB of L span and EI flexural rigidity is subjected to M_B moment at B. Then the rotation at A is given by

(a) $\Theta_A = \frac{ML}{6EI}$ (b) $\Theta_A = \frac{ML}{4EI}$

(c) $\Theta_A = \frac{ML}{3EI}$ (d) $\Theta_A = \frac{ML}{2EI}$

APPSC AEE MAINS 2016

Ans. (a) : $M = \frac{6EI}{L}\theta$

the rotation at A –

$\theta = \frac{ML}{6EI}$

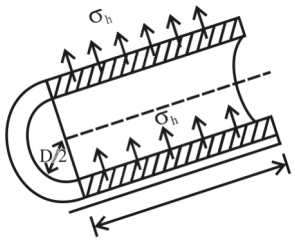
1493. What will be the circumferential stress developed in case of thin cylindrical pipe with diameter 'd' and thickness 't' subjected to internal pressure 'P'.

(a) $\frac{Pd}{4t}$ (b) $\frac{P}{t}$

(c) $\frac{Pd}{2t}$ (d) $\frac{P}{2t}$

GPSC Poly. Tech. Lect. 2016
ESE 2003

Ans. (c) : Hoop Stress of circumferential stress is given by:



$\sigma_h = \frac{Pd}{2t}$ (tensile)

1494. A thin cylinder of diameter d and thickness t with closed ends is subjected to an internal pressure p. If shear stress is restricted to q, what is the maximum permissible value of p?

(a) $\frac{2qt}{d}$ (b) $\frac{4qt}{d}$

(c) $\frac{8qt}{d}$ (d) $\frac{16qt}{d}$

UPSC JWM 2017

Ans. (c)

Maximum shear stress $= \frac{\sigma_h - \sigma_L}{2} = \frac{\frac{PD}{2t} - \frac{PD}{4t}}{2} = \frac{PD}{8t}$

$\tau_{max} = \frac{PD}{8t}$, $q = \frac{pD}{8t}$ $[\because \text{shear stress } \tau_{max} = q]$

$p = \frac{8qt}{D}$

1495. A spherical steel pressure vessel 400 mm in diameter with a wall thickness of 20 mm, is coated with a brittle lacquer that cracks when strain exceeds 200 × 10^{-7}. What internal pressure will cause the lacquer to develop cracks?
E = 200 GPa, μ = 0.3.

(a) 0.57 MPa (b) 1.14 MPa
(c) 2.28 MPa (d) 4.56 Mpa

TANGEDCO AE 2018

Ans. (b) :

$\varepsilon_L = \varepsilon_h = \frac{Pd}{4tE}(1-\mu)$

$200\times10^{-7} = \frac{P\times400}{4\times20\times200\times10^3}(1-0.3)$

P = 1.14 MPa

1496. A thin spherical shell of diameter d and thickness t is subjected to internal pressure p, the tensile stress in the shell will be

(a) $\frac{pd}{2t}$ (b) $\frac{pd}{4t}$

(c) $\frac{pt}{2d}$ (d) $\frac{pt}{4d}$

TSGENCO_2015

Ans. (b) : Tensile stress or longitudinal stress in thin spherical shell

$\sigma_t = \frac{Pd}{4t}$

d = dia of spherical shell
t = thickness
P = Internal pressure

1497. A 0.5 m internal diameter shell contains fluid at a limiting pressure of 500 N/m^2. If the thickness of the shell is 20mm, then what is the tensile stress (N/m^2) exerted on the shell?

(a) 4250 (b) 5500
(c) 6250 (d) 7200

UPRVUNL AE 2016

Ans.: (c) Internal diameter of shell d=0.5 meter
pressure P = 500 N/m^2
thickness of the shell t = 20 mm =0.020 m.

Tensile stress $= \frac{P.d}{2.t}$

$= \frac{500\times0.50}{2\times0.020} = 6250 N/m^2$

1498. If 'p' is the pressure on the cylinder, 'd' and 't' are the diameter and wall thickness, E is the Young's Modulus and m is the Pisson's Ratio, the volumetric strain (ε_v) in thin wall cylinder can be given by

(a) $\frac{pd}{2tE}(5-4\mu)$ (b) $\frac{pd}{4tE}(5-4\mu)$

(c) $\frac{pd}{6tE}(5+4\mu)$ (d) $\frac{pd}{8tE}(5-4\mu)$

GPSC AE Dec 2018

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Ans. (b) : $\varepsilon_v = 2\varepsilon_c + \varepsilon_l$

$$= 2\left[\frac{\sigma_c}{E} - \frac{\mu\sigma_l}{E}\right] + \left[\frac{\sigma_l}{E} - \frac{\mu\sigma_c}{E}\right]$$

$$= 2\left[\frac{pd}{2tE} - \frac{\mu pd}{4tE}\right] + \left[\frac{pd}{4tE} - \frac{\mu pd}{2tE}\right]$$

$$\varepsilon_v = \frac{5pd}{4tE} - \frac{\mu pd}{tE} = \frac{pd}{4tE}[5 - 4\mu]$$

1499. If 'p' is the pressure on the cylinder, 'd' and 't' are the diameter and wall thickness respectively, the maximum in-plane shear stress in thin wall of cylinder can be given by

(a) $\frac{pd}{2t}$ (b) $\frac{pd}{4t}$

(c) $\frac{pd}{6t}$ (d) $\frac{pd}{8t}$

GPSC AE Dec 2018

Ans. (d) :

$$\tau_{max} = \frac{\sigma_c - \sigma_l}{2}$$

$$= \frac{\frac{pd}{2t} - \frac{pd}{4t}}{2} = \frac{pd}{8t}$$

1500. The maximum bending moment at the center of laminated spring due to load W is given by:

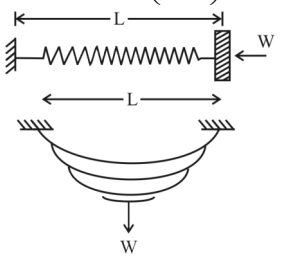
(a) $\frac{WL}{2}$ (b) 2WL

(c) WL (d) None of the above

PTCUL AE 2016

Ans. (d) : Maximum bending moment in spring for load

Bending moment maximum $\left(\frac{WL}{4}\right)$



1501. Cylinder are said to be thin, if their wall thickness t is given by formula if d is dia of cylinder:

(a) $t > \frac{d}{20}$ (b) $t < \frac{d}{20}$

(c) $t = \frac{d}{20}$ (d) None of the above

PTCUL AE 2016

Ans. (d) : Thin cylinder if $\frac{t}{d} < \frac{1}{20}$

Thick cylinder if $\frac{t}{d} > \frac{1}{20}$

t = thickness of cylindrical shell

d = cylinder diameter

1502. A long thin walled cylindrical shell, closed at both the ends, is subjected to an internal pressure. The ratio of the circumferential stress (hoop stress) to longitudinal stress developed in the shell is:

(a) 0.5 (b) 1.0

(c) 2.0 (d) 4.0

PTCUL AE 2016, UJVNL AE 2016 UPPCL AE 2014, ESE 1996

Ans. (c) : Hoop stress $(\sigma_{hoop}) = \frac{Pd}{2t}$

longitudinal stress $(\sigma_{long}) = \frac{Pd}{4t}$

$$\frac{\sigma_{hoop}}{\sigma_{long}} = \frac{Pd}{2t}\left(\frac{4t}{Pd}\right) = 2$$

1503. A thin circular plate of radius r and thickness t is subjected to radial stress σ throughout its circumference. The unit volume change of the entire plate is

(a) $\frac{\sigma}{E}(1-\mu)$ (b) $\frac{2\sigma}{E}(1-\mu)$

(c) $\frac{\sigma}{E}(1-2\mu)$ (d) $\frac{2\sigma}{E}(1-2\mu)$

WBPSC AE 2007, PTCUL AE 2016

Ans. (c) : $\frac{\Delta V}{V} = \frac{\sigma_x + \sigma_y + \sigma_z}{E}(1-2\mu)$

$$\frac{\Delta V}{1} = \frac{\sigma}{E}(1-2\mu)$$

$$\Delta V = \frac{\sigma}{E}(1-2\mu)$$

1504. A thin walled spherical shell is subjected to an internal pressure. If the radius of the shell is increased by 1% and the thickness is reduced by 1%, with the internal pressure remaining the same, the percentage change in the circumferential (hoop) stress is:

(a) 0 (b) 1

(c) 1.08 (d) 2.02

PTCUL AE 2016

Ans. (d) : Hoop stress $(\sigma_h) = \frac{pr}{2t}$ for sphere

$$(\sigma_\theta)_f = \frac{p.(1.01r)}{2(0.99t)}$$

$$\sigma_{\theta_f} = 1.0202\frac{pr}{2t}$$

$$\therefore \%\text{change in Hoop stress} = \frac{1.0202\frac{pr}{2t} - \frac{pr}{2t}}{\frac{pr}{2t}} \times 100$$

$$= 2.02$$

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1505. Wahl's correction factor (K) is given by the relation where S = Spring Index

(a) $K = \frac{3S-1}{3S-4} + \frac{0.615}{S}$ (b) $K = \frac{4S-1}{4S-4} + \frac{0.615}{S}$

(c) $K = \frac{5S-1}{5S-4} + \frac{0.615}{S}$ (d) $K = \frac{6S-1}{6S-4} + \frac{0.615}{S}$

APPSC (PHE) AE Paper-2 2012

Ans. (b) : Wahl's correction factor (k) = $\frac{4S-1}{4S-4} + \frac{0.615}{S}$

where;

$S = \frac{D}{d}$ = spring index

D = mean diameter of the coil

d = diameter of the wire of the coil

1506. The bursting pressure for a cold drawn seamless steel tubing of 60 mm inside diameter with 2 mm wall thickness is (the ultimate strength of steel is 380 MN/m^2)

(a) 25.33 MN/m^2 (b) 24.33 MN/m^2

(c) 26.33 MN/m^2 (d) 50.66 MN/m^2

APPSC (PHE) AE Paper-2 2012

Ans. (a) : For thin cylinder

$\sigma_C = \frac{Pd}{2t}$

P = bursting pressure = ?

d = Inside diameter of steel tube = 60 mm

t = wall thickness of seamless steel tube = 2 mm

σ_c = circumferential stress or brushing stress = ultimate strength of steel = 380 MN/m^2

$$P = \frac{\sigma_c \times 2t}{d} = \frac{380\ \frac{MN}{m^2} \times 2 \times 2mm}{60\,mm}$$

P = 25.33 MN/m^2

1507. The stresses in a thick cylinder subjected to uniform pressure vary proportional to

(a) r (b) 1/r

(c) r^2 (d) 1/r^2

APPSC (PHE) AE Paper-2 2012

Ans. (d) : From the Lame's equation for thick cylinder–

$\sigma_r = \frac{v}{r^2 - a}$

and $\sigma_c = \frac{b}{r^2} + a$

where; σ_r = radial stress

σ_c = circumferential stress

a & b is constant

if a = 0; then

$$4\left\{\begin{matrix}\sigma_r \propto \frac{1}{r_2} \\ \sigma_C \propto \frac{1}{r_2}\end{matrix}\right\}$$

1508. A close-coiled helical spring absorbs 80 N mm of energy while extending by 4 mm. The stiffness of the spring is

(a) 5 N/mm (b) 10 N/mm

(c) 16 N/mm (d) 20 N/mm

APPSC (PHE) AE Paper-2 2012

Ans. (b) : For a closed - coiled helical spring–

Absorbs energy (V) = $\frac{1}{2}$W.δ ..... (i)

Given; δ = deflection of coil under the load W = 4 mm

V = Absorbs energy = 80 N-mm

$\therefore$ $W = \frac{2V}{\delta}$ [From equation (i)]

$W = \frac{2 \times 80\ N-mm}{4\,mm} = 40\ N$

$\therefore$ Stiffness of spring (K) = $\frac{W}{\delta} = \frac{40N}{4mm} = 10$ N/mm

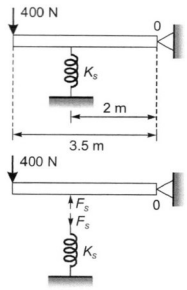
1509. A wooden diving board of length 3.5 m is hinged at one end and supported 2 m from this end by a spring with a constant of 35 kN/m. How much will the spring deflect if a young man welding 400 N stands at the end of the board?

(a) 0.002 m (b) 0.2 m

(c) 0.02 m (d) 2.0 m

[HPPSC AE 2016-II]

Ans. (c) :



$\sum M_o = 0$

$400 \times 3.5 = F_s \times 2$

F_s = 700 N

K_s =35 kN/m

$\Sigma M_A = 0$

$400 \times 3.5 = F_s \times 2 \Rightarrow F_s = 700N$

K_s = 35 KN/m

Deflection $\delta = \frac{F_s}{K_s} = \frac{700N}{35\,KN/m} = 0.02$ m

1510. In a closed coiled helical spring subjected to axial load, other quantities remaining the same, if the wire diameter is doubled, then the stiffness of the spring when compared to the original one, will become:

(a) Twice (b) 4 times

(c) 8 times (d) 16 times

KPSC AE 2016

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Ans. (d) : Stiffness of spring (k) $= \frac{Cd^4}{64R^3n}$

Now,

$$k_1 = \frac{Cd_1^4}{64R^3n} \text{ ..... (i)}$$

As per question,

$$d_2 = 2d_1$$

So,

$$k_2 = \frac{Cd_2^4}{64R^3n}$$

$$k_2 = \frac{C(2d_1)^4}{64R^3n} = \frac{16Cd_1^4}{64R^3n}$$

$$\boxed{k_2 = 16k_1} \text{ ..... [from eq}^{n} \text{ (i)]}$$

1511. Circumferential and longitudinal strains in a cylindrical boiler under steam pressure are ε_1 and ε_2 respectively. Change in volume of the boiler cylinder per unit volume will be

(a) $\varepsilon_1 + 2\varepsilon_2$ (b) $\varepsilon_1\varepsilon_2^2$

(c) $2\varepsilon_1 + \varepsilon_2$ (d) $\varepsilon_1^2\varepsilon_2$

APPSC (PHE) AE Paper-2 2012

Ans. (c) : Given, for cylindrical boiler steam pressure

ε_1= Circumferential strains

ε_2= Longitudinal strains

$\frac{\delta V}{V} = ?$

Where; δV = change in volume

V = initial volume of cylinder.

For a cylinder, volume (V) = $\frac{\pi}{4}d^2L$

$\therefore$ Change in the volume (δ_V) = $\frac{\pi}{4}d^2.dL + 2\frac{\pi}{4}Ld.\delta d$

$$\frac{\delta V}{V} = \frac{\delta L}{L} + 2.\frac{\delta d}{d}$$

$$= \varepsilon_1 + \varepsilon_2$$

hence;

Change in the volume of boiler cylinder per unit volume $= 2\varepsilon_1+\varepsilon_1$

1511. A body having weight of 1000 N is dropped from a height of 10 cm over a close coiled helical spring of stiffness 200 N/cm. The resulting deflection of spring is nearly

(a) 5 cm (b) 16 cm

(c) 35 cm (d) 100 cm

APPSC (PHE) AE Paper-2 2012

Ans. (b) : Work done by the falling weight = P (h+δ)

P (h+δ) = energy stored in the spring due to static load.

$\left[P(h+\delta) = \frac{1}{2}w\delta\right]$

[W= Kδ]

$P(h+\delta) = \frac{1}{2}(K\delta)\delta$

$2Ph + 2P\delta = K\delta^2$

$K\delta^2 - 2P\delta - 2Ph = 0$

$200\,\delta^2 - 2000\,\delta - 20000 = 0$

$\delta^2 - 10\delta - 100 = 0$

δ = 16 cm

1512. If p is the internal pressure in a thin cylinder of diameter d and thickness t, the developed hoop stress is:

(a) pd/3t (b) pd/4t

(c) pd/t (d) 2pd/t

(e) pd/2t

Haryana PSC AE 2018 Paper–I

Ans. (e) : Hoop's stress $(\sigma_h) = \frac{pd}{2t}$

'p' is internal fluid pressure.

Longitudinal stresses $(\sigma_L) = \frac{pd}{4t}$

't' = Thickness of cylinder

'd' = internal dia of cylinder.

1513. In closely coiled helical spring carrying an axial load 'W', diameter of the rod of the spring 'd', and 'R' is the mean radius of the coil. Every section of the rod is subjected to a torque 'WR'. Maximum shear stress at any section of the rod is ....., spring consists of 'n' turns.

(a) $\frac{16WR}{\pi d^3}$ (b) $\frac{64WR^3n}{cd^4}$

(c) $2\pi Rn$ (d) $\frac{60WR}{\pi d^3}$

GPSC R&B AE 2016

Ans. (a) : Torsional shear stress

$$= \frac{\tau_r}{J} = \frac{\frac{WD}{2}\cdot d/2}{\pi/32\,d^4} = \frac{8WD}{\pi d^3} = \frac{16WR}{\pi d^3}$$

Direct shear stress = $\frac{W}{A} = \frac{4W}{\pi d^2}$

R = mean radius of coil.

d = diameter of coil

W = axial load

1514. A thin cylindrical shell of internal diameter D and thickness t is subjected to internal pressure p, E and v are respectively the Elastic modulus and Poisson's ratio. The change in diameter is

(a) $\frac{pD^2}{4tE}(1-2\nu)$ (b) $\frac{pD^2}{4tE}(2-\nu)$

(c) $\frac{pt^2}{4DE}(2-\nu)$ (d) $\frac{pt^2}{4DE}(1-2\nu)$

APPSC AEE (CIVIL/MECHANICAL) 2019

ESE 2002

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Ans. (b) : Circumferential strain in thin cylinder

$$\in_h = \frac{(\sigma_h - \mu\sigma_l)}{E}$$

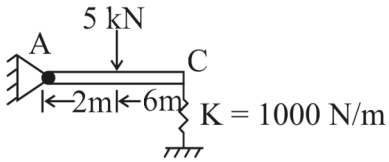
$$\in_h = \frac{\sigma_h}{E}\left(1 - \frac{\mu}{2}\right)$$

$$\left(\text{where } \sigma_h = \frac{PD}{2t}\right)$$

$$\therefore \quad \frac{\delta D}{D} = \frac{PD}{2tE}\left(\frac{2-\mu}{2}\right)$$

$$\therefore \quad \delta D = \frac{PD^2}{4tE}(2-\mu)$$

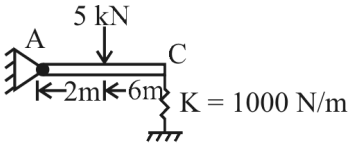
1515. Rigid bar AC shown in figure is supported by spring at C and loaded at B. Deflection of point B will be :



(a) 0.3125 mm (b) 0.4167 mm
(c) 1.25 mm (d) 0.626 mm

HPPSC Poly. Lect. 2016

Ans : (a)



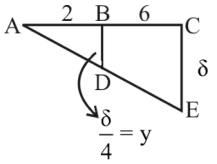
$\Sigma M_A = 0$

$5 \times 2 = R_C \times 8$

$R_C = 1.25$ kN

$1250 \text{ N} = k \times \delta$

$$\delta = \frac{1250}{1000} = 1.25 \text{ mm}$$



$$\therefore \ y = \frac{\delta}{4} = \frac{1.25}{4} = 0.3215 \text{ mm}$$

1516. A helical spring is subjected to an axial load W and corresponding deflection in the spring is δ. Now if the mean diameter of the spring is made half of its initial diameter keeping the material, number of turns and wire cross-section same, the deflection will be

(a) $\frac{\delta}{2}$ (b) $\frac{\delta}{8}$
(c) $\frac{\delta}{4}$ (d) 2δ

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : Deflection in spring $(\delta) = \frac{8WD^3n}{Gd^4} \Rightarrow \delta \alpha D^3$

$$\therefore \quad \frac{\delta_2}{\delta_1} = \left(\frac{D_2}{D_1}\right)^3 = \left(\frac{\frac{D}{2}}{D}\right)^3$$

$$\Rightarrow \quad \delta_2 = \frac{\delta}{8}$$

1517. The stiffness of a closed coiled spring is more when _____.

(a) wire diameter, rigidity modulus are more and mean radius, number of turns are lesser
(b) number of turns, rigidity modulus are more and mean radius, number of turns are lesser
(c) number of turns, rigidity modulus are more and mean radius, wire diameter, number of turns are lesser
(d) rigidity modulus is more and wire diameter, mean radius, number of turns are lesser

TSPSC Deputy Engineer 2015

Ans. (a) : Stiffness of spring (K)

$$= \frac{Gd^4}{64R^3n}$$

$$K \propto G, \ K \propto d^4, \ K \propto \frac{1}{R^3}, \ K \propto \frac{1}{n}$$

So, stiffness of a closed coiled spring is more when wire diameter rigidity modulus are more and mean radius, number of turns are lesser.

1518. A thin cylinder is subjected to an internal pressure of 40 MPa. If the longitudinal stress is 60 MPa, then the circumferential stress is

(a) 120 MPa (b) 30 MPa
(c) 20 MPa (d) 10 MPa

TSPSC AEE 2017

Ans. (a) : $\sigma_L = \frac{Pd}{4t} = \frac{Pd}{2 \times 2t} = \frac{1}{2} \times \sigma_h$

$$60 = \frac{1}{2}\sigma_h \ \Rightarrow \ \sigma_h = 120 \text{ MPa}$$

1519. For a thin spherical shell subjected to internal pressure, the ratio of volumetric strain to diametral circumferential strain is

(a) 1.25 (b) 1.5
(c) 2.0 (d) 3.0

Chandigarh AE 2017

Ans. (d) : Volumetric strain = 2 hoop strain + longitudinal strain = $2\varepsilon_c + \varepsilon_\ell$

In case of spherical shell both strain volume = $2\varepsilon_c + \varepsilon_c = 3\varepsilon_c$

$$\frac{\text{Volumetric strain}}{\text{Circumferential strain}} = \frac{3\varepsilon_c}{\varepsilon_c} = 3$$

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05.

Structural Analysis

1. Fundamental Structure, Determinacy & Indeterminacy of Structure

1. The degree of static indeterminacy of a rigid jointed space frame is:

(a) m + r – 2j (b) m + r – 3j
(c) 3m + r – 3j (d) 6m + r – 6j
(e) 2m + r – 3j

Haryana PSC AE 2018 Paper–I
APPSC AEE 2012
UPPSC State Eng. A.E. 2003

Ans. (d) : For a pin jointed plane structure to be statically determinate, $D_S = 0$
m + r = 2j → 2D Truss/Pin jointed plane
m + r = 3j → 3D Truss/Pin jointed plane
m + r = 3j → 2D Rigid Frame
3m + r = 6j → 3D Rigid Frame

2. The number of independent equations to be satisfied for static equilibrium of a space structure is :

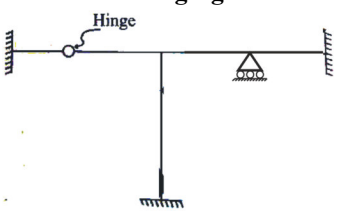
(a) 1 (b) 2
(c) 3 (d) 6

OPSC AEE 2019 Paper–I
Haryana PSC AE Paper-I 2018
UPPSC State Eng. A.E. 2007 (I)
ESE 2011

Ans. (d) : For space frames number of equation of equilibrium is 6.
Equations of equilibrium for space frames

$\Sigma f_x = 0$, $\Sigma M_x = 0$
$\Sigma f_y = 0$, $\Sigma M_y = 0$
$\Sigma f_z = 0$, $\Sigma M_z = 0$

3. The degree of static indeterminacy of the frame shown in the following figure is

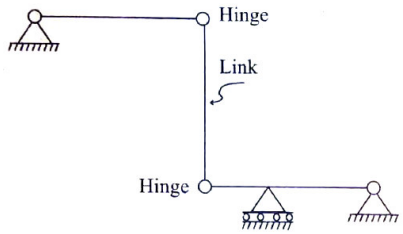


(a) 2 (b) 4
(c) 6 (d) 8

GPSC AE Dec 2018

Ans. (c) : Static indeterminacy = 3+3+3+1–1–3 = 6.

4. The degree of static indeterminacy of the structure shown in the following figure is



(a) 0 (b) 1
(c) 2 (d) 3

GPSC AE Dec 2018
ESE 2015

Ans. (a) : Degree of static indeterminacy = [No. of equation] – [Static equilibrium equation] – [Number of internal hinges]
$D_s = 5 – 3 – 2$
$[D_s = 0]$

5. The degree of static indeterminacy of a rigid-jointed space frame is :

(a) m+r-2j (b) m + r-3j
(c) 3m + r-3j (d) 6m + r- 6j

Punjab MCP SDO 2015

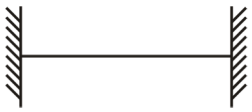
Ans. (d) : The degree of static indeterminacy of a rigid-jointed space frame $D_{si} = 6m + r - 6j$
The degree of static indeterminacy of a rigid-jointed plane frame $D_s = (3m + r) – 3j$

6. What is the value of kinematic indeterminacy and statical indeterminacy of a fixed beam respectively?

(a) 3, 1 (b) 1, 2
(c) 2, 0 (d) 0, 3

OPSC AEE 2019 Paper–I

Ans. (d)



At fixed support, DOF = 0
Kinematic indeterminacy = 0
$D_s = D_{sc} = r – S = 6 – 3 = 3$

7. Consider the following statements:

1. When the number of members (n) and joints (j) are such that the equation n=(2j–3) is satisfied, the framed structure is said to be a perfect structure.
2. In a redundant frame, the number of members is less than that required for a perfect frame.
3. If, in a framed structure, the number of members provided is more than that required for a perfect frame, it is called as a deficient frame.

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Which of the above statements is /are *correct?*

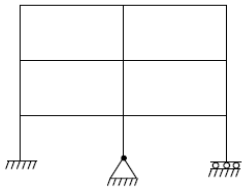
(a) 1, 2 and 3 (b) 1 only
(c) 2 only (d) 3 only

ESE 2017

Ans. (b)

- n = (2j–3) → Perfect frame.
  n > (2j–3) → Redundant frame
  n< (2j –3) → Deficient frame (Unstable)
- Perfect frame are the frames which can be analysed to get the internal member forces and external support reaction by using the three conditions of static equilibrium.
- Deficient frame is also called as unstable frame i.e. the frame will deform if the external load is applied.

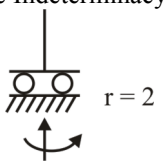
8. The kinetic indeterminacy of the structure shown in the figure is equal to



(a) 14 (b) 15
(c) 16 (d) 17

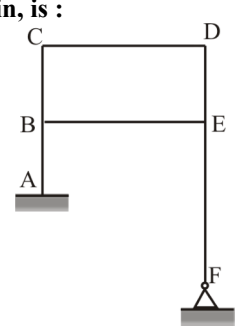
ESE 2017

Ans. (a) : Kinematic Indeterminacy 'D_k' = 3j – (r+m).



j = Number of joints = 12
r = Reaction = 3 + 2 + 2 = 7
m = Number of members = 15
$D_k = 3j - r - m$
$D_k = 3 \times 12 - 7 - 15$
$\boxed{D_k = 14}$

9. The degree of kinematic indeterminacy of a plane structure shown in the figure neglecting axial strain, is :



(a) 4 (b) 5
(c) 6 (d) 7

Karnataka PSC AE 2017 Paper - II

Ans. (d) :

$$D_k = 3j - m - r = 3 \times 6 - 6 - (3+2)$$
$$= 18 - 6 - 5 = 7$$

10. The number of unknown reactions to be found at a fixed support of a beam, during analysis is/are :

(a) 1 (b) 2
(c) 3 (d) 4

TNPSC AE 2015

Ans. (c) :

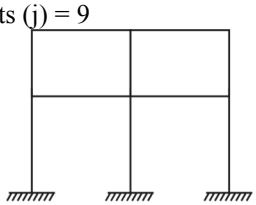
Type of support	No. of unknown reaction
fixed	3
hinged	2
roller	1

11. What are static and kinematic indeterminacies, respectively, of a two storey two bay frame having fixed supports at base ?

(a) 9, 15 (b) 9, 12
(c) 6, 15 (d) 12, 18

Rajasthan JEN (Degree) Shift - II 2016

Ans. (d) : Number of external reaction (Re) = 9
Number of members (m) = 10
Number of joints (j) = 9



Two storey two bay frame

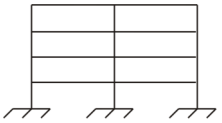
Degree of static indeterminacy (D_s)–

$D_s = 3m - 3j + Re$
$= 3 \times 10 - 3 \times 9 + 9$
$= 12$

Degree of kinematic indeterminacy (D_k) –

$D_k = 3j - Re$
$= 3 \times 9 - 9$
$= 27 - 9 = 18$

12. Total degree of kinematic indeterminacy for the plane frame shown in the figure is given by:



(a) 27 (b) 36
(c) 54 (d) 24

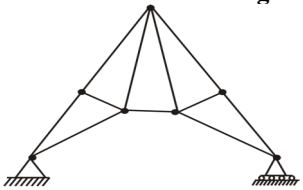
JPSC AE (Pre) 2019

Ans. (b) :

Degree of kinematic indeterminacy for a rigid plane,

$D_K = 3J - r_e - m$
$= (3 \times 15) - 9$
$= 36$ [∴ Total no, of indeterminate member, m=0]

13. The kinematic indeterminacy of the plane of the plane truss shown in the figure is :



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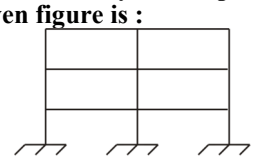
(a) 11 (b) 8
(c) 3 (d) 0

JPSC AE (Pre) 2019

Ans. (a) : Number of joints (J) = 7
For rigid joint plan truss kinematic indeterminacy
$= 2j - R$
$D_K = 2\times 7 - (2+1) = 14 - 3$
$[D_K = 11]$

14. The total (both internal and external) degree of static indeterminacy of the plane frame shown in the given figure is :

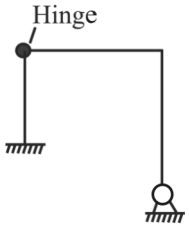


(a) 14 (b) 16
(c) 18 (d) 20

JPSC AE (Pre) 2019

Ans. (c) :
$D_S = 3m + Re - 3j - r_r$
$= (3\times 15) + (3+3+3) - (3\times 12) - 0 = 18$

15. For the plane frame as shown in the figure

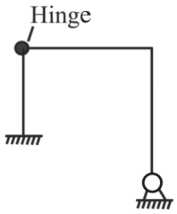


The degree of kinematic indeterminacy neglecting axial deformation, is -

(a) 3 (b) 5
(c) 7 (d) 9

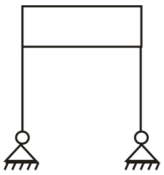
RPSC AE 2018
ESE 2014

Ans. (b) :



Joint = 4
Reaction = 4
$D_k = 3J - R_e - m' = 3\times 4 - (4+3) = 12 - 7 = 5$

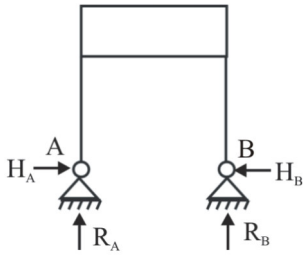
16. What is the degree of Kinematic indeterminacy of the frame shown in figure? Neglect axial deformation.



(a) 14 (b) 12
(c) 10 (d) 8

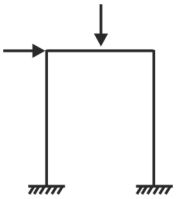
TNPSC AE 2018

Ans. (d) :



Degree of kinematic indeterminacy = 3j – m – r
j = no. of joints ; m = no. of members; r = no. of support reactions. j = 6; m = 6; r = 2 + 2
Degree of kinematic indeterminacy
$= 3j - m - r = 3\times 6 - 6 - (2+2) = 8$

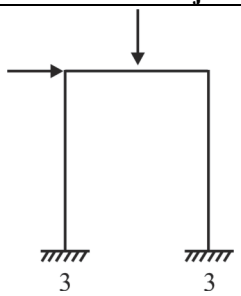
17. What is the possible number of independent mechanism in the portal frame given below



(a) 3 (b) 4
(c) 2 (d) 1

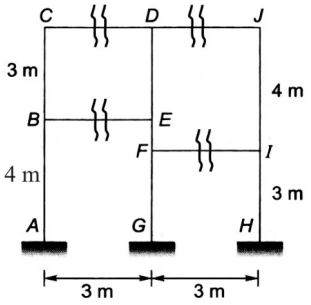
Gujrat AE (Class 1 & 2) 2019

Ans : (c) :



Number of possible locations where of plastic hinges can be formed = n = 4.
Static indeterminacy (D_s) = r
r = 5 – 3 = 2
Number of independent mechanism
S = n – r
[S = 4 – 2 = 2]

18. Determine the degree of static and kinematic indeterminacy of the frame structure as shown in the figure-



(a) 15, 8 (b) 12, 12
(c) 12, 10 (d) 15, 9

Maharashtra PSC (Mains) 2017 Paper–I

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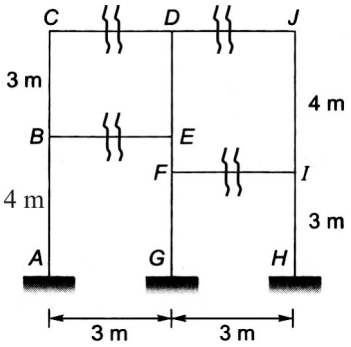
Ans. (c) :



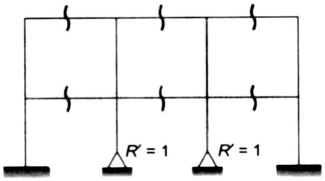
Kinematic indeterminacy is
$D_k = 3j–r–m$
$= 3 \times 10 – 9 – 11 = 10$
Static indeterminacy
$Ds = 3C – R$
$= 3 \times 4 – 0$
$= 12$
j = No. of joint
r = No. of force
m = No of member
C = no of cuts
R' = No of restrained

19. A rigid jointed frame with three bays and two stories has two fixed column supports and two hinged. The degree of static indeterminacy of structure is-

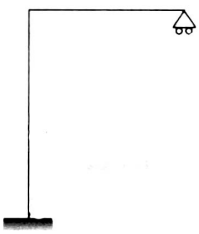
(a) 25 (b) 27
(c) 18 (d) 16

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : $D_S = 3C – R^1$
C = No. of cuts to make structure determinate R^1 = No. of restraints applied to make all joint rigid.
C = 6
R' = 2
$D_S = 3 \times 6 – 2$
$D_s = 16$



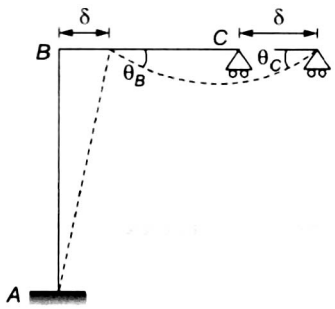
20. The Kinematic Indeterminacy of a frame as shown is-



(a) $KI = 1$ (b) $KI = 2$
(c) $KI = 3$ (d) $KI = 5$

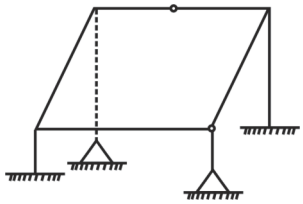
Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) :



$KI = 3\,(\delta, \theta_B \text{ and } \theta_C)$
$KI = 3j – R – m = 3 \times 3 – 4 – 2 = 3$

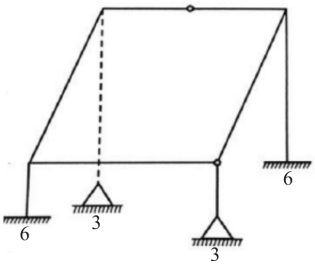
21. The statical indeterminacy for the given 3D frame is?



(UPRVUNL AE 2015)

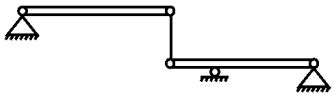
(a) 8 (b) 6
(c) 9 (d) 12

Ans : (c)



$D_S = D_{se} + D_{si}$
$D_{se} = r_E – 6$
$r_E = 6 + 3 + 3 + 6 = 18$
$[D_{se} = 12]$
$D_{si} = 6C – r_R = 6 \times 1 – 3 \sum(M'–1)$
$= 6 – 3 [(2–1) + (3–1)]$
$[D_{si} = 6–9 = –3]$
$\therefore\ D_s = 12 + (–3)$
$D_s = 9$

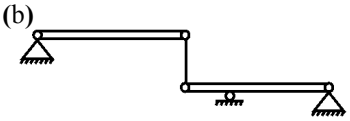
22. Number of degree of static indeterminacy for the beam shown below is:



(a) -1 (b) 0
(c) 1 (d) 2

[UPPSC AE 12.04.2016 Paper–I]

Ans : (b)



$D_s = 5 – [3 + 2] = 0$

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23. If there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints then the degree of static indeterminacy of a pin jointed plane frame is given by:

(a) m+r+2j (b) m–r+2j
(c) m+r–2j (d) m+r–3j

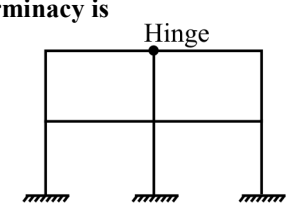
[UPPSC State Eng. A.E. 2008]

Ans : (c) If there are 'm' member in the pin jointed frame, then there will be m internal reactions i.e, axial force in each member L_t, r_e be the total external support reactions and at each joint in the truss there are two equilibrium conditions.

($\Sigma f_x = 0$ and $\Sigma f_y = 0$. Let there are j number of joint. Hence total equilibrium conditions at all joint will be 2j.

$$D_s = (m + r_e) - 2j = m + r_e - 2j$$

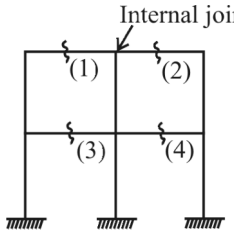
24. For the rigid frame shown in figure the statical indeterminacy is



(a) 8 (b) 10
(c) 12 (d) 14

[UPPSC State Eng. A.E. 2011]
[UPPSC State Eng. A.E. 2007 (II)]

Ans : (b)



Static indeterminacy of frame $(D_s) = 3C - r_r$

C = No. of Cut to make structure like free structure

r_r = No. of Additional Released.

= (m'-1) (case of joint Hinge)

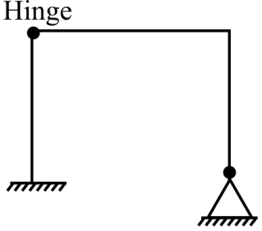
m'= No. of member meeting at joint

$$\left.\begin{matrix} C = 4 \\ m' = 3 \end{matrix}\right\}$$

$$D_s = 3 \times C - (m' - 1) = 3 \times 4 - (3 - 1) = 12 - 2 = 10$$

$$\boxed{D_s = 10}$$

25. The kinematic indeterminacy (Degree of Freedom) of the frame given below is:



(a) 4 (b) 6
(c) 8 (d) 10

Arunachal Pradesh Poly. Tech. Lect. 2019
PTCUL AE 2016
UPPSC State Eng. A.E. 2011
ESE 2012

Ans : (c) $3m + r - 3j$
$= 3 \times 3 + 5 - 3 \times 2$
$= 8$
Or
$D_K = 3j - r_E + r_R$
$D_K = 3 \times 4 - 5 + 1 = 8$

26. A rigid jointed plane frame is stable and statically determinate if:

(a) (m+r)=2j (b) (m+r)=3j
(c) (3m+r)=3j (d) (m+3r)=3j

Where
m = number of unknown member forces
r= unknown reaction components
j = number of joints

[UPPSC State Eng. A.E. 2008]

Ans : (c) A structure is determinate and stable if

$$D_s = 0$$

for rigid jointed plane frame D_s is given by

$$D_s = 3m + r - 3j$$

Where,
m = number of unknown member forces.
r = unknown reaction components
j = number of joints

$$D_s = 0$$

$$\boxed{3m + r = 3j}$$

27. The degree of static indeterminacy of a pin-jointed space frame is given by:

(a) m+r–2j (b) m+r–3j
(c) 3m+r–3j (d) m+r+3j

m = number of unknown member forces
r = unknown reaction components
j = number of joints

[UPPSC State Eng. A.E. 2008
WBPSC AE (PHE) 1998

Ans : (b) The degree of static indeterminacy of a pin-jointed space frame is given by

$$\boxed{D_S = (m+r) - 3j}$$

28. A plane frame is statically determinate if

(a) 3m+r = 3j+C (b) 3m+C=3j+r
(c) 3m+C<3j+r (d) 3m+C>3j+r

Where
m= No. of members
j = no. of joints
r= no. of reactions
C= no. of equations of conditions

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (a) A plane frame is determinate if degree of static indeterminacy is zero.

P_S = Total unknown reactions - Total equation of equilibrium - additional condition of equilibrium like r_r

$Ds = (3m + r) - 3j - C$

for frame to be determinate

$D_S = 0$

$3m + r = 3j + C$

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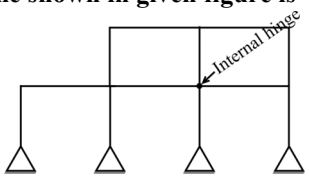
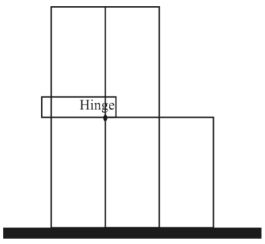
29. The total degree of static indeterminacy of the plane frame shown in given figure is



(UPPSC State Eng. A.E. 2003)
ESE 1999

Or

What is the static indeterminacy of the 2D frame given below?



(a) 10 (b) 11
(c) 12 (d) 15

(LMRC AM 2017 I–Shift)
LMRC AM 2015

Ans : (c) $D_s = 3m + r_e - 3j - r_r$

$m = 12 \quad j = 11 \quad r_e = 3+3+3+3 = 12$

r_r = Reaction Released

$\sum(M-1) = (4-1)$

$= 3$

$D_s = 3\times12 + 12 - 3\times11 - 3$

$= 12$

Alternative approach

$D_s = D_{se} + D_{si}$

$D_{se} = r_e - 3$

$= 12 - 3 = 9$

$D_{si} = 3C - r_r$

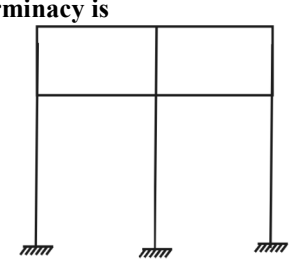
$= 3\times2 - 3$

$= 6 - 3 = 3$

$D_s = D_{se} + D_{si}$

$= 9 + 3 = 12$

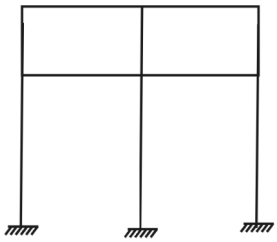
30. Find the frame Shown in the figure below, the indeterminacy is



(a) 6 (b) 9
(c) 12 (d) 15

(JPSC Combined AE Paper V 2013)

Ans : (c)



No. of Reaction (r) = 9
No. of member (m) = 10
Joint (j) = 9
Degree of static indeterminacy

$D_{si} = 3m + r - 3j$

$= 3\times10 + 9 - 3\times9 = 12$

$\boxed{D_{Si} = 12}$

31. If m is number of members; r is reactions; and j is number of joints then in case of a planar structure, 3m + r < 3j leads to

(a) stable structure
(b) determinate structure
(c) unstable structure
(d) indeterminate structure

[MPPSC AE 2016]

Ans : (c) Planar structure $\boxed{\alpha_s = 3m + r - 3J}$

Stable structure $\alpha_s > 0, 3m + r > 3J$

Unstable structure $\alpha_s < 0, 3m + r < 3J$

32. Degree of static indeterminacy of a rigid jointed plane frame having 15 members, 3 reaction components and 14 joints is-

(a) 2 (b) 3
(c) 6 (d) 8

GOA PSC Assistant Prof. 2020
Maharashtra PSC Mains 2018 Paper-1
Haryana PSC AE 2018 Paper–I
Karnataka PSC AE 2017 Paper - II
BPSC AE 2012 Paper–V

Ans : (c) For plane frame (Rigid jointed)
No. of Member (m) = 15
No. of Reaction (r) = 3
No. of Joints (j) = 14
Degree of static indeterminacy

$Ds = 3m + r - 3j$

$= 3\times15 + 3 - 3\times14$

$= 6$

33. Which of the followings are indeterminate structure?

(a) 3-hinged arch (b) continuous beam
(c) redundant frame (d) both (b) and (c)

[UPPSC AE 12.04.2016 Paper–I]

Ans : (d) Both (b) and (c) are indeterminate structure.

34. The kinematic indeterminacy of the plane frame shown in figure is (disregarding the axial deformation of the members)

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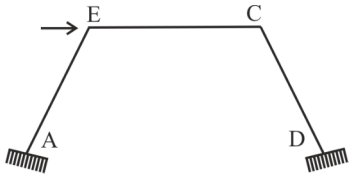


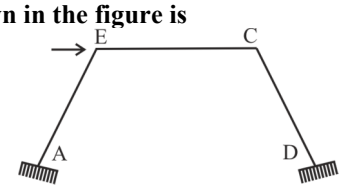
(a) 4 (b) 3
(c) 2 (d) 0

RPSC POLY. TECH. LECT. 2014
UPPCL AE 2014

Ans : (b) 2D-Frame (disregarding the axial deformation)

$D_k = 3J - R_e$
$D_{KF} = 3(J - J') - R_e + R_r - m$
$D_K = 3J - R_e + R_r$
$D_K = 3J - R_e - m$
$J = 4, R_e = 3+3 = 6, m = 3$
$D_K = 3\times4-6-3 = 12 - 9 = 3$
$\boxed{D_k = 3}$

35. The degree of kinematic indeterminacy of the rigid frame with clamped ends at A and D as shown in the figure is



(a) 7 (b) 5
(c) 6 (d) 4

RPSC POLY. TECH. LECT. 2014

Ans : (c) 2D – frame kinematic indeterminacy

$D_k = 3J - R_e$
$J = 4 \quad R_e = 3+3 = 6$
$D_k = 4\times3 - 6 = 12 - 6 = 6$
$\boxed{D_k = 6}$

36. In a statically determinate plane frame the relationship between member of bars and joints can be expressed as

(a) $j = 2n-3$ (b) $n = 2j-3$
(c) $j = 3j-2$ (d) $n = 3j-2$

Where n = number of bars, and J = number of joints

RPSC A En. 2013

Ans : (b) 2D – frame/ Truss, statically determinate

$n + 3 - 2J = 0$
$\boxed{n = 2J - 3}$

37. The number of equations required to obtain axial force in the members of a statically determinate plane frame is

(a) 2 (b) 3
(c) 4 (d) 6

RPSC A En. – 2013

OR

The number of independent equations to be satisfied for static equilibrium of a plane structure is

(a) 1 (b) 3
(c) 2 (d) 6

(UPPSC State Eng. A.E. 2003)

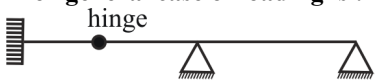
Ans : (b) For static equilibrium of plane structure following three equation must be satisfied:

(i) $\Sigma F_x = 0$
(ii) $\Sigma F_y = 0$
(iii) $\Sigma M_z = 0$

For static equilibrium of space (3D) structure following (6) six equation must be satisfied:

(i) $\Sigma F_x = 0$ $\Sigma M_x = 0$
(ii) $\Sigma F_y = 0$ $\Sigma M_y = 0$
(iii) $\Sigma F_z = 0$ $\Sigma M_z = 0$

38. The degree of indeterminacy of the following beam for general case of loading is :



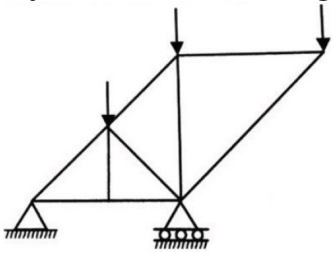
(a) 1 (b) 2
(c) 3 (d) 4

RPSC VPITI 2012

Ans : (c) $D_s = D_{si} + D_{se}$

$D_{si} = 3C - R_r$
$C = 0, R_r = 1$
$D_{Si} = 3 \times 0 - 1 = -1$
$D_{si} = -1$
$D_{se} = R_e - 3$
$= (3 + 2 + 2) - 3$
$= 4$
$D_s = D_{si} + D_{se}$
$= -1 + 4 = 3$
$\boxed{D_s = 3}$

39. The pin-jointed frame shown in figure is



(a) A perfect frame
(b) A redundant frame
(c) A deficient frame
(d) Both redundant and deficient frame

UPRVUNL AE 2015

Ans. (a) : $M = 2j - 3$ (for perfect frame)
given, $j = 6, M = 9$
So,
$M = 2\times6 - 3 = 9$
Therefore it will be perfect frame.

40. For any structure, when both ends have roller supports, then the structure will be

(a) stable
(b) unstable
(c) (a) and (b) of the above
(d) none of the above

(UK Combined AE Paper– I, 2012)

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Ans : (b)

For any structure, when both ends have roller supports then the structure will be in unstable condition. Roller supports will move in horizontal direction.

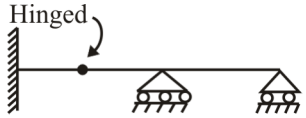
41. The necessary condition for equilibrium of body is

(a) $\Sigma H = 0$
(b) $\Sigma V = 0$
(c) $\Sigma M = 0$
(d) All of the above

RPSC ACF 2011

Ans : (d) The three static equilibrium condition are–
$\Sigma V = 0,\ \Sigma H = 0,\ \Sigma M = 0$

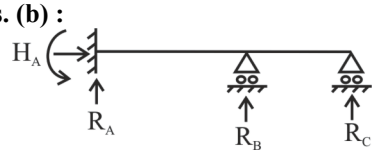
42. The degree of static indeterminacy of the beam given below is



(a) zero (b) one
(c) two (d) three

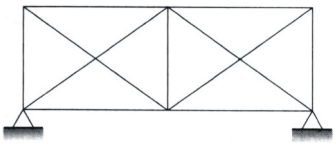
TNPSC AE 2018
ESE 2002

Ans. (b) :



Number of reactions = 3+1+1=5
Equilibrium equations available = 3+1 (hinge) = 4
static indeterminacy = 5 – 4 = 1
Or
$D_s = r_e + 3m - r_r - 3(j - j')$
$r_e = 3 + 1 + 1 = 5,\ m = 3,\ j = 3,\ j' = 1,\ r_r = 1$
$\therefore\ D_s = 5 + 9 - 1.0 - 3(3 + 1) = 14 - 1 - 12$
$[D_s = 1.0]$

43. In the pin-jointed truss shown in the figure, the static degree of indeterminacy is-

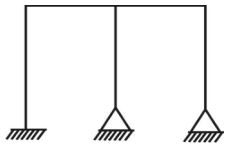


(a) 2 (b) 1
(c) 3 (d) 4

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (c) : static degree of indeterminacy
$D_S = M + r - 2j$
$= 11 + 4 - 2 \times 6 = 3$
Where, M = Number of member
r = Number of Reaction force
j = Number of joint

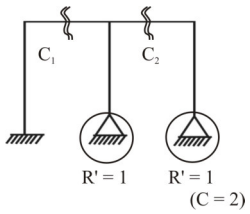
44. The degree of static indeterminacy N_S and degree of kinematic indeterminacy, N_K for the plane frame as shown axial deformation are given by–



(a) $N_S = 6, N_K = 11$ (b) $N_S = 4, N_K = 6$
(c) $N_S = 6, N_K = 6$ (d) $N_S = 4, N_K = 4$

(UTTRAKHAND AE 2013)

Ans : (b)



$N_S = 3C - R'$
$= 3 \times 2 - (1+1) = 4$
$N_K = 3J - r - m$
$= 3 \times 6 - (3+2+2) - 5 = 6$

45. The number of unknowns to be determined in the stiffness method is equal to

(a) static indeterminacy
(b) kinematic indeterminacy
(c) difference of static and kinematic indeterminacy
(d) sum of static and kinematic indeterminacy
(e) static indeterminacy/kinematic indeterminacy

CIL MT 2017
Chandigarh AE 2017
UPRVUNL AE 2016
CGPSC AE 2014 Shift–I
ESE 2005

Ans. (b) : Since, stiffness method is displacement method, so in this method unknown are displacement. Therefore, number of unknown is equal to kinematics indeterminacy.

46. A two-dimensional structure in general is classified as a statically indeterminate structure if it cannot be analyzed by which of the following equations of equilibrium

(a) Sum of vertical forces is zero
(b) Sum of horizontal forces is zero
(c) Sum of moments of all the forces about any point in the plane is zero
(d) i, ii & iii above
(e) i & iii above

CGPSC AE 2014 Shift–I

Ans. (d) : If structure is determinate it is analyzed by equilibrium equations only - $\Sigma f_x = 0,\ \Sigma f_y = 0,\ \Sigma M = 0$.
If structure is not analyzed by using above equation only, then structure is called statically indeterminate structure.

47. The following structures is statically indeterminate to the degree of

(a) Zero (b) One
(c) Two (d) Three
(e) Five

CGPSC AE 2014 Shift–I

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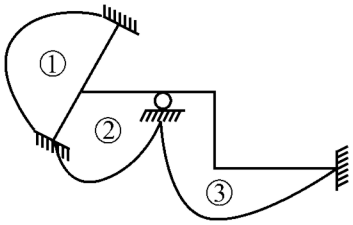
Ans. (d) :



$D_S = 3C - R'$

$C \rightarrow$ Closed loop

$R' \rightarrow$ No. of constrained added to make the given joint right.

$= 3(3) - (2+2+1+1)$

$= 9-6$

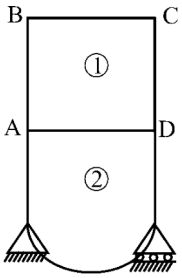
$\boxed{D_S = 3}$

48. The adjacent frame is indeterminate to the degree of

(a) One (b) Two
(c) Three (d) Five
(e) Zero

CGPSC AE 2014 Shift–I

Ans. (c) :



$D_S = 3C - R'$

$= 3(2) - (1+2)$

$= 6-3$

$\boxed{D_S = 3}$

49. Static equilibrium equation is

(a) algebraic sum of all the vertical forces are equal to zero
(b) algebraic sum of all the horizontal forces are equal to zero
(c) algebraic sum of all the moments are equal to zero
(d) all of these

TNPSC AE 2012

Ans. (d) : All three are static equilibrium equations :

$\begin{bmatrix} \sum H = 0 \\ \sum V = 0 \\ \sum M = 0 \end{bmatrix}$ $\Rightarrow$ If all these condition are satisfied then member is in equilibrium condition

50. For a statically determinate structure, the degree of indeterminacy is equal to

(a) one (b) more than one
(c) zero (d) none of these

TNPSC AE 2012

Ans. (c) The structure for which the reaction at the supports and internal forces in the members can be found out by the conditions of static equilibrium. The degree of indeterminacy is equal to zero.

51. The degree of indeterminacy for a fixed beam is

(a) 1 (b) zero
(c) 3 (d) 2

TNPSC AE 2012

Ans. (d) The degree of indeterminacy for a fixed beam is 2.

Ds = No. of unknown reaction– static equation

$= 4 - 2 = 2$

52. A redundant frame satisfies the equation

(a) $m = 2j - 3$ (b) $m < 2j - 3$
(c) $m > 2j - 3$ (d) none of these

TNPSC AE 2012

Ans. (a) : $m = 2j - 3$ –perfect frame

$m > 2j - 3$ –redundant

$m < 2j - 3$ – deficient

m = Number of member

j = Number of joints

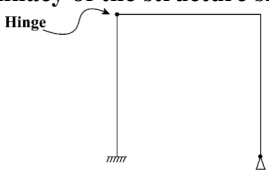
53. A statically indeterminate structure :

(a) can be analyzed using equations of compatibility only
(b) can be analyzed using equations of statics and compatibility equation
(c) can be analyzed using equations of statics only
(d) cannot be analyzed at all

APPSC Poly. Tech. Lect. 2020
Maharashtra PSC Mains 2019
GPSC AE (CLASS 1 & 2) 2019
ESE-2008

Ans. (b) : A statically indeterminate structure can be analyzed using equations of static equilibrium and compatibility equation.

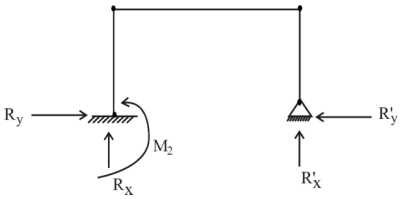
54. What is the Total, External and Internal indeterminacy of the structure shown below?



(a) Total 1; External 2; Internal (-1)
(b) Total 1; External 2; Internal (1)
(c) Total 2; External 1; Internal (1)
(d) Total 2; External 1; Internal (-1)

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) :



Total = 1

External = 2

internal = –1

Total indeterminacy $D = (3m + r_e) - 3j - \Sigma(m' - i)$

$r_e = 5$

$m = 3$

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J= 4
m' = 2
D = (9+5) – 3 ×4 – (2–1)
D = 1
External indeterminacy E = r_e – 3
= 5 – 3 = 2
Internal indeterminacy (I) = D – E
= 1 – 2 = – 1

55. What will be the degree of static and kinematic indeterminacy for a beam having one end hinge and other end roller?
(a) Zero and three (b) Zero and zero
(c) Three and three (d) Two and two

LMRC AM 2019

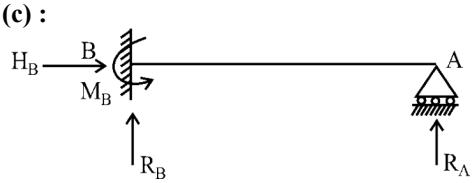
Ans. (a)

D_s = r_e – r
r_e = Total Number of independent support reaction.
r = Total number of available equations of static equilibrium.
r_e = 3, r = 3
D_s = Static Indeterminacy
D_s = 3–3 = 0
D_k = Kinematic Indeterminacy
D_k = 3j – r_e
r_e = No. of available support reactions which generate of restrained joint displacement
j = no of joints
D_k = 3×2 – 3
D_k = 3

56. The static indeterminacy of a propped cantilever beam is equal to :
(a) 3 (b) 2
(c) 1 (d) 0

KPSC AE 2020

Ans. (c) :



Static indeterminacy (D_s) = R- r
R = Total no. of support reaction
R = 4
r = No. of equations of static equilibrium available
r = 3
D_s = 4-3 = 1

57. The degree of static indeterminacy of a pin jointed plane frame is
Where J - number of unknown joints
r = number of unknown reactions
(a) j – 2r (b) 3j – 2r
(c) 2j – r (d) 3j– r

TANGEDCO AE 2018
Nagaland PSC CTSE 2017 Paper–I
ISRO Scientist/Engineer 2015
UPRVUNL AE 2015

Ans. (c) : Degree of static indeterminancy for pin jointed plane frame.

$$D_s = 2j - r$$

j = no. of unknown joints
r = no. of unknown reaction

58. If a deformable structure in equilibrium under the action of a system of loads is given a small virtual deformation, then the virtual work done by the external forces is equal to the virtual work done by the internal forces. This principle is known as :
(a) Principle of virtual displacements
(b) Principle of virtual work
(c) Principle of complimentary work
(d) Principle of complimentary displacements

KPSC AE 2020

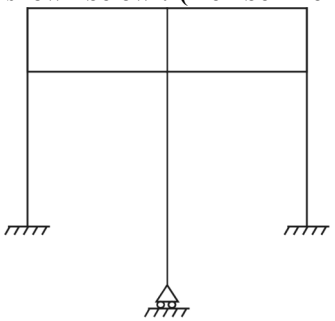
Ans. (b) : If a deformable structure in equilibrium under the action of a system of loads is given a small virtual deformation, then the virtual work done by the external forces is equal to the virtual work done by the internal forces. This principle is known as principle of virtual work.

59. Which of the following is true example of statically determinate beam ?
(a) One end is fixed and other end is simply supported
(b) Both the ends are fixed
(c) The beam over hangs over two supports
(d) The beam is supported on three supports

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : An overhanging beam is statically determinate.

60. What is the kinematics indeterminacy for the frame shown below ? (member inextensible) :



(a) 6 (b) 11
(c) 12 (d) 21

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : Degree of kinematic indeterminacy for a plane rigid frame having in extensible member is given by D_K = 3j – r_e – M.
Here, joint 'j' = 9
Reaction 'r_e' = 3 + 1 + 2 = 6
member = 10
D_K = 3 × 9 – 6 – 10
[D_K = 11]

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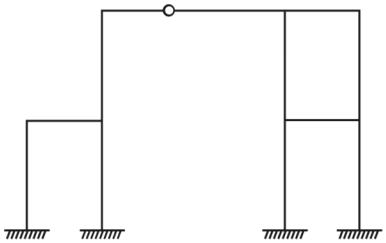
61. What is the statical indeterminacy for the frame shown below ?

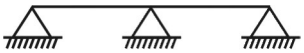


(a) 12 (b) 15
(c) 11 (d) 14

Nagaland PSC CTSE 2017 Paper–I
PTCUL AE 2016
ESE 2008

Ans. (c) : Ds = 3C – R'
C → no. of cuts.
R' → no. of restrained required to make rigid frame.
Ds = 3(4) –1
= 12 – 1
$\boxed{D_s = 11}$

62. What is the degree of kinetic indeterminacy of the beam shown below :

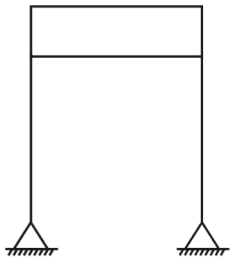


(a) 2 (b) 3
(c) 4 (d) 5

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) :
$D_k = 3j - r_e - m$
= 3 (3) – (2+2+2) – (1)
= 9 – 7
$\boxed{D_k = 2}$

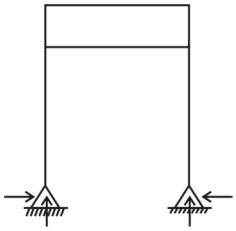
63. What is the degree of kinetic indeterminacy of the frame shown below if axial deformation is neglected :



(a) 6 (b) 8
(c) 10 (d) 12

Nagaland PSC CTSE 2017 Paper–I
UPPSC State Eng. A.E. 2008, 2003

Ans. (b) :



$\boxed{D_k = 3j - r_e - m}$

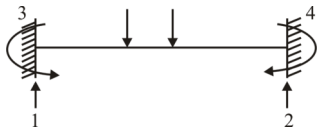
$D_k \rightarrow$ Degree of kinematic indeterminacy
$i \rightarrow$ no. of joint
$r_e \rightarrow$ no. of support reaction
$m \rightarrow$ member are Axially Rigid/inextensible,
$\therefore D_k = 3(6) - 4 - 6$
$= 18 - 4 - 6$
$\boxed{D_k = 8}$

64. Which of the following are examples of indeterminate structures?
1. Fixed beam
2. Continuous beam
3. Two-hinged arch
4. Beam overhanging on both sides

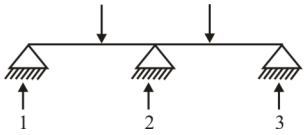
Select the ***correct*** answer using the codes given below:
(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

ESE 2017

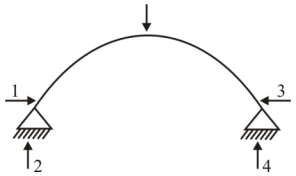
Ans. (a)



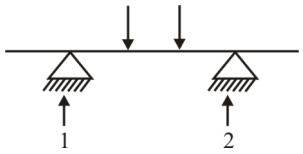
Fixed Beam
Reaction = 4
$D_s = (r-2)$
$\boxed{D_s = 2}$



Continuous Beam
Reaction = 3
$\boxed{D_s = 1}$



Two hinged arch
Reaction = 4
$\boxed{D_s = 2}$



Over hanged beam
Reaction = 2
$\boxed{D_s = 0}$

65. In a braced cantilever truss, there are 9 members and 5 joints, find out the degree of static indeterminacy
(a) 1 (b) 5
(c) 4 (d) 2

DSSSB AE 2019

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Ans. (d) Degree of static equilibrium:

$D_{si} = M + r_e - 2J$

$= 9 + 3 - 2 \times 5$

$= 2$

66. In a frame there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints. What is the degree of static indeterminacy of pin jointed frames?

(a) m+r-3j (b) m+r+2j

(c) m+r-2j (d) m-r+2j

Rajasthan AE Nagar Nigam Shift-3, 2016
ISRO Scientist/Engineer 2015

Ans. (c) : m = no. of unknown member forces
r = unknown reaction components
j = no. of joints
Therefore, degree of static indeterminacy of pin jointed frame = m+r-2j.

67. A pin jointed plane frame of *m* members and *j* number of joints is unstable if

(a) $(m + r) < 2j$ (b) $(m + r) = 2j$

(c) $(m + r) > 2j$ (d) None of the above

where r = number of reaction components.

WBPSC Poly Lect. 2019

Ans. (a) : A pin jointed plane frame is unstable if;
$(m + r) < 2j$
m = no. of members
j = no. of joints
r = no. of reaction components.

68. When equations of statics are not sufficient to determine all the reactive forces at the supports, such beams are called

(a) statically indeterminate
(b) statically determinate
(c) Imperfect
(d) Defective

APPSC AEE MAINS 2016

Ans. (a) : When equations of statics are not sufficient to determine all the reactive forces at the supports such beams are called statically indeterminate.

69. Structures having more reactions than that required for necessary and sufficient conditions are

1. Hyperstatic
2. Determinate
3. Indeterminate
4. Hypostatic

(a) Only 1 (b) 1 and 3

(c) Only 2 (d) 2 and 4

GPSC AE March 2018

Ans. (b) : Hyperstatic and indeterminate structure both have the no. of unknown reactions greater that the known reactions (3 equilibrium conditions).

70. Statistically indeterminate beam can be solved by :

1. Displacement method
2. Energy Method
3. Matrix Method
4. Four moment equation Method

(a) 1 and 2 (b) 2 and 3

(c) 1, 2 and 3 (d) 1, 2, 3 and 4

GPSC AE March 2018

Ans. (c) : Indeterminate structures can be solved by methods-
1. Flexibility method
2. Slope deflection method
3. Moment distribution method
4. Matrix method
5. Energy method
6. Displacement method

71. It can be stated in general that if a statically indeterminate frame has R redundancy, then it will become statically determinate if the number of plastic hinge formed is

(a) R (b) R–1

(c) R+ 1 (d) 2R

UPSC JWM 2017

Ans. (a) : Indeterminacy = R
determinate Str = 0
No. of plastic hinge = R-S
= R-0
= R

72. Study the followings statements:

1. The displacement method is more useful when degree of kinematic indeterminacy is greater than the degree of static indeterminacy.
2. The displacement method is more useful when degree of kinematic indeterminacy is less than the degree of static indeterminacy.
3. The force method is more useful when degree of static indeterminacy is greater than the degree of kinematic indeterminacy.
4. The force method is more useful when degree of static indeterminacy is less than the degree of kinematic indeterminacy.

(a) 1 and 3 (b) 2 and 3

(c) 1 and 4 (d) 2 and 4

GOA PSC Assistant Prof. 2020

Ans. (d) : (1) The displacement method is more useful when degree of kinematic indeterminacy is less than the degree of static indeterminacy.
(2) The force method is more useful when degree of static indeterminacy is less than the degree of kinematic indeterminacy.

73. Which one of the following is the true example of a statically determinate beam?

(a) One end is fixed and other end is simply supported
(b) Both ends are fixed
(c) The beam overhangs over two supports
(d) The beam is supported on three supports

JPSC AE 2009
ESE 2004

Ans. (c) Over hanging beam supports are not provided at the ends. An over hangs beam are statically determinate beam.

74. For a determine pin-jointed plane frame, the relation between the number of joints j and members is given by-

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(a) $m = 2j - 3$ (b) $m = 3j - 6$
(c) $m > 2j - 3$ (d) $m > 3j - 6$

JPSC AE 2009

Ans. (a) : For a pin-jointed plane frame-
Static indeterminacy $\delta_i = 0$

$$\boxed{\delta_i = m - (2j - 3)}$$

$\delta_i = 0$ so, $\boxed{m = 2j - 3}$
joint j and member m.

75. If 2j > m + r in a plane truss with number of points 'j' number of members 'm' and number of reactions 'r' the truss is :
(a) Redundant (b) Determinate
(c) Stable (d) Unstable

OPSC Civil Services Pre 2011

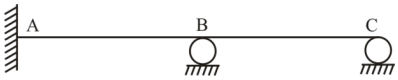
Ans. (d) Static indeterminacy $D_S = M + \gamma_e - 2j$.
If $2g > m + \gamma$ then static indeterminacy is negative it means structure is unstable.

76. When a body is in equilibrium undergoes an infinitely small displacement, work imagined to be done, is known as:
(a) Imaginary work (b) Negative work
(c) Virtual work (d) None of the above

OPSC Civil Services Pre 2011

Ans. (c) : Explanation virtual work – Virtual work is the total work done by the applied forces and the internal forces of a mechanical system as its waves through a set of virtual displacement.
Negative work– Negative work is done when an object moves in opposite direction of the direction of application of forces.

77. The degree of redundancy of the structure shown is :



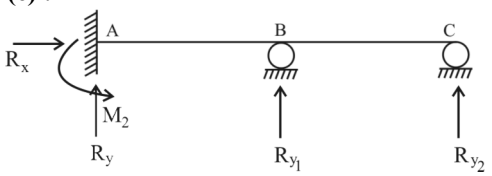
TRB Poly. Lect. Tech. 2017

Or

What is the degree of indeterminacy of a two span continuous beam, if the end support is fixed and the other two supports are simply supported?
(a) 3 (b) 1
(c) 2 (d) 4

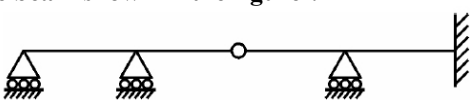
AP (VSS-1) PART - A 2017

Ans. (c) :



Degree of redundancy (D_S) = R - r
$D_S = 5 - 3 = 2$

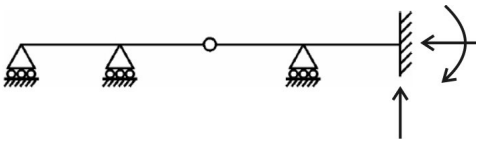
78. What is the degree of static indeterminacy of the beam shown in the figure ?



(a) 1 (b) 2
(c) 3 (d) 4

UKPSC AE (Urban) 2011

Ans. (b) :



DS = R – r – r^1
= 6 – 3 – 1
DS = 2

79. In a indeterminate structure, when there is no lack of fit, the partial derivative of strain energy with respect to any redundant
(a) Is zero
(b) Will give deflection in the direction of the redundant
(c) Will give slope in the direction of the redundant
(d) Is a maximum

Punjab PSC SDE 2016

Ans. (a) :

$$R = \frac{\sum \mu_i \lambda_i}{\sum \frac{\mu^2 L_i}{A_i E_i}}$$

where, λ_i = lack of fit in any member
R = Redundant

$\frac{\mu_i L_i}{A_i E_i}$ = strain energy

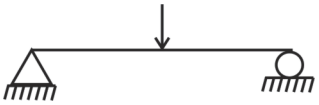
$$\frac{\sum \mu^2 L_i}{A_i E_i} = \frac{\sum \mu_i \lambda_i}{R}$$

when $\lambda_i = 0$ (No lack of fit)
Then derivative becomes zero.

80. A horizontal beam with both the ends hinged will be statically determinate for the following type of loading:
(a) Purely vertical loading
(b) Purely inclined loading
(c) Any general loading
(d) Inclined load with moment

Punjab PSC SDE 2016
JUVNL AE 2016

Ans. (a) : The vertical load on a beam does not affect its static or kinematic indeterminacy at all.



For statically determinate structure, degree of static indeterminacy should be zero. No stresses due to lack of fit or differential settlement.

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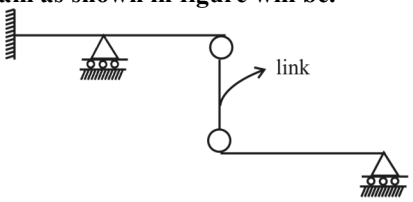
81. The degree of static indeterminacy for the beam as shown in figure will be.



(a) 0 (b) 1
(c) 2 (d) 3

UJVNL AE 2016

Ans. (a) : $D_s = D_{se} + D_{si}$
D_{se} = No. of reaction – No. of equilibrium reactions
= 5 – 3 = 2
$D_{si} = -2$
$[D_s = 2 - 2 = 0]$

82. Which one of the following condition has to be satisfied for both elastic and plastic method of analysis of indeterminate structures?

(a) Yield condition
(b) Equilibrium condition
(c) Mechanism condition
(d) Compatibility condition

UJVNL AE 2016

Ans. (c) Equilibrium condition has to be satisfied for both elastic and plastic method of analysis of indeterminate structure.

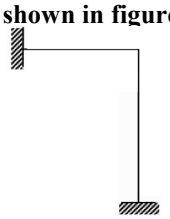
83. In a spherical dome subjected to concentrated load at crown or uniformly distributed load, the meridional force is always

(a) Zero (b) Tensile
(c) Compressive (d) None of these

APPSC AEE 2012

Ans. (c) : In a spherical dome subjected to concentrated load at crown or uniformly distributed load, the meridional force is always compressive. The primary stresses induced in the dome are a compressive force acting along the meridian, called the meridional thrust.

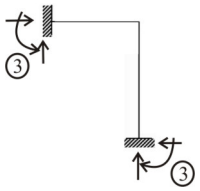
84. What is the degree of static indeterminacy of the structure shown in figure?



(a) 1 (b) 2
(c) 3 (d) 4

APPSC AEE 2012

Ans. (c) :



Degree of static = R – E
indeterminacy = 6 – 3 = 3

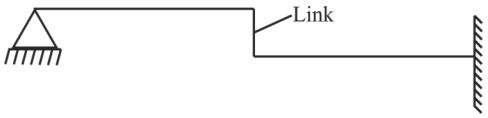
85. The number of independent equations to be satisfied for static equilibrium of a plane structure is

(a) 1 (b) 2
(c) 3 (d) 4

APPSC AEE 2012
ESE 2011

Ans. (c) : The number of independent equation to be satisfied for static equilibrium of a plane structure is three

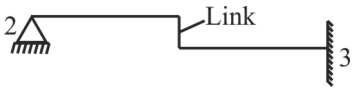
86. The structure is



(a) unstable determinate
(b) unstable indeterminate
(c) stable determinate
(d) stable indeterminate externally

BPSC Poly. Tech. Lect. 2016

Ans. (c) :



$D_{se} = r_e - r_r$
= 5 – 3 – 2 = 0
D_{si} = m – (2j – 3) = 3 – (2×4 – 3)
3 – 9 = –6
stable and determinate.

87. Unequal settlements in the support of a statically indeterminate structure develop

(a) member forces
(b) reactions at support(s)
(c) strains in the members
(d) no reactions

BPSC AE Aug. 2019 Paper-V

Ans. (b) : For indeterminate structures, due to unequal settlements of supports, redundant forces will develop and it can be solved by removing redundant forces as converting it to determinate structure.

88. If in a Pin jointed plane frame (m+r)> 2j, then the frame is

(a) Stable and statically indeterminate
(b) Stable and statically determinate
(c) Unstable
(d) Unstable and statically indeterminate

GPSC AE June 2019

Ans. (a) : A a pin jointed plane frame (m+r) > 2J, then the frame is stable and statically indeterminate.

89. Consider the following:

1. In addition to equilibrium equations, compatibility equations are also required for solving indeterminate structure.
2. A fixed beam (two ends are fixed) is a kinematically determinate structure.

(a) Both 1 and 2 are correct
(b) Both 1 and 2 are wrong
(c) Only 1 is correct
(d) Only 2 is correct

APPSC AEE (CIVIL/MECHANICAL) 2019

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Ans. (a) : For analysing statically indeterminate structures both equilibrium and compatibility equations are required.
Fixed beam has no degree of freedom.
∴ It is kinematically determinate beam.

90. Find out the Static indeterminacy of the beam in the figure below

(a) 0 (b) 2
(c) 3 (d) 6

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : If not given in a problem consider general loading on a beam.
The beam is indeterminate by = 2 degree
Or
D_s = R – release – equilibrium equation
D_s = 6 – 1 –3 = 2

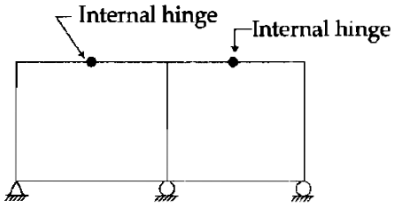
91. In 3-dimension, how many degrees of freedom does a fixed support have?

(a) 0 (b) 2
(c) 3 (d) 6

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : For a fixed support DOF = 0

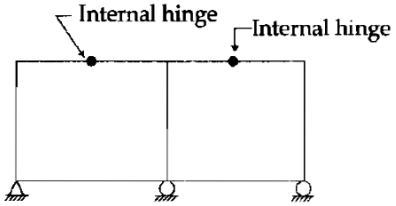
92. Degree of static indeterminacy for the frame shown below is



(a) 8 (b) 7
(c) 6 (d) 5

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a)



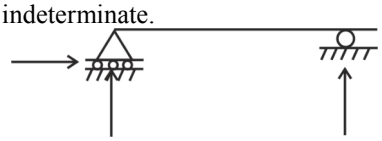
D_S = 3C – R – 2 (m-1)
D_S = 3 × 4– (1+2+3) – 2× (2 – 1)
D_S = 5
C = number of cut of open (tree like structure)
R = number of restrains to make joint rigid fixed
m = number of member joint in hinge

93. Which of the following is NOT an example of indeterminate structure ?

(a) Continuous beam
(b) Two hinged arch
(c) Simply supported beam with overhanging on both the sides
(d) Fixed beam

BHEL ET 2019

Ans. (c) : If the reaction forces can be calculated using equilibrium equations alone, the it called as statically determinate.
If the reaction forces can't be calculated using equilibrium equation alone, then it is called as statically indeterminate.



No. of reaction = 3
No. of equilibrium equation = 3
structure is determinate .

94. A perfect frame should satisfy the relation

(a) m = 2j – 3 (b) m = 2j – 4
(c) m = 3j – 2 (d) m = 3j – 3

APPSC (PHE) AE Paper-2 2012

Ans. (a) : A perfect frame should satisfy the relation– m = 2j–3

95. Which of the following are the statically determinate beams?

(a) Cantilevers
(b) Simply Supported Beam
(c) Overhanging beams
(d) All of the above

APPSC (PHE) AE Paper-2 2012

Ans. (d) : A beam is said to be statically determined if all its reaction components can be calculated by applying three condition of statical equilibrium.
i.e, ΣV = 0, ΣH = 0 and ΣM=0
Ex. Cantilever beam, simply supported beam (SSB) and overhanging beams etc.

96.

What is the degree of kinematic indeterminacy of the beam shown in fig.1 above

(a) 2 (b) 3
(c) 5 (d) 9

MPPSC AE 2014

Ans. (c)

$$D_k = \begin{bmatrix} \text{total possible} \\ \text{DOF (total unrestrained} \\ \text{DOF possible)} \end{bmatrix} - \begin{bmatrix} \text{total} \\ \text{Restrained} \end{bmatrix}$$

$$\therefore D_K = [3\times 3] - [2+1+1]$$

$$\boxed{D_K = 5}$$

97. What is the degree of indeterminacy for the beam shown in Figure ?

(a) 4 (b) 6
(c) 3 (d) 7

CIL MT 2020

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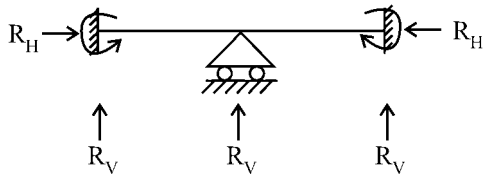
Ans. (a) :



R = 3 + 1 + 3 = 7
indeterminacy = R – 3
= 7–3
= 4

98. In Deflection of determinate structure, the principle of virtual work and energy states that
(a) Work of internal forces / work of external loads = 0
(b) work of external loads is equal to work of internal forces
(c) work of external loads is not equal to work of internal forces
(d) work of external loads/ work of internal forces = 0

DMRC AM 2020

Ans. (b) : The principle of virtual work and energy states that :
work of external load = work of internal load.

99. Ordinary moment resisting frames should not be used in seismic :
(a) zones IV, V
(b) Zone II
(c) zone III with importance factor greater than 1
(d) zone II, III
(e) can used in all 5 zones

CGPSC AE 2017

Ans. (a) Ordinary moment resisting frames should not be used in seismic : zones IV, V

100. A continuous beam has a fixed end at support A and roller support at B and C. What will be the degree of redundancy of the structure?
(a) 3 (b) 1
(c) 2 (d) 4
(e) 6

CGPSC AE 2017

Ans. (c)
Total no. of reaction (r) = 3+1+1 = 5
Total no. of equilibrium equation (e) = 3
degree of redundancy = r - e
= 5 -3
= 2

101. A statically determinate structures:
(a) Cannot be analysed without the correct knowledge of modulus of elasticity
(b) Must necessarily have roller support at one of its ends
(c) Requires only statical equilibrium equations for its analysis
(d) None of the above

Haryana PSC AE 2017
ESE 2008

Ans. (c) : A statically determinate structures requires only statical equilibrium equations for its analysis.

102. If a structure has three degrees of indeterminacy, then what would be the number of plastic hinges that would be formed at complete collapse?
(a) 1 (b) 2
(c) 3 (d) 4

UPRVUNL AE 2014

Ans. (d) : Number of plastic hinges = degree of indeterminacy + 1 = 3 + 1 = 4

103. Consider the following:
1. In addition to equilibrium equations, compatibility equations are also required for solving indeterminate structure.
2. A fixed beam (two ends are fixed) is a kinematically determinate structure.
(a) Both 1 and 2 are correct
(b) Both 1 and 2 are wrong
(c) Only 1 is correct
(d) Only 2 is correct

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : For analysing statically indeterminate structures both equilibrium and compatibility equations are required.
Fixed beam has no degree of freedom.
∴ It is kinematically determinate beam.

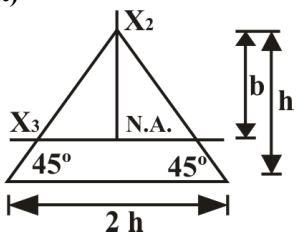
104. Find out the Static indeterminacy of the beam in the figure below

(a) 0 (b) 2
(c) 3 (d) 6

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : If not given in a problem consider general loading on a beam.
The beam is indeterminate by = 2 degree

105. The beam of triangular cross-section as shown in the figure below, is subjected to pure bending. If a plastic hinge develops at a section, determine the location of neutral axis (distance b from top) at that section. The beam material is elastic-perfectly plastic (i.e., yield stress is constant)



(a) $\frac{h}{\sqrt{3}}$ (b) $\frac{h}{2}$
(c) $\frac{h}{\sqrt{2}}$ (d) $\frac{h}{3}$

APPSC AEE (CIVIL/MECHANICAL) 2019

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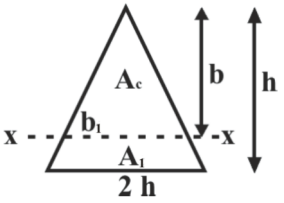
Ans. (c) :



$A_c = A_t$

$$\frac{1}{2}\times b\times b_1 = \frac{1}{2}\left[\frac{1}{2}\times 2h\times h\right]$$

$$b_1 = \frac{h^2}{b}$$

from similar triangles

$$b \to b_1$$

$$h \to 2h$$

$b \times 2h = b_1 h$

$$b = \frac{b_1}{2}$$

$$b = \frac{h^2}{b\times 2}$$

$$b^2 = \frac{h^2}{2}$$

$$b = \frac{h}{\sqrt{2}}$$

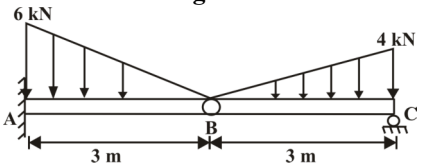
106. In 3-dimension, how many degrees of freedom does a fixed support have?

(a) 0 (b) 2
(c) 3 (d) 6

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : For a fixed support DOF = 0

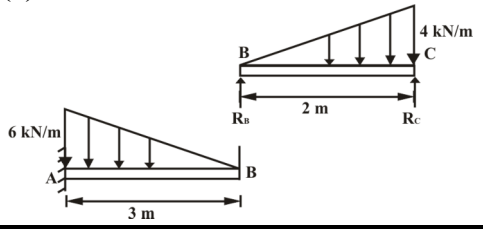
107. Calculate the shear force and bending moment at point B for the beam AB subjected to linearly varying load as shown in the figure. The value of the linearly varying load at the point is 6 kN/m and 4 kN/m, respectively. Point B is an internal hinge.



(a) 2.67 kN and 0 kN-m
(b) 4 kN and 0 kN-m
(c) 4 kN and 1.33 kN-m
(d) 1.33 kN and 0 kN-m

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) :

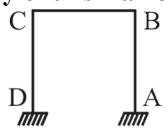


Consider BC part

$$SF_B = R_B = \frac{(4)(2)}{6} = 1.33\ kN$$

$BM_B = 0$ (at hinge moment is zero)

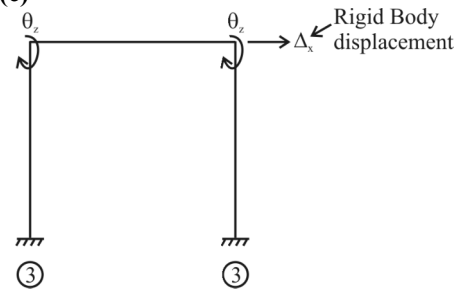
108. A portal frame ABCD is fixed at supports A and D as shown in figure. By neglecting axial deformation, determine the kinematic indeterminacy of this frame :



(a) 1 (b) 2
(c) 3 (d) 6

HPPSC Poly. Lect. 2016

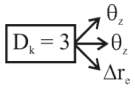
Ans : (c)



$r_e = 6, m = 3, j = 4$

$D_k = 3j - r_e - m$

$= 3 \times 4 - 6 - 3 = 12 - 9$



109. If structure cannot be analyzed for external and internal reactions using static equilibrium conditions alone, such a structure is:

(a) Concrete structures
(b) Indeterminate structure
(c) Determinate structure
(d) Steel structures

IRCON AE 2017

Ans. (b) : A structure is statically indeterminate when the static equilibrium equation (force and moment equilibrium conditions) are insufficient for determining the internal forces and reactions on that structure.

110. The moment need to produce a unit rotation at one end of a member if the other end of the member is fixed is known as:

(a) Rotational Stiffness (b) Deflection
(c) Flexural Stiffness (d) Deformation

IRCON AE 2017

Ans. (a) : Rotational stiffness is the ratio of the applied moment to the angular rotation. For unit rotation, the moment is equal to the rotational stiffness if the other end of member is fixed. $\boxed{K = \frac{M}{\theta}}$

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111. "The first partial derivative of the total internal energy in a structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of the its line of action". This principle is known as :

(a) Castigliano's First Theorem
(b) Castigliano's Second Theorem
(c) Castigliano's Third Theorem
(d) None of the above

GPHC AE 2018
[UPPSC AE 12.04.2016 Paper–I]
[UPPSC State Eng. A.E. 2008]

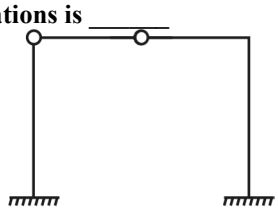
Ans : (b) Castigliano's second theorem : The first Partial derivation of the total Internal Energy in structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action.

$$\frac{\partial E}{\partial F} = \Delta$$

– Castigliano's second theorem :

Where : $\frac{\partial E}{\partial F}$ = Partial derivative of Internal Energy W.R.T. force applied.
Δ = Deflection at the point.

112. The kinematic indeterminacy of the 2-D portal frame shown in Fig. 3 including axial deformations is _____



(a) 6 (b) 11
(c) 8 (d) 9

TSPSC Deputy Engineer 2015

Ans. (d) : For axial deformation –
$= 3j - r$
$= 3 \times 5 - 6$
$= 9$

2. Arches & Suspension Bridges

113 . The horizontal thrust due to rise in temperature in a semi-circular two hinged arch of radius R is proportional -

(a) R (b) R^2
(c) I/R (d) I/R^2

Rajasthan VP ITI 2018
PTCUL AE 2016
APPSC AEE 2012

Ans. (d) : In semi circular two hinged arch, the horizontal thrust due to rise in temperature is proportional to $1/R^2$

$$H = \frac{4EI \propto T}{\pi R^2}$$

$$H \propto \frac{1}{R^2}$$

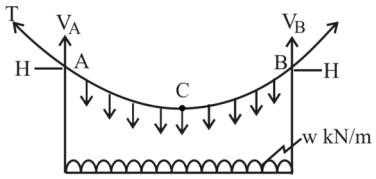
Where,
H = Horizontal thrust
EI = Flexural Rigidity
$\propto$ = Temperature coefficient of material
T = Temperature
R = Radius of arch

114. The maximum tension in a cable occurs

(a) at the highest point in the cable
(b) at the lowest point in the cable
(c) at the centre of the cable
(d) at all point in the cable

TNPSC AE 2019

Ans. (a)



•The maximum tension in cable occur at the height point of the cable i.e. point A and B

$$\left[T_{max} = \frac{WL}{2}\sqrt{1+\frac{L^2}{16d^2}}\right]$$

•The minimum tension in cable occur at the lowest point at 'C'

$$\left[T_{min} = T_C = H\right]$$

115. A 3 hinged circular arch hinged at springing and crown points has a span of 40 m and a central rise of 8 m. It carries a uniformly distributed load 20 kN/m over the left half of the span together with a concentrated load of 100 kN at 30 m away from left end support. Find the horizontal component of reaction developed at left end support.

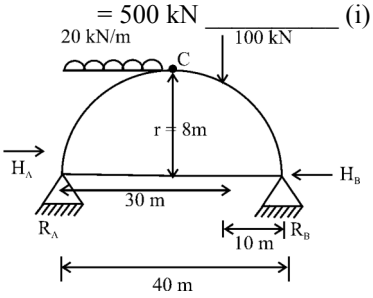
(a) 364.5 kN (b) 235.5 kN
(c) 371.5 kN (d) 312.5 kN

UPPCL AE 2019

Ans. (d) : $\sum V = 0$

$R_A + R_B = 400 + 100$

$= 500$ kN _____ (i)



$\sum M_B = 0$

$R_A \times 40 - 400 \times 30 - 100 \times 10 = 0$

$R_A \times 40 - 12000 - 1000 = 0$

$R_A \times 40 = 13000$

$\boxed{R_A = 325kN}$

from equation (i)

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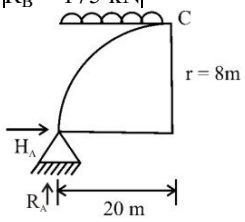
$R_A + R_B = 500$ kN

$\boxed{R_B = 175 \text{ kN}}$



taking moment about C

$H_A \times 8 - 325 \times 20 + 400 \times 10 = 0$

$H_A \times 8 = 2500$

$H_A = \frac{2500}{8}$

$\boxed{H_A = 312.5 \text{ kN}}$

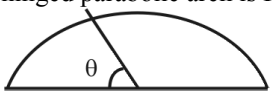
116. The influence line for horizontal thrust in two-hinged parabolic arch is

(a) parabolic (b) cubic

(c) triangular (d) rectangular

BPSC AE 2019 Paper (V) Section-I
GPSC AE June 2019

Ans. (a) : The influence line diagram for Horizontal thrust in a two hinged parabolic arch is Parabolic.



Horizontal thrust (H) = $\frac{W \sin^2 \theta}{\pi}$

117. Maximum sag (or) Dip of the cable varies from_____of horizontal span.

(a) $\frac{1}{5}$ to $\frac{1}{10}$ (b) $\frac{1}{10}$ to $\frac{1}{15}$

(c) $\frac{1}{15}$ to $\frac{1}{20}$ (d) $\frac{1}{20}$ to $\frac{1}{25}$

TNPSC AE 2019

Ans. (b) Dip or Sag of Cable Varies from $\left(\frac{1}{10} \text{ to } \frac{1}{15}\right)$ of Horizontal span

118. If the span and dip of a parabolic cable are *L* and *h* respectively. then the length of the cable is approximately equal to-

(a) $L + \frac{3}{8}h$ (b) $L + \frac{3}{8}h$

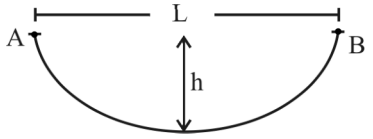
(c) $L + \frac{3}{8}\frac{h^2}{L}$ (d) $L + \frac{8}{3}\frac{h^2}{L}$

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (d) :

Length of the cable (both ends at the same level) ;

$y = kx^2$



At, A, $x = \frac{L}{2}$ and $y = h$

$K = \frac{y}{x^2} = \frac{h}{(L/2)^2} = \frac{4h}{L^2}$

$\therefore y = \frac{4h}{L^2}x^2$

$\therefore \frac{dy}{dx} = \frac{8h}{L^2}x$

Consider an element of length ds of the curve having co-ordinates x an y. The total length S of the curve is given by :

$$S = \int_0^L ds = 2\int_0^{L/2}\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{1/2}$$

$$S = 2\int_0^{L/2}\left[1 + \frac{64h^2}{L^4}x^2\right]^{1/2} dx$$

Expending $\left[1 + \frac{64h^2}{L^4}x^2\right]^{1/2}$ by binomial theorem and neglecting higher power of $\frac{h^2}{L^4}x^2$, we get:

$$S = 2\int_0^{L/2}\left[1 + \frac{1}{2}\frac{64h^2}{L^4}x^2 + ...\right]dx$$

$$= 2\left[x + \frac{32h^2}{3L^4}x^3\right]_0^{L/2}$$

$$= 2\left[\frac{L}{2} + \frac{4}{3}\frac{h^2L^3}{L^4}\right]$$

$$\boxed{S = L + \frac{8h^2}{3L}}$$

119. Which of the following statement(s) is/are true?

(i) Magnification factor is minimum at resonance

(ii) The maximum value of amplification factor increases as damping factor decreases

(iii) The maximum value of amplification factor increases as damping factor increases

(iv) Magnification factor is maximum at resonance

(a) (i) and (ii) (b) (i) and (iii)

(c) (ii) and (iv) (d) Only (ii)

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) : Magnification factor is maximum at resonance

- The maximum value of amplification factors increase as damping factor increases.

120. Bending moment in a parabolic arch when subjected to a uniformly distributed load along its horizontal span is:

(a) sagging always (b) hogging always

(c) zero always (d) sagging or hogging

[UPPSC State Eng. A.E. 2007 (I)]

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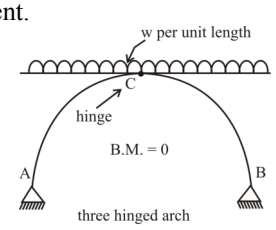
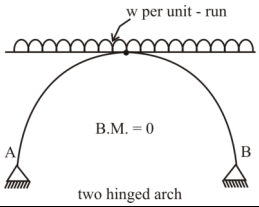
Ans : (c) A parabolic arch either two hinged or three hinged subjected to UDL over entire span is free from bending moment.



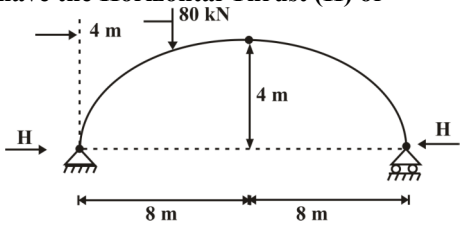


121. A three-hinged parabolic arch of span 16 m and rise 4 m is subjected to a vertical downward concentrated load of 80 KN quarter span. The horizontal reaction at A will be

(JPSC Combined AE Paper V 2013)

OR

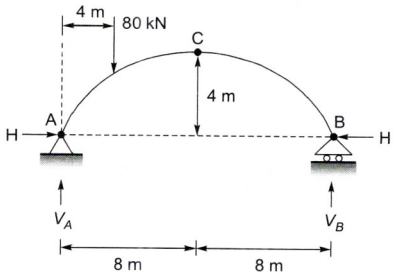
The three hinged arch shown in figure will have the Horizontal Thrust (H) of



(a) 20 kN (b) 30 kN
(c) 40 kN (d) 50 kN

TNPSC AE 2018
ESE 2018
OPSC Civil Services Pre 2011

Ans. (c) :



$\sum M_A = 0$

$80 \times 4 - V_B \times 16 = 0$

$V_B = 20$ kN

$\sum M_C = 0$ from right half

$V_B \times 8 - H \times 4 = 0$

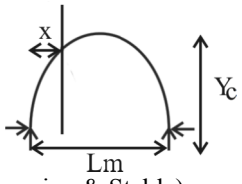
$H = 40$ kN

122. A three hinged parabolic arch of span 'L_m' and central rise 'Y_c' carries a point load W at 'x'm horizontally from the left hinge. The normal thrust 'T' at the section under the load if the horizontal thrust is H and vertical reaction V is

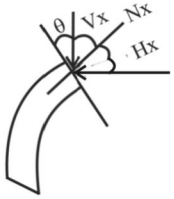
(a) $H\cos\theta - V\sin\theta$ (b) $H\sin\theta + V\cos\theta$
(c) $H\cos\theta + V\sin\theta$ (d) $H\sin\theta - V\cos\theta$

TNPSC AE 2019

Ans. (c) Parabolic – 3 Hinge → BM & radial shear is always (zero).



Ds = 3–3=0 (Determine & Stable)



For Normal thrust:- (T)

$T = V_x \sin\theta + N_x \cos\theta$

For Radial Shear (R)

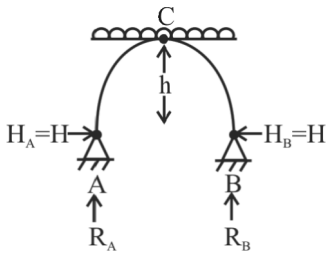
$R = V_x \cos\theta - H_x \sin\theta$

123. A three hinged arch (parabolic) of span 'L' and crown rise 'h' carries a uniformly distributed super imposed load of intensity 'W' per unit length. The hinges are located on two abutments at the same level. The third hinge is at a quarter span from the left abutment. The horizontal thrust on the abutment is

(a) $\frac{WL^2}{4h}$ (b) $\frac{WL^2}{6h}$
(c) $\frac{WL^2}{8h}$ (d) $\frac{WL^2}{12h}$

TNPSC AE 2019
ESE 2003

Ans. (c)



3-hinged arch-parabolic,

$R_A = R_B = \frac{wl}{2}$

$\sum M_C = 0$

$H \times h - \frac{wl}{2} \times \frac{l}{2} + \frac{wl^2}{8} = 0$

$H = \frac{wl^2}{8h}$

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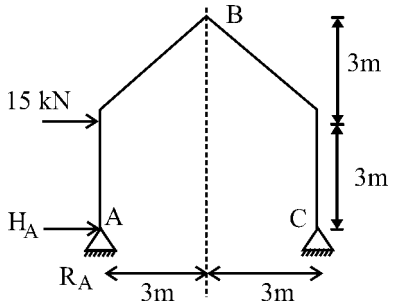
124. For a pin jointed frame at points A, B and C, the horizontal component of reaction at A is

(a) 11.25 kN acting towards right
(b) 11.25 kN acting towards left
(c) 15 kN acting towards right
(d) 15 kN acting towards left
(e) 7 kN acting towards right

CGPSC AE 2014 Shift–I

Ans. (b) :



$\sum M_C = 0$

$\Rightarrow R_A \times 6 - 15\,(3) = 0$

$\Rightarrow R_A = 7.5$ kN

$\sum M_B = 0$ of left part of frame

$R_A \times 3 + 15 \times 3 + H_A \times 6 = 0$

$H_A = -\dfrac{7.5\times3+15\times3}{6}$

$H_A = -11.25$ kN

i.e. direction of horizontal reaction is towards left.

125. Radial Shear (S_x) for three hinge arch is given by:

(a) $R_A.x - H_y$ (b) $R_A\cos\theta - H\sin\theta$

(c) $R_A\sin\theta + H\cos\theta$ (d) $R_A\sin\theta - H\cos\theta$

APPSC Poly. Tech. Lect. 2020

Ans. (b) Radial shear $(S_x) = R_A\cos\theta - H\sin\theta$

and Normal shear $(N_x) = R_A\sin\theta + H\cos\theta$

R_A and H are important for the design of the cross-section of the arch.

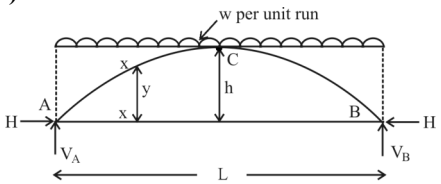
126. A two-hinged parabolic arch is subjected to u.d.l. w over entire span. Then the horizontal thrust is

(a) $\dfrac{wl^2}{3h}$ (b) $\dfrac{wl^2}{4h}$

(c) $\dfrac{wl^2}{6h}$ (d) $\dfrac{wl^2}{8h}$

GPSC Poly. Tech. Lect. 2016
GPSC R&B AE 2016
Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d)



Taking moment at point A

$$-V_B \times L + wL \times \frac{L}{2} = 0$$

$$V_B = \frac{wL}{2}$$

$$V_A = \frac{wL}{2}$$

$$M_x = \frac{wl}{2}\times x - \frac{wx^2}{2}$$

$$y_x = \frac{4h}{l^2}x(l-x)$$

$$\text{Horizontal thrust (H)} = \frac{\int_0^l \mu y\,dx}{\int_0^l y^2\,dx} \quad \{\mu = M_x\}$$

$$= \frac{\int_0^l \left(\frac{wlx}{2} - \frac{wx^2}{2}\right)\frac{4h}{l^2}x(l-x)\,dx}{\int_0^l \left[\frac{4h}{l^2}x(l-x)\right]^2 dx}$$

$$= \frac{\frac{4h}{l^2}\times\frac{w}{2}\int_0^l \left(lx - x^2\right)x(l-x)\,dx}{\left(\frac{4h}{l^2}\right)^2 \int_0^l \left[x(l-x)\right]^2 dx}$$

$$= \frac{\frac{w}{2}\left(\frac{4h}{l^2}\right)\int_0^l \left(lx - x^2\right)dx}{\left(\frac{4h}{l^2}\right)^2 \int_0^l \left(lx - x^2\right)}$$

$$\boxed{H = \frac{wl^2}{8h}}$$

127. A three-hinged symmetrical parabolic arch is subjected to a u.d.l. of w per unit run over the whole span. Then the bending moment is zero at

(a) supports (b) quarter spans
(c) crown (d) All the three above

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d) The bending moment in an arch throughout the span will be zero, if–

- The arch is parabolic.
- The arch carries UDL throughout the span.

128. A three hinged arch is hinged at supports and :

(a) at one quarter span (b) at the crown
(c) anywhere in arch (d) none of the above

(UKPSC AE (Paper–I) 2007)

Ans : (b) A three hinged arch is hinged at supports and at the crown.

129. A three hinged parabolic arch is subjected to a udl of 30 kN/m on the left half of the span. It has a span of 16 m and central rise of 3 m. The horizontal thrust at the left support will be :

(a) 80 kN (b) 120 kN
(c) 182.3 kN (d) 160 kN

KPSC AE 2020

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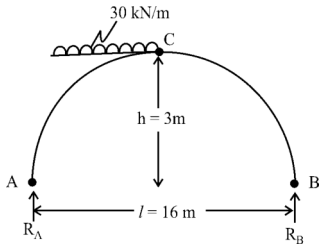
Ans. (d) :



Taking moment about B

$\Sigma M_B = 0$

$R_A \times 16 - 30 \times 8 \times 12 = 0$

$R_A \times 16 = 30 \times 8 \times 12$

$R_A = \frac{30 \times 8 \times 12}{16}$

$R_A = 180$ kN

Taking moment about C

$\Sigma M_C = 0$

$-H_A \times 3 + R_A \times B - 30 \times 8 \times 4 = 0$

$H_A \times 3 = R_A \times 8 - 30 \times 8 \times 4$

$H_A \times 3 = 180 \times 8 - 30 \times 8 \times 4$

$H_A \times 3 = 1440 - 960$

$H_A = 480/3$

$H_A = 160$ kN

130. An arch resists loads by developing :

(a) axial thrust only

(b) axial thrust and shear

(c) shear and bending moment

(d) axial thrust, shear and bending moment

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (d) Primarily the arches are compression members but is often subjected to BM and shear along with axial thrust.

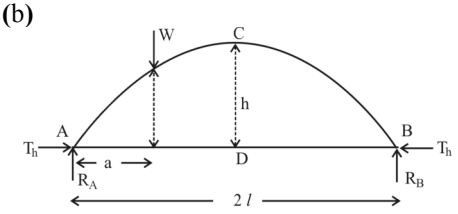
131. A point load 'W' is acting at a distance 'a' from the left support of a three hinged arch of span 2ℓ and rise 'h' hinged at the crown. The horizontal reaction at the support is:

(a) Wa/h (b) Wa/2h

(c) 2W/ha (d) 2h/Wa

[UPPSC AE 12.04.2016 Paper–I]

Ans : (b)



$\Sigma M_A = 0$

$W \times a - R_B \times 2l = 0$

$\left[R_B = \frac{Wa}{2l}\right]$

For horizontal thrust, moment at crown 'C'

$T_h \times h - R_B \times l = 0$

$T_h \times h - \left(\frac{Wa}{2l}\right) l = 0$

$T_h = \frac{Wa}{2h}$

132. A horizontal semi-circular beam of radius 'R' is fixed at the ends and carries a uniformally distributed load 'w' over the entire length. The bending moment at the fixed supports is:

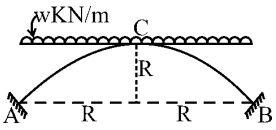
(a) $\frac{wR^2}{4}$ (b) $\frac{wR^2}{3}$

(c) $\frac{wR^2}{2}$ (d) wR^2

[UPPSC AE 12.04.2016 Paper–I]

TRB Poly. Tech. Lect. 2012

Ans : (c)



B.M. at fixed support $= \frac{WR^2}{2}$

133. The horizontal component of reaction at both lower end of a 3-hinged arch is known as

(a) thrust (b) pull

(c) bending (d) shear

[UPPSC State Eng. A.E. 2008]

Ans : (a) The horizontal component of reaction at supports of an arch either three hinge, two hinge or fixed arch is known as thrust.

134. A two hinged semi circular arch of radius, R, carries a concentrated load, N, at the crown. The horizontal thrust is:

WBPSC Poly Lect. 2019

TRB Poly. Lect. Tech. 2017

OR

A two-hinged semi-circular arch of radius 'a' carrying a concentrated load 'P' at the crown develops horizontal thrust equal to:

UPPSC State Eng. A.E. 2008

Maharashtra PSC (Mains) 2019 Paper–I

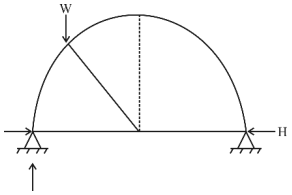
OR

A two hinged semicircular arch of radius R carries a concentrated load W at the crown, the horizontal thrust is

(a) $\frac{W}{2\pi}$ (b) $\frac{4W}{3\pi}$ (c) $\frac{2W}{3\pi}$ (d) $\frac{W}{\pi}$

(UPPSC State Eng. A.E. 2003)

Ans : (d) For two hinged semi-circular arch with load W applied at any section, the radius vector corresponding the horizontal



$H = \frac{W}{\pi} \sin^2 \alpha$

when load is at crown,

$\alpha = 90^0$

$\therefore \quad H = \frac{W}{\pi}$

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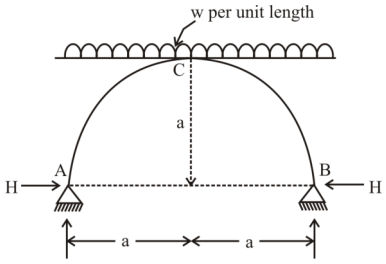
135. A three-hinged semi-circular arch of radius 'a' is subjected to uniformly distributed load of 'w' per unit length on the whole span 'L'. The horizontal thrust will be given by :

(a) $\frac{wL}{2}$ (b) wL

(c) $\frac{wa}{2}$ (d) wa

Maharashtra PSC (Mains) 2017 Paper–I
UPPSC State Eng. A.E. 2008
ESE 1997

Ans : (c)



By symmetry $R_A = R_B = wa$

$$M_c = 0$$

$$R_A \times a - H \times a - w \times a \times \frac{a}{2} = 0$$

$$a\left(R_A - H - \frac{wa}{2}\right) = 0$$

$$\text{or } R_A - H - \frac{wa}{2} = 0$$

$$H = R_A - \frac{wa}{2} = \frac{wa}{2}$$

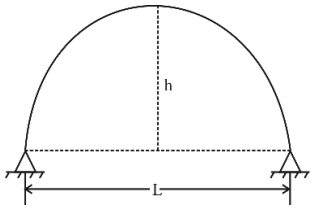
$$H = \frac{wa}{2}$$

136. A two hinged parabolic arch is subjected to a temperature rise of t^oC, then the horizontal thrust at support will:

(a) Increases
(b) decrease
(c) remains same
(d) first increase then decrease

[UPPSC State Eng. A.E. 2011]

Ans : (a) Two hinge arch is an indeterminate structure if two hinge arch is unloaded then there will be no horizontal thrust, but on heating horizontal thrust will produce increase in horizontal thrust.



$$H = \frac{\ell \propto t}{\int \frac{y^2 ds}{EI}} = \frac{\alpha t l EI}{\int_0^l y^2 ds}$$

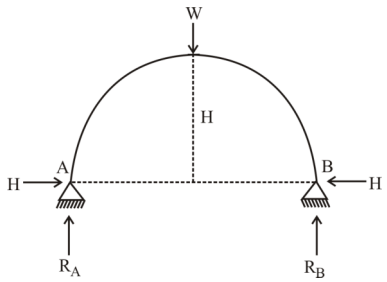
137. A two-hinged parabolic arch of span 'L' and rise 'h' carrying a concentrated load 'P' at the crown, will have horizontal thrust equal to:

(a) $\frac{128}{25}\frac{P.h}{L}$ (b) $\frac{25}{128}\frac{P.h}{L}$

(c) $\frac{128}{25}\frac{PL}{h}$ (d) $\frac{25}{128}\frac{PL}{h}$

LMRC AM 2019
[UPPSC State Eng. A.E. 2008]

Ans : (d)



We know the horizontal thrust in two hinged parabolic arch subjected to concentrated load W at crown in which moment of Inertia $(I) = I_0 \sec\theta$ is given by

$$\boxed{H = \frac{25}{128}\frac{wl}{h}}$$

$$w = p$$

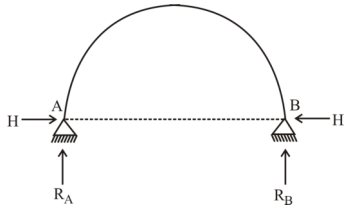
$$H = \frac{25}{128} \times \frac{PL}{h}$$

138. A two-hinged arch is :

(a) statically determinate
(b) statically indeterminate by one degree
(c) statically indeterminate by 2 degrees
(d) geometrically unstable

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (b)



For two hinged arch

$$D_S = \gamma_e - 3$$

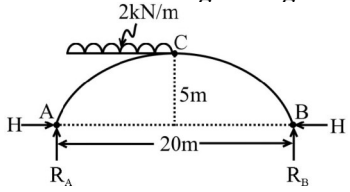
$$\gamma_e = 4$$

$$D_S = 4 - 3$$

$$D_S = 1$$

∴ A two hinged arch is statically indeterminate to one degree.

139. The horizontal thrust at support A in a three hinged arch shown in the given figure is



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(a) 5kN (b) 10kN
(c) 15kN (d) 20kN

(UPPSC State Eng. A.E. 2003)

Ans : (b) $\Sigma F_Y = 0$;

$R_A + R_B = 2 \times 10$

$\Sigma M_B = 0$;

$R_A \times 20 - 2 \times 10 \times 15 = 0$

$R_A = 15\text{kN}(\uparrow)$

$R_B = 5\text{kN}(\uparrow)$

$M_C = 0$ from left

$R_A \times 10 - H \times 5 - 10 \times 2 \times 5 = 0$

$15 \times 10 - 10 \times 2 \times 5 = 5H$

$5H = 150 - 100 = 50$

$H = 10\text{kN}(\rightarrow)$

140. Three hinged arches are statically.....structure.

(a) determinate (b) indeterminate
(c) stoic (d) none of these

(BPSC AE Paper V 2001)

Ans : (a)

→ Three hinged arches are statically determinate structure.

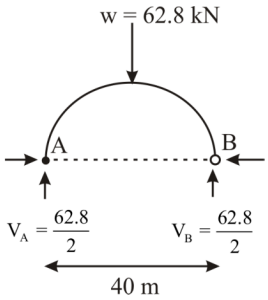
→ Two hinged arches are statically indeterminate structure.

141. A two hinged arch of span 40m carries a point load of 62.8 kN at its crown. The thrust in the arch is (in kN)

(a) 20 (b) 40
(c) 31.0 (d) 62.8

(BPSC AE Paper V 2001)

Ans : (a)



Horizontal thrust at each support–

$$H = \frac{W}{\pi} = \frac{62.8}{3.14} = 20\text{KN}$$

$\boxed{H = 20\text{KN}}$

142. In three hinged arch, maximum hogging moment occurs when, the point load is at:

(a) springing (b) crown
(c) quarter span (d) the section itself

[UPPSC AE 12.04.2016 Paper–I]

Ans : (b) In three hinged arch, maximum hogging moment occurs when, the point load is at crown.

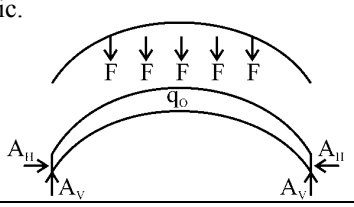
143. The line of thrust in a parabolic arch is:

(a) Parabolic (b) Circular
(c) Triangular (d) Funicular Polygon

Nagaland PSC CTSE 2017 Paper–I

UPPSC AE 12.04.2016 Paper–I

Ans : (a) The line of thrust in a parabolic arch is parabolic.

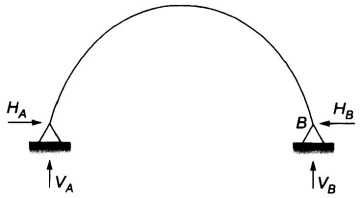


144. In two-hinged arch, how many unknown forces exist?

(a) One unknown (b) Two unknown
(c) Three unknown (d) Four unknown

MPPSC AE 2017

Ans. (d) :



The two-hinged arch is statically indeterminate of first degree. It can be transformed into a primary structure either by removing the horizontal reaction component or by introducing a hinge at crown.

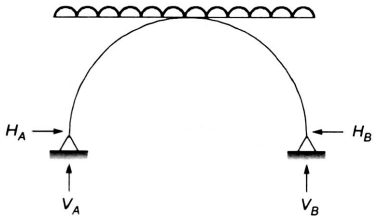
Four unknown reactions – H_A, H_B, V_A and V_B.

145. If load system applied on the arch is entirely vertical the horizontal thrust at each of the supports must be-

(a) Same (b) Different
(c) Zero (d) None of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (a) :



Only vertical loading, no horizontal loading,

$$\Sigma H = 0$$

$$H_A - H_B = 0$$

$$H_A = H_B$$

146. For a single-point load W moving on symmetrical 3-hinged parabolic arch of span *L*, the maximum sagging moment occurs at a distance *x* from ends. The value of *x* is

(a) $0.211L$ (b) $0.25L$
(c) $0.34L$ (d) $0.5L$

BPSC AE 2019 Paper (V) Section-I

Ans. (a) : Considering a three hinged parabolic arch of span l and subjected to a moving point load W, the position of the point load for,

→ Maximum negative bending moment is 0.25 l from end supports.

→ Maximum positive bending moment is 0.211 l from end supports.

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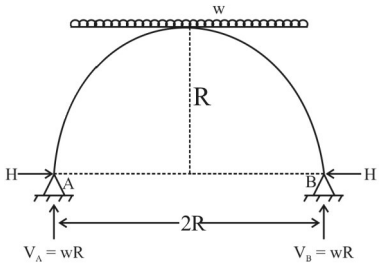
147. A symmetrical two-hinged parabolic arch when subjected to a uniformly distributed load on the entire horizontal span, is subjected to

(a) radial shear alone
(b) normal thrust alone
(c) normal thrust and bending moment
(d) normal thrust, radial shear and bending moment

BPSC AE 2019 Paper (V) Section-I
Punjab PSC SDE 2016
MPPSC AE 2014
UPPSC State Eng. A.E. 2003

Ans. (b) : A two hinged parabolic arch carries a UDL of w per unit run on entire span then it is subjected to normal thrust alone.



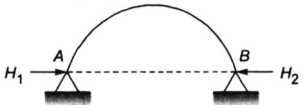
normal thrust (H) = $\frac{4}{3}\frac{WR}{\pi}$

148. For a 2-hinged arch, if one of the supports settles down vertically, then the horizontal thrust

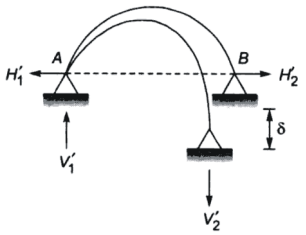
(a) is increased (b) is decreased
(c) remains unchanged (d) becomes zero

BPSC AE 2019 Paper (V) Section-I

Ans. (b) : 1. When the abutments are at same level



2. When the abutments settle down vertically



V_1' and V_2' will try to rotate the beam in clockwise direction. So, that the horizontal thrust H_1' and H_2' will developed in such a way that it will rotate the beam in anticlockwise rotation.
So, net horizontal thrust:

$H_A = H_1 - H_1'$

$H_B = H_2 - H_2'$

149. The Eddy's theorem is valid for

(a) Vertical loads only
(b) Horizontal loads only
(c) Dynamic loads only
(d) All loads

APPSC AE 2016 (Mains)

Ans. (a) : Eddy's theorem is valid only for vertical load. For 3-hinged arch its states that the bending moment at any section an arch is proportional to the vertical intercept between the linear arch and the centre line of actual arch.

150. In a two-hinged arch, an increase in temperature induces

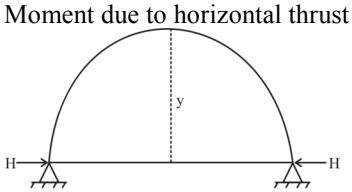
Or

A temperature rise in a two hinged symmetric and parabolic arched rib causes

(a) no bending moment in the arch rib
(b) the uniform bending moment in the arch rib
(c) the maximum bending moment at the crown
(d) the minimum bending moment at the crown

BPSC AE 2019 Paper (V) Section-I
APPSC AE 2016 (Mains)

Ans. (c) Increase in temperature in a two hinged arch (degree of indeterminacy one) will cause horizontal thrust only. Moment due to horizontal thrust is 'Hy'.



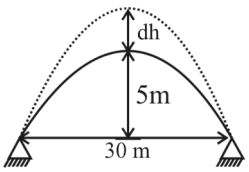
So, maximum B.M. will be at crown as crown has maximum value of y.

151. A parabolic arched rib, of span 30 m, is hinged at the springing and crown and is having a central rise of 5 m. If the coefficient of thermal expansion for the arch material is 12×10^{-6} per ^{0}C, the effect of a temperature rise of 300C is

(a) to cause thermal stresses
(b) to cause thermal stresses as well as a central rise of 18 mm
(c) to cause a central rise of 18 mm
(d) to cause no effect on the structure

APPSC AE 2016 (Mains)

Ans. (c) :



Given, $\alpha = 12\times10^{-6}$ /^{o}C, h = 5m, $\Delta T = 300\ ^oC$, l = 30m
dh = Rise due to temperature change

$$= \frac{4h^2 + l^2}{4h} \times \alpha \times \Delta T$$

$$= \frac{100 + 900}{20} \times 12 \times 10^{-6} \times 300$$

$$= 0.18\text{m} = 180\text{mm}$$

152. The reaction locus for a two hinged semicircular arch is

(a) a horizontal line parallel to the span
(b) a parabolic curve
(c) a cycloid
(d) a elliptical curve

TNPSC AE 2018

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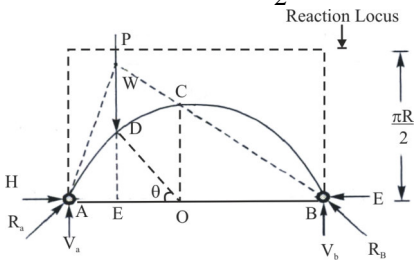
Ans. (a) : Reaction locus of two hinged semicircular arches:

- Reaction locus is straight line parallel to the line joining abutments and height at $\frac{\pi R}{2}$



$\Sigma M_B = 0$

$V_a \times 2R = WR(1 + \cos\theta)$

$$\boxed{V_a = \frac{w}{2}(1+\cos\theta)}$$

$$H = \frac{w}{\pi}\sin^2\theta$$

$$PE = \frac{\pi R}{2}(\text{constant})$$

Reaction locus of semicircular two hinged arch is straight line.

153. Which method is suitable for analyzing a beam when the degree of kinematic indeterminacy is less than degree of static indeterminacy?

(a) Strain energy method
(b) Virtual work method
(c) Slope deflection method
(d) Castigliano's theorem (method of least work)

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) : When ($D_k < D_s$) i.e., Kinematic indeterminacy is less than static indeterminacy then it is displacement stiffness method:-

(1) Moment distribution method
(2) Slope deflection method
(3) Potential energy
(4) Kani's method

154. For a redundant frame if number of members is *m* and number of joints is *j* then which of the following relations will be satisfied?

(a) $m > (2j - 3)$ (b) $m < (2j - 3)$
(c) $m < 2(j - 3)$ (d) $m > 2(j - 3)$

Kerala PSC AE 2015

Ans. (a) : Degree of static indeterminacy = $m + r - 2J$

D_{se}= Degree of external indeterminacy = r–3

Where,

Three is known reaction ($\Sigma F_x = 0$ $\Sigma F_y = 0$ and $\Sigma M = 0$)

$D_s = D_{si} + D_{se}$

$m + r - 2j = D_{si} + (r - 3)$

$D_{si} = m - (2j + 3)$

Condition–

M > (2J–3) – Redundant truss

M = (2J–3) – Balanced, perfect truss

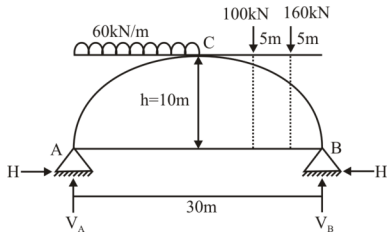
M < (2J–3) – Deficient truss

155. A three-hinged arch has a span of 30 m and a rise of 10 m. The arch carries UDL of 60 kN/m on the left half of its span. It also carries two concentrated loads of 160 kN and 100 kN at 5 m and 10 m from the right end. The horizontal thrust will be nearly

(a) 446 kN (b) 436 kN
(c) 428 kN (d) 418 kN

ESE 2019

Ans. (c)



Taking moment about 'B' :

$\Sigma M_B = 0$

$V_A \times 30 - 60\times15\times22.5 - 100\times10 - 160\times5 = 0$

$30V_A = 1000 + 800 + 20250$

$30V_A = 22050$

$[V_A = 735\text{kN}]$

$\Sigma M_C = 0$ (LHS)

$V_A\times15 - 60\times15\times7.5 - H_A\times10 = 0$

$735\times15 - 60\times15\times7.5 - H_A\times10 = 0$

$10H_A = 11025 - 6750$

$10H_A = 4275$

$H_A = 427.5\text{kN}$

$$\boxed{H_A = 428\text{kN}}$$

156. Which of the following statements are correct in respect of temperature effect on a load-carrying three-hinged arch?

1. No stresses are produced in a three-hinged arch due to temperature change alone.
2. There is a decrease in horizontal thrust due to rise in temperature
3. There is an increase in horizontal thrust due to rise in temperature.

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 only (d) 3 only

ESE 2019

Ans. (a)

- A three hinged arch has two hinges at abutments and one hinge at the crown. It is a statically determinate structure.
- Increase in temperature causes increase in the length of the arch since the hinge at crown is not connected to any permanent object. Increase in the rise of the arch equals

$$\left[\int \frac{\ell^2 + 4h^2}{4h}\alpha T\right]$$

- Stresses are not produced in the arch due to the temperature change alone.
- The decrease in the horizontal thrust due to the rise in temperature is given by :

$$dH = -\frac{dh}{h}(H)$$

Where,

dh = Increase in rise of arch due to rise in temperature.
h = Rise of arch before the temperature increase.
H = Horizontal thrust due to loading before rise in temperature.

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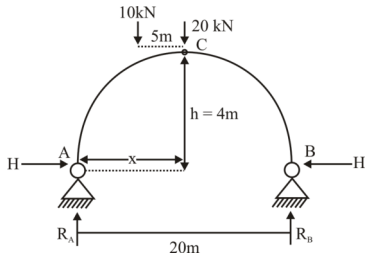
157. A three-hinged parabolic arch ABC has a span of 20 m and a central rise of 4.0 m. The arch has hinges at the ends and at the crown. A train of two point loads of 20 kN and 10 kN, 5m apart, crosses this arch from left to right, with 20 kN load leading. The maximum thrust induced at the supports is

(a) 25.0 kN (b) 28.13 kN
(c) 31.25 kN (d) 32.81 kN

ESE 2017

Ans. (c)



- So, $\Sigma M = 0$ at A

$\Rightarrow R_B \times 20 = 20x + 10\,(x-5)$

$20R_B = 20x + 10x - 50$

$20R_B = 30x - 50$

$[R_B = 1.5x - 2.5]$

Taking moment about 'C'

$\Sigma M_C = 0$

$R_B \times 10 = H \times 4$

$$\left[H = \frac{10}{4}(1.5x - 2.5)\right]$$

H is linear function of x.

So, value of H will be maximum when x = 0.

or at x = 10 (max value of x).

at x = 10

$$H = \frac{10}{4}(1.5\times 10 - 2.5)$$

$$H = \frac{10}{4}(15 - 2.5)$$

$$\boxed{H = 31.25\text{kN}}$$

158. Consider the following statements in respect of arched construction made of voussoirs:

(a) The superimposed load is transferred to the sidewalls only by the strength of cohesion of the mortar between the voussoirs.
(b) The arch may fail under crushing when the compressive stress or thrust in it exceeds the safe crushing strength of the voussoir material.
(c) Every element in the arch is subjected to compression only.
(d) Failure of the arch due to the sliding of any voussoir past the adjacent one due to transverse shear can be avoided by reducing the height of the voussoirs.

Which of the above statements are ***correct?***

(a) 1 and 4 only (b) 1 and 3 only
(c) 2 and 4 only (d) 2 and 3 only

ESE 2017

Ans. (d)

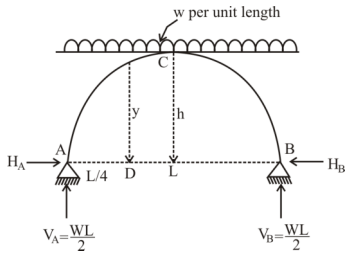
- The load transfer in an arch is not only by cohesion between mortar particles but other parameters are also to be considered.
- In the arch there will be only axial compression therefore the failure is by crushing of voussoir material.

159. A three hinged arch of span 'L' and central rise 'h' is subjected to a UDL of KN/m throughout its span the bending moment at point L/4 from left support will be:

(a) $\frac{wL^2}{4H}$ (b) $\frac{3wL^2}{32}$
(c) $\frac{wL^2}{8h}$ (d) zero

DSSSB AE 2019

Ans. (d)



$$y = \frac{4h}{L^2}x(L-x)$$

at $x = \frac{l}{4}$

$$y = \frac{4h}{L^2}\cdot\frac{L}{4}\left(L - \frac{L}{4}\right) = \frac{4h}{L^2}\cdot\frac{L}{4}\times\frac{3L}{4}$$

$$y = \frac{3h}{4}$$

$$H_A = H_B = \frac{wL^2}{8h}$$

Moment at section x-x from A

$$M_D = V_A.x - \frac{Wx^2}{2} - Hy$$

$$= \frac{WL}{2}\cdot\frac{L}{4} - \frac{w}{2}\left(\frac{L}{4}\right)^2 - \frac{WL^2}{8h}\times\frac{3h}{4}$$

$$= \frac{4WL^2}{32} - \frac{WL^2}{32} - \frac{3WL^2}{32}$$

$$\boxed{M_D = 0}$$

160. A suspension bridge with two hinged stiffening girder is:

(a) statically determinate
(b) statically indeterminate of 1 degree
(c) statically indeterminate of 2 degree
(d) statically indeterminate of 3 degree

Arunachal Pradesh Poly. Tech. Lect. 2019
Nagaland PSC CTSE 2017 Paper–I
UPSC JWM 2017
UPPSC State Eng. A.E. 2011
ESE 2012, 2008

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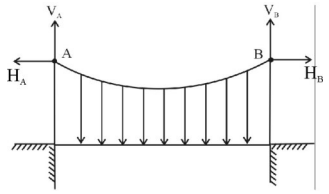
Ans : (b)



$D_s = r_e - 3$

$r_e = 4$

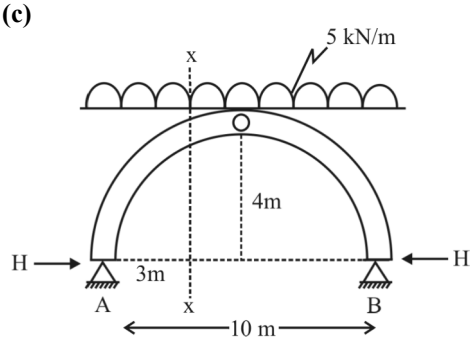
$D_S = 4 - 3 = 1\,\text{Degree}$

161. What is the value of Bending Moment in kN-m at 3m from the left support of a three-hinged parabolic arch of span 10m and rise 4m which carries a uniformly distributed load of 5 kN/m over the whole span?

(a) 27.5 (b) 120
(c) 0 (d) 225

OPSC AEE 2019 Paper–I

Ans. (c)



$H = \frac{w\ell^2}{8h}$, $y = \frac{4hx(l-x)}{l^2}$, $x = 3; l = 10$

$y = \frac{4\times4\times3\times7}{100} = 3.36$, $V_A = \frac{WL}{2} = \frac{5\times10}{2} = 25$

Moment at section x–x from A

$M_D = V_A \times 3 - \frac{5\times(3)^2}{2} - \frac{5\times(10)^2}{8\times4}\times 3.36$

$M_D = 25\times3 - \frac{45}{2} - \frac{1680}{8\times4}$

$M_D = 75 - 22.5 - 52.5$

$[M_D = 0]$

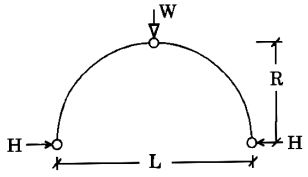
162. Influence line for horizontal thrust in a three hinged parabolic arch is

(a) parabolic (b) circular
(c) triangular (d) none of the above

(UKPSC AE 2012 Paper–I)

Ans : (c) Influence line for horizontal thrust in a three hinged parabolic arch is triangular.

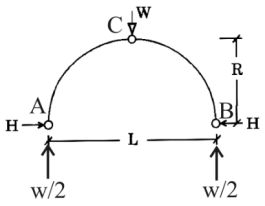
163. For a three-hinged parabolic arch, what will be the ratio L/R to satisfy H = W?



(a) 0.50 (b) 1.50
(c) 2.00 (d) 4.00

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) :



$H = w$

$\frac{wL}{4R} = w \Rightarrow \frac{L}{R} = 4$

$V_A = V_B = w/2$

$BM_C = 0$ (– ve) (+ ve)

$\frac{w}{2}\times\frac{L}{2} - H\times R = 0$

$H = \frac{wL}{4R}$

164. Match the following:

A. Three-hinged arch	1. Statically indeterminate to third degree
B. Two-hinged arch	2. Statically indeterminate to first degree
C. Hingeless arch	3. Statically determinate

	A	B	C
(a)	1	2	3
(b)	3	2	1
(c)	2	1	3
(d)	2	3	1

Maharashtra PSC Mains 2018 Paper-1

Ans. (b) :

Three hinged arch → statically determinate

Two hinged arch → statically indeterminate to first degree

Hingeless arch → statically indeterminate to third degree.

165. If L = span of the suspension bridge, h = dip, t = change in the temperature and α = coefficient of thermal expansion of cable material, then the sag in suspension cable due to change in temperature can be given as

(a) $\frac{3}{16}\times\frac{L}{h}\times\alpha\times t$ (b) $\frac{3}{16}\times\frac{L^2}{h}\times\alpha\times t$

(c) $\frac{16}{3}\times\frac{L}{h}\times\alpha^2\times t$ (d) $\frac{16}{3}\times\frac{L}{h}\times\alpha\times t^2$

GPSC AE March 2018

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Ans. (a) : Sag in suspension cable due to change in temperature = $\frac{3}{16}\times\frac{L}{h}\times\alpha\times t$

166. A linear arch has
(a) Normal thrust only
(b) Shear force only
(c) Bending moment only
(d) Normal thrust and shear force

Vizag steel MT 2017
APPSC AEE 2012

Ans. (a) : Only Normal thrust acts on linear arch.

167. In fixed arch structure, ... nos. of hinges are necessary.
(a) one (b) three
(c) two (d) zero

GPSC R&B AE 2016

Ans. (d) : In case of three hinged arch, we have three hinges:two at the support and one at the crown thus making it statically determinate structure.

168. If cable in transmission line tower, move or deflect downward direction due to it's own self weight, the profile (shape) achieved in elevation is .....
(a) straight line profile
(b) cubic parabola profile
(c) catenary profile
(d) parabolic profile

GPSC R&B AE 2016

Ans. (c) : Catenary profile achieved in elevation when the cable is left freely in a transmission line and move downward due to its own weight.
Catenary curve has s U-like shape, superficially similar in appearance to a parabolic arch. It has minimal surface.

169. For a two-hinged arch if one of the supports settles down vertically, then the horizontal thrust
(a) is increased (b) is decreased
(c) remains unchanged (d) becomes zero

GOA PSC Assistant Prof. 2020

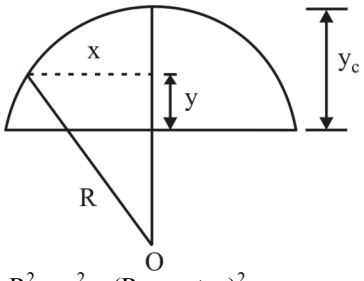
Ans. (c) : For a two-hinged arch if one of the supports settles down vertically, then the horizontal thrust remains unchanged.

170. If a three-hinged parabolic arch carries a uniformly distributed load over the entire span, then any section of the arch is subjected to
(a) normal thrust only
(b) normal thrust and shear force
(c) normal thrust and bending moment
(d) normal thrust, shear force and bending moment

OPSC AEE 2019 Paper–I
JPSC AE 2009

Ans. (a) : If a three-hinged parabolic arch carries a uniformly distributed load over the entire span, then any section of the arch is subjected to normal thrust only.

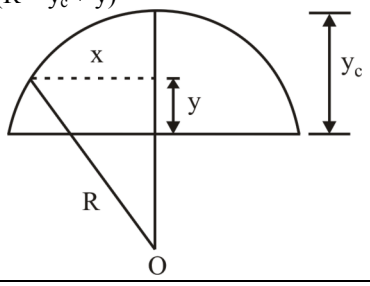
171. In case of circular arches, the following equation will be used :



(a) $R^2 = x^2 - (R - y_c + y)^2$
(b) $R^2 = x^2 + (R - y_c + y)^2$
(c) $R^2 = x^2 + (R + y_c - y)^2$
(d) $R^2 = x^2 + (R - y_c - y)^2$

TRB Poly. Lect. Tech. 2017

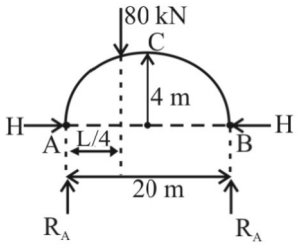
Ans. (b) : In case of circular arches,
$R^2 = x^2 + (R - y_c + y)^2$



172. A three hinged parabolic arch subjected to a load w at L/4 distance from left support. The rise of the crown is 4 m, span is 20 m and w is 80 kN, the horizontal thrust at the supports is
(a) 40 kN (b) 25 kN
(c) 50 kN (d) 80 kN

TRB Poly. Tech. Lect. 2012

Ans. (c) :



Verified R_{x_n} at Right hinged support (B)

$$R_B = \frac{80\times 5}{20}$$

$$R_B = 20\,kN$$

Assuming Horizontal R_{x_n} at the hinge support is (H) taking moment about crown C

BM at C = 0

$H \times 4 + R_B\,(10) = 0$

$$\Rightarrow H = \frac{20(10)}{4}$$

$$\boxed{H = 50kN}$$

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173. The simply supported bending moment at the central hinge of a three hinged arch is 36 kN-m. The span and rise at the hinge are 12 m and 3 m respectively. The horizontal thrust in the arch is

(a) 3 kN (b) 6 kN
(c) 9 kN (d) 12 kN

TRB Poly. Tech. Lect. 2012

Ans. (d) : For simply supported $(M) = \frac{wl^2}{8}$

$36 = \frac{w \times 12^2}{8}$

$w = \frac{36 \times 8}{12^2}$

$w = 2$ kN

Horizontal thrust for three hinge arch

$H = \frac{2 \times 12^2}{8 \times 3}$

$H = 12$ kN

174. The shape of the cable under horizontal UDL is

(a) parabolic (b) catenary
(c) circular (d) triangular

TRB Poly. Tech. Lect. 2012

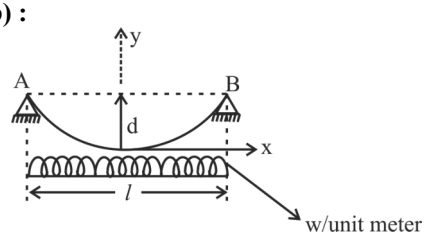
Ans. (a) : The shape of the cable under horizontal UDL is parabolic.

175. A cable of span *l* and central dip d is subjected to uniform load w per unit horizontal length. The horizontal component of tension in the cable is

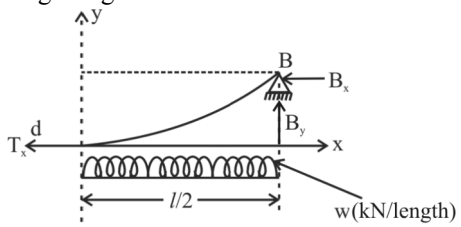
(a) $\frac{wl^2}{4d}$ (b) $\frac{wl^2}{8d}$
(c) $\frac{wl^2}{12d}$ (d) $\frac{wl^2}{16d}$

TRB Poly. Tech. Lect. 2012

Ans. (b) :



FBD of right segment



$\sum M_B = 0$

$T_x(d) - w\left(\frac{\ell}{2}\right) \times \frac{\ell}{4} = 0$

$\Rightarrow \boxed{T_x = \frac{w\ell^2}{8d}}$

176. An arch is used in long span bridges because there is

(a) considerable reduction in horizontal thrust
(b) considerable reduction in bending moment
(c) reduction in shear force and bending moment both
(d) no change in shear force and bending moment

UKPSC AE (Urban) 2011

Ans. (b) : Arch is used in long span bridge because there is considerable reduction in bending moment and it will be helpful.

177. The equation of a parabolic area of span L and central rise h is given by

(a) $y = \frac{h}{L^2}.x(L-x)$ (b) $y = \frac{2h}{L^2}.x(L-x)$
(c) $y = \frac{3h}{L^2}.x(L-x)$ (d) $y = \frac{4h}{L^2}.x(L-x)$

UPPCL AE 2014

UKPSC AE (Urban) 2011

Ans. (d) : The equation of parabolic area of span L and central rise h is given by

$y = \frac{4h}{L^2}.x(L-x)$

178. A three-hinged symmetrical parabolic arch is hinged at the springing and at the crown. The span and rise are 40 m and 10 m respectively. The left half of the arch is loaded with a UDL of 3 t/m. The horizontal thrust at the springing will be

(a) 15 t (b) 20 t
(c) 30 t (d) 40 t

Punjab PSC SDE 2016

Ans. (c) : When the half of Parabolic Arch is loaded by UDL, then

$$H = \frac{wl^2}{16h}$$

w = weight in half span

l = lenght of span

h = height of rise

$$= \frac{3 \times 40^2}{16 \times 10} = 30\,\text{t}$$

179. If a three-hinged parabolic arch carries a uniformly distributed load throughout, every section of the arch resists only

(a) normal thrust (b) shear force
(c) bending moment (d) All of the above

BPSC AE Aug. 2019 Paper-V

Ans. (a) : Three hinged parabolic arch subjected to uniformly distributed load is not subjected to bending moment at any cross section. It support the load in pure compression or normal thrust.

180. A cable supported at two ends primarily resists external loads by

(a) Bending (b) Compression
(c) Tension (d) Bending and compression

UPRVUNL AE 2016

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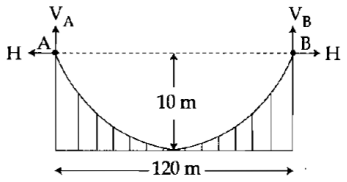
Ans.: (c) On both ends, comes the compression force on applying the external load, but cable particles will oppose compression, that is, they will apply tensile stress.

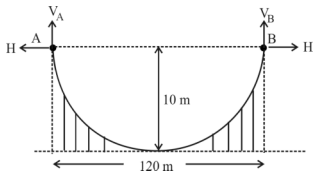
181. A cable of span 120 m and dip 10 m carries a load of 6 kN/m of horizontal span. The maximum tension in the cable is .....



(a) 1238.42 kN (b) 1138.42 kN
(c) 1038.42 kN (d) 1338.42 kN

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (b) :



Horizontal Pull –

$$H = \frac{WL^2}{8h} = \frac{6\times(120)^2}{8\times10}$$

$$H = \frac{8640}{8} = 1080 \text{ kN}$$

Vertical reaction of supports :

$$V_A = V_B = \frac{WL}{2} = \frac{6\times120}{2}$$

$$\left[V_A = V_B = 360 \text{ kN}\right]$$

Maximum Tension in cable –

$$T_{max} = \sqrt{H^2 + V^2}$$

$$T_{max} = \sqrt{(1080)^2 + (360)^2}$$

$$T_{max} = \sqrt{1166400 + 129600}$$

$$T_{max} = \sqrt{1296000}$$

$$\boxed{T_{max} = 1138.42 \text{ kN}}$$

182. The cable and arch are subjected to axial forces respectively as, ...

(a) Tensile and Compressive
(b) Compressive and Tensile
(c) Tensile and Tensile
(d) Compressive and Compressive

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a) : The cable and arch are subjected to axial forces respectively as, tensile and compressive.

183. A parabolic three hinged arch ABC is supporting Uniformly Distributed Load of 500 N/m over its entire span of 100 m. The center point 'B' is vertically 25 m high from supports A and C. The reactions shall be .....

(a) 50 kN horizontal and vertical reactions at each support
(b) 25 kN horizontal and 50 kN vertical reactions at each support
(c) 50 kN horizontal and 25 kN vertical reactions at each support
(d) 25 kN horizontal and vertical reactions at each support

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) :

$$H = \frac{wl^2}{8h} = \frac{500\times100^2}{8\times25}$$

$$H = 25 \text{ kN}$$

$$V = \frac{wl}{2} = \frac{500\times100}{2} = 25000 \text{ N} = 25\text{kN}$$

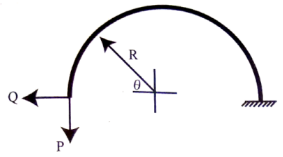
184. The triangle space formed between the extrados and the horizontal line drawn through the crown of an arch is known as

(a) haunch (b) spandrel
(c) voussoirs (d) skewbacks

GPSC AE Dec 2018

Ans. (b) : The triangle space formed between the extrados and the horizontal line drawn through the crown of an arch is known as spandrel.

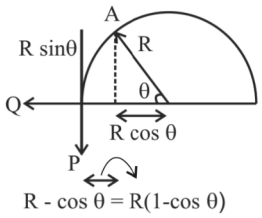
185. The moment equation as a function of angle θ for a built-in arch shown in the following figure can be given as



(a) $M = QR\sin\theta - PR(1-\cos\theta)$
(b) $M = QR(1-\cos\theta) - PR\sin\theta$
(c) $M = QR\cos\theta - PR(1-\sin\theta)$
(d) $M = QR(1-\sin\theta) - PR\cos\theta$

GPSC AE Dec 2018

Ans. (a) :



R - cos θ = R(1-cos θ)

$M_A = Q \times R \sin\theta - P \times R(1 - \cos\theta)$

186. The span of the bridge is said to be economical when

(a) the cost of supporting system of one span is equal to cost of one pier
(b) the cost of supporting system of one span is equal to cost of one abutment
(c) the cost of pier is equal to half the cost of abutment
(d) the cost of supporting system of one span is equal to half the cost of pier

GPSC AE Dec 2018

Ans. (a) : The span of the bridge is said to be economical when the cost of supporting system of one span is equal to cost of one pier.

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187. For a two hinged arch having constant EI, the horizontal thrust H at a point (x, y) on the arch, in terms of beam moment M is given by : (E : modulus of elasticity, I – moment of inertia)

(a) $\dfrac{\int M\, y \dfrac{dy}{dx}}{\int y^3 \dfrac{ds}{EI}}$ (b) $\dfrac{\int M\, y^2 \dfrac{dy}{dx}}{\int y^2 \dfrac{ds}{EI}}$

(c) $\dfrac{\int M\, y \dfrac{dy}{dx}}{\int y^2 \dfrac{ds}{EI}}$ (d) $\dfrac{\int M\, y \dfrac{dy}{dx}}{\int y \dfrac{ds}{EI}}$

BHEL ET 2019

Ans. (c) : The horizontal displacement in a curved member can be evaluated by utilizing either castiglion's second theorem or the unit load method.

$$I_1 = \int_A^B M \frac{\partial M}{\partial H} \frac{dx}{EI} = \int_A^B M.y \frac{dx}{EI}$$

Similarly $I_2 = = \int_A^B y^2 \frac{dx}{EI}$

So, $\left[H = \frac{I_1}{I_2} \right]$

$$\left[H = \frac{\int M.y \frac{dx}{EI}}{\int y^2 \frac{dx}{EI}} \right]$$

188. A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $Y = 2x - 0.1\, x^2$ where Y is the height of the arch in meters. The maximum possible height of the arch is:

(a) 8 m (b) 10 m

(c) 12 m (d) 14 m

PTCUL AE 2016

Ans. (b) : $\boxed{Y = 2x - 0.1\, x^2}$ → eqn of height of Arch

for maximum height

$$\frac{dy}{dx} = 0 = 2 - 0.2x \Rightarrow 2 - 0.2x = 0 \Rightarrow 0.2\, x = 2$$

$x = 10$ m

$Y = 2\times 10 - 0.1 \times 10^2 = 10$

$\boxed{Y = 10 \text{ m}}$

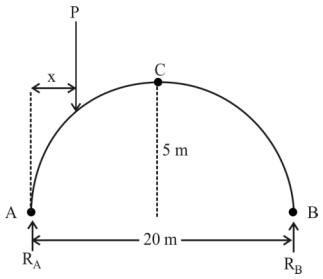
189. A parabolic arch, symmetrical, with hinges at centre and ends, carries a point load P at distance x from left support. The arch has a span of 20 m and rise of 5 m. What is the value of x if the left hinge reaction is inclined with a slope of two vertical on one horizontal?

(a) 8 m (b) 5 m

(c) 4 m (d) None of the above

UJVNL AE 2016

Ans. (c) :



$R_A + R_B = P$

$H_A = H_B$

$\Sigma M_A = 0$

$Px - 20\, R_B = 0$

$$R_B = \frac{Px}{20}$$

$$R_A = P - \frac{Px}{20}$$

$\Sigma M_C = 0$

$-10\, R_B + 5\, H_B = 0$

$$5H_B = 10 \times \frac{Px}{20} = \frac{Px}{2}$$

$$H_B = \frac{Px}{10}$$

$$H_A = \frac{Px}{10}$$

$$\tan\theta = \frac{P - Px/20}{Px/10}$$

$$\tan\theta = \frac{1 - x/20}{x/10}$$

$$\alpha = \frac{1 - \frac{x}{20}}{x/10}$$

$$\Rightarrow \frac{2x}{10} = 1 - \frac{x}{20}$$

$$\Rightarrow \frac{x}{5} = 1 - \frac{x}{20}$$

$$\Rightarrow \frac{4x + x}{20} = 1 \Rightarrow 4\text{m}$$

190. For given two hinged semi circular arches A,B and C of radii 5m, 7.5m and 10m respectively and carrying concentrated crown load 'W', ratio of horizontal thrust at their support will be –

(a) 2 : 1½ : 1 (b) 1 : 1½ : 2

(c) 1 : 1½ : 3 (d) 1 : 1 : 1

UPPCL AE 2014

Ans. (d) : For two hinged semi-circulation arches with load 'W' applied at any section.

$$\left[\text{Horizontal thrust} = H = \frac{W}{\pi} \right]$$

Which is independent of radius of semi-circular arch.

So, for radius, R_1, R_2 and R_3 the H is in the 1 : 1 : 1 ratio.

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191. A symmetrical parabolic arch of span 20 m and rise 5 m is hinged at the springings. If supports uniformly distributed load of 2 tonnes per meter run of the span. The horizontal thrust in tonnes at each of the springings is

APPSC AEE 2012

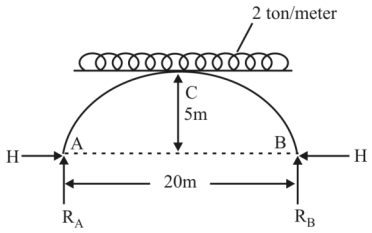
Or

A three hinged symmetric parabolic arch of span 20 meters and rise 5 meters is hinged at the crown and springing. It supports a uniformly distributed load of 2 tonnes per meter run of the span. The horizontal thrust in tonnes at each of the springing is

(a) 8 (b) 16
(c) 20 (d) Zero

MPPSC AE 2014

Ans. (c)



$$R_A = R_B = \frac{WL}{2} = \frac{2\times 20}{2} = 20\,\text{ton}$$

B.M at C = 0

$$R_A \times \frac{L}{2} - H \times h - w\frac{L}{2}\times\frac{L}{4} = 0$$

$$= \frac{WL^2}{4} - \frac{WL^2}{8} = Hh$$

$$\Rightarrow H = \frac{WL^2}{8h} \qquad \text{(given h = 5 m)}$$

$$= \frac{2\times 20^2}{8\times 5}$$

$$\boxed{H = 20\,\text{ton}}$$

192. Among the following, identify the correct statement regarding Linear arch:

(a) Linear arch is one which represents the centre line of an arch.
(b) Linear arch is one which represents variation of bending moment.
(c) Linear arch is one which represents thrust line.
(d) Linear arch is one which represents the variation of shear force.

UPRVUNL AE 2014
ESE 2005

Ans. (c) : Linear arch is one which represents thrust line.

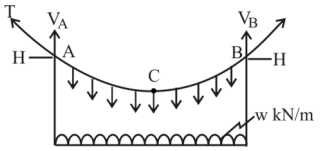
193. The cable carrying UDL load of wkN/m of horizontal span is stretched between supports *l* meter apart. The supports are at same level and central dip is h m. The maximum tension in cable is

(a) $\frac{wl}{2}\sqrt{1+\left(\frac{l}{h}\right)^2}$ (b) $\frac{wl}{2}\sqrt{1+\left(\frac{l}{2h}\right)^2}$

(c) $wl\sqrt{1+\left(\frac{l}{4h}\right)^2}$ (d) $\frac{wl}{2}\sqrt{1+\left(\frac{l}{4h}\right)^2}$

WBPSC AE 2014

Ans. (d)



•The maximum tension in cable occur at the height point of the cable i.e. point A and B

$$\left[T_{max} = \frac{WL}{2}\sqrt{1+\frac{L^2}{16d^2}}\right]$$

•The minimum tension in cable occur at the lowest point at 'C'

$$\left[T_{min} = T_C = H\right]$$

194. Which one is not a redundant structure?

(a) Completely closed ring
(b) Fixed Portal frame
(c) Three-hinged arch
(d) Propped cantilever

WBPSC AE 2003

Ans. (c) : The excess members or reaction of an indeterminate structure are called redundants.
Three hinged arch is not a redundant structures.

195. In a three hinged parabolic arch, the third hinge is located at the crown. The beam moment coinciding with H-moment indicates _____

(a) Arch all along carries only radial shear.
(b) Arch all along carries only beam moment.
(c) Arch all along carries only normal thrust.
(d) Arch all along carries only H-moment.

TSPSC Deputy Engineer 2015

Ans. (c) : In a three hinged parabolic arch, the third hinge is located at the crown. The beam moment coinciding with H-moment indicates arch all along carries only normal thrust.

196. A horizontal thrust 'H' is developed in a two-hinged semi-circular arch of radius 'R' carrying a concentration load 'W' at the crown. What will be the horizontal thrust if radius of the arch is doubled?

(a) H (b) 2H
(c) H/2 (d) H/4

TSPSC AEE 2017

Ans. (a) : For two hinged semi-circulation arches with load 'W' applied at any section.

$$\left[\text{Horizontal thrust} = H = \frac{W}{\pi}\right]$$

So, the horizontal thrust is equal to H

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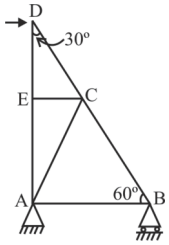
3. Trusses

197. Member(s)n of the frame shown below which carries/carry zero forces is/are:

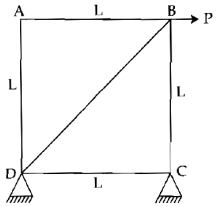


(a) EC only (b) BC and AB
(c) EC and AC (d) EC, AC and AB

Haryana PSC AE 2017
ESE 2003

Ans. (c) : The inspection of truss shows that member EC has zero force as there is no horizontal load at joint E. And member AC is zero force member.

198. What is the force in member AB of the pin-jointed frame as shown below?

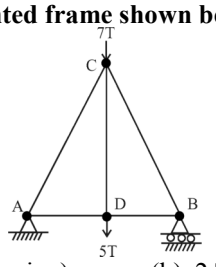


(a) P (tension) (b) P (compression)
(c) $\frac{P}{\sqrt{2}}$ (compression) (d) zero

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) When in perpendicular members no force is acting on the same then in that member the force is zero. Hence AB and AD are perpendicular members and no force is acting on joint hence force in AB is zero.

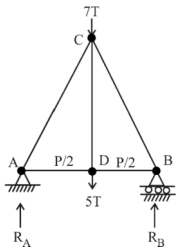
199. What is the force in the vertical member CD of the pin-jointed frame shown below?



(a) 12 T (tension) (b) 2 T (compression)
(c) 7 T (compression) (d) 5 T (tension)

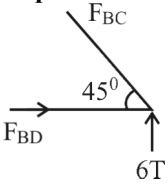
Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (d) :



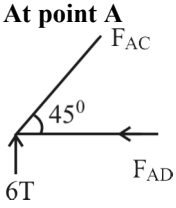
$R_A + R_B = 12$ T
$\sum M_B = 0$
$R_A = 6T$
$R_B = 6T$

At point B



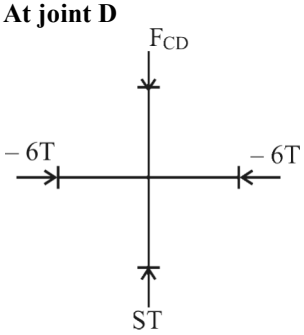
$F_{BC} \sin 45^0 = 6T$
$F_{BD} = -F_{CB} \cos 45^0$
$F_{BD} = -6T$

At point A



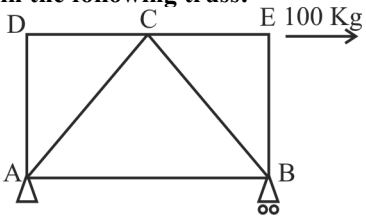
Similarly
$F_{AD} = -6T$

At joint D



$F_{CD} = -5T$
[F_{CD} = 5 T tensile]

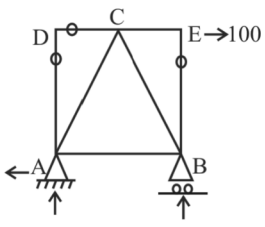
200. Identify the members with zero member force in the following truss:



(a) AD, DC and EB (b) AD, DC and CE
(c) AD, AB and AC (d) None of the above

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) :



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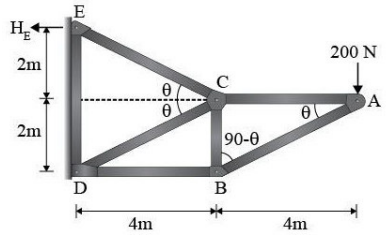
F_{EB} = 0 (because member BE and force 100 are perpendicular)

- At joint E two collinear force member meet then third member force will have (zero force) (F_{BE} = 0)
- AD, DC are meeting at joint (D) where no load act then both member carry zero force

201. Consider the frame as shown in the figure.

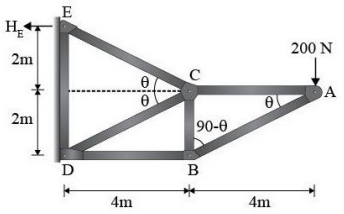


The magnitude of the horizontal support reaction at E is

(a) 400 kN (b) 300 kN
(c) 250 kN (d) 200 kN

ESE 2019

Ans. (a)



Taking moment about D

$\Sigma M_D = 0$

$-H_E \times 4 + 200 \times 8 = 0$

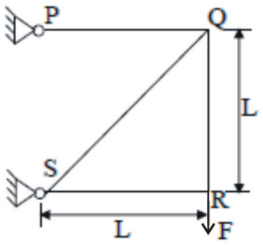
$200 \times 8 = H_E \times 4$

$$H_E = \frac{1600}{4}$$

$[H_E = 400N \text{ or } 4 \times 10^{-1}kN]$

Note–In the above diagram, load is given in 'Newton'.

202.



For the truss shown in the figure, the force in the member *PQ* is

(a) F (b) $\frac{F}{\sqrt{2}}$

(c) $\sqrt{2}F$ (d) $2F$

ESE 2018

Ans. (a) : Considering the moment about; [S = 0]

$F_{pQ} \times L = F \times L$

$\boxed{F_{PQ} = F}$

So, $[F_{PQ} = F \text{ (Tension)}]$

203. Zero load Test indicate where a plane truss is

(a) statically indeterminate (extremely)
(b) statically indeterminate (internally)
(c) statically determinate
(d) geometrically unstable

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d) Zero load test is done to check the geometrical unstable of a statically determinate truss obtained by adding triangles to simple truss.

204. Deflection in a truss depends upon

(a) axial rigidity
(b) flexural rigidity
(c) torsional rigidity
(d) axial and flexural rigidity

GUJRAT PSC AE - II 2017

[UPPSC AE 12.04.2016 Paper–I]

Ans : (b) Deflection of a truss is depends on flexural rigidity

$\left[\delta = \frac{NL}{AE}\right]$ Axial deformation

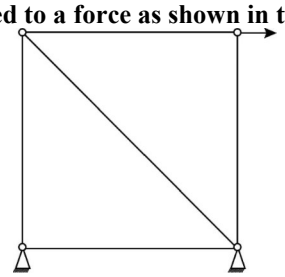
$\left[\Delta = \sum n \frac{NL}{AE}\right]$ Deflection

$N \rightarrow$ Applied load $E \rightarrow$ Modulus of elasticity

$A \rightarrow$ Area $n \rightarrow$ Member

$L \rightarrow$ Length

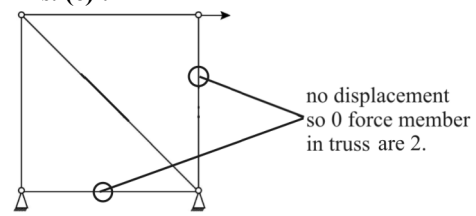
205. Number of zero force members in the truss subjected to a force as shown in the figure is



(a) 0 (b) 1
(c) 2 (d) 3

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) :



206. The truss can be analysed by the methods of joints when the number of unknowns at a joint is equal to

(a) 1 (b) 2
(c) either (a) or (b) (d) 3

TNPSC AE 2012

Ans. (c) Method of joints – In a determinate structure, the reaction components can be worked out using equation of equilibrium. The analysis can be commenced from a joint where there are only two unknown member there are only two unknown member forces utilizing the equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

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207. Struts are the load carrying members of a frame which are subjected to

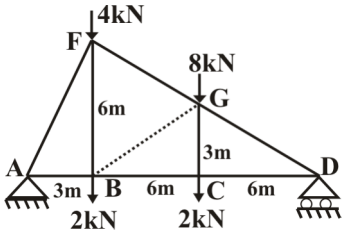
(a) Axial tension loads
(b) Axial compression loads
(c) Torsional loads
(d) Transverse loads

ISRO Scientist/Engineer 2018

Ans : (b)

- A structural member subjected to axial compressive force is called strut.
- Strut may be vertical, horizontal or inclined.
- The cross-sectional, dimensions of strut are small.
- Struts are used in roof truss and bridges trusses.
- When strut is vertical it is known as column it can heavy compressive loads.

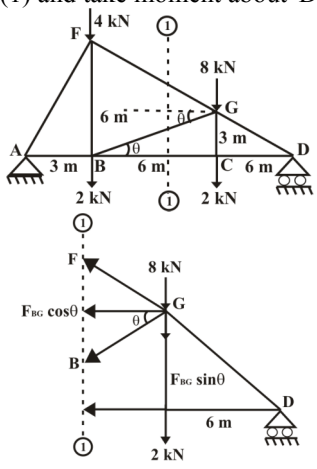
208. Determine the force in the member BG in the given truss



(a) 11.18 kN Tension
(b) 14.4 kN tension
(c) 11.18 kN Compression
(d) 14.40 kN compression

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : Take section 1 - 1 as shown in figure.
To find the force in BG, consider the right part of section (1) - (1) and take moment about 'D'.



$$\cos\theta = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\sin\theta = \frac{3}{3\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$\sum M_D = 0$ ($\curvearrowright$+ ve, $\curvearrowleft$- ve)

$-F_{BG}\cos\theta \times 3 - F_{BG}\sin\theta \times 6 - 2 \times 6 - 8 \times 6 = 0$

$$-F_{BG} \times \frac{6}{\sqrt{5}} \times F_{BG} \times \frac{6}{\sqrt{5}} - 60 = 0$$

$$-2\left[F_{BG} \times \frac{6}{\sqrt{5}}\right] = 60$$

$F_{BG} = -5\sqrt{5}$ kN (compressive)

$F_{BG} = 11.18$ kN (compressive)

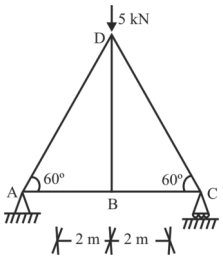
209. The moment distribution method is best suited for

(a) Indeterrminate pin jointed truss
(b) Space frames
(c) Eccentric columns
(d) Rigid frames
(e) Trussed beams

CGPSC AE 2014 Shift–I

Ans. (d) : Moment distribution method is a displacement method.
→ It is a trial & error method, in which final moment are calculated by distributing moment that why best suited for rigid frames.
→ It is time consuming method.

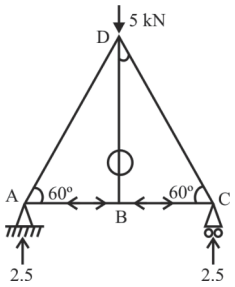
210. The force in the member BD is



(a) 5 kN
(b) $5\sqrt{2}$ kN
(c) 10 kN
(d) 0 kN

TNPSC AE 2019

Ans. (d)



Force in BD = ?
As at joint (B) → Two forces
(F_{BA} & F_{BC}) are linear & meeting at B point & then third force must be zero
$F_{BD} = 0$

211. If the slope of the truss is 15^0, the live load on truss is taken as:

(a) 0.5
(b) 0.55
(c) 0.6
(d) 0.65

APPSC Poly. Tech. Lect. 2020

Ans. (d) : 0.65
Live load for $\theta \le 10^0$
Live load = 0.75 kN/m^2
if $\theta > 10^0$ then
LL = $0.75 - (\theta - 10^0) \times 0.02$
LL = $0.75 - (15^0 - 10^0) \times 0.02$
LL = 0.65 kN/m^2

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212. Consider the following assumptions in the analysis of a plane truss :

(A) The individual members are straight.

(B) The individual members are connected by frictionless hinges.

(C) The loads and reactions act only at the joints.

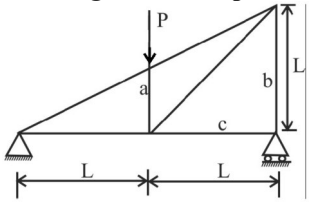
of these assumptions.

(a) (A) and (B) are valid
(b) (A) and (C) are valid
(c) (B) and (C) are valid
(d) (A), (B) and (C) are valid

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : Analysis of trusses :
Assumptions – (1) the centroidal axis of each member coincides with the line connecting the centre of the Adjacent member & members only carry Axial force.
(2) All members are connected only at their end by frictionless hinges in plane trusses.
(3) All loads and support reactions are applied only at the joint.

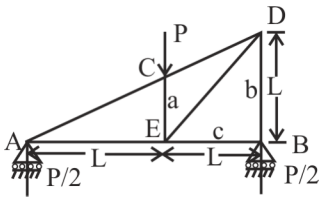
213. The forces in members a, b, c in the truss as shown in the figure are, respectively



(a) P, P/2, 0 (b) P/2, P, 0
(c) P, P, P (d) P/2, P/2, 0

RPSC POLY. TECH. LECT. 2014

Ans : (a)

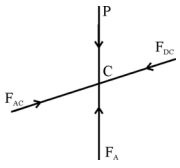


$R_A = p/2$
$R_B = p/2$

at point B–

$\sum V = 0$
$F_b - p/2 = 0$
$F_b = p/2$ compressive
$\Sigma H = 0$
$F_C = 0$

at point C



$F_{AC} = -F_{DC}$
$F_a = P$
$\therefore F_a, F_b, F_c \rightarrow P, P/2, 0$

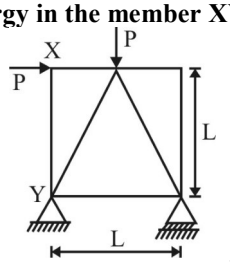
214. Lack of fit between members of a truss causes

(a) no deflections
(b) vertical deflection only
(c) horizontal deflection only
(d) horizontal & vertical deflections both

(UKPSC AE 2012 Paper–I)

Ans : (d) Lack of fit between members of a truss causes horizontal & vertical deflection both.
Internal forces can develop in a redundant frame (with external loads) due to lack of fit.
$M = 2J - 3$ Stable perfect truss
$M < 2J - 3$ Insufficient or unstable truss
$M > 2J - 3$ Internally Redundant truss
Truss pitch $\Rightarrow 25^0$ to 60^0

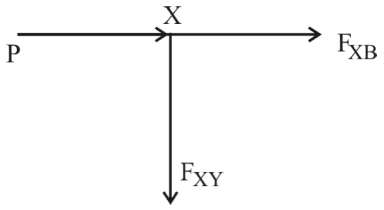
215. A truss is shown in the figure. The cross sectional area of each member is 'A' and the modulus of elasticity of the material is E. The strain energy in the member XY is given by :



(a) $P^2L/2AE$ (b) $P^2L/6AE$
(c) $P^2L/3AE$ (d) Zero

(UKPSC AE 2012 Paper–I)

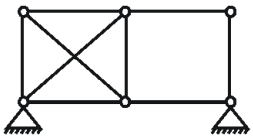
Ans : (d) Member do not have any force, therefore strain energy in it is zero.
Consider joint 'X'



$[F_{XB} = -P] \therefore \sum F_X = 0$
$[F_{XY} = 0] \therefore \sum F_Y = 0$
So, strain energy stored in the member XY –

$$\left[U_{XY} = \int (F_{XY})^2 \frac{dx}{2AE} = 0\right]$$

216. In the truss shown below which statement is correct?



(a) Externally unstable
(b) Internally unstable
(c) Statically determinate structure
(d) Statically indeterminate structure

[UPPSC AE 12.04.2016 Paper–I]

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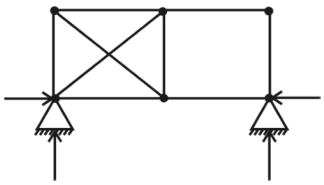
Ans : (b)



For given truss,

External Indeterminacy (α_{se}) = R – 3

= 4 – 3 = 1

Statically Indeterminacy (α_s) = m+R- 2j

= 9+4 – 2 × 6 =1

$\boxed{\alpha_s=1}$

Internally Indeterminacy = $\alpha_s - \alpha_{se}$ = 1–1 =0

The given truss is Internally unstable.

217. The stress carried by the King-Post of a King-Post roof truss is

(a) Tensile

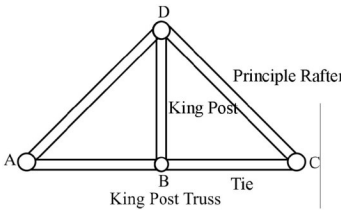
(b) Compressive

(c) Tensile and bending

(d) Compressive and bending

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (b)



The king post (BD) of the king post truss is a compression member. So the stress carried by the king-post of a king post roof truss is compressive in nature.

218. In an internally indeterminate truss, if the area of cross-section of a redundant member is doubled, then

(a) The force in that member will be doubled

(b) The force in that member will be half

(c) The force in that member will increase but remain less than twice the previous value

(d) The force in that member will decrease out remain more than half the previous value

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c) Force in redundant member is given by

$$x = \frac{\Sigma \frac{PkL}{AE}}{\Sigma \frac{k^2L}{AE}}$$

When area of redundant member is double then term $\Sigma \frac{PkL}{AE}$ remain unchanged.

Since P force in redundant in zero. But term $\Sigma \frac{K^2L}{AE}$ decrease. Hence x will be increase.

219. All compression members in trusses are called

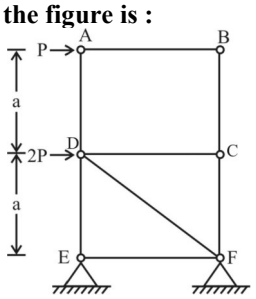
(a) struts (b) ties

(c) stanchions (d) purlins

(BPSC AE Paper V 2001)

Ans : (c) Under compressive loads, a member deflects in direction perpendicular to that of the load. These are all type of compression members : column, strut, post, stanchion and top chords of truss.

220. The force in the member CD of the truss as shown in the figure is :



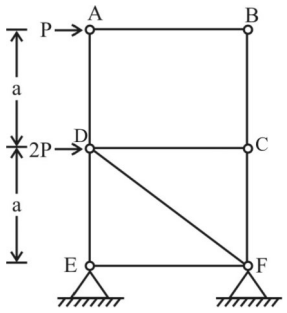
(a) zero (b) 2P compressive

(c) P compressive (d) P tensile

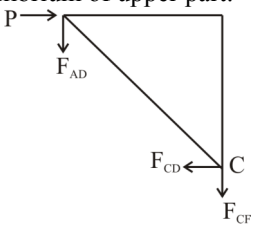
(UKPSC AE (Paper-I) 2007)

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans : (d)



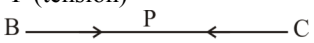
Cutting X-X through member AD, DC and CF and considering equilibrium of upper part.



For horizontal equilibrium

$\Sigma F_H = 0$

F_{CD} = P (tension)



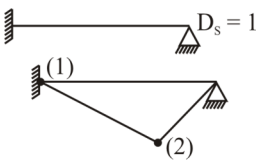
221. In plastic design of structures, if degree of static indeterminacy is 'j', the numbers of hinges 'p' required to convert the structure into mechanism is given by–

(a) P = j + 3 (b) P = j + 2

(c) P = j + 1 (d) None of above

(UTTRAKHAND AE 2013)

Ans : (c)



2 plastic hinges for collapse

Hence option 'C' $\boxed{P = J+1}$

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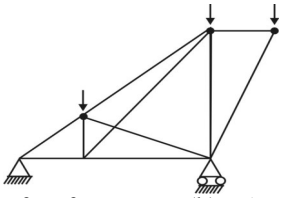
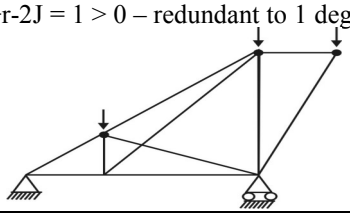
222. The pin jointed frame as shown in following figure is–



(a) A perfect frame (b) A redundant frame
(c) A deficient frame (d) None of above

(UTTRAKHAND AE 2013)

Ans : (b) M = 10, r = 3, J = 6
M+r-2J = 1 > 0 – redundant to 1 degree

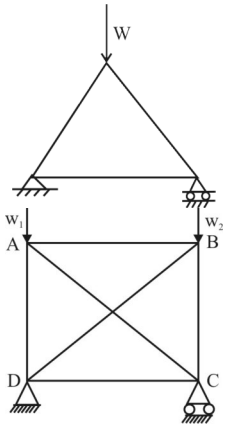


223. For a statically indeterminate pin jointed plane frame, the relation between number of member 'm' and number of joints 'j' is expressed as–

(a) m = 3j – 6 (b) m > 2j – 3
(c) m = 2j – 3 (d) m > 3j – 6

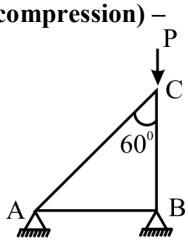
OPSC AE - 2016 (I)
(UTTRAKHAND AE 2013)

Ans : (b) For determinate truss
M = 2J – 3 in 2D
and M = 3J – 6 in 3D



IF M > 2j–r, then the number of unknowns is more than the number of equation, A redundant frame is indeterminate, If m < 2j–r, the number of equation is more, than the number of unknown.

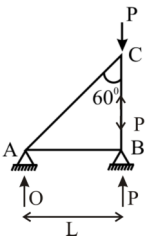
224. The forces in the members AC and AB for the truss shown below will be respectively (+tension,–compression) –



(a) –P/cos 60 and + P tan 60
(b) –P/cos 60 and zero–P
(c) zero and + P tan 60
(d) zero and zero

(UTTRAKHAND AE 2013)

Ans : (d)



$\Sigma M_A = 0$
$R_B \times L = P \times L$
$R_B = P$
At joint B,
$F_{BC} = P$
$F_{AB} = 0$
At joint A,
$F_{AC} = 0$

225. If k_i is the stiffness of i^{th} member at a joint, the distribution factor for the member is-

(a) k_i (b) $\frac{k_i}{\Sigma k_i}$
(c) Σk_i (d) $\left(\Sigma k_i - k_i\right)$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) : If the stiffness factor of the i^{th} member is K_i then the moment contributed by the member is $M_i = K_i$, then the distribution factor for the i^{th} member is –

$$Df_i = \left(\frac{\text{Stiffness of } i^{th} \text{ member}}{\text{Total stiffness of joint}}\right) = \frac{K_i}{\Sigma K_i}$$

226. A truss deflection is to be calculated for a point '*X*' using unit load method. Find the contribution to deflection by a truss member having details as below :
Length = 8.00 m
Area = 11.25×10^{-3} m^2
Internal force resisted member under actual loading condition = 450 kN (compression) modulus of elasticity = 200 × 10^6 kN/m^2 internal force after unit load application = 0.0 kN-

(a) -0.5 mm (b) +0.5 mm
(c) -0.6 mm (d) Zero

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : Deflection using unit load method is given as

$$\delta = \frac{P_u L}{AE}$$

$$= \left(\frac{450 \times 0 \times 8}{11.25 \times 10^{-3} \times 200 \times 10^6}\right) = 0$$

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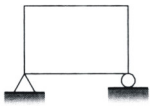
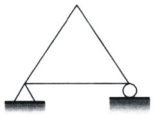
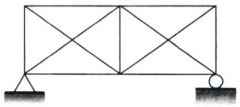
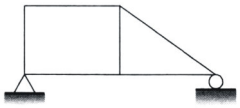
227. Which truss is the perfect out of the following?

(a) 

(b) 

(c) 

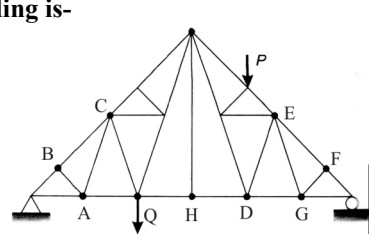
(d) 

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : A perfect truss is one in which the following equation is satisfied $m = (2j - 3)$

1.	m = 4, j = 4,	m= 2j – 3 ⇒ 4 = 5	No
2.	m = 3, j = 3	3 = 6 – 3 = 03	Yes
3.	m = 11, j = 6	11 = 9	No
4.	m = 6, j = 5	6 = 7	No

228. For the plane truss shown in the figure, the number of zero force members for the given loading is-



(a) 4 (b) 8
(c) 11 (d) 13

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Ans. (b) : Zero force member concept : If 3 members meet at a point and two of them are collinear, then force in 3rd member is 0.

229. A given determinate truss is loaded with gravity loads. Under these loads different nodes undergo deflection horizontally and vertically. Thereafter the truss is subjected to a temperature drop of 50°C in the lower chord only. The coefficient of expansion or contraction $\alpha = 117 \times 10^{-6}$/°C. Which of the following statements is true?

(a) Vertical and horizontal deflection along lower chord nodes remains the same
(b) Vertical and horizontal deflections along lower chord nodes shall change
(c) Horizontal deflection along lower chord nodes shall change but vertical deflection shall be not change
(d) None of the above

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : Vertical and horizontal deflection along lower chord nodes shall change.
given values,
$\Delta T = 50^0 C$
$\alpha = 117 \times 10^{-6}\ ^0C$
Due to temperature change, change in length of members is given as
$\Delta = \alpha \Delta T L$

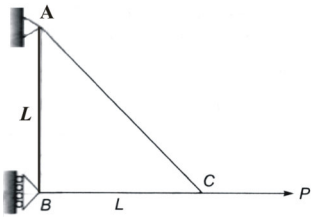
230. In the force method of analysis of indeterminate trussed, If the truss is indeterminate to degree one, the change in length of redundant member due to unit force is found by using the formula-
where,
A is cross-sectional area-
***I*- Moment of Inertia**
***n*- force in the member due to unit load application**
***N*- force in the member due to actual load**
***E*- Modulus of Elasticity**

(a) $\Sigma \frac{n\,NL}{EI}$ (b) $n\Sigma \frac{NL}{AI}$
(c) $\Sigma \frac{n\,NL}{AE}$ (d) $\Sigma \frac{NL}{AE}$

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (c) : $\delta = \sum n \frac{NL}{AE}$
It is derived from unit load method.

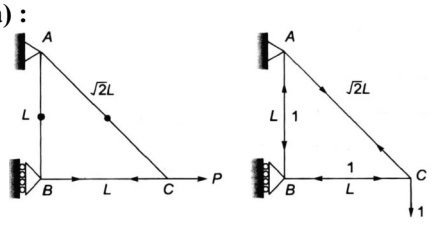
231. A simple truss *ABC* is supported at *A* and *B* as shown in the figure. If a point load (P) along *BC* is applied at joint *C* in horizontal direction, then what will be the vertical deflection at *C*?
Assume same $\frac{C}{5}$ area and same materials (i.e., *A, E, I* same for all members).



(a) $\frac{PL}{AE}(\uparrow)$ (b) $\frac{2PL}{AE}(\downarrow)$
(c) $\frac{PL}{AE}(\downarrow)$ (d) $\frac{2PL}{3AE}(\downarrow)$

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (a) :



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$$\delta_C = \frac{\sum PKL}{AE}$$

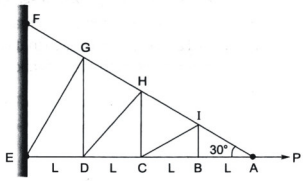
$$\delta_c = \left(\frac{\text{PKL}}{\text{AE}}\right)_{AB} + \left(\frac{PKL}{AE}\right)_{AC} + \left(\frac{PKL}{AE}\right)_{BC}$$

$$= 0 + 0 + \frac{\text{P}\times(-1)\times\text{L}}{\text{AE}}$$

$$= -\frac{\text{PL}}{\text{AE}}$$

Negative sign indicates deflection of the joint is opposite to applied force of 1 KN i.e. upwards.

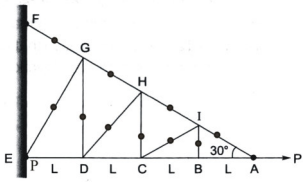
232. A cantilever truss as shown in the figure is subjected to a horizontal load 'P' at joint A. The total number of zero force members in the truss is-



(a) 6 (b) 4
(c) 9 (d) 10

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (d) :



For truss joint with two member which have no external load at that joint is not support reaction, the two members are zero force member, zero member force – BI, IC, CH, HD, DG, GE, FG, GH, HI, IA. We also know three member meet a point and two of them are collinear then force in 3rd member is zero.

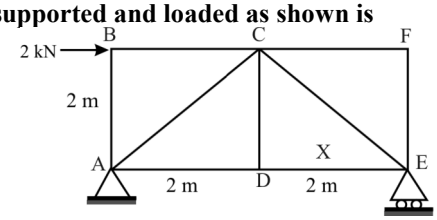
233. In which of the following truss members the stress depends upon whether the load is moving on top chord or bottom chord?

(a) Top chord and bottom chord
(b) Verticals
(c) Diagonals
(d) Verticals and diagonals

BPSC AE 2019 Paper (V) Section-I

Ans. (c) : The top and the bottom members are connected by vertical or diagonal members called web members.
In diagonal members the stress depends on whether the load is moving on top chord or bottom chord.

234. The force in the member 'X' of the truss supported and loaded as shown is



(a) $\sqrt{2}$ kN, Tension (b) $\sqrt{2}$ kN, Compression
(c) 1 kN, Tension (d) 1 kN, Compression

APPSC AE 2016 (Mains)

Ans. (c) : Members AB, CD, CF and EF are zero member forces

$H_A = 2kN$

balance the moment about roller support E

$V_A \times 4 + 4 = 0$

$V_A = -1$

$V_E = +1$

$F_{AC} \cos\theta = 1$

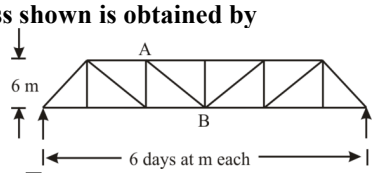
$F_{AC} = \sqrt{2}$ (Tension)

$F_{AD} = F_{AC} \cos\theta = \sqrt{2} \times \frac{1}{\sqrt{2}} = 1kN$

At joint D

$F_{AD} = F_{DE} = 1$ kN (Tension)

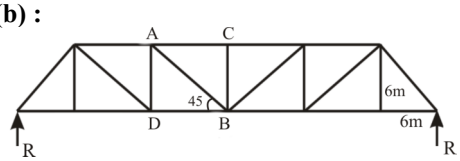
235. Influence line for member force in AB of the truss shown is obtained by



(a) $\sqrt{2}$ times the ordinates of influence line for bending moment at A
(b) $\sqrt{2}$ times the ordinates of influence line for shear in 3rd panel from left
(c) $\sqrt{2}$ times the ordinates of influence line for bending moment at B
(d) $s\sqrt{2}$ times the ordinates of influence line for shear in 3rd panel from right

APPSC AE 2016 (Mains)

Ans. (b) :



Now use $\sum f_x = 0$ at Joint B

$\tan\theta = 45^0$

$\therefore \theta = 45^0$

$F_{AB} \cos\theta + F_{DB} = 0$ (Due to symmetry analyze only left portion)

$F_{AB} = -\sqrt{2}F_{BD}$

$\therefore$ i.e., $\sqrt{2}$ force in 3rd panel.

236. Which of these are the assumptions made in the force analysis of simple trusses?

(i) All members have same cross-sectional area
(ii) The bending resistance of all the members is small in comparison with their axial force resistance
(iii) All the external loads are applied directly or indirectly at the joints
(iv) All joints are idealized to be frictionless hinges

(a) (i) (ii) and (iv) (b) (ii), (iii) and (iv)
(c) (i), (ii) and (iii) (d) (iii) and (iv)

GPSC AE (CLASS 1 & 2) 2019

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Ans : (b) : The assumptions of force-analysis in simple truss are:-

(1) The load or any external load applied is always to be at joint.
(2) In force method, condition of equilibrium is enough to analyze the structure so BM & SF is quite less and hence negligible as compare to axial load in truss.
(3) In simple truss, BM & SF (Shear force) are independent of cross section area.
(4) All joints in truss are hinged end i.e. (BM=0) truss is designed only for Axial force

237. An unstable vibratory motion due to combined bending and torsion which occurs in flexible plate like structures is called

(a) Galloping (b) Ovalling
(c) Flutter (d) Oscillation

ESE 2019

Ans. (c) : Flutter is an oscillatory motion that results from the coupling of aerodynamic forces with the elastic deformation of a structure. It is often the result of combined bending and torsion. It effects plate like structures. Ex. Sign boards and suspension bridges decks etc.

238. For a plane truss member, the length is 2 m, E = 200 GPa and area of cross- section is 200 mm^2. The stiffness matrix coefficient K_{11} with reference to its local axis is

(a) 200 N/m (b) 2×10^7 N/m
(c) 4×10^7 N/m (d) 400 N/m

ESE 2018

Ans. (b)

Length = 2m Area = 200mm^2
E = 200 GPa

Stiffness matrix co-efficient $\left[K_{11}=\frac{AE}{L}\right]$

$$K_{11}=\begin{bmatrix}\frac{AE}{L} & \frac{-AE}{L}\\ -\frac{AE}{L} & \frac{AE}{L}\end{bmatrix}$$

$$K_{11}=\frac{AE}{L}=\frac{200\times200\times10^3}{2}$$

$$\boxed{K_{11}=2\times10^7\ N/m}$$

239. The proof of Muller-Breslau principle can be established using.

(a) Principle of virtual work
(b) Moment area method
(c) Conjugate method
(d) Castingliano's theorem

DSSSB AE 2019

Ans. (a) The proof of the Muller-Breslau principle can be established using the principle of virtual work. Recall that work is the product of either a linear displacement and force in the direction of the displacement or a rotational displacement and moment in the direction of the displacement.

240. Independent displacement at each joint of a rigid-jointed plane frame are :

(a) Three linear movements
(b) Two linear movements and one rotation
(c) One linear movement and two rotations
(d) Three rotations

OPSC AEE 2019 Paper–I

Ans. (b) : Independent displacement components or degree of freedom for different types of joints :

Type of Joint	Degree of freedom
(i) Pin joint of plane frame	2 (Translations)
(ii) Pin joint of space frame	3 (Translations)
(iii) Rigid joint of a plane frame	3 (One rotation and two translations)
(iv) Rigid joint of a space frame	6 (3 rotations and 3 translations)

241. Degree of kinematic indeterminacy of a pin-jointed plane frame is given by (where j is number of joints and r is unknown reactions) :

(a) 2j – r (b) j – 2r
(c) 3j – r (d) 2j + r

OPSC AEE 2019 Paper–I

Ans. (a) :

- The degree of kinematic indeterminacy of a pin-jointed plane frame is given by the equation

$[D_K= 2j–r]$

- For a pin-jointed space frame the degree of Kinematic indeterminacy is given by the equation.

$[D_K= 3j–r]$

- For a rigid jointed space frame the degree of kinematic indeterminacy is

$[D_K= 6j–r]$

242. A beam of length "L", fixed at A and B of uniform flexural rigidity carries two loads "P" at one-third interval of span length C and D respectively. The maximum bending moment will occur :

(a) At C and D and will be equal to 2PL/9
(b) Between C and D and will be equal to PL/9
(c) At A and at B will be equal to 2PL/9
(d) Between A and C and also between B and D will be equal PL/9

OPSC AEE 2019 Paper–I

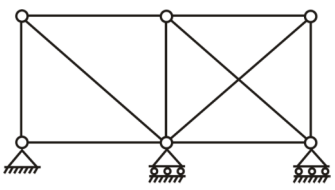
Ans. (b) : The maximum bending moment will occur between C and D and will be equal to PL/9

243. The basic perfect frame is a :

(a) Triangle (b) Rectangle
(c) Square (d) Hexagon

OPSC AEE 2019 Paper–I
TNPSC AE 2012

Ans. (a)



The basic perfect frame is a triangle.

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244. Method of joints is applicable only when the number of unknown forces at the joint under consideration is not more than :

(a) One (b) Two
(c) Three (d) Four

OPSC AEE 2019 Paper–I

Ans. (b) : At every joint two equations of equilibrium $\Sigma V = 0$, $\Sigma H = 0$ are available and so at one joint two unknowns can be evaluated. Method of joints is suitable only when two unknown forces are at a joint.

245. The degree of static indeterminacy of a rigid-jointed space frame is where m, r and j are unknown as member forces, r is unknown as reaction components and j is number of joints :

(a) M + r – 2j (b) M + r – 3j
(c) 3m + r – 3j (d) 6 m + r – 6j

OPSC AEE 2019 Paper–I

Ans. (d)

D_S = (6m + r) – 6j, for rigid joined space frame
m = Number of member forces
r = Reaction components

246. Castigliano's first theorem is applicable

(a) For statically determinate structures only
(b) When the system behaves elastically
(c) Only when principle of superposition is valid
(d) For statically indeterminate structures only

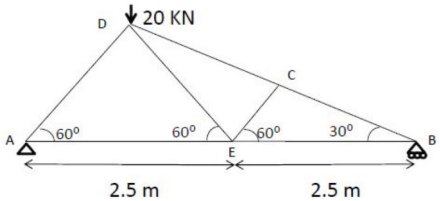
OPSC AEE 2019 Paper–I

Ans. (b) : Castigliano's first theorem is applied to both elastic and inelastic material behaviour.

Castigliano's first theorem is useful in analyzing statically indeterminate structure. Castigliano's second theorem :

- useful for finding the deflections of statically determinate structures.
- Applicable only when principle of super position is valid.

247. The axial forces in the members CE and DE of the plane truss shown in the figure below are equal to



(a) 10, 10 in KN (b) 20, 20 in KN
(c) 8.7, 8.7 in KN (d) 0, 0

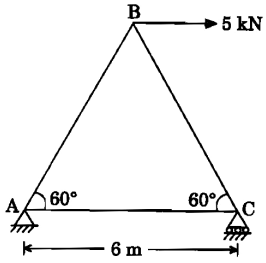
APPSC AEE MAINS 2016

Ans. (d) : • If a joint has only two non-collinear members and there is no external load or support reaction at that joint, then those two member are zero-force members.

• In this figure members CE and DE are zero force member.

So, CE = 0, DE = 0

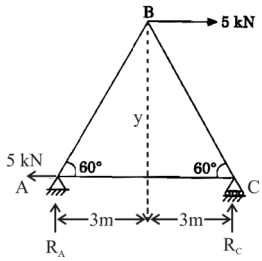
248. Force in the member BC of the truss shown in the figure below is:

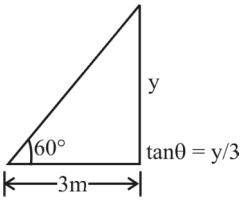


(a) 5 KN (tensile)
(b) Zero
(c) 2.88 KN (compressive)
(d) 5 KN (compressive)

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) :





$y = 3 \tan 60^\circ$

$y = 3\sqrt{3}$ m

Taking moment about 'A'

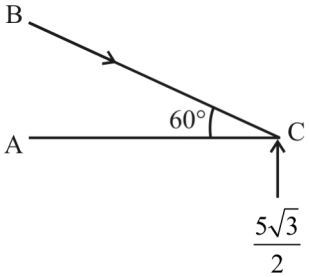
$\Sigma M_A = 0$

$R_C \times 6 - 5 \times 3\sqrt{3} = 0$

$$R_C = \frac{15\sqrt{3}}{6}$$

$$[R_C = \frac{5\sqrt{3}}{2} \text{ kN}]$$

At joint 'C'



since, $\Sigma v = 0$

so, $F_{BC} \sin 60^\circ + \frac{5\sqrt{3}}{2} = 0$

$F_{BC} = -5$ KN

[F_{CC} = 5 KN (Compression)]

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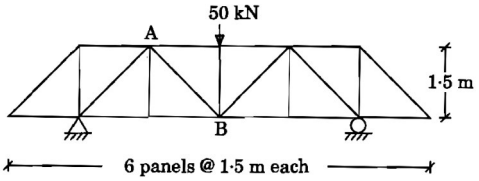
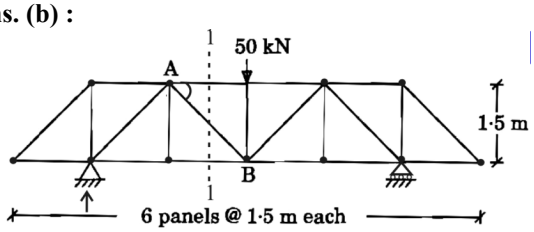
249. The force in member AB of the truss shown in the figure below is



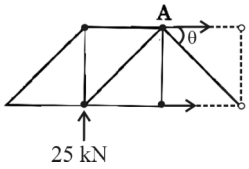
(a) 25 kN (compression)
(b) $25\sqrt{2}$ kN (tension)
(c) $25\sqrt{2}$ kN (compression)
(d) 25 kN (tension)

Maharashtra PSC Mains 2018 Paper-1
UPPCL AE 2014

Ans. (b) :



Using method of section



$\Sigma V = 0$

$F_{AB} \sin 45° = 25$

$F_{AB} = 25\sqrt{2}$ kN (Tension)

250. Depending on the transmission of loads to joints, bridges are classified as

(a) Deck type (b) Pratt type
(c) Warren type (d) Howe type

GPSC AE March 2018

Ans. (a) : The trusses and girders in bridge of deck type transmit all loads received by the joints to the abutments or supporting piers.

251. Williot-Mohr diagram is used to find

(a) displacement in a structure
(b) settlement of a structure
(c) strain energy in a structure
(d) principal stresses in a structure

GPSC AE March 2018

Ans. (a) : Williot Mohr Diagram is a graphical method presented in the following paragraphs provides a means for obtaining the deflection of statically determinate truss structure. The deflections in truss arise from axial extensions or shortenings of members.

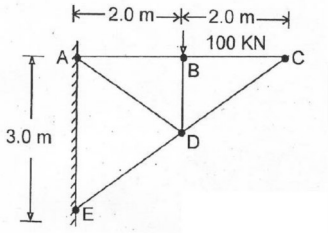
252. A truss is said to be determinate when

(a) Degree of static indeterminacy (Ds) is zero Ds = 0
(b) Ds < 0
(c) Ds > 0
(d) None of the above

GPSC Poly. Tech. Lect. 2016

Ans. (a) : A truss is said to be determinate when degree of static indeterminacy is zero.
Where M = 2j – 3, the truss is said to be statically determinate, because the (M+3) internal member forces and support reactions can then be completely determined by '2j' equilibrium equations, once we know the external loads and the geometry of the truss.

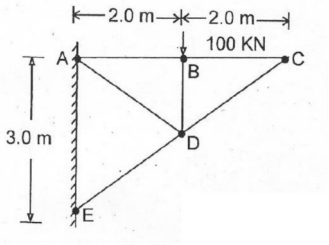
253. In the truss shown fig. below the force in member CD is :



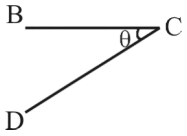
(a) 100 kN compressive (b) 100 kN tensile
(c) Zero (d) Intermediate

OPSC Civil Services Pre 2011

Ans. (c)



In the truss shown fig. below the force in member CD is Zero.
At joint 'C'



$\Sigma V = 0$

$F_{CD} \text{ Sin } \theta = 0$

$[F_{CD} = 0]$

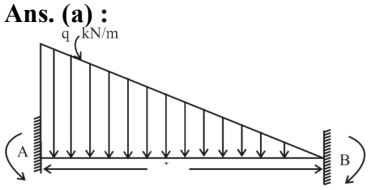
If two members meet a joint where no load is acting, then the forces in those members are zero.

254. The fixed end moment at left end of a beam subjected '0' uniformly varying load of intensity 'q' a left end and zero intensity at right end is given by :

(a) $\frac{-qL^2}{20}$ (b) $\frac{-qL^2}{30}$
(c) $\frac{-qL^2}{24}$ (d) $\frac{-qL^2}{16}$

TRB Poly. Lect. Tech. 2017

Ans. (a) :



$(FEM)_{AB} = \frac{qL^2}{20}$ $(FEM)_{BA} = \frac{qL^2}{30}$

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255. The displacement of the joint of a truss can be obtained directly from the

(a) Space diagram
(b) Force diagram
(c) Williot-Mohr diagram
(d) Funicular diagram

Punjab PSC SDE 2016

Ans. (c) : The Williot-Mohr diagram is used to find deflection or displacement of a truss. It is a graphical method.
Williot Mohr Diagram is a graphical method presented in the following paragraphs provides a means for obtaining the deflection of statically determinate truss structure. The deflections in truss arise from axial extensions or shortenings of members.

256. The truss can be analyzed by the

(a) 1 (b) 2
(c) 3 (d) 4

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (c) The truss can be analysed by 3-methods
(a) Method of joints
(b) Method of sections
(c) Graphical method

257. In a plane truss, if M is the number of member. R is the number of reactions and J is the number of joints, then for this truss to be determinate

(a) J = M + R (b) J = 2M + R
(c) 3J = M + 2R (d) 2J = M + R

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (d) In a plane truss if M is the number of member, R is number of reaction and J is the joint then for this truss to be determinate and stable.

$$\boxed{2J = M + R}$$

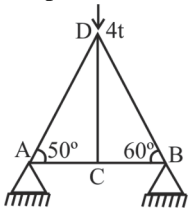
258. Analysis of space trusses can be carried out using.....

(a) Mohr's method
(b) Combination of section method and joint method
(c) Shape factor and joint method
(d) Only shape factor

SJVNL ET 2019

Ans. (b) : Analysis of space trusses can be carried out using combination of section method and joint method.

259. In a pin-jointed plane frame

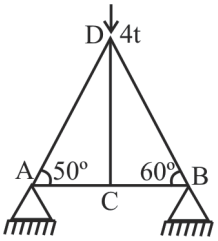


the force in the member DC is

(a) 4t (b) $4\sqrt{3}t$
(c) zero (d) $4\sqrt{2}t$

BPSC Poly. Tech. Lect. 2016

Ans. (c) :



Taking joint (c). Here at joint C as force f_{CA} and f_{CB} are collinear, therefore third member carry zero force.
• If a joint has only two non-collinear members and there is no external load or support reaction at that joint, then those two member are zero-force members.

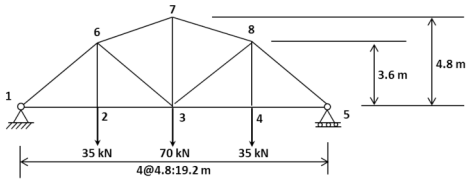
260. The primary unit of a truss is

(a) trapezium (b) parallelogram
(c) rectangle (d) triangle

BPSC AE Aug. 2019 Paper-V

Ans. (d) : A truss is an assembly of beams or other elements that creates a rigid structure.
Trusses typically comprise five or more triangular units constructed with straight members whose ends are connected at joints referred as nodes.

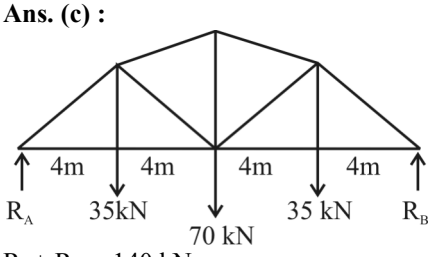
261. Vertical reactions at the supports for the given truss are



(a) 35 kN, 35 kN (b) 35 kN , 70 kN
(c) 70 kN, 70 kN (d) 70 kN, 35 kN

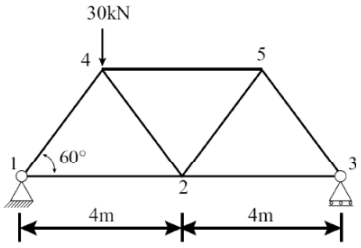
GPSC AE June 2019

Ans. (c) :



$R_A + R_B = 140$ kN
$\sum M_A = 0$
$R_B \times 16 = 35 \times 12 + 70 \times 8 + 35 \times 4$
$\boxed{R_B = 70 \text{ kN}}$, $\boxed{R_A = 70 \text{ kN}}$

262. The force in member 2-5 in the truss shown in fig is



(a) 22.5 kN compression
(b) 7.5 kN tension

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(c) 8.66 kN compression
(d) 8.66 kN tension

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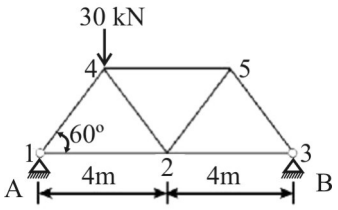
Ans. (c) :



$R_A + R_B = 30$ kN

$\sum M_B = 0$

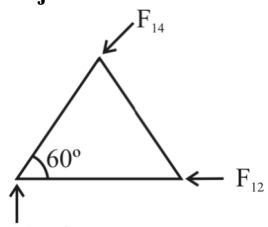
$R_A \times 8 - 30 \times 6 = 0$

$8R_A = 180$

$R_A = 22.5$ kN

$R_B = 7.5$ kN

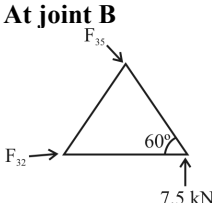
At joint A :



$F_{14} \sin 60° = 22.5$ $\quad F_{12} + F_{14} \cos 60° = 0$

$F_{14} = 26$ kN $\quad F_{12} = -13$ kN

At joint B

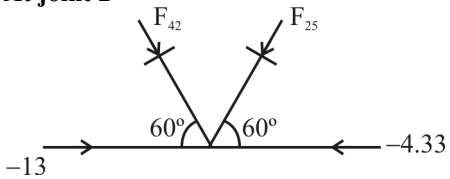


$F_{35} \sin 60° = 7.5$

$F_{35} = 8.66$

$F_{32} = -4.33$

At joint 2



$F_{42} \sin 60° + F_{25} \sin 60° = 0$

$\boxed{-F_{42} = +F_{25}}$

$F_{42} \cos 60° - 13 = F_{25} \cos 60° - 4.33$

$\boxed{F_{42} = -8.66 \text{ kN}}$

$\boxed{F_{25} = 8.66 \text{ kN}}$ Compression

263. A member in a truss can take

(a) axial force and bending moment
(b) only axial force
(c) only bending moment
(d) bending moment and shear force

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : Truss members take only axial forces (tension or compression). They cannot take shear force, bending moment and torsion.

264. Regarding the ability of a truss, the condition $m + r > 2j$ is

(a) Necessary
(b) Sufficient
(c) Necessary and Sufficient
(d) Sufficient but not necessary

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Given condition

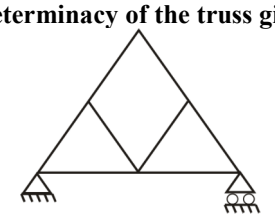
$m + r > 2j$

$m > 2j - r$

$\therefore$ The member is structurally stable but not necessarily have $m > 2j - r$. It is just sufficient to have

$m = 2j - r$

265. Choose the correct combination of the stability and indeterminacy of the truss given.



(a) Statically indeterminate and stable
(b) Statically determinate and stable
(c) Unstable
(d) Statically determinate

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Ans. (c & d) : No. of member, $m = 8$

No. of joints, $j = 6$

$m < 2j - 3$

$\therefore$ The truss is internally unstable but determinate

$\therefore$ Both the options (c) and (d) are correct

266. Hudson's formula gives the dead weight of a truss bridge as a function of

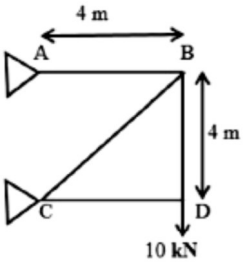
(a) bottom chord area
(b) top chord area
(c) middle chord area
(d) effective span of bridge

GPSC AE Dec 2018

Ans. (a) : Hudson's formula– It estimates quite satisfactorily the dead weight of the truss girder bridges. In this the dead weight is assumed to be a function of bottom chord area:

$\boxed{W = 0.785 A}$

267. For the truss shown in figure (supports A & C are hinge type), the force in member AB and BC are respectively :



(a) 10 (compressive); $\frac{10}{\sqrt{2}}$ (tensile)

(b) zero; $10\sqrt{2}$ (compressive)

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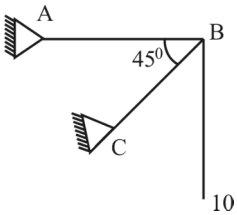
(c) $10\sqrt{2}$ (tensile); $\frac{10}{\sqrt{2}}$ (compressive)

(d) 10 (tensile); $10\sqrt{2}$ (compressive)

BHEL ET 2019

Ans. (d) :



At joint D number BD and forced 10 kN are on the same line hence force in CD = 0

Apply $\sum V = 0$ at B

$F_{BC} \sin 45^0 = 10$

$F_{BC} = 10\sqrt{2}$ kN (compressive)

apply $\sum H = 0$ at B

$F_{AB} = F_{BC} \cos 45^0$

$= 10\sqrt{2} \times \frac{1}{2} = 10$ (Tensile)

268. In the king post truss the member used to prevent the sagging of beam and connected with the ridge is called:

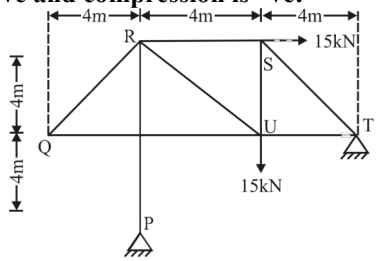
(a) Collar (b) Rafter

(c) King post (d) Strut

PTCUL AE 2016

Ans. (c) : In the post truss, the member used to prevent the sagging of beam and connected with ridge is called king post.

269. A pin-jointed 2-D truss is loaded with a horizontal force of 15 KN at joint S and another 15 KN vertical force at joint U as shown in figure. Find the force in member RS in (KN) and report your answer taking tension as +ve and compression is –ve:

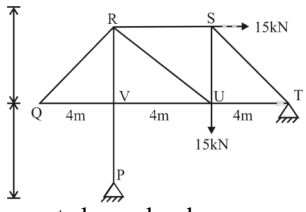


(a) $\frac{1}{\sqrt{2}}$ (b) 1

(c) 0 (d) $\sqrt{2}$

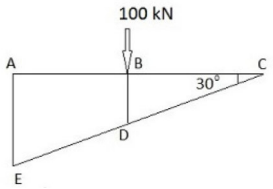
PTCUL AE 2016

Ans. (c) :



In figure Given not show clearly.

270. In the truss shown below, force in member BC is –



(a) 100 N tensile (b) zero

(c) 100 N compressive (d) 50 N tensile

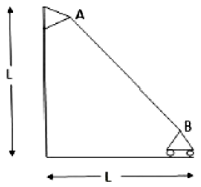
UPPCL AE 2014

Ans. (b) : AB, BC, AE, DE, DC all are zero force member because 100 kN load is vertically applied at B so all load will be taken by member BD.

$\boxed{F_{BC} = 0}$

- If a joint has only two non-collinear members and there is no external load or support reaction at that joint, then those two member are zero-force members.

271. Horizontal stiffness coefficient (K11) of a bar 'AB' is given by –



(Where A is cross section area and E is Young's Modulus)

(a) AE/L (b) AE/√2L

(c) 2AE/L (d) AE/2L

UPPCL AE 2014

Ans. (b) : $K_{11} = \frac{AE}{L_{AB}} = \frac{E}{\sqrt{2}L}$

272. If a plane truss satisfies the condition m = 2j – 3 where m is number of member, and j is number of joints, then it is .....

(a) Determinate internally

(b) Determinate externally

(c) Indeterminate internally

(d) Indeterminate externally

GPSC AE January 2018

Ans. (a) : Internal degree of indeterminacy

$(D_{si}) = m - (2j - 3)$

here, m = no. of member = 2j – 3

j no. of joints

$D_{si} = m - (2j - 3)$

$D_{si} = (2j - 3) - (2j - 3) = 0$

So, internally determinate structure.

273. How is a truss, which undergoes rigid body translation for an arbitrary load, classified as?

(a) Determinate structure

(b) Geometrically unstable structure

(c) Statically unstable structure

(d) Structurally unstable structure

GPSC AE (CLASS 1 & 2) 2019

UPRVUNL AE 2015

Ans : (b) : When a rigid body translate it means it is geometrically unstable structure.

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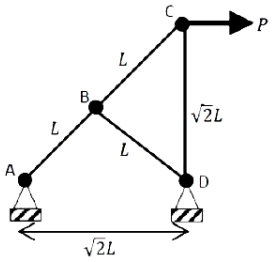
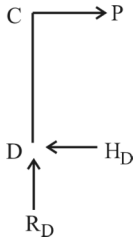
274. In the pin jointed truss shown below: members AB, BC are of length L, while member CD is length $\sqrt{2}L$. Joints A, B and C are in a straight line. All members have the same AE. A force P acts joint C as shown. What is the magnitude of the horizontal reaction at support D?



(a) P (b) Zero
(c) 2P/3 (d) 3P/2

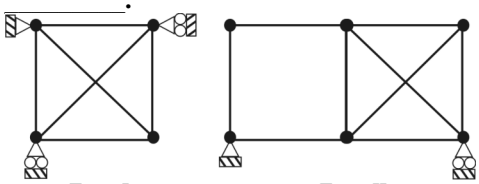
UPPCL AE 2018

Ans. (b)



At point B no force is acting
So, force in member AB, BC and BD = 0
$F_{DC} = 0$
$H_D = 0$

275. Amongst the following two pin joined trusses

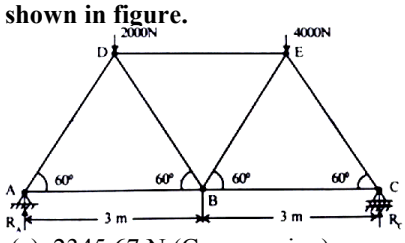


(a) Truss I is stable, but Truss II is unstable
(b) Both Truss I and Truss II are unstable
(c) Both Truss I and Truss II are stable
(d) Truss I is unstable, but Truss II is stable

UPPCL AE 2018

Ans. (b) : Both truss I and truss II are unstable as they will deform when load is applied.

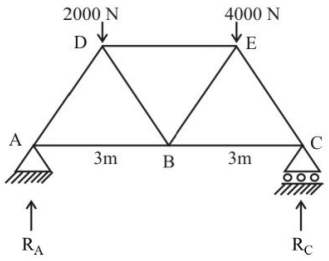
276. Find the force in member AD of the truss shown in figure.



(a) 2345.67 N (Compression)
(b) 1890.37 N (Compression)
(c) 2886.75 N (Compression)
(d) 1456.98 N (Compression)

UPPCL AE 2019

Ans. (c) :



from $\sum f_y = 0$

$R_A + R_C = 2000 + 4000$

$R_A + R_C = 6000$ N ..... (i)

taking moment about C

$(\sum M_c = 0)$

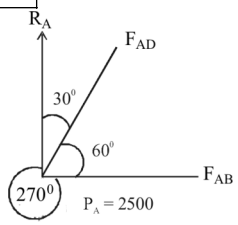
$R_A \times 6 - 2000 \times 4.5 - 4000 \times 1.5 = 0$

$R_A \times 6 = 15000$

$\boxed{R_A = 2500}$ N

from equation (i)

$\boxed{R_C = 3500}$ N



Now, apply Lami's theorem at point A

$$\frac{R_A}{\sin 60^0} = \frac{F_{AD}}{\sin 270^0}$$

$$\frac{2500}{\sqrt{3}/2} = \frac{F_{AD}}{-1}$$

$F_{AD} = -2886.751$ N

or $\boxed{F_{AD} = 2886.751(\text{compression})}$ N

277. A compound truss may be formed by connecting two simple rigid frames by:
(a) Two bars
(b) Three parallel bars
(c) Three bars
(d) Three bars intersecting at a point
(e) All of the above options

Haryana PSC AE 2018 Paper–I

Ans. (c) : For Truss Minimum 3 members or bars are required.
A compound truss is a truss made by connecting two or more simple trusses.

278. Consider the following given parameters and choose the one upon which the economic spacing of a roof truss depends.
(a) Cost of purlins and cost of roof covering
(b) Cost of roof covering and dead loads
(c) Dead loads and live loads
(d) Live loads and cost of purlins

UPRVUNL AE 2014

Ans. (a) : The economic spacing of a roof truss depends cost of purlins and cost of roof covering.

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279. Among the following mentioned assumptions which are considered in the force analysis of simple trusses?

(1) All members have same cross-sectional area.
(2) The bending resistance of all the members is small in comparison with their axial force resistance
(3) All the external loads are applied directly or indirectly at the joints.
(4) All joints are idealized to be frictionless hinges.

Choose the correct answer from the options given below.

(a) 1, 2 and 4 (b) 2, 3 and 4
(c) 1, 2 and 3 (d) 3 and 4

UPRVUNL AE 2014

Ans. (b) : Assumption in the force Analysis of simple trusses :
- The bending resistance of all the members is small in comparison with their axial force resistance.
- All the external loads are applied directly or indirectly at the joints.
- All joints are idealized to be frictionless hinges.

280. A member in a truss can take

(a) axial force and bending moment
(b) only axial force
(c) only bending moment
(d) bending moment and shear force

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : Truss members take only axial forces (tension or compression). They cannot take shear force, bending moment and torsion.

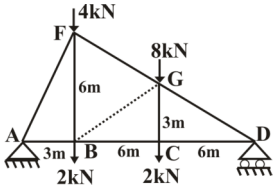
281. Regarding the ability of a truss, the condition m + r > 2j is

(a) Necessary
(b) Sufficient
(c) Necessary and Sufficient
(d) Sufficient but not necessary

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Given condition
$m + r > 2j$
$m > 2j - r$
$\therefore$ The member is structurally stable but not necessarily have $m > 2j - r$. It is just sufficient to have
$m = 2j - r$

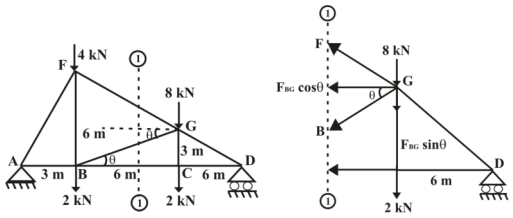
282. Determine the force in the member BG in the given truss



(a) 11.18 kN Tension
(b) 14.4 kN tension
(c) 11.18 kN Compression
(d) 14.40 kN compression

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : Take section 1 - 1 as shown in figure.
To find the force in BG, consider the right part of section (1) - (1) and take moment about 'D'.



$$\cos\theta = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\sin\theta = \frac{3}{3\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$\sum M_D = 0$ (↷ + ve, ↶ - ve)

$-F_{BG}\cos\theta \times 3 - F_{BG}\sin\theta \times 6 - 2 \times 6 - 8 \times 6 = 0$

$$-F_{BG} \times \frac{6}{\sqrt{5}} \times F_{BG} \times \frac{6}{\sqrt{5}} - 60 = 0$$

$$-2\left[F_{BG} \times \frac{6}{\sqrt{5}}\right] = 60$$

$F_{BG} = -5\sqrt{5}$ kN (compressive)

$F_{BG} = 11.18$ kN (compressive)

283. Analysis of trusses is based on the following assumption. The correct combination of assumptions is _____.

i. All loads act at joints
ii. All joints are smooth friction less pin joints
iii. Truss is made with material which is homogenous, elastic and isotropic.
iv. The axis of all members is straight and their self weights are ignored

(a) i, ii and iv (b) i, ii, iii
(c) i, iv (d) i, iii, iv

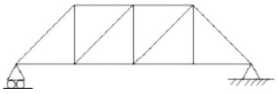
TSPSC Deputy Engineer 2015

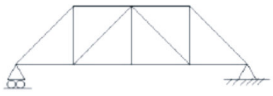
Ans. (a) : Assumption in the force Analysis of simple trusses : • The bending resistance of all the members is small in comparison with their axial force resistance.
• All the external loads are applied directly or indirectly at the joints.
• All joints are idealized to be frictionless hinges.

284. Which of the following truss is unstable -

(a)

(b)

(c) 

(d) 

UPPCL AE 2014

Ans. (a) :

It is a unstable truss.

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4. Frame

285. Which of the following case comes under perfect frame?

(a) Number of members =1, number of joints = 2
(b) Number of members = 7, number of joints = 5
(c) Number of members = 3, number of joints = 3
(d) Number of members = 4, number of joints = 3

UPPCL AE 2019

Ans. (c)
Perfect Truss -

$m = 2j - 3$
m = number of members
j = number of joints

Redundant truss : The number of members in this truss is more than necessary

$m > 2j-3$

(3) Deficient Truss : The number of members in this truss is less than the required $m < 2j - 3$

286. The deflection at any point of a perfect frame can be obtained by applying a unit load at the joint in-

(a) Vertical direction
(b) Horizontal direction
(c) Inclined direction
(d) The direction in which the deflection is required

OPSC AEE 2015 PAPER–I
Sikkim PSC AE 2017
APPSC AEE 2016

Ans. (d) : A frame is called perfect frame if the numbers of members of the frame is just enough to keep it in equilibrium.
Deflection of structure is given by equation–

$$\left[\Delta = \int P'edv\right]$$

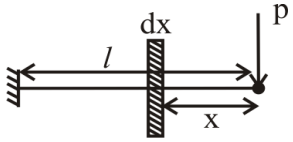
where, Δ = is the displacement at the point and in the direction of unit load applied e = strain, P = stress.
The deflection at any joint/point of a perfect frame can be obtained by applying a unit load at the joint/point in the direction in which the deflection is required.

287. The Strain energy stored due to bending for a cantilever beam of span *l*, Modulus of elasticity E, moment of inertia I and a point load P at free end is

(a) $P^2l^3/(6EI)$ (b) $P^2l^5/(40EI)$
(c) $P^2l^3/(96EI)$ (d) $P^2l^5/(240EI)$

APPSC AEE 2016

Ans. (a) :

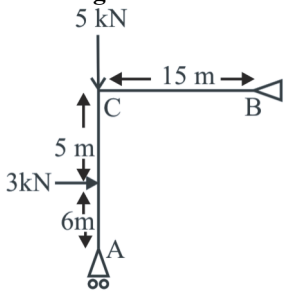


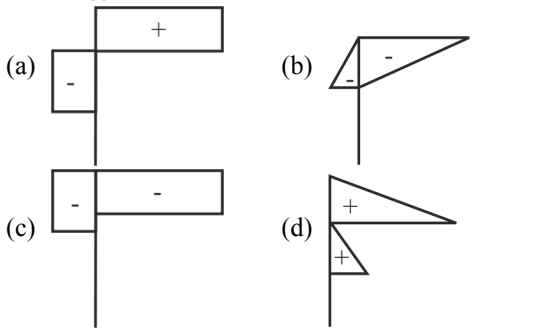
Moment $M_x = px$

$$\text{Strain energy} = \int Ux = \int_{\partial}^{l} \frac{(px)^2}{2EI} dx$$

$$U = \frac{p^2l^3}{6EI}$$

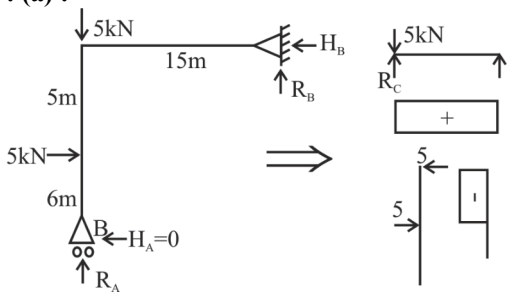
288. Select the Shear Force Diagram for the following beam:





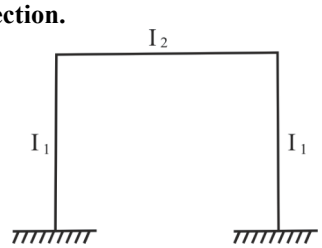
GPSC AE (CLASS 1 & 2) 2019

Ans : (a) :



289. The rigid portal frame shown in figure will not have any side sway, if

I_1= the moment of inertia of the column cross-section.
I_2= the moment of inertia of the beam cross-section.



(a) it is subjected to vertical loading only
(b) $I_1 = 2I_2$
(c) the loading is symmetrical about its centre line
(d) loaded in any manner.

GPSC AE (CLASS 1 & 2) 2019
TNPSC AE 2012
ESE 1996

Ans. (c) The frame is symmetrical in shape and size about a central vertical line. Therefore symmetrical loading about the centre line will not cause any side sway.

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290. Degree of freedom available for space-rigid frame is–

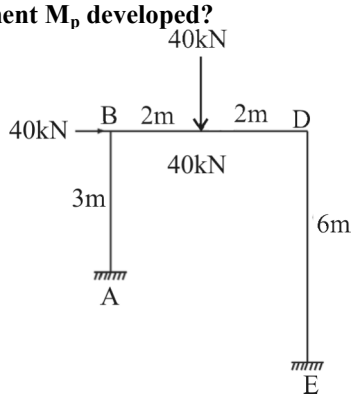
(a) 2 (b) 3
(c) 4 (d) 6

APPSC Poly. Tech. Lect. 2020

Ans. (d) :

– Degree of freedom (DOF) : The Number of independent parameters (Measurements) that are needed to uniquely finds its position in space and at instant of time.
– Each joint of space rigid jointed frame has 6 degree of freedom.
– Each joint of plane rigid jointed frame has 3 degree of freedom.
– Each joint of plane pin-jointed frame has 2 degree of freedom.
– Each joint of space pin-jointed frame has 3 degree of freedom.

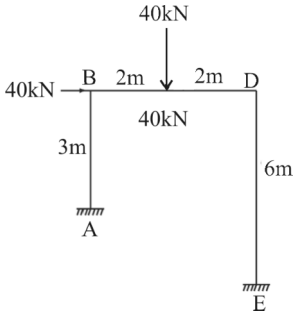
291. If only a sway mechanism is possible in the frame given below, what will be the plastic moment M_p developed?



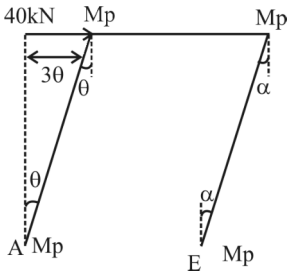
(a) 10 kNm (b) 20 kNm
(c) 40 kNm (d) 80 kNm

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) :



M_p = constant for each member.
only sway mechanism



External work done = internal work done

$40 \times 3\theta = M_p.\theta + M_p.\theta + M_p.\alpha + M_p.\alpha$

$= 2M_p.\theta + 2M_p.\alpha$

$120\theta = 2M_p(\theta + \alpha)$

Beam (BD) is inextensible so.

$3\theta = 6\alpha$

$\theta = 2\alpha$

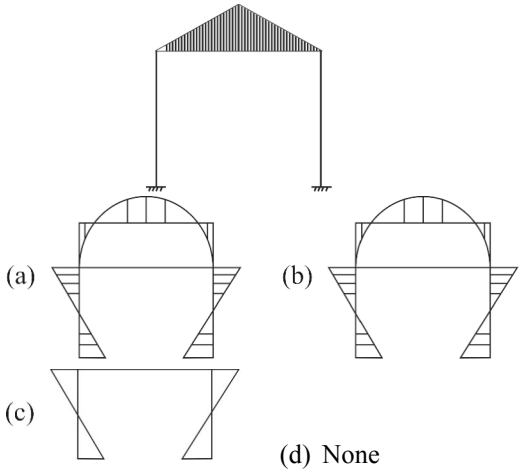
$\alpha = 0.5\theta$

$120\theta = 2M_p(\theta + 0.5\theta)$

$60\theta = Mp \times 1.5\theta$

$M_p = 40$ kNm

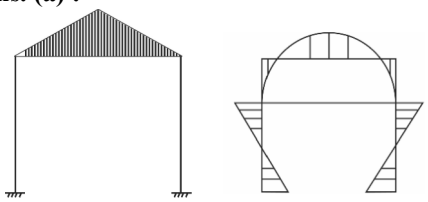
292. Which of the following is a correct bending moment diagram of the structure shown below?



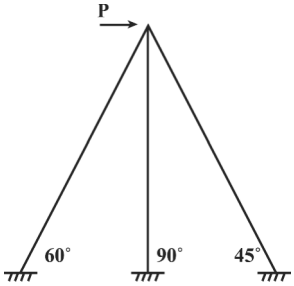
(d) None

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) :



293. Which is the most appropriate method to analyze the structure shown below ?



(a) Slope Deflection method
(b) Stiffness method
(c) Flexibility method
(d) Strain Energy Method

Gujarat PSC AE (N.W.R.) 2020
ESE 2007

Ans. (d) : Strain Energy Method is the most appropriate method to analyze the given structure.

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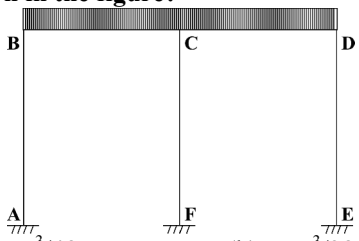
294. What is the moment at joint F, in a case where there is no side sway and the member CF is along the axis of symmetry while a UDL of W kN/m is applied all along the length BD as shown in the figure?

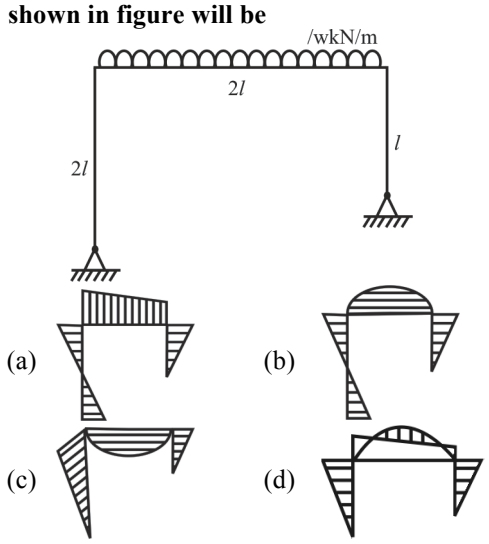


(a) $WL^3/48\ EI$ (b) $WL^3/384\ EI$
(c) 0 (d) None of the above

Gujarat PSC AE (N.W.R.) 2020

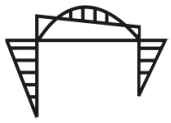
Ans. (c) : The symmetry of the loading and frame does not allow any bending moment in column M_{CF}. Due to symmetry, zero moment in member CF.

295. The bending moment diagram of the frame shown in figure will be

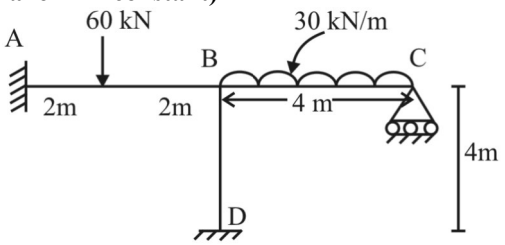


TNPSC AE 2019
ESE 1995

Ans. (d) Due to unsymmetrical nature of frame, the lateral translation will occur towards left. Therefore horizontal force equal and opposite nature will develop on the supports.



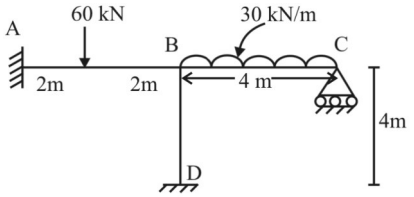
296. The slope deflection equation at B of member BC for the frame shown in the figure is --------. (Take EI = constant)



(a) $M_{BC} = 60 + EI\theta_B$
(b) $M_{BC} = -40 + 0.5EI\theta_B + EI\theta_C$
(c) $M_{BC} = -40 + EI\theta_B + 0.5EI\theta_C$
(d) $M_{BC} = -60 + EI\theta_B$

TRB Poly. Lect. Tech. 2017

Ans. (c) :



$$M_{FBC} = \frac{-wl^2}{12} = \frac{-30\times16}{12}$$
$$= -40\ \text{kN-m}$$
$$M_{BC} = -M_{FBC} + \frac{2EI}{l}\left(2\theta_B + \theta_C - \frac{\delta}{l}\right)$$
$$= -40 + \frac{2EI}{4}(2\theta_B + \theta_C) \quad [\because \delta = 0]$$
$$M_{BC} = -40 + EI\,\theta_B + 0.5\,EI\theta_C$$

297. The number of plastic hinges which will cause the overall total collapse of a structure is :

(a) one more than the order of static indeterminacy
(b) equal to order of static indeterminacy
(c) one less than the order of static indeterminacy
(d) not determinable

(UKPSC AE (Paper–I) 2007)
ESE-2007

Ans : (a) The number of plastic hinges which will cause the overall total collapse of a structure is one more than the order of static indeterminacy.
If an indeterminate structure has redundancy 'r' the insertion of 'r' plastic hinges make it statically determinate. For overall collapse of structure the number of plastic hinges required are (r +1)

298. The analysis of statically indeterminate structures by the unit load method is based on

(a) consistent deformation
(b) consistent force
(c) stiffness
(d) none of the above

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (a) The analysis of statically indeterminate structure by the unit load method is based on consistent deformation method.
The method consistent deformations development in the proceeding sections for analyzing structures with a single degree of indeterminacy can easily be extend to the analysis of structures with multiple degree of indeterminacy.

299. The horizontal deflection at *C* for the frame loaded and supported as shown above is (*EI* is constant)

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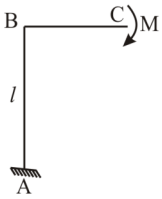
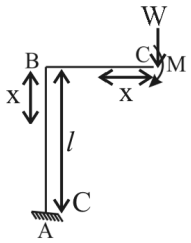


(a) $\frac{Ml^2}{(4EI)}$ (b) $\frac{Ml^2}{(2EI)}$

(c) $\frac{Ml^2}{(3EI)}$ (d) $\frac{2Ml^2}{(EI)}$

APPSC AE 2016 (Mains)
(JPSC Combined AE Paper V 2013)

Ans. (b) :



For member BC

$Mx = Wx + M$

$\frac{\partial Mx}{\partial W} = x$

$\delta v = \int Mx \frac{\partial Mx}{\partial W} . \frac{dx}{EI}$

$\delta v = \int_0^l (Wx + M) x . \frac{dx}{EI}$

$\therefore$ W = 0 (Factious load)

$\delta Lv = \frac{M}{EI} \int_0^l x.ds$

$\left[\delta Lv = \frac{Ml^2}{2EI}\right]$

300. The number of independent displacement components at each joint of a rigid jointed space frame is:

(a) 1 (b) 2

(c) 3 (d) 6

[UPPSC State Eng. A.E. 2008]

Ans : (d) • Each joint of a rigid-jointed plane frame has three independent displacement components because it has two linear movement and one rotation.

• Each joint of a rigid-jointed space frame has six (6) independent displacement component.

301. If a framed structure is statistically indeterminate to the second degree, then the minimum number of plastic hinges required for the complete collapse of the structure will be

(a) 1 (b) 2

(c) 3 (d) 4

(JPSC Combined AE Paper V 2013)
ESE 2003

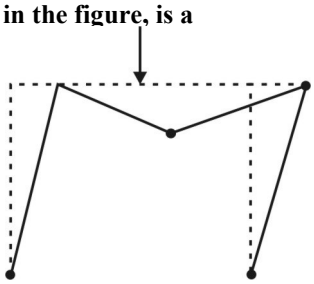
Ans : (c) If a framed structure is statically indeterminate to the second degree, then the minimum number of plastic hinges, for the complete collapse of the structure will be 3.

Number of plastic hinges

Degree of indeterminacy +1

= 2 + 1 = 3

302. A portal frame has a collapse mechanism as shown in the figure, is a



(a) Pure portal mechanism

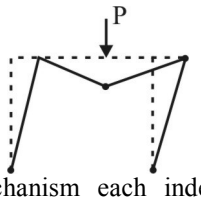
(b) Panel mechanism

(c) Combined mechanism

(d) Beam mechanism

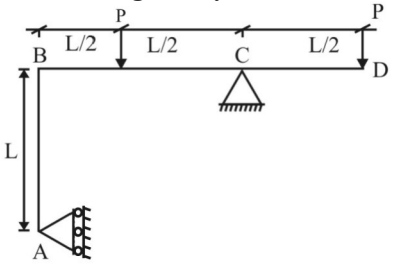
(UKPSC AE (Paper-I) 2007)
ESE 2008

Ans : (c)



In combined mechanism each independent collapse mechanism provides an independent equilibrium equation.

303. A frame ABCD is supported by a roller and a hinge as shown in figure : The reaction at the roller end A is given by

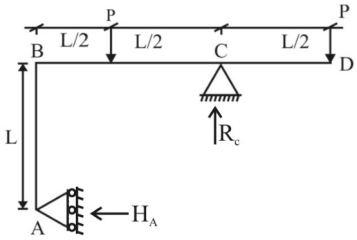


(a) P (b) 2P

(c) P/2 (d) Zero

(UKPSC AE 2012 Paper–I)
ESE 2013

Ans : (d)



$\Sigma V = 0$

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$R_C = P + P = 2P$

$R_C = 2P$ ----------(i)

Taking moment at C point

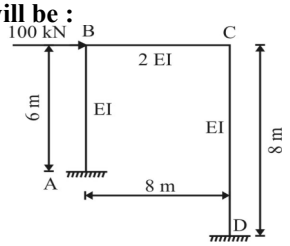
$H_A \times L + P \times L/2 = P \times L/2$

$H_A \times L = 0$

$\boxed{H_A = 0}$

Hence, Point A the reaction will be zero.

304. The slope deflection equation at the end B of member BC for the frame shown in figure below will be :



(a) $M_{BC} = \frac{4EI}{8}[2\theta_c - \theta_B]$

(b) $M_{BC} = \frac{4EI}{8}[2\theta_B - \theta_C]$

(c) $M_{BC} = \frac{4EI}{8}[2\theta_c - \theta_B]$

(d) $M_{BC} = \frac{4EI}{8}[2\theta_B + \theta_C]$

(UKPSC AE 2012 Paper–I)
ESE 1999

Ans : (d) $M_{BC} = \frac{2(2EI)}{\ell}[2\theta_B + \theta_C]$

$M_{BC} = \frac{4EI}{8}[2\theta_B + \theta_C]$

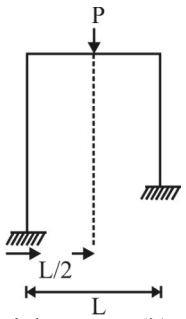
305. Which one of the following methods is more accurate for the analysis of rigid frames?

(a) Substitute frame method
(b) Portal method
(c) Factor method
(d) Slope deflection method

(UPRVUNL AE 2015)

Ans : (d) Slope deflection methods is more accurate for the analysis of rigid frames.

306. A rigid jointed plane frame as shown below will have–

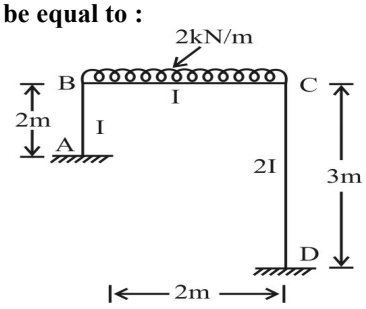


(a) Sway to right (b) Sway to left
(c) Not sway (d) None of above

(UTTRAKHAND AE 2013)

Ans : (b) The rigid joint plane frame will sway to left.

307. In the portal frame shown in the figure, the ratio of sway moments in column AB and CD will be equal to :



(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $\frac{9}{8}$ (d) $\frac{13}{8}$

[HPPSC 2016-II]
(UKPSC AE (Paper-I) 2007)
ESE 2000

Ans : (c) Sway moment in column,

$$AB = \frac{6EI(\Delta)}{(2)^2} = \frac{6EI(\Delta)}{4} = \frac{3EI(\Delta)}{2}$$

Sway moment in column,

$$CD = \frac{6EI(2I)(\Delta)}{(3)^2} = \frac{12EI(\Delta)}{9} = \frac{4EI(\Delta)}{3}$$

The ratio of sway moment in column AB and CD–

$$R = \frac{AB}{CD} = \frac{\frac{3}{2}EI(\Delta)}{\frac{4}{3}EI(\Delta)}$$

$$\text{Ratio}\left(\frac{AB}{CD}\right) = \frac{9}{8}$$

308. Substitutive frame method is used to analyse the building frame if the frame is subjected to _____.

(a) Lateral loads due to wind
(b) Lateral loads due to earthquake
(c) Vertical live load only
(d) Vertical dead load and live loads

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) :

⇒ Substitutive frame method for analysis of multi-storey frame can be handy an approximate and quick analysis. This method has been applied only for vertical loading conditions.

⇒ The method assumes that the moments in the beams of any floor are influenced by loading on that floor alone. The influence of loading on the lower or upper floor is ignored all together.

⇒ The process involves the division of multistoried structure into smaller frames. These sub frame are known as equivalent frames or substitutive frames.

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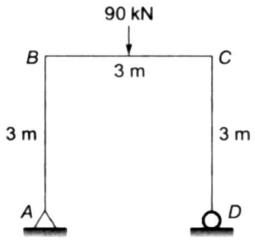
309. A rigid frame ABCD with vertical members AB and CD is pinned at A and roller supported at 'D'. Horizontal member BC is loaded with vertical point load of 90 kN at the centre of BC. AB = BC = CD = 3 m. If support A rotates by $\frac{1}{300}$ rad. and also sinks downward by 30 mm. The reactions at A and D-

(a) shall be affected due to rotation and sinking
(b) shall not be affected due to rotation and sinking
(c) reaction at A shall change but reaction at D shall not change due to rotation and sinking
(d) None of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) :



Structure is determinate :

⇒ Since structure is determinate, reaction of A and D shall not be affected due to rotation and sinking, because determinate structures can allow rotation and sink without generating additional reactions. It is because they are not stiff enough to generate resistance to rotation and translations of joints.

310. A rigid frame detailed to provide good ductility and support for both lateral and gravity loads by flexural action is called-

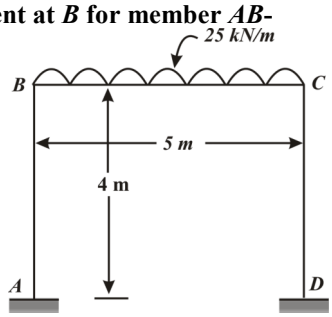
(a) Ordinary moment resisting frame
(b) Intermediate moment resisting frame
(c) Special moment resisting frame
(d) All of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : A rigid frame detailed to provide good ductility and support for both lateral and gravity loads by flexure action is known as moment resisting frame. It can be classified further as-

⇒ Ordinary moment resisting frame.
⇒ Intermediate moment resisting frame.
⇒ Special moment resisting frame.

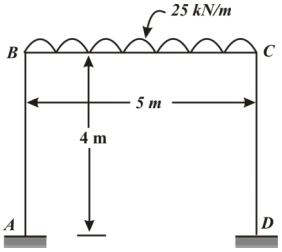
311. A fixed base symmetrical frame is shown in the figure write slope deflection equation for moment at *B* for member *AB*-



(a) $M_{BA} = 0.5EI.\theta_B$ (b) $M_{BA} = EI.\theta_B$
(c) $M_{BA} = 0.5EI.\theta_A$ (d) $M_{BA} = EI.\theta_A$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) :



$M_{FBA} = M_{FAB} = 0$
$M_{FCD} = M_{FDC} = 0$

$$M_{FBC} = \frac{-25\times5^2}{12} = -52 \text{ kNm}$$

$$M_{FCB} = \frac{+25\times5^2}{12} = 52 \text{ kNm}$$

For span *AB*, $L_{AB} = L = 4$ m
Slope deflection equation

$$M_{BA} = M_{FBA} + \frac{2EI}{L}\left(2\theta_B + \theta_A - \frac{3\delta}{L}\right)$$

$\delta = 0;\ \theta_A = 0$

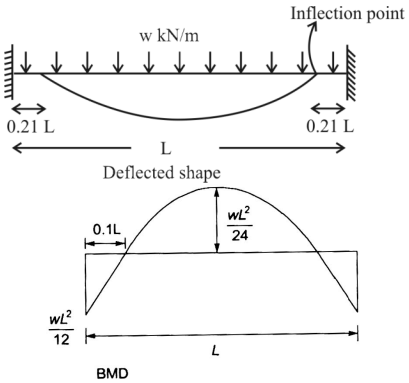
$$M_{BA} = \frac{2EI}{4}.(2\theta_B) = EI\theta_B$$

312. For approximate analysis of building frames under vertical loads, the point of inflection is assumed at-

(a) Centre of each beam
(b) One-tenth of the span length from each end of the beam
(c) Centre of each column
(d) Both (a) and (c)

OPSC AEE 2015 PAPER–I

Ans. (b) :



• For fixed beam, subjected to vertical loads the point of inflection or point of zero moment occurs at 0.21L from the both ends of the support.
• For a building frame, the support provided by the columns is neither fixed nor simply supported. For the purpose of approximate analysis the inflection point is assumed to occur at $\left(\frac{0+0.21L}{2}\right) \simeq 0.1\ L$ from the support.

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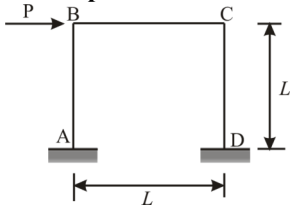
313. Independent displacement components at each joint of a rigid jointed plane frame are-

(a) Three linear movements
(b) One linear movement and two rotations
(c) Two linear movements and one rotation
(d) Three rotations

OPSC AEE 2015 PAPER–I

Ans. (c) : For a rigid jointed plane frame, at each joint, 3 displacements are possible –
(i) linear movement in Δ_x in x direction
(ii) Linear movement in Δ_y in y direction
(iii) Rotation about Z axis, θ_z.

314. What is the shear equation in slope deflection method for the portal frame shown below?



(a) $\frac{M_{AB}+M_{BA}}{L}+\frac{M_{CD}+M_{DC}}{L}+P=0$

(b) $\frac{M_{AB}+M_{BA}}{L}+\frac{M_{BC}+M_{CB}}{L}+P=0$

(c) $\frac{M_{BC}+M_{CB}}{L}+\frac{M_{CD}+M_{DC}}{L}+P=0$

(d) $\frac{M_{BC}+M_{CB}}{L}+P=0$

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (a) : Shear equation is given by

$$\Sigma H=0$$

i.e. $H_A+H_D+P=0$

$$H_A=\frac{M_{AB}+M_{BA}}{L}$$

and $$H_D=\frac{M_{DC}+M_{CD}}{L}$$

Hence shear equation is

$$\frac{M_{AB}+M_{BC}}{L}+\frac{M_{CD}+M_{DC}}{L}+P=0$$

315. A perfect frame should satisfy the relation

(a) m = 3j – 3 (b) m = 3j – 2
(c) m = 2j – 4 (d) m = 2j – 3

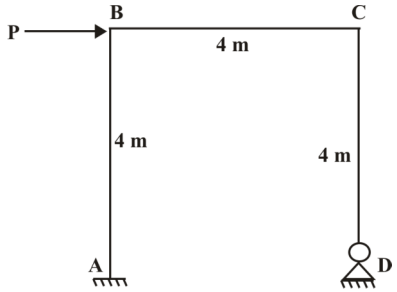
TNPSC AE 2018

Ans. (d) : Internal indeterminacy for a perfect frame is zero.

$$D_s=D_{se}+D_{si}$$
$$m+r-2j=r-3+D_{si}$$
$$D_{si}=m-(2j-3)=0$$

or $m=2j-3$

316. For the portal frame shown in figure, the shear equation is



(a) $\frac{M_{BA}+M_{AB}}{4}+\frac{M_{CD}}{4}+P=0$

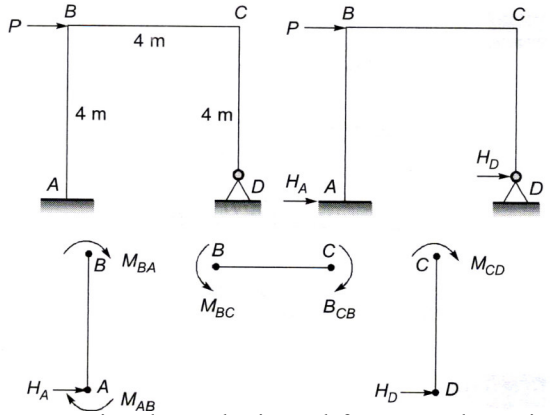
(b) $\frac{M_{BC}+M_{CB}}{4}+P=0$

(c) $\frac{M_{BA}+M_{AB}}{4}+P=0$

(d) $\frac{M_{CD}}{4}+P=0$

TNPSC AE 2018
Maharashtra PSC (Mains) 2017 Paper–I
ESE 1996

Ans. (a) :



Shear equation due to horizontal force P and reaction in horizontal at A and D as

$$H_A+H_D+P=0$$
$$\Sigma M_B=0$$
$$\Rightarrow \quad M_{BA}+M_{AB}=H_A\times 4$$
$$\Rightarrow \quad H_A=\frac{M_{BA}+M_{AB}}{4}$$
$$\Sigma M_C=0$$
$$\Rightarrow \quad M_{CD}=4\times H_D$$
$$\Rightarrow \quad H_D=\frac{M_{CD}}{4}$$

Putting value of H_A and H_D in shear equation

$$\frac{M_{BA}+M_{AB}}{4}+\frac{M_{CD}}{4}+P=0$$

317. A fixed beam of length L is subjected to concentrated load W mid-span, the collapse load is (plastic moment = M_P; length of beam = L)-

(a) $\frac{6M_P}{L}$ (b) $\frac{8M_P}{L}$

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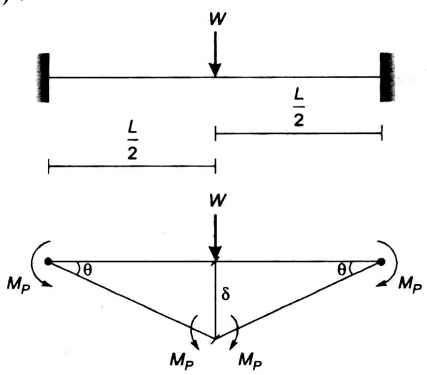
(c) $\frac{16M_P}{L}$ (d) $\frac{4M_P}{L}$

MPPSC AE 2017

Ans. (b) :



External work = Internal work

$4M_P\theta = W\delta$

$4M_P\theta = W\frac{L}{2}\theta$

$M_P = \frac{WL}{8}$

$W = \frac{8M_P}{L}$

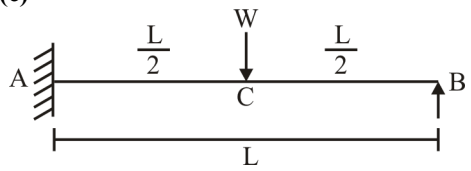
318. A propped cantilever beam of span *l* and constant plastic moment capacity M_P carries a concentrated load at mid-span. The load at collapse will be

(a) $\frac{2M_p}{\ell}$ (b) $\frac{4M_p}{\ell}$

(c) $\frac{6M_p}{\ell}$ (d) $\frac{8M_p}{\ell}$

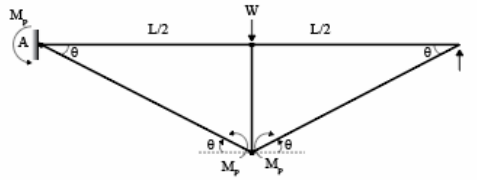
ESE 2019

Ans. (c)



■ Static indeterminacy 'D_s' = r–s.
r = Number of support reactions = 3 ---- (V_B, V_A & M_A)
s = Number of equilibrium equations = 2
D_s = 3–2 = 1

Number of possible plastic hinges 'N' = 2 [At A & C]



■ External work done 'W_e' = Load × Displacement under load.

$[W_e = W_c \times \delta]$

$\tan\theta \cong \theta = \frac{\delta}{L/2}$

$\left[\delta = \frac{L}{2}\theta\right]$

Internal work done (W_i) = Moment × Rotation

$W_i = M_p\theta + M_p 2\theta$

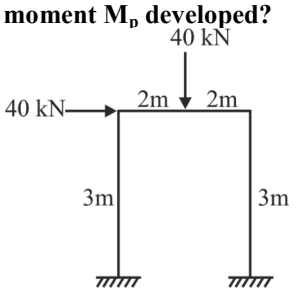
$[W_i = 3M_p\theta]$

Equating External Work = Internal Work

$W_c \times \frac{L}{2}\theta = 3M_p\theta$

$\boxed{W_c = \frac{6M_p}{L}}$

319. If only beam mechanism is possible in the frame given below, what will be the plastic moment M_p developed?

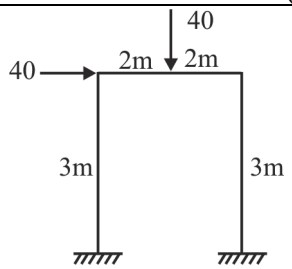


(a) 10 kNm (b) 20 kNm

(c) 40 kNm (d) 80 kNm

GPSC AE (Class 1 & 2) 2019

Ans : (b) :



External = Internal

$w \times \Delta = M_P(\theta_1 + \theta_2) + M_P\theta_1 + M_P\theta_2$

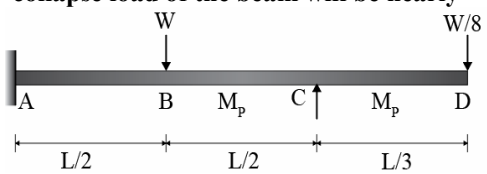
$40 \times \frac{l}{2} \times \theta = M_P(2\theta_2) + M_P\theta_2 + M_P\theta_2$

$40 \times \frac{l}{2} \times \theta = 4M_p\theta_2 \qquad \therefore \ \Delta = \frac{2\times\theta_1 = 2\times\theta_2}{\theta_1 = \theta_2}$

$40 \times 2 \times \theta_2 = 4M_p\theta_2$

$\boxed{Mp = 20KN - M}$

320. A propped cantilever ABCD is loaded as shown in figure. If it is of uniform cross-section, the collapse load of the beam will be nearly



(a) $6.5\frac{M_p}{L}$ (b) $5.6\frac{M_p}{L}$

(c) $4.7\frac{M_p}{L}$ (d) $3.8\frac{M_p}{L}$

ESE 2019

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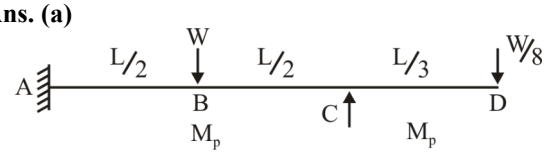
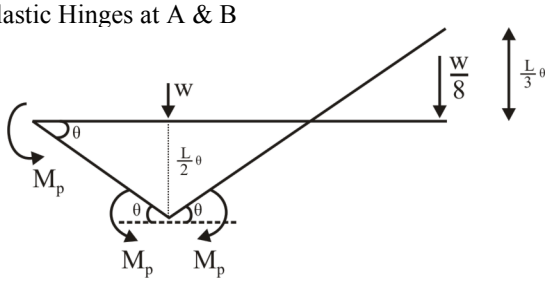
Ans. (a)



Plastic Hinges at A & B



$$-M_p\theta - M_p\theta - M_p\theta = -\frac{WL}{2}\theta + \frac{W}{8}\times\frac{L}{3}\theta$$

$$-3M_p\theta = -WL\frac{\theta}{2} + \frac{WL}{24}\theta$$

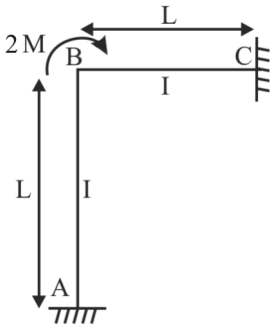
$$-3M_p\theta = \frac{-11WL\theta}{24}$$

$$W = \frac{24M_p\theta}{11L\theta}\times 3$$

$$W = \frac{72}{11}\frac{M_p}{L}$$

$$\boxed{W = 6.5\frac{M_p}{L}}$$

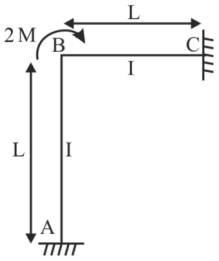
321. Members AB and BC shown in the figure below are identical. Due to a moment 2M applied at B, what is the value of axial force in the member AB?



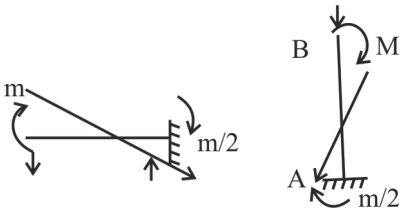
(a) M/L (Compression)
(b) M/L (Tension)
(c) 1.5 M/L (Compression)
(d) 1.5 M/L (Tension)

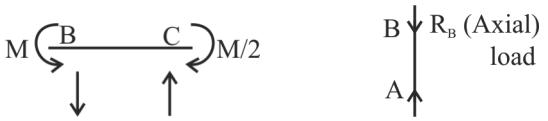
GPSC AE (CLASS 1 & 2) 2019

Ans : (d) :



From carry over factor of 1/2 member end moment at C = M/2





$$\boxed{R_B = \frac{M + M/2}{l} = \frac{1.5M}{l}}$$ Tension

(this R_B will act Axially (AB) member)

322. Assertion (A): The principle of superposition for deflection of beams subjected to a number of loads can be applied in the case of large deformations.
Reason (R): In the principle of superposition, the resultant deflection due to all the loads will be the algebraic sum of the deflections due to each load acting separately.

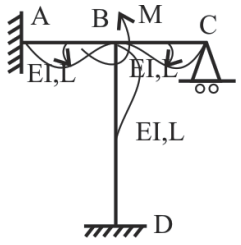
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

GPSC AE (CLASS 1 & 2) 2019

Ans : (d) : Superposition method follows only for small deflection.

- The principle of superposition states that resultant deflection due to all the loads will be the Algebraic sum of individual deflection.

323. All members of the frame shown below have the same flexural rigidity EI and length L. If a moment M is applied at the joint B, then the rotation of the joint is



(a) ML/12EI (b) ML/11EI
(c) ML/8EI (d) ML/7EI

Maharashtra PSC Mains 2018 Paper-1

GPSC AE (Class 1 & 2) 2019

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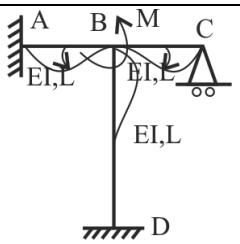
Ans : (b) :



The moment M is distributed to three member meeting at joint B according to flexural rigidity length and end condition.

$$K = \frac{4EI}{L}(\text{for end fixed})$$

$$K = \frac{3EI}{L}(\text{for end hinged})$$

The rotation 'θ' of all member meeting at joint B is same.

$$M = \frac{4EI}{L}\theta + \frac{4EI}{L}\theta + \frac{3EI}{L}\theta = \frac{11EI}{L}, \qquad \left[\theta = \frac{ML}{11EI}\right]$$

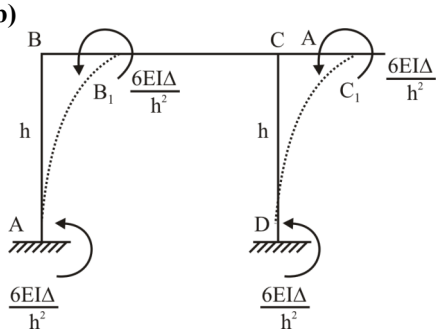
324. A single –bay portal frame of height h fixed at the base is subjected to a horizontal displacement Δ at the top. With constant EI, the base moment developed is proportional to

(a) $\frac{1}{h}$ (b) $\frac{1}{h^2}$

(c) $\frac{1}{h^3}$ (d) $\frac{1}{h^4}$

CIL MT 2020
ESE 2017, 2013

Ans. (b)



Base Moment 'M' = $\frac{6EI\Delta}{h^2}$

$$\boxed{M \propto \frac{1}{h^2}}$$

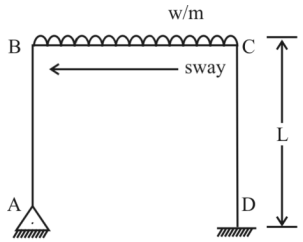
So, the base moment developed is inversely proportional to 'h^2'

325. A single bay single storey portal frame has hinged at left support and fixed at right support. It is loaded with uniformly distributed load on the beam. Which one of the following statement is true with regard to the deformation of the frame

(a) It would sway to the left side
(b) It would sway to the right side
(c) It would not sway at all
(d) It would sway both ways

CIL MT 2020
BPSC AE Aug. 2019 Paper-V
ISRO Scientist/Engineer 2014

Ans : (a)



Stiffness of member BA, and CD.

$$K_{BA} = \frac{3EI}{L}, \qquad K_{CD} = \frac{4EI}{L}$$

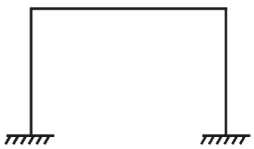
The frame will sway towards the column of low stiffness. Therefore, the frame would sway towards left side.

326. In a single bay, single storey frame with fixed ends and columns have the same values of moment of inertia, the displacement factor for the column is

(a) –0.75 (b) –0.50
(c) –1.0 (d) +1.0

AP (VSS-1) PART - A 2017

Ans : (a)



Distribution factor = $\frac{\frac{4EI}{L}}{\frac{4EI}{L}+\frac{4EI}{L}}$

$$\left[DF = \frac{4}{8} = 0.5\right]$$

Displacement factor (Δ) = –1.5 × D.F

$\Delta = -1.5 \times 0.5$

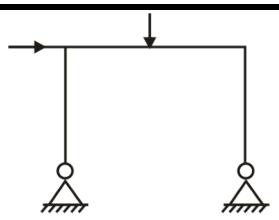
$\boxed{\Delta = -0.75}$

327. How many independent mechanism can be formed for a symmetrical portal frame of single bay and single storied, statically indeterminate to first degree and subjected to a concentrated load on the beam portion and one at the beam-column junction?

(a) 1 (b) 4
(c) 3 (d) 2

OPSC AEE 2019 Paper–I

Ans. (d)



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Static indeterminacy (D_S)

$D_s = D_{sc} + D_{si}$

where,

D_{sc} = External indeterminacy

D_{sc} = r–s, for a plane frame

where,

s = Number of equilibrium equations = 3

r = Reaction components = 4

D_{si} = Internal Indeterminacy

D_{si} = 3C for rigid jointed plane frame

C = Number of cuts required for obtaining an open configuration (or) number of closed boxes

$D_s = D_{se} + D_{si} = 1$

Number of independent mechanism (I) = N – D_s

number of possible plastic hinges (N) = 3

I = 3 – 1 = 2

328. If the number of members (m) are more than that required by the equation m = 2j – 3 (j is number of joints) such frames are called

(a) Redundant frame (b) Weak frame

(c) Perfect frame (d) Ideal frame

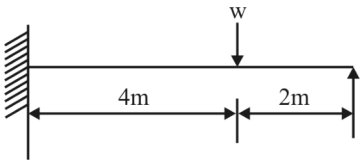
Punjab PSC SDE 2016
APPSC AEE MAINS 2016
APPSC AEE 2016

Ans. (a) : m = 2j – 3 (Perfect frame)

m > 2j – 3 (Imperfect redundant frame)

m < 2j – 3 (Deficient frame)

329.



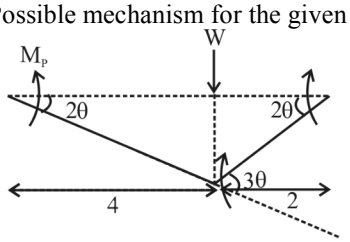
A propped cantilever beam is shown in the above diagram. What would be the collapse load, if plastic moment capacity is given as M_p?

(a) 1.25 M_p (b) 1.5 M_p

(c) M_p (d) 2 M_p

UPRVUNL AE 2014

Ans. (c) : Possible mechanism for the given beam is



By principle of virtual work

$$W \times 4\theta = M_p\theta + M_p \times 3\theta$$

$$W = M_P$$

330. Which of the following assumption is not made while analysis a perfect frame / truss :

(a) All members are pin jointed

(b) All members are loaded at the centre

(c) Self weight of the members is neglected

(d) All members are subjected either tension or compression

PSPCL AE 2012

Ans. (b) : For the perfect frame or truss the assumption are–

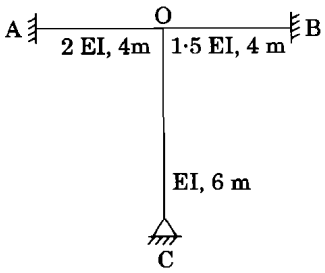
1) Firstly they should be loaded at joints not at centre.

2) Self weight of truss/frame member is neglected

3) The joints, should be free i.e, hinged

4) Perfect frame/truss or being called because they are stable i.e, donot collapse under load.

331. The distribution factor for the members OA, OB and OC are



(a) 0.125, 0.375, 0.5 (b) 0.375, 0.5, 0.125

(c) 0.5, 0.125, 0.375 (d) 0.5, 0.375, 0.125

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) : $DF_{OA} = \dfrac{k_{OA}}{\Sigma k_o}$

$DF_{OB} = \dfrac{k_{OB}}{\Sigma k_o}$, $DF_{OC} = \dfrac{k_{OC}}{\Sigma k_o}$

$\Sigma k_o = k_{OA} + K_{OB} + K_{OC}$

$$= \frac{4(2\,EI)}{4} + \frac{4(1.5\,EI)}{4} + \frac{3EI}{6}$$

$$= 2\ EI + 1.5\ EI + 0.5\ EI$$

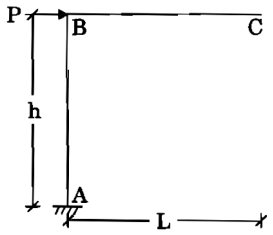
$\Sigma k_o = 4\ EI$

$$DF_{OA} = \frac{k_{OA}}{\Sigma k_o} = \frac{2\ EI}{4EI} = 0.5$$

$$DF_{OB} = \frac{k_{OB}}{\Sigma k_o} = \frac{1.5\ EI}{4EI} = 0.375 \Big\} DF = 1$$

$$DF_{OC} = \frac{k_{OC}}{\Sigma k_o} = \frac{0.5\ EI}{4EI} = 0.125$$

332. A rigid cantilever frame ABC is loaded and supported as shown in the figure below. The horizontal displacement of point C is



(a) $\dfrac{2\,Ph^3}{3EI}$ (b) $\dfrac{Ph^2(2h+L)}{2EI}$

(c) $\dfrac{Ph^3}{3EI}$ (d) $\dfrac{Ph^2(h+L)}{3EI}$

Maharashtra PSC Mains 2018 Paper-1

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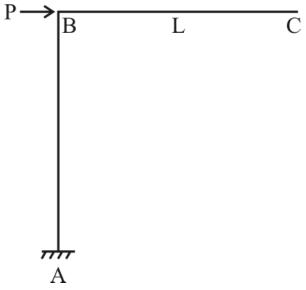
Ans. (c) :

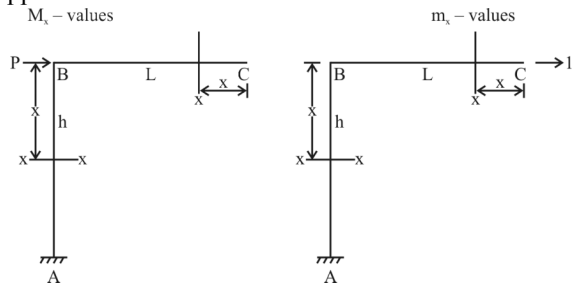


Horizontal displacement at 'C'

$$\delta_{HC} = \int_{D}^{L} \frac{M_x\, m_x}{EI} dx$$

M_x = BM at a section x – x due to real loads

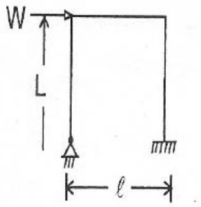
m_x = BM at a section x – x due to virtual unit load applied where we want to find the deflection.



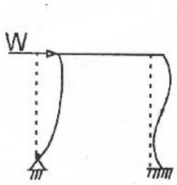
Member	M_x–values	m_x–values	$\int_0^L \frac{M_x m_x}{EI} dx$
CB	**0**	**0**	**0**
BA	**–Px**	**–x**	$\int_0^h \frac{(-Px)(-x)}{EI} dx$

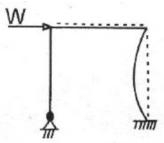
$$\delta_{HC} = \int_0^h \frac{(-Px)(-x)}{EI} dx = \frac{ph^3}{3EI}$$

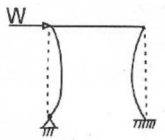
333. A portal frame is shown in fig. below. The deflection shape of the frame for constant EI is as per :

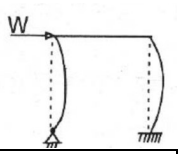


OPSC Civil Services Pre 2011
ESE 1999

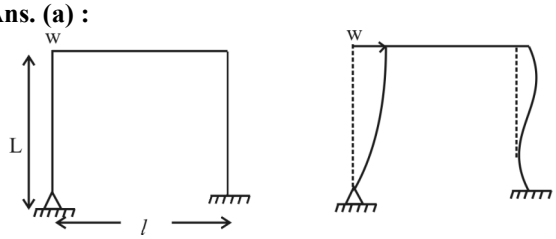
(a) 

(c) 

(b) 

(d) 

Ans. (a) :



The horizontal member will not be horizontal after deflection unless it is a rigid member but it is given that EI is constant.

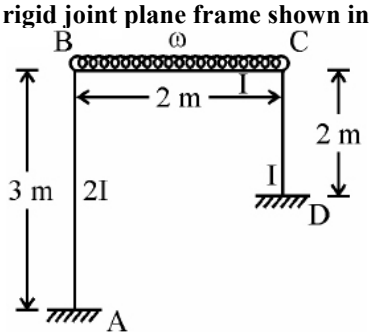
334. Compatibility conditions are essentially required to solve

(a) substitute frame (b) complex truss
(c) redundant frame (d) compound truss

TRB Poly. Tech. Lect. 2012

Ans. (c) : If the number of member in a frame are more then (2j – 3), then frame is called redundant frame. So, far analysis we use additional condition called compatibility condition to find out unknowns.

335. The rigid joint plane frame shown in figure will



(a) sway to right (b) sway to left
(c) not sway at all (d) none of the above

UKPSC AE (Urban) 2011

Ans. (a) : The rigid joints place from shown in figure will be sway to left.

336. A propped cantilever of span L carries a vertical concentrated load at the mid-span. If the plastic moment capacity of the section is Mp, the magnitude of the collapse load is :

(a) 8 Mp/L (b) 6 Mp/L
(c) 4 Mp/L (d) 2 Mp/L

JPSC AE (Pre) 2019

Ans. (b) :

$$-M_P\theta - M_P\theta - M_P\theta + \frac{PL}{2}\theta = 0$$

$$P \quad \frac{6M_P}{L}$$

337. The sway correction for a frame will be necessary

(a) Only when the frame is unsymmetrical
(b) Only when the loading is unsymmetrical

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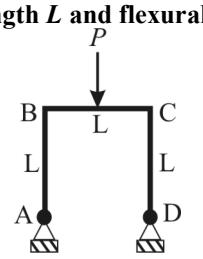
(c) Only when both the loading and the frame are unsymmetrical
(d) When either the loading or the frame is unsymmetrical

Punjab PSC SDE 2016

Ans. (d) : When the frame is unsymmetrical or the loading is unsymmetrical, there is sway correction for a frame.

338. For the frame shown below : supports A and D are hinged, joints B and C are rigid joints. The beam DC is under a concentrated load P acting and mid-span of the beam. All members have the same length *L* and flexural rigidity *EI*.

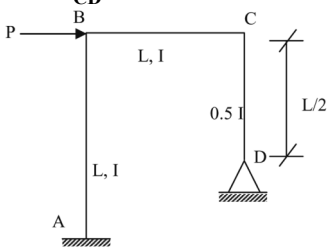


Now consider the following two statements:
Statement I: Horizontal reactions at supports A and D are zero
Statement I: Vertical reactions at supports A and D are *P*/2
Which of the following is CORRECT?
(a) Statement I is false, but Statement II is true
(b) Both Statement I and Statement II are true
(c) Both Statement I and Statement II are false
(d) Statement I is true, But Statement Ii is false

UPPCL AE 2018

Ans. (a) : statement I is false, statement II is true.

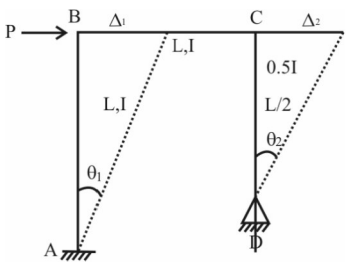
339. Figure shows a portal frame with one end fixed and other hinged. The ratio of the fixed end moments $\frac{M_{BA}}{M_{CD}}$ due to side sway will be



(a) 1.0 (b) 2.0
(c) 2.5 (d) 3.0

Haryana PSC AE 2017
APPSC AEE 2012
ESE 1995

Ans. (a) :



$\Delta_1 = \Delta_2 = \Delta$

$\Delta_1 = L\theta_1$

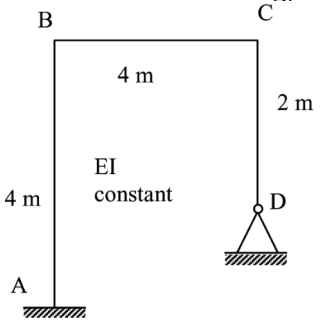
$\Delta_2 = \frac{L}{2}\theta_2$

$M_{BA} = \frac{6EI}{L^2}\Delta$

$M_{CD} = \frac{3E(0.5I)}{(L/2)^2}\Delta = \frac{6EI\Delta}{L^2}$

$\frac{M_{BA}}{M_{CD}} = 0.5$

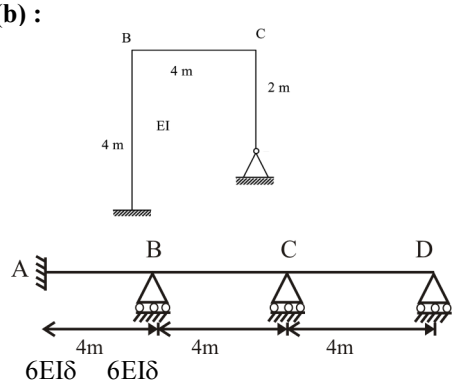
340. In the frame shown in the figure, the support 'D' settles by 'δ'. The fixed end moment in the horizontal member of the frame will be (other symbols have the usual meaning)



(a) $\frac{EI\delta}{8}$ (b) $\frac{6EI\delta}{16}$
(c) $\frac{3EI\delta}{16}$ (d) $\frac{EI\delta}{16}$

APPSC AEE 2012

Ans. (b) :



$M_{CD} = \frac{6EI\delta}{L^2} = \frac{6EI\delta}{4^2}$

$M_{DC} = \frac{-6EI\delta}{4^2}$

At B $\begin{cases} BC = \frac{4EI}{4} = 0.5 \\ BA = \frac{4EI}{4} = 0.5 \end{cases}$ 2EI

B $\begin{cases} CD = \frac{4EI}{2} = .33 \\ CB = \frac{4EI}{4} = 0.67 \end{cases}$ 3EI

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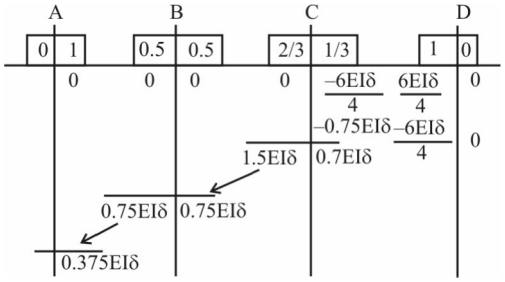


$M_{FED} = 0.375EI\delta$

or

$M_{FED} = \frac{6}{16}EI\delta$

341. In a linear elastic structural element

(a) stiffness is directly proportional to flexibility
(b) stiffness is inversely proportional to flexibility
(c) stiffness is equal to flexibility
(d) stiffness and flexibility are not related

BPSC AE Aug. 2019 Paper-V

Ans. (b) : Linear stiffness $(K) = \frac{P}{\Delta}$ $\because$ P : reaction

Δ : displacement

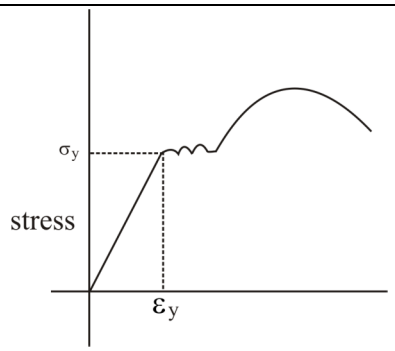
So, $K = \frac{1}{C}$ Linear stiffness is inversely proportional to linear flexibility.

342. The energy stored in a structural member is called resilience when the member is deformed:

(a) within the limit of proportionality
(b) upto elastic limit
(c) till fracture point
(d) half of elastic limit

[UPPSC State Eng. A.E. 2008]

Ans : (b) Resilience is defined as the total strain energy which can be stored in the given volume of the metal and can be released after unloading. It is equal to area under load deflection curve within elastic limit.



Modulus of resilience = Elastic strain energy per unit volume up to elastic limit.

$$= \frac{1}{2}\sigma_E \in_E$$

[$\because$ at elastic limit $\sigma_E = \sigma_y$ and $\in_E = \frac{\sigma_y}{E}$]

$\therefore$ modulus of resilience $= \frac{1}{2}\sigma_y \times \frac{\sigma_y}{E} = \frac{\sigma_y^2}{2E}$

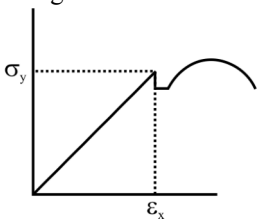
343. A beam of span 'L' of flexural rigidity 'EI' and subjected to bending moment 'M', then strain energy stored is equal to:

(a) $\int_0^L \frac{M}{2EI}dx$ (b) $\int_0^L \frac{M^2}{EI}dx$

(c) $\int_0^L \frac{M}{EI}dx$ (d) $\int_0^L \frac{M^2}{2EI}dx$

[UPPSC State Eng. A.E. 2008]
[UPPSC State Eng. A.E. 2007 (I)]

Ans : (d) When load deformation is within elastic limit, the stress - strain diagram will be linear.



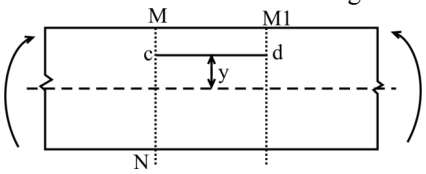
Then the strain energy per unit volume is given by

$$U = W = \frac{1}{2} \times \text{stress} \times \text{strain}$$

$$= \frac{1}{2} \times \sigma \times \in$$

$$U = \frac{1}{2}\sigma \times \frac{\sigma}{E} = \frac{\sigma^2}{2E} \quad \text{..... (i)}$$

consider small element dx as shown in figure.



Let M_x be the BM at section

$\therefore$ $\sigma = \frac{M_x y}{I}$

$\therefore$ strain energy of the elementary volume of elemental length Δx and area ΔA is

$\therefore$ $\delta u = \frac{\sigma^2}{2E} \times dx \times \Delta A$

$$\delta u = \frac{M_x^2 y^2}{2I^2 E}.dx.\Delta A$$

$$\delta_U = \sum \frac{M_x^2 dx}{2I^2 E} \times d_x.\Delta A$$

$$\delta U = \frac{M_x^2 dx}{2I^2 E} \times \sum \Delta A Y^2$$

$$\delta U = \frac{M_x^2 dx}{2EI} \quad \left(\text{since } \sum dA.Y^2 = I\right)$$

$\therefore$ Total strain energy

$$U = \frac{1}{2EI}\int_0^L M_x^2 dx$$

$$= \int \frac{M^2 dx}{2EI}$$

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344. In column analogy method, the width of the column is equal to

(a) half the depth (b) depth
(c) width of the beam (d) unity

BPSC AE Aug. 2019 Paper-V

Ans. (d) : Column analogy method is used to find end moments and fixed and moment in statically indeterminate beams.
An analogous column is a short column with width as 1/EI or unity.

345. The shear force carried by the intermediate column compared to end column in a multi-bay portal frame is

(a) half (b) double
(c) equal (d) triple

BPSC AE Aug. 2019 Paper-V

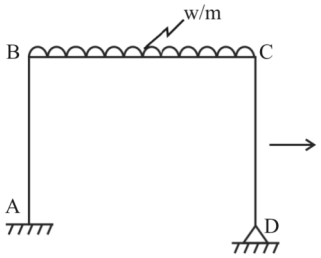
Ans. (b) : Since, an internal column (intermediate) of multi bay frame represents two portal legs where as exterior column (end) column represents only one leg portal. Intermediate column has twice shear force as compared to end column.

346. A single bay single storey portal frame has a fixed left support and hinged right support. It is loaded with a uniformly distributed load of w/ m length on the beam. Which one of the following statements is true with regard to the deformation o the frame ? (Assume all the members have equal length).

(a) Insufficient data
(b) It would sway to the left side
(c) It would sway to the right side
(d) It would not sway at all

BHEL ET 2019

Ans. (c) :

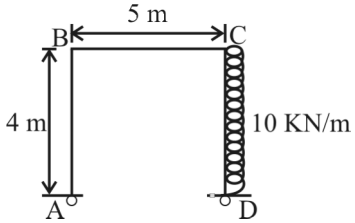


It would sway to the right side because there is fixed support in the left side so, it is free to move in the right side.

Stiffness of BA = $\frac{4EI}{L}$ and

Stiffness CD = $\frac{3EI}{L}$

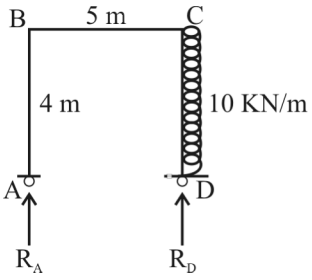
347. What is the value of vertical reaction at A for the frame shown:



(a) 0 KN (b) 40 KN
(c) 10 KN (d) 16 KN

PTCUL AE 2016

Ans. (d) :



$\Sigma M_D = 0$

$R_A \times 5 - (4 \times 10) \times 2 = 0$

$R_A = 80/5$

$\boxed{R_A = 16 \text{ KN}}$

348. In the analyses of plane frame, the stiffness factor at a joint is taken equal to the sum of stiffness factors the members connected to the joint by using

(a) Principle of virtual work
(b) Principle of angular momentum
(c) Principle of super position
(d) Principle minimum work done

APPSC AEE 2016

Ans. (c) : For analysis of plane frame the stiffness factor at a joint is taken equal to the sum of stiffness factors the member connected to the joint by using principle of super position.

349. A method that cannot not be used to determine the forces in the members of a frame is :

(a) moment distribution method
(b) method of sections
(c) graphical method
(d) method of joints

DMRC AM 2020

Ans. (a) Moment distribution method is a displacement method as its is not used to determine the forces in the member of a frame.

350. Read the following statements about the assumptions made in finding out the forces in a frame :
Statement 1 : The frame is a perfect frame.
Statement 2 : The frame carries load at the joints.
Select the appropriate option from the below.

(a) Statement 1 is false and statement 2 is true
(b) Both the statements 1 and 2 are true
(c) Statement 1 is true and statement 2 is false
(d) Both the statements 1 and 2 are false

DMRC AM 2020

Ans. (b) : Both the statements are individually correct as the frame is a perfect frame and frame carries load at the joints.

351. A portal frame may sway due to :
a) Eccentric loading on the portal frames.
b) Non-uniform section of the member.
Select the true statement.

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(a) Both the statements (a) and (b) are true
(b) Statement (a) alone is true
(c) Statement (b) alone is true
(d) Both the statements (a) and (b) is false
(e) Information insufficient

CGPSC AE 2017

Ans. (a) A portal frame may sway due to
(a) Eccentric loading on the portal frames
(b) Non- uniform section of the number

352. Match list I (Collapse load for beam) with List II (Values) and select the correct answer using the codes given below in the lists if M_p is the fully plastic moment in beam (as specified below) of length L :

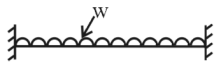
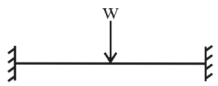
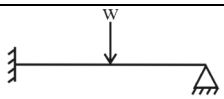
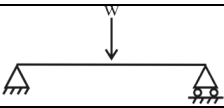
List-I (Collapse Load for beam)	List-II (Values)
A. Fixed beam with a uniformly distributed load of intensity 'w'	1. $\frac{4M_p}{L}$
B. Fixed beam with a central point load	2. $\frac{6M_p}{L}$
C. Propped cantilever with a central point load	3. $\frac{8M_p}{L}$
D. Simply supported beam with a central point	4. $\frac{16M_p}{L^2}$

Code :

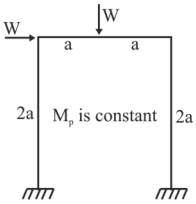
	A	B	C	D
(a)	1	2	4	3
(b)	2	1	3	4
(c)	4	3	2	1
(d)	1	2	3	4

HPPSC Poly. Lect. 2016
ESE 2000

Ans : (c)

1		→	Collapse load = $\frac{16M_p}{L^2}$
2		→	Collapse load = $\frac{8M_p}{L}$
3		→	Collapse load = $\frac{6M_p}{L}$
4		→	Collapse load = $\frac{4M_p}{L}$

353. What is the ultimate load for the frame shown in the figure below?

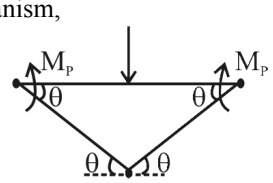


(a) M_p/a (b) $2M_p/a$
(c) $3M_p/a$ (d) $4M_p/a$

Haryana PSC AE 2017

Ans. (b) : There are five possible locations at which the plastic hinges can form statical indeterminacy of the frame is 3. Total number of independent mechanism i = n – r = 5 – 3 = 2

- Beam mechanism,



$-M_P\theta - M_P\theta - M_P\theta - M_P\theta + W.a.\theta = 0$

$$\left[W = \frac{4M_p}{a}\right]$$

- Combined mechanism,

$-M_P\theta - M_P\theta - M_P\theta - M_P\theta - M_P\theta - M_P\theta + 2W \times 2.a.\theta - Wa\theta = 0$

$3Wa\theta = 6M_P\theta$

$$\left[W = \frac{2M_p}{a}\right]$$

The collapse load given by combined mechanism is minimum hence it is the ultimate load for the frame.

354. Which of the following is the condition for a rigid-jointed plane frame to be stable and statically determinate?
(Where n is number of member, j is number of joint and r is reaction component)
(a) n + r = 2j (b) n+r = 3j
(c) 3n + r = 3j (d) n + 3r = 3j

UPRVUNL AE 2014

Ans. (c) : Condition for a rigid = jointed plane frame to be stable and statically determinate,
3 n + r = 3j

355. Ties are load carrying members of a frame which are subjected to
(a) transverse loads
(b) axial tension loads
(c) axial compressive loads
(d) torsional loads

WBPSC AE 2007

Ans. (b) : Ties are those members that are subjected to axial tension forces only. Load is applied to ties only at the ends. Ties cannot resist flexural forces.

356. Compatibility conditions are essentiality required to solve
(a) substitute frame
(b) Complex truss
(c) Redundant frame
(d) Compound truss

Chandigarh AE 2017

Ans. (c) : Redundant sectional forces are required for compatibility condition in statically in determinant system.

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5. Influence Line Diagram and Rolling Loads

357. Maxwell's reciprocal theorem is valid
(a) for all statically determinate structure
(b) for all structures
(c) for all elastic structures
(d) for all structures with linear force displacement relation

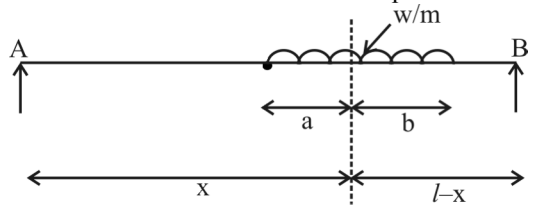
[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d) Maxwell's reciprocal theorem is valid for any structure which is linear elastic in which hook's law is valid, temperature is constant and supports are unyielding.

358. When a uniformly distributed load, shorter than the span of the girder, moves from left to right, then the conditions for maximum bending moment at a section is that
(a) The head of the load reaches the section.
(b) The tail of the load reaches the section.
(c) The load position should be such that the section divides it equally on both side.
(d) The load position should be such that the section divides the load in the same ratio as it divides the span

Arunachal Pradesh Poly. Tech. Lect. 2019
Nagaland PSC CTSE 2017 Paper–I
MPPSC AE 2014, ESE 2011

Ans. (d) : Bending moment at section is maximum when load is placed such that the section divide the load in the same ratio as it divides the span.



For maximum bending moment at section 'C'

$$\left[\frac{a}{b}=\frac{x}{l-x}\right]$$

359. The reciprocal theorem of Maxwell is valid for structure :
(a) having linear force-displacement relation
(b) that are elastic
(c) that are plastic
(d) irrespective of their type

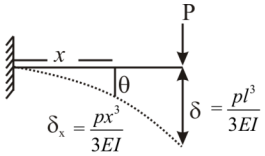
CIL MT 2017

Ans. (a) : The reciprocal theorem of Maxwell is valid for structures having linear force displacement relation.

360. The influence line for deflection at the free end of a cantilever is
(a) a triangle with zero ordinate at fixed end and maximum ordinate at free end
(b) a constant line
(c) a cubic parabola with zero ordinate at fixed end and maximum ordinate at free end
(d) a parabola with zero ordinate at fixed end and maximum ordinate at free end

APPSC AE 2016 (Mains)

Ans. (c) :



$$\theta \propto x^3$$

In a cantilever, the influence line diagram for deflection at the free end is same as elastic Curve of the beam due to unit load placed at free end.

Deflection due to point load placed on cantilever is given as $Px^3/3EI$ is cubic parabola and hence ILD for deflection for a cantilever will be also be cubic parabola.

361. The variation of influence line for stress function in a statically determinate structure is:
(a) parabolic (b) bilinear
(c) linear (d) uniformly rectangular

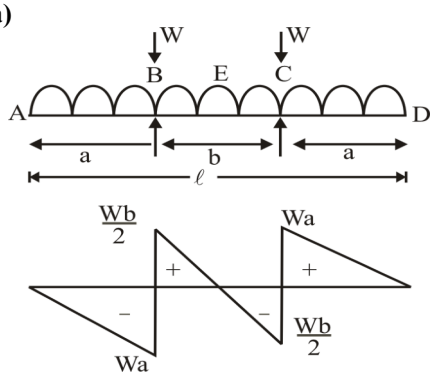
(UKPSC AE (Paper-I) 2007)
ESE 2007

Ans : (c) The variation of influence line for stress function in a statically determinate structure is linear.

362. Two parallel rails are running on railway sleepers. The centre-to-centre distance between the rails is 'b' with the sleepers projecting by an amount 'a' at each end beyond the rails. When the train passes over the rails, the reaction exerted by the ground can be taken as uniformly distributed over the sleeper. The ratio $\frac{b}{a}$ for the condition that the maximum bending moment is as small as possible is
(a) 2.83 (b) 2.90
(c) 2.50 (d) 3.00

ESE 2017

Ans. (a)



Shear force diagram

- Maximum (+) bending moment will occur at the mid span of central span E and maximum (–ve) B.M. will occur at the supports B and C.
- For sagging moment is equal to hogging moment the overhang distance; a = 0.207ℓ

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The distance between two supports is,

$b = \ell - 2a$

$$\frac{b}{a} = \frac{\ell - 2\times 0.207\ell}{0.207\ell}$$

$$\left[\frac{b}{a} = 2.83\right]$$

363. The Muller-Breslau principle in structural analysis is used for,

Maharashtra PSC Mains 2018 Paper-1
OPSC AEE 2015 PAPER–I
ISRO Scientist/Engineer 2013
ESE 2012

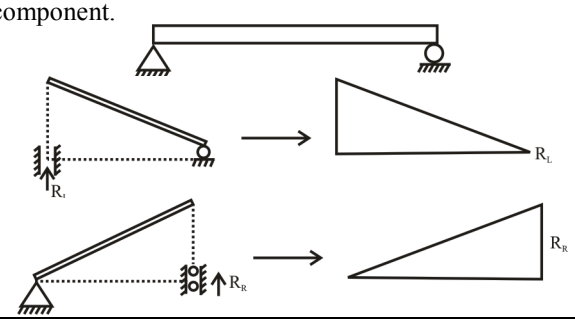
OR

Muller-Breslau's principle for obtaining influence lines is applicable to

Arunachal Pradesh Poly. Tech. Lect. 2019
Nagaland PSC CTSE 2017 Paper–I

(a) Drawing influence line diagram for any force function
(b) Super-position of load effects
(c) Writing virtual work equation
(d) Calculating strain energy

Ans : (a) Muller-Breslau Principle : It states that, for drawing the influence line for any force quantity (like thrust, shear, and reaction support moment or bending moment) in a structure. If an internal stress component or reaction is considered to act through some small distance and thereby to deflect or displaced the structure. The curve of the deflected structure will be at some scale of influence line for the stress or reaction component.

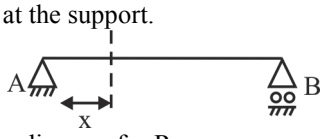


364. The influence line diagram for reaction at a support of a simply supported beam is

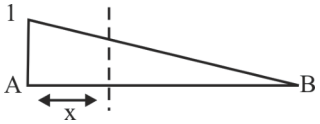
(a) triangle with ordinate 1 at that support
(b) a triangle with ordinate 1 at the support
(c) a rectangle with ordinate of 1
(d) a rectangle with ordinate of $\frac{1}{2}$

TNPSC AE 2018
UPPSC State Eng. A.E. 2007 (II)
UPPSC State Eng. A.E. 2003

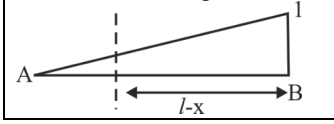
Ans. (a) : The influence line diagram for reaction at a support of simply supported beam is a triangle with ordinate 1.0 at the support.



Influence line diagram for R_A



Influence line diagram for R_B



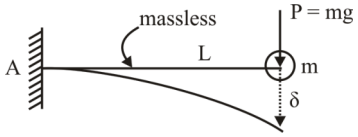
365. The natural frequency of a mass m at the end of the cantilever beam of negligible mass with usual notations will be :

(a) $\frac{1}{2\pi}\left(\frac{3EI}{mL^3}\right)^{\frac{1}{2}}$ (b) $\frac{1}{\pi}\left(\frac{6EI}{mL^3}\right)^{\frac{1}{2}}$

(c) $\frac{1}{2\pi}\left(\frac{6EI}{mL^3}\right)^{\frac{1}{2}}$ (d) $\frac{1}{\pi}\left(\frac{3EI}{mL^3}\right)^{\frac{1}{2}}$

ESE 2020

Ans. (a)



Deflection at free end due to load 'mg'

$$\delta = \frac{(mg)L^3}{3EI} \quad \text{..... (i)}$$

$\because$ spring stiffness (k) = $\frac{P}{\delta}$

$\therefore$ Beam transverse stiffness (k_t) = $\frac{P}{\delta}$

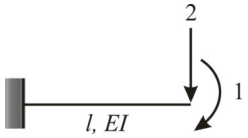
$$k_t = \frac{mg}{\frac{(mg)L^3}{3EI}} = \frac{3EI}{L^3}$$

Now,

Natural frequency (f_n) = $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ Hz

$$\boxed{f_n = \frac{1}{2\pi}\sqrt{\frac{3EI}{mL^3}}\ \text{Hz}}$$

366. For the structure shown. The elements of the flexibility matrix are



(a) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;

$f_{22} = \frac{l^3}{(3EI)}$

(b) $f_{11} = \frac{l^3}{(3EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;

$f_{22} = \frac{l}{(EI)}$

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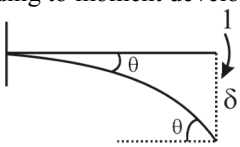
(c) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(EI)}$; $f_{12} = \frac{l^2}{(EI)}$;
$f_{22} = \frac{l^3}{(3EI)}$

(d) $f_{11} = \frac{l}{(EI)}$; $f_{21} = \frac{l^2}{(2EI)}$; $f_{12} = \frac{l^2}{(2EI)}$;
$f_{22} = \frac{l^3}{(4EI)}$

APPSC AE 2016 (Mains)

Ans. (a) :

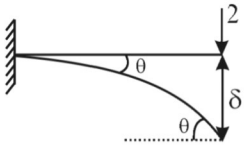
Development of flexibility matrix :
According to moment developed :



$\delta = \frac{ML^2}{2EI} = \frac{L^2}{2EI} = F_{21}$

$\theta = \frac{ML}{EI} = \frac{L}{EI} = F_{11}$

According to force developed :



$\delta = \frac{WL^3}{3EI} = \frac{L^3}{3EI} = F_{22}$

$\theta = \frac{WL^2}{2EI} = \frac{L^2}{2EI} = F_{12}$

Flexibility matrix = $\begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix}$

367. In moment distribution method, the sum of distribution factors of all members meeting at any joint is always-

(a) Less than one (b) Zero
(c) One (d) Greater than one

WBPSC Poly Lect. 2019
TNPSC AE 2018
Nagaland PSC CTSE 2017 Paper–I
BPSC AE 2012 Paper–V
UKPSC AE (Urban) 2011
OPSC AEE 2015 PAPER–I

Ans. (c) : The distribution factor for a member at a joint is the ratio of the stiffness of the member to the total stiffness of all the members meeting at that joint.

$DF_i = \left(\frac{\text{Stiffness i}}{\text{Total Stiffness of joint}}\right)$

$\sum DF = \frac{k_1 + k_2 + ...k_n}{\sum k} = 1$

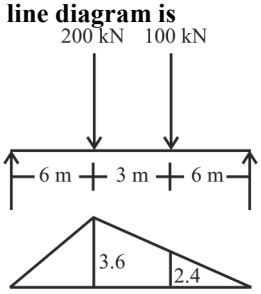
368. The ordinate of influence line diagram for bending moment always will be:

(a) Force (b) Force/length
(c) Length (d) Force × length

DSSSB AE 2019
(BPSC AE 2012 Paper–V)

Ans. (c) The ordinate of influence line diagram for bending moment always will be dimension of length.

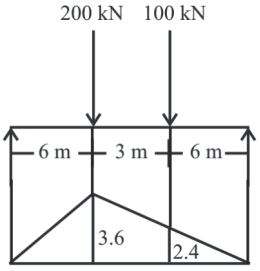
369. The maximum bending moment for given influence line diagram is



(a) 240 kNm (b) 720 kNm
(c) 960 kNm (d) 1060 kNm

TNPSC AE 2019
TRB Poly. Lect. Tech. 2017

Ans. (c) Maximum bending moment at a section occur if the point load is at section.



$\left(\frac{ab}{l}\right) = \frac{6 \times 9}{15} = 3.6$

maximum BMD = w(loading) × Area of BMD
= 200 × 3.6 + 100 × 2.4
= 720 + 240 = 960 kN-m

370. The Muller-Breslau principle can be used to
A. Determine the shape of the influence line
B. Indicate the parts of the structure to be loaded to obtain the maximum effect
C. Calculate the ordinates of the influence lines
Which of the statement are correct?

(a) Only A (b) Only A and B
(c) Only B and C (d) All three

UPRVUNL AE 2015

Ans. (d) : The Muller- Breslau principle can be used for drawing ILD of any force function not only for statically determinate structures but also of indeterminate structure. It uses deflection profile of the basis for ILD

371. Two wheel loads 80 kN and 200 kN respectively spaced 2 m apart, move on a girder of span 16 m. Any wheel load can lead the other. The maximum negative shear force at a section 4 m from the left end will be

(a) – 50 kN (b) – 60 kN
(c) – 70 kN (d) – 80 kN

ESE 2019
CGPSC AE 2014 Shift–I

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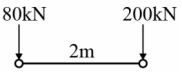
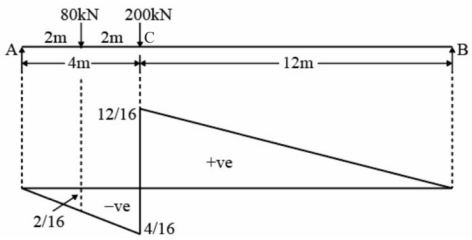
Ans. (b)



Note : Any wheel load can lead the other.



Influence Line Diagram for shear force

To get maximum negative shear force at a section 'C' place the greater load i.e. 200kN just left to the section 'C' and 80 kN load on span 'A' maximum negative shear force at section 'C'

$$SF = -80 \times \frac{2}{16} - 200 \times \frac{4}{16}$$

$$SF = -10 - 50$$

$$\boxed{SF = -60kN}$$

372. The Influence Line Diagram for bending moment for determinate structure is–

(a) Curved
(b) Linear
(c) Combination of curved and linear
(d) Irregular

APPSC Poly. Tech. Lect. 2020

Ans. (b) : Linear : The influence line diagram (ILD) of shear force and bending moment B.M. is linear for determinate structures and will be curved for indeterminate structures.

373. If a structure is subjected to moving loads, the variation in bending moment and shear is best described using_____.

(a) Shear force diagram
(b) Influence load diagram
(c) Influence line diagram
(d) Bending moment diagram

LMRC AM 2019

Ans. (c) Influence line diagram best describe the variation in bending moment and shear when structure is subjected to moving loads.

374. In the displacement method of structural analysis the basic unknowns are :

(a) Displacements
(b) Force
(c) Displacement and forces
(d) None of these

Nagaland PSC CTSE 2017 Paper–I

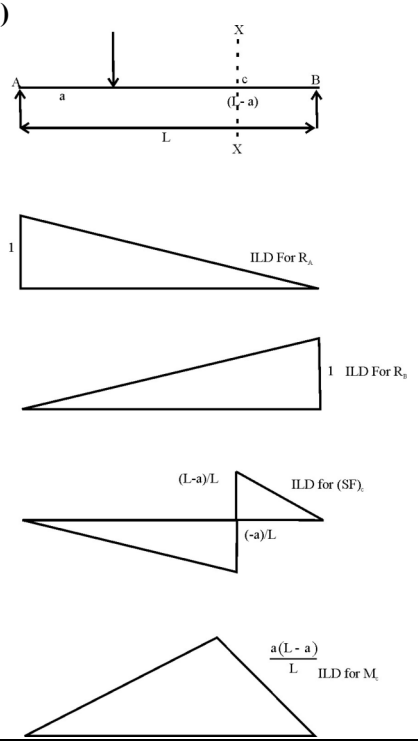
Ans. (a) : Displacement Method –
(1) Unknowns taken are Displacement and forces.
(2) To find unknown displacement joint equilibrium conditions are written.
(3) The no. of equilibrium conditions needed is equal to degree of kinematic indeterminacy.

375. Influence line diagram for bending moment in a simply supported beam is a:

(a) straight line (b) parabola
(c) triangle (d) None of these

[UPPSC AE 12.04.2016 Paper–I]

Ans : (c)

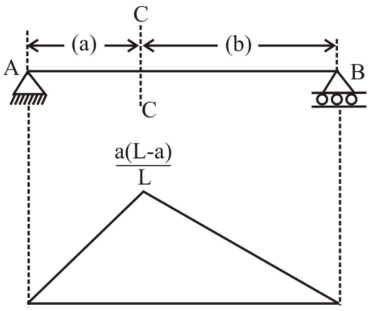


376. The maximum bending moment due to single moving point on a simply supported beam occurs at

(a) quarter span
(b) mid span
(c) any where in the beam
(d) a section under the load

[UPPSC State Eng. A.E. 2011]
[UPPSC State Eng. A.E. 2007 (II)]

Ans : (d)



ILD for m_c

From the ILD for B.M. at section C-C it is clear that the maximum bending moment due to single moving point load on simply supported beam occurs at section under the load.

377. For stable structures, one of the important properties of flexibility and stiffness matrix is that the elements on the main diagonal

(1) must be positive of stiffness matrix
(2) must be negative of stiffness matrix

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(3) must be positive of flexibility matrix
(4) must be negative of flexibility matrix
The correct answer is–
(a) 1 and 3 (b) 2 and 3
(c) 1 and 4 (d) 2 and 4

(BPSC AE 2012 Paper–V)

Ans : **(a)** 1 and 3
For stable structure, important properties of flexibility and stiffness matrix.
(1) Must be positive of stiffness matrix.
(2) Must be positive of flexibility matrix.

378. The Muller-Breslau principle can be used to-
(1) determine the shape of the influence line
(2) indicate the parts of the structure to be loaded to obtain the maximum effect
(3) calculate the ordinates of influence lines
The correct answer is-
(a) 1 only (b) Both 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

OPSC AEE 2019 Paper–I
(BPSC AE 2012 Paper–V)

Ans : **(d)** The Muller–Breslau principle can be used to–
(a) Determine slope of the influence line.
(b) The structure to be loaded to obtain the maximum effect.
(c) Calculate the ordinates of influence lines.

379. The maximum ordinate of the influence line diagram for bending moment at the fixed support in cantilever beam will occur at:
(a) fixed support
(b) mid span
(c) quarter span from fixed end
(d) free end

[UPPSC State Eng. A.E. 2007 (II)]
[UPPSC State Eng. A.E. 2011]

Ans : (d) The maximum ordinate of the influence line diagram for bending moment at the fixed support in cantilever beam will occur at free end.

380. In a frame the point of inflection occurs in
(a) beam only
(b) column only
(c) beam column joint
(d) anywhere in the frame

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d) Point of inflection, where bending moment changes sign, it depends on loading so it may occur anywhere in the frame.

381. Influence line for a forcing function gives its variation at
(a) mid span (b) support
(c) given section (d) every where in the beam

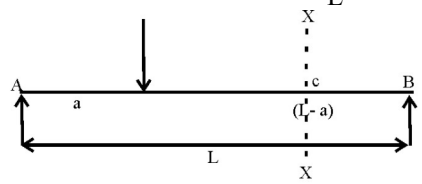
UK Combined A.E. Paper Ist 2012)
UPPSC State Eng. A.E. 2004 (I)
OPSC AEE 2015 PAPER–I

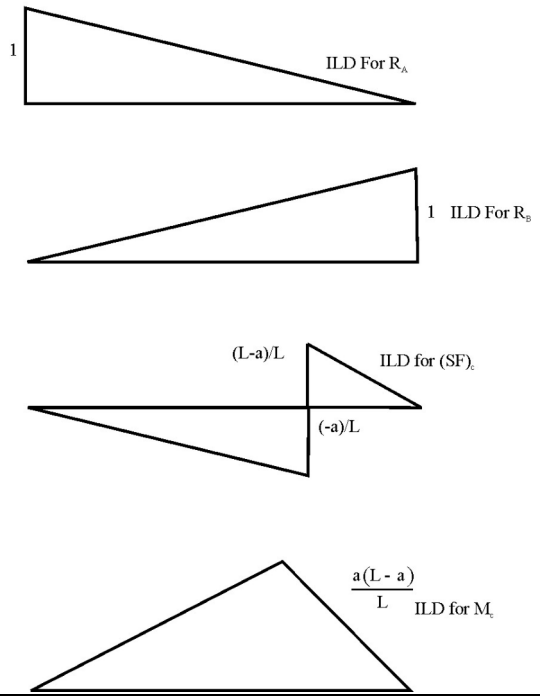
Ans : (c) Influence line represents variation of a stress function such as shear force, bending moment, reaction, slope or deflection at a section when unit concentrated load moves from one end to the other end.

382. The peak ordinate of Influence line diagram for bending moment at 'a' from left support of simply supported beam of span 'L' is given by
(a) aL (b) $\frac{L(a-L)}{a}$
(c) $\frac{a(L-a)}{a}$ (d) $\frac{a(L-a)}{L}$

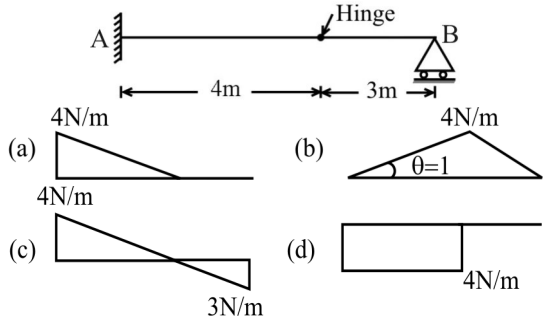
(BPSC AE Paper V 2006)

Ans : **(d)** The peak ordinate of influence line diagram for bending moment at 'a' from left support of simply supported beam of span 'L' is $\frac{a(L-a)}{L}$.





383. The influence line for moment at point 'A' of the beam shown in fig. is.



[UPPSC State Eng. A.E. 2004 (I)]

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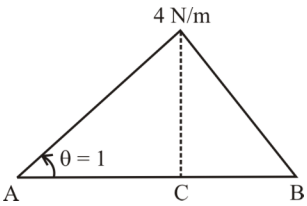
Ans : (b)



According to Muller-Breslau principle "the ILD for any stress function in structure is represented by its deflected shape obtained by removing the restraint offered by that stress junction and introducing a direct reaction generating unit displacement in the direction of that stress function."
So that obtained ILD for moment of A provide hinge at A and give unit displacement.

384. Written below are some types of structure

1. Beams
2. Frames
3. Trusses

Muller – Breslau's principle is applicable to which type of structures

Select correct answer from the following–

(a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

RPSC VPITI 2012

OR

Muller-Breslau principle is applicable for:

(a) Determinate structures
(b) Indeterminate structures
(c) Influence lines
(d) All of the above

[UPPSC State Eng. A.E. 2007 (II)]
Karnataka PSC AE 2017 Paper - II
[UPPSC State Eng. A.E. 2007 (I)]

Ans : (d) Muller-Breslau principle gives qualitative ILD for any structure either determinate or indeterminate.

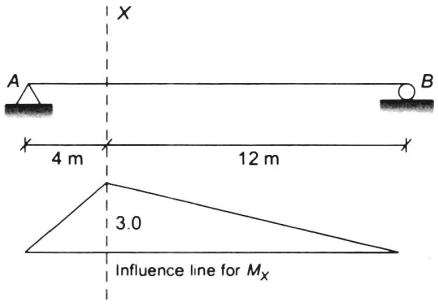
385. Influence line for redundant structures can be obtained by

(a) Castigliano's theorem
(b) Muller-Breslau principle
(c) Unit load method
(d) Maxwell-Betti reciprocal theorem

(JPSC Combined AE Paper V 2013)

Ans : (b) Influence line for redundant structures can be obtained by Muller-Breslau principle.

386. The influence line for bending moment at section $X(M_X)$ at a distance of 4 m from the left support of a simply supported girder AB is shown in figure below. A uniformly distributed load of intensity *2 t/m* longer than the span crosses the girder from left to right-

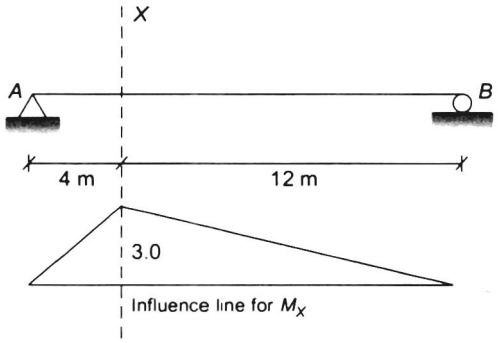


The max bending moment at section *X* is equal to-

(a) 12 *tm* (b) 24 *tm*
(c) 48 *tm* (d) 96 *tm*

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) : For udl longer than span, it is on full length of the beam. To calculate maximum value of bending moment at section X, we will have to calculate the area under ILD.



$$M = \frac{1}{2}\times 4\times 3\times 2 + \frac{1}{2}\times 12\times 3\times 2$$

M = 12 + 36

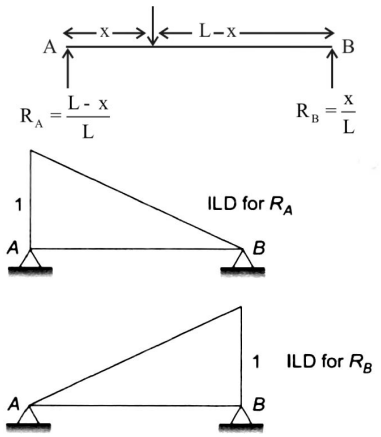
M = 48 *tm*

387. If unit load moves from left to right on Simply Supported beam the nature of ILD for reaction is-

(a) Parabolic
(b) Hyperbolic
(c) Horizontal Straight Line
(d) Inclined Straight Line

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : ILD for reaction on simply supported beam is given as



i.e. inclined straight line.

388. Free body diagram is an-

(a) Isolated joint with only body forces acting on it
(b) Isolated joint with all the forces internal as well as external, acting on it
(c) Isolated joint with internal acting on it
(d) None of these

OPSC AEE 2015 PAPER–I

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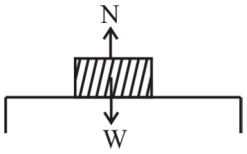
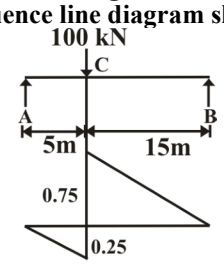
Ans. (b) :



Free body diagram is a graphical illustration used to visualise the applied forces, moments and resulting reactions on a body in a given condition.

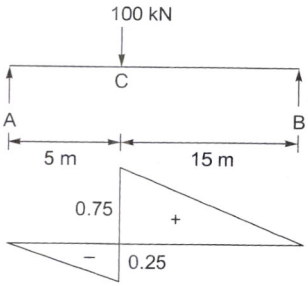
389. The positive and negative shear force (kN) of a given influence line diagram shown below:



(a) 75, 50 (b) 75, 75
(c) 75, 25 (d) 25, 75

TNPSC AE 2018

Ans. (c) :



for maximum +ve shear 100 KN is placed just right to C maximum +ve shear at C = 100 × 0.75 = 75 KN
for maximum–ve shear 100 KN is placed just left to C
Maximum – ve shear at C = 0.25×100=25 KN

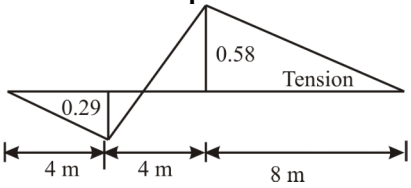
390. Which of the following method is a displacement method?

(a) Column analogy method
(b) Three moment equation
(c) Euler's method
(d) Moment distribution method

TNPSC AE 2018

Ans. (d) : In displacement method forces are known and displacement is calculated.
There are three types of displacement methods:-
(1) Slope deflection method
(2) Moment distribution method
(3) Stiffness matrix method.

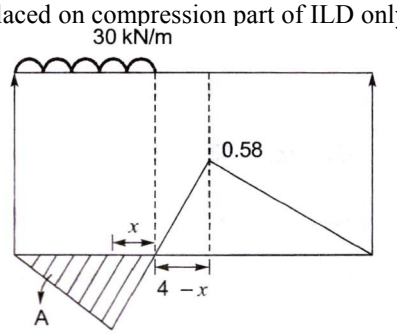
391. The Influence Line Diagram (ILD) for a force in a truss member is shown in the figure. If a Uniformly Distributed Load (UDL) of intensity of 30 kN/m longer than the span traverses, then the maximum compression in the member is :



(a) 92.8 kN (b) 23.186 kN
(c) 17.4 kN (d) 69.66 kN

Karnataka PSC AE 2017 Paper - II

Ans. (b) : Maximum compression will occur when the UDL is placed on compression part of ILD only.



$$\frac{x}{4-x}=\frac{0.29}{0.58}$$

$$x=\frac{4-x}{2}$$

$$x=\frac{4}{3}\text{m}$$

Maximum compression in the member

$$=\frac{1}{2}\times 0.29\times\left(4+\frac{4}{3}30\text{kN}\right)$$

$$= 23.186 \text{ kN}$$

392. What is/are the use(s) of influence lines?

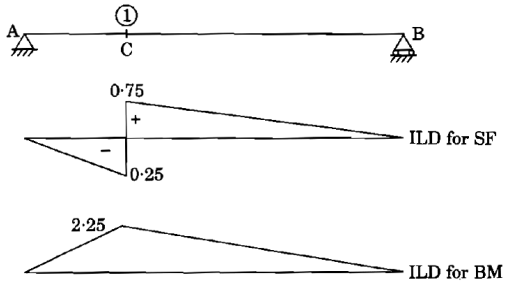
(i) To study the effect of moving loads on the structure
(ii) To calculate the value of stress function with the critical load condition
(iii) To find the position of live load for a maximum value of particular stress function

(a) Only (i) (b) (i) and (ii)
(c) (ii) and (iii) (d) (i), (ii) and (iii)

GPSC AE (CLASS 1 & 2) 2019
(UKPSC AE 2012 Paper–I)

Ans : (d) : • Influence lines are important in the design of structures that resist large line loads.
• If a structure is subjected to a line or moving load the variation in shear and moment is best described using influence lines.
• Constructing an influence line is completely different from constructing a shear or moment diagram.

393. The given figure shows ILD for SF and BM at section 1
AC = 3 m, BC = 9 m



The value of SF and BM at 1 due to concentrated load of 20 kN at mid span will be

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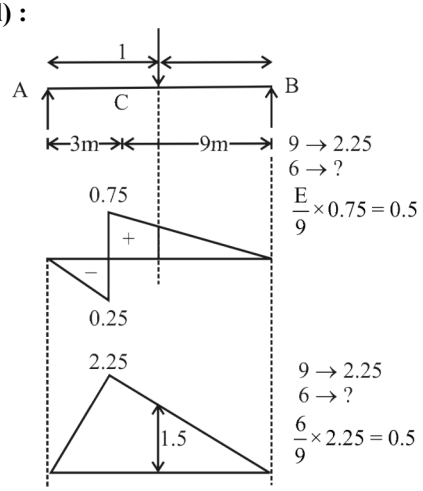
(a) 0.75 kN and 2.25 kN-m
(b) 5 kN and 5 kN-m
(c) 7.5 kN and 10 kN-m
(d) 10 kN and 30 kN-m

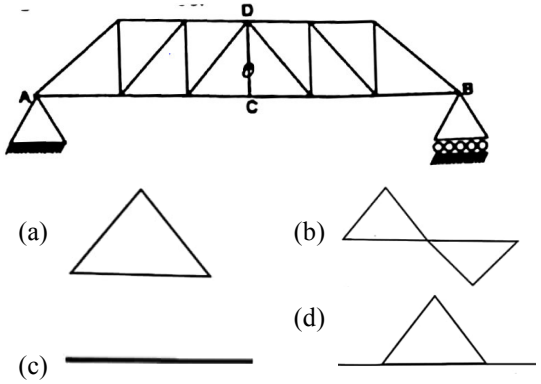
Maharashtra PSC Mains 2018 Paper-1

Ans. (d) :



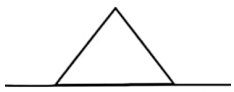
shear force = 20 × 0.5 = 10 kN
Bending moment = 20 × 1.5 = 30 kN-m

394. Influence line diagram for member DC in given truss will be.



Vizag steel MT 2017

Ans. (d) : Influence line diagram



The ILD for top chord members may also be drawn in similar lines as that of bottom chord members.

395. The influence line for structural function is used for obtaining the maximum value due to :
(a) Single point load only
(b) Uniformly distributed load only
(c) Several point loads
(d) All of the above

OPSC Civil Services Pre 2011

Ans. (d) The influence line diagram is drawn for maximum value due to structural element is –
- Single point load
- Uniformly distributed load
- Several points load

396. SFD and BMD represents effect of fixed load at all points. Influence line represents effect of moving load at
(a) center of a beam
(b) along the moving load
(c) critical point under load
(d) specified point only

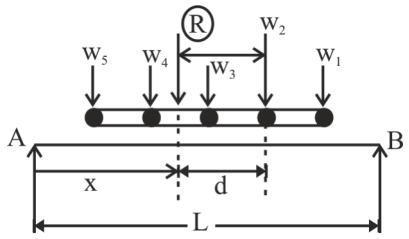
Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (d) : Influence line represents effect of moving load at any given point on the beam.

397. The maximum bending moment under a particular load moving among the several moving loads on a simply supported beam occurs when that point load is placed
(a) at mid-span
(b) at one third-point
(c) at quarter point
(d) at a point when CG of all loads and particular load are equidistant from mid-span

TRB Poly. Tech. Lect. 2012

Ans. (d) : Let a train of concentrated load w_1, w_2, w_3 ..... move on S.S Beam. AB from left to right an shown in figure. Now, let the condition the condition for moment to be maximum under wheel load w_2 is required



Let R be the resultant of all loads and its distance from w_2 be (d) and support (x)

Now, $R_A = \frac{R(L-x)}{L}$

therefore, moment under load w_2,

$$M = R_A (x + d) - R \times d$$

$$= R\left[\frac{L-x}{L}\right](x+d) - R\times d$$

$$= \frac{R}{L}(Lx - x^2 + Ld - xd] - R\times d$$

For the moment to be maximum

$$\frac{dM}{dx} = 0 = \frac{R}{L}[L - 2x - d]$$

or $x = \frac{L}{2} - \frac{d}{2}$

distance of w_2 from A

$$= x + d = \frac{L}{2} - \frac{d}{2} + d$$

$$= \frac{L}{2} + \frac{d}{2}$$

thus, we can conclude, for moment to be maximum under any particular load, the load and the resultant should be equidistant from the mid span.

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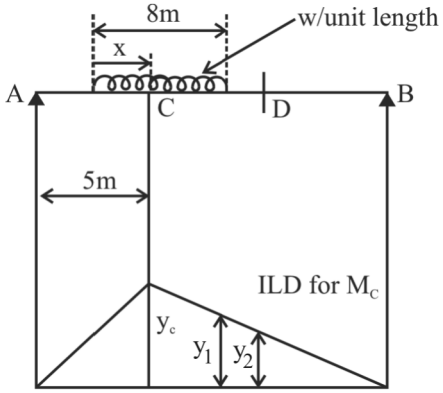
398. A uniformly distributed load of length 8 m passes over a simply supported beam of 20 m span. If there is a maximum bending moment at the left quarter point then the distance between the CG of load and centre of span will be

(a) 0 (b) 2 m
(c) 3 m (d) 4 m

TRB Poly. Tech. Lect. 2012

Ans. (c) :



Position of load for maximum moment

D is mid point - maximum B.M. will developed at 'C' when load is partly to left of section and partly to right of section. The position of section is shown in figure. By figure

$$M_c = w.x\left(\frac{y_1 + y_c}{2}\right) + w(8-x)\left(\frac{y_c + y_2}{2}\right)$$

For M_c to be maximum

$$\frac{dM_c}{dx} = 0 = \frac{w(y_1 + y_c)}{2} - \frac{w(y_c + y_2)}{2}$$

This moment at (c) will be maximum when co-ordinate at ILD for M_c at head and tail of UDL are equal.

Now, $y_1 = y_2$

$$\left(\frac{5-x}{5}\right)y_c = \frac{(20-5)-(8-x)}{(20-5)}y_c$$

$$\left(\frac{5-x}{5}\right) = \frac{7+x}{15}$$

$$(5-x)3 = (7+x)$$

$15 - 3x = 7 + x$

$8 = 4x$

$\boxed{x = 2}$

CG of load will be at 2m from (c) and distance b/w mid span and (c) = 5m so distance b/w point at CG of total load and mid span = 5 – 2 = 3m

Note:-B.M. at section is maximum when load is so placed that the section divides the load in the same ratio as it divide the span.

399. Influence lines can be drawn for

(a) statically determinate beam only
(b) statically indeterminate beam only
(c) statically determinate truss only
(d) any structure

UKPSC AE (Urban) 2011

Ans. (d) : Influence lines can be generated by independently applying a unit load at several points on a structure and determining the value of the function due to this load i.e. shear, axial and moment at desired location.

400. An influence line diagram shows variation of

(a) shear force only
(b) bending moment only
(c) shear force and bending moment
(d) the function for which it is drawn

UKPSC AE (Urban) 2011

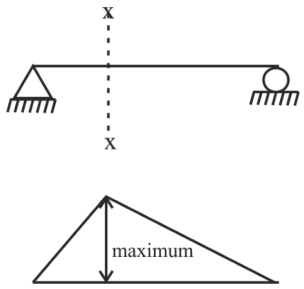
Ans. (d) : An influence line for a given function, such as reaction, axial force, shear force is a graph that shows the variation of that function for which it is drawn.

401. The influence line diagram for BM at a section of a simply supported beam will be a

(a) Triangle with maximum ordinate at the section and zero at the two supports
(b) Triangle with maximum ordinate at the support nearer to the section and zero at the other supports
(c) Rectangle with unit ordinate
(d) Triangle with maximum ordinate of unity at the section and zero at both the supports

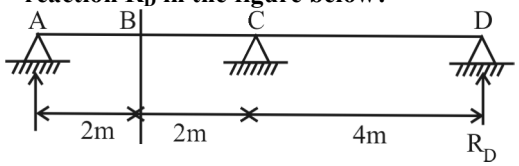
Punjab PSC SDE 2016

Ans. (a) :



ILD for BM is triangular with maximum ordinate at the section and zero at two supports.

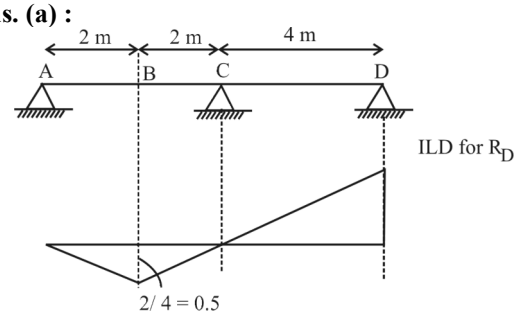
402. What is the ordinate of influence line at B for reaction R_D in the figure below?



(a) 0.5 (b) 0.4
(c) 0.2 (d) zero

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (a) :



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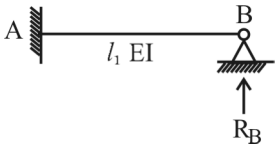
403. What is the area of influence line diagram for the reaction at the hinged end of a uniform propped cantilever beam of span L?

(a) L/ 8 (b) L/ 2

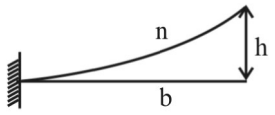
(c) L/ 4 (d) 8 L/ 8

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (c) :



Area of cubic parabola $\frac{bh}{n+1} = \frac{L}{4}$



404. The absolute maximum bending moment in a simply supported beam of span 20 m due to a moving udl of 4 t/m spanning over 5 m is

(a) 87.5 t-m at the support

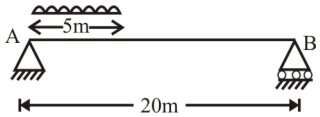
(b) 87.5 t-m near the mid point

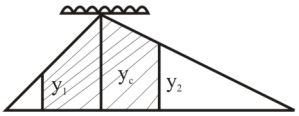
(c) 3.5 t-m at the mid point

(d) 87.5 t-m at the mid point

APPSC AEE 2012

Ans. (d) :





$$y_c = \frac{Z(L-Z)}{L} = \frac{10(20-10)}{20}$$

$y_c = 5$

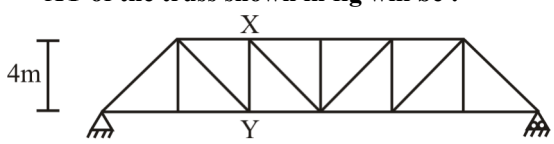
$$\frac{x}{5-x} = \frac{10}{10} \qquad y_1 = \frac{10-2.5}{10} \times 5$$

$x = 2.5$ m $\qquad y_1 = 3.75\text{m} = y_2$

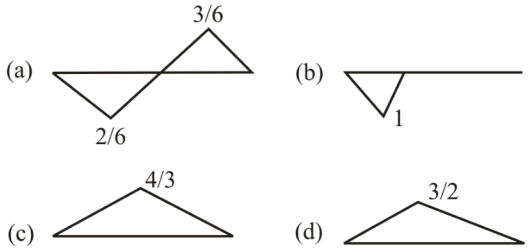
$$\text{Maximum moment} = 4 \times \left[\frac{3.75+5}{2} \times 2.5\right] \times 2$$

= 87.5 t–m at mid point

405. The influence line diagram for force in member XY of the truss shown in fig will be :

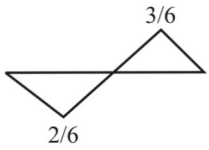


Panels at 4m each = 24m



TRB Poly. Lect. Tech. 2017

Ans. (a) :



406. Influence line for any stress function are used for obtaining maximum value because of :

(a) Uniform live load only

(b) Multiple point loads

(c) Single point load, uniform live load and multiple point loads

(d) Unit point load only

UPPCL AE 2015

Ans. (c) : An influence line graphs the variation of a function at a specific point on a frame or truss caused by unit load placed at any point along structure.

407. Muller-Breslau principle is applicable to get influence line for which of the following?

(a) Reaction at ends of a simple beam

(b) Bending moment at a section

(c) Shear force at a section

(d) Forces and moments at any section

GPSC AE June 2019

Ans. (d) : Muller-Breslau Principle is a method to determine influence lines. The principle states that influence lines of an action (Force or moment) assumes the sealed form of deflection displacement.

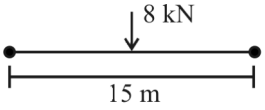
408. A single rolling load of 8 kN rolls along a girder of 15m span. The absolute maximum bending moment will be

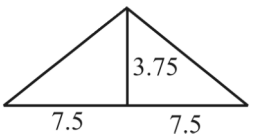
(a) 8 kN-m (b) 15 kN-m

(c) 30 kN-m (d) 60 kN-m

GPSC AE June 2019

Ans. (c) :





Absolute bending moment –

M = 8 × 3.75

M = 30 kN-m

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409. The maximum bending moment at a section due to moving loads in a simple beam occurs when :

(a) load covers the entire span

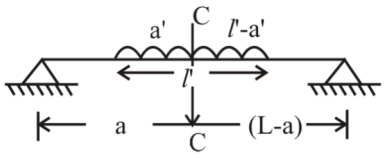
(b) load extends to one side of the section

(c) the section divides the load in the same ratio as it divides the span

(d) the load is placed on both sides of the section in 50:50 ratio

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (c) Maximum B.M. at section C-C will occur when average loading just to the left of C is equals to average loading just to right of C, it means section C-C divides load in same ratio as it divide to the span.



410. Absolute maximum bending moment under any given wheel in a beam due to several point loads occurs when:

(a) this wheel is at mid-span

(b) this wheel and centre of gravity of the total load system are equidistant from the ends of the beam

(c) this wheel and the maximum load wheels are equidistant from the ends of the beam

(d) None of the above

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (d) Absolute maximum bending moment occurs when position of load is such that centre of span is mid way between C-G of load group and load under consideration.

411. For simply supported beam of span 10 m, influence line diagram is drawn for bending moment at a section 4 m from left had support. The maximum bending moment at the section due to moving point load of 160 kN is equal to .....

(a) 640 kN-m (b) 960 kN-m

(c) 384 kN-m (d) 400 kN-m

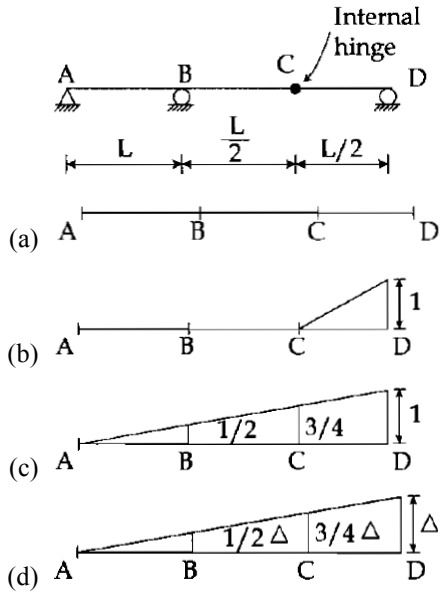
Maharashtra PSC (Mains) 2019 Paper–I

Ans. (c)

Max. BM at C = $P\left(\frac{ab}{l}\right)$

$= 160 \times \left(\frac{4 \times 6}{10}\right) = 384$ kNm

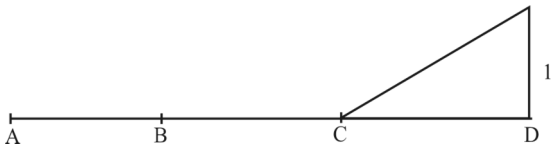
412. For the continuous beam shown in figure, the ILD for reaction at D is .....



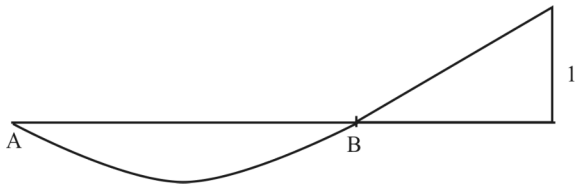
Maharashtra PSC (Mains) 2019 Paper–I

Ans. (b) : Give unit deflection at D.

Since beam is determinate So, ILD will be straight line.



If there is no hinge at C then beam will be indeterminate and ILD will be curved.



413. Identify the incorrect statement with respect to Muller-Breslau principle by which influence lines for structures can be found.

(a) Law of superposition holds good for Muller-Breslau principle

(b) Muller-Breslau principle is applicable only when the material is within the elastic limit

(c) Muller-Breslau principle is applicable only for determinate structures

(d) Muller-Breslau principle gives an excellent experimental approach to determine the influence lines of structures

UPPCL AE 2019

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Ans. (c) Muller - Breslau has application in the Analysis of both statically determinate and indeterminate structures, especially for the purpose of drawing influence line.

As per Muller Breslau principle, influence line for any stress function line for any stress function at any section may be obtained by remaining the restraint in that direction and applying unit displacement in that direction. The deflected shape gives the desired ILD.

414. Identify the incorrect statement about Muller Breslau principle.

(a) It gives excellent experimental approach to determine the influence lines for indeterminate structures

(b) It is applicable only for indeterminate structures

(c) It gives analytical method to find Influence line diagram

(d) It is applicable when the material is within the elastic limit and obeys Hooke's law

CIL MT 2020

Ans. (b) : Muller - Breslau has application in the analysis of both statically determinate and indeterminate structures, especially for the purpose of drawing influence lines.

415. A method that is used to draw influence lines for various quantities pertaining to a structure is :

(a) Muller-Breslau principle

(b) moment distribution theorem

(c) unit load theorem

(d) Castigliano's theorem

DMRC AM 2020

Ans. (a) : Muller-Breslau principle is used to draw influence line for various quantities pertaining to a structure.

416. What is the stiffness for a far end fixed beam?

(a) $K = \frac{3EI}{L}$ (b) $K = \frac{4EI}{L}$

(c) $K = \frac{12EI}{L}$ (d) $K = \frac{16EI}{L}$

(e) $K = \frac{20EI}{L}$

CGPSC AE 2017

Ans. (b) Stiffness for a far end fixed beam :

Stiffness for a far end hinged beam

417. A single load of 100 kN rolls along a girder of 20 m span. What will be the maximum positive shear force and bending moment?

(a) 100 kN, 1,000 kN-m

(b) 200 kN, 800 kN-m

(c) 100 kN, 500 kN-m

(d) 400 kN, 1,200 kN-m

(e) 400 kN, 1,000 kN-m

CGPSC AE 2017

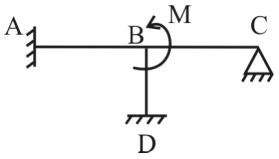
Ans. (c)

SF_A = 1 kN (due to unit load)

$SF_A = 1 \times 100$ kN = 100 kN (for 100 kN load)

$\frac{\text{ILD for BM}}{\text{due to 100 kN}} = 5 \times 100 = 500$ kN-m

418. All members of the frame shown below have the same flexural rigidity EI and length L. If a moment M is applied at joint B, the rotation of the joint is :

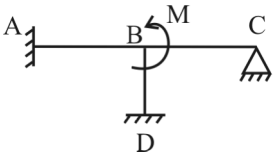


(a) $\frac{ML}{12EI}$ (b) $\frac{ML}{11EI}$

(c) $\frac{ML}{8EI}$ (d) $\frac{ML}{7EI}$

HPPSC Poly. Lect. 2016

Ans : (b)



$$M = \frac{4EI\theta}{L} + \frac{4EI\theta}{L} + \frac{3EI\theta}{L}$$

$$M = \frac{11EI\theta}{L}$$

then, $\boxed{\theta = \frac{ML}{11EI}}$

419. Maxwell's reciprocal theorem in structural analysis-

(a) Is true for any structure obeying Hooke's law

(b) can be applied to the rotations caused by flexure, shear or torsion

(c) is useful in analyzing indeterminate structures

(d) All of these

RPSC AE 2018

Ans. (d) Maxwell's reciprocal theorem in structural analysis is true for all the above options.

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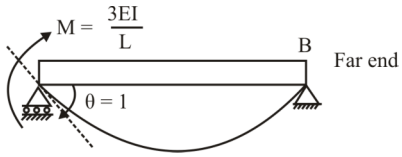
6. Method of Structural Analysis

420. The moment required to rotate the near end of a prismatic beam of length 'L' through a unit angle without translation, the far end being simply supported, is given by (Where El is the flexural rigidity of beam)

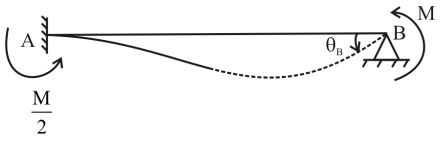
(a) 3EI/L (b) 2EI/L
(c) 4EI/L (d) EI/L

OPSC AEE (Paper–I) 2019
Maharashtra PSC (Mains) (Paper–I) 2019, 2015
UJVNL AE 2016
BPSC AE (Paper–V) 2012

Ans. (a) : The moment required to rotate the near end of prismatic beam through a unit angle, without translation, the far end being simple is $\frac{3EI}{L}$



- When the far end being fixed, then moment = $\frac{4EI}{L}$



421. The carry over factor in a prismatic member whose far end is hinged is

(a) zero (b) 3/4
(c) 1/2 (d) 1

OPSC AEE 2019 Paper–I
UJVNL AE 2016

Ans. (a) : Carry over factor is defined as the ratio of the moment at the far end to the moment at the rotating near end.

- If far end is hinged –

$$\left[\text{COF} = \frac{0}{M} = 0\right]$$

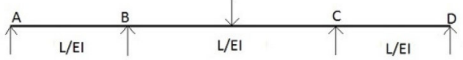
- If far end is fixed–

$$\left[\text{COF} = \frac{M/2}{M} = \frac{1}{2}\right]$$

- If far end is guided roller–

$$\left[\text{COF} = \frac{-M}{M} = -1\right]$$

422. For a symmetric continuous beam shown below, which is the correct distribution at B –



(a) δ_{BA}: δ_{BC} = 3:8 (b) δ_{BA}: δ_{BC} = 3:4
(c) δ_{BA}: δ_{BC} = 1:2 (d) δ_{BA}: δ_{BC} = 3:2

UPPCL AE 2014

Ans. (b) : $\delta_{BA} = \frac{3EI}{l}$

$\delta_{BC} = \frac{4EI}{l}$

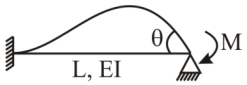
$\frac{\delta_{BA}}{\delta_{BC}} = \frac{\frac{3EI}{l}}{\frac{4EI}{l}} = \frac{3}{4}$

423. The absolute stiffness of prismatic member with one end hinged is

(a) $\frac{2EI}{l}$ (b) $\frac{4EI}{l}$
(c) $\frac{3EI}{l}$ (d) $\frac{EI}{l}$

APPSC AE (Mains) 2016
OPSC AEE (PAPER–I) 2015

Ans. (b) :



We know,

$$\text{Stiffness} = \frac{M}{\theta}$$

$$\therefore \quad \theta = \frac{ML}{4EI}$$

$$K = \frac{M}{ML/4EI} = \frac{4EI}{L}$$

424. The displacement method is also referred to as _____?

(a) Minimum strain energy method
(b) Maxwell Mohr method
(c) Consistent deformation method
(d) Slope-deflection method

GPSC AE June 2019
UPPCL AE 2014

Ans. (d) : In the slope deflection method displacement (slope and deflection) are treated as unknown. Thus the slope deflection method is a displacement method. In this it is assumed that deformation are caused due to bending moment only.

425. Slope deflection method is suitable for analysis of

(a) Continuous beam
(b) Simply supported beam
(c) Fixed beam
(d) Cantilever beam

GPSC AE January 2018

Ans. (a) : Slope deflection method is the classical method of analysis of indeterminate beams and frames. For determinate structure, slope deflection method is suitable for continuous beam.

426. In an elastic system, the external work done by a force p_i during the deformations caused by another p_j is equal to the external work done by the force p_j during the deformations caused by p_i is stated by

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(a) Maxwell (b) Betti
(c) Maxwell-Betti's (d) Euler

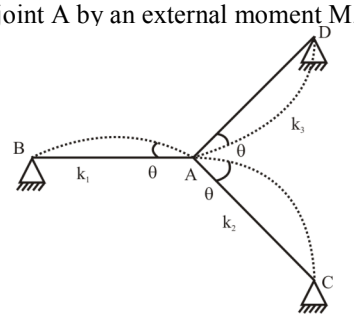
TRB Poly. Tech. Lect. 2012

Ans. (c) : Maxwell-Bettis– State that, in an elastic system, the external work done by force Pi during the deformation caused by another force Pj is equal to the external work done by the force Pj during the deformations caused by Pi.

427. At a joint where several members meet, any applied moment will be distributed among the various members
(a) Equally
(b) In proportion to their stiffness
(c) Equally among such members whose far ends are fixed
(d) Equally among such members whose far ends are hinged

Punjab PSC SDE 2016

Ans. (b) : A moment which tends to rotate without translation a joint to which several members are connected to be divided amongst the connected member in proportion to their stiffness.
If BA, CA, DA beams connected to A and a rotation θ is given to joint A by an external moment M.



- Rotation of each member at A to θ.
- M_{AB}, M_{AC}, M_{AD} will be in the ratio k_1, k_2, k_3

$$M_{AB} = \frac{k_1}{\sum k} m \quad M_{AC} = \frac{k_2}{\sum k} m \quad M_{AD} = \frac{k_3}{\sum k} m$$

The moments are distributed to different members according to their relative stiffness at a joint in moment distribution method.

428. Which one of the following statements is true with regard to the flexibility method of analysis?
(a) The method is used to analyze determinate structures
(b) The method is used only for manual analysis of indeterminate of flexible structure
(c) The method is used only for analysis of flexible structure
(d) The method is used for analysis of indeterminate structures with lesser degree of static indeterminacy

Punjab PSC SDE 2016
UTTRAKHAND AE 2013

Ans. (d) : Flexibility method is used for analysis of redundant structure with lesser degree of static indeterminacy. .
Note-Stiffness methods is used where kinetic indeterminacy is less than static indeterminacy.

429. The flexibility of an element can be defined as
(a) Flexural moment per unit rotation
(b) Rotation per unit moment
(c) Flexibility for unit translation
(d) None of these

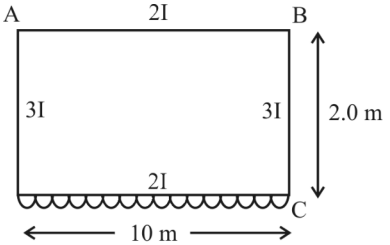
Punjab PSC SDE 2016

Ans. (b) : Flexibility of a element is defined as displacement caused by a unit force or rotation caused by unit moment.

$$\boxed{f = \frac{\delta}{R}} \text{ or } \boxed{f = \frac{\theta}{M}}$$

where = flexibility
δ = displacement
R = force
θ = rotation, M = moment

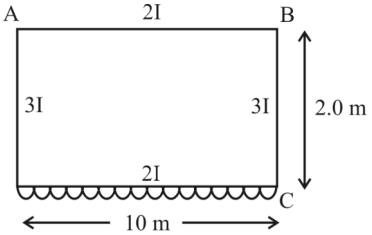
430. For the box section, the distribution factor for members AB and BC are.



(a) 1/2, 1/2 (b) 2/17, 15/17
(c) 15/17, 2/17 (d) 1/3, 2/3

UJVNL AE 2016

Ans. (b) :



Relative stiffness –

$$K_{AB} = \frac{2EI}{10}; \; K_{BC} = \frac{3EI}{2}$$

Distribution factors –

$$D_{AB} = \frac{K_{AB}}{K_{AB} + K_{AD}} = \frac{2EI/10}{\frac{2EI}{10} + \frac{3EI}{2}} = \frac{2}{17}$$

$$D_{BC} = \frac{K_{BC}}{K_{AB} + K_{BC}} = \frac{\frac{3EI}{2}}{\frac{2EI}{10} + \frac{3EI}{2}} = \frac{15}{17}$$

431. A moment, which tends to rotate a joint without translation, will be divided amongst the connecting members at the joint in the proportion to their
(a) flexural rigidity (b) stiffness
(c) relative stiffness (d) moment of inertia

BPSC AE Aug. 2019 Paper-V

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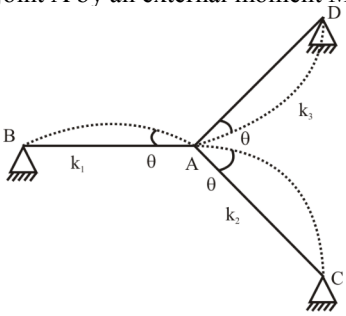
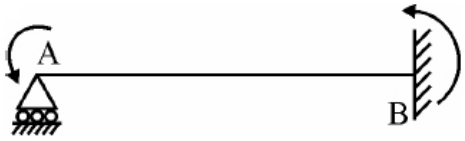
Ans. (c) : A moment which tends to rotate without translation a joint to which several members are connected to be divided amongst the connected member in proportion to their stiffness.
If BA, CA, DA beams connected to A and a rotation θ is given to joint A by an external moment M.



- Rotation of each member at A to θ.
- M_{AB}, M_{AC}, M_{AD} will be in the ratio k_1, k_2, k_3

$$M_{AB}=\frac{k_1}{\sum k}m \quad M_{AC}=\frac{k_2}{\sum k}m \quad M_{AD}=\frac{k_3}{\sum k}m$$

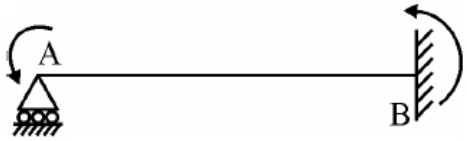
432. When a unit rotation is given at end A of beam shown in figure, the moment produced at end B will be



(a) 2EI/L (b) 4EI/L
(c) –4EI/L (d) 6EI/L

UKPSC AE (Urban) 2011

Ans. (a) :



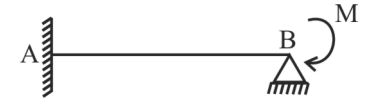
A unit rotation is given at end A of beam then the moment produced at end B will be that 2EI/L

433. A beam is hinged at one end A and fixed at other end B. If a moment M is applied at A, what is the moment developed at the end B.

(a) –M (b) M
(c) $\frac{M}{2}$ (d) $-\frac{M}{2}$

JPSC AE 2009
ESE 2004

Ans. (c) :



Carryover factor $=\frac{1}{2}$

$$M_A=\frac{M}{2}$$

434. For stable structures, one of the important properties of flexibility and stiffness matrices is that the elements on the main diagonal

1. of a stiffness matrix must be positive
2. of a stiffness matrix must be negative
3. of a flexibility matrix must be positive
4. of a flexibility matrix must be negative

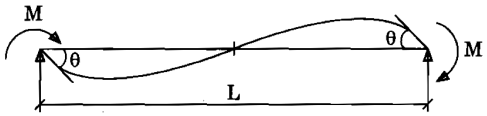
The correct answer is

(a) 1 and 3 (b) 2 and 3
(c) 1 and 4 (d) 2 and 4

GOA PSC Assistant Prof. 2020

Ans. (a) : Elements of main diagonal of stiffness matrix and flexibility are always positive because at a point, displacement will always occur in the direction of force applied at that point.

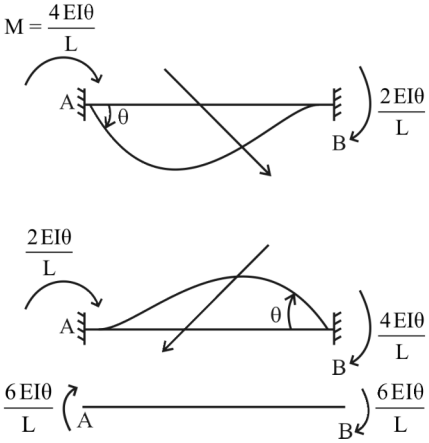
435. A beam EI-constant of span L is subjected to clockwise moments M at both the ends A and B. The rotation of end A works out to be



(a) $\frac{ML}{2EI}$ (b) $\frac{ML}{3EI}$
(c) $\frac{ML}{4EI}$ (d) $\frac{ML}{6EI}$

Maharashtra PSC Mains 2018 Paper-1

Ans. (d) : $M=\frac{4EI\theta}{L}$

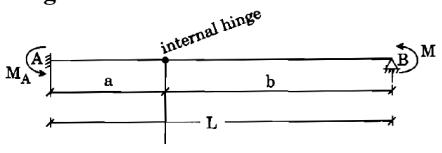


Rotational at A due to moment at $A=\frac{ML}{3EI}$

Rotational at A due to moment at $B=\frac{-ML}{6EI}$

$$M=\frac{6EI\theta}{L}\Rightarrow\theta=\frac{ML}{6EI}$$

436. Carry-over factor C_{BA} for the beam shown in the figure below is



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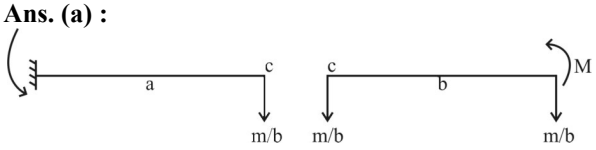
(a) a/b (b) 3/4
(c) a/L (d) 1/2

Maharashtra PSC Mains 2018 Paper-1

Ans. (a) :



$$\text{Carry over factor (COF)} = \frac{\text{Induced moment}}{\text{Applied moment}}$$

$$\text{COF} = \frac{\frac{M \times a}{b}}{M} = \frac{a}{b}$$

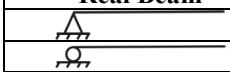
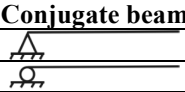
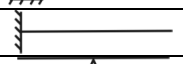
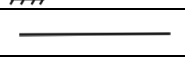
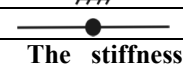
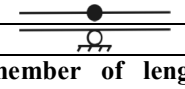
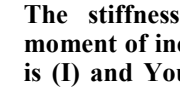
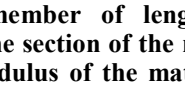
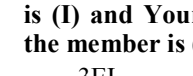
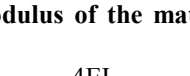
437. The fixed support condition in a real beam becomes in the conjugate beam a condition as

(a) fixed (b) roller
(c) hinged (d) free

GPSC AE June 2019
BPSC AE Aug. 2019 Paper-V
BPSC AE 2019 Paper (V) Section-I
[UPPSC State Eng. A.E. 2008]

Ans. (d) When the real beam is fixed supported both the slope and displacement are zero. Here the conjugate beam has a free end, since at this end there is zero shear and zero moments.
Supports are converted as-

Real beam	Conjugate beam
Hinged or pinned	Hinged or pinned
Fixed	Free
Free	Fixed
Intermediate rollers	Hinged or pinned
Intermediate pinned	Roller
Roller	Roller

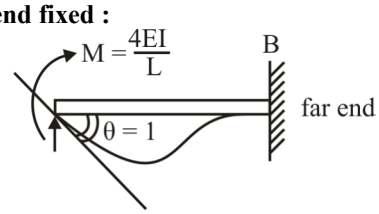
Real Beam	Conjugate beam
	
	
	
	
	

438. The stiffness of a member of length (L), moment of inertia of the section of the member is (I) and Young's Modulus of the material of the member is (E) is :

(a) $\frac{3EI}{L}$ (b) $\frac{4EI}{L}$
(c) $\frac{2EI}{L}$ (d) $\frac{4EI}{L^2}$

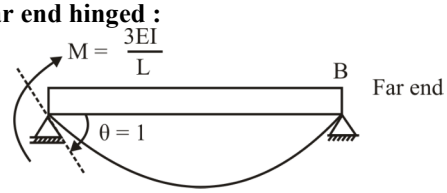
OPSC AEE 2019 Paper–I

Ans. (a) :
(A) Far end fixed :



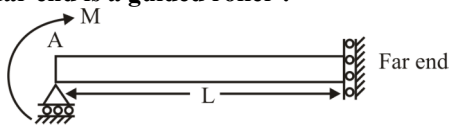
$$K = \frac{M}{\theta} = \frac{4EI}{L}$$

(b) Far end hinged :



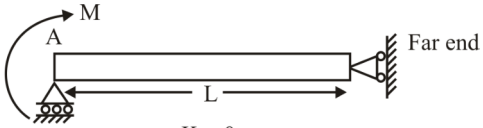
$$K = \frac{M}{\theta} = \frac{3EI}{L}$$

(c) If far end is a guided roller :

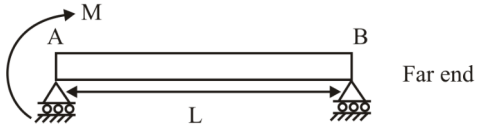


$$K = \frac{EI}{L}$$

(d) If far end is a roller :



$$K = 0$$



$$K = \frac{3EI}{L}$$

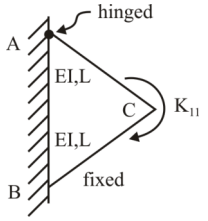
439. Assertion (A): The slope deflection method is a stiffness method in which the joint displacements are found by applying the equilibrium conditions at each joint.
Reason (R): The displacement at a joint of a member are independent of the displacements of the member at the far end of the joint.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

GPSC AE (CLASS 1 & 2) 2019
ESE 2004

Ans : (b) : In the stiffness method ($D_k < D_s$) where joints are taken as unknown.
The displacement at member ends are independent but moment or reaction is dependent on displacement components at both ends.

440. The rotational stiffness coefficient indicated as K_{11} for the frame with details as shown as



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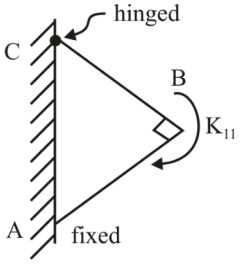
(a) $\frac{9EI}{L}$ (b) $\frac{8EI}{L}$

(c) $\frac{7EI}{L}$ (d) $\frac{6EI}{L}$

ESE 2017

Ans. (c) : For the rotational stiffness coefficient



- $K_{BA} = \frac{4EI}{L}$
- $K_{BC} = \frac{3EI}{L}$

$K_{11} = K_{BA} + K_{BC}$

$K_{11} = \frac{4EI}{L} + \frac{3EI}{L}$

$\boxed{K_{11} = \frac{7EI}{L}}$

441. The size of the basic stiffness matrix for a plane truss member, in member coordinate system is

(a) 2×2 (b) 3×3

(c) 4×4 (d) 6×6

(JPSC Combined AE Paper V 2013)

Ans : (a) The size of the basic stiffness matrix for a plane truss member in member co-ordinate system is 2×2.

442. Which one of the following is a force method of structural analysis?

(a) Slope deflection method

(b) Kani's method

(c) Moment distribution method

(d) Strain energy method

GPSC AE (CLASS 1 & 2) 2019

Ans : (d) : Force method of analysis are :

- Method of consistent deformation.
- Strain Energy method
- Three moment theorem : clapeyron.
- Column Analogy Method : Hardy cross.
- Elastic Centre Method.
- Maxwell – Mohr equations.
- Castigliano's theorem of minimum strain energy.
- Flexibility Matrix method.

Slope deflection method, Kani's method, Moment distribution method are all displacement method.

443. Flexibility method of structural analysis starts with :

(a) Compatible deformation

(b) Equilibrium condition

(c) Force deformation relation

(d) Equilibrium state of internal forces

JPSC AE (Pre) 2019

Ans. (d) : Structures which are highly kinematically determinate represent convenient starting point for the analysis of structures by the stiffness method. In the flexibility method the primary structure is the released from, where unknown actions, either internal or external, have been released.

444. Equilibrium method is also known as :

(a) compatibility method

(b) flexibility method

(c) force method

(d) displacement method

UKPSC AE (Paper-I) 2007

Ans : (d) Equilibrium method is known as stiffness coefficient method or equilibrium method. In this method displacement are chosen as unknown and equilibrium equations used for solutions.

445. The Castigliano's theorem is essentially a

(a) Force method

(b) Displacement method

(c) Strain method

(d) Equilibrium method

UPRVUNL AE 2016

Ans.: (a) The castigliano's Method, named after Carlo Alberto Castigliano, is a method for determining the displacement of a linear-elastic system based on the partial derivatives of the energy.

446. The moment required to rotate end of a prismatic beam through the angle without translation, when the end is fixed, is

(a) EI/L (b) 2 EI/L

(c) 3 EI/L (d) 4 EI/L

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (d) : Stiffness of beam AB when far end B is fixed

$R_{AB} = \frac{4EI}{l}$

Stiffness = $M/\theta = \frac{4EI}{l}$

for unit rotation, $\theta = 1$, ie moment $M = \frac{4EI}{l}$

447. In the displacement method of structural analysis, the basic unknowns are

(a) displacements

(b) force

(c) displacements and force

(d) None of the above

OPSC AEE 2015 PAPER–I

BPSC AE 2019 Paper (V) Section-I

APPSC AEE 2012

Ans. (a) : In the displacement method of structural analysis, the basic unknowns are displacements

448. If one end of the prismatic beam AB with fixed ends is given a transverse displacement 'Δ' without any rotation, then the transverse reactions at A or B due to displacement is

(a) $\frac{6EI\Delta}{l^2}$ (b) $\frac{6EI\Delta}{l^3}$

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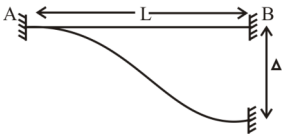
(c) $\frac{12EI\Delta}{l^2}$ (d) $\frac{12EI\Delta}{l^3}$

APPSC AEE 2012

Ans. (d) :



At point A $M_{AB} = M_{BA} = \frac{6EI\Delta}{L^2}$

$-M_{AB} + R_B \times L = M_{BA}$

$R_B = \frac{2M_{AB}}{L}$

$R_B = \frac{12EI\Delta}{L^3}$

449. Which method of structural analysis is a Force Method of analysis?

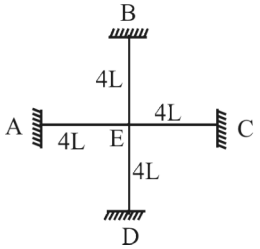
(a) Moment distribution method
(b) Slope deflection method
(c) None of these
(d) Both of these

OPSC AEE 2019 Paper–I

Ans. (c) : It is also known as flexibility coefficient method or compability method.
Force method of analysis are :

- Method of consistent deformation.
- Three moment theorem : clapeyron.
- Column Analogy Method : Hardy cross.
- Elastic Centre Method.
- Maxwell – Mohr equations.
- Castigliano's theorem of minimum strain energy.
- Flexibility Matrix method.

450. Distribution factor BE in given figure is.



EI is constant

(a) 1/4 (b) 1.24
(c) 3/14 (d) 0.4

UJVNL AE 2016

Ans. (a) : Stiffness of BE, AE, CE and DE = $\frac{4EI}{l}$

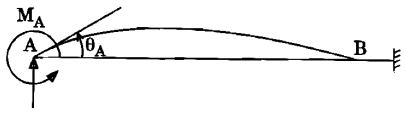
$= \frac{4EI}{4L} = \frac{EI}{L}$

Distribution factor of BE $= \frac{\text{Stiffness of BE}}{\text{Total Stiffness}}$

$= \frac{EI/L}{\frac{EI}{L}+\frac{EI}{L}+\frac{EI}{L}+\frac{EI}{L}}$

$= \frac{EI/L}{4EI/L} = \frac{1}{4}$

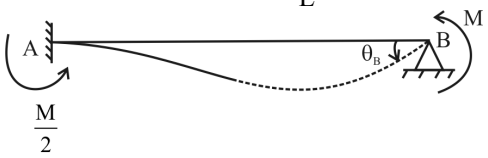
451. For the given figure, the moment at A, whose far end is fixed, M_A is



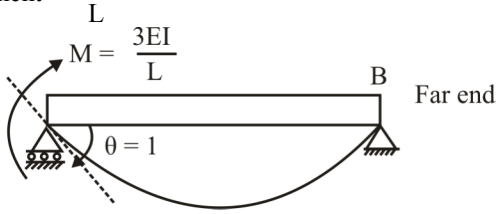
(a) $\frac{3EI}{l}.\theta_A$ (b) $\frac{4EI}{l}.\theta_A$
(c) $\frac{2EI}{l}.\theta_A$ (d) $\frac{6EI}{l}.\theta_A$

Maharashtra PSC Mains 2018 Paper-1
RPSC VPITI 2012
UPPSC State Eng. A.E. 2003

Ans. (b) : The moment required to rotate the near end of prismatic beam through a unit angle, without translation, the far end is fixed $\frac{4EI}{L}$



- When the far end being simply supported, then moment = $\frac{3EI}{L}$



452. The distribution factor is:

(a) Ratio of stiffness of member and member
(b) Ratio of stiffness of near joint and far joint
(c) Ratio of stiffness of member and joint (sum of member stiffness)
(d) Ratio of stiffness of joint and member

Maharashtra PSC Mains 2018 Paper-1

Ans. (c) :
Distribution factor (Df) =

$$\frac{\text{Stiffness of a member}}{\text{Total stiffness of all member at joint}}$$

$Df = \frac{k}{\Sigma k}$

453. Which of the following does not apply to statically determinate structures?

(a) Stress are caused due to temperature variations
(b) no stresses are caused due to lack of fit
(c) bending moment at a section is independent of cross-sectional area of the components
(d) Force in a component is independent of the mat
(e) Conditions of the equilibrium are sufficient to fully analyze the structure

CGPSC AE 2014 Shift–I

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Ans. (b) : Due to lack of fit (shorter or longer member)
Member undergo either compressive/tensile stresses
If member is short → compressive stress
If member is longer → tensile stress

454. Slope deflection method is used when :
(a) $D_K = D_S$ (b) $D_K = 0, D_S = 0$
(c) $D_K < D_S$ (d) $D_S < D_K$

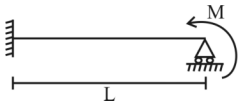
APPSC Poly. Tech. Lect. 2020

Ans. (c) : $D_K < D_S$
slope deflection method is displacement method. It is generally used
When $D_K < D_S$
Example of displacement method are
– moment distribution method
– stiffness matrix method
– Kani's method

455. The carryover factor in a prismatic member whose far end is fixed is
(a) 0 (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) 1

GPSC AE June 2019
TNPSC AE 2019
RPSC POLY. TECH. LECT. 2014

Ans. (b) The carry over factor is always (1/2) for members of constant moment of inertia (prismatic section).
If the end is hinged/pin connected the carry over factor is zero.



$$\text{Carry over factor} = \frac{\text{carry over moment at farther end}}{\text{Applied moment at near end}}$$
$$= \frac{1}{2} \text{ when far end is fixed}$$
= 0 when far end is hinged.

456. The order of the flexibility matrix for a structure is
(a) equal to the number of redundant force
(b) more than the number of redundant force
(c) less than the number of redundant force
(d) equal to the number of redundant force plus three

TNPSC AE 2019

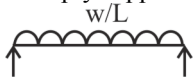
Ans. (a) The order of the flexibility matrix for a structure is equal to the number of redundant forces. Flexibility method is used for analysis of redundant structure with lesser degree of static indeterminacy.

457. The ratio of load carrying capacity of fixed beam to that of simply supported beam having same loading of uniformly distributed load throughout the span
(a) 0.33 (b) 1
(c) 1.5 (d) 3

TNPSC AE 2019

Ans. (c) Let the intensity of UDL far fixed beam and simply supported be w_1 and w_2 per unit length respectively. But maximum bending moment for fixed beam = Maximum B.M. for simply supported.

$$m_1 = \frac{w_1 l^2}{12}$$



$$m_2 = \frac{w_2 l^2}{8}$$

$$m_1 = m_2$$
$$\frac{w_1 L^2}{12} = \frac{w_2 L^2}{8}$$
$$\frac{w_1}{w_2} = \frac{12}{8} = 1.5$$

458. Statically indeterminate structures can be solved by
(a) Using equations of statics alone
(b) Equation of compatibility alone
(c) Ignoring all deformations and assuming the structure to be rigid
(d) Using the equations of statics and the necessary numbers of compatibility equations

TNPSC AE 2019

Ans. (d) For indeterminate structure analysis only static equilibrium is not enough we need compatibility condition i.e., displacement condition or stiffness condition.

459. Out of the following methods used for structural analysis
(1) Moment distribution method
(2) Slope deflection method
(3) Euler's method
(4) Method of joint
Those include in the analysis of indeterminate structure are
(a) 1 and 2 (b) 1, 2 and 3
(c) 1, 2, 3 and 4 (d) 1 and 4

TNPSC AE 2019

Ans. (a) For indeterminate structure analysis only static equilibrium is not enough we need compatibility condition i.e., displacement condition or stiffness condition
(1) MDM (2) SDM (3) Kani Method (4) Potential Method

460. For a linear elastic frame, if stiffness matrix is doubled with respect to the existing stiffness matrix, the deflection of the resulting frame will be
(a) twice the existing value
(b) half the existing value
(c) the same as existing value
(d) indeterminate value

Gujarat PSC AE 2017

Ans : (b) Stiffness (k) is defined as the force per unit deflection.

$$k = \frac{P}{\delta}$$

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$$\boxed{\delta = \frac{P}{k}}$$

If stiffness matrix is doubled

i.e. k' = 2k

then,

$$\delta' = \frac{P}{k'}$$

$$\delta' = \frac{P}{2k}$$

$$\left[\delta' = \frac{\delta}{2}\right]$$

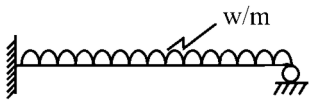
So, If the stiffness matrix is doubled with respect to the existing stiffness matrix, the deflection of the resulting frame will be half the existing value.

461. The flexibility coefficient for a propped cantilever beam (EI=constant throughout the length) loaded with a uniformly distributed load w/m is given by

(a) $L^3/3EI$ (b) $L^3/3EI$
(c) $L^3/2EI$ (d) $L^3/4EI$
(e) $L^3/6EI$

CGPSC AE 2014 Shift–I

Ans. (a)



$$f = \frac{1 \times L^3}{3EI} = \frac{L^3}{3EI}$$

462. Bending moment at any section in a conjugate beam gives in the actual beam

(a) slope (b) curvature
(c) deflection (d) bending moment

CGPSC AE 2014 Shift–I

Ans. (c) • Bending moment at any section of the conjugate beam is equal to the deflection of the elastic curve at corresponding section of the actual beam.
• Shear force at any section of the conjugate beam is equal to the slope of elastic curve at corresponding section of the actual beam.

463. In column analogy method, the area of an analogous column for a fixed beam of span length L and flexural rigidity EI is

(a) 0.25L/EI (b) 0.5 L/EI
(c) 0.75 L/EI (d) 1.25 L/EI
(e) L/EI

CGPSC AE 2014 Shift–I
MPPSC AE 2014

Ans. (e) : Direct result : In column analogy method, the area of an analogous column for a fixed beam of span length (L) & flexural rigidity (EI) is $\frac{L}{EI}$.

464. The Clapeyron's method formulates:

(a) loading conditions
(b) equilibrium conditions
(c) moment conditions
(d) compatibility conditions

APPSC Poly. Tech. Lect. 2020

Ans. (d) : Clapeyron provides a condition tool for analyzing continuous beam. The three-moment equation represent in general form, the compatibility condition that the slope of the elastic cause be continuous at an interior support of the continuous beam.

465. In the moment distribution method, the distribution factor for the end span, on the fixed end side is–

(a) 0 (b) 1/4
(c) 1/2 (d) 1

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) : In the moment distribution method, the distribution factor for the end span. On the fixed end side is zero.
Distribution factor to a cantilever is zero as its stiffness is zero.

466. Maximum number of unknown forces that can be determined in concurrent force system under equilibrium is

(a) zero (b) 2
(c) 3 (d) 6

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (b) The body is said to be in equilibrium if the resultant of all forces acting on it is zero.
Concurrent planer forces as we have two equilibrium equations, two unknown can be determined.

467. Which of the following methods is NOT a force method?

(a) Slope deflection method
(b) column analogy method
(c) Virtual work method
(d) Three moment theorem

LMRC AM 2019

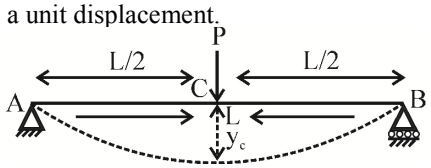
Ans. (a) In slope deflection method, displacement are unknown quantities, so it is displacement method not force method.

468. For a simply supported beam of span 'L' with point load 'P' at the centre, the elastic stiffness will be given by____.

(a) $384EI/5L^4$ (b) $48EI/L^3$
(c) $48EI/5L^3$ (d) $48EI/L^4$

LMRC AM 2019

Ans. (b) Stiffness is defined as the force required to produce a unit displacement.



As we know,

$$y_c = \frac{WL^3}{48EI}$$ From standard formula

$$\frac{W}{y_c}(\text{Elastic Stiffness}) = \frac{48EI}{L^3}$$

469. Kani's method can be considered as an extension of the :

(a) Moment-area method
(b) Slope deflection method

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(c) Moment distribution method
(d) Column-analogy method

KPSC AE 2020

Ans. (b) : Kani's method can be considered as an extension of the slope deflection method.

470. At fully plastic section, infinite rotation can occur at :
(a) Zero moment
(b) Constant elastic moment
(c) Constant plastic moment
(d) All of the above

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : At the plastic hinge an infinite large rotation can occur under a constant moment equal to the plastic moment of the section.

471. Which of the following conditions is to be satisfied both in elastic and plastic analysis ?
(a) Yield condition
(b) Equilibrium condition
(c) Plastic moment condition
(d) Mechanism condition

Gujarat PSC AE (N.W.R.) 2020
Nagaland PSC CTSE 2017 Paper–I
CIL MT 2017

Ans. (b) :
$\Rightarrow$ Conditions to be satisfied both in elastic analysis of Indeterminate structures.
(I) yield condition
(II) equilibrium condition
$\Rightarrow$ Conditions to be satisfied in plastic analysis of indeterminate structure
(I) Mechanism condition
(II) Equilibrium condition

472. The structures made of rigid curved surfaces are known as :
(a) Surface structure (b) Frame structure
(c) Shell structure (d) Space structure

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : The shell structures are rigid curved surface structures. The shell structures are built of sheets or plates.

473. Which of the following methods of structural analysis is a force method ?
(a) Slope deflection method
(b) Column analogy method
(c) Moment Distribution method
(d) None of these

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : Column analogy method of structural analysis falls in the category of force method.

474. The conjugate beam method falls in the category of :
(a) Force method
(b) Stiffness method
(c) Displacement - method
(d) None of the above

Nagaland PSC CTSE 2017 Paper–I
Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (a) : The conjugate beam method falls in the category of force method.

475. The stiffness method in structural analysis is also known as :
(a) Unit load method
(b) Consistent deformation method
(c) Force method
(d) Displacement method

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : The stiffness method (also known as the displacement method) is the primary method used in matrix analysis of structure.

476 . The units of flexural stiffness are :
(a) Radians per unit rotation
(b) Moment per unit rotation
(c) Force per unit deflection and rotation
(d) Extension per unit force

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : Stiffness – The stiffness is defined as the force required to produce a unit displacement.

$\boxed{K = \frac{P}{\delta}}$ OR $\boxed{K = \frac{M}{\theta}}$

P = force, M= moment, δ = displacement,
θ = rotation, K = stiffness

477. The method of moment distribution in structural analysis is :
(a) An iterative method
(b) An exact method
(c) An appropriate method
(d) None of these

Nagaland PSC CTSE 2017 Paper–I
Punjab PSC SDE 2016

Ans. (a) : The moment distribution method in mathematical terms can be demonstrated as the process of solving a set of simultaneous equations by means of iteration.

478. In conjugate beam, the loading is equal to
(a) Shear force diagram of actual beam
(b) Bending moment diagram of actual beam
(c) Loading of actual beam
(d) M/EI diagram of actual beam

RPSC POLY. TECH. LECT. 2014

Ans : (d) The length of a conjugate beam is always equal to the length of the actual beam. The load on the conjugate beam is $\frac{M}{EI}$ diagram of actual beam.

479. Shear force at any section of a conjugate beam shows the following in the actual beam as :
(a) deflection (b) Slope
(c) Bending moment (d) Shear force

[UPPSC State Eng. A.E. 2008]

Ans : (b) According to conjugate beam method the shear force at any section in conjugate beam will give slope at that section in real beam.

480. The deflection at any section of a beam, shows the following in the same section of a conjugate beam as
(a) Shear force (b) Load

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(c) Bending moment (d) Deflection

[UPPSC State Eng. A.E. 2008]

Ans : (c) According to conjugate beam method the bending moment at any section in conjugate beam will give deflection at that point in real beam.

481. When an end of continuous beam is fixed, in Kani's method, the rotation contribution will be

(a) Zero (b) EI/L
(c) 2EI/L (d) 3 EI/L

RPSC POLY. TECH. LECT. 2011

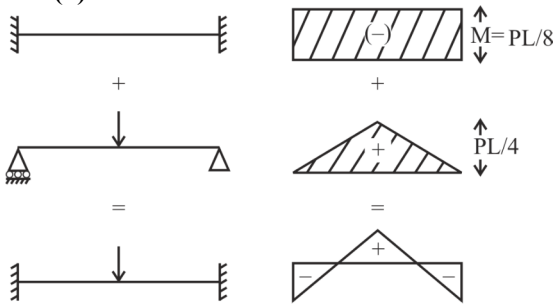
Ans : (a) Kani's Method to continuous beam with fixed end noting that the rotation contribution at the end is zero, since the end being fixed, its rotation is zero.

482. The fixed end moment of uniform beam of span L and fixed at the ends subjected to central point load P is

(a) PL/2 (b) PL/8
(c) PL/12 (d) PL/16

RPSC POLY. TECH. LECT. 2014

Ans : (b)



Fixed end beam–

$$A_f + A_s = 0$$

$$M\times\frac{L}{2}+\frac{1}{2}\times\frac{PL}{4}\times\frac{L}{2}=0$$

$$\boxed{M=\frac{-PL}{8}}$$

483. The number of simultaneous equations to be solved in the slope deflection method is equal to

(a) The degree of static indeterminacy
(b) The degree of kinematic indeterminacy
(c) The number of joints in the structure
(d) The number of member in the structure

RPSC POLY. TECH. LECT. 2014
JPSC Combined AE Paper V 2013

Ans : (b) In slope deflection method, the unknown are slope & deflections {i.e. independent displacement} so, if kinematic indeterminacy or degree of freedom, is less then use slope deflection method.

484. Match List–I with List–II and select correct answer using the codes given in the list.

List-I	List-II
A. Slope deflection method	1. Force method
B. Moment distribution method	2. Displacement method
C. Method of three moments	
D. Castigliano's second theorem	

	A	B	C	D
(a)	2	2	1	1
(b)	1	2	2	1
(c)	2	1	1	2
(d)	1	1	2	2

RPSC POLY. TECH. LECT. 2014

Ans : (a)

Methods of Analysis for rigid frames

Exact Method		Approximate Method
Force method/Flexibility Method/ Method of consistent deformation/ compatibility method **Example -** **1) Strain Energy Method** **2) Minimum Potential Energy method** **3) Unit Load Method** **4) Castigliano's Theorem** **5) Three Moment Theorem** **6) Virtual work method** **7) Column Analogy method** **8) Flexibility method**	**Displacement method/ stiffness method/ equilibrium method** **Example -** **1) Moment Distribution method** **2) Hardy Cross Method** **3) Method of successive approximation** **4) slope and deflection method** **5) Kani's method** **6) Stiffness matrix method**	**Example -** **1) Cantilever method** **2) Factor Method** **3) Portal Method**

485. In a linear elastic structural element

(a) Stiffness is directly proportional to flexibility
(b) Stiffness is inversely proportional to flexibility
(c) Stiffness is equal to flexibility
(d) Stiffness and flexibility are not related

RPSC POLY. TECH. LECT. 2014

Ans : (b) A linear elastic structural element, stiffness is inversely proportional to flexibility.

$$\boxed{\text{Stiffness} \propto \frac{1}{\text{Flexibility}}}$$

486. The ratio of the stiffness of a beam at the near end when the far end is hinged to the stiffness of the beam at the near end when the far end is fixed is

(a) 1/2 (b) 3/4
(c) 1 (d) 4/3

RPSC POLY. TECH. LECT. 2014

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Ans : (b) The stiffness of a beam at the near end when the far and is hinged.

$$K_1 = \frac{3EI}{L}$$

The stiffness of the beam at the near end when far end is fixed.

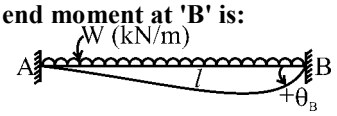
$$K_2 = \frac{4EI}{L}$$

According to Question

$$\frac{K_1}{K_2} = \frac{\frac{3EI}{L}}{\frac{4EI}{L}} = \frac{3}{4}$$

$$\boxed{\frac{K_1}{K_2} = \frac{3}{4}}$$

487. For a fixed beam loaded as shown below, if the support, B rotates + θ_B radian anticlockwise, the fixed end moment at 'B' is:

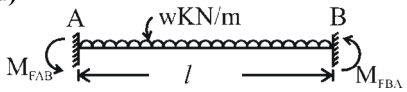


(a) $-\frac{W\ell^2}{12} + \frac{2EI\theta_B}{\ell}$ (b) $-\frac{W\ell^2}{12} + \frac{4EI\theta_B}{\ell}$

(c) $+\frac{W\ell^2}{12} + \frac{2EI\theta_B}{\ell}$ (d) $+\frac{W\ell^2}{12} - \frac{4EI\theta_B}{\ell}$

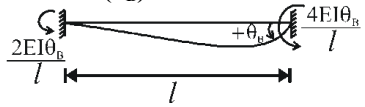
[UPPSC AE 12.04.2016 Paper–I]

Ans : (d)



$$M_{FAB} = -\frac{wl^2}{12}, \quad M_{FBA} = +\frac{wl^2}{12}$$

Effect of rotation **(θ_B)-**



Fixed end moment, $M_{BA} = M_B = M_{FBA} - \frac{4EI\theta_B}{l}$

$$\boxed{M_B = \frac{wl^2}{12} - \frac{4EI\theta_B}{l}}$$

488. Column Analogy method may be used to analyse:

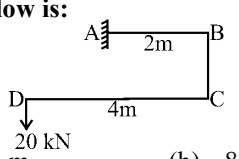
(a) Fixed beam (b) Portal frame

(c) Box frame (d) All of the above

[UPPSC AE 12.04.2016 Paper–I]

Ans : (d) Column analogy method is used to analyse fixed beam, portal frame, box frame etc.

489. The fixed end moment at A for the structure shown below is:

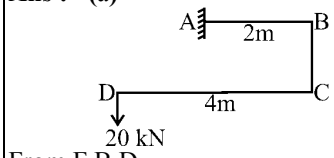


(a) 40kN-m (b) 80kN-m

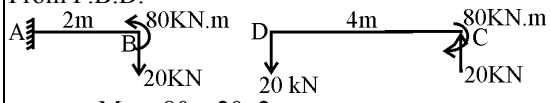
(c) 100kN-m (d) 120kN-m

[UPPSC AE 12.04.2016 Paper–I]

Ans : (a)



From F.B.D.



$M_A = 80 - 20 \times 2$

$M_A = 40$ KNm

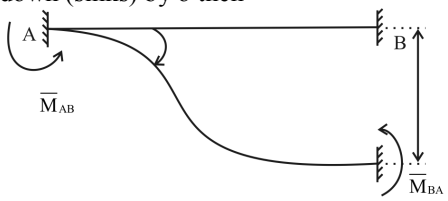
490. The effect of sinking of supports by 'δ' is to create a bending moment equal to:

(a) $\frac{2EI\delta}{\ell^2}$ (b) $\frac{6EI\delta}{\ell^2}$

(c) $\frac{3EI\delta}{\ell^2}$ (d) $\frac{EI\delta}{\ell^2}$

UPPSC AE 12.04.2016 Paper–I
UPPSC State Eng. A.E. 2008, 2003

Ans : (b) If both the supports are fixed and support B settles down (sinks) by δ then

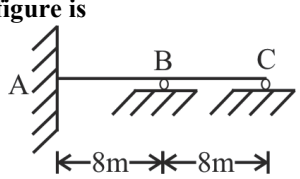


$$\overline{M}_{AB} = \frac{+6EI\delta}{L^2}$$

$$\overline{M}_{BA} = \frac{-6EI\delta}{L^2}$$

if settlement of supports causes rotation of member in clockwise direction the fixed end moment developed will be in anticlockwise direction vice-versa.

491. The stiffness matrix for the beam shown in the following figure is

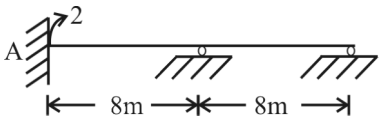


(a) $\frac{EI}{4}\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ (b) $\frac{EI}{4}\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$

(c) $\frac{EI}{2}\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ (d) $\frac{EI}{2}\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

[MPPSC AE 2016]

Ans : (b)



$$k = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}$$

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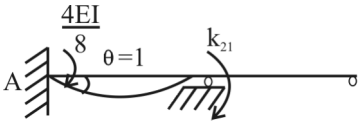
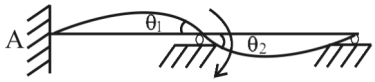


1st Column

$$k_{22} = \frac{4EI}{8}$$

$$k_{12} = \frac{2EI}{8}$$



2nd Column

$$k_{21} = \frac{2EI}{8}$$

$$k_{11} = \frac{4EI}{8} + \frac{4EI}{8} = EI$$

or $\boxed{K = \frac{EI}{4}\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}}$

492. Which one of the following method does not fall under the category of force method?

(a) Method of consistent deformation
(b) Column analogy method
(c) Equilibrium method
(d) Three moment equation

[UPPSC AE 12.04.2016 Paper–I]

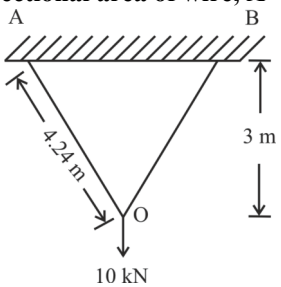
Ans : (c) Method used as force Method

1- Method of consistent deformation
2- Column Analogy Method
3- Three Moment Method
4- Virtual work/unit load Method
5- Elastic centre method
6- Castigliano's theorem of minimum
7- Maxwell-Mohr equation

Method of displacement

1- Stiffness Method
2- Equilibrium Method
3- Slope deflection method
4- Moment distribution method
5- Minimum potential energy method

493. Using strain energy, the vertical deflection at O is (take modulus of elasticity, $E = 2\times10^5$ N/mm^2 , cross–sectional area of wire, A = 100 mm^2).

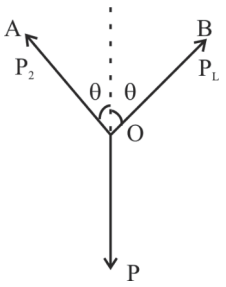


(a) 2.12 mm (b) 21.2 mm
(c) 1.50 mm (d) 15.0 mm

[MPPSC AE 2016]

Ans : (a) Given that,

$A = 100\ mm^2$
$E = 2\times10^5\ N/mm^2$
$L = 4.24$ m.
$P = 10$ kN



$\Sigma f_x = 0$

$P_2 \sin\theta = P_1 \sin\theta$

$\boxed{P_1 = P_2}$

$\Sigma F_v (P_1 + P_2)\cos\theta = P$

$$\boxed{P_1 = \frac{P}{2\cos\theta}}$$

Total strain energy $U = U_{OA} + U_{OB}$

$$= \frac{P_1^2 L}{2AE} + \frac{P_2^2 L}{2AE}$$

$$= \frac{P_1^2 L}{AE}$$

$$\Delta_o = \frac{dU}{dP} = \frac{2P_1 L}{AE}\left(\frac{dP_1}{dP}\right)$$

$$= \frac{2\times P}{2\cos\theta} \times \frac{L}{AE} \times \frac{1}{2\cos\theta}$$

$$\Delta_o = \frac{2\times\left(10\times10^3\right)\times\left(1000\times4.24\right)}{2^2\times\left(\frac{3}{4.24}\right)^2\times100\times2\times10^5}$$

(Where $\cos\theta = \frac{3}{4.24}$)

$\boxed{\Delta_o = 2.12\,mm}$

494. The three-moment equation is applicable only, when-

(a) the beam is prismatic
(b) there is no settlement of supports
(c) there is no discontinuity such as hinges within the span
(d) the span are equal

APPSC AEE 2012
BPSC AE 2012 Paper–V

Ans : (c) The three – moment equation is applicable only when no discontinuity equation such as hinges within the span.

495. Choose the correct statements from the following :

(1) A stable and statically determinate real beam will have a stable and statically determinate conjugate beam
(2) A statically indeterminate real beam will have unstable conjugate beam
(3) A statically indeterminate conjugate will unstable real beam

The correct answer is-

(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) 1, 2 and 3

(BPSC AE 2012 Paper–V)

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Ans : (c)

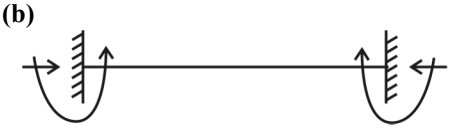
1. A stable and statically determinate real beam will have a stable and statically determinate conjugate beam.
2. A statically indeterminate real beam will have unstable conjugate beam.

496. A fixed beam is an example of:

(a) 3 dimensional structure
(b) 2 dimensional structure
(c) 1 dimensional structure
(d) Determinate structure

[UPPSC State Eng. A.E. 2007 (II)]

Ans : (b)



Since all component of reaction are in 2-D plane and beam is indeterminate to 3rd degree.
Hence it is 2-D indeterminate structure.

497. A continuous beam ABCDE is 12 m long, and contains 4 spans of 3m each. The beam is loaded with uniformly distributed load of 4,000 kgf/m, throughout its length. The bending moment at A and E will be equal to:

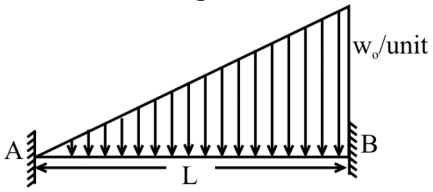
(a) zero (b) 6,000kgf-n
(c) 12,000kg-f-m (d) 3,000kg f-m

[UPPSC State Eng. A.E. 2008]

Ans : (a)

Since support A and E are hinged support,
hence $M_A = 0$
$M_E = 0$

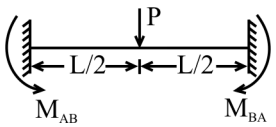
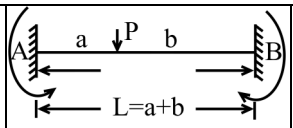
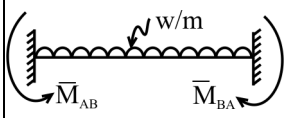
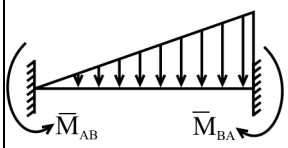
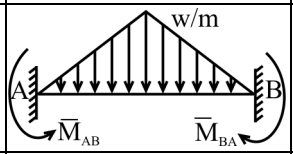
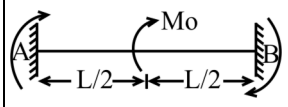
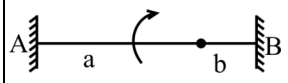
498. The fixed end moment at the end of A of the beam shown in figure is



(a) $\frac{w_o L^2}{30}$ (b) $\frac{w_o L^2}{12}$
(c) $\frac{w_o L^2}{24}$ (d) $\frac{w_o L^2}{10}$

[UPPSC State Eng. A.E. 2008]

Ans : (a)

Sr. No.	Loading	fixed and moments
1.		$\bar{M}_{AB} = \frac{-PL}{8}$ $\bar{M}_{BA} = \frac{+PL}{8}$
2.		$\bar{M}_{AB} = \frac{-Pab^2}{L^2}$ $\bar{M}_{BA} = \frac{+Pa^2b}{L^2}$
3.		$\bar{M}_{AB} = \frac{-wL^2}{12}$ $M_{BA} = \frac{+wL^2}{12}$
4.		$\bar{M}_{AB} = \frac{-wL^2}{30}$ $\bar{M}_{BA} = \frac{+wL^2}{20}$
5.		$\bar{M}_{AB} = \frac{-5}{96}wL^2$ $M_{BA} = \frac{+5}{96}wL^2$
6.		$\bar{M}_{AB} = \frac{+Mo}{4}$ $\bar{M}_{BA} = \frac{+Mo}{4}$
7.		$\bar{M}_{AB} = \frac{M_o b(3a-L)}{L^2}$ $\bar{M}_{BA} = \frac{M_o a(3b-L)}{L^2}$

499. Rotation contribution at the fixed end of a member is:

(a) 0.70 (b) 0.50
(c) zero (d) 1.00

[UPPSC State Eng. A.E. 2008]

Ans : (c) If an end of a member is fixed, the rotation at the end being zero, the rotation contribution at the end is also zero.

500. The stiffness of propped cantilever is equal to:

(a) $\frac{2EI}{L}$ (b) $\frac{3EI}{L}$
(c) $\frac{4EI}{L}$ (d) $\frac{EI}{L}$

[UPPSC State Eng. A.E. 2008]

Ans : (c) The stiffness at a joint for a member is equal to moment required to produce unit rotation at that joint.

$\theta_B = \frac{ML}{4EI}$,

$M = \frac{4EI}{L}\theta_B$

If $\theta = 1$ then $M = K$ (stiffness)

$K = \frac{4EI}{L}$

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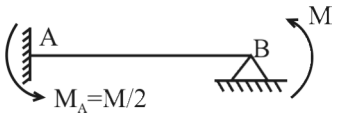
501. In a beam AB (end A is fixed), the reacting moment M/2 at the end A due to application of moment M at B is known as

(a) fixed end moment
(b) stiffness moment
(c) carry over moment
(d) distribution moment

UPPSC State Eng. A.E. 2008

Ans : (c)



$$\text{carry over factor} = \frac{\text{carry over moment at farther end A}}{\text{Applied moment at near end B}}$$

$$\text{COF} = \frac{M_A}{M} = \frac{1}{2}$$

502. The ratio of stiffness of any member to that of total stiffness of all members meeting at a joint is called:

(a) stiffness factor (b) distribution factor
(c) flexibility factor (d) rotation factor

[UPPSC State Eng. A.E. 2008]

Ans : (b) Distribution factor of any member at any joint,

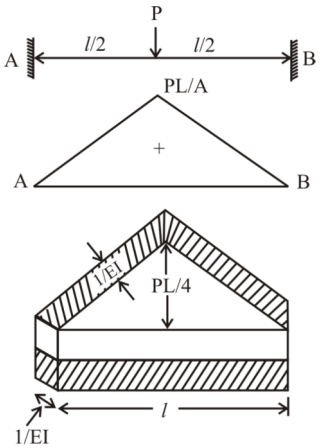
$$\text{Df} = \frac{\text{stiffness of member}}{\text{total stiffness of all member meeting at joint}}$$

503. The width of the analogous column in the method of column analogy is

(a) $\frac{2}{EI}$ (b) $\frac{1}{EI}$
(c) $\frac{1}{2EI}$ (d) $\frac{1}{4EI}$

[UPPSC State Eng. A.E. 2008]

Ans : (b) In the method of column analogy, the thickness of analogous column is taken as $\frac{1}{EI}$

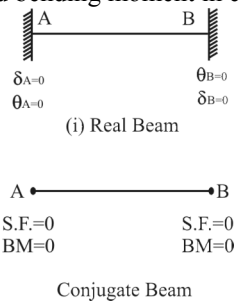


504. The conjugate beam for a fixed beam is :

(a) a beam hinged at both ends
(b) a beam simply supported at both ends
(c) a beam free at both ends
(d) a beam fixed at both ends

[UPPSC State Eng. A.E. 2007 (II)]
[UPPSC State Eng. A.E. 2011]
[UPPSC State Eng. A.E. 2007 (I)]

Ans : (c) The end conditions of conjugate beam will be such that slope and deflection in real beam corresponds to shear force and bending moment in conjugate beam



505. Clapeyron's theorem is also known as the theorem of

(a) 3-moments (b) 2-moments
(c) single-moment (d) no-moment

[UPPSC State Eng. A.E. 2008]

Ans : (a) Clapeyron provides a condition tool for analyzing continuous beam. The three-moment equation represent in general form, the compatibility condition that the slope of the elastic cause be continuous at an interior support of the continuous beam.

506. The value of axial strain energy of the member shown in figure is

A = area of cross-section.
E = modulus of elasticity

(a) $\frac{W^2L}{5AE}$ (b) $\frac{W^2L}{3AE}$
(c) $\frac{W^2L}{2AE}$ (d) $\frac{W^2L}{AE}$

[UPPSC State Eng. A.E. 2011]
[UPPSC State Eng. A.E. 2007 (II)]
(UPPSC State Eng. A.E. 2003)

Ans : (c) Strain energy stored,

$$U = \text{Work done} = \frac{1}{2} \times P \times \Delta$$

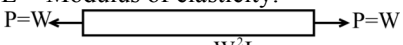
$$U = \frac{1}{2} \times P \times \Delta$$

We know that

$$\Delta = \frac{PL}{AE}$$

$$U = \frac{1}{2} \times P \times \frac{PL}{AE} = \frac{P^2L}{2AE}$$

Where, P = Load, A = Area of cross-section
E = Modulus of elasticity.



$$U = \frac{W^2L}{2AE}$$

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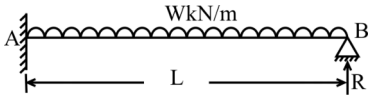
507. The propped reaction of the propped cantilever beam with UDL 'w' kN/m on entire span 'l' is

(a) $\frac{8}{3}wl$ (b) $\frac{3}{8}wl$

(c) $\frac{5}{8}wl$ (d) $\frac{8}{5}wl$

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (b)



By compatibility,

Net deflection at B = 0

$\frac{wL^4}{8EI}$ (downward deflection due to UDL)

$-\frac{RL^3}{3EI}$ (upward deflection due to R)

$$\frac{wL^4}{8EI}-\frac{RL^3}{3EI}=0$$

$$\frac{wL}{8}=\frac{R}{3}$$

$$R=\frac{3wL}{8}$$

508. Which one of the following is force method?

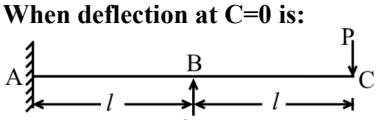
(a) Slope deflection method
(b) Strain energy method
(c) Moment-distribution method
(d) Stiffness method

[UPPSC State Eng. A.E. 2011, 2007 Ist Paper]

Ans : (b)

Force Method	Displacement Method
(i) Redundant forces are considered as un-known	(i) Displacement are considered as unknown
(ii) As Redundant forces are unknown, static indeterminacy is a key parameter in selection of this method	(ii) D_k or degrees of freedom is the governing parameter in selection of this methods.
(iii) As compatibility equations are required to solve force-displacement relationship, hence this method is also called compatibility method	(iii) As equilibrium equations are required to find displacement components, hence it is also called equilibrium method.
(iv) Examples of force method are:	(iv) Examples of displacement method are
• Clapeyron's three moment method	• Moment distribution method
• Catigliano's theorem/ strain energy method	• Slope deflection method
• Flexibility matrix method	• Kani's method
• Column analogy method	• Stiffness matrix method

509. For the beam shown below the ratio of P/Q. When deflection at C=0 is:

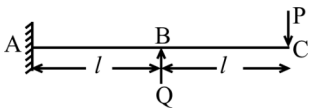


(a) 3/8 (b) 5/8

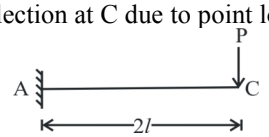
(c) 3/16 (d) 5/16

[UPPSC State Eng. A.E. 2011]
(UPPSC State Eng. A.E. 2003)

Ans : (d)



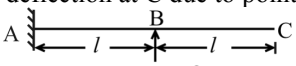
Downward deflection at C due to point load P at C



$$\delta_C=\frac{P(L_{AC})^3}{3EI}=\frac{P(2\ell)^3}{3EI}$$

$$=\frac{8P\ell^3}{3EI}(\downarrow)$$

Upward deflection at C due to point load Q at B



$$\delta_C=\delta_B+\theta_B\times\ell$$

$$\delta_C=\frac{Q\ell^3}{3EI}+\frac{Q\ell^2}{2EI}\times\ell$$

$$\delta_C=\frac{5Q\ell^3}{6EI}(\uparrow)$$

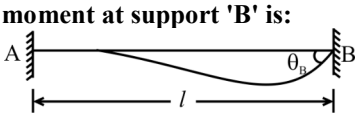
When deflection at C = 0

$$\frac{8P\ell^3}{3EI}=\frac{5Q\ell^3}{6EI}$$

$$16P=5Q$$

$$\boxed{\frac{P}{Q}=\frac{5}{16}}$$

510. In a fixed beam shown below, the fixed end moment at support 'B' is:



(a) $-\frac{2EI\theta_B}{\ell}$ (b) $+\frac{2EI\theta_B}{\ell}$

(c) $-\frac{4EI\theta_B}{\ell}$ (d) $+\frac{4EI\theta_B}{\ell}$

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (d)

$$\theta_B=\frac{ML}{4EI}$$

$$M=+\frac{4EI}{L}\theta_B(\downarrow)$$

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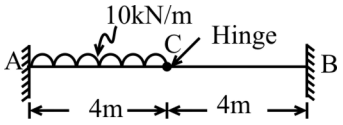
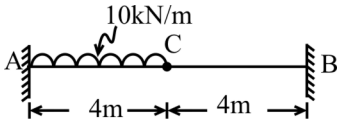
511. The reaction at C of the beam shown is:



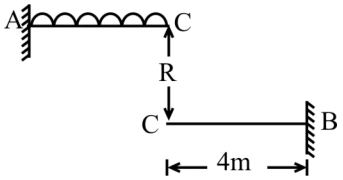
(a) 5.5kN (b) 6.5kN
(c) 7.5kN (d) 8.5kN

[UPPSC State Eng. A.E. 2011]

Ans : (c)



Let R be the reaction at C



by compatibility, deflection of hinge will be same for portion AC and BC

$$(\Delta C)_{AB} = (\Delta C)_{CB}$$

$$\frac{w(L_{AC})^4}{8EI}(\downarrow) - \frac{R(L_{AC})^3}{3EI}(\uparrow) = \frac{R(L_{CB})^3}{3EI}(\downarrow)$$

$$\frac{10\times 4^4}{8EI} - \frac{R\times 4^3}{3EI} = \frac{R\times 4^3}{3EI}$$

$$\frac{10\times 4^4}{8EI} = \frac{2R\times 4^3}{3EI}$$

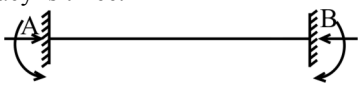
$$R = \frac{10\times 4\times 3}{8\times 2} = 7.5KN$$

512. The size of stiffness matrix for a beam is:

(a) 3×3 (b) 4×4
(c) 6×6 (d) 9×9

[UPPSC State Eng. A.E. 2007 (II)]

Ans : (a) For a 2-D beam maximum degree of static indeterminacy is three.



$D_S = r_e - 3$

$D_S = 6 - 3 = 3$

Therefore to analyze beam there can be three maximum possible coordinates. If there are N coordinates the stiffeners or flexibility will be N × N size.

Hence size of matrix for beam is [3×3].

513. If an any structural element a load is applied gradually and then the same load is applied suddenly as impact, the stresses developed have a ratio of stresses of gradually applied load to sudden applied load is:

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.25

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (a) We know that, stress developed due to gradually applied load is given by,

$$\sigma_g = \frac{P}{A}$$

and stress developed due to impact loading is given by,

$$\sigma_s = \frac{2P}{A}$$

$$\frac{\sigma_g}{\sigma_s} = \frac{P/A}{2P/A} = 0.5$$

514. In continuous beams, the moment over the supports are:

(a) sagging (b) hogging
(c) zero (d) none of the above

[UPPSC State Eng. A.E. 2007 (I)]

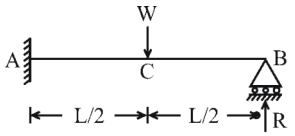
Ans : (d) For the general loading on the continuous beams, sagging moment occurs in the middle regions of the spans and hogging moment occurs over the intermediate supports.

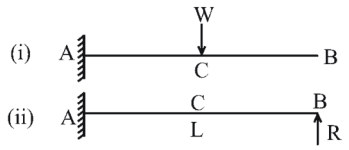
515. A level cantilever beam of uniform section propped at free end on unyielding support carries a point load W at mid span. The reaction at prop is:

(a) 3W/8 (b) 5W/8
(c) 3W/16 (d) 5W/16

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (d)





By compatibility net deflection at point B = 0

Downward deflection due to W - upward deflection due to R

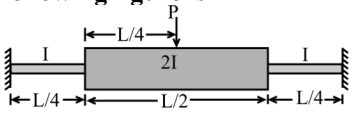
$$\left[\frac{W\left(\frac{L}{2}\right)^3}{3EI} + \frac{W\left(\frac{L}{2}\right)^2}{2EI}\times\frac{L}{2}\right] - \frac{R(L)^3}{3EI} = 0$$

$$\frac{5WL^3}{48EI} - \frac{RL^3}{3EI} = 0$$

$$\frac{RL^3}{3EI} = \frac{5WL^3}{48EI}$$

$$R = \frac{5}{16}W(\uparrow)$$

516. Bending moment at the fixed ends of the beam built in ends and centrally loaded as shown in following figure is



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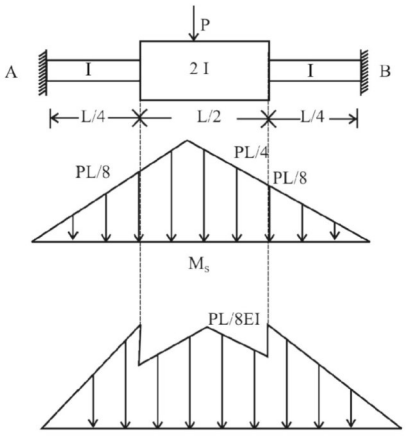
(a) $\frac{PL}{8}$ (b) $\frac{3PL}{48}$

(c) $\frac{5PL}{48}$ (d) $\frac{7PL}{48}$

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c)



M_S/EI diagram

Total load an analogous column = volume of the load diagram

$$=\frac{1}{2}\times L\times\frac{PL}{4}\times\frac{1}{2EI}+2\left[\frac{1}{2}\times\frac{L}{4}\times\frac{PL}{8}\times\frac{1}{2EI}\right]=\frac{5PL^2}{64EI}$$

Area of the analogous column

$$=\frac{1}{2EI}\times\frac{L}{2}+2\times\frac{L}{EI}\times\frac{L}{4}=\frac{3}{4}\times\frac{L}{EI}$$

Uniform stress on the column section $=\frac{\text{load}}{\text{area}}=\frac{5}{48}PL$

stress at A= stress at $B=\frac{5}{48}PL$

fixed end moment at A = fixed end moment at B

$$=M_s-M_i=0-\frac{5}{48}PL$$

$$\overline{M}=\frac{5}{48}PL$$

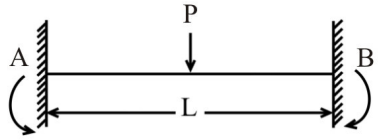
517. An 8m beam fixed both ends carries a central point load of 10kN. The value of fixed end moment at the right end will be.

(a) 12kN-m (b) 10kN-m

(c) 8kN-m (d) 6kN-m

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (b)



$$\overline{M}_{BA}=+\frac{PL}{8}$$

$$\overline{M}_{BA}=+\frac{10\times8}{8}=10\text{kN-m}$$

518. At a hinge the bending moment is

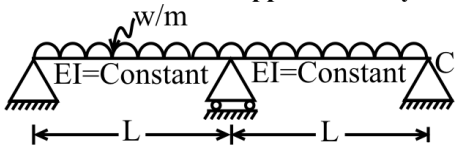
(a) always sagging (b) always hogging

(c) always zero (d) always maximum

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c) At a hinge the bending moment is always zero.

519. If a continuous beam shown in fig. below is subjected to uniformly distributed load of w/m run, the value of central support reaction becomes zero if central support sinks by

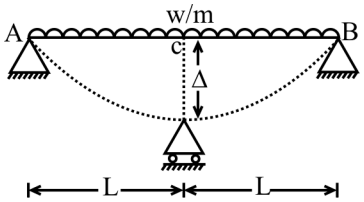


(a) $\frac{5wL^2}{24EI}$ (b) $\frac{5wL^4}{384EI}$

(c) $\frac{10wL^4}{384EI}$ (d) $\frac{wL^4}{4.8EI}$

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d)



$$\Delta=\frac{5}{384}\frac{WL_{AB}^4}{EI}$$

$$\Delta=\frac{5}{384}\times\frac{W\times(2L)^4}{EI}=\frac{5\times16WL^4}{384EI}$$

$$\Delta=\frac{WL^4}{4.8EI}$$

0hence the value of central support reaction becomes zero if central support sink by $\frac{WL^4}{4.8EI}$

520. The force method of structural analysis starts with

(a) equilibrium field

(b) compatible deformation

(c) equilibrium and compatibility conditions

(d) force deformation relation only

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c) In force method unknowns are taken as forces or reactions. To find unknowns, equilibrium conditions $\sum fx=0,\ \sum fy=0,\ \sum Mz=0$ and compatibility conditions are written.

521. The theorem of three moment equation for a continuous beam expresses the condition of

(a) equilibrium at an intermediate support

(b) slope compatibility at an intermediate support

(c) zero support settlement

(d) None of the above

[UPPSC State Eng. A.E. 2004 (I)]

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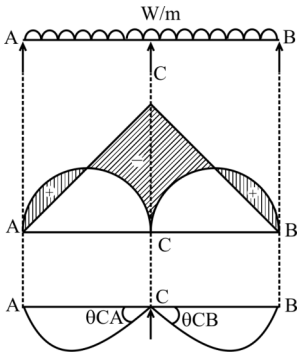
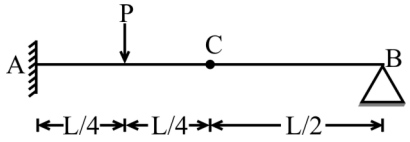
Ans : (b)



At intermediate support $\theta_{CA} = \theta_{CB} = 0$

Thus the three moment theorem for continuous beam expressed condition of slope compatibility at an intermediate support.

522. For the propped cantilever shown in figure, the vertical reaction at the fixed end is

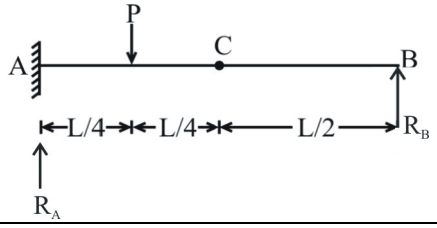


(a) $\frac{4P}{3}$ (b) P

(c) $\frac{3P}{4}$ (d) $\frac{P}{2}$

(UPPSC State Eng. A.E. 2003)
ESE 2008

Ans : (b)



$M_c = 0$

$\therefore \quad -R_B \times \frac{L}{2} = 0$

$\therefore \quad R_B = 0$

Also $\quad \Sigma F_y = 0$

$R_A + R_B = P$

$R_A + 0 = P$

$R_A = P$

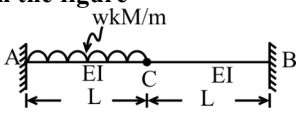
523. The maximum bending moment when a train of point loads crossing a simply supported beam occurs when the load is

(a) At one third span
(b) At mid span
(c) The point load under consideration and the C.G. of train of point loads equidistant from the mid span
(d) At one quarter span

(UPPSC State Eng. A.E. 2003)

Ans : (c) Absolute maximum bending moment occurs when position of load train is such that centre of span is mid way between C.G. of load system and load under consideration.

524. What is the reaction on the hinge C for a beam as shown in the figure

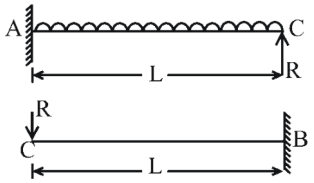


(a) $\frac{3}{8}wL$ (b) $\frac{1}{2}wL$

(c) $\frac{1}{4}wL$ (d) $\frac{3}{16}wL$

(UPPSC State Eng. A.E. 2003)

Ans : (d) Let R be the reaction at hinge C



By compatibility

$$(\Delta_C)_{AC} = (\Delta_C)_{CB}$$

$$\frac{WL^4}{8EI} - \frac{RL^3}{3EI} = \frac{RL^3}{3EI}$$

$$\frac{2RL^3}{3EI} = \frac{WL^4}{8EI}$$

$$R = \frac{3}{16}WL$$

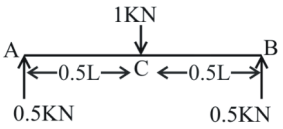
525. A simply supported beam of span L and flexural rigidity EI, carries a unit point load at its centre. The strain energy in the beam due to bending is

(a) $\frac{L^3}{48EI}$ (b) $\frac{L^3}{192EI}$

(c) $\frac{L^3}{96EI}$ (d) $\frac{L^3}{16EI}$

(UPPSC State Eng. A.E. 2003)

Ans : (c) Strain energy due to bending $= \int \frac{M_x^2 dx}{2EI}$



$M_x = 0.5x$

$$U = 2U_{AC} = 2\int_0^{L/2} \frac{M_x^2 dx}{2EI} = 2\int_0^{L/2} \frac{\left(\frac{1}{2}x\right)^2 dx}{2EI}$$

$$U = \frac{2}{EI} \times \frac{1}{4}\int_0^{L/2} x^2 dx = \frac{1}{4EI}\left[\frac{x^3}{3}\right]_0^{\frac{1}{2}}$$

$$U = \frac{L^3}{96EI}$$

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Alternative Approach

We know, Strain energy,

$$U = \text{work done}$$

$$U = \frac{1}{2} \times P \times \Delta$$

$$U = \frac{1}{2} \times P \times \frac{PL^3}{48EI}$$

$$U = \frac{P^2L^3}{96EI}$$

Here P=1 KN

$$U = \frac{L^3}{96EI}$$

526. The number of plastic hinges required to cause collapse of a simply supported beam is

(a) 0 (b) 1

(c) 2 (d) 3

(BPSC AE Paper V 2001)

Ans : (b) The number of plastic hinges required to cause collapse of a simply supported beam is 1.

527. Sum of distribution factors at a joint is-

(a) -0.50 (b) +1.0

(c) +0.50 (d) -1.0

ISRO Scientist/Engineer 2015

(BPSC AE Paper V 2001)

Ans : (b) The value of sum of distribution factor = +1.0

528. Stiffness approach of solution of indeterminate structures intends to solve for:

(a) unknown force

(b) unknown action

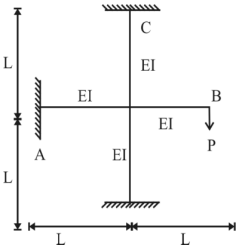
(c) unknown displacement

(d) none of the above

(BPSC AE Paper V 2006)

Ans : (a) Stiffness approach of solution of indeterminate structure intends to solve for unknown forces.

529. The given figure shows frame with a single concentrated load P the fixed end moment developed at joint A will be



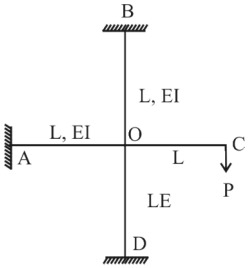
(a) $\frac{PL}{8}$ (b) $\frac{PL}{6}$

(c) $\frac{PL}{4}$ (d) $\frac{PL}{3}$

(JPSC Combined AE Paper V 2013)

ESE 1995

Ans : (b)



Stiffness of OA = OD = OB = $\frac{4EI}{L}$

Moment at point force 'O' = PL

Sum of stiffness = $\frac{4EI}{L} + \frac{4EI}{L} + \frac{4EI}{L} = \frac{12EI}{L}$

$\theta = 1$ rad.

$$\boxed{PL = \frac{12EI}{L} \times 1}$$

Fixed end moment developed at point 'A' (M_A) = $\frac{2EI}{L}$

$$= \frac{2EI}{L} \times \left(\frac{6}{6}\right) = \left(\frac{12EI}{L}\right) \times \frac{1}{6}$$

$$PL \times \frac{1}{6} = \frac{PL}{6}$$

$$\boxed{M_A = \frac{PL}{6}}$$

530. A rigid frame ABCD has AB member as overhang, with end 'A' being free. Supports C and D are fixed. DB is vertical member and BC is horizontal member. DB = BC = 5 m. If the moment equations are given by-

$M_{DB} = 0.4\,\theta_B$ $M_{BD} = 0.8\,EI\,\theta_B$

$M_{BA} = 54$ $M_{BC} = -135 + 1.6\,EI\,\theta_B$

$M_{CB} = 135 + 1.6\,EI\,\theta_B$ **then**

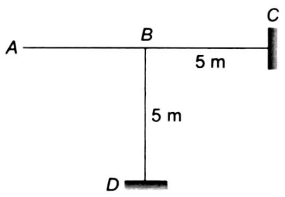
θ_B rotation at *B* is given by-

(a) $\frac{-112.5}{EI}$ (b) $\frac{33.75}{EI}$

(c) Zero (d) None of the above

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (b) :



Applying moment equilibrium at joint *B*

$M_{BA} + M_{BC} + M_{BD} = 0$

$54 + (-135 + 1.6EI\,\theta_B) + 0.8\,EI\,\theta_B = 0$

$$EI\theta_B = \frac{135 - 54}{2.4}$$

$$\theta_B = \frac{33.75}{EI}$$

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531. Three prismatic members AB, BC and BD meet at a joint for a rigid frame to be analyzed using moment distribution method. The distribution factors for members AB and BC are 0.5 and 0.3 respectively. The distribution factor for member BD shall be-

(a) 0.2 (b) 1.67
(c) 0.6 (d) 0.15

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (a) : We know that sum of distribution factors for various members at a joint is always 1. Hence

$DF_{BA} + DF_{BC} + DF_{BD} = 1$
$0.5 + 0.3 + DF_{BD} = 1$
$DF_{BD} = 0.2$

532. A structure is said to be statically indeterminate when-

(a) the number of unknown reaction components exceeds the number of equilibrium conditions
(b) the number of equilibrium conditions exceeds the number of unknown reaction components
(c) the number of equilibrium conditions equal to the number of unknown reaction components
(d) None of the above

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (a) : In statics and structural mechanics, a structure is statically indeterminate when the static equilibrium equations are insufficient for determining the internal forces and reaction on that structure.

533. Castigliano's first theorem is applicable-

(a) For statically determinate structure only
(b) When the system behaves elastically
(c) Only when principle of superposition is valid
(d) None of these

OPSC AEE 2015 PAPER–I
UJVNL AE 2012

Ans. (c) : Castigliano's first theorem is applicable only when principle of superposition is valid.

Castigliano's first theorem : The first partial derivative of the total internal energy (strain energy) in a structure with respect to any particular deflection component at a point is equal to the force applied at that point and in the direction corresponding to that deflection component.

This first theorem is applicable to linearly or non-linearly elastic structures in which the temperature is constant and the supports are unyielding.

Castigliano's second theorem : The first partial derivative of the total internal energy (strain energy) in a structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action.

The second theorem of Castigliano's is applicable to linearly elastic (Hookean material) structures with constant temperature and unyielding supports.

534. If the displacement at coordinate *i* due to unit force at coordinate *j* is δ_{ij} and displacement at coordinate *j* due to unit force at coordinate *i* is δ_{ji} then according to Maxwell's Reciprocal Theorem-

(a) $\delta_{ij} = \delta_{ji}$ (b) $\delta_{ij} > \delta_{ji}$
(c) $\delta_{ij} < \delta_{ji}$ (d) $\delta_{ij} \neq \delta_{ji}$

OPSC AEE 2015 PAPER–I

Ans. (a) : According to Maxwell's reciprocal theorem

$$\delta_{ij} = \delta_{ji}$$

535. To generate the j^{th} column of flexibility matrix-

(a) A unit force is applied at coordinate *j* and the displacement are calculated at all coordinates
(b) A unit force is applied at coordinate *j* and the forces are calculated at all coordinates
(c) A unit displacement is applied at coordinate *j* and the forces are calculated at all coordinates
(d) A unit displacement is applied at coordinate *j* and the forces are calculated at all coordinates

OPSC AEE 2019, 2015 PAPER–I
ISRO Scientist/Engineer 2015

Ans. (a) : To generate flexibility matrix,

$$f = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$$

$$\text{Flexibility} = \frac{\Delta}{F} = \frac{\text{Deflection}}{\text{Force}}$$

To generate the j^{th} column of the flexibility matrix, we apply a unit force at co-ordinate 'j' only and compute displacement at all the co-ordinates.

536. Which of the following methods of structural analysis is a force method?

(a) Column analogy method
(b) Slope deflection method
(c) Moment distribution method
(d) None of these

BPSC AE 2019 Paper (V) Section-I
OPSC AEE 2015 PAPER–I

Ans. (a) : Force method of structural analysis are as follows :

- Column analogy method
- Theorem of least work
- Method of consistent deformation

537. The Castigliano's 2^{nd} theorem can be used to compute deflections

(a) in statically determinate structures only
(b) for any type of structure only
(c) at the point under the load only
(d) for beam and frames only

Haryana PSC AE 2018 Paper–I
BPSC AE 2019 Paper (V) Section-I
GPSC AE June 2019

Ans. (b) : The first partial derivative of the total internal energy in a structure with respect to the force applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action.

The above statement is Castigliano's 2^{nd} theorem and is applicable for any type of structure to be linearly elastic.

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538. Which of the following methods of structural analysis is a displacement method?

(a) Three-moment equation
(b) Column analogy method
(c) Moment distribution method
(d) None of the above

BPSC AE 2019 Paper (V) Section-I

Ans. (c) : In displacement method, the displacement is assumed to be unknown.
This method is preferable when structure has less kinematic indeterminacy.
Moment Distribution method, slope deflection method, Kani's method, stiffness matrix method are the example of displacement method.

539. The width of the analogous column in the method of column analogy is

(a) $\frac{2}{EI}$ (b) $\frac{1}{EI}$
(c) $\frac{1}{2EI}$ (d) $\frac{1}{4EI}$

BPSC AE 2019 Paper (V) Section-I
APPSC AEE 2016

Ans. (b) : Analogous column is a short column with cross-section width L and breath $\frac{1}{EI}$, where L is the length of beam and E is young's modulus of beam material and I is second moment of inertia of beam cross-section.

540. A statistically indeterminate structure is the one which

(a) cannot be analysed using equations of statics alone
(b) cannot be analysed at all
(c) is not stable for general loading
(d) can be analysed with the equations of statics alone

APPSC AE 2016 (Mains)

Ans. (a) : A statically indeterminate structure can not be analysed by using equation of statics alone, we use compatibility equation also to find all the unknowns. When the static equilibrium (force and the moment) are insufficient for determining the internal forces and reaction on that structure.

541. Statically indeterminate structures are also called as

(a) unstable structures
(b) serviceable structures
(c) redundant structures
(d) determinate structures

TNPSC AE 2018

Ans. (c) : A redundant or indeterminate structure more members are present than necessary.

542. A deformable body is under the action of external forces (F_i). The external forces satisfy the following conditions with respect to an inertial frame:

(i) $\Sigma F_i = 0$
(ii) $\Sigma r_i \times F_i = 0$

These conditions are:

(a) necessary and not sufficient for equilibrium
(b) sufficient for equilibrium
(c) necessary and sufficient for equilibrium
(d) none of the above

[HPPSC AE 2016]

Ans. (a) : For equilibrium of a deformable body, we need a minimum three equilibrium equation i.e. $\Sigma F_x = 0$, $\Sigma F_y = 0$ and $\Sigma M = 0$
As only two are given the equations are necessary but not sufficient.

543. Two different sets of forcing systems are said to statically equivalent if,

(i) They produce same shear force and bending moment in a particular section.
(ii) The require same set of external forces to reduce each system to equilibrium.
(iii) They generate identical reactions with respect to direction and magnitude.
(iv) They produce same deflection in any given section of the beam

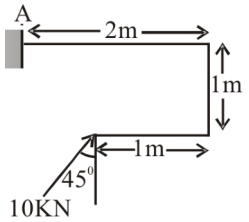
Which of the above are true?

(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) and (iv) (d) (iii) and (iv)

[HPPSC AE 2016]

Ans. (a) : Two different sets of forcing system are said to be statically equivalent if they produce same resultant force and same resultant moment.

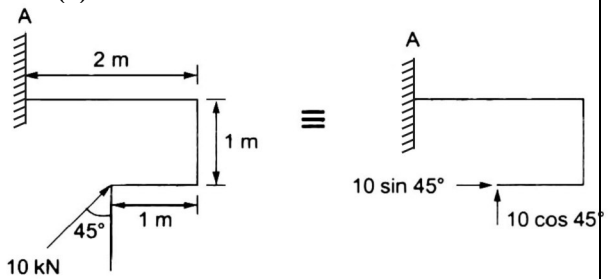
544. For the structure shown in the figure, the fixed end moment at support 'A' is :



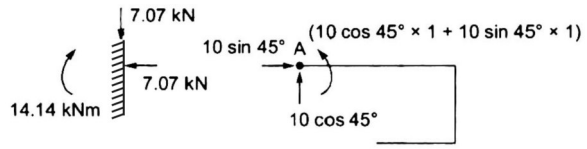
(a) 14.14 kN-m (b) 10 kN-m
(c) zero (d) 3 kN-m

Karnataka PSC AE 2017 Paper - II

Ans. (a) :



Transferring these forces to point A and making F_{BD}



Hence $M_A = 10\cos 45^\circ \times 1 + 10\sin 45^\circ \times 1$
$= 7.07 \text{ kNm} + 7.07 \text{ kNm} = 14.142 \text{ kNm}$

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545. The conditions required to be satisfied for the analysis of indeterminate structure are -

(a) Equilibrium
(b) Compatibility
(c) Force-displacement relationship
(d) All of these

RPSC AE 2018

Ans. (d) To solve an indeterminate structure it is necessary to satisfy equilibrium, compatibility and force-displacement requirement of structure.
The additional equation required to solve indeterminate structure are obtained by the conditions of compatibility and/or force displacement relations.

546. In slope deflection method, the joints are considered rigid when -

(a) no change in value of the angles between members
(b) 90^0 angle between the members in frame
(c) 180^0 angle between the member in beams
(d) All of these

RPSC AE 2018

Ans. (a) The slope-deflection method the rotations of the joints are treated as unknown. For any one member bounded by two joints the end moment can be expressed in terms of rotations.
In this method all joints are considered rigid, i.e. the angle between members at the joints are considered not to change in value as loads are applied.

547. The shape factor of an isosceles triangle should be -

(a) 1.5 (b) 1.7
(c) 2.34 (d) 2

RPSC AE 2018

Ans. (c) :

Shape)	Shape factor
Triangle	2 . 346 ≈ 2 . 34
Diamond Section	2
I-Section	1 . 12 to 1 . 2
T-Section	1 . 90 to 1 . 95
Square	1 . 5
Circular	1 . 7

548. The three moment equation in structural analysis is basically a:

(a) Stiffness method (b) Displacement method
(c) Energy method (d) Flexibility method

GPSC AE Class (1&2) Paper - 2 2017
PTCUL AE 2016
UJVNL AE 2012

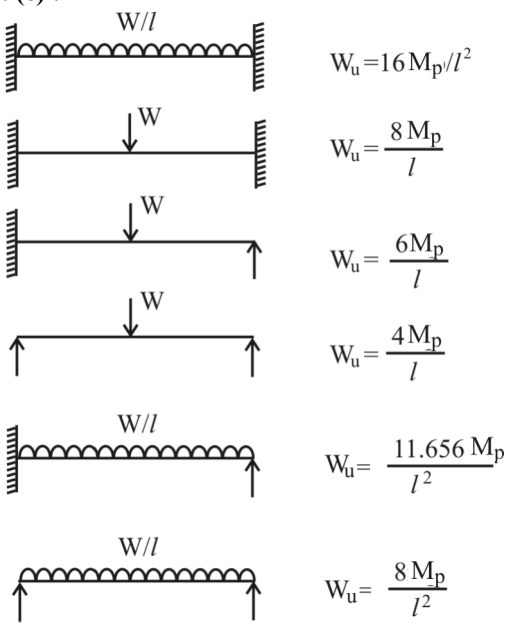
Ans. (d) : The three moment equation in structural analysis is basically a flexibility method.
In structural engineering, the flexibility method also called the method of consistent deformations is the traditional method for computing member forces and displacement in structural systems.

549. In plastic analysis, collapse load for a fixed uniform beam with central point load is :
(Note : M_P is plastic moment capacity and L is the span of the beam)

(a) $2M_p/L$ (b) $4M_p/L$
(c) $8M_p/L$ (d) $16Mp/L$

GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) :



550. For a linear elastic structural analysis system, minimization of potential energy yields

(a) compatibility conditions
(b) constitutive relations
(c) equilibrium equations
(d) strain-displacement relations

GPSC AE (CLASS 1 & 2) 2019

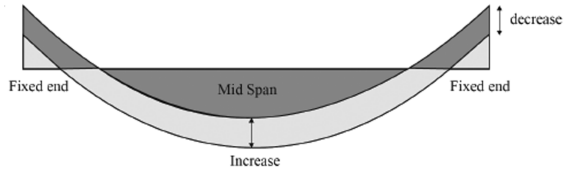
Ans : (a) : For a linearly elastic structure, the minimization of potential energy yields compatibility condition

551. Statement (I): If degree of fixity at supports is lessened, the maximum hogging moment at the ends will decrease.
Statement (II): If degree of fixity at supports is lessened, the maximum sagging moment at mid-span decreases.

(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.
(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I
(c) Statement-I is true but Statement-II is false.
(d) Statement-I is false but Statement-II is true.

ESE 2019

Ans. (c) Due to redistribution of moments as the fixed end moment decreases the sagging moment at midspan increases.



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552. Statement (I): Slope-deflection method is a displacement method of analysis.
Statement (II): Forces are the basic unknowns in the slope-deflection method.

(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.
(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I
(c) Statement-I is true but Statement-II is false.
(d) Statement-I is false but Statement-II is true.

ESE 2017

Ans. (c)
- The slope deflection method was widely used for more than a decade until the moment distribution method was developed.
- The moment distribution method is a structural analysis method for statically indeterminate beams and frames. In this every joint of the structure to be analyzed is fixed so as to develop the fixed end moments.
- The displacement and rotation (δ, θ) are the basic, unknowns in the slope deflection method.
- The slope deflection equation can also be written using the stiffness factor $\left[K = \frac{I_{ab}}{L_{ab}}\right]$ and chord rotation $\left[\psi = \frac{\Delta}{L_{ab}}\right]$

553. Match List-I with List - II and select the correct answer using the code given below the lists :

List - I	List - II
A. Axel Bendixen	1. The mathematical theory of elasticity
B. Hardy Cross	2. Theory of curved bars
C. Winkler	3. Slope-deflection method
D. St. Venant	4. Moment distribution

(a) A B C D 1 2 3 4 (b) A B C D 3 4 2 1
(c) A B C D 1 2 4 3 (d) A B C D 3 2 4 1

ISRO Scientist/Engineer 2013

Ans : (b)

A. Axel Bendixen	3. Slope-deflection method
B. Hardy Cross	4. Moment distribution
C. Winkler	2. Theory of curved bars
D. St. Venant	1. The mathematical theory of elasticity

554. A cantilever beam having a constant depth carries a uniformly distributed load of w per unit length over the whole span. To make the maximum bending stress at all sections of the beam to be the same, the breadth of the section at a distance x from the free end should be proportional to :

(a) x^3 (b) x
(c) x^2 (d) $x^{0.5}$

CIL AE 2020

Ans. (c) : At a distance x from free end

$M_x = \frac{wx^2}{2}$

and $Z_x = \frac{b_x d^2}{6}$

Maximum bending stress

$f = \frac{M_x}{Z_x}$

$\therefore b_x = \frac{3wx^2}{fd^2}$

$\left[b_x \propto x^2\right]$

555. Match the following and pick the correct series:

(i) Slope deflection method	(1) Rotation factor
(ii) Moment distribution method	(2) Flexibility coefficient
(iii) Kani's method	(3) Kinematic redundant
(iv) Matrix method	(4) Carryover factor

	(i)	(ii)	(iii)	(iv)
(a)	(1)	(2)	(3)	(4)
(b)	(3)	(4)	(2)	(1)
(c)	(2)	(3)	(4)	(1)
(d)	(3)	(4)	(1)	(2)

AP (VSS-1) PART - A 2017

Ans : (d)

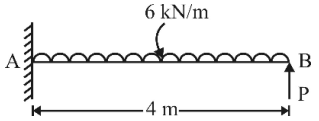
(i) Slope deflection method	Kinematic redundant
(ii) Moment distribution method	Carryover factor
(iii) Kani's method	Rotation factor
(iv) Matrix method	Flexibility coefficient

556. A cantilever beam of length 4 metres carries a UDL of 6 kN/m throughout its length. It is propped at the end such that the levels of the free end is the same as that of the fixed end. What is the magnitude of the propped reaction?

(a) 3 kN (b) 6 kN
(c) 9 kN (d) 12 kN

AP (VSS-1) PART - A 2017

Ans : (c) :



From figure,
Deflection at B = 0

or $\frac{6\times4^4}{8EI} - \frac{P\times4^3}{3EI} = 0$

or $\frac{6\times4}{8} = \frac{P}{3}$

$\boxed{P = 9\,kN}$

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557. The element of flexibility matrix of a structure :

(a) Are dependent on the choice of coordinates
(b) Are independent on the choice of coordinates
(c) Are always dimensionally homogeneous
(d) Both (A) and (C) are correct

OPSC AEE 2019 Paper–I

Ans. (d) : The elements of a flexibility matrix are not necessarily dimensionally homogeneous. They represent either a translation or a rotation due to a unit load or a couple.

558. For stable structures, one of the important properties of flexibility and stiffness matrices is that the element of main diagonal :

(a) Of a stiffness and flexibility matrix must be negative.
(b) Of a stiffness matrix must be negative and flexibility matrix must be positive.
(c) Of a stiffness matrix must be positive and flexibility matrix must be negative.
(d) Of a stiffness and flexibility matrix must be positive.

OPSC AEE 2019 Paper–I

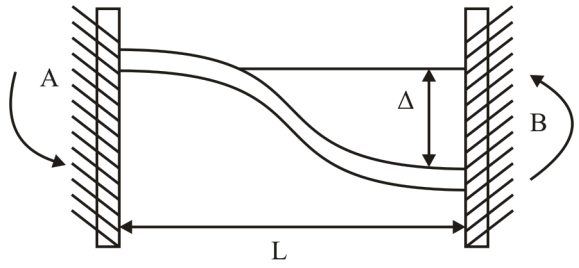
Ans. (d) While the other elements of the flexibility matrix may be positive or negative, the elements lying on the leading diagonal are always positive.

559. If the sinking of a support of a fixed beam causes the beam to rotate in the clockwise direction, then the moments induced at both the ends of the beam will be

(a) In anticlockwise direction and of equal magnitude
(b) In clockwise direction and of different magnitude
(c) In opposite directions and of equal magnitude
(d) In opposite directions and of different magnitude

OPSC AEE 2019 Paper–I

Ans. (a) The moment induced at both the ends of the beam will be in anticlockwise direction and of equal magnitude.



$(FEM)_{AB} = -\frac{6EI\Delta}{L^2}$ $\quad (FEM)_{BA} = -\frac{6EI\Delta}{L^2}$

560. The stiffness matrix of a beam elements is $(2EI/L)\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ which one of the following is its flexibility matrix?

(a) $(L/2EI)\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ (b) $(L/6EI)\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$

(c) $(L/4EI)\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ (d) $(L/6EI)\begin{bmatrix}1 & -1\\ -2 & 1\end{bmatrix}$

JPSC AE (Pre) 2019
ESE 2005

Ans. (b) The stiffness and flexibility matrix are inverse of each other and both are symmetrical.

Stiffness matrix, $K = \frac{2EI}{L}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$

Flexibility matrix $= [F]$

$[K][F] = I \Rightarrow [F] = [K]^{-1}$

Flexibility matrix is equal to the inverse of stiffness

$$|K| = \left(\frac{2EI}{L}\right)^2 (4-1) = 3\left(\frac{2EI}{L}\right)^2$$

$$\text{Adjoin } (K) = \frac{2EI}{L}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

$$\text{So, } [K]^{-1} = \frac{\text{Adj}(K)}{|K|}$$

$$[K]^{-1} = \frac{\frac{2EI}{L}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}}{3\left(\frac{2EI}{L}\right)^2}$$

$$= \frac{\frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}}{\frac{2EI}{L}}$$

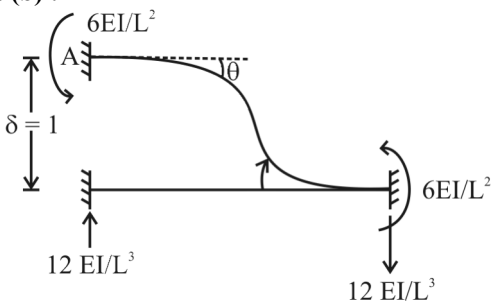
$$[K]^{-1} = \frac{L}{6EI}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

561. A stiffness matrix is to be generated for beam AB as horizontal flexural member. As per the method adopted for calculation of stiffness matrix, if end A is given translation displacement in vertically upward direction, the end forces generated at end 'B' are

(a) No forces at end B
(b) $-12\ EI/L^3$ vertical force and $6EI/L^2$ moment
(c) $-6\ EI/L^2$ vertical force and $2EI/L$ moment
(d) $-6\ EI/L^2$ vertical force and $4EI/L$ moment

Maharashtra PSC Mains 2018 Paper-1

Ans. (b) :



Vertical force is $12\ EI/L^3$
Moment is $6\ EI/L^2$ At support 'A'

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562. If the stiffness matrix of beam element is given as $\frac{2EI}{L}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$, then the flexibility matrix is:

(a) $\frac{L}{6EI}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ (b) $\frac{L}{2EI}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$

(c) $\frac{L}{3EI}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ (d) $\frac{L}{6EI}\begin{bmatrix}-1 & 2\\ 2 & -1\end{bmatrix}$

Maharashtra PSC Mains 2018 Paper-1

Ans. (a) : The stiffness and flexibility matrix are inverse of each other and both are symmetrical.

Stiffness matrix, $K = \frac{2EI}{L}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$

Flexibility matrix = [F]

$[K][F] = I \Rightarrow [F] = [K]^{-1}$

Flexibility matrix is equal to the inverse of stiffness

$$|K| = \left(\frac{2EI}{L}\right)^2 (4-1) = 3\left(\frac{2EI}{L}\right)^2$$

$$\text{Adjoin (K)} = \frac{2EI}{L}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$$

$$\text{So, } [K]^{-1} = \frac{\text{Adj}(K)}{|K|}$$

$$[K]^{-1} = \frac{\frac{2EI}{L}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}}{3\left(\frac{2EI}{L}\right)^2}$$

$$= \frac{\frac{1}{3}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}}{\frac{2EI}{L}}$$

$$[K]^{-1} = \frac{L}{6EI}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$$

563. What is true for flexibility and stiffness matrix?

A. They are square matrix

B. The diagonal elements are non-zero and having positive values

C. Element ij = Element ji

D. They are inverse of each other

Answer Options:

(a) A and B (b) All of the above

(c) C and D (d) A, C and D

Maharashtra PSC Mains 2018 Paper-1

Ans. (b) : Some of true points about flexibility and stiffness matrix

1. They are square matrix
2. The diagonal elements are non-zero and having positive values
3. Element ij = Element ji
4. They are inverse of each other

564. "The partial derivative of the total internal energy in a beam, with respect to the load applied at any point is equal to the deflection at that point." This is the statement of:

(a) Moment area theorem

(b) Castigliano's second theorem

(c) Conjugate beam theorem

(d) Muller – Breslau's influence theorem

Maharashtra PSC Mains 2018 Paper-1

Ans. (b) : Castigliano's second theorem–The first partial derivative of total strain energy with respect to the load applied at any point is equal to the deflection at the point of application of that force in the direction of its line of action

$$\frac{\partial \mu}{\partial P} = \delta$$

565. Flexibility method is also called as :

1. force method

2. compatibility method

3. consistent deformation method

(a) Only 1 (b) 1 and 2

(c) 1 and 3 (d) 1, 2 and 3

GPSC AE March 2018

Ans. (d) : The flexible method is also called as method of consistent deformations is the traditional method for computing member forces and displacement in structural systems.

It is also known as Matrix force method and compatibility method.

566. Matrix stiffness method

1. forms the basis for computerization

2. yields the displacements and forces in one go

3. can be used to analyse both determinate and indeterminate structure

(a) 1 and 2 (b) 1 and 3

(c) 2 and 3 (d) 1, 2 and 3

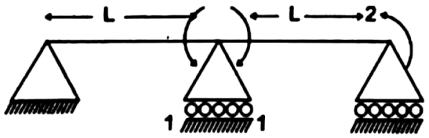
GPSC AE March 2018

Ans. (d) : Matrix stiffness method is suited for computer automated analysis of complex structures.

It is used for computing forces and displacement in structures.

It is used for analysing complex structures including indeterminate structures.

567. For the given structure produce flexibility matrix for coordinate 1 and 2.



EI-Constant

(a) $\frac{L}{3EI}\begin{bmatrix}4 & 1\\ 1 & 2\end{bmatrix}$ (b) $\frac{L}{2EI}\begin{bmatrix}3 & 2\\ 2 & 4\end{bmatrix}$

(c) $\frac{L}{6EI}\begin{bmatrix}4 & 1\\ 1 & 2\end{bmatrix}$ (d) $\frac{L}{3EI}\begin{bmatrix}3 & 2\\ 2 & 4\end{bmatrix}$

Vizag steel MT 2017

Ans. (c) : Flexibility matrix

$$\frac{L}{6EI}\begin{bmatrix}4 & 1\\ 1 & 2\end{bmatrix}$$

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568. Consider the following methods of analysis:
1. Method of consistent deformation
2. Column analogy method
3. Equilibrium method
4. Three-moment equation
Which of the above methods come under the category of force method?
(a) 1, 2 and 3 only (b) 3 and 4 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 4

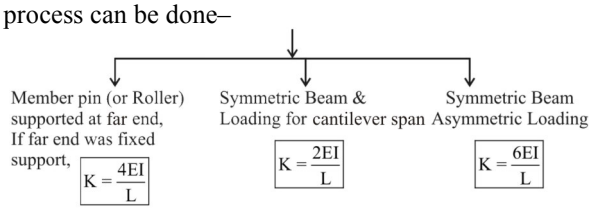
UPSC JWM 2017

Ans. (c) : Methods under the category of force method
1. Method of consistent depression
2. Column analogy method
3. Three-moment equation
4. Castigliano's theorem
5. Unit load method

569. If the far end of the beam is fixed, the stiffness of beam with usual notation is
(a) $\frac{2EI}{L}$ (b) $\frac{3EI}{L}$
(c) $\frac{4EI}{L}$ (d) $\frac{4EI}{L^2}$

GPSC R&B AE 2016

Ans. (c) : Three cases where moment distribution process can be done–



570. The number of unknowns to be determined in stiffness method is equal to-
(a) static indeterminacy
(b) kinematic indeterminacy
(c) the sum of kinematic and static indeterminacy
(d) two times the number of support

JPSC AE 2009

Ans. (b) : In the stiffness method, the unknown quantities will be the joint displacements. Hence, the no. of unknown is equal to the degree of kinetic indeterminacy of the structure.

571. Match the List-I with List-II and select the correct answer using the codes given below the lists :

	List-I (Members)	List-II (Strain energy)
A.	Axial loaded members	$K\int_0^l \frac{V^2}{2GA}dx$
B.	Members under bending	$\int_0^l \frac{T^2}{2GJ}dx$
C.	Members under shearing	$\int_0^l \frac{M^2}{2EI}dx$
D.	Circular members under section	$\int_0^l \frac{P^2}{2EA}dx$

Codes :

	A	B	C	D
(a)	1	4	2	3
(b)	2	1	3	4
(c)	3	2	4	1
(d)	4	3	1	2

OPSC Civil Services Pre 2011

Ans. (d) :

572. The stiffness matrix for a beam element is $k = \frac{EI}{63}\begin{bmatrix} 4 & -9 \\ -9 & 36 \end{bmatrix}$. The corresponding flexibility matrix is
(a) $\frac{63}{EI}\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}$ (b) $\frac{1}{EI}\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}$
(c) $\frac{63}{EI}\begin{bmatrix} -4 & 9 \\ 9 & -36 \end{bmatrix}$ (d) $\frac{1}{EI}\begin{bmatrix} 4 & 9 \\ 9 & 36 \end{bmatrix}$

TRB Poly. Tech. Lect. 2012

Ans. (a) :
The stiffness and flexibility matrix are inverse of each other and both are symmetrical.

Stiffness matrix, $K = \frac{EI}{63}\begin{bmatrix} 4 & -9 \\ -9 & 36 \end{bmatrix}$

Flexibility matrix $= [F]$

$[K][F] = I \Rightarrow [F] = [K]^{-1}$

Flexibility matrix is equal to the inverse of stiffness

$$|K| = \left(\frac{EI}{63}\right)^2 (144-81) = 63\left(\frac{EI}{63}\right)^2$$

$$\text{Adjoin (K)} = \frac{EI}{63}\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}$$

$$\text{So, } [K]^{-1} = \frac{\text{Adj}(K)}{|K|}$$

$$[K]^{-1} = \frac{\frac{EI}{63}\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}}{63\left(\frac{EI}{63}\right)^2}$$

$$= \frac{\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}}{\frac{EI}{63}}$$

$$K = \frac{63}{EI}\begin{bmatrix} 36 & 9 \\ 9 & 4 \end{bmatrix}$$

573. The force required to cause a rotation '□' at the propped end of propped cantilever beam of span *l*, Young's modulus E, moment of Inertia I is
(a) $\frac{12EI}{L^3}\theta$ (b) $\frac{2EI}{L}\theta$
(c) $\frac{4EI}{L}\theta$ (d) $\frac{6EI}{L^2}\theta$

TRB Poly. Tech. Lect. 2012

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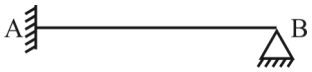
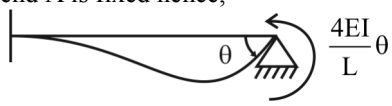
Ans. (c) :



When far end is fixed then required force/moment for θ rotation is $\frac{4EI\theta}{L}$ ——— (A)

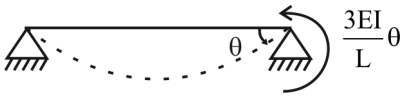
& when far end is hinged then required for θ rotation is $\frac{3EI\theta}{L}$ ———(B)

Here far end A is fixed hence,



$\because \boxed{M = \frac{4EI\theta}{L}}$

Case (B) will be like



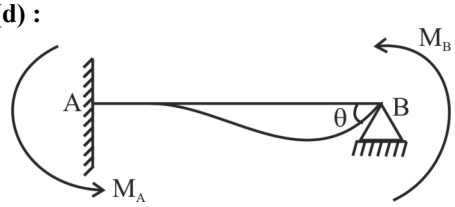
574. If the hinged end of a propped cantilever of span L and flexural rigidity EI undergoes a rotation, then the shear force in the beam will be

(a) $\frac{EI}{L^2}\theta$ (b) $\frac{2EI}{L^2}\theta$

(c) $\frac{3EI}{L^2}\theta$ (d) $\frac{6EI}{L^2}\theta$

TRB Poly. Tech. Lect. 2012

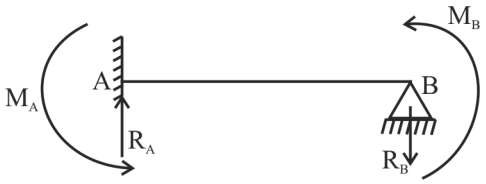
Ans. (d) :



Far end fixed and rotation θ given at hinged (B)

$\because \quad M_B = \frac{4EI}{L}\theta$ (because far end is fixed)

$\because \quad M_A = \frac{2EI}{L}\theta \left(= \frac{M_B}{2} \text{ carry over transfer}\right)$



$\therefore \quad R_A = R_B = \frac{M_A + M_B}{L} = \frac{4EI\theta}{L} + \frac{2EI\theta}{L}$

$\boxed{R_A = R_B = \frac{6EI}{L^2}\theta}$ shear in beam

575. The moment required to rotate the near end of a prismatic beam through a unit angle without translation, when the far end is pinned is given by

(a) 3 EI/L (b) 4 EI/L

(c) 2 EI/L (d) EI/L

TSGENCO_2015

Ans. (a) : The moment required to rotate the near end of a prismatic beam through a unit angle without translation, when the far end is pinned = 3 EI/L.

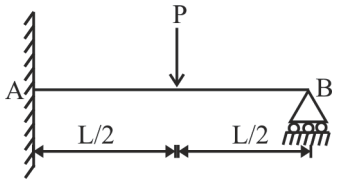
576. What is the magnitude of fixed end moment of a propped cantilever beam having length 'L' carrying a concentrated load 'P' at the mid-span? Take EI = constant

(a) $\frac{3PL}{8}$ (b) $\frac{3PL}{16}$

(c) $\frac{5PL}{16}$ (d) $\frac{PL}{4}$

TSGENCO_2015

Ans. (b) :



Deflection at B due to load P

$$= \frac{P(L/2)^3}{3EI} + \frac{P(L/2)^2}{2EI} \times \frac{L}{2}$$

$$= \frac{PL^3}{24EI} + \frac{PL^3}{16EI}$$

$$= \frac{5PL^3}{48EI}$$

$(\Delta_{\text{due to load}}) - (\Delta_{\text{due to reaction}}) = 0$

$$\frac{5}{48}\frac{PL^3}{EI} - \frac{R_B \times L^3}{3EI} = 0$$

$$R_B = \frac{5}{16}P$$

Moment at A $= \frac{5P}{16} \times 1 - \frac{P \times L}{2}$

$$= \frac{3PL}{16}$$

577. Kani's rotation contribution method is advantageous over Moment distribution method since

(a) Kani's method is iterative

(b) Any arithmetic error that creeps in will automatically get corrected

(c) It involves actual solution of simultaneous equations

(d) None of these

Punjab PSC SDE 2016

Ans. (b) : Kani method is more illustrative as any arithmetic error that comes will automatically get corrected.

578. In Kani's method, an overhang can be conveniently dealt with by regarding it as a member with

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(a) Infinite length (b) Zero length
(c) Unit length (d) None of these

Punjab PSC SDE 2016

Ans. (a) : Kani's method is displacement method, which easily deal with overhang portion by regarding it as a member with infinite length.

579. The analysis of a structure is

(a) Deciding the material of the member
(b) Deciding the dimension of the member
(c) Calculating the magnitude and nature of various straining actions at salient points of the structure
(d) Planning of the structure

Punjab PSC SDE 2016

Ans. (c) : The analysis of a structure is calculating the magnitude and nature of various straining actions at salient points of the structure.

580. The number of simultaneous equation to be solved in the slope deflection method is equal to

(a) the degree of statically indeterminacy
(b) the degree of kinematics
(c) the number of joints in the structure
(d) the number of spans in the structure

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (c) : The number of simultaneous equation to be solved in the slope deflection method is equal to the number of joints in the structure.

581. The moment distribution method in structural analysis can be treated as:

(a) Force method
(b) Displacement method
(c) Flexibility method
(d) None of the above

Arunachal Pradesh Poly. Tech. Lect. 2019
UPPCL AE 2018
Nagaland PSC CTSE 2017 Paper–I
UJVNL AE 2012

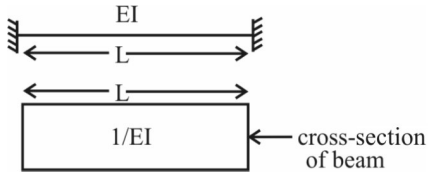
Ans. (b) : The moment distribution method in structural analysis can be treated as displacement method.
Example of displacement method are
- moment distribution method
- slope deflection method
- Kani's method
- Stiffness matrix method

582. In column analogy method, the area of analogous column for a fixed beam of span 'L' and flexural rigidity EI is taken as

(a) $\frac{L}{EI}$ (b) $\frac{L}{2EI}$
(c) $\frac{L}{3EI}$ (d) $\frac{L}{4EI}$

APPSC AEE 2012

Ans. (a) :



$$\text{Area of cross-section beam} = L \times \frac{1}{EI}$$

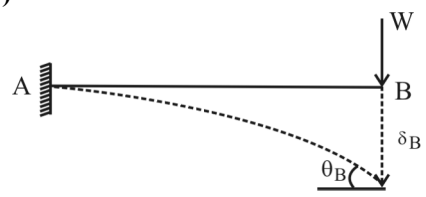
$$= \frac{L}{EI}$$

583. A cantilever of span 'l' carries a load 'W' at the free end. The flexibility of the beam is given by:

(a) $\frac{L^2}{3EI}$ (b) $\frac{L^3}{3EI}$
(c) $\frac{L}{3EI}$ (d) $\frac{L^3}{8EI}$

TRB Poly. Lect. Tech. 2017

Ans. (b)



$$\Delta_B = \frac{WL^3}{3EI}$$

$$\frac{\Delta_B}{W} = \frac{L^3}{3EI}$$

$$\text{flexibility} = \frac{L^3}{3EI}$$

584. Flexibility matrix for a beam element is [F] = $\frac{1}{EI}\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$

(a) $S = \frac{EI}{50}\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$

(b) $S = \frac{EI}{50}\begin{bmatrix} 6 & -2 \\ -2 & 9 \end{bmatrix}$

(c) $S = \frac{EI}{50}\begin{bmatrix} 2 & 9 \\ 6 & 2 \end{bmatrix}$

(d) $S = \frac{EI}{50}\begin{bmatrix} 6 & 2 \\ 9 & 2 \end{bmatrix}$

TRB Poly. Lect. Tech. 2017

Ans. (a) : The stiffness and flexibility matrix are inverse of each other and both are symmetrical.

$$\text{Stiffness matrix, } K = \frac{1}{EI}\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$$

Flexibility matrix = [F]

$[K][F] = I \Rightarrow [F] = [K]^{-1}$

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Flexibility matrix is equal to the inverse of stiffness

$$|K| = \left(\frac{1}{EI}\right)^2 (54-4) = 50\left(\frac{1}{EI}\right)^2$$

$$\text{Adjoin (K)} = \frac{1}{EI}\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$$

$$\text{So, } [K]^{-1} = \frac{\text{Adj}(K)}{|K|}$$

$$[K]^{-1} = \frac{\frac{1}{EI}\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}}{50\left(\frac{1}{EI}\right)^2}$$

$$= \frac{\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}}{50\left(\frac{1}{EI}\right)}$$

$$[K]^{-1} = \frac{EI}{50}\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$$

585. The Final moment at the end A, in a beam AB due to rotations θ_A, θ_B and downward settlement of support at B is given by :

(a) $M_{AB} = M_{FAB} + \frac{2EI}{l}\left[2\theta_A + \theta_B - \frac{3\delta}{L}\right]$

(b) $M_{AB} = M_{FAB} - \frac{2EI}{l}\left[2\theta_A + \theta_B - \frac{3\delta}{L}\right]$

(c) $M_{AB} = M_{FAB} + \frac{4EI}{l}\left[2\theta_A + \theta_B - \frac{3\delta}{L}\right]$

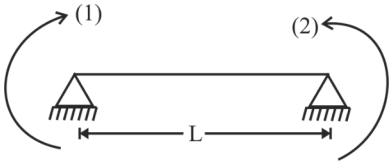
(d) $M_{AB} = M_{FAB} + \frac{2EI}{l}\left[\theta_A + 2\theta_B - \frac{3\delta}{L}\right]$

TRB Poly. Lect. Tech. 2017

Ans. (a) : Final moment at the end A –

$$M_{AB} = M_{FAB} + \frac{2EI}{l}\left[2\theta_A + \theta_B - \frac{3\delta}{L}\right]$$

586. The flexibility matrix for the simply supported beam, as shown below, with reference to the coordinates, is

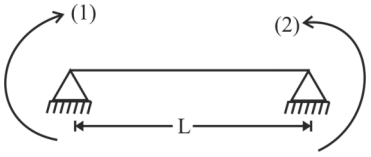


(a) $\frac{L}{6EI}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\frac{L}{3EI}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(c) $\frac{L}{6EI}\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ (d) $\frac{L}{3EI}\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

BPSC Poly. Tech. Lect. 2016

Ans. (a) :



$$\frac{L}{6EI}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

587. If a single-span beam fixed at both ends sinks by Δ at one end, the fixed-end moment developed is

(a) 0 (b) $\frac{6EI\Delta}{l^2}$

(c) $\frac{3EI\Delta}{l^2}$ (d) $\frac{12EI\Delta}{l^2}$

BPSC AE Aug. 2019 Paper-V

Ans. (b) : Let us suppose or given as beam of length l having supports A & B as fixed. Now, if B supports only settles by Δ, then a counterbalance moment is developed at both end of beams.

$$M_{fixed} = \frac{6EI\Delta}{l^2}$$

588. In the moment-area method, the difference in slope between any two sections is equal to

(a) area of M/EI diagram between these two sections

(b) moment of M/EI diagram between these two sections

(c) 0.5 time area of M/EI diagram between these two sections

(d) 0.5 time moment of M/EI diagram between these two sections

BPSC AE Aug. 2019 Paper-V

Ans. (a) : The moment area method is based on two theorems. The first one is the slope change between two points of the beam and second one is related to deflection.

'The change of slope between any two points of the elastic curve is equal to the area of the bending moment diagram between these points divided by EI.

589. The Muller-Breslau principle is the straight application of

(a) Kani's theorem

(b) Maxwell reciprocal theorem

(c) Moment area method

(d) Unit load method

BPSC AE Aug. 2019 Paper-V

Ans. (b) : The Muller-Breslau principle is a method to determine influence lines. This principle states that the influence line of an action assumes the scaled form of displacement or deflection. Muller's Breslau principle is straight application of Maxwell reciprocal theorem.

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590. The flexibility method is also known as

(a) force method
(b) displacement method
(c) energy method
(d) equilibrium method

BPSC AE Aug. 2019 Paper-V
UKPSC AE 2012 Paper–I

Ans. (a) : Flexibility method is also known as matrix force method due to its use of member forces as the primary unknowns.

591. The stiffness matrix of a beam is given as:

$$K \times \begin{bmatrix} 12 & 4 \\ 4 & 5 \end{bmatrix}$$

Calculate the flexibility matrix.
Flexibility matrix will be .....

(a) $\frac{K}{44}\begin{bmatrix} 12 & -4 \\ -4 & 5 \end{bmatrix}$ (b) $\frac{K}{44}\begin{bmatrix} 12 & 4 \\ 4 & 5 \end{bmatrix}$

(c) $\frac{1}{44K}\begin{bmatrix} 12 & -4 \\ -4 & 5 \end{bmatrix}$ (d) $\frac{1}{44K}\begin{bmatrix} 5 & -4 \\ -4 & 12 \end{bmatrix}$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) The stiffness and flexibility matrix are inverse of each other and both are symmetrical.

Stiffness matrix, $F = K\begin{bmatrix} 12 & 4 \\ 4 & 5 \end{bmatrix}$

Flexibility matrix $= [S]$

$[F][S] = I \Rightarrow [S] = [F]^{-1}$

Flexibility matrix is equal to the inverse of stiffness

$|F| = (K)^2(60-16) = 44(K)^2$

Adjoin (F) $= K\begin{bmatrix} 5 & -4 \\ -4 & 12 \end{bmatrix}$

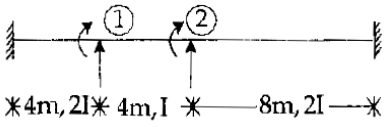
$$\text{So, } [F]^{-1} = \frac{\text{Adj}(F)}{|F|}$$

$$[F]^{-1} = \frac{K\begin{bmatrix} 5 & -4 \\ -4 & 12 \end{bmatrix}}{44(K)^2}$$

$$= \frac{\begin{bmatrix} 5 & -4 \\ -4 & 12 \end{bmatrix}}{44K}$$

$$[F]^{-1} = \frac{1}{44K}\begin{bmatrix} 5 & -4 \\ -4 & 12 \end{bmatrix}$$

592. Displacement coordinators for the beam are as shown in figure. The stiffness matrix is given by:



(a) $EI\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $EI\begin{bmatrix} 3 & -0.5 \\ -0.5 & 2 \end{bmatrix}$

(c) $EI\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ (d) $EI\begin{bmatrix} 3 & 0.5 \\ 0.5 & 2 \end{bmatrix}$

Maharashtra PSC (Mains) 2019 Paper–I
ESE 2000

Ans. (d) : Stiffness matrix can be obtained by making second co-ordinate i.e. rotation in the direction of 2 as zero and considering unit rotation in the direction of co-ordinate 1.

The beam AB has rotational stiffness

$\frac{4E(2I)}{4} = 2EI$ and beam BC has rotational stiffness

$\frac{4E(2I)}{4} = EI$ So moment required in the direction of 1 of produce unit rotation will be 2EI + EI = 3 EI

Thus K11= 3EI

The moment generated at point C in the direction of co-ordinate 2 is $\frac{1}{2} \times EI = 0.5EI$ as the carry over factor is half, So $K_{21} = 0.5$ EI

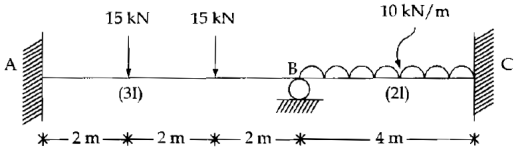
Similarly make $D_1 = 0$ and $D_2 = 1.0$

$$K_{22} = \frac{4EI}{4} + \frac{4E(2I)}{8} = 2EI$$

$$K_{12} = 0.5\,EI$$

$$\text{Stiffness matrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} = \begin{bmatrix} 3EI & 0.5EI \\ 0.5EI & 2EI \end{bmatrix}$$

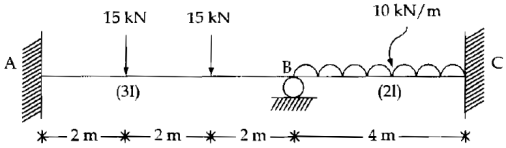
593. A two span continuous beam ABC is as shown in figure below. The distribution factors at joint B are:



(a) 0.4, 0.6 (b) 0.6, 0.4
(c) 0.5, 0.5 (d) 0.55, 0.45

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (c) :



Joint	Member	Stiffness	Total Stiffness	D.F.
B	BA	$\frac{4E(3I)}{6} = 2I$	4 EI	$\frac{2}{4} = 0$

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	BC	$\frac{4E(2I)}{4}=2I$		$\frac{2}{4}=0$

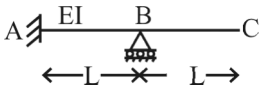
594. The absolute stiffness of a prismatic member with one end fixed is:

(a) $\frac{4EI}{L}$ (b) $\frac{2EI}{L}$

(c) $\frac{4EI}{\sqrt{2L}}$ (d) $4\sqrt{2}\frac{EI}{L}$

PTCUL AE 2016

Ans. (a) :



Stiffness $K_{AB} = \frac{4EI}{\ell}$

$K_{BC} = \frac{3EI}{L}$

595. Which of the following methods of structural analysis is a force method

(a) Slope deflection method
(b) Moment distribution method
(c) Three moment equation
(d) None of the above

UJVNL AE 2016

Ans. (c) : Force method of structure analysis areas follows :

- Strain energy method
- Castigliano's method
- Flexibility method
- Three moment equation

596. In slope deflection equations, the deformations are considered to be caused by

(a) Axial force (b) Shear force
(c) Bending moment (d) All the above

APPSC AEE 2016

Ans. (c) : By forming slope deflection equations and applying joint and shear equilibrium conditions, the rotation angles are calculated. Substituting them back into the slope deflection equations, member and moment are ready determined. Deformation of member is due to bending moment.

597. Virtual work means

A. Work done by real forces due to hypothetical displacements

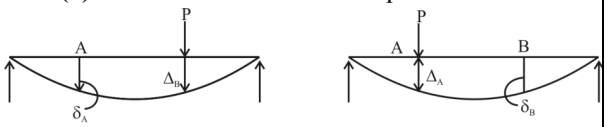
B. Work done by hypothetical forces during real displacements

Which of these statement are correct?

(a) Both A and B are correct
(b) A is correct and B is wrong
(c) A is wrong and B is correct
(d) Both A and B are wrong

UPRVUNL AE 2015

Ans. (a) : Maxwell theorem or Reciprocal deflection



Deflecltion at A due to load at B

Deflecltion at B due to load at A

598. Which of the following is not the displacement method ?

(a) Equilibrium method
(b) Column analogy method
(c) Moment distribution method
(d) Kani's method

MPPSC AE 2014

Ans. (b) : Column analogy method is a force method.

599. If DL denotes dead load and EL denotes earthquake load, then the value of the load factor *a* is the load combination *a*DL+1.5EL used in structural design is_____:

(a) 1.0 (b) 2.0
(c) 0.9 (d) 1.2

UPPCL AE 2018

Ans. (c) : Given,
DL = Dead load
EL = Earthquake load
So, load combination is given as
0.9 DL +1.5 EL

600. Consider the following two statements:

Statement I: In a statically determinate structure, influence line diagrams for internal forces are always composed of straight line segments.

Statement II: Muller-Breslau's principle is not applicable for statically indeterminate structures.

Which of the following is CORRECT?

(a) Statement I is true, but Statement II is false
(b) Both Statement I and Statement II are true
(c) Both Statement I and Statement II are false
(d) Statement I is false, but Statement II is true

UPPCL AE 2018

Ans. (a) :
Statement I: In a statically determinate structure, influence line diagrams for internal forces are always composed of straight line segments.
Statement II: Muller-Breslau's principle is not applicable for statically indeterminate structures.

601. Which of the following theorem states that "The displacement at Point A due to load at B is same as displacement at point B due to the same load acting at point A, the displacements being measured in the direction of loads"?

(a) Maxwell's theorem of reciprocal deflection.

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(b) Castigliano's theorem of reciprocal deflection
(c) Rayleigh Ritz theorem of reciprocal deflection
(d) Muller-breslau theorem of reciprocal deflection

UPPCL AE 2019

Ans. (a) : Maxwell's reciprocate theorem state that in a linearly elastic structure, the deflection at any point. A due to a load applied at some other point B will be equal to the deflection at B when the same load is applied at A.

602. In moment distribution method, the distribution factor at any point can be calculated by, using which of following expression; where, k is stiffness of the member, M is moment in member due to loads.

(a) $k/\Sigma k$ (b) $(k/\Sigma k) \times M$
(c) $(k/M) \times \Sigma k$ (d) $\Sigma k/k$

LMRC AM 2018

Ans. (a) : In moment distribution method, the distribution factor at any point can be calculated by using $k/\Sigma k$.
where, k is stiffness of the member.

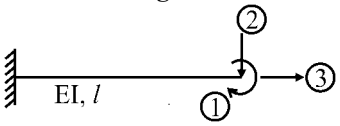
603. Flexibility matrix method is also known as:

(a) Equilibrium method
(b) Displacement method
(c) Compatibility method
(d) Stiffness method

CIL MT 2020

Ans. (c) : Compatibility method. Flexibility matrix method is also known as compatibility method.

604. Stiffness matrix with reference to coordinates 1, 2 and 3 as shown in fig. below is:



(a) $[K] = \begin{bmatrix} 4EI/\ell & -6EI/\ell^2 & 0 \\ -6EI/\ell^2 & 12EI/\ell^3 & 0 \\ 0 & 0 & AE/\ell \end{bmatrix}$

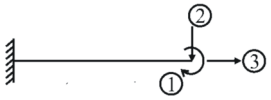
(b) $[K] = \begin{bmatrix} -6EI/\ell^2 & 4EI/\ell & 0 \\ 4EI/\ell & 12EI/\ell^3 & 0 \\ 0 & 0 & AE/\ell \end{bmatrix}$

(c) $[K] = \begin{bmatrix} 0 & 4EI/\ell & -6EI/\ell^2 \\ 4EI/\ell & 0 & 12EI/\ell^3 \\ 0 & 0 & AE/\ell \end{bmatrix}$

(d) $[K] = \begin{bmatrix} 0 & 4EI/\ell & -6EI/\ell^2 \\ 4EI/\ell & 0 & 12EI/\ell^3 \\ AE/\ell & 0 & 0 \end{bmatrix}$

[UPPSC AE 12.04.2016 Paper–I]

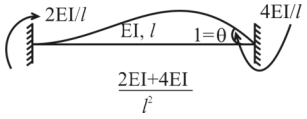
Ans : (a)



Stiffness matrix–

$$[K] = \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix}$$

from matrix Ist column



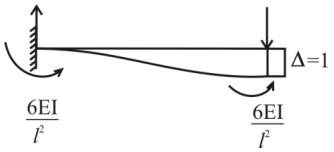
$$K_{11} = \frac{4EI}{\ell}$$

$$K_{21} = \frac{-\left(\frac{4EI}{\ell} + \frac{2EI}{\ell}\right)}{\ell}$$

$$K_{21} = \frac{-6EI}{\ell^2}$$

$$K_{31} = 0$$

from matrix IInd column-



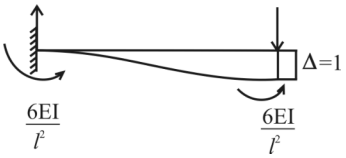
$$K_{12} = \frac{-6EI}{\ell^2}$$

$$K_{22} = \frac{\left(\frac{6EI}{\ell^2} + \frac{6EI}{\ell^2}\right)}{\ell}$$

$$K_{22} = \frac{12EI}{\ell^3}$$

$$K_{32} = 0$$

from matrix 3rd column-



$$K_{13} = 0, K_{23} = 0, K_{33} = \frac{AE}{\ell}$$

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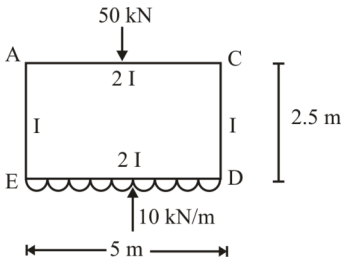
$$[K]=\begin{bmatrix} 4EI/\ell & -6EI/\ell^2 & 0 \\ -6EI/\ell^2 & 12EI/\ell^3 & 0 \\ 0 & 0 & AE/\ell \end{bmatrix}$$

605. The distribution factors members AE and AC of the box section are :



(a) 0.5 and 0.5 (b) 0.25 and 0.75
(c) 0.6 and 0.4 (d) 0.63 and 0.37

TRB Poly. Lect. Tech. 2017

Ans. (a) :
Stiffness of member AC,

$$K_{AC}=\frac{4E(2I)}{L}$$

$$K_{AC}=\frac{4E(2I)}{5}=\frac{8}{5}EI$$

Stiffness of member AE,

$$K_{AE}=\frac{4EI}{2.5}=\frac{8}{5}EI$$

Distribution factor for member AE,

$$D_{AE}=D_{AE}=\frac{K_{AE}}{K_{AE}+K_{AC}}=0.5$$

Distribution factor for member AC,

$$D_{AC}=1-0.5=0.5$$

606. A beam element which is fixed at support A and roller at Support B. What will be slope defection equation at support A?

(a) $M_{AB}=M_{FAB}+\frac{2EI}{l\left[\theta_A+2\theta_B+\frac{3\delta}{l}\right]}$

(b) $M_{AB}=M_{FAB}+\frac{4EI}{l\left[2\theta_A+\theta_B+\frac{3\delta}{l}\right]}$

(c) $M_{AB}=M_{FAB}+\frac{2EI}{l\left[2\theta_A+\theta_B+\frac{3\delta}{l}\right]}$

(d) $M_{AB}=M_{FAB}+\frac{EI}{l\left[2\theta_A+\theta_B+\frac{3\delta}{l}\right]}$

(e) $M_{AB}=M_{FAB}+\frac{EI}{l\left[2\theta_A+\theta_B+\frac{6\delta}{l}\right]}$

CGPSC AE 2017

Ans. (c) $M_{AB}=M_{FAB}+\frac{2EI}{l\left[2\theta_A+\theta_B+\frac{3\delta}{l}\right]}$

607. Stiffness matrix method is also known as :

(a) flexibility method
(b) force method
(c) compatibility method
(d) equilibrium method
(e) conjugate beam method

CGPSC AE 2017

Ans. (d) Stiffness matrix method is also known as equilibrium method.

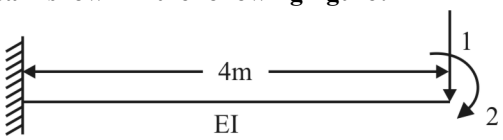
608. Identify the correct statement among the following.

(a) At the location of the plastic hinge of a deformed structure curvature is infinite.
(b) At the location of the plastic hinge of a deformed structure radius of curvature is infinite.
(c) At the location of the plastic hinge of a deformed structure moment is infinite
(d) At the location of the plastic hinge of a deformed structure flexural stress is infinite

UPRVUNL AE 2014

Ans. (a) : At the location of the plastic hinge of a deformed structure curvature is infinite.

609. What would be the flexibility matrix of the beam shown in the following figure?



(a) $\begin{bmatrix} \frac{64}{EI} & -\frac{8}{EI} \\ -\frac{8}{EI} & \frac{8}{3EI} \end{bmatrix}$ (b) $\begin{bmatrix} \frac{64}{3EI} & \frac{8}{EI} \\ \frac{8}{EI} & -\frac{64}{3EI} \end{bmatrix}$

(c) $\begin{bmatrix} \frac{64}{3EI} & \frac{8}{EI} \\ \frac{8}{EI} & \frac{4}{EI} \end{bmatrix}$ (d) $\begin{bmatrix} \frac{64}{3EI} & \frac{8}{EI} \\ \frac{4}{EI} & \frac{8}{EI} \end{bmatrix}$

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ESE 2000

Ans. (c) : Elements of flexibility matrix can be obtained by applying unit force in the direction of anyone co-ordinate and calculating displacement in the direction of both co-ordinates.

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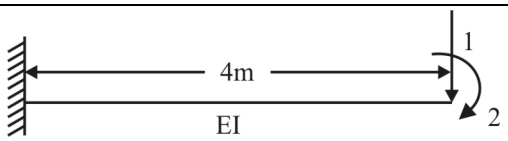

Applying unit load in the direction of a.

Deflection at B,

$$f_{11} = \frac{1\times 4^3}{3EI} = \frac{64}{3EI}$$

Rotation at B $f_{21} = \frac{1\times 4^2}{2EI} = \frac{8}{EI}$

The positive sign is used when the displacement is in the direction of co-ordinate, otherwise negative sign is used.

Applying unit moment in the direction of 2

Deflection at B

$$f_{12} = \frac{1\times 4^2}{2EI} = \frac{8}{EI}$$

Rotation at B,

$$f_{22} = \frac{1\times 4}{EI} = \frac{4}{EI}$$

So, flexibility matrix $= \begin{bmatrix} \frac{64}{3EI} & \frac{8}{EI} \\ \frac{8}{EI} & \frac{4}{EI} \end{bmatrix}$

610. Among the following mentioned statements regarding the stiffness method of analysis, identify those which are correct.

1. **Statically indeterminate structures alone are considered.**
2. **Both statically determinate and indeterminate structures can be considered.**
3. **It is necessary to identify the redundant.**
4. **The displacements are considered as the unknowns.**
5. **To calculate displacements and establish the equations of equilibrium, it will be convenient to develop the stiffness matrix for the structure.**

choose the correct answer from the options given below.

(a) 1,3 and 4 (b) 1,3 and 5
(c) 2,4 and 5 (d) 2,3,4 and 5

UPRVUNL AE 2014

Ans. (c) : Regarding the stiffness method of analysis :

- Both statically determinate and indeterminate structure can be considered.
- The displacement considered as the unknowns.
- To calculate displacements and establish the equations of equilibrium, it will be convenient to develop the stiffness matrix for the structure.

611. The_____ method of structural analysis is/are force method/s:

(a) Column analogy
(b) Moment distribution
(c) Slope deflection
(d) both slope deflection and moment distribution

UPRVUNL AE 2014

Ans. (a) : Examples of force method :

- Strain energy method or minimum potential energy method.
- Castigliano's theorem
- Unit load method
- Virtual work method
- Column analogy method
- Three moment method
- Flexibility method

612. On the actual beam, bending moment at any section in a conjugate beam gives-

(a) slope (b) curvature
(c) deflection (d) bending moment

JPSC AE 2009

Ans. (c) : On the actual beam, bending moment at any section in a conjugate beam method gives-

B.M. → deflection

Shear force → slope.

613. Match List 1 with List 2 and select the correct answer using the code given below the list:

List 1	List 2
A. Maxwell's diagram	1. Displacement of joints in a truss
B. Hinesburg's method	2. Deflection by numerical procedure
C. Newmark's method	3. Forces in bar members
D. Williot-Mohr diagram	4. concept of substitute members

	A	B	C	D
(a)	1	2	4	3
(b)	3	4	2	1
(c)	3	2	4	1
(d)	1	4	2	3

Chandigarh AE 2017

Ans. (b) :	
A. Maxwell's diagram	Forces in bar members
B. Hinesburg's method	concept of substitute members
C. Newmark's method	Deflection by numerical procedure
D. Williot-Mohr diagram	Displacement of joints in a truss

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06.

Design of Cement Concrete and Pre-stressed Concrete

1. Plain and Reinforced Cement Concrete

1. The striping time for the Reinforced Concrete slab spanning less than 4.5 m and using fly ash as an admixture should be–

(a) 7 days (b) 10 days
(c) 14 days (d) 21 days

Gujarat PSC AE (N.W.R.) 2020

Ans. (b) : The striping time for the Reinforced Concrete slab spanning less than 4.5 m and using fly ash as an admixture should be 10 days.

2. As per IS 456 : 2000, minimum period before striking formwork for vertical surface of the columns

(a) 1 days (b) 7 days
(c) 14 days (d) 28 days

Gujarat PSC AE 2017

Ans : (a)

Type of formwork	Minimum period before striking formwork
1. Vertical sides of the slabs, beams, columns, and walls	16 to 24 hours
2. Soffit formwork to slabs (props to be refixed immediately after removal of formwork)	3 days
3. Soffit formwork to beams (props to be refixed immediately after removal of formwork)	7 days

3. Consider the following strength of concrete

(a) Cube strength
(b) Cylinder strength
(c) Split tensile strength
(d) Modulus of rupture

The correct sequence in increasing order of these strength.

(a) 3-4-1-2 (b) 4-3-1-2
(c) 3-4-2-1 (d) 4-3-2-1

TNPSC AE 2019

Ans. (d)
Cylinder strength = 0.80 cube strength
Split tensile strength → Modulus of rupture → cylinder strength → cube strength

4. The effect of air entrainment in concrete is to

1. increase resistance to freezing and thawing
2. improve workability
3. decrease strength

Which of these statements is/are correct?

(a) 1 alone (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3

TNPSC AE 2019

Ans. (d) Air entrainment in a concrete increase resistance to freezing and thawing provide moisture i.e., water therefore increase workability and strength decrease.

$$\uparrow \frac{w}{c}\text{ratio} \propto \text{workability} \uparrow \propto \frac{1}{\text{strength} \downarrow}$$

5. The minimum and maximum percentages of longitudinal reinforcement in columns according to IS 456-2000 are_____ respectively.

(a) 0.8% and 0.3% (b) 1% and 8%
(c) 0.8% and 4% (d) 0.8% and 6%

TNPSC AE 2019, 2012

Ans. (d) According to code IS 456 : 2000 clause 26.5.3.1 (a) the cross-sectional area of longitudinal reinforcement, shall be not less than 0.8% and not more than 6% of the gross cross-sectional area of column.

6. Maximum area of compression reinforcement for beams is

(a) 0.04 bD
(b) 0.85 bd/fy
(c) 0.1 percent of web area
(d) 0.5 percent of web area

Gujarat PSC AE 2017

Ans : (a) longitudinal reinforcement:
⇒ Minimum percentage of reinforcement should be

$$A_s = \frac{0.85\,bd}{f_y}$$

b = width of beam
d = effective depth
f_y = characteristics strength of reinforcement (N/mm^2)
⇒ Maximum percentage of reinforcement (both compression and tension steel) should be 4% of cross-sectional area of beam

$$A_s = 0.04\ bD$$

7. As per IS 456 : 2000, minimum grade of concrete for plain concrete exposed to sea coast is

(a) M 20 (b) M 30
(c) M 40 (d) M 25

Gujarat PSC AE 2017

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Ans : (a) As per clause 8.2.8 of IS 456 : 2000 concrete in sea water or exposed directly along the sea coast shall be atleast M20 grade in the case of plain concrete and M 30 in case of reinforced concrete.

8. According to IS 456 : 2000, the characteristic strength of concrete is defined as the strength of material below which not more than *x* percent of the test results are expected to fall. The value of *x* is :

(a) 0.75 (b) 0.05
(c) 10.00 (d) 7.50
(e) 5.0

[UPPSC State Eng. A.E. 2008]
CGPSC AE 2014 Shift–I

Ans. (e) : As per IS 456-2000,
Characteristic strength of concrete is the value of strength of concrete below which not more than 5% of test results are expected to fall.

9. Which of the following loads need not be considered to be acting simultaneously?

(a) Wind and Seismic
(b) Wind and dead
(c) Wind and shrinkage
(d) Seismic and temperature
(e) Temperature and shrinkage

CGPSC AE 2014 Shift–I

Ans. (a) : As per IS 456-2000, load combination as follows :
1- D.L. + L.L.
2- D.L. +(W.L.) OR (E.L.)
(i) For normal case, DL + WL (or EL)
(iii) for checking stability against overturning/ stress reversal. DL + WL (0.6 L)
3. DL + LL + WL (or) EL

10. According to IS 1343, the flexural strength of a concrete member having a measured compressive strength of 50 MPa will approximately be equal to

(a) 5 MPa (b) 7 MPa
(c) 7.5 MPa (d) 10 MPa
(e) 4.5 MPa

CGPSC AE 2014 Shift–I

Ans. (a) : Flexural strength of concrete = $0.7\sqrt{f_{ck}}$
$= 0.7\sqrt{50} = 4.49 \simeq 5\text{MPa}$

11. The nominal cover requirements for meeting the durability requirement of mild, very severe, severe, moderate, extreme types of exposure are respectively in mm

(a) 20, 30, 45, 50, 75 (b) 20, 45, 30, 50, 75
(c) 20, 50, 45, 30, 75 (d) 20, 75, 50, 30, 45
(e) 20, 30, 60, 75, 90

CGPSC AE 2014 Shift–I

Ans. (c) : As per IS 456 : 2000

Exposure	Nominal cover to meat durability requirement (mm)
Mild	20 mm
Moderate	30 mm
Severe	45 mm
Very severe	50 mm
Extreme	75 mm

12. As per IS 456, the weighing or batching accuracy (±) of cement, aggregate, water and admixture shall respectively be

(a) 2, 2, 2, 2 (b) 2, 3, 3, 3
(c) 2, 2, 3, 3 (d) 2, 2, 3, 3
(e) 3, 2, 2, 2

CGPSC AE 2014 Shift–I

Ans. (b) : Batching of material

Material	Accuracy of measurement % of batch quantity
Cement	± 1 to 3
Aggregate	± 2 to 3
Water	± 1 to 3
Admixture	± 3 to 5

13. As per IS 456, nominal mix concrete may be used for concrete of grades upto

(a) 7.5 MPa (b) 10 MPa
(c) 15 MPa (d) 20 MPa
(e) 25 MPa

CGPSC AE 2014 Shift–I

Ans. (d) : As per IS code 456-2000
Nominal mixture concrete may be used for concrete of grade up to M-20.

14. Poisson's ratio for concrete

(a) remains unchanged
(b) increases with richer mixes
(c) decreases with richer mixes
(d) none of these

TNPSC AE 2012
[UPPSC State Eng. AE. 2008]
GESCOM (KARNATAKA) AE 2011

Ans. (c) Poisson's ratio of concrete decrease with richer mixes as you can judge it from ranges given below-
⇒ 0.1 for high strength concrete
⇒ 0.15 for standard concrete
⇒ 0.2 for low strength concrete

15. Modulus of elasticity of steel shall be taken as :

(a) 200 kN/mm^2 (b) 2×10^3 N/mm^2
(c) 2×10^4 N/mm^2 (d) 2×10^2 N/mm^2

TNPSC AE 2015

Ans. (a) As per IS 456- 1978 the modulus of elasticity of steel, shall be taken as 200 kN/mm^2.

16. The grade of concrete generally not used in the reinforced concrete is :

(a) M40 (b) M25
(c) M20 (d) M10

TNPSC AE 2015

Ans. (d) Minimum grade of concrete for RCC is M20 which is the minimum grade of concrete for 1 part of cement, 1.5 parts of fine aggregate and 3 parts of coarse aggregate for RCC work dose not use M10 concrete.

17. In M20 & M25, the number is characteristic compressive strength in N/mm^2 at _____ days:

(a) 7 (b) 14
(c) 21 (d) 28

TNPSC AE 2015

Ans. (d) In M20 & M25, the number is characteristic compressive strength in N/mm^2 at 28 days.

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18. In the designation of concrete mix, the letter 'M' and the number stands for :

(a) Mix and characteristic compressive strength of 100 mm cube at 28 days
(b) Mix and characteristic compressive strength of 75 mm cube at 28 days
(c) Mix and characteristic compressive strength of 150 mm cube at 28 days
(d) Mix and characteristic compressive strength of 125 mm cube at 28 days

TNPSC AE 2015

Ans. (c) The compressive strength of concrete is given in terms of the characteristic compressive strength of 150 mm size cubes tested after 28 days (f_{ck}). The characteristic of the concrete below which not more than 5% of the test results are expected to fall.

19. The relation between modulus of rupture (f_{cr}) and characteristic compressive strength (f_{ck}) is given by :

(a) $f_{cr} = 0.7\ f_{ck}$ (b) $f_{cr} = 0.7\sqrt{f_{ck}}$
(c) $f_{cr} = 0.75\ f_{ck}$ (d) $f_{cr} = 0.7/\sqrt{f_{ck}}$

Vizag steel MT-2017
TNPSC AE 2015

Ans. (b) Flexural strength (f_{cr}) = $0.7\sqrt{f_{ck}}$ N/mm^2 where f_{ck} is the characteristic cube compressive strength of concrete in N/mm^2.

20. A structural member is generally designed so that the material is stressed up to :

(a) Yield stress (b) Ultimate stress
(c) Breaking stress (d) Working stress

TNPSC AE 2015

Ans. (d) A structural member is generally designed so that the material is stressed to working stress.

21. What will be the minimum yield strength (MPa) of Fe 500 steel used as reinforcement in concrete?

(a) 550 (b) 500
(c) 415 (d) 250

CIL MT 2017

Ans. (b) :

Property	Min. value (Fe 500)
Yield stress (N/mm^2)	500
Ultimate tensile stress	545
UTS/YS Ratio	1.08
Elongation	12

UTS = ultimate tensile stress
YS = yield stress

22. What will be the estimated value of modulus of elasticity (GPa) for a concrete of compressive strength 25 MPa?

(a) 25 (b) 27.5
(c) 30 (d) 32.5

CIL MT 2017

Ans. (a) Modulus of Elasticity (E_c) –

$$\left[E_c = \frac{10^5}{2.2 + \frac{34.7}{f_{ck}}}\right]$$

$f_{ck} = 25\text{MPa}$

$$E_c = \frac{10^5}{2.2 + \frac{34.7}{25}}$$

$$E_c = \frac{10^5}{3.588}$$

$E_c = 2.76 \times 10^4$
$E_c = 27.6$ GPa

23. Concrete exposed to severe freezing-thawing condition shall be classified as exposed to :

(a) Moderate (b) Extreme
(c) Severe (d) Very severe

CIL MT 2017

Ans. (d) : Concrete exposed to severe freezing-thawing condition shall be classified as exposed to very severe.

Exposure Category	Description
Mild	Protected against weather or aggressive conditions, except those located in coastal area.
Moderate	Sheltered from severe rain or freezing whilst wet, or exposed to condensation and rain, or continuously under water, or in contact with or buried under non-aggressive solid or ground water, or sheltered from saturated 'salt air' in coastal area.
Severe	Exposed to severe rain, alternate wetting and drying or occasional freezing whilst wet or severe condensation, or completely immersed in sea water, or exposed to coastal environment.
Very severe	Exposed to sea water spray, corrosive fumes or severe freezing, whilst wet, or in contact with or buried under aggressive sub-soil or ground water.
Extreme	Members in tidal zone, or members in direct contact with liquid/solid aggressive chemicals.

24. In reinforced concrete construction, the structures in which marked changes in plane dimension take place abruptly shall be provided with :

(a) contraction joints (b) vertical cut
(c) construction joints (d) expansion joints

CIL MT 2017

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Ans. (d) : In reinforced concrete construction, the structure in which marked changes in plane dimensions take place abruptly shall be provided with expansion joints.
Expansion joints : Expansion joints are provided in the concrete structure due to volume change tendency of concrete for various reasons like temperature.
Construction joints : Construction joints are provided whenever the construction works stops temporarily.
These are vertical saints cut into the wall that allows the concrete to shrink without noticeable harm.
Contraction joints : Contraction joints are the joints which are given in any concrete structure due to the shrinkage of concrete.

25. In reinforced cement concrete construction; lap splices are not recommended to be used for rebars when the bar diameter is more than :
(a) 32 mm (b) 30 mm
(c) 36 mm (d) 25 mm

CIL MT 2017

Ans. (c) : In reinforced cement concrete construction, Lap splices are not recommended to be used for rebars. when the bar diameter is more than 36 mm.

26. Which of the following statements is/are true for a natural building material like bamboo during an earthquake?
(i) It will result in a flexible structure, however needs to be designed stiff enough to resist earthquake forces
(ii) Since it has a good specific strength (MPa/kg), the earthquake resistance is better
(iii) Being light in weight, it will experience lesser earthquake forces
(a) Only (i) (b) (i) and (ii)
(c) (ii) and (iii) (d) (i), (ii) and (iii)

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) : For a natural building material like bamboo during an earthquake -
- It will result in a flexible structure, however needs to be designed stiff enough to resist earthquake forces.
- Since it has a good specific strength (MPa/kg), the earthquake resistance is better.
- Being light in weight, it will experience lesser earthquake forces.

27. In a reinforced concrete section, as the percentage of steel increases-
(a) the depth of neutral axis decreases
(b) the depth of neutral axis increases
(c) the lever arm increases
(d) both depth of the neutral axis and the lever arm decreases

Gujarat PSC AE (N.W.R.) 2020
RPSC POLY. TECH. LECT. 2014

Ans. (b) : In a reinforced concrete section, as the percentage of steel increases the depth of neutral axis increases.
Force of compression = force of tension
$\Rightarrow 0.36\ \sigma_{ck} b x = 0.87\ \sigma_y A_t$
$\Rightarrow \boxed{x = \dfrac{0.87\ \sigma_y A_t}{0.36\ \sigma_{ck}.b}}$

28. If the permissible compressive and tensile stresses in a singly reinforced beam are 20 MPa and 250 MPa respectively, and if the neutral axis depth is limited to 0.3 times the effective depth of the section, then the percentage area of the steel required for an economic section is
(a) 0.56% (b) 0.76%
(c) 0.86% (d) 0.96%

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) : In case of balanced section :
Compression force = Tension force
$0.36\ f_{ck}\ .b.x_u = 0.87\ f_y\ .A_{st}$

$$\frac{A_{st}}{b.d}\times 100 = \frac{0.36 f_{ck}}{0.87 f_y}\left(\frac{x_u}{d}\right)\times 100$$
$$= \frac{0.36\times 20\times 0.3}{0.87\times 250}\times 100$$
$$= 0.99\%$$
$$\simeq 0.96$$

29. Effective moment of inertia for short term deflection of a reinforced concrete section does not depend on–
(a) the maximum moment under the service load
(b) depth of the neutral axis
(c) breadth of the compression face
(d) ultimate shrinkage strain of concrete.

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) Effective moment of inertia for short term deflection of a reinforced concrete section depend on **:**
(a) The maximum moment under the service load.
(b) Depth of the neutral axis.
(c) Breadth of the compression face.

30. The cover of longitudinal reinforcing bar in a beam subjected to sea spray should not be less than which one of the following?
(a) 25 mm (b) 40 mm
(c) 70 mm (d) 90 mm

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) : The cover of longitudinal reinforcing bar in a beam subjected to sea spray should not be less than 70 mm. Minimum cover require is 50 mm. most appropriate answer would be 70 mm.

31. Which of the following statements is correct?
(a) The factor of safety for steel is lower than that for concrete
(b) The factor of safety for steel is higher than that for concrete
(c) The factor of safety for steel and concrete are the same
(d) None of the above

TNPSC AE 2012
GESCOM (KARNATAKA) AE 2011

Ans. : (a) The partial safety factor for steel (1.15) is comparatively lower than that of concrete (1.5) because the steel for reinforcement is produced in steel plant and commercially available in specific diameters with expected better quality control than that of concrete.

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32. The extent of carbon in wrought iron is up to

(a) 0% (b) 0.25%
(c) 0.5% (d) 1%

GESCOM (KARNATAKA) AE 2011

Ans. : (b) Wrought Iron–It's the purest form of iron and contains carbon to the extend of 0.25%.
Cast Iron (Pig Iron)–
It contains 2-5% carbon along with traces of other impurities like sulphur, phosphorous, Manganese etc.
Steel–It contains 0.5 to 1.5% of carbon along with varying amount of other elements.

33. In construction terminology, what does RCC stand for?

(a) Reinforced Concrete Cement
(b) Reinforced Cement Concrete
(c) Reused Cement Concrete
(d) Reused Concrete Cement

GESCOM (KARNATAKA) AE 2011

Ans. : (b) : R.C.C. in construction terminology, stands for Reinforced Cement Concrete

34. As per IS-456 : 2000, Side face reinforcement shall be provided along the two faces of a reinforced concrete beam, when the depth of the web exceeds :

(a) 500 mm (b) 750 mm
(c) 250 mm (d) 400 mm

Rajasthan AE (Nagar Nigam) 2016
CGPSC AE 2014 Shift–I

Ans. (b) : As per IS 456:2000 clause 26.5.1.3 gives the guidelines for the provision of side face reinforcement in beam in which web thickness is greater than 750mm.
Side face reinforcement is provided to ensure lateral stability as well as confinement to increase the ductility of beam member.

35. In a cantilever beam, tensile or main reinforcement is provided :

(a) On the bottom or lower face of the beam
(b) On the top or upper face of the beam
(c) On both the top and bottom of the beam
(d) In the middle of the beam

Rajasthan AE (Nagar Nigam) 2016

Ans. (b) : In cantilever beam, the upper fibres of the section are subjected to tensile stress and lower fibres are subjected to compressive stress. Hence the main reinforcement is provided at the upper face and not at the lower face.

36. As per IS 456-2000, When HYSD bars are used, the minimum reinforcement in either direction of slabs is :

(a) 0.35% of the total cross sectional area
(b) 0.12% of the total cross sectional area
(c) 0.80% of the total cross sectional area
(d) 0.25% of the total cross sectional area

Rajasthan AE (Nagar Nigam) 2016

Ans. (b) : Minimum reinforcement for slab –
$A_{st(min)}$ = 0.15% × Gross sectional Area [Fe-250 bar]
$A_{st(min)}$ = 0.12% × Gross Sectional Area (HYSD Bar]

37. At the time of bending of a beam, a layer is neither elongated nor contracted is.

(a) Axis of symmetry (b) Neutral axis
(c) Axis of moment (d) Axis of load

LMRC AM 2019

Ans. (b) At the time of bending of a beam, a layer is neither elongated nor contracted is Neutral axis.

38. The maximum compressive strain in concrete in axial compression should be taken as _____.

(a) 0.02 (b) 0.002
(c) 2 (d) 0.2

LMRC AM 2019

Ans. (b) The maximum compressive strain in concrete in axial compression is taken as 0.002

39. For an under reinforced beam section, the neutral axis lies _____.

(a) At the same level of critical neutral axis of balanced section.
(b) Below the critical neutral axis of balanced section
(c) At mid depth of beam
(d) Above the critical neutral axis of balanced section

LMRC AM 2019

Ans. (d) For under reinforced beam. The neutral axis lies above the critical neutral axis of balanced section.

40. Deformation in concrete due to sustained loading is :

(a) Shrinkage (b) Creep
(c) Segregation (d) Bleeding

KPSC AE 2020

Ans. (b) : Deformation in concrete due to sustained loading is creep.
Segregation : Separation of the constituent material of concrete i.e cement, sand and aggregate because they have different specific gravity.
Bleeding : Bleeding is the tendency of water to the surface of freshly concrete. It is caused by the inability of solid constituents of the mixing water as they settle down.

41. Quality of reinforcing steel is evaluated by :

(a) Compressive strength
(b) Tensile strength, ductility
(c) Yield strength, ductility
(d) None of the above

KPSC AE 2020

Ans. (c) : Quality of reinforcement steel is evaluated by yield strength, ultimate strength and elongation (ductility).

42. The additional cover thickness to be provided in reinforced concrete member that are total immersed in sea water is :

(a) 25 mm (b) 30 mm
(c) 35 mm (d) 40 mm

Nagaland PSC CTSE 2017 Paper–I
(ESE 2012)

Ans. (a) For reinforcement concrete members totally immersed in sea water, the cover shall be 40 mm or more than that. For the reinforced concrete members, periodically immersed in sea water or subjected to sea spray, the cover of concrete shall be 50 mm more than specified.

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43. Grade of steel is designated as Fe 415 if :
(a) The upper yield stress of the steel in 415 N/mm^2
(b) The ultimate stress of the steel is 415 N/mm^2
(c) The Partial safety factor is 1.15
(d) The characteristics strength is 415 N/mm^2

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : Grade of steel Fe 415
Here, 415 is the characteristics strength of steel in (N/mm^2)

44. The minimum grade of concrete to be used in reinforced concrete as per IS 456-2000 is

(UKPSC AE 2012 Paper–I)

OR

Grade of concrete for reinforced concrete shall not be lower than :
(a) M 30 (b) M 25
(c) M 20 (d) M 15

(UKPSC AE (Paper-I) 2007)

Ans : (c) According to IS 456 – 2000 minimum grade of concrete for reinforced cement concrete is M20, for pre-tensioned member it is M40 and for post tensioned it is M30.
For M 20 ratio of cement, sand and coarse aggregate is $1:1\frac{1}{2}:3$ this concrete is used in water tanks, retaining walls, bridge etc.

45. In case of reinforced concrete, what is the minimum grade of concrete that should be used in sea water or when exposed directly along the sea–coast?
(a) M15 (b) M20
(c) M25 (d) M30

(UPRVUNL AE 2016
(ESE 2005, 2012)

Ans : (d) According to IS 456 – 2000 minimum grade of concrete for structures in sea water or when exposed directly along sea coast should be M30.

46. The factor used to account for the difference in strength indicated by cube test and the strength of concrete in structures–
(a) 0.67 (b) 0.87
(c) 1.15 (d) 1.87

UKPSC AE 2013 Paper–I

Ans : (a) The 0.67 factor is introduced to account for difference in strength indicated by a cube test and the strength of concrete in structure.
Maximum compressive strength of concrete without partial safety factor is taken as 0.67 f_{ck}, where f_{ck} is characteristics strength indicated by test.

47. Strength of cement concrete mainly depends upon :-
(a) Quality of water
(b) Quantity of aggregate
(c) Quantity of cement
(d) Water cement ratio

UKPSC A E Paper II 2013]

Ans : (d) Strength of cement concrete depends upon several factors such size and shape of aggregate, fineness of cement, quality of water and water-cement ratio. Among these, water-cement ratio effects the strength very much, if water-cement ratio is increased the strength decreases.

48. In general, the coefficient of thermal expansion of concrete does not depend on which of the following factors
(a) Type of cement
(b) Type of aggregate
(c) The cement content
(d) The quality of water

(RRB SSE (Shift-III), 03.09.2015)

Ans : (d) Physical properties of concrete such as thermal expansion, etc depends upon type and quantity of cement. Physical properties does not depend upon quality of water.

49. Which of the following is not a non-destructive test used for concrete?
(a) Rebound hammer (b) Pull-out
(c) Ultra-sonic (d) Direct tensile test

(RRB SSE (Shift-III), 03.09.2015)

Ans : (d) Tests of concrete in which the specimen are not destroyed are known as non-destructive tests. Non-destructive test are –
(i) **Rebound hammer**→ To find compressive strength of concrete.
(ii) **Pull off test**→ To find compressive strength
(iii) **Ultrasonic test**→ To check quality of concrete and aggregates.
Direct tensile test is a destructive test on small specimens to find tensile strength. Direct tensile strength of concrete is difficult to determine, recourse is often taken to the determination of flexural strength or the splitting tensile strength and computing the direct tensile.

50. The size of the aggregate accepted as suitable for R.C.C. construction work is about:-
(a) 5 mm to 20 mm (b) 3 mm to 5 mm
(c) 1 mm to 3 mm (d) 0.5 mm to 1.0 mm

[UKPSC A E Paper II 2013]

Ans : (a) Size of coarse aggregate is 10 mm to 20 mm being used in concrete. In given options 5 to 20 mm is best suited.

51. Liquefaction of sand is–
(a) Sand enhance its shear strength due to oscillatory motion
(b) Sand looses its shear strength due to oscillatory motion
(c) Sand enhances its bearing capacity due to oscillatory motion
(d) None of above

UKPSC AE 2013 Paper–I

Ans : (b) Liquefaction is phenomenon in which sand loses its shear strength due to oscillatory motion. The structures resting on such soils sink. The soils most susceptible to liquefaction are the saturated, fine and medium sands of uniform, particle size.

52. Good quality sand is never obtained from
(a) River (b) Drain
(c) Sea (d) Gravel powder

(U.K. Combined A E Paper II 2012)

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Ans : (c) Sand from sea should not be used for constructional purpose as they contain large quantity of salts which are harmful and reduce strength of mix considerably.

53. With increase in moisture content the bulking of sand
(a) increases
(b) decreases
(c) First increases to certain maximum value and then decreases
(d) first decreases to a certain minimum value and then increases

(U.K. Combined A E Paper II 2012)

Ans : (c) Fine sands bulk more than coarse sand and for moisture content 5 to 6% the increase in volume may vary from 20% to 35%.
If we further increase moisture content, then volume decreases due to breaking of films formed by capillary action in sands.

54. According to IS 456, if the maximum aggregate size is increased from 20 mm to 40 mm, the minimum cement content requirement changes (in kg/cum) by :
(a) –20 (b) 20
(c) –30 (d) 30

(RRB SSE (Shift-III), 01.09.2015)

Ans : (c) Size of aggregate is increased from 20 mm to 40 mm the specific surface area decrease, due to which lesser amount of cement paste is required.
Adjustments to minimum cement content for aggregates other than 20 mm nominal maximum size aggregate as per IS 456 : 2000.

Nominal maximum aggregate size (mm)	Adjustment to minimum cement content (kg/m^3)
10	+40
20	0
40	–30

55. According to IS 456, under normal circumstances, the maximum cement content in kg/cum (including fly ash and slag) allowed is
(a) 500 (b) 400
(c) 450 (d) 600

(RRB SSE (Shift-II) Online 01.09.2015)

Ans : (c) Taking durability in consideration, according to IS : 10262 the maximum cement content is limited to 450 kg/m^3.

56. Slump test is used for
(a) Workability (b) Consistency
(c) Strength (d) Durability

(Uttarakhand Combined State AE Paper II 2007/ (UPPSC State Eng. AE 2011)

Ans : (a) Slump test is used to find workability. Workability is the ease with which concrete can be mixed. Workability is affected by water content, size of aggregate, cement percentage. To measure workability IS:456, IS:1199 is used.

57. Compaction factor apparatus is used to measure which of the following properties of concrete?
(a) Consistency
(b) Compatibility
(c) Compatibility with admixture
(d) Cohesiveness

(UP RVNL AE 2016)

Ans.: (a) Compaction factor is used to find consistency of concrete. In this test coarse aggregate of size greater than 38 mm should not be present in concrete.

$$\text{Compaction factor} = \frac{\text{wt. of concrete due to partial compaction}}{\text{weight of concrete due to complete compaction}}$$

Compaction factor value is 0.85 for low workability, 0.92 for medium workability and 0.95 for good workability concrete.

58. A concrete mix of grade M40 is to be used for pavements using paver machine. The recommended slump range as per IS:456-2000 is:
(a) 25-75 mm (b) 50-100 mm
(c) 75-100 mm (d) 100-150 mm

(LMRC AE 2017 I–Shift)

Ans : (a) According to I.S. 456–2000 slump value for different types of structures–

Type of work	Slump range (mm)
Road pavement	25-75
under water concrete work	100-150
column and foundation	25–75
bridges	25–75
Mass concrete work	25–50
Dense concrete work	50–100

59. The recommended slump for pumped concrete is :
(a) 1-2 inches (b) 2-4 inches
(c) 3-4 inches (d) 4-6 inches

(RRB SSE (Shift-I) Online 03.09.2015)

Ans : (d) Slump test used to find workability of concrete slump value for pumped concrete work is 4 to 6 inches.
If coarse aggregate of size greater than 38mm are used, then slump test is not conducted. For low water content mix slump value is zero.

60. The recommended slump of concrete for hand-placed pavements is :
(a) 2-4 inches (b) 3-4 inches
(c) 4-6 inches (d) 1-3 inches

(RRB SSE (shift-III), 02.09.2015)

Ans : (d)
- For hand placed pavement slump value should be between 1 – 3 inches (25 – 75 mm).
- For trench fill – 100 - 150 mm
- For heavily reinforced section in slab, beam, columns – 50- 100 mm

61. Which of the following properties is not required for concrete mix design, as per the guidelines given by the Indian Standard?
(a) Specific gravity of cement
(b) Density of aggregate
(c) Maximum size of aggregate
(d) Characteristic strength of concrete

(UP RVNL AE 2016)

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Ans.: (b) Information required for design of mix–
(1) Grade of concrete
(2) Type of cement
(3) Minimum cement content
(4) Water cement ratio
(5) Degree of workability
(6) Exposure condition, type of admixture, method of placing
Density of aggregate is not required for mix design.

62. Which of the following is not used in the design of concrete mixes as per the relevant Indian standard?
(a) Air content
(b) Water content
(c) Admixture content
(d) Bulk density of cement

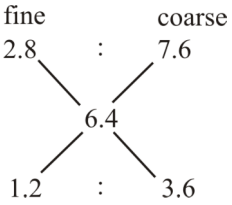
(RRB SSE (Shift-III), 01.09.2015)

Ans : (d) In design of concrete, air content, water content and admixture content are taken into consideration. Bulk density of cement is not used in mix design.

63. If in a concrete mix the fineness modulus of coarse aggregate is 7.6., the fineness modulus of fine aggregate is 2.8 and the economic value of the fineness modulus of combined aggregate is 6.4, then the proportion of fine aggregate is
(a) 25% (b) 33.3%
(c) 50% (d) 66.6%

(UPRVUNL AE 2015)

Ans : (b)



Therefore for combined mixture, proportion of fine aggregate = $\frac{1.2}{3.6}\times 100 = 33.3\%$

64. The batching tolerance for aggregates as per IS 456 is
(a) ±1% (b) ±1.5%
(c) ±2% (d) ±3%

(RRB SSE (shift-II), 02.09.2015)

Ans : (d) According IS–456 tolerance limit for aggregate is ± 3%. Tolerance for water is also ± 3% and tolerance for cement is ± 2%.

65. How are concrete mixers specified?
(a) by the number of cement bags used in a batch
(b) by the nominal volume of concrete that can be mixed in a batch
(c) by the volume of water used
(d) by the volume of aggregate used

(UPRVUNL AE 2015)

Ans : (b) To prepare concrete of good strength we use proper ratio of coarse and fine aggregates, amount of cement used and water cement ratio in definite proportion, this is known as concrete mix.
According to IS 456-2000, concrete mix is prepared by two types–
(a) Design mix concrete – The concrete produced under quality control keeping in view the strength, durability and workability is called the design mix.
(b) Nominal mix concrete– In nominal mix design we take the ingredients of mix in rough ratio and we do not check them before use. This is also known as general mix concrete. In this the ratio of aggregates is shown in volume or weight.

66. A flat slab of span length of 6.45 m was cast today afternoon. As per the recommendations of IS 456, the minimum time after which the props should be removed is:
(a) 7 days (b) 10 days
(c) 14 days (d) 21 days

(LMRC AE 2017 I–Shift)

Ans : (c) According to I.S. 456–2000, minimum time after which props are removed.

Sn.	Type of formwork	Minimum period before striking formwork.
1.	Vertical formworks to columns, wall beam	16 – 24 hours
2.	Soffit formworks to slabs	3 days
3.	Soffit formworks to beams	7 days
4.	Props to slabs– (a) Spanning up to 4.5 m (b) Spanning over 4.5 m	 7 days 14 days
5.	Props to beam and arches (a) Spanning upto 6 m (b) Spanning over 6 m	 14 days 21 days

67. The minimum time before striking the props to slabs (less than 4.5 m) as per IS 456 is :
(a) 14-day (b) 7-day
(c) 3-day (d) 16-24 h

(RRB SSE (Shift-III), 01.09.2015)

Ans : (b) According to I.S. 456–2000, minimum time after which props are removed.

S. No.	Type of formwork	Minimum period before striking formwork.
1.	Vertical formworks to columns, wall beam	16 – 24 hours
2.	Soffit formworks to slabs	3 days
3.	Soffit formworks to beams	7 days
4.	Props to slabs– (a) Spanning up to 4.5 m (b) Spanning over 4.5 m	 7 days 14 days
5.	Props to beam and arches (c) Spanning upto 6 m (d) Spanning over 6 m	 14 days 21 days

68. The minimum time before striking the props to arches (less than 6 m) as per IS 456 is :
(a) 14-day (b) 7-day
(c) 3-day (d) 16-24 h

(RRB SSE (Shift-II) Online 01.09.2015)

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Ans : (a) According to I.S. 456–2000, minimum time after which props are removed.

S. No.	Type of formwork	Minimum period before striking formwork.
1.	Vertical formworks to columns, wall beam	16 – 24 hours
2.	Soffit formworks to slabs	3 days
3.	Soffit formworks to beams	7 days
4.	Props to slabs–	
	(a) Spanning up to 4.5 m	7 days
	(b) Spanning over 4.5 m	14 days
5.	Props to beam and arches	
	(e) Spanning upto 6 m	14 days
	(f) Spanning over 6 m	21 days

69. The minimum time before striking the form for columns as per IS 456 is :

(a) 14 – day (b) 7 – day
(c) 3 – day (d) 16 –24 hr.

(RRB SSE Secundrabad (Shift-I), 01.09.2015)

Ans : (d) According to I.S. 456–2000, minimum time after which props are removed

S.No.	Type of formwork	Minimum period before striking formwork.
1.	Vertical formworks to columns, wall beam	16 – 24 hours
2.	Soffit formworks to slabs	3 days
3.	Soffit formworks to beams	7 days
4.	Props to slabs–	
	(a) Spanning up to 4.5 m	7 days
	(b) Spanning over 4.5 m	14 days
5.	Props to beam and arches	
	(g) Spanning upto 6 m	14 days
	(h) Spanning over 6 m	21 days

70. Creep deformation is a property of any material to deform under the influence of

(a) Thermal stresses
(b) Moisture related stresses
(c) Mechanical stresses
(d) Stresses due to wind loads

(UP RVNL AE 2016)

Ans.: (c) Creep is the tendency of a solid material to move slowly or deform permanently under the influence or persistent mechanical stresses. It can occur as a result of long-term exposure to high levels of stress that are still below the yield strength of the material.

71. Which of these statements are correct for bleeding of concrete leads?

A. Drying up of concrete surface
B. Formation of pores inside
C. Segregation of aggregate
D. Decrease in strength

Select the correct answer given below code.

(a) Only A (b) Only A and B
(c) Only A and C (d) Only B and D

(UPRVUNL AE 2015)

Ans : (d) Bleeding occurs when excess water comes up at the surface of concrete resulting in several small pores and less durability.

72. Which of the following roles fly ash does not play in concrete

(a) Improving the workability
(b) Accelerating the strength gain
(c) Delaying the setting time of concrete
(d) Helps in long-term strength gain

(RRB SSE (Shift-II) Online 01.09.2015)

Ans : (b) Mixing fly ash in concrete improves its workability, helps in long term strength gain and delaying the setting time of concrete.

73. Which of the following is not a component of an expansion joint used in jointed plain cement concrete?

(a) Dowel bars (b) Joint filler
(c) Joint sealer (d) Tie bar

(LMRC AE 2017 I–Shift)

Ans : (d) Tie bars are not part of expansion joint
According to BIS, Minimum length of structure for development length is as follows–
(i) Walls → 30 m
(ii) open slabs → 6 to 12 m
(iii) framed structures → 30 m
(iv) parapet → 6 to 12 m

74. Shrinkage in cement concrete may be reduced by:-

(a) Proper curing
(b) Adding minimum water
(c) Adding more coarse aggregate
(d) Adding mild steel bar

[UKPSC A E Paper II 2013]

Ans : (a) Proper curing can prevent concrete shrinkage loss of moisture by evaporation causes volumetric changes of concrete structures, which is known as concrete shrinkage or shrinkage of concrete. It is kind of deformation which reduces the volume of concrete without the impact of external forces.

75. The flexural tensile strength of M-25 grade of concrete in (N/mm^2) as per IS 456:2000 is

(a) 3 N/mm^2 (b) 3.5 N/mm^2
(c) 4 N/mm^2 (d) 4.5 N/mm^2

RPSC POLY. TECH. LECT. 2014
(ESE 2005, 2009)

Ans : (b) According to IS 456 : 2000 flexural tensile strength of concrete is

$$\boxed{f_{cr} = 0.7\sqrt{f_{ck}}}$$

Given, $f_{ck} = 25 N/mm^2$

$$F_{cr} = 0.7\sqrt{25} = 0.7 \times 5 = 3.5$$

$$\boxed{f_{cr} = 3.5 N/mm^2}$$

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76. The modulus of elasticity of concrete (E_c) can be assumed as

RPSC GWD A.EN.– 2014

Or

The modulus of elasticity of cement concrete can be assumed as following as per IS 456 : 2000

(a) $2000\sqrt{f_{ck}}$ (b) $3000\sqrt{f_{ck}}$

(c) $5000\sqrt{f_{ck}}$ (d) $4000\sqrt{f_{ck}}$

[UPPSC State Eng. A.E. 2007(I)]

[UPPSC State Eng. A.E. 2003]

Or

If f_{ck} is the characteristic strength of concrete then as per the Indian Standard (IS) 456: 2000, the modulus of elasticity of the concrete is

(a) $5700\sqrt{f_{ck}}$

(b) $5200\sqrt{f_{ck}}$

(c) $5000\sqrt{f_{ck}}$

(d) None of these are correct

MPPSC AE 2016

[JPSC Combined AE Paper I 2013]

ESE 2006

Ans : (c) The modulus of elasticity of concrete can be assumed as follows:

$$E_c = 5000\sqrt{f_{ck}}$$

Where, E_c is the short term static modulus of elasticity in N/mm^2.

Note:

The elasticity is primarily influenced by the elasticity properties of the aggregate and to a lesser extent by the conditions of curing, age of the concrete, the mix proportions and type of concrete.

The modulus of elasticity is normally related to the compressive strength of concrete.

77. Concrete in sea water shall be at least grade in case of reinforced concrete

(a) M20 (b) M25

(c) M30 (d) M40

RPSC GWD A. En. - 2014

Ans : (c) Minimum grade of concrete for structure in sea water should be M30. Minimum grade of concrete depends upon atmospheric conditions.

Sr. No.	Exposure	Min Grade of concrete	Min Cement Content (kg/m^3)	Max Free w/c ratio
1	Mild	M 20	300	0.55
2	Moderate	M 25	300	0.50
3	Severe	M 30	320	0.45
4	very severe	M 35	340	0.45
5	Extreme	M 40	360	0.40

78. The 28 days characteristic compressive strength of concrete of grade M-20 is

(a) $5N/mm^2$ (b) $7N/mm^2$

(c) $10N/mm^2$ (d) $20N/mm^2$

[UPPSC State Eng. A.E. 2011]

Ans : (d) The compressive strength of 28 day:

Grade of concrete	min compressive strength (7day) (N/mm^2)	min compressive strength (28day) (N/mm^2)
M15	10	15
M20	13.5	20
M25	17	25
M30	20	30
M35	23.5	35

79. A beam shall be deemed to be a deep beam when the ratio of effective span to overall depth is less than for a continuous beam :

(a) 2.5 (b) 3

(c) 3.5 (d) 4

RPSC POLY. TECH. LECT. 2014

Ans : (a) A beam shall be deemed to be a deep beam when the ratio of effective span to overall depth *l*/D is less than –

(i) 2.0 for a simply supported beam and

(ii) 2.5 for a continuous beam

80. Target strength for the mix proportioning (f_{ck}) is given by

(a) $f_{ck}–1.65\sigma$ (b) $f_{ck}+1.65\sigma$

(c) $f_{ck}–2.65\sigma$ (d) $f_{ck}–3.65\sigma$

RPSC GWD A. En.– 2014

Ans : (b) Target mean design strength–

A concrete mix is designed for a target mean design strength given by following expression :

$$\boxed{f_t = f_{ck} + 1.65\sigma}$$

f_{ck} = characteristic strength

f_t = standard deviation to measure the precision of production of concrete.

Grade of concrete	Standard deviation (MPa)
M 10 –M 15	3.5
M 20 – M 25	4.0
M 30 – M 60	5.0

81. Nominal mix M20 concrete represent the nearest ingredients ratio is

(a) 1 : 3 : 6 (b) 1 : 2 : 4

(c) 1 : 1.5 : 3 (d) 1 : 1 : 2

RPSC LECTURE - 2011

Ans : (c) Nominal mix ratio for different grade of concrete–

M 10 $\Rightarrow$ 1 : 3 : 6

M 15 $\Rightarrow$ 1 : 2 : 4

M 20 $\Rightarrow 1 : 1\frac{1}{2} : 3$

M 25 $\Rightarrow$ 1 : 1 : 2

82. The modulus of ruptures of concrete gives

(a) The direct shear strength of the concrete

(b) The direct compressive strength of the concrete

(c) The tensile strength of the concrete under bending

(d) None of above

RPSC LECTURE - 2011

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Ans : (c) Tensile strength in flexural is its strength just before the failure of specimen in tension. Modulus of rupture represents the tensile strength in flexure.

83. Individual variation between tensile strength of sample should not be more than :

(a) +5% of the average
(b) + 10 % of the average
(c) + 15 % of the average
(d) + 20 % of the average

RPSC ACF - 2011

Ans : (c) According to IS : 456, the individual variation should must be greater than ± 15% of the mean otherwise the test results of the sample are invalid.

84. According to Indian standard specification, the temperature for curing is

(a) 5^0C (b) 10^0C
(c) 27^0C (d) 42^0C

RPSC POLY. TECH. LECT. 2011

Ans : (c) About 30% of the strength can be lost by premature drying out of the concrete. Curing of concrete should be done for 28 days and the temperature should be near about 27^oC.

85. In case of framed RCC construction, the form work to be removed first are

(a) Bottom of beams
(b) Bottom of roof slab
(c) Sides of beams
(d) Any of Bottom of beams or Bottom of roof slab

RPSC VIPITI 2012

Ans : (c) Formwork is removed in steps and should be removed easily without sudden jerks. first of all sides of beam are removed.

86. The minimum cement content in moderately exposed reinforced concrete with normal weight aggregates of 20 mm nominal maximum size is

(a) 220 kg/m^3 (b) 240 kg/m^3
(c) 280 kg/m^3 (d) 300 kg/m^3

RPSC A En. 2013

Ans : (d) Minimum cement content and maximum w/c ratio for durability

Exposure	Min. Grade	RCC min. cement (kg/m^3)	w/c
Mild	M 20	300	0.6
Moderate	M 25	300	0.6
Severe	M 30	320	0.5
Very severe	M 35	340	0.45
Extreme	M 40	360	0.4

87. For lightly reinforced sections in slabs, beams, columns etc. the slump should be

(a) 15-25 mm (b) 25-75 mm
(c) 50-100 mm (d) 75-100 mm

RPSC A En. 2013

Ans : (b) Workability for different placing conditions–

Placing condition	Degree of workability	Slump (mm)
Binding concrete, shallow section	Very Low	15-25
Mass concrete, lightly reinforced section in slabs, beams, walls, column, floors, Hand placed pavements, canal lining, strip footing	Low	25-75
Heavily reinforced sections in slabs, beam, walls, columns.	Medium	50-100
Slip form work, pumped concrete	Moderate	75-100
Trench fill, In-situ piling	High	100-150
Tremie concrete	Very high	150-200

88. The height of the mould (cone) used for slump test is

(a) 450 mm (b) 300 mm
(c) 200 mm (d) 100 mm

RPSC VIPITI 2012

Ans : (b) Slump test is used to measure workability of concrete. Slump cone is a vessel in form of a frustum of a cone with base and top diameter 200 mm and 100 mm respectively. Height of vessel is 300 mm.

89. The strength of concrete is directly proportional to :

(a) Water Cement Ratio
(b) Cement water ratio
(c) Sand cement ratio
(d) Water aggregate ratio

RPSC AE (GWD) 2014

Ans : (b) Strength of concrete is directly proportional to cement water ratio, whose relation is as follows-

$$P = \frac{948}{7^x}$$

p = strength of concrete after 28 days curing
x = water – cement ratio
In the given equation strength is inversely proportional to the water cement ratio, therefore, it is directly proportional to cement-water ratio.

90. For most of the applications, water to cement ratio should be between

(a) 0.4 and 0.5 (b) 0.5 and 0.55
(c) 0.55 and 0.60 (d) 0.60 and 0.65

RPSC A. En. 2013

Ans : (a) In concrete a w/c ratio of 0.24 is required for hydration, but minimum 0.4 w/c ratio is required for it to be workable. For mild concrete works it can be taken upto 0.6 but the strength decrease rapidly with increase in w/c ratio. In given options 0.4 to 0.5 is best suited.

91. The ratio of 28 days strength of cube to that of a standard cylinder is

(a) 0.8 (b) 1.25
(c) 0.67 (d) 1.00

RPSC POLY. TECH. LECT. 2011

Ans : (b) For compression testing moulds of size 150×150×150 mm^3 are used. When cylindrical shape moulds are used, the test result show strength nearly 20% lesser than that from cube test.

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Let strength from cube test = P
Therefore, strength from cylinder test = 0.8P
$\text{Ratio} = \frac{P}{0.8P} = 1.25$

92. In slump test, each layer of concrete is compacted by a tamping rod with
(a) 20 blows (b) 25 blows
(c) 30 blows (d) 40 blows

RPSC POLY. TECH. LECT. 2011

Ans : (b) In slumps test the mould is filled with freshly mixed concrete in four layers each layer is tamped 25 times.

93. Specific gravity of most of the coarse aggregate in concrete lies between
(a) 1.5 to 2.0 (b) 2.0 to 2.5
(c) 2.5 to 3.0 (d) 3.0 to 3.5

[RPSC LECTURE - 2011]

Ans : (c) In concrete we use sand and gravel as fine and coarse aggregate. For a good mix specific gravity of coarse aggregate should be between 2.5 to 3.0.

94. The flakiness index of coarse aggregate is the percentage by weight of particles having least dimension (thickness) less than of their mean dimension.
(a) One-fifth (b) Two-fifth
(c) Three-fifth (d) Four-fifth

(RPSC A.En. 2013)

Ans : (c) Flakiness index is the percentage by which particles in it, whose least dimension (i.e., thickness) is less than three-fifths of its mean dimension.

95. Aluminium powder in concrete is used as ____ .
(a) retarder (b) air entraining agent
(c) accelerator (d) plasticizer

[RPSC VPITI-2012]

Ans : (b) Air entraining agents are used to increase workability of mix without increasing the water content so that their resistance to freezing and thawing is good. Air entraining agents improve workability and durability, but they decrease strength of concrete. Zinc and aluminium powder are used as air entraining agents.

96. The top diameter, bottom diameter and the height of a slump mould are .....
(a) 10 cm, 20 cm, 30 cm
(b) 10 cm, 30 cm, 20 cm
(c) 20 cm, 10 cm, 30 cm
(d) 20 cm, 30 cm, 10 cm

(RPSC LECTURE - 2011)

Ans : (a) Slump test is used to measure workability of concrete. Dimensions of slump vessel is 10 cm top diameter, 20 cm bottom diameter and 30 cm height.

97. As per IS 456: 2000, normally structures are designed with one or more expansion joints exceeding
(a) 45 m in length (b) 60 m in length
(c) 75 m in length (d) 80 m in length

[UPPSC State Eng. A.E. 2011]

Ans : (a) According to IS 456 : 2000, if the length of structure is greater than 45 m, then we use expansion joint. Expansion joints are designed taking in consideration atmospheric and temperature condition.

98. Modulus of rupture of concrete is a measure of
(a) Compressive strength
(b) Direct tensile strength
(c) Flexural tensile strength
(d) Split tensile strength

RPSC ACF - 2011

Ans : (c) Tensile strength in flexural is its strength just before the failure of specimen in tension. Modulus of rupture represents the tensile strength in flexure.

99. The following code is used to find out earthquake forces in a building
(a) IS 456 : 2000 (b) IS 875 : 1984
(c) IS 1893 : 2002 (d) None of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (c) IS 456 : 2000 plain and reinforced concrete
IS 875 : 1984 code for wind load and Earthquake load
IS 1893 : 2002 earthquake resistant design.

100. Long term elastic modulus in terms of creep coefficient (θ) and 28-day characteristic strength (f_{ck}) is given by:
(a) $\frac{5000\sqrt{f_{ck}}}{1+\theta}$ MPa (b) $\frac{50000\sqrt{f_{ck}}}{1+\theta}$ MPa
(c) $\frac{5000 f_{ck}}{1+\sqrt{\theta}}$ MPa (d) $\frac{5000\sqrt{f_{ck}}}{\sqrt{1+\theta}}$ MPa

[UPPSC State Eng. A.E. 2007(I)]
(ESE 2007)

Ans : (a) Long term modulus of elasticity:
It considers the effect of creep and is defined as

$$E_{long} = \frac{E_c}{1+\theta} = \frac{5000\sqrt{f_{ck}}}{1+\theta}$$

$$\theta = \text{creep coefficient} = \frac{\text{Ultimate creep strain}}{\text{Elastic strain}}$$

101. As per IS 456 : 2000 the minimum grade of concrete shall not be less than the following in reinforced concrete work :
(a) M15 (b) M20
(c) M25 (d) M30

[UPPSC State Eng. A.E. 2007(I)]

Ans : (b) As per IS 456 : 2000 the minimum grade of concrete shall not be less than the M 20 in reinforced concrete work.

S.No.	Exposure	Minimum grade
1	Reinforced concrete	M 20
2	water tank	M 30
3	pre-tensioning	M 40
4	post-tensioning	M 30

102. The workability of concrete required for the heavily reinforced sections in slab, beam and column is:
(a) very low (b) very high
(c) low (d) medium

[UPPSC State Eng. A.E. 2007(I)]

Ans : (d) The workability of concrete required for the heavily reinforced sections in slab, beam and column is medium.

As per IS 456 : 2000

Reinforcement	Workability
Light reinforcement	Low
Heavly reinforcement	Medium
Trench fill or pilling	High

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103. If σ_{cbc} is permissible compressive stress in flexural compression in N/mm² in service, the modular ratio is:

(a) $\frac{280}{3\sigma_{cbc}}$ (b) $\frac{280}{4\sigma_{cbc}}$

(c) 19 (d) 13

[UPPSC State Eng. A.E. 2003]

Ans : (a) Modular ratio is the ratio of modulus of elasticity of steel to modulus of elasticity of concrete.
In working stress design,

$$M = \frac{280}{3\sigma_{cbc}}$$

where σ_{cbc} = permissible stress in extreme compression fiber in concrete.

104. In M–15, 15 stands for.....days compressive strength of concrete in N/mm²

(a) 28.5 (b) 10

(c) 20 (d) None of these

(BPSC AE Paper V 2001)

Ans : (d) In M-15, 15 shows characteristic compressive strength of concrete in N/mm² for a cube of 15 cm size tested at 27 days (f_{ck}).

105. Young's modulus of concrete for M-25 grade concrete is.....kN/mm²

(a) 28.5 (b) 18

(c) 20 (d) none of these

(BPSC AE Paper V 2001)

Ans : (d)
M-25, $f_{ck} = 25$ N/mm²
According to IS 456 : 2000, the Young's modulus of elasticity

$$E_c = 5000\sqrt{f_{ck}}$$
$$= 5000 \times \sqrt{25}$$
$$\boxed{E_c = 25 \text{ kN/mm}^2}$$

106. The Modular ratio for M-20 grade concrete is given by

(a) 10 (b) 18

(c) 15 (d) 13

[UPPSC State Eng. A.E. 2007(I)]
(BPSC AE Paper V 2001)

Ans : (d) modular ratio $(m) = \frac{280}{3\sigma_{cbc}}$

where σ_{cbc} = Permissible stress in bending compression
For M-20 $\sigma_{cbc} = 7 \text{N/mm}^2$

$$m = \frac{280}{3 \times 7}$$
$$m = 13.33 \simeq 13$$

107. The permissible bending stress for M15 grade concrete is (in N/mm²)

(a) 15 (b) 30

(c) 5 (d) 7.5

(BPSC AE Paper V 2001)

Ans : (c) In M-15 15 shows characteristic compressive strength of concrete in N/mm² for a cube of side 15 cm. Standard deviation for M-15 grade is 4-MPa. Permissible bending stress for M-15 is 5 MPa.

108.. The permissible shear stress in concrete depends upon

(a) grade of concrete and % steel in compression
(b) grade of concrete only
(c) % steel in compression only
(d) grade of concrete and % steel in tension

(UKPSC AE 2012 Paper–I)

Ans : (d) Permissible shear stress in concrete depends upon grade of concrete and percent steel in tension.

109. According to IS–456, the estimated flexural strength of concrete of grade M25 (f_{ck} = 25 MPa) would be:

(a) 35 kg/sq cm (b) 50 kg/sq cm

(c) 75 kg/sq cm (d) 25 kg/sq cm

(LMRC AE 2017 I–Shift)

Ans : (a) According to IS-456, the estimated flexural strength of concrete is given by –

(i) $f_{cr} = 0.7\sqrt{f_{ck}} \text{ N/mm}^2$

given, $f_{ck} = 25$ MPa

$\therefore$ $f_{cr} = 0.7\sqrt{25} = 3.5 \text{ N/mm}^2$

$f_{cr} = 35 \text{ kg/cm}^2$

110. The characteristic compressive strength of M30 grade concrete as per IS: 456-2000 is

(a) 3.83 MPa (b) 5.47 MPa

(c) 21.23 MPa (d) 30.00 MPa

(UKPSC AE 2012 Paper–I)

Ans : (d) Characteristic compressive strength for M 30 grade concrete is 30 MPa.

111. If the permissible compressive stress in concrete in bending for M-20 grade of concrete is 7 N/mm² then modular ratio will be

(a) 18.66 (b) 13.33

(c) 10.98 (d) 9.33

(UKPSC AE 2012 Paper–I)

Ans : (b) Given, $\sigma_{cbc} = 7 \text{ N/mm}^2$

$$\text{modular ratio (m)} = \frac{280}{3\sigma_{cbc}} = \frac{280}{3 \times 7} = 13.33$$

112. As per IS : 456 – 2000 actual measure values of modulus of elasticity E_c may be from $E_s = 5000\sqrt{f_{ck}}$ (with usual notation)

(a) ± 25% (b) ± 5%

(c) ± 10 % (d) ± 20%

(UKPSC AE (Paper-I) 2007)

Ans : (d) According to IS : 456 – 2000 modulus of elasticity $E_c = 5000\sqrt{f_{ck}}$ can vary ± 20% from its actual value.

113. The estimated flexural strength (MPa) of a concrete having a measured compressive strength of 64 MPa would be (as per IS 456)

(a) 7.2 (b) 6.4

(c) 5.6 (d) 4.8

(UPRVUNL AE 2016)

Ans.: (c) Flexural strength $= 0.7\sqrt{f_{ck}}$

$= 0.7\sqrt{64} = 5.6$ MPa

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114. The yield strength of Fe 415 steel in MPa would be a minimum of

(a) 250.0 (b) 380.0
(c) 415.0 (d) 500.0

(UP RVUNL AE 2016)

Ans.: (c) Fe – 415 has minimum yield strength = 415 N/mm^2

$$\text{working strength} = \frac{\text{yield strength}}{1.78}$$

$$= \frac{415}{1.78} = 233 \approx 230\text{MPa}$$

For working stress we take factor of safety 1.78 and for limit state we take 1.15.

115. Steel of yield strength 400 MPa has been used in a structure. What is the value of the maximum allowable tensile strength?

(a) 240 MPa (b) 200 MPa
(c) 120 MPa (d) 96 MPa

(UPRVUNL AE 2015)

Ans : (a) Tensile strength = $\frac{\text{yield strength}}{\text{factor of safety}}$

$$= \frac{400}{1.78}$$

$$= 225 \approx 240\text{MPa}$$

116. The minimum grade of reinforced concrete in sea water or exposed directly along the sea-coast shall be at least

(a) M-10 (b) M-15
(c) M-20 (d) M-25

(BPSC AE Paper V 2006)

Ans : (d) Minimum grade of concrete in sea water or exposed directly along sea coast is taken as M 30. Since M 30 is not given, we will choose nearest value give i.e., M 25.

117. As per the Indian Standard (IS) 456 : 2000, if f_{ck} is the characteristic strength of concrete, the tensile strength of the concrete is-

(a) $0.7\sqrt{f_{ck}}$ (b) $0.5\sqrt{f_{ck}}$
(c) $0.87\sqrt{f_{ck}}$ (d) $0.46\sqrt{f_{ck}}$

MPPSC AE 2017

UPPSC AE 12.04.2016 Paper–I

Ans. (a) : According to clause 6.2.2 of IS 456 : 2000, the flexural strength, for concrete $f_{cr} = 0.7\sqrt{f_{ck}}\ N/mm^2$

Where f_{ck} is the characteristic cube compressive strength of concrete in N/mm^2.

Flexural strength is measure of the tensile strength of concrete. It is a measure of an un-reinforced concrete beam or slab to resist failure in bending.

118. The time dependent in elastic deformation of steel resulting from sustained stress and a function there of is known as _____.

(a) Shrinkage (b) Creep
(c) Deformation (d) Relaxation

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : The time dependent in elastic deformation of steel resulting from sustained stress is known as relaxation.

119. Which of the following is a method of mechanical ventilation?

(a) Plenum System
(b) Bleeding System
(c) Segregation System
(d) Natural Ventilation System

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (a) Plenum system of ventilation – a system of mechanical ventilation in which fresh air is forced into the spaces to be ventilated from a chamber (plenum chamber) at a pressure slightly higher than atmospheric pressure, so as to expel foul air.

120. As per IS 456, the minimum grade of concrete for the design of reinforced concrete structure in moderate exposure condition is-

(a) M 20 (b) M 25
(c) M 15 (d) M 30

OPSC AEE 2015 PAPER–I

Ans. (b) : According to IS Code 456 : 2000

Exposure Condition	Minimum grade of RCC required
Mild	M20
Moderate	M25
Severe	M30
Very Severe	M35
Extreme	M40

121. Whenever the earthquake or wind loading is considered in design of a member, the permissible stresses may be increased by-

(a) 25% (b) 30%
(c) 35% (d) 33.33%

OPSC AEE 2015 PAPER–I

Ans. (d) :

- When the effect of wind or seismic load is taken into account, the permissible stress in steel are increased by $33\frac{1}{3}\%$
- For rivets, bolts and tension rods, the permissible stress are increased by 25%, when the effect of wind or seismic load is taken in to account.

122. The minimum clear covers (in mm) to the main steel bars in slab, beam, column and footing are respectively

(a) 10, 15, 20, 25 (b) 15, 25, 40, 75
(c) 20, 25. 30, 40 (d) 20, 35, 40, 75

BPSC AE 2019 Paper (V) Section-I

(ESE 1995, 2011)

Ans. (b) : Clear cover to main reinforcement are-

- Footings – 75 mm
- Column – 40 mm
- Shear wall – 25 mm
- Beams – 25 mm
- Slab – 15 mm
- Flat slab – 20 mm
- Retaining wall – 20-25 mm
- Staircase – 15 mm

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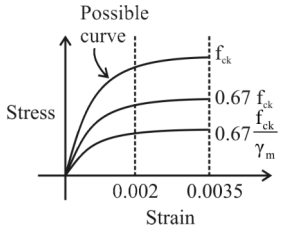
123. The maximum compressive stress in concrete for design purposes is based on a partial safety factor of

(a) 1.55 (b) 1.50
(c) 1.85 (d) 2.20

BPSC AE 2019 Paper (V) Section-I

Ans. (b) :



Where,
f_{ck} = characteristic compressive strength of concrete.
For design purpose, the compressive strength of concrete in the structure shall be assumed to be 0.67 times the characteristic strength. The partial safety factor $\gamma_m = 1.5$ shall be applied in addition to this.

124. The creep strains are

(a) caused due to dead loads only
(b) caused due to live loads only
(c) caused due to both dead load and live loads
(d) independent of loads

BPSC AE 2019 Paper (V) Section-I

Ans. (a) : Creep is defined as the deformation of structure under sustained or dead load. This deformation usually occurs in the direction the force is being applied. Subsequent strain produced in material or structure is called creep strain.

125. Concrete in sea-water or exposed directly along the sea-coast shall be at-least..... in case of reinforced concrete.

(a) M 20 (b) M 30 (c) M 25 (d) M 40

[HPPSC 2016-II]

Ans. (b) : For severe exposure conditions such as concrete immersed in sea water, concrete exposed to coastal environment, exposed to severe rain, alternate wetting and drying etc, we use minimum grade of concrete M30 in RCC work and M20 in PCC works

126. Grades of concrete for reinforced concrete shall not be lower than :

(a) M10 (b) M15
(c) M20 (d) M7.5

Karnataka PSC AE 2015 Paper - II

Ans. (c) Minimum grade of concrete required for different purpose:

Plain cement concrete	–	M 15
Reinforced cement concrete	–	M 20
Post Tensioned PSC	–	M 30
Pre Tensioned PSC	–	M 40

127. The yield strength of HYSD and CTD bars :

(a) 250 MPa (b) 415 MPa
(c) 450 MPa (d) 500 MPa

Karnataka PSC AE 2015 Paper - II

Ans. (b) Fe 415 is known as HYSD bars. They are formed by cold working process.
HYSD– High yield strength deformed bars.
CTD – Cold Twisted deformed bars.

128. According to IS 456-2000, the minimum grade of concrete with maximum free water to cement ratio of 0.5 and minimum cement content of 300 kg/m^3 is :

(a) M 20 (b) M 30
(c) M 25 (d) M 35

Karnataka PSC AE 2017 Paper - II

Ans. (c) : From table 5 of IS 456 : 2000, for RCC we have

Exposure	Minimum cement content(kg/m^3)	Maximum free w/c ratio	Minimum grade of concrete
Mild	300	0.55	M20
Moderate	300	0.5	M25
Severe	320	0.45	M30
Very severe	340	0.45	M35
Extreme	360	0.4	M40

129. Which of the following grades of concrete in R.C.C. is recommended for severe exposure condition by BIS ?

(a) M20 (b) M25
(c) M30 (d) M15

Rajasthan JEn (Degree) 2016

Ans. (c) :

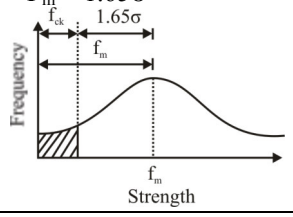
Exposure Condition	Grade of Concrete
Mild	M20
Moderate	M25
Severe	M30
Very Severe	M35
Extreme Server	M40

130. Which is a correct formula for characteristic strength (F_{ck}) where F_m is mean strength and σ is standard deviation?

(a) $F_{ck} = F_m + 1.55\sigma$ (b) $F_{ck} = F_m + 1.65\sigma$
(c) $F_{ck} = F_m + 1.75\sigma$ (d) $F_{ck} = F_m + 1.85\sigma$

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : $F_{ck} = F_m + 1.65\sigma$



131. Relation between Young's modulus and cube strength of concrete is:

(a) $E_c = 500\sqrt{f_{ck}}$ (b) $E_c = 5700\sqrt{f_{ck}}$
(c) $Ec = 5000\sqrt{f_{ck}}$ (d) $E_c = 700\sqrt{f_{ck}}$

Kerala PSC AE 2015

Ans. (b) : Young's modulus of concrete is related with characteristic strength of concrete as per IS code by the following formula

$E_c = 5700\sqrt{f_{ck}}$ As per – I.S 456–1978

Modulus of elasticity $= \frac{\text{Stress}}{\text{Strain}}$

$E_c = 5000\sqrt{f_{ck}}$ As per – I.S 456–2000

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132. The height to diameter ratio of cylindrical specimen for uniaxial compression test of concrete is:

(a) 0.50 (b) 0.30
(c) 0.25 (d) 2.00

Kerala PSC AE 2015

Ans. (d) Test specimens recommended for compressive strength test are 150 ×150 × 150 mm cubes or cylinders of 150 mm diameter and 300 mm height.

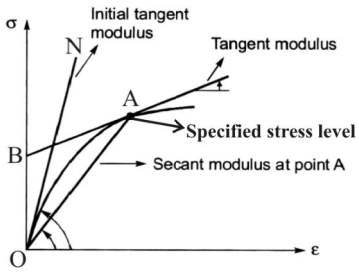
As per question = $\frac{300}{150} = 2$

133. Which of the following is measure of dynamic modulus of elasticity of concrete?

(a) Tangent modulus
(b) Secant modulus
(c) Initial tangent modulus
(d) All the above

GPSC AE March 2018
Kerala PSC AE 2015

Ans. (c) :



Modulus of elasticity of concrete is taken as secant modulus at a stress of around 0.33 f_{ck} for all practical purpose.

→ Secant modulus is slope of AO line.
→ Tangent modulus – Slope of AB line.
→ Initial tangent modulus– Slope of NO line.

134. Minimum period before striking soffit formwork to slabs:

(a) 21 days (b) 7 days
(c) 3 days (d) 1 day

Kerala PSC AE 2015

Ans. (c) :

Type of Formwork	Maximum Period before striking formwork
• Soffit, formwork to slabs	3 days
• Soffit formwork to beams	7 days
• Vertical formwork to columns walls beams.	16-24 hours
• Props to slabs : (i) Spanning up to 4.5 m (ii) Spanning over 4.5 m	 7 days 14 days

135. Stainless steel is an alloy of which among the following?

(a) chromium, nickel and iron
(b) manganese, copper and iron
(c) copper, carbon and iron
(d) copper, tin and zinc

Kerala PSC AE 2015

Ans. (a) : Stainless steel is a metal alloy made up of steel mixed with elements such as chromium, nickel, molybdenum, silicon, aluminium carbon, Iron mixed with carbon to produce steel is the main component of stainless steel.

Stainless steel contains 16-26% chromium but generally provides 18% chromium and 8%-10% nickel increasing for corrosion resistance.

136. The decrease of stress in steel at constant strain is termed as

(a) creep (b) fatigue limit
(c) relaxation (d) endurance limit

TNPSC AE 2018

Ans. (c) : Relaxation of steel is defined as the decrease in stress with time under constant strain. Due to the relaxation of steel, the prestress in the tendon is reduced with time.

Initial stress	Relaxation loss (MPa)
0.5 f_p	0
0.6 f_p	35
0.7 f_p	70
0.8 f_p	90

137. The permissible or allowable compressive stress f_{ac} of brick masonry does not depend on

(a) Type and strength of bricks
(b) Efflorescence of bricks
(c) Strength of mortar
(d) Slenderness ratio

ESE 2019

Ans. (b) : Efflorescence is a deposit of water soluble salts formed on the surface of brick masonry due to movement of water through passes. When water gets evaporated, efflorescence is formed as the dissolved salts gets deposited on the surface.

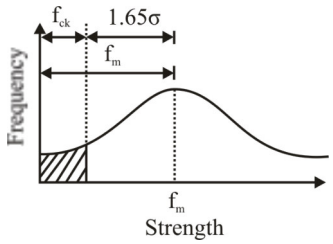
So, efflorescence of bricks does not affect the strength of brick or brick masonry.

138. The desired characteristic strength of a mix is 20 N/mm^2. The standard deviation is 4 N/mm^2 for 150 mm size of concrete cubes; and K = 1.645. The average strength of the cubes will be nearly

(a) 38.2 N/mm^2 (b) 32.4 N/mm^2
(c) 26.6 N/mm^2 (d) 22.8 N/mm^2

ESE 2019

Ans. (c)



$f_m = f_{ck} + 1.645\sigma$
$f_m = 20 + 1.645 \times 4$
$f_m = 26.58 N/mm^2$

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139. Fatigue in RCC beams will not be a problem if the number of cycles is less than

(a) 20,000 (b) 25,000
(c) 30,000 (d) 35,000

ESE 2019

Ans. (a) The distinction depends on whether the repeated loading causes a failure at less than or more than an arbitary number of cycles, usually considered in the range of 100-1000 cycles.

140. Poisson's ratio of concrete μ can be determined using the formula

(a) $\left(\frac{V}{2nL}\right)=\frac{(1-\mu)}{(1-2\mu)(1+\mu)}$

(b) $\left(\frac{V}{2nL}\right)=\frac{(1+\mu)}{(1-2\mu)(1+\mu)}$

(c) $\left(\frac{V^2}{2nL}\right)=\frac{(1-\mu)}{(1-2\mu)(1+\mu)}$

(d) $\left(\frac{V^2}{2nL}\right)=\frac{(1-\mu^2)}{(1-2\mu)(1+\mu)}$

Where

V is pulse velocity, in mm/s,
n is resonant frequency of longitudinal vibration, in Hz
L is distance between transducers, in mm

ESE 2019

Ans. (*) This dynamic Poisson's ratio is calculated by a test based on the principle of the ultrasonic test.

$$\left(\frac{V}{2nL}\right)^2=\frac{(1-\mu)}{(1-2\mu)(1+\mu)}$$

where V, n and L are known from test

141. The creep strain of cement attains its terminal value by

(a) 1 year (b) 2 years
(c) 5 years (d) 6 months

ESE 2019

Ans. (c) : Creep is a time dependent phenomenon and creep strain decreases with time. Although it doesn't stop for a long period, but creep strain at 5 years are taken as terminal values.

142. The value of creep coefficient of concrete at an age of 28 days loading is:

(a) 3.5 (b) 1.6
(c) 2.2 (d) 1.1

DSSSB AE 2019

Ans. (b)

$$\text{creep coefficient}\,(\phi)=\frac{\text{ultimate creep strain}}{\text{Elastic strain at age of loading}}$$

Values of creep coefficient

Age of loading	creep coefficient (ϕ)
7 days	2.2
28 days	1.6
1 year	1.1

143. What is the flexural strength of M25 grade concrete?

(a) 3.5 MPa (b) 4.56 MPa
(c) 1.27 MPa (d) 4.27 MPa

DSSSB AE 2019

Ans. (a) Flexural strength $=0.7\sqrt{f_{ck}}$

$=0.7\sqrt{25}=0.7\times5$
$=3.5$ MPa

144. Given below are the statements associated with concrete.
Statement 1 : As the compaction factor increases slump increases.
Statement 2 : Slump test helps in qualitatively understanding the strength of concrete.

(a) Statement 1 and statement 2 are false
(b) Statement 1 is true and statement 2 is false
(c) Statement 1 is false and statement 2 is true
(d) Statement 1 and statement 2 are true

DSSSB AE 2019

Ans. (b) In the compaction factor test, the degree of workability of concrete is measured in terms of internal energy required to compact the concrete thoroughly.

(CF) compaction factor

$$=\frac{\text{Mass of partially compacted conerete}}{\text{Mass of fully compacted conerete}}$$

In general

CF	< 0.85	low workability
CF	0.86–0.92	medium workability
CF	0.93–0.95	high workability

as increasing the value of compaction factor, workability of concrete is increases, hence the value of slump is also increase.
There is no relationship b/w the workability and the strength of the concrete.

145. The strength parameter of concrete used in design of plain jointed cement concrete pavement is

(a) Flexural strength
(b) Compressive strength
(c) Tensile strength
(d) Shear strength

ISRO Scientist/Engineer 2019

Ans : (a) :

- Flexural strength of the concrete is required for the purpose of design of concrete slab and this flexural strength is taken for 90 days insist of 28 days because initial repletion are very low and it can be obtained by multiplying factor 1.1.

$$\left[F_{cr}=1.1\times0.7\sqrt{f_{ck}}\right]$$

146. As per BIS 456–2000 (reaffirmed 2005), the minimum of concrete, with normal weight aggregate of 20 mm nominal maximum size, specified for a severe exposure condition is

(a) M 20 (b) M 25
(c) M 30 (d) M 35

ISRO Scientist/Engineer 2018

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Ans : (c)

- Severe exposure condition means concrete surfaces exposed to severe rain, alternate wetting and drying or occasional freezing while wet or severe condensation. Concrete completely immerged in sea water concrete exposed to coastal environment. So, as per IS 456-2000, the minimum grade of concrete with normal weight aggregates of 20mm nominal maximum size, for a severe exposure condition in–
- Plain Concrete – M20
- Reinforced Concrete – M30

147. The modulus of Rupture of concrete in terms of its characteristic cube compressive strength (f_{ck}) in MPa according to IS 456-2000 is

(a) $5000\ f_{ck}$ (b) $0.7\ f_{ck}$

(c) $5000\sqrt{f_{ck}}$ (d) $0.7\sqrt{f_{ck}}$

ISRO Scientist/Engineer 2014

Ans : (d) f_{cr} = modulus of rupture

characteristic cube compressive strength in MPa = f_{ck}

$$f_{cr} = 0.7\sqrt{f_{ck}}$$

- The modulus of elasticity of concrete ⇒

$$E_c = 5000\sqrt{f_{ck}}\ N/mm^2$$

148. Which one of the following is not used in concrete mix design?

(a) Degree of quality control of concrete

(b) Workability of concrete

(c) Characteristic compressive strength of concrete at 28 days

(d) Initial setting time of cement

APTRANSCO 2017

Ans : (d) The information required for concrete mix-design–

- Weather condition
- Exposure conditions of concrete
- Batching and mixing methods
- Quality of material
- Special requirement of concrete
- Quality control of site
- Workability and characteristic strength of concrete.

149. The modulus of elasticity, $E = 5000\sqrt{f_{ck}}$ where f_{ck} is the characteristic compressive strength of concrete, specified in IS : 456–2000 is based on

(a) Tangent modulus

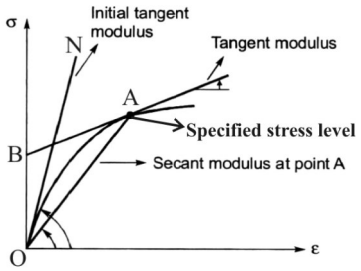
(b) Initial tangent modulus

(c) Secant modulus

(d) Chord modulus

APTRANSCO 2017

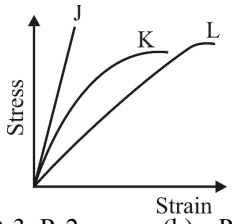
Ans : (b) :



Initial tangent modulus of concrete. It is a slope to the curve from origin. Initial tangent modulus is applicable when material behaves linearly elastic. So, in case when behavior of concrete will be linear then in that case elastic modulus of concrete will be evaluated by using initial tangent modulus.

150. Group I contains representative stress-strain curves as shown in the figure, while Group II gives the list of materials. Match the stress-strain curves with the corresponding materials.

Group I	Group II
P. Curve J	**1.** Cement paste
Q. Curve K	**2.** Coarse aggregate
R. Curve L	**3.** Concrete

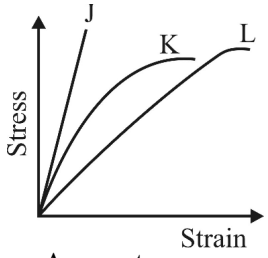


(a) P-1, Q-3, R-2 (b) P-2, Q-3, R-1

(c) P-3, Q-1, R-2 (d) P-3, Q-2, R-1

AP TRANSCO AEE 2017

Ans : (b) :



Where J = Coarse Aggregate

K = Concrete

L = Cement paste

151. Approximate value of shrinkage strain in concrete is :

(a) 0.003 (b) 0.0003

(c) 0.00003 (d) 0.03

OPSC AEE 2019 Paper–I

TNPSC AE 2015

Ans. (b) As per IS 456 : 2000 (clause 6.2.4.1) in the absence of data, the approximate value of total shrinkage strain for design may be taken as 0.0003.

152. For walls, columns and vertical faces of all structural members, the form work is generally removed after :

(a) 24 to 48 hours (b) 3 days

(c) 7 days (d) 14 days

OPSC AEE 2019 Paper–I

Ans. (a) : Form work–form work is the term given to either temporary or permanent moulds into which concrete or similar materials are poured.

⇒ For walls, columns and vertical faces of all structural members, the form work is generally removed after 24 to 48 hours.

153. The maximum compressive stress in concrete for design purposes is based on a partial safety factor of :

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(a) 1.15 (b) 1.50
(c) 1.85 (d) 2.20

OPSC AEE 2019 Paper–I

Ans. (b) :

Material	Limit state		
	Collapse	**Deflection**	**Cracking**
Concrete	1.5	1.0	1.3
Steel	1.15	1.0	1.0

154. Yield line theory results in :
(a) Elastic solution
(b) Lower bound solution
(c) Upper bound solution
(d) Unique solution

OPSC AEE 2019 Paper–I

Ans. (c) : The yield line analysis is an upper bound method in which the predicted failure load of a slab for given moment of resistance (capacity) may be higher than the true value. Thus, the solution of the upper bound method (yield line analysis) may result into unsafe design if the lowest mechanism could not be chosen.

155. In the limit state design method of RCC structures, the partial safety factor for steel according to IS : 456-2000 is :
(a) 1.15 (b) 0.87
(c) 1.00 (d) 1.5

PSPCL AE 2012

Ans. (a) : As per I.S. 456 : 2000 in LSM(limit state method)
Partial safety is factor for steel → 1.15
Partial safety factor for concrete → 1.5

156. Expansion joints are provided if the lengths of the concrete structure exceeds
(a) 35 m (b) 45 m
(c) 15 m (d) 25 m

GPSC AE March 2018

Ans. (b) They are placed in concrete to prevent expansive cracks formed due to temperature change. Expansion joints are provided in slabs, pavements buildings, bridges, sidewalks, railway tracks, piping systems, ships and other structures.
In interval of two consecutive joints should not exceed 45m length of concrete structure.

157. In the fourth amendment May 2013 of IS 456-2000, M60 grade has been shifted to
(a) standard concrete from high strength concrete
(b) high strength concrete from high strength standard concrete
(c) standard concrete from an ordinary concrete
(d) ordinary concrete from a standard concrete

GPSC AE March 2018

Ans. (a) : In IS : 456-2000,
In fourth amendment in May 2013, M-60 grade of concrete is specified as standard concrete whereas, it is as named before as high strength concrete.

158. In a uniaxial tension test on a mild steel bar, the Lueders' line will be
(a) inclined at 45° to the direction of tensile stress applied
(b) perpendicular to the direction of tensile stress applied
(c) along the direction of tensile stress
(d) perpendicular to the resultant compressive stress

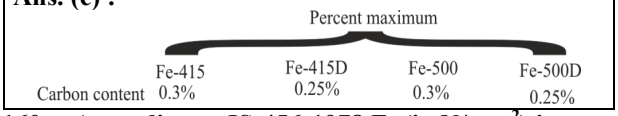
GPSC AE March 2018

Ans. (a) : Lueders' line elongated surface markings or depressions, often visible to an unaided eye, that from along the length of a tension specimen at an angle of approximately 45° to longitudinal axis.

159. Maximum carbon content in high strength deform bars should not be more than
(a) 0.60% (b) 0.06%
(c) 0.30% (d) 0.03%

GPSC AE March 2018

Ans. (c) :



160. According to IS-456-1978 E_c (in N/mm^2) is
(a) $E_c = 5700\sqrt{f_{ck}}$ (b) $E_c = 5000\sqrt{f_{ck}}$
(c) $E_c = 700\sqrt{f_{ck}}$ (d) $E_c = 5700\sqrt{f_{ck}}$

Vizag steel MT 2017

Ans. (a) : As per IS-456-1978, $E_c = 5700\sqrt{f_{ck}}$ N/mm^2
As per IS-456-2000, $E_c = 5000\sqrt{f_{ck}}$ N/mm^2

161. Flexural strength of M25 concrete is
(a) 5 N/mm^2 (b) 17.5 N/mm^2
(c) 3.0 N/mm^2 (d) 3.5 N/mm^2

GPSC Poly. Tech. Lect. 2016

Ans. (d) : Flexural strength $F_{cr} = 0.7\sqrt{F_{ck}}$
$= 0.7\sqrt{25}$
$= 3.5$ N/mm^2
f_{ck} = characteristic strength

162. In case of pre-stressed concrete tensile strength of concrete in flexure is f_{cr} = ...
(a) $5000\sqrt{f_{ck}}$ (b) $0.7\sqrt{f_{ck}}$
(c) $5700\sqrt{f_{ck}}$ (d) None of the above

GPSC Poly. Tech. Lect. 2016

Ans. (b) :
- Modulus of elasticity of concrete $\boxed{E_c = 5000\sqrt{f_{ck}}}$
- Tensile strength of concrete in flexure,

$\boxed{F_{cr} = 0.7\sqrt{f_{ck}}}$

where, F_{ck} = characteristic strength of concrete.

163. The diameter of reinforcing bar shall not exceed
(a) one eight of total thickness of slab
(b) ¼ × depth of slab
(c) ½ × depth of slab
(d) None of the above

GPSC Poly. Tech. Lect. 2016

Ans. (a) : Maximum diameter of reinforcing bars shall not exceed 1/8th of the total thickness of slab.

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164. Creep strain is
(a) Caused due to live load only
(b) Caused due to cyclic load only
(c) Caused due to dead load only
(d) independent of load

GPSC Poly. Tech. Lect. 2016

Ans. (c) : Creep is time dependent phenomena of loads being applied. So, as for calculation of moments for creep (pre-stressed) only dead loads are used.

165. The partial factor of safety for concrete as per IS 456-2000 is
(a) 1.15 (b) 0.87
(c) 0.446 (d) 1.5

GPSC Poly. Tech. Lect. 2016

Ans. (d) : Clause 36.4.2 of IS 456-2000 the partial safety factors is–concrete = 1.5, steel = 1.15

166. As per IS 456 - 2000, concrete in sea water or exposed directly along the sea coast shall be at least …….. grade
(a) M 30 (b) M 20
(c) M 10 (d) M 40

GPSC Poly. Tech. Lect. 2016

Ans. (b) : As per clause 8.2.8 of IS 456 : 2000 concrete in sea water or exposed directly along the sea coast shall be atleast M20 grade in the case of plain concrete and M 30 in case of reinforced concrete.

167. Loss of stress with time at constant strain in steel is called :
(a) relaxation (b) creep
(c) shrinkage (d) ductility

Punjab MCP SDO 2015

Ans. (a) : Relaxation is the reduction in stress with time at constant strain, decrease in the stress is due to the fact that some of the initial elastic strain is transformed into inelastic strain under constant strain.

168. According to Whitney's theory, ultimate strain of concrete is assumed to be :
(a) 0.03% (b) 0.10%
(c) 0.30% (d) 3.00%

Punjab MCP SDO 2015

Ans. (c) : According to Whitney's theory, ultimate strain of concrete is assumed to be 0.3%.

169. With the increase in rate of loading during testing, compressive strength of concrete
(a) increases (b) decreases
(c) remains same (d) none of the above

Punjab MCP SDO 2015

Ans. (a) : With the increase in rate of loading during testing, compressive strength of concrete increases.

170. The split tensile strength of M15 grade concrete when expressed as a percentage of its compressive strength is
(a) 10 to 15% (b) 15 to 20%
(c) 20 to 25% (d) 25 to 30%

Punjab PSC Sub divisional Engineer 2014

Ans. (a) : Modulus of rupture for M15 grade concrete is $= 0.75\sqrt{15} = 2.71$ N/mm^2.
Split tensile strength is found to be 50 to 75% of modulus of rupture.
So, split tensile strength for M15 grade of concrete.
$\Rightarrow$ 50 to 75% of 2.71 $\Rightarrow$ 10 to 15% of f_{cr}

171. As per IS : 456 - 2000, for mild exposure, for M20 grade of concrete, minimum cement content for reinforced concrete is .....
(a) 220 kg/m^3 (b) 250 kg/m^3
(c) 360 kg/m^3 (d) 300 kg/m^3

GPSC R&B AE 2016

Ans. (d) : As per IS-456 : 2000, for mild exposure, for finished concrete–

Grade of concrete	Minimum cement content (kg/m^3)
M-20	300 kg/m^3
M-25	300 kg/m^3

Maximum cement content for M-20 grade concrete is 450 kN/m^3.

172. Reinforced cement concrete is equally strong in taking
(a) tensile, compressive and shear stresses
(b) tensile and shear stresses
(c) tensile and compressive stresses
(d) compressive and shear stresses

GWSSB AAE 2015

Ans. (a) The reinforced cement concrete consists is a cement is which reinforcement is embedded. The reinforced cement concrete is equally strong to taking tensile, compressive and shear stresses.

173. Light weight concrete is used for
(a) air conditioned buildings
(b) non-load bearing walls
(c) reducing thickness of structures
(d) heat resistance

GWSSB AAE 2015

Ans. (b) : Light weight concrete is used for non-load bearing walls.
* Density of this concrete is considerably low (300 kg /m^3 to 1850 kg/m^3) when compared to normal concrete.
* 28 days compressive strength of more than 17 MPa and 28 days dry density not exceeding 1840 kg/m^3.

174. In adverse circumstances, the reinforced concrete member immersed in sea water or subjected to sea spray, the maximum permissible cover for the reinforcing bars, should not exceed :
(a) 40 mm (b) 50 mm
(c) 60 mm (d) 75 mm

KPTCL AE 2016

Ans. (d) : In adverse circumstances, the reinforced concrete member immersed in sea water or subjected to sea spray, the maximum permissible cover for the reinforcing bars, should not exceed – 70 to 75 mm.

175. The development length of each bar of three bars bundled together is increased by :
(a) 0.1 (b) 0.2
(c) 0.33 (d) 0.5

KPTCL AE 2016

Ans. (b) : Development length of bar increase.
2 bars – 10%
3 bars – 20%
4 bars – 33%

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176. Permissible bending tensile stress in high yield strength deformed bars of grade 415 N/mm^2 in a beam is :

(a) 190 N/mm^2
(b) 230 N/mm^2
(c) 140 N/mm^2
(d) None of the other three options

KPTCL AE 2016

Ans. (b) Permissible bending tensile stress in high yield strength deformed bars of grade 415 N/mm^2 in a beam is 230 N/mm^2

177. The characteristic strength of concrete is defined as that compressive strength below which not more than :

(a) 10% of results fall
(b) 5% of results fall
(c) 2%of results fall
(d) None of the other three options

KPTCL AE 2016

Ans. (b) The compressive strength of concrete is given in terms of the characteristic compressive strength of 150 mm size cube tested at 28 days (f_{ck}), the characteristic strength is defined as the strength of the concrete below which not more than 5% of the test results are expected to fall.

178. According to the concept of Limit State Design as per IS456: 2000, the probability of failure of a structure is :

(a) 0.097 (b) 0.079
(c) 0.067 (d) 0.078

KPTCL AE 2016

Ans. (a)
Probability of not failing under load = 95% = 0.95
Probability of not failing under material strength criteria = 95% = 0.95
Overall probability of not failing is = 0.95 × 0.95
= 0.9025
Therefore the overall probability of failure of a structure as per IS : 456 - 2000
$\Rightarrow$ 1 – 0.9025 = 0.0975

179. As per IS 456-2000 which one of the following represents the modulus of elasticity of concrete in MPa?

(a) $0.7\sqrt{f_{ck}}$ (b) $5000\sqrt{f_{ck}}$
(c) $5700\sqrt{f_{ck}}$ (d) $500\sqrt{f_{ck}}$

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (b) : As per IS 456-2000
Modulus of elasticity of concrete $(E_c) = 5000\sqrt{f_{ck}}$
As per IS 456-1978, $(E_c) = 5700\sqrt{f_{ck}}$

180. If the factor of safety is applied to a load, then it is called as

(a) Safe load (b) Margin of safety
(c) Safety factor (d) Load factor

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (d) : Load factor–Load factor determines the capability of load utilization i.e. how-long it is utilizing the power.

$$\text{Load factor} = \frac{\text{Factored load}}{\text{Applied load}}$$

181. Which design philosophy uses multiple safety factor format?

(a) Ultimate stress method
(b) Working stress method
(c) Limit state method
(d) Load resistance method

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (c) : Multiple or partial safety factor helps to provide safety at ultimate loads and adequate serviceability.

182. As per Indian standards, the formwork from of column can be removed

(a) Within 30 minutes of casting
(b) About 24 hours after casting
(c) Minimum 2 days after casting
(d) At 28 days after casting

Sikkim PSC AE 2017

Ans. (b) : According to IS 456-2000, Deshuttering period of different part of structural work are following-
For column, wall, beam sider = 16-24 hours

183. A bar is made of steel and it is 25 cm long. Its diameter is 5 mm. It is heated from 20°C to 50°C, while it is free to expand. The bar will develop

(a) Tensile stress (b) No stress
(c) Compressive stress (d) Shear stress

Sikkim PSC AE 2017

Ans. (b) : When bar is heated and free to expand, there will no stress is develop in bar because for stress generation restrain is mandatory.

184. In R.C.C. steel is used because it can provide good:

(a) Tensile strength
(b) Compressive strength
(c) Bond strength
(d) Shear strength

UJVNL AE 2012

Ans. (a) : In R.C.C. steel is used because it can provide good Tensile strength.

185. Loss of stress with time at constant strain in steel is called

(a) relaxation (b) creep
(c) ductility (d) shrinkage

APPSC AEE 2012

Ans. (a) : Relaxation is the reduction in stress with time at constant strain decrease in the stress is due to the fact that some of the initial elastic strain is transformed in to inelastic strain under constant strain.

186. As per IS 456-2000, the ultimate load in an RC structure in which effect of wind and earthquake load is neglected, should be :

(a) 1.5 DL + 0.5 LL (b) 1.2 DL + 2.5 LL
(c) 1.5 DL + 2.2 LL (d) 1.5 DL + 1.8 LL

SJVNL ET 2019

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Ans. (c) : As per IS 456-2000, the ultimate load in an RC structure in which effect of wind and earthquake load is neglected, should be 1.5 DL + 2.2 LL.

187. Bangalore falls under which seismic zone?
(a) Zone III (b) Zone II
(c) Zone IV (d) Zone V

SJVNL ET 2019

Ans. (b) : Bangalore falls under zone II seismic zone.

188. Which of the following is NOT considered as dead load
(a) Erection load
(b) Self-weight of the floor slab
(c) Self-weight of a beam
(d) Self-weight of a column

SJVNL ET 2019

Ans. (a) : All loads required to be carried by the structure or any part of it due to storage or positioning of construction material and erection equipment including all loads due to operation of such equipment, shall be considered as erection loads.

189. The usual mix adopted for cement concrete for RCC pipes is
(a) 1 : 4 : 8 (b) 1: 3 : 6
(c) 1 : 2 : 4 (d) 1 : 1.5 : 3

BPSC Poly. Tech. Lect. 2016

Ans. (d) : The usual mix adopted for cement concrete for RCC pipes is 1 : 1.5 : 3.
When this ratio is used it mean 1 part of cement, 1.5 part of fine aggregate 3 part of coarse aggregate, this ratio gives high strength and recommended for work such as in heavy loads. Important RCC structure such as columns beams, slabs, cantilever, chajja, porch, balcony.

190. Approximately, what would be the estimated value of elastic modulus (GPa) of concrete of M50 grade as per IS 456-2000?
(a) 27.5 (b) 30
(c) 32.5 (d) 35

UPPCL AE 2015

Ans. (d) : M50

$\boxed{E_c = 5000\sqrt{f_{ck}}}$ As per IS 456 : 2000

$= 5000\sqrt{50}$

$= 35355.34\ N/mm^2$ $\quad 1\ GPa = 10^3\ N/mm^2$

$\boxed{E_c = 35.35\ GPa}$

191. Why is minimum tension steel provided in a reinforced concrete beam?
(a) To arrest attack width
(b) To control excessive deflection
(c) To prevent hair-line crack formation
(d) To prevent sudden failure

UPPCL AE 2015

Ans. (c) : Minimum tension steel is provided in a reinforced concrete beam due to prevent hair-line crack formation.

192. As per IS 456, what is the minimum nominal concrete cover (mm) to reinforcement for reinforced concrete structures in exposed to severe exposure?
(a) 50 (b) 45
(c) 30 (d) 20

UPPCL AE 2015

Ans. (b) : As per IS 456, the minimum nominal concrete cover is 45 mm to reinforcement in exposed to severe exposure.

193. A building is to be constructed along the shore of a sea. The concrete for foundation is to be designed. What exposure class should be considered for the same?
(a) Moderate (b) Severe
(c) Extreme (d) Very severe

UPPCL AE 2015

Ans. (c) : Extreme exposure condition is adopted when a building is to be constructed along the shore of a sea.

194. As per IS 456, what would be the estimated value of flexural strength (MPa) for M25 grade of concrete?
(a) 4.5 (b) 2.5
(c) 3.5 (d) 3

UPPCL AE 2015

Ans. (c) : Flexural strength for M25

$f_{cr} = 0.7\sqrt{f_{ck}}$

$= 0.7\sqrt{25}$

$\boxed{f_{cr} = 3.5\ N/mm^2}$

195. What is the maximum permissible acid soluble chloride content (kg/cum) for reinforced concrete?
(a) 1.5 (b) 0.6
(c) 0.4 (d) 3

UPPCL AE 2015

Ans. (b) : If the chloride content increase from its permissible limit, the reinforced concrete becomes corrosive and decaying. So, 0.6 kg/cum is fixed limit for chloride content of reinforced concrete.

196. In the rebend test of reinforcing steel bars, after initial bending and boiling in water for 30 min, the bar is again bent to an angle of :
(a) 125^o (b) 157.5^o
(c) 180^o (d) 130^o

UPPCL AE 2015

Ans. (b) : In Re-bend test, reinforcing steel is bent 45^o at 100^oC for 30 minutes and the bent lock back 22.5^o
Bent angle = $180^o - 22.5^o$
$= 157.5^o$

197. The approximate ratio of direct tensile strength to characteristic strength is
(a) 0.25 (b) 0.50
(c) 0.33 (d) 0.10

BPSC AE Aug. 2019 Paper-V

Ans. (d) : Tensile strength of concrete is $\frac{1}{10}$th of the characteristic strength of concrete.
If M - 20 concrete is opted.

$\left.\begin{matrix}\text{characteristic strength} = 20\text{MPa} \\ \text{its tensile strength} = 2\text{MPa}\end{matrix}\right\}$ experimental value

$= \frac{2}{20} = 0.10$

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198. In an under-reinforced section

(a) concrete is not fully stressed up to permissible stress
(b) reinforcement is not fully stressed up to permissible stress
(c) Either (A) or (B)
(d) Both (A) and (B)

BPSC AE Aug. 2019 Paper-V

Ans. (a) : In an under reinforced section, the steel yields or tends to yield to its permissible limit before the concrete yield. So, that the structure fails due to steel yielding in under reinforce section.

199. The strength of concrete is maximum in

(a) compression (b) shear
(c) twisting (d) fatigue

BPSC AE Aug. 2019 Paper-V

Ans. (a) : The strength of concrete is maximum in compression generally, the maximum compressive strength than can possibly be obtained as 51.71N/mm^2 for a 28 day curing.

200. The shrinkage of concrete is due to change in

(a) length (b) cross-sectional area
(c) volume (d) the surface area

BPSC AE Aug. 2019 Paper-V

Ans. (c) : The volumetric changes of concrete structure due to the loss of moisture by evaporation is known as concrete shrinkage.

201. A site engineer wants to quickly pour a circular column to support a G+1 building. there are no other major loading considerations, but the building is in seimic zone-1, what should be the minimum number of longitudinal rebars provided?

(a) 3 (b) 4
(c) 5 (d) 6

UPRVUNL AE 2016

Ans.: (d)

202. A foundation is to be designed for a column supporting a G+11 building. This foundation is resting on a sound rock. What should the minimum cover (mm) be of provided to the reinforcement?

(a) 40 (b) 50
(c) 60 (d) 70

UPRVUNL AE 2016

Ans.: (b)

203. The yield strength of Fe 415 steel in MPa would be a minimum of

(a) 250.0 (b) 380.0
(c) 415.0 (d) 500.0

UPRVUNL AE 2016

Ans.: (c) Yield strength of Fe 415 = 415 MPa or 415 N/mm^2.

204. An engineer wants to design a reinforced cement concrete (RCC) structure in sea- water. What should be the minimum grade of concrete, he should use, while following IS 456?

(a) M20 (b) M25
(c) M30 (d) M40

UPRVUNL AE 2016

Ans.: (c) The minimum grade of concrete used for structure in sea water should be M30.
As per IS 456-2000, if the RCC structure is in exposure of sea water spray then the minimum grade of concrete to be used is M35 and if the structure is sub merged or fully merged is sea water then the minimum grade of concrete will be M40.

205. A project manager intends to design a concrete mix proportion for M30 grade of concrete. From his previous experience. he knows that the standard deviation is 3 MPa. to validate the same, he asks his Materials engineer to cast 40 samples, the standard deviation for which is found to be 4 MPa in lab. If the concrete mix is to be designed failure rate of 1 in 100 (t= 2.33) then the target mean strength for proportioning the mixture would approximately be:

(a) 37 (b) 39
(c) 40 (d) 42

UPRVUNL AE 2016

Ans.: (b)

206. Creep deformation is a property of any material to deform under the influence of

(a) Thermal stresses
(b) Moisture related stresses
(c) Mechanical stresses
(d) Stresses due to wind loads

UPRVUNL AE 2016

Ans.: (c) Creep: The property by virtue of which materials undergoes additional deformation (over and above that due to applied load) with passage of time under sustained loading within elastic limit is called creep.

207. The cracking moment of a beam made of M-25 concrete having a size of 600 mm × 300 mm is

(a) 63×10^6 MPa (b) 64×10^6 MPa
(c) 65×10^6 MPa (d) 61×10^6 MPa

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Ans. (a) :

208. Which one of the following types of steels is used in the manufacture of rails?

(a) mild steel (b) manganese steel
(c) cast steel (d) bessemer steel

GPSC AE Dec 2018

Ans. (b) : Manganese steel is used in the manufacture of rails.

209. Modulus of elasticity of concrete can be given as

(a) $2000\sqrt{f_{ck}}$ (b) $3700\sqrt{f_{ck}}$
(c) $5000\sqrt{f_{ck}}$ (d) $5700\sqrt{f_{ck}}$

GPSC AE Dec 2018

Ans. (c) : As per IS 456 : 2000
$E_c = 5000\sqrt{f_{ck}}$
where f_{ck} = characteristic compressive strength.

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210. When the building does not have closely-spaced modes of vibration, then ..... method has good accuracy in estimating the peak response quantities like member forces, storey forces, storey shears and base reactions.

(a) Complete Quadratic Combination
(b) Square Root of Sum of Squares
(c) Absolute Sum
(d) Double Sum Combination

GPSC AE Dec 2018

Ans. (b) : When the building does not have closely-spaced modes of vibration, then Square Root of Sum of Squares method has good accuracy in estimating the peak response.

211. In Ultimate Load Method, the load used in the design of R.C.C. structures refers to

(a) total collapse load
(b) working load multiplied by a load factor
(c) working load divided by a load factor
(d) plastic load

GPSC AE Dec 2018

Ans. (b) : In Ultimate Load Method, the load used is equal to working load multiplied by a load factor.

212. In 4th amendment of IS 456-2000, M60 grade has been shifted

(a) to standard concrete from high strength concrete
(b) to high strength concrete from high standard concrete
(c) to standard concrete from an ordinary concrete
(d) to ordinary concrete from a standard concrete

GPSC AE Dec 2018

Ans. (a) : In 4th amendment of IS 456-2000, M60 grade has been shifted to standard concrete from high strength concrete.

213. As per IS 456 : 2000, the minimum grade of concrete to be used for plain concrete and reinforced concrete in concrete works exposed directly along the sea coast are respectively :

(a) M15, M20 (b) M15, M25
(c) M20, M25 (d) M20, M30

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Ans. (d) : As per IS 456 : 2000, the minimum grade of concrete to be used for plain concrete and RCC in concrete works exposed directly along the sea coast are M20 and M30 respectively.

214. As per IS 1893 : 2002, mass irregularity shall be considered to exist when the seismic weight of any storey is more than ..... of that of its adjacent storey's.

(a) 250% (b) 200%
(c) 300% (d) 150%

BHEL ET 2019

Ans. (b) : As per Is 1893 : 2002 mass irregularity shall be considered to exist when the seismic weight of any storey is more than 200% of that of its adjacent storey's.

215. Match the items in List 1 (Type of materials for repair of structures) with those in List 2 (Use/characteristics). Choose the best answer using the codes given in options.

List 1	List 2
A. Carbon fibre reinforced polymeric composite	1. Flowable, shrinkage free, high early strength concrete.
B. Fibre Reinforced polymeric (FRP) Composite	2. Repair of column
C. Micro concrete	3. Replacement of defective/corroded reinforcement
D. High performance concrete	4. Heavy duty floors with congested reinforcement
E. Carbon aramid meshes	

(a) A–4, B–2, C–3, D–1, E–2
(b) A–2, B–3, C–1, D–4, E–3
(c) A–3, B–2, C–3, D–4, E–1
(d) A–3, B–2, C–4, D–1, E–2

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Ans. (b) : A–2, B–3, C–1, D–4, E–3

216. Match the information given in Group-I with those in Group-II :

Group-I		Group-II	
P.	Factor to decrease ultimate strength to design strength	1.	Upper bound on ultimate load
Q.	Factor to increase working load to ultimate load for design	2.	Lower bound on ultimate load
R.	Statically method of ultimate load analysis	3.	Material partial safety factor
S.	Kinematical mechanism method of ultimate load analysis	4.	Load factor

(a) P-1; Q-2; R-3; S-4 (b) P-2; Q-1; R-4; S-3
(c) P-3; Q-4; R-2; S-1 (d) P-4; Q-3; R-2; S-1

AP TRANSCO AEE 2017

Ans. (c)

- Factor to decrease ultimate strength to design strength is material partial safety factor.
- Factor to increase working load to ultimate load for design is load factor.
- Statical method of ultimate load analysis is lower bound on ultimate load.
- Kinematical mechanism method of ultimate load analysis is upper bound on ultimate load.

217. Which one of the following is not used in concrete mix design ?

(a) Degree of quality control of concrete
(b) Workability of concrete
(c) Characteristic compressive strength of concrete at 28 days
(d) Initial setting time of cement

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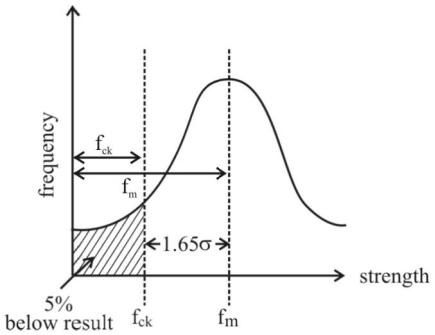
Ans. (d) : Factors used in concrete mixed design
(i) Quality control
(ii) Durability of concrete
(iii) Workability of concrete
(iv) Maximum nominal size of aggregate
(v) Grading and type of aggregate
(vi) Characteristic compressive strength of concrete

218. The target mean strength f_m for concrete mix design obtained from the characteristic strength f_{ck} and standard deviation σ, as defined in IS : 456-2000, is
(a) $f_{ck} + 1.35\sigma$ (b) $f_{ck} + 1.45\sigma$
(c) $f_{ck} + 1.55\sigma$ (d) $f_{ck} + 1.65\sigma$

AP TRANSCO AEE 2017

Ans. (d) : As per IS 456 : 2000
Target mean strength (f_m) = $f_{ck} + 1.65\sigma$



219. The modulus of elasticity, E = 5000 $\sqrt{f_{ck}}$ where f_{ck} is the characteristic compressive strength of concrete, specified in IS : 456–2000 is based on
(a) Tangent modulus
(b) Initial tangent modulus
(c) Secant modulus
(d) Chord modulus

AP TRANSCO AEE 2017

Ans. (b) : Initial tangent modulus of concrete. It is a slope to the curve from origin. Initial tangent modulus is applicable when material behaves linearly elastic. So, in case when behavior of concrete will be linear then in that case elastic modulus of concrete will be evaluated by using initial tangent modulus.

220. For reinforced concrete structures to be constructed along sea coast minimum grade of concrete should be:-
(a) M15 (b) M20
(c) M30 (d) M25

UPPCL AE 2014

Ans. (c) : For reinforced concrete structures to be constructed along sea coast minimum grade of concrete should be M-30.

221. Consider the following strengths of concrete:-
1. Cube strength
2. Cylinder strength
3. Split tensile strength
4. Modulus of rupture
The correct sequence in increasing order of these strength is:-
(a) 3-4-1-2 (b) 4-3-1-2
(c) 3-4-2-1 (d) 4-3-2-1

UPPCL AE 2014

Ans. (d) : Modulus of Rupture < Split Tensile strength < Cylinder strength < Cube strength.
Cube strength is maximum
Cylinder strength is 80% of Cube strength.
Modulus of Rupture in case of concrete is minimum.

222. The anchorage value of a hook is assumed four times the diameter of the bar if the angle of the bend is:-
(a) 60º (b) 30º
(c) 45º (d) 40º

UPPCL AE 2014

Ans. (c) : If the angle of bed is 45º, anchorage value of hook is assumed four times the diameter.

223. What is the value of flexural strength of M 25 concrete:-
(a) 1.75 MPa (b) 3.0 MPa
(c) 4.0 MPa (d) 3.5 MPa

UPPCL AE 2014

Ans. (d) : Flexural strength = $0.7\sqrt{f_{ck}}$
= $0.7\sqrt{25}$
= 3.5 MPa.

224. Which one of the following sections performs better on the ductility criterion:-
(a) Under reinforced section
(b) Balanced section
(c) Non prismatic section
(d) Over reinforced section

UPPCL AE 2014

Ans. (a) : Under reinforced section–Section in which tension steel also reaches yield strain at loads lower then the load at which concrete reaches the failure strain in bending.

225. For moderate exposure, the minimum nominal cover to reinforcement to meet durability requirements as specified by IS 456 : 2000 is
(a) 50 mm (b) 30 mm
(c) 75 mm (d) 20 mm

GPSC AE January 2018

Ans. (b) : As per IS:456-2000, the term clear cover is replaced by nominal cover. Nominal cover is the distance between the exposed concrete surface to the nearest surface of the reinforcement bar.
Nominal cover ≥ 300 mm or diameter of bar.

Condition	Nominal cover
Mild	20 mm
Moderate	30 mm
Severe	45 mm
Very severe	50 mm

226. In nominal mix of concrete designated by 1:2:4, the values indicates the proportion of
(a) Course aggregate : Fine aggregates : Cement
(b) Cement : fine aggregates : coarse aggregates
(c) Cement : water : aggregates
(d) Aggregate : cement : water

GPSC AE January 2018

Ans. (b) : A nominal mix of concrete as 1:2:4.
1 part of cement : 2 part of fine aggregate : 4 part of coarse aggregate.

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227. For RC braced frames maximum redistribution of moment allowed is

(a) 30% (b) 40%
(c) 20% (d) 10%

GPSC AE January 2018

Ans. (a) According to IS code : 456 :2000 (Clause 37.1.1). The elastic moment at any section in the member due to a particular combination of load shall not be reduced by more than 30% of the elastic maximum moment.

228. Steel of yield strength 400 MPa has been used in a structure. What is the value of the maximum allowable tensile strength?

(a) 240 MPa (b) 200 MPa
(c) 120 MPa (d) 96 MPa

UPRVUNL AE 2015

Ans. (a) :

$$\text{maximum allowable tensile strength} = \frac{\text{yield strength}}{1.67}$$

$$= \frac{400}{1.67} = 239.52 \approx 240\,\text{MPa}$$

229. Creep strains in concrete structures are usually _____:

(a) independent of loads
(b) caused by wind loads
(c) caused by earthquake loads
(d) caused by sustained loads

UPPCL AE 2018

Ans. (d) : Creep strain in concrete structures are usually caused by sustained loads.

230. Ratio of modulus of elasticity of two different materials is called as _____.

(a) Factor of safety (b) Poisson's ratio
(c) Stress ratio (d) Modular ratio

UPPCL AE 2019

Ans. (d) Ratio of modulus of elasticity of two different material is called as modular ratio.

$$\text{Modular ratio} = \frac{E_S}{E_C} = \frac{2\times10^5}{5000\sqrt{f_{ck}}}$$

E_S = Modulus of elasticity of steel = 2×10^5 N/mm^2
f_{ck} = characteristic compression strength of concrete.

231. According to IS 456, the estimated flexural strength of concrete of grade M25 (f_{ck} = 25 MPa) would be:

(a) 35 kg/sq cm (b) 50 kg/sq cm
(c) 75 kg/sq cm (d) 25 kg/sq cm

LMRC AM 2015

Ans (a) Flexural stength (f_{cr}) = $0.7\sqrt{f_{ck}}$ N/mm^2

f_{ck} = 25 MPa

$$f_{cr} = 0.7\sqrt{25} = 3.5\ \text{N/mm}^2 = 3.5\times\frac{100}{10}\ \text{kg/cm}^2$$

f_{cr} = 35 kg/cm^2

232. When the permissible compressive stress of concrete in bending is 8.5 MPa for the concrete grade M25, the value of m (modular ratio) is :

(a) 9.33 (b) 13.33
(c) 8.11 (d) 10.98

DMRC AM 2020

Ans. (d) : Modular ratio $= \frac{280}{3\sigma_{cbc}}$

$$= \frac{280}{3\times8.5} = 10.98$$

233. As per IS 456 : 2000, the flexural strength of concrete at 28 days of ageing is related to compressive strength by the equation :

(a) $0.7\sqrt{f_{ck}}$ (b) $0.50\sqrt{f_{ck}}$
(c) $1.25\sqrt{f_{ck}}$ (d) $1.55\sqrt{f_{ck}}$
(e) $2.75\sqrt{f_{ck}}$

CGPSC AE 2017

Ans. (a) As per IS 456 : 2000, the flexural strength of concrete at 28 days of ageing is related to compressive strength by the equation

Flexural strength of concrete = $0.7\sqrt{f_{ck}}$

234. The creep coefficient at 28 days loading is :

(a) 2.2 (b) 1.1
(c) 1.6 (d) 3.2
(e) 5.7

CGPSC AE 2017

Ans. (c) The creep coefficient at 28 days loading is 1.6.

$$\text{creep coefficient} = \frac{\text{Ultimate creep strain}}{\text{Elastic strain at age of loading}}$$

Value of creep coefficient :

Age at loading	Creep coefficient
7 days	2.2
28 days	1.6
1 year	1.1

235. Time-dependent strain is termed as :

(a) elastic strain (b) ultimate creep strain
(c) creep (d) shrinkage
(e) plastic shrinkage

CGPSC AE 2017

Ans. (c) Time-dependent strain is termed as creep.
Creep : The property by virtue of which material undergoes additional deformation (ever and above that due to applied load) with passage of time under sustained loading within elastic limit is called creep.

236. For determination of allowable stress in axial compression, Bureau of Indian Standards has adopted:

(a) Engesser formula
(b) Secant formula
(c) Euler's formula
(d) Rankine formula
(e) None of the above options

Haryana PSC AE 2018 Paper–I

Ans. (b) : For determination of allowable stress in axial compression, Bureau of Indian Standards has adopted Secant formula.

237. According to the Indian standard, IS : 800 (2007), the coefficient of expansion of steel per degree centrigrade per unit length is taken as:

(a) 0.000006 (b) 0.000008
(c) 0.000010 (d) 0.000012
(e) 0.000014

Haryana PSC AE 2018 Paper–I

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Ans. (d) : As per IS 800:2007
(Steel) Co-efficient of Expansion
$(\alpha) = 0.000012/^{o}C$
(Concrete) Co-efficient of Expansion
$(\alpha) = 10 \times 10^{-6}/^{o}C$

238. According to the Indian Standard (IS) code of practice, the unit weight of concrete is recommended to be taken as:
(a) 1000 kg/m^3 (b) 1500 kg/m^3
(c) 2000 kg/m^3 (d) 2500 kg/m^3
(e) 3000 kg/m^3

Haryana PSC AE 2018 Paper–I

Ans. (d) : Unit weight of concrete = 2500 kg/m^3 -
R.C.C. = 2400 kg/m^3 -
P.C.C. = 2400 kg./m^3.

239. The mix proportion by volume for nominal mix concrete of M20 grade made of 20 mm nominal maximum size of aggregate is:
(a) 1:2:3 (b) 1:2:4
(c) $1:1\frac{1}{2}:2$ (d) $1:1\frac{1}{2}:3$
(e) $1:1\frac{1}{2}:4$

Haryana PSC AE 2018 Paper–I

Ans. (d) :

Grade of concrete	Mix portion	Perspective characteristic strength (N/mm^2)
M10	1:3:6	10
M15	1:2:4	15
M20	1:1.5:3	20
M25	1:1:2	25

240. If q is the punching shear resistance per unit area a, is the side of a square footing for a column of side b, carrying a weight W including the weight of the footing, the depth (D) of the footing from punching shear consideration, is:
(a) $D = \frac{W(a-b)}{4a^2bq}$ (b) $D = \frac{W(a^2-b^2)}{4a^2bq}$
(c) $D = \frac{W(a^2-b^2)}{8a^2bq}$ (d) $D = \frac{W(a^2-b^2)}{4abq}$

Haryana PSC AE 2017

Ans. (b) : Punching shear is a type of failure of reinforced concrete slabs subjected to high localized forces, in flat slab structures, this occurs at column supports points. The failure is due to shear. This type of failure is critical because no visible signs are shown prior to failure.

241. Long term elastic modulus in terms of creep coefficient (θ) and 28 days characteristics strength (f_{ck}) is given by:
(a) $\frac{5000\sqrt{f_{ck}}}{1+\theta}MPa$ (b) $\frac{50000\sqrt{f_{ck}}}{1+\theta}MPa$
(c) $\frac{5000\sqrt{f_{ck}}}{1+\sqrt{\theta}}MPa$ (d) $\frac{5000\sqrt{f_{ck}}}{\sqrt{1+\theta}}MPa$

Haryana PSC AE 2017

Ans. (a) : Long term modulus of elasticity–
$$\frac{E_c}{1+\theta} = \frac{5000\sqrt{f_{ck}}}{(1+\theta)}\text{MPa}$$

242. Assertion (A) : In the case of mild steel, the tensile strength (expressed as per unit area) of smaller diameter bars are more than that of larger diameter bars.
Reason (R): In case of smaller diameter mild steel bars, the ratio of outer hard core to total area (outer hard core + inner soft core) is more.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not a correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Haryana PSC AE 2017

Ans. (a) :

243. A beam curved in plan is designed for -
(a) Bending moment and shear
(b) Bending moment and torsion
(c) Shear and torsion
(d) Bending moment, shear and torsion

Rajasthan VP ITI 2018

Ans. (d) A beam curved in plan is designed for bending moment, shear and torsion.

244. The pin of a rocker bearing in a bridge is design for-
(a) Bearing and Shear
(b) Bending and Shear
(c) Bearing and Bending
(d) Bearing, Shear and Bending

Rajasthan VP ITI 2018

Ans. (d) : A rocker bearing is type of expansion that comes in a great variety. It typically consists of a pin at the top that facilitates rotations and a curved surface at the bottom that accommodates the translational moments. Rocker and pin bearings are used in steel bridges. rocker bearing in a bridge is design for bearing, shearing and bending.

245. The principle of virtual work states that, for a body to be in equilibrium, the virtual work should be:
(a) Any value between zero and one
(b) maximum
(c) Zero
(d) minimum

CIL MT 2020

Ans. (c) : The principle of virtual work state that, for a body to be in equilibrium, the virtual work should be zero.

246. Strength of commonly used concrete, for constructing low rise residential building is:
(a) 300 psi (b) 8000 psi
(c) 15000 psi (d) 25000 psi

RRB SSE Secunderabad : 21-12-2014

Ans : (c) Strength of commonly used concrete, for constructing low rise residential building is 15000 psi.

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247. The expected ultimate tensile strength of Fe-415 steel is expected to be at least:

(a) 485 (b) 455
(c) 425 (d) 435

IRCON AE 2017

Ans. (a) : Yield strength for Fe-415 as 415 N/mm^2 or 60 kpsi with ultimate tensile strength (UTS) as 485 N/mm^2 or ≈ 70 kpsi and minimum elongation of 14.5%.

248. Loss of stress due to elastic deformation of concrete depends upon:

(a) Relaxation of steel
(b) Modular ratio
(c) Friction and anchorage slip
(d) Shrinkage strain

IRCON AE 2017

Ans. (b) : Loss of stress due to elastic deformation–

$$\sigma_{Ls} = f_c m$$

f_c = average stress is concrete

m = modular radio = $\frac{E_S}{E_C}$

E_S and E_C = Young's modulus of steel and concrete respectively.

249. According to IS 456:2000, the dosages of retarders, plasticizer and super plasticizer by weight of cementations materials respectively are restricted to:

(a) 0.5%, 0.5% and 1.0%
(b) 0.5%, 1.0% and 2.0%
(c) 1.0%, 0.5% and 2.0%
(d) 1.1%, 0.5% and 2.0%

JUVNL AE 2016

Ans. (b) : According to IS 456:2000.

Material	Limit as weight of cementations materials
Retarders	0.5%
Plasticizers	1.0%
Super Plasticizers	2.0%

250. Usually for the work of M 25 grade of concrete, the nominal mix will be approximately

(a) 1 : 1 : 2 mix (b) 1 : 2 : 4 mix
(c) 1 : 3 : 6 mix (d) 1 : 2.5 : 3 mix

JUVNL AE 2016

Ans. (a) : Grade of concrete = M25
nominal mix = 1 : 1 : 2 mix.
(cement : sand : aggregate)

251. The ultimate shrinkage strain value of concrete as per IS:456 can be taken as:

(a) 0.0004 (b) 0.0005
(c) 0.0003 (d) 0.0002

JUVNL AE 2016

Ans. (c) : The ultimate shrinkage strain value of concrete as per IS:456 can be taken as 0.0003.

252. The concrete grade M-20 means concrete cube having crushing strength of ..... at 28 days period.

(a) 20 kg/cm^2 (b) 2 ton/mm^2
(c) 20 N/mm^2 (d) 200 N/cm^2

GPHC AE 2018

Ans : (c) M20 grade concrete denotes a concrete whose mix is so designed as to generate a characteristic strength of 20 MPa or 20 N/mm^2 at 28 days period.

253. Strength of cubes for M30 grade concrete at 7 days shall be :

(a) 30 kg/cm^2 (b) 13 kg/cm^2
(c) 20 kg/cm^2 (d) 25 kg/cm^2

GPHC AE 2018

Ans : (c)

Grade of concrete	Minimum compressive strength (N/mm^2) (7 days)
M20	13.5
M25	17
M30	20
M35	23.5

254. The main reinforcement in a cantilever beam shall be at :

(a) Bottom of the beam (b) Top of the beam
(c) Middle of the beam (d) Non of three

GPHC AE 2018

Ans : (b) The tension side of cantilever beam is always at top, therefore reinforcement should be provided to top of beam.

255. In case of small sized column with minimum dimension of 200 mm or under whose reinforcing bar does not exceed 12 mm; the recommended cover is :

(a) 25 mm (b) 30 mm
(c) 35 mm (d) 40 mm

GPHC AE 2018

Ans : (a) Nominal cover for different member :
Beam → 25 mm to 30 mm
Slab → 20 mm to 30 mm
Column → 25 mm to 40 mm

256. Additional length to be provided for estimating length of RCC bar for one 45° bent up is :

(a) 0.35d (b) 0.45d
(c) 0.55d (d) 0.90d

GPHC AE 2018

Ans : (b) The additional length to provided for estimating length of R.C.C. bar for one 45° bend up is 0.45 D.
Where : d → is dia of bar.

257. The horizontal distance between two parallel main bars shall not be less than the following :

(a) Diameter of bars if bars are of same diameter.
(b) Diameter of larger bar if diameters are unequal.
(c) 5 mm more than nominal maximum size of coarse aggregate.
(d) All the above

GPHC AE 2018

Ans : (d) The Horizontal distance b/w two Parallel main bars shall not be less than :
1. Diameter of bars if bars are of same diameter.
2. Diameter of bars (Large bar) if diameter are unequal.
3. 5 mm more than nominal maximum size of coarse aggregate.

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258. Percentage of carbon content in mild steel is

(a) less than 0.25 (b) between 0.25 and 0.7
(c) between 0.7 and 1.5 (d) greater than 1.5

WBPSC AE 2016

Ans. (a) : Percentage of carbon content in mild steel is less than 0.25.

259. Characteristics load means that value of the load

(a) Which is not expected to fall by more than 5%
(b) Which is not expected to fall by more than 5%
(c) Which has 95% probability of not being exceeded during the life of the structure
(d) Which has 85% probability or not being exceeded during the life of the structure

WBPSC AE 2001

Ans. (c) : Characteristics load means that value of the load which has 95% probability of not being exceeded during the life of the structure.

260. Steel is used as reinforcement instead of other metals in reinforced concrete because

(a) concrete and steel have the same modulus of elasticity
(b) steel is cheaper compared to other metals
(c) concrete and steel have the same coefficient of linear thermal expansion
(d) steel is readily available in large quantity

WBPSC AE (PHE) 1998

Ans. (c) : Steel is used as reinforcement instead of other metals in reinforced concrete because concrete and steel have the same coefficient of linear thermal expansion.

261. For purpose of determining the seismic force in structure, India has been classified into several zones an Calcutta falls in

(a) Zone I (b) Zone III
(c) Zone IV (d) Zone V

WBPSC AE (Common) 1998

Ans. (b) : Calcutta falls in Zone III of seismic zone in India.

262. Which of the following materials has the highest carbon percentage?

(a) Cast iron (b) Mild steel
(c) High carbon steel (d) Stainless steel

WBPSC AE (Common) 1998

Ans. (a) : Cast iron typically contains more than 2% carbon while cast steel often contains between 0.1–0.5 percent carbon.

263. The sulphate content as SO_4 in water to be used for concreting should not exceed

(a) 1000 mg/L (b) 750 mg/L
(c) 500 mg/L (d) 1250 mg/L

WBPSC AE (Common) 1998

Ans. (d) :

264. Nominal mix concrete can be used for concrete of grade

(a) M 35 (b) M 30
(c) M 25 (d) M 20

TSPSC AEE 2017

Ans. (d) : Nominal mixes for grades of concrete such as M20 are generally used for small scale construction.

265. For compressive strength determination the minimum number of cubes required in a sample are

(a) 2 (b) 3
(c) 4 (d) 5

TSPSC AEE 2017

Ans. (b) : A sample consist of three cubes/specimens it is for the same concrete sample cast more than three cubes, so that any defective cube should not be tested or any doubt full test result due to testing may be rejected and not included in the average of three cubes.

266. The minimum grade of concrete for reinforced concrete in sea water and exposing to coastal environment is:

(a) M 20 (b) M 25
(c) M 30 (d) M 35

TSPSC AEE 2017

Ans. (d) : Concrete surfaces exposed to sea water spray corrosive fumes or severe freezing conditions whist wet, M35 grade concrete use.

267. The degree of workability of concrete for heavily reinforced sections of slab should be

(a) Very high (b) High
(c) Low (d) Medium

TSPSC AEE 2017

Ans. (d) : The degree of workability of concrete for heavily reinforced sections of slab should be medium.

268. Bond strength between reinforcement and concrete is affected by

(a) steel properties (b) concrete properties
(c) both (a) and (b) (d) none of these

Chandigarh AE 2017

Ans. (d) :

269. Rebound hammer is used to determine to find

(a) in-situ tensile strength
(b) in-situ compressive strength
(c) in-situ flexural strength
(d) all of the above

Chandigarh SDE 2017

Ans. (b) :

270. The individual variation between test strength of sample should not be more than

(a) ±5% of average (b) ±10% of average
(c) ±15% of average (d) ±20% of average

Chandigarh SDE 2017

Ans. (c) : Variation of individual samples strength should not be more than ±15% of average.

271. Flexural strength of M25 grade concrete will be _____.

(a) 4.5N/mm^2 (b) 3.5 N/mm^2
(c) 5.5 N/mm^2 (d) 2.5 N/mm^2

KPSC AE 2020
LMRC AM 2019

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Ans. (b) (f_r) Flexural strength of concrete

$$= 0.7\sqrt{f_{ck}}$$

f_{ck} = characteristic compressive strength of concrete

f_{ck} of M25 is 25 N/mm^2

Flexural strength of M25 $= 0.7\sqrt{f_{ck}}$

$$= 0.7\sqrt{25} = 3.5 \text{ N/mm}^2$$

272. The pH value of water for quality concrete shall not be less than:

[RPSC LECTURE - 2011]

Or

The pH value of water for construction should not be less than

(a) 2 (b) 4

(c) 6 (d) 12

RPSC GWD A. En. – 2014

Ans : (c) Water with pH range 6 to 8.5 is used in concrete works. Limit of impurities in water used for concreting–

Organic	200 mg/ltr
Inorganic	3000 mg/ltr
Sulphates	400 mg/ltr
chlorides	R.C.C. – 500 mg/ltr P.C.C. – 2000 mg/ltr
Suspended solids	2000 mg/ltr

273. As per IS 456 : 2000 the limit of suspended matter in water to be used for construction is

(a) 200mg/*l* (b) 300mg/*l*

(c) 2000mg/*l* (d) 3000mg/*l*

[UPPSC State Eng. A.E. 2007(I)]

[UPPSC State Eng. A.E. 2003]

Ans : (c) IS 456 : 2000 the limit of suspended matter in water to be used for construction is 2000 mg/*l*.

Permissible limit for solids

Constituents	**Permissible limit (max.)**
1. organic	200 mg/*l*
2. inorganic	3000 mg/*l*
3. sulphate	400 mg/*l*
4. chloride	2000 mg/*l* (without steel) 500 mg/*l* (with steel)
5. suspended matter	2000 mg/*l*

274. The main element for the initial hardening of cement paste is:

(a) Tetra calcium aluminate

(b) Dicalcium silicate

(c) calcium oxide

(d) Tricalcium silicate

[UPPSC State Eng. A.E. 2007(I)]

Ans : (d) The main element for the initial hardening of cement paste is Tricalcium silicate (C_3S).

Tricalcium silicate :

- Higher C_3S content increases rate of strength gain.
- Higher C_3S increase heat evolution.
- Rapid hardening cement have high specific surface area.

⇒ Tetracalcium Alumino ferrite (C_4AF) :

- Poorest cementing value.

⇒ Tricalcium Aluminate (C_3A) :

- It rapidly reacts with water and responsible for flash set.
- Gypsum is added to regulate rapidity of reaction.
- Mainly responsible for this the initial setting.
- Excess of it produces high heat of hydration and has greater tendency to volume change causing cracking
- It weaken resistance to sulphate attack.

⇒ Dicalcium silicate (C_2S)

- It hydrates slowly and takes long time to add to the strength (after a year or more).
- It is responsible for ultimate strength.
- It imparts resistance to chemical attack.

275. As per IS 456-2000, the permissible limit for sulphates in water to be used for making concrete is :

(a) 10 mg/lit (b) 200 mg/lit

(c) 400 mg/lit (d) 500 mg/lit

GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) : Permissible limit of following in water IS 456-2000

i). Sulphate – 400 mg/*l*

ii). Chlorine– 500 mg/*l* (without steel)

iii). Total dissolve solid – 2000 mg/*l*

276. What is the nominal cover to rebar in reinforced cement concrete exposed to very severe conditions?

(a) 75 mm (b) 40 mm

(c) 60 mm (d) 50 mm

CIL MT 2017

Ans. (d) :

Exposure category	**Minimum cover (mm)**
mild	20
moderate	30
sever	45
very sever	50
extreme	75

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2. Fundamentals of Working Stress Method

277. In under-reinforced section

(a) the depth of the actual neutral axis is less than the balanced neutral axis depth
(b) the depth of the actual neutral axis is more than the balanced neutral axis depth
(c) the concrete reaches maximum permissible stress first
(d) both concrete and steel reach maximum permissible stresses simultaneously

WBPSC AE (Common) 1998

Ans. (a) : In under-reinforced section the depth of the actual neutral axis is less than the balanced neutral axis depth.

278. The vertical deflection limits may generally be assumed to be satisfied provided that the span to depth ratios are not greater than.....for simply supported beam for span upto 10 m.

(a) 7 (b) 10
(c) 20 (d) 26

GUJRAT PSC AE - II 2017

Ans : (c)

	Type of Beam/slab	Span to depth ratio
1.	Simply supported	20
2.	Continuous	26/40
3.	Cantilever	7/12
4.	One way slab	30
5.	Two way slab	35

279. Design load is equal to

(a) Characteristic load × partial factor of safety for materials (r_m)
(b) Characteristic load × partial factor of safety for load (r_f)
(c) $\dfrac{\text{Characteristic load}}{r_m}$
(d) $\dfrac{r_f}{\text{Characteristic load}}$

TNPSC AE 2019

Ans. (b) In design load we design for (F.O.S.) to characteristic load so as maximum loading condition could be satisfy to.

280. In a simply supported 9 m long beam, the effective depth should be a minimum of (approximately) d mm is required for assuming the vertical deflection to be satisfactory

(a) 1286 (b) 346
(c) 450 (d) 600
e) 360

CGPSC AE 2014 Shift–I

Ans. (c) : As per IS 456 : 2000
The vertical deflection limit may generally be satisfied if :
(a) Basic span to effective depth ratio for span up to 10 m is
for cantilever $\rightarrow 7$
for S.S. beam $\rightarrow 20$

$$\therefore \frac{\text{span}}{\text{effective depth}\,(\text{mm})} = 20$$

$$\Rightarrow \frac{9\times 1000\,(\text{mm})}{\text{depth}} = 20$$

$$\Rightarrow \text{effective depth (depth)} = \frac{9000}{20} = 450 \text{ mm}$$

281. In the working stress method, the modular ratio, m has a value of 280/(3×permissible compressive stress). This expression for m

(a) partially takes into a account the long-term effects such as creep
(b) fully takes into account the long-term effects such as creep
(c) does not take into account the long-term effects such as creep
(d) is the same as the modular ratio based on the value of the elastic modulus of the structural member
(e) takes into account twice the long-term effect like creep

Ans. (a) : Modular Ratio $(m) = \dfrac{280}{3\sigma_{cbc}}$
the creep & any other long term effect going on continuously deforming concrete structure, the modular ratio partially taken into account long term effect of shrinkage & creep.
$\therefore$ We use $m = \dfrac{280}{3\sigma_{cbc}}$

282. The application of elastic theory to the beams is based on the assumption at

(a) any cross-section, plane sections before bending remain plane after bending
(b) all tensile stresses are taken up by reinforcement alone and none by the concrete
(c) steel reinforcement is free from initial stresses when it is embedded in concrete
(d) all of these

TNPSC AE 2012

Ans. (d) The elastic theory of bending or simply straight line theory forms the basis of working stress method of design.

- The section which is plane before bending remains plane after bending.
- The concrete and steel reinforcement are perfectly bonded.
- The stress - strain relationship of steel and concrete under working loads is a straight line.

283. A beam of 400 mm effective depth with a neutral axis constant of 0.39, the value of lever arm is :

(a) 250 mm (b) 300 mm
(c) 348 mm (d) 358 mm

TNPSC AE 2015

Ans. (c) Lever arm $= d - \dfrac{x}{3}$

$x = 0.39 \times 400$

$x = 156$ mm

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Lever arm $= 400 - \frac{156}{3} = 400 - 52$

Lever arm = 348 mm

284. The moment of resistance offered by balanced section in W.S.M. is given by (As per IS 456-2000)

(a) $\frac{1}{2}.\sigma_{cbc}.b.x^2$ (b) $\frac{1}{2}.\sigma_{cbc}.b.d^2$

(c) $\frac{1}{2}.\sigma_{cbc}.b.d\left(1-\frac{x}{3}\right)$ (d) $\frac{1}{2}.\sigma_{cbc}.b.x.\left(d-\frac{x}{3}\right)$

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d) The resistance moment for balanced section is $M_r = \frac{1}{2}.\sigma_{cbc}.b.x.\left(d-\frac{x}{3}\right)$.

The total compressive force acts at a depth equal to the one third of the depth of critical axis taken from top extreme fiber.

285. Which of the following is primary cause of flexural collapse in over-reinforced beams

(a) Primary tension failure
(b) Failure of bond
(c) Primary compression failure
(d) Relative movement between steel and concrete

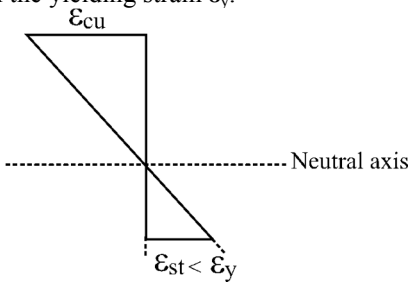
CIL MT 2017

Ans. (c) : Primary compression failure is a primary cause of flexural collapse in over-reinforced beams

Primary compression failure : Failure of concrete without yielding of steel.

Secondary compression failure : Failure of concrete after yielding of steel.

An 'over reinforced section in one in which the area of tension steel is such that at the ultimate limit state, the ultimate compressive strain in concrete is reached, however the tensile strain in the reinforcing steel is less than the yielding strain ε_y.



Thus, the concrete fails in compression before the steel reaches its yield point and this type of failure is called primary compression failure.

286. According to the basic consideration of Working Stress method for reinforced concrete design, the stress-strain relationship of steel and concrete under working loads is taken as :

(a) parabolic (b) quadratic
(c) logarithmic (d) linear

CIL MT 2017

Ans. (d) According to the basic considerations of working stress method of for reinforced concrete, design the stress-strain, relationship of steel and concrete under working loads is taken as linear.

287. The end of side covers for a steel bar in RCC work may be taken as :.....

(a) 1.2 cm to 2 cm (b) 2.5 cm to 5 cm
(c) 4 cm to 5 cm (d) 3 cm to 7 cm

APPSC Poly. Tech. Lect. 2020

Ans. (c) : The end of side covers for a steel bar in RCC work may be taken as 4 cm to 5 cm.

288. The permissible concrete stress in tension due to bending of M20 in calculation relating to resistance to cracking as per IS 3370-1965 part II is:

(a) 20kg/cm^2 (b) 17 kg/cm^2
(c) 14 kg/cm^2 (d) 10kg/cm^2

KPSC AE 2020

Ans. (a) : $f_{cr} = 0.7\sqrt{f_{ck}}$

Split tensile strength $= 0.66\ f_{cr}$

$= 0.66 \times 0.7\sqrt{20}$

$= 2.066\ N/mm^2 = 20\ kg/cm^2$

289. A _____ section is one in which the area of tension steel is such that the ultimate limit state, the ultimate compressive strain in concrete is reached, However the tensile strain in the reinforcing steel is less than the yield strain:

(a) balanced (b) under reinforced
(c) over reinforced (d) composite

KPSC AE 2020

Ans. (c) : A over reinforced section is one in which the area of tension steel is such that the ultimate limit state, the ultimate compression strain in concrete is reached however the tensile strain in the reinforcing steel is less than the yield strain.

Balanced section : It is a section in which both steel and concrete reaches their maximum permissible value simultaneously.

Under reinforced section ;

When the section is under-reinforced, tensile steel reaches the maximum permissible value prior to concrete. In under reinforced section, failure is tensile failure or ductile failure

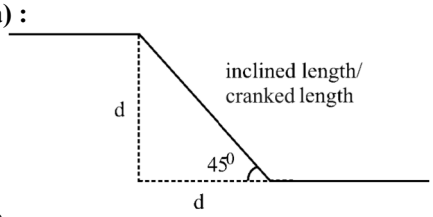
Over reinforced section : Concrete reaches its maximum permissible value prior of steel failure is a compression failure/brittle failure.

290. For 45^0 cranked or bent up bar of dia ''d'', the additional length of 2 bent up is approximately :

(a) 0.9 d (b) 0.45 d
(c) 0.3 d (d) 0.8 d

KPSC AE 2020

Ans. (a) :



For 45^0 cranked or bent up bar the additional length for one bent up is = difference in length of hypotenuse and base

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$\therefore$ Additional length $= \dfrac{d}{\sin 45^0} - d$

$= \dfrac{d}{0.707} - d$

$= 1.42\,d - d$

$= 0.42\,d = 0.45d$ (approx.)

Now, for 2 bent up additional length is equal to $2 \times 0.45\,d = 0.9\,d$

So, $\boxed{\text{Additional length} = 0.9d}$

291. The neutral axis of the reinforced beam passes through

(a) C.G. of the concrete section
(b) Metacentre of the concrete section
(c) Centroid of the transformed concrete section
(d) Centroid of the concrete section

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : The neutral Axis of the Reinforced beam passes through the centroid of the concrete section.

292. In RCC work in the adhesion of the concrete to steel is not perfect within elastic limits, then :

(a) Steel will be overstressed
(b) Steel reinforcement may break
(c) Concrete will be overstressed
(d) Steel will bear more stress than concrete

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : In RCC work if the Adhesion of concrete to steel is not perfect within elastic Limits, then concrete will be overstressed, because we know, concrete is weak in tension, so reinforcement is provided to provide strength to concrete in tension side.

293. Equivalent area of a reinforced cement concrete section is :

(a) $m\,A_c + A_{sc}$ (b) $A_c + m\,A_{sc}$
(c) $A_c + A_{sc}$ (d) $(A_c + m\,A_{sc})\sigma_c$

(UKPSC AE (Paper-I) 2007/ Uttarakhand AE 2016)

Ans : (b) Equivalent area $\boxed{A = A_c + m.A_{sc}}$

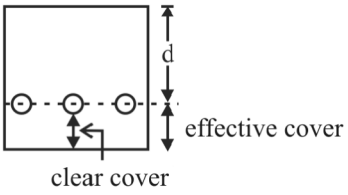
A_c = area of concrete in compression
A_{sc} = Area of steel in compression
m = modular ratio

294. Effective cover of reinforcement is :

(a) Total depth – 1.5 × bar diameter
(b) Total depth – 2 × bar diameter
(c) Clear cover + bar diameter
(d) Clear cover $+\dfrac{1}{2}\times$ bar diameter

(UKPSC AE (Paper-I) 2007)

Ans : (d)



effective cover = clear cover $+\dfrac{1}{2}\times$ dia of bar

295. A RCC beam of 200 mm width and 300 mm effective depth is reinforced with Fe 415 grade steel. The grade of concrete used is M20. The limiting moment carrying capacity of this beam is

(a) 47.88 kNm (b) 53.28 kNm
(c) 49.68 kNm (d) None of the above

(UKPSC AE 2012 Paper–I)

Ans : (c)

Here $\sigma_{ck} = 20\ N/mm^2$
$b = 200$ mm
$d = 300$ mm

Moment of resistance $M = 0.138\,\sigma_{ck}\,bd^2$
$= 0.138 \times 20 \times 200 \times 300^2$
$\boxed{M = 49.680\ kNm}$

296. In a under–reinforced section the value of actual neutral axis x_a and the critical neutral axis x_c are related as :

(a) $x_a = x_c$
(b) $x_a > x_c$
(c) $x_a > 2x_c$
(d) none of the above

(UKPSC AE (Paper-I) 2007)

Ans : (d) In under reinforced section neutral axis depth is less than critical neutral axis.

$\boxed{x_a < x_c}$

297. Economical percentage of steel in RCC section is :

(a) $\dfrac{50x^2}{md(d-x)}$ (b) $\dfrac{50x}{md(d-x)}$

(c) $\dfrac{50x^2}{2md(d-x)}$ (d) $\dfrac{100x^2}{2md(d-x)}$

Where symbols have their usual meanings.

(UKPSC AE (Paper-I) 2007)

Ans : (a) Economical percentage of steel in cement concrete is given by

$$\boxed{p = \frac{50x^2}{md(d-x)}}$$

Maximum percent of steel is 4% i.e. 0.04 bD (beam).

298. Moment of resistance for a under – reinforced beam section is :

(a) $M_r = \sigma_{st} A_{st}\left(d - \dfrac{n}{3}\right)$

(b) $M_r = \sigma_{st} A_{st}\left(d + \dfrac{n}{3}\right)$

(c) $M_r = b\,n\dfrac{\sigma_{cb}}{2}\left(d - \dfrac{n}{3}\right)$

(d) $M_r = b\,n\dfrac{\sigma_{st}}{2}\left(d - \dfrac{n}{3}\right)$

UKPSC AE (Paper-I) 2007

Ans : (a) An under reinforced beam will fail in tension failure of steel, therefore, moment of resistance

$$\boxed{M_r = \sigma_{st}.A_{st}\left(d - \frac{n}{3}\right)}$$

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299. Minimum tension steel in RCC beam needs to be provided to :
(a) control excessive deflection
(b) control surface cracks
(c) prevent sudden failure
(d) none of the above

(UKPSC AE (Paper-I) 2007)

Ans : (c) To avoid sudden failure collapse we use steel in tension. Minimum amount of steel is taken 0.85 bd/f_y.

300. Reinforced concrete beam sections in which the steel reaches yield strain at loads lower than the load at which the concrete reaches failure strain are called
(a) Under-reinforced sections
(b) Balanced sections
(c) Over-reinforced sections
(d) Prestressed sections

(UP RVNL AE 2016)

Ans.: (a) In under reinforced section, first of all steel in tension reaches its yield strength and fails.
In over-reinforced section, concrete reaches its maximum compressive strength and fails.
In balanced section both steel and concrete fail simultaneously.

301. The working stress method of design of RC members is known as–
(a) Non–deterministic approach
(b) Deterministic approach
(c) Probabilistic approach
(d) None of the above

UKPSC AE 2013 Paper–I

Ans : (b) Working stress method is deterministic approach to design an RC member. The basis of method is that the permissible stress for concrete and steel are not exceeded any where in the structure when it is subjected to worst combination of working loads.

302. IS 456 : 1978, recommends to provide certain minimum steel in a RCC beam
(a) To ensure compression failure
(b) To avoid rupture of steel in case a flexural failure occurs
(c) To hold the stirrups steel in position
(d) To provide enough ductility to the beam

RPSC POLY. TECH. LECT. 2014

Ans : (d) To provide enough ductility to the beam, IS : 456 : 1978 recommends minimum percentage of steel as

$$A_o = \frac{0.85bd}{f_y}$$

303. When quantity of steel in an RC beam section is less than required, the section is called:
(a) over reinforced
(b) balanced
(c) under reinforced
(d) critical

[UPPSC State Eng. A.E. 2007(I)]

Ans : (c) Under - reinforced section :
When the section is under reinforced
(a) Tension steel reaches the maximum permissible value prior to concrete.
(b) $X < X_{cr}$ where
X = Actual depth of neutral axis
X_{cr} = Depth of neutral axis corresponding to balance section.
(c) Area of tension steel is less than that into balanced section ($A_{st} < A_{st\ bal}$).
(d) In under reinforced section, failure is tensile failure or ductile failure.

Over reinforced section :
(a) concrete reaches its maximum permissible value prior of steel.
(b) $X > X_{cr}$
(c) $A_{st} > A_{st,\ bal}$
(d) Failure is a compression failure/brittle failure
(e) IS code does not recommend over-reinforced section.

Balanced section :
It is a section in which both steel and concrete reaches their maximum permissible value simultaneously, i.e. when $f_{st} = \sigma_{st}$ (permissible tensile stress in steel), $f_{cbc} = \sigma_{cbc}$ (permissible compression stress in steel).

304. An under reinforced section means:
(a) reinforcing steel reaches its yield stress first
(b) concrete reaches its maximum stress first
(c) reinforcement provided is equal to maximum reinforcement
(d) None of the above

UPPSC AE 12.04.2016 Paper–I

Ans : (a) In an under reinforced section, the stress in steel first reaches it permissible value, while the concrete is under stressed .
The actual neutral axis will shift upwards i.e, $n_c > n$.

305. In a RCC beam compression failure occurs in a/an
(a) balance section
(b) over reinforced section
(c) under reinforced section
(d) none of the above

BPSC AE 2012 Paper–V

Ans : (b) In an over reinforced section failure occurs in compression fibre. In these section the reinforcement is more than required therefore when the load exceeds the capacity of section then concrete in compression starts failing first.

306. In a critical section of a beam
(a) The full strength of steel in tension is not being utilized
(b) The full strength of concrete in compression is not being utilized
(c) The stresses developed in steel and concrete will simultaneously be the same as assumed in design
(d) The concrete will reach its maximum allowable stress earlier than steel

RPSC AE 2013

Ans : (c) In a critical section of beam, the ratio of steel to concrete in a beam in such a way that the maximum strain in the two material reach simultaneously, a sudden failure would occur with less alarming deflection. Such a beam is referred as balanced reinforced beam.

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307. For M-30 grade concrete, the 28 days characteristic compressive strength will be

(a) 30 kN/m^2 (b) 30N/m^2

(c) 30N/cm^2 (d) 30 N/mm^2

(JPSC Combined AE Paper V 2013)

Ans : (d) For M-30 grade characteristic compressive strength for 28 days is 30 MPa or 30 N/mm^2.

308. The total length of tie bar of 1 cm diameter embedded in a cement concrete pavement with allowable working stress in steel in tension equal to 1400 kg/cm^2 and allowable bond stress in deformed bars in concrete 24.6 kg/cm^2 is-

(a) 18.87 cm (b) 113.82 cm

(c) 56.9 cm (d) 28.45 cm

[MPSC 2015 Paper-II Mains]

Ans. (d) : For developed in bar in concrete = Applied pull $\left(A_{st} \times \sigma_{st}\right) = \left(\pi \phi L \tau\right)$

$$\frac{\pi}{4}(1)^2 \times 1400 kg/cm^2 = \pi \times 1cm \times L \times 24.6 kg/cm^2$$

$$L = \frac{1400 \times \frac{\pi}{4}(1)^2}{\pi \times 24.6 \times 1} = 14.227\ cm$$

Total length required = $2 \times L$

$= 2 \times 14.227 = 28.45$ cm

309. Maximum reinforcement provided in a beam section shall not exceed_____.

(a) $\frac{0.85}{f_y}(bd)$ (b) $\frac{0.87}{f_y}(bd)$

(c) 0.04 (bd) (d) 0.04 (bD)

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : According to clause 26.5.1.1 of IS 456 : 2000, the maximum area of compression reinforcement shall not exceed 0.04 bD.

310. If a rectangular under-reinforced section is subjected to bending moment equal to its moment carrying capacity and the stress in steel and extreme compression fiber of concrete at this moment are σ_s and σ_c respectively then which of the following is correct?

(a) $\sigma_c = f_c$ (b) $\sigma_s = f_y$

(c) $\sigma_s = 0.5 f_y$ (d) $\sigma_s = f_y$ and $\sigma_c = f_c$

Where f_y and f_c are the permissible stresses in steel and concrete respectively.

[HPPSC 2016-II]

Ans. (b) : In an under reinforced section the failure of member is due to yielding of reinforcement i.e. steel bars in tension reaches their yield strength.

$\sigma_s = f_y$

311. In a simply supported rectangular beam loaded transversely, the maximum tensile bending stress occurs at :

(a) top fiber

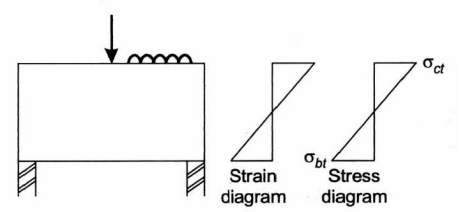
(b) bottom fiber

(c) neutral axis

(d) between top fiber and neutral axis

[HPPSC 2016-II]

Ans. (b) :



Maximum bending moment will be at mid-section stress due to BM $= \frac{M_y}{P}$

At bottom tensile and at top maximum compressive stress will occur.

312. Flexural collapse in over-reinforced beams is due to :

(a) primary compression failure

(b) secondary compression failure

(c) primary tension failure

(d) bond failure

Karnataka PSC AE 2015 Paper - II

(UKPSC AE 2012 Paper–I)

Ans. (a) :

(i) For over reinforced beams failure is due to crushing of concrete – primary compression failure.

(ii) For under reinforced beams, failure is due to yielding of steel– secondary compression failure.

313. A rectangular beam section of 300 mm width and 500 mm effective depth has tensile reinforcement of 1346 mm^2. The beam is constructed using M20 grade concrete and Fe 415 steel. Calculate the neutral axis.

(a) 225 mm (b) 2.22 mm

(c) 135 mm (d) 240 mm

Karnataka PSC AE 2017 Paper - II

Ans. (a) : Equalizing tensile and compressive forces $0.36\ f_{ck}\ b\ x_u = 0.87\ f_y\ A_{st}$

$$x_u = \frac{0.87 \times 415 \times 1346}{0.36 \times 20 \times 300} = 225 mm$$

314. A reinforced cement concrete beam section has a size of 300 mm width and 400 mm effective depth, the grade of concrete used is M20 and grade of steel is Fe 415. If the ultimate bending moment of 150 kN-m acts at the section, the beam has to be designed as :

(a) Singly reinforced beam

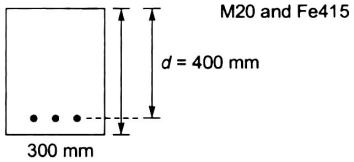
(b) Doubly reinforced beam

(c) Beam cannot be designed

(d) Balanced beam

Karnataka PSC AE 2017 Paper - II

Ans. (b) :



Limiting MOR for singly reinforced beam

$= 0.138\ f_{ck}\ bd^2$

$= 0.138 \times 20 \times 300 \times (400)^2$

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$= 132480000\,N - mm$

$M_{u,lim} = 132.48 kNm$

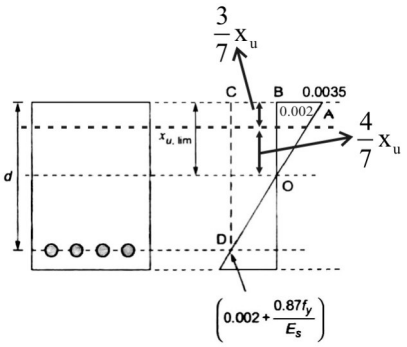
Now $M_{u,lim} < M_{design}$, hence we will have to design it as doubly reinforced beam.

315. As per the stress block defined is IS 456-2000, the limiting depth of neutral axis in a flexural member having effective depth 'd' reinforced with Fe 550 grade steel in tension side is :

(a) 0.48 d (b) 0.53 d
(c) 0.42 d (d) 0.44 d

Karnataka PSC AE 2017 Paper - II

Ans. (d) : For calculation of limiting depth of neutral axis:



Now, from similarity of triangle OAB and ACD

$$\frac{BA}{OB} = \frac{AC}{DC}$$

$$\frac{0.0035}{x_{u,lim}} = \frac{0.0055 + \frac{0.87f_y}{E_s}}{d}$$

$$\therefore x_{u,lim} = \frac{0.0035}{0.0055 + \frac{0.87f_y}{E_s}} d$$

For Fe550 grade, $f_y = 550\ N/mm^2$

$E_s = 2 \times 10^5\ N/mm^2$

$$x_{u,lim} = \frac{0.0035}{0.0055 + \frac{0.87 \times 550}{2 \times 10^5}} d$$

$$x_{u,lim} = 0.44d$$

316. Creep coefficient at the age of 7-days loading is:

(a) 2.2 (b) 1.6
(c) 1.1 (d) 0.9

Rajasthan JEN (Degree) Shift - II 2016

Ans. (a) Creep coefficient

Age of loading	Creep coefficient (ϕ)
7 days	2.2
28 days	1.6
1 year	1.1

317. Assertion A : According to IS : 456; over reinforced sections are not permitted.
Reason R : There is ductile failure of over reinforced section.
Select your answer based on the coding system given below -

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

RPSC AE 2018

Ans. (c)
- In an over reinforced section the percentage of steel provided is greater than the balanced section.
  In this sudden failure occur
- In an under reinforced section, the percentage of steel provided is less than that provided in balanced section.
  In this ductile failure occur.

318. If the shape factor of a section is 1.5 and the factor of safety to be adopted in 2, then the load factor will be :

(a) 0.75 (b) 1.5
(c) 2.5 (d) 3.0

GPSC AE Class (1&2) Paper - 2 2017

Ans. (d) : Load factor = factor of safety × shape factor

$(\lambda) = f_s \times \alpha$

$\lambda = 2 \times 1.5$

$\lambda = 3.0$

319. As per ductility criterion, which of the following section is preferred?

(a) Over reinforced section
(b) Under reinforced section
(c) Balanced section
(d) Doubly reinforced section

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : Under reinforced section in which steel first goes to its last limit (yield) that than of concrete, which give time to safety for instant collapse as in brittle material

320. The factor of safety for flexural compression in case of working stress method is

(a) 1.5 (b) 2
(c) 3 (d) 3.5

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) :
In WSM

Concrete $= \frac{f_{ck}}{fos}$ fos = 3

Steel $= \frac{f_y}{fos}$ fos = 1.8

In LSM
Steel = 1.15
Concrete = 1.5

321. The temperature at which ductility test is conducted is

(a) 25ºC (b) 50ºC
(c) 20ºC (d) 27ºC

TNPSC AE 2018

Ans. (d) : The ductility test is conducted at 27ºC.

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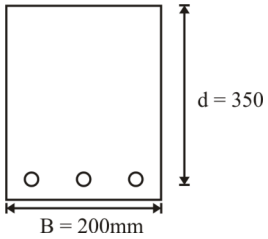
322. A rectangular beam 200 mm wide has an effective depth of 350 mm. It is subjected to a bending moment of 24,000 Nm. The permissible stresses are c = 5 N/mm², t = 140 N/mm²; and m is 18. The required area of tensile reinforcement will be

(a) 688 mm² (b) 778 mm²
(c) 864 mm² (d) 954 mm²

ESE 2019

Ans. (b)



Bending moment = 24 kNm
c = 5N/mm²
t = 140N/mm² (steel)
modular ratio 'm' = 18
Design Constant

$$K = \frac{mc}{t+mc} = \frac{18\times5}{140+18\times5} = \frac{90}{230}$$

$$[K = 0.39 \simeq 0.4]$$

$$J = 1-\frac{K}{3} = 1-\frac{0.4}{3} = 0.87$$

$$Q = \frac{1}{2}CJK = \frac{1}{2}\times5\times0.87\times0.4$$

[Q = 0.87]
Moment of Resistance of Balance section.
$M_r = Qbd^2$
$M_r = 0.87\times200\times(350)^2$
$M_r = 21.315\times10^6$ N–mm
[Mr = 21.315 kN–m]
[Bending Moment > Moment of Resistance]
So, Over reinforced section is required
Actual Depth of Neutral Axis (X_a)

$$\text{Bending Moment} = \frac{1}{2}cb\,X_a\left(d-\frac{x_a}{3}\right)$$

$$24\times10^6 = \frac{1}{2}\times5\times200X_a\left(300-\frac{X_a}{3}\right)$$

$$350X_a - \frac{X_a^2}{3} = 48000$$

$$\frac{X_a^2}{3} - 350X_a + 48000 = 0$$

$$[x_a = 162.2\text{mm}]$$

$$\frac{C}{X_a} = \frac{\frac{T_a}{m}}{d-x_a}$$

$$\frac{5}{162.19} = \frac{\frac{T_a}{18}}{350-162.20}$$

$[T_a = 104.2\text{N/mm}^2]$
Area of tension steel required

$$C = T$$

$$\frac{1}{2}Cbx_a = T_aA_{st}$$

$$104.2A_{st} = \frac{1}{2}\times5\times200\times162.19$$

$$104.2\,A_{st} = 81095$$

$$\boxed{A_{st} = 778.26\text{mm}^2}$$

323. The Permissible bending compressive strength for M 25 grade of concrete is 8.5 N/mm². Its short-term and long-term modular ratios are, nearly

(a) 8 and 11 (b) 8 and 8
(c) 11 and 11 (d) 11 and 6

ESE 2018

Ans. (a)

- Short term modular ratio = $\frac{E_s}{E_c} = \frac{2\times10^5}{5000\sqrt{25}} = 8$
- Long term modular ratio = $\frac{280}{3\sigma_{abc}} = \frac{280}{3\times8.5}$

$$[\text{Long term} = 10.98 \simeq 11]$$

324. What is the ratio of flexural strength (f_{cr}) to the characteristic compressive strength of concrete (f_{ck}) of M25 grade concrete?

(a) 0.08 (b) 0.11
(c) 0.14 (d) 0.17

ISRO Scientist/Engineer 2020

Ans. (c) : $f_{cr} = 0.7\sqrt{f_{ck}}$

$$\frac{f_{cr}}{f_{ck}} = \frac{0.7\sqrt{25}}{25} = 0.14$$

325. The characteristic load means the value of the load:

(a) Below which not more than 5% of the results are expected to fall
(b) Which has a 95% probability of not being exceeded during the life of the structure
(c) Which has been factored with partial safety factor
(d) Which has a probability of being exceeded during the life of the structure

ISRO Scientist/Engineer 2020

Ans. (b) : Characteristic load or ultimate load is that value of load which has a 95 percent probability of not being exceeded during the life time of the structure.
In absence of any data, loads given in various standards shall be assumed as characteristic loads. The following standards may be used for this purpose.
IS 875 (Part 1) – For dead loads
IS 875 (Part 2) – For imposed loads
IS 875 (Part 3) – For wind loads
IS 875 (Part 4) – For snow loads
IS 1893 (Part 1) – For earthquake loads

326. Minimum spacing between horizontal parallel reinforcement bars of different diameters should not be less than:

(a) One diameter of thinner bar

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(b) One diameter of a thicker bar
(c) Sum of the diameters of the thinner and thicker bars
(d) Twice the diameter of the thinner bar

ISRO Scientist/Engineer 2020

Ans. (b) : The horizontal distance between two parallel main reinforcing bars shall usually be not less than the greatest of the following :
- The diameter of the bar if the diameters are equal.
- The diameter of the larger bar if the diameter are unequal.
- 5 mm more than the nominal maximum size of coarse aggregate.

327. The section in which concrete is not fully stressed to its permissible value when stress in steel reaches its maximum value, is called
(a) Under reinforced section
(b) Over reinforced section
(c) Critical section
(d) Balanced section

ISRO Scientist/Engineer 2017

Ans : (a) : Reinforced concrete beam sections in which the steel reaches yield strain at loads lower than the load at which the concrete reaches failure strain are called under reinforced section.

328. If ϕ is the nominal diameter of reinforcing bar, f_x is compressive stress in the bar and f_{bd} is design bond stress of concrete, the anchorage length L_a of straight bar in
(a) $L_a = \frac{\phi f_x}{f_{bd}}$ (b) $L_a = \frac{\phi f_x}{2f_{bd}}$
(c) $L_a = \frac{\phi f_x}{\pi f_{bd}}$ (d) $L_a = \frac{\phi f_x}{4f_{bd}}$

ISRO Scientist/Engineer 2014

Ans : (d) The required length, necessary to develop full resisting force, is called anchorage length in case of axial tension (or compression) and development length in the case of flexural tension and is designated by symbol L_a.
- Force developed in bar in concrete
= Applied pull surface area of bar $\times L_a \times f_{bd} = T$

$\pi \phi L_a . f_{bd} = \frac{\pi}{4}(\phi)^2 f_x$

$\left[La = \frac{\phi f_x}{4f_{bd}} \right]$

329. The maximum percent of moment redistribution allowed in RCC beams is
(a) 10% (b) 20%
(c) 30% (d) 40%

APTRANSCO 2017

Ans : (c) : IS-456:2000 the ultimate moment of resistance at any section should not be less than 70% of the factored moment at that section as detained from the elastic moment envelope (considering all loading combinations)

330. The load carrying capacity of a column designed by working stress method is 500 kN. The ultimate collapse load of the column is
(a) 500 kN (b) 662.5 kN
(c) 750 kN (d) 1100 kN

APTRANSCO 2017

Ans : (c) : Given
Capacity of Column Load (P) = 500 kN
The ultimate collapse load of the column (P_σ) = 1.5 × P
$P_\sigma = 1.5 \times 500 = 750$ kN

331. According to IS : 456-1978, the modulus of elasticity of concrete E_c (in N/mm^2) can be taken as :
(a) $E_c = 5000\sqrt{f_{ck}}$ (b) $E_c = 570\sqrt{f_{ck}}$
(c) $E_c = 5700\sqrt{f_{ck}}$ (d) $E_c = 700\sqrt{f_{ck}}$

OPSC AEE 2019 Paper–I

Ans. (c) : As per IS : 456–1978, the modulus of elasticity of concrete is $E_c = 5700\sqrt{f_{ck}}$

332. The effect of creep on modular ratio is :
(a) To decrease it
(b) To increase it
(c) Either to decrease or to increase it
(d) To keep it unchanged

OPSC AEE 2019 Paper–I

Ans. (b) : The effect of creep on modular ratio is to increase it.

333. As per IS : 456 for a double reinforced section, the equivalent area of concrete is given by :
(a) $\frac{mA_{sc}}{2}$ (b) $\frac{mA_{sc}}{4}$
(c) $\frac{3mA_{sc}}{4}$ (d) $\frac{5}{2mA_{sc}}$

PSPCL AE 2012

Ans. (a) : As per IS : 456 for a double reinforced section, the equivalent area of concrete is given by $0.5mA_{sc}$

334. The moment of resistance of a rectangular section depend upon
(a) ultimate strain in concrete
(b) area of high tensile tendons
(c) tensile strength in concrete
(d) compressive stress in concrete

Maharashtra PSC Mains 2018 Paper-1

Ans. (b) :

335. In working stress design, if the modular ratio is 'm', steel ratio is 'r', the critical neutral axis constant 'k' is given by
(a) $\frac{m}{m-r}$ (b) $\frac{m}{m+r}$
(c) $\frac{m+r}{m}$ (d) $\frac{m-r}{m}$

GPSC AE March 2018

Ans. (b) : Depth of critical neutral axis = kd
where, k = critical neutral axis constant $\boxed{k = \frac{m}{m+r}}$

$m = \frac{280}{3\sigma_{cbc}}$ (Long term modular ratio)

r : steel ratio.

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336. The tensile strength of concrete to be used in the design of reinforced concrete member is

(a) $0.2\ f_{ck}$ (b) $0.1\ f_{ck}$

(c) $0.7\sqrt{f_{ck}}$ (d) zero

GPSC AE March 2018

Ans. (c) : Tensile strength of concrete = $0.7\sqrt{f_{ck}}$

(By split cylinder test).

Characteristic strength (f_{ck}) of concrete is the value of strength of concrete below which not more than 5% of test results are expected to fall.

337. As per IS-456 2000 for a singly reinforced rectangular section which one of the following is correct?

(a) $\left(x_{u_{max}}/d\right)$ for Fe-415 steel is 0.85

(b) $\left(x_{u_{max}}/d\right)$ for Fe-415 steel is 0.48

(c) the depth of the centroid of compression is $0.43\left(x_{u_{max}}\right)$

(d) the depth of the rectangular position of the stress block is $0.38\ x_{u_{max}}$

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (b) : $\left(x_{u_{max}}/d\right)$ for Fe-415 steel is 0.48

Steel grade	$\left(x_{u_{max}}/d\right)$
Fe 250	0.53
Fe 415	0.48
Fe 500	0.46

338. Limit state method of structural designing is practised as

(a) semiprobabilstic aaproach

(b) deterministic approach

(c) Probabilistic approach

(d) Both 1 and 3

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (a) : Limit state method of structural designing is practised as semiprobabilstic aaproach.

339. Widely used factor of safety is the ratio of

(a) Allowable load to Allowable stress

(b) Ultimate stress to Allowable stress

(c) Allowable stress to Ultimate load

(d) Ultimate stress to Ultimate load

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (b) : Widely used factor of safety is the ratio of Ultimate stress to Allowable stress.

$$\text{Factor of safety} = \frac{\text{ultimate stress}}{\text{working stress}}$$

340. When a steel bar is subjected to tensile load, after yield point and large strain, the material undergoes its atomic and crystalline structure. This is named as

(a) Strain hardener (b) Strain expansion

(c) Strain shrinkage (d) Strain smoothner

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (a) : When a metal is strained beyond the yield point. An increasing stress is required to produce additional plastic deformation and the metal apparently becomes stronger and more difficult to form.

341. Which material fails first in RCC over-reinforced beams?

(a) Concrete

(b) Steel

(c) Neither steel nor concrete fails

(d) Both steel and concrete fail simultaneously

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (a) : Over Reinforced section–Section in which, tension steel also reaches yield strain at loads higher then the load at which concrete reaches the failure strain in bending.

Under Reinforced section–Section in which tension steel also reaches yield strain at loads lower then the load at which concrete reaches the failure strain in bending.

342. In a reinforced concrete beam the stress in the concrete at the level of steel is measured as 6 N/mm^2. The modular ratio between steel and concrete is 18. What is the stress in steel?

(a) 118 N/mm^2 (b) 108 N/mm^2

(c) 180 N/mm^2 (d) 60 N/mm^2

TRB Poly. Tech. Lect. 2012

Ans. (b) : $f_c = \frac{f_s}{m}$ m = 18 (given)

f_c = 6 N/mm^2 (given)

$\Rightarrow f_s = f_c \times m = 18 \times 6$

$\boxed{f_s = 108\,\text{N/mm}^2}$

343. Working stress of mild steel is determined from the

(a) upper yield stress (b) ultimate stress

(c) fracture stress (d) lower yield stress

TRB Poly. Tech. Lect. 2012

Ans. (d) : As we know,

$$FOS = \frac{f_y(yield\ stress)}{f(working\ stress)}$$

And yield stress (f_y) is taken corresponding to lower yield point.

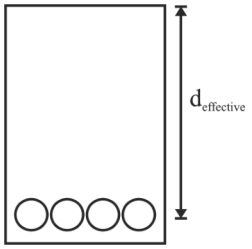
344. In reinforced concrete beam design, the distance between the centroid of the area of tension reinforcement and the maximum compressive fibre is known as

(a) Overall depth (b) Lever arm

(c) Effective depth (d) Depth of neutral axis

TSGENCO 2015

Ans. (c) : Effective depth is the distance between the centroid of the area of tension reinforcement and the maximum compressive fibre.



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